

**THE KINGDOM OF DARAVEK**  
c. 1300 - schematic map of the worked example (chapter 12)  
Shown: the capital, 3 cities, all 38 towns and 26 of the ~180 market towns; the ~7,700 villages appear only as texture.

**KEY**

**Settlements**  
★ Capital, 40,000  
● City (3), 10-12,000  
● Town (38), 2-8,000  
● Walled town  
● County town  
● Duchy earldom seat  
● Market town (26)  
● Villages (texture)

**Church**  
✠ Cathedral (13)  
✠ Archbishop  
✠ Abbey, priory  
✠ Military order  
● Great shrine

**Castles**  
■ Great castle (5)  
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■ Royal residence  
■ Beacon chain (10)

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⚙ Silver, lead mines  
⚙ Salt pans  
⚙ Quarry, lime  
⚙ Fulling mills  
⚙ Vineyards  
⚙ Shipyard  
⚙ Other industry  
Sept. Fair: months in label

**Routes**  
— Royal highway  
— Regional road  
— Old road, "Street"  
— Pilgrims' Way  
... Mule path  
... Sea lane  
✚ Bridge  
✚ Ford  
✚ Ferry  
✚ Pass  
⚓ Head, member port  
⚓ Lighthouse  
— Land and water  
— Navigable river  
— Head of navigation  
— Stream  
— Mountains  
— Hills, downs, wolds  
— Forest  
— Royal forest bound  
— Marsh, fen  
— Heath

**Borders**  
— Realm  
— Duchy, earldom  
— Church liberty  
— Frontier band  
— Debatable Land  
— Step 10 local map

**Other layers (grey)**  
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Based on real history, c. 1000–1500, with notes to 1650 · continent, kingdom and local maps

# HOW TO USE THIS GUIDE

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1. **Start with [Chapter 0: Step by Step](#).** It tells you what to draw, in order, from a blank continent down to a single valley. Each step links to the chapter that explains it.
2. **Keep [Chapter 13: Quick Reference](#) open while you draw.** It has every key number and the master legend of map symbols.
3. **Read [Chapters 1–11](#) when you want the reasons,** the real-world examples and the fine detail.
4. **See it done in [Chapter 12: Worked Example](#),** which builds a whole kingdom with the guide's numbers and shows the finished map.
5. **Check your map against [Chapter 11: Common Mistakes](#)** when you are done.

## THE FOUR KINDS OF NOTE BOX

### MAP TIP

What to draw, where, and which symbol to use.

### RULE OF THUMB

A short rule you can remember while drawing.

### LATER ERA (1500s+)

What changes after the Middle Ages.

### FANTASY TWIST

How magic, monsters or other races change the rule.

Distances are given in kilometres and miles. Populations are ranges, because historians' estimates differ. Blue text is a link: click it to jump to that chapter or section.

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## CHAPTER 0

# STEP BY STEP: FROM BLANK MAP TO LIVING WORLD

This chapter is the master method of the guide. It puts the other chapters in order: what to decide first, what to show at each scale, what to place next and how to test each result. It assumes your terrain is already drawn: coasts, mountains, rivers, forests, marshes and climate. Every historical number is copied from the chapter section linked in that step or table.

## IN THIS CHAPTER

Step 0: Choose the date, realm type and scale

What to draw at each scale

The order of work

Part A: Continent and world scale

Part B: Kingdom and region scale

Part C: Local area scale

Finishing: fantasy and the final check

Printable checklist

Quick summary

Sources and further reading

## Step 0: Choose the date, realm type and scale

## RULE OF THUMB

Make four decisions before you draw a single town: the date, the kind of realm, the scale and the amount of magic. Each one changes the numbers you use.

**Why:** The same kingdom looks very different in 1100 and in 1600. A strong kingdom with one capital looks different from a loose empire of princes. And the scale decides what

you can draw at all: on a kingdom map a whole village is smaller than a pen dot.

### Choose the date

Pick one year and keep to it. These four dates give four quite different maps. Every feature has a first date, so check [Mixing eras without reason](#) before you add anything new.

| Date           | People and towns                                                                                                                                                                                               | Castles and war                                                                                                                           | Church and learning                                                                                                                                         | Rulers and links                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>c. 1100</b> | Growing, but thin. England had about 1.7 million people in 1086, about a third of its 1300 peak. Fewer and smaller towns.                                                                                      | Mostly mottes (man-made earth mounds) and ringworks of earth and timber; a few great stone keeps (the White Tower, London, c. 1075–1100). | No friaries yet (they arrive in the 1210s–1220s). Universities are only beginning (Bologna, late 11th century). The Cistercian order is new (founded 1098). | Most western kings travel: draw 5–15 royal residences, not one capital. Watermills, but no windmills (first certain record 1185). |
| <b>c. 1300</b> | The crowded peak: England about 4.75 million. Most towns and markets have been founded (1,746 English markets recorded by 1300, many tiny).                                                                    | Stone castles, concentric castles and town walls.                                                                                         | Friaries in every big town; about 20 universities in Europe.                                                                                                | Fixed capitals in the strongest kingdoms (Paris, Westminster and London). Windmills common in flat, dry country.                  |
| <b>c. 1450</b> | A third to a half fewer people after the Black Death (1347–51) and later plagues; England's low point was about 1.9 million c. 1450. Deserted villages and more sheep pasture; many small markets are failing. | Many 12th-century castles half-ruined; tower houses in raided borders; by the mid-1400s cannon decide sieges.                             | Universities spreading (60–80 by 1500).                                                                                                                     | The first canal pound locks (Low Countries, northern Italy).                                                                      |
| <b>c. 1600</b> | Back near the 1300 level (England about 4.1 million). Capitals and ocean ports 2–4 times bigger than in 1500 (London about 200,000).                                                                           | Star forts at frontiers, capitals and main ports; interior castles become ruins or palaces.                                               | In Protestant lands most monasteries are closed (England 1536–1541): draw abbey ruins, and show each faith as a soft colour tint, not a new border line.    | Fixed capitals everywhere (Madrid from 1561); post roads with stations every 20–40 km (12–25 mi); ocean trade.                    |

All figures come from the guide's chapters: [Change over time](#), [When and why capitals became fixed](#) and [What Changes After 1500](#).

### Choose the realm type

Most real realms sit somewhere between the two columns below. France c. 1200, for example, had a royal core around Paris and six great fiefs that were nearly independent. Decide which end your realm is closer to.

|                      | Centralised kingdom                                                                                                      | Fragmented realm                                                                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Real examples</b> | England; France (more loosely) by about 1300                                                                             | The Holy Roman Empire (Germany); the city-states of northern Italy                                                                                                                         |
| <b>Capital</b>       | One fixed capital that holds 1–2% of the realm's people and is 3–7 times the size of the second city                     | No real capital, or a court in a middling town. Several rival cities of similar size. The largest city holds about 0.5% of the people or less (Cologne in Germany c. 1300: about 0.4–0.5%) |
| <b>Inner borders</b> | A regular network of counties, mostly 50–70 km (30–45 mi) across, each with a king's officer (a sheriff in England)      | A patchwork of princes, bishops and free cities, with enclaves (the Empire had about 400 imperial estates in 1521)                                                                         |
| <b>Castles</b>       | Fewer, mostly royal or great baronial: about 4 active castles per 1,000 km <sup>2</sup> (390 sq mi) in England and Wales | Many small lords' castles and toll castles: in hilly German-type lands about 25 castle sites per 1,000 km <sup>2</sup> , 8–12 of them active (estimate)                                    |
| <b>Mints</b>         | 3–4, controlled by the king                                                                                              | One in every princely seat, many bishops' cities and the big free towns                                                                                                                    |

**Read:** [Capital is not always the biggest city](#); [Kinds of states and how they look on a map](#); [How many mints?](#)

### Choose the scale and the amount of magic

| Decide       | Options                          | What it changes                                                                                                       |
|--------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Scale</b> | Continent, kingdom or local area | What you draw and how big it is on paper: see <a href="#">What to draw at each scale</a> , then Part A, B or C below. |
| <b>Magic</b> | None, a little or a lot          | Nothing at first. Use the real rules, then adjust in <a href="#">Step 17</a> .                                        |

If you draw only one kingdom, do Part A for that kingdom, then Parts B and C.

**Check:** Can you write one line under the title, such as "The Kingdom of X, Year 1312: a strong kingdom, 1:1,000,000, little magic"?

#### MAP TIP

Write the date in the title box ("The Kingdom of X, Year 1312") and draw a scale bar first. If you are unsure which date to choose, choose c. 1300: most of this guide's numbers are for that date.

## What to draw at each scale

#### RULE OF THUMB

Divide the scale number by 1,000 to get metres per millimetre. At 1:1,000,000, 1 mm on paper is 1,000 m (1 km) on the ground. If a thing is smaller than about 1 mm at your scale, draw a symbol; if it is bigger, you can draw its outline.

**Why:** A map cannot show everything. As the scale gets smaller, mapmakers leave features out and replace the rest with symbols (this is called *generalisation*). The chapters of

this guide each give advice for their own features. This section puts it in one table, so each map shows the right things in the right numbers.

#### Scale arithmetic: how big things are on paper

A scale of 1:50,000 means that 1 cm on the map stands for 50,000 cm (500 m) on the ground. A **large-scale** map has a small number after the colon and shows more detail; a **small-scale** map has a big number and shows less.

| Map              | Typical scale          | 1 mm on paper = | 1 cm =                                        | One A3 sheet (297 × 420 mm, 11.7 × 16.5 in) covers                                                                       | How big things are                                                                                                                                                                           |
|------------------|------------------------|-----------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Continent</b> | 1:10,000,000           | 10 km           | 100 km (about 1 inch = 160 mi)                | 3,000 × 4,200 km (1,850 × 2,600 mi): about the whole Mediterranean, which is about 4,000 km (2,500 mi) from west to east | An England-sized kingdom (130,000 km <sup>2</sup> , 50,000 sq mi) is only about 36 × 36 mm. A great city 2–4 km across is 0.2–0.4 mm: a dot. A county is 5–7 mm.                             |
| <b>Kingdom</b>   | 1:1,000,000            | 1 km            | 10 km (about 1 inch = 16 mi)                  | 300 × 420 km (185 × 260 mi), about 125,000 km <sup>2</sup> : one England-sized realm only if it were square              | A great city (2–4 km) is 2–4 mm; a city (1–2 km) 1–2 mm; a town (0.5–1 km) 0.5–1 mm. Use symbols. A county is 50–70 mm across. Market towns are 9–16 mm apart; villages only 1.5–4 mm apart. |
| <b>Local</b>     | 1:50,000–<br>1:100,000 | 50–100 m        | 0.5–1 km (1:100,000 is about 1 inch = 1.6 mi) | 15 × 21 km (9 × 13 mi) to 30 × 42 km (18 × 26 mi)                                                                        | A village (250–450 m) is 2.5–9 mm; a town (0.5–1 km) 5–20 mm; a great abbey precinct (350–600 m) 3.5–12 mm. Draw outlines. A moated manor (40–80 m) is under 2 mm, so use a symbol.          |

Real realms are rarely square. England is about 500 km (310 mi) from London to Berwick alone, so a whole England-sized kingdom fits on one A3 sheet only at about 1:2,000,000 (1 mm = 2 km). At that scale even a great city is a 1–2 mm symbol.

The sizes of places come from [How big to draw it](#) and [Footprints](#).

#### The feature filter: what to show and how many

#### RULE OF THUMB

Draw what a traveller at that scale would notice. A continent map shows the few great places, a kingdom map shows every town, and a local map shows every village and church.

The counts are for c. 1300 and an English-style realm unless the table says otherwise. They are averages: real features cluster on good land and along routes. For areas, 100 km<sup>2</sup> is 39 sq mi and 1,000 km<sup>2</sup> is 386 sq mi. The local column fits maps up to about 20 × 20 km (12 × 12 mi). On a bigger local map, such as a 50 × 50 km (30 × 30 mi) county, draw villages as dots and leave out mills and hamlets.

| Feature                                | Continent (1:10,000,000)                                                                                             | Kingdom (1:1,000,000)                                                                                                                                                                                   | Local (1:50,000–1:100,000)                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Realms and fiefs</b>                | All realms and great fiefs; at least three kinds of state                                                            | The realm border, 4–8 duchies or great fiefs, 1–3 special zones (a march, a palatinate, a church territory)                                                                                             | Only a border that crosses the map                                                                                                                                            |
| <b>Counties and districts</b>          | —                                                                                                                    | Counties 50–70 km (30–45 mi) across: 30–40 in an England-sized realm (England had 39 shires)                                                                                                            | Hundreds (districts within a county) 12–15 km (7–9 mi) across; parish boundaries                                                                                              |
| <b>Great cities (50,000+)</b>          | 3–8 on the whole continent, almost all on the sea or a navigable river                                               | All, as large symbols                                                                                                                                                                                   | Full outline with suburbs, 20–80 mm across                                                                                                                                    |
| <b>Cities and towns (2,000–50,000)</b> | Capitals and the main cities only, as dots                                                                           | Every town of 2,000+, as a symbol                                                                                                                                                                       | Outline with walls, gates and market place                                                                                                                                    |
| <b>Market towns (500–2,000)</b>        | —                                                                                                                    | The most important third or so: 30–40 of about 90 in a 60,000 km <sup>2</sup> realm of 1.2 million                                                                                                      | All, 9–16 km (6–10 mi) apart                                                                                                                                                  |
| <b>Villages</b>                        | —                                                                                                                    | Only as texture in crowded lowland                                                                                                                                                                      | All: 1.5–4 km (1–2.5 mi) apart, about 7 per 100 km <sup>2</sup> in lowland                                                                                                    |
| <b>Hamlets and farmsteads</b>          | —                                                                                                                    | —                                                                                                                                                                                                       | 0–5 per 100 km <sup>2</sup> in open-field lowland; 20+ in dispersed uplands                                                                                                   |
| <b>Castles and forts</b>               | The greatest fortresses, citadels and naval arsenals: about 1 per 20,000–50,000 km <sup>2</sup> (8,000–19,000 sq mi) | Royal and major castles: about 1 per 1,000 km <sup>2</sup> on average, many more on a contested border                                                                                                  | All: in England about 12 castle sites and 4 active castles per 1,000 km <sup>2</sup> , so about 5 sites and 1–2 active castles on a 20 × 20 km map; 3–5 times more in a march |
| <b>Routes</b>                          | Sea lanes, river spines and long trade routes; 1–3 main passes per mountain range                                    | Main roads: about 30 km per 1,000 km <sup>2</sup> (48 mi per 1,000 sq mi); an inn or village every 15–30 km (10–19 mi); bridges about 20 km (12 mi) apart on a lowland river; minor passes and hospices | All roads, lanes and footpaths: 2–3 roads out of each village                                                                                                                 |
| <b>Church</b>                          | 1–2 great shrines on the whole continent; archbishops' seats; universities (only about 20 in all Europe c. 1300)     | Cathedrals: 5–10 per 50,000 km <sup>2</sup> in an English-style church, 25–100 in an Italian-style one; great abbeys; 1–3 shrines; 0–2 universities                                                     | Every parish church (about 7 per 100 km <sup>2</sup> , or 73 per 1,000 km <sup>2</sup> ); abbeys, friaries, hospitals, leper houses and gallows                               |
| <b>Industry and resources</b>          | Mining-district symbols on 3–6 mountain areas; one or two great fair regions                                         | 1–3 mining districts, 1–3 salt sources, one cloth region, 1–2 fair towns and several ports                                                                                                              | Shafts, quarries and saltpans; mills, about 8–11 per 100 km <sup>2</sup> (roughly one per village)                                                                            |
| <b>Manors and parks</b>                | —                                                                                                                    | Royal forests, as green dash-dot boundaries                                                                                                                                                             | Moated sites (4–5 per 100 km <sup>2</sup> ), deer parks (2–3 per 100 km <sup>2</sup> ), fishponds and warrens                                                                 |
| <b>Ruins and older layers</b>          | Only famous ruins and the old empire's roads                                                                         | Big ruins and long straight old roads                                                                                                                                                                   | All: hillforts (1–3 per 100 km <sup>2</sup> in hilly country), deserted villages and earthworks                                                                               |

**Example: an England-sized kingdom (130,000 km<sup>2</sup>, 50,000 sq mi) on one sheet.** Draw about 6 cities of 10,000+, roughly 50–60 towns of 2,000–10,000, 30–40 counties, 13–26 cathedrals in the English style (England had 17), about 130 royal and major castles and about 3,900 km (2,400 mi) of main road. Leave the 9,500 villages and parish churches for local maps.

**Read:** Map symbols and labels; How many to draw; How much main road to draw; What to show at each scale; How many features: England c. 1300 averages; Mining districts on the map; Drawing it all: symbols, density and scale

**Check:** Multiply each "per area" count by the area of your map. Is every count close to the target, or can you say why not (a march, a desert, a crowded Flanders-like plain)?

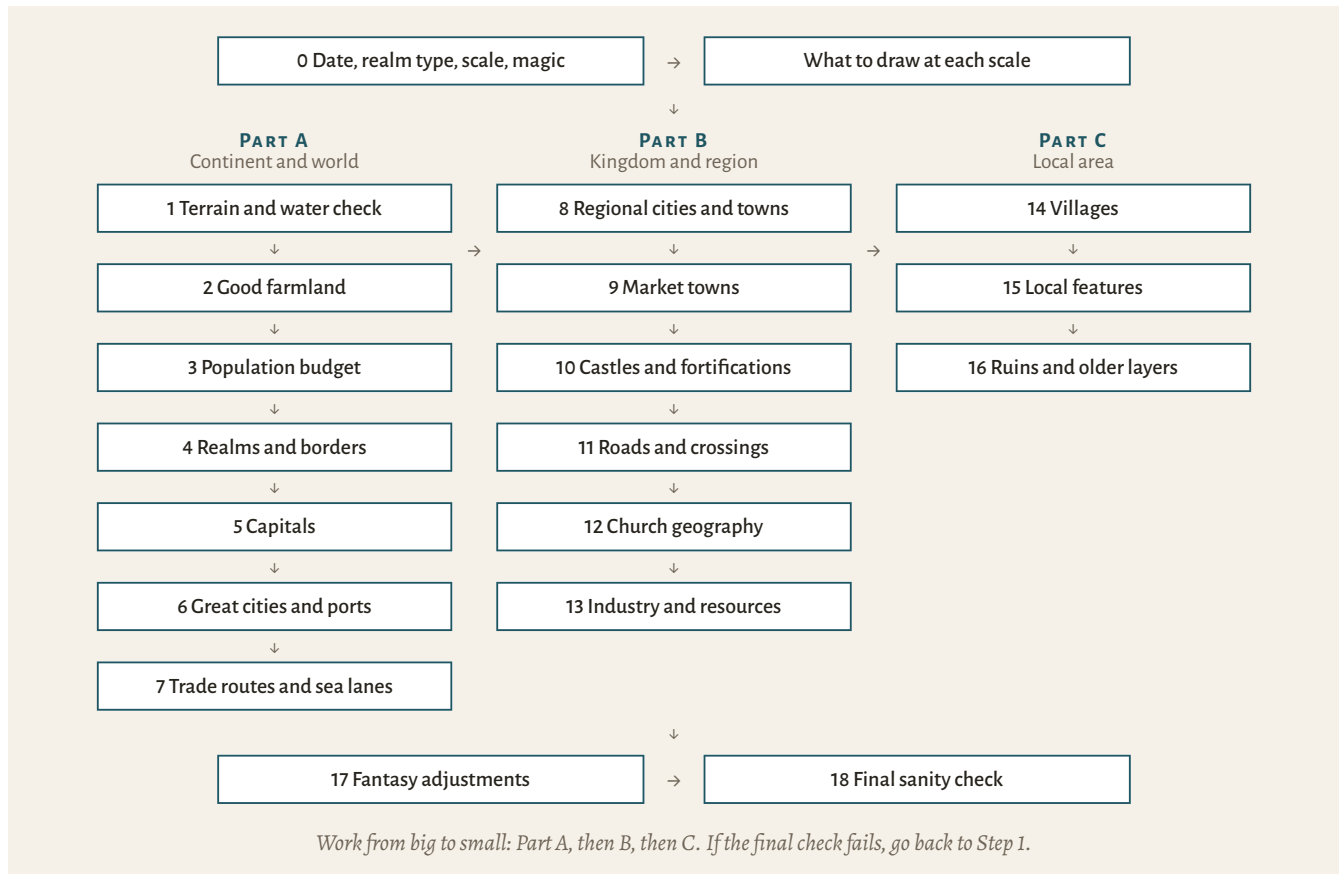
#### MAP TIP

Copy the column for your scale into the margin of your sketch, write the target number next to each feature and tick them off as you place them. A count far from the target is not always wrong, but it needs a reason you could write on the map.

## The order of work

### RULE OF THUMB

Work from big to small, and from nature to people: water and food first, then people, then power, then everything that serves them.



Each step says what to **place**, gives the key **rules**, links to the section to **read** and ends with a **check**. Fix a failed check before you go on: early mistakes spread into every later step.

### MAP TIP

Work in pencil or on separate layers, so you can redraw an early step without erasing everything after it.

## Part A: Continent and world scale

Here you decide where the people are, who rules them and how goods move.

### Step 1: Check the terrain and water

**Place:** rivers, springs, marshes, fords, and on each big river the tidal limit (where the tide stops) and the head of navigation (the highest point boats can reach).

#### Rules:

- Mark which stretches are **navigable** (deep enough for boats) and where they stop: falls, rapids or shallows.

- Where the lowest bridge, the tidal limit and the highest point for sea ships come together is the best spot on the map for a big town.
- Rivers join as they flow downhill. They almost never split, except in deltas and marshes.
- Mark strong sites too: hills, river loops, islands and spurs (ridge ends with cliffs on two or three sides).

**Read:** [Placing settlements on your map: a method](#)

**Check:** Can you point to the head of navigation on every big river, and does no river split outside a delta or marsh?

### Step 2: Find the good farmland

**Place:** light shading for crowded lowland, ordinary farmland, thin hill country and empty land (mountain, marsh, forest, desert).

#### Rules:

- A person needs about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land once pasture, meadow and woodland are counted.
- A village needs arable (ploughed fields), meadow (hay), pasture and woodland, so the best land is where these meet: river valleys, plains and gentle hills.
- Good lowland holds 30–50 people per km<sup>2</sup> (80–130 per sq mi) at its peak. Hill country (farmed valleys with upland grazing) holds 8–15 (21–39 per sq mi) at its peak;

high uplands, moors and mountains hold only 2–5 (5–13 per sq mi).

#### Read: Population density

**Check:** Can you name the reason for every empty zone: mountain, marsh, heath, forest or desert?

### Step 3: Work out the population budget

**Place:** nothing yet. Work out the numbers per region now, and add up each realm after Step 4.

#### Rules:

1. **Measure** each zone. A hex's area is about  $0.866 \times (\text{width across the flat sides})^2$ ; a 30 km hex is about 780 km<sup>2</sup>.
2. **Multiply by a density:** rich lowland 30–40 people per km<sup>2</sup>, average land 15–25, hill country 8–15, high uplands, mountains, forest and marsh 2–5, desert 0. (Multiply by 2.59 for people per sq mi.)
3. **Split the total** by an urban profile:

| Per 1 million people  | Frontier (Poland, Hungary, Scotland, Scandinavia) | Average (England, France, Germany c. 1300) | Highly urban (Flanders, Lombardy, Tuscany) |
|-----------------------|---------------------------------------------------|--------------------------------------------|--------------------------------------------|
| Cities of 10,000+     | 0–1                                               | 1–2                                        | 5–10                                       |
| Towns of 2,000–10,000 | 8–12                                              | 12–18                                      | 30–40                                      |
| Small market towns    | 40–60                                             | 60–90                                      | 80–100                                     |
| Villages              | 3,000–4,500                                       | 2,500–3,500                                | 2,000–2,500                                |

1. **Compare** with real realms: a medium European kingdom c. 1300 had 1–5 million people, and a realm the size of England (130,000 km<sup>2</sup>, 50,000 sq mi) should have 2–5 million. After the plague, use a third to a half fewer.

#### Read: Step-by-step population calculator

**Check:** Is each realm's total within the range of a real kingdom of the same size?

### Step 4: Draw realms and borders

**Place:** realm outlines, great fiefs (duchies and large counties) and frontier zones.

#### Rules:

- Use at least three kinds of state: one or two feudal kingdoms, a fragmented zone (city-states or a loose empire) and one unusual kind (a town league, church state, sea empire or nomads).
- A kingdom covers 10,000–450,000 km<sup>2</sup> (4,000–175,000 sq mi). Direct rule reaches roughly 500–700 km (300–450 mi) from the capital, 8–12 days for a messenger. Farther out, a realm needs viceroys, great vassals, sea routes or a relay post, or it splits.

- Borders are lines in settled farmland and wide bands in mountains, forests, marshes and war zones. Draw a march (a militarised border province) as hatching 10–50 km (6–30 mi) wide. Add an enclave or a disputed strip.

#### Read: Borders and frontiers; How big can a realm be

**Check:** Can you say what every border segment follows: a crest, a river, a forest, an old district line or a treaty?

### Step 5: Place the capitals

**Place:** one capital per realm, or a set of royal residences if the court travels.

#### Rules:

- A capital needs a rich core region, river or sea access, a defensible site, loyal land and legitimacy (an old royal or sacred site). Centrality matters less: Paris, London and Kraków all lay off-centre.
- In a strong, centralised kingdom the capital holds about 1–2% of the realm's people and is 3–7 times the size of the second city. In a loose realm the largest city holds about 0.5% or less.

- Before about 1200 most western kings had no fixed capital: draw 5–15 royal residences about a day's travel apart instead.
- Around a fixed capital, draw 3–6 residences within about 60 km (37 mi), a royal forest and a burial church, and perhaps a coronation town 50–150 km (30–90 mi) away.

**Read:** What makes a capital; Around the capital: the royal landscape

**Check:** Can you say why each capital is where it is, with a better reason than "it is in the middle"?

#### LATER ERA (1500s+)

Fixed capitals win everywhere (Madrid from 1561). For a 1600-style map, draw the capital 2–4 times bigger than its medieval size. See [Population and bigger cities](#).

### Step 6: Place the great cities and ports

**Place:** the few great cities (50,000+ people) and the main ports.

#### Rules:

- Around 1300, Europe, the Middle East and North Africa together had only about 22 cities of 50,000+ and 8 of 100,000+. Mark only 3–8 great cities on a continent map.
- A city of 50,000 almost always needs river or sea supply (rare exception: Granada on its irrigated plain). A city of 100,000 is only possible by water.
- Best sites: the lowest bridging point, the head of navigation, a confluence, an estuary or a superb harbour with shelter, deep water and a route inland. On tidal coasts, big ports often sat 50–120 km (30–75 mi) up an estuary.

**Read:** [The biggest cities](#); [What makes a good harbour](#)

**Check:** Is every great city on the sea or a navigable river (a rare exception, such as Granada, sits in a rich irrigated plain), with a clear reason to be great (a court, a strait, a delta, an export trade)?

#### FANTASY TWIST

A griffon carries only a pack horse's load (100–120 kg). Only magic that moves bulk, such as a portal passing shiploads of grain every day, frees great cities from the water rule. See [Teleport circles and portals](#).

### Step 7: Draw the trade routes and sea lanes

**Place:** sea lanes, river routes, main mountain passes and long land routes.

#### Rules:

- Land : river : sea transport cost about 8 : 4 : 1 (England c. 1300). Bulk goods (grain, timber, stone, salt, wine) follow water; they go more than 30–50 km (19–31 mi) by land only if valuable, or in war or famine.
- A continent usually has one or two bulk sea networks, several river spines and one or two long luxury land routes. Copy a real one (the Hanse, Venice and Genoa, the Silk Roads).
- Sea lanes follow coasts and island chains. On a galley coast, put a watering port every 100–300 km (60–190 mi).
- Give each mountain range 1–3 main passes and 2–5 minor ones. Desert routes need water every 30–40 km (19–25 mi).

**Read:** [Real trade networks to copy](#); [Mountain passes](#)

**Check:** For each region, can you write "exports ...", imports ..." and trace each export to water or a market?

#### MAP TIP

A continent map shows realms and great fiefs, 3–8 great cities, sea lanes, main passes, great shrines, mining districts in 3–6 mountain areas and only the greatest fortresses (about 1 per 20,000–50,000 km<sup>2</sup>, 8,000–19,000 sq mi). Leave counties, villages and ordinary castles for the kingdom map.

## Part B: Kingdom and region scale

Now zoom in to one realm and fill it in.

### Step 8: Place regional cities and towns

**Place:** counties and their seats, cities, and towns of 2,000–10,000 people.

#### Rules:

- Divide the realm into 4–8 duchies or great fiefs, then into counties 50–70 km (30–45 mi) across, each with a seat near its middle. An England-sized kingdom has 30–40 counties.
- Put towns at confluences, heads of navigation, gaps in hill ranges, the feet of passes, harbours and crossings of main roads.

- Cities of 10,000+ stand 60–90 km (37–56 mi) apart in very urban regions, 130–170 km (80–105 mi) in average ones and 260–400 km (160–250 mi) in thinly urban ones. For each, expect about 10 towns and 50–100 market towns.
- A town of 10,000 needs farmland within 21–27 km (13–17 mi) on good land. Draw that circle faintly; if two circles overlap heavily, shrink one town or put it on water.

**Read:** How far apart: the numbers; Political units and their sizes

**Check:** Can you write a "site" note (the ground it stands on) and a "situation" note (its place among routes and regions) for every town?

### Step 9: Fill in the market towns

**Place:** small market towns of 500–2,000 people, each with a weekly market and a yearly fair.

**Rules:**

- Space them about 9–16 km (6–10 mi) apart, so a peasant can walk to market and home in one day (6–10 km, 4–6

mi, each way). In thinly settled land they can be up to about 25 km (15 mi) apart.

- Put them on rivers and road junctions. Two market towns 3 km (2 mi) apart compete; make one a village.
- Christaller's central places form neat hexagons only on a flat, even plain. Stretch the pattern along rivers and roads, and squeeze it on rich land.

**Read:** Central place theory, made simple

**Check:** Can every village reach a market town and get home the same day?

### Step 10: Place castles and fortifications

**Place:** castles, town walls, watchtowers and beacon chains.

**Rules:**

- Every fortification controls something. Write its job in one word (*road, bridge, town, border, estate, refuge or watch*); the job gives the site.
- Peaceful core: a working castle every 20–30 km (12–19 mi). Contested border: one every 4–6 km (2.5–4 mi). Conquest chain: one every day's march, 20–40 km (12–25 mi), each reachable by water.

| Zone, per 1,000 km <sup>2</sup> (about 390 sq mi), c. 1300 | All castle sites ever built | Active castles |
|------------------------------------------------------------|-----------------------------|----------------|
| Peaceful lowland core                                      | 4–8                         | 1–3            |
| England and Wales average                                  | ~12                         | ~4             |
| March or contested border                                  | 35–60                       | 12–20          |

- Put the castle at the edge of its town, with the market at its gate. Wall cities, regional capitals, frontier towns and raided ports; leave most market towns open.
- Peacetime garrisons are only 5–20 men. Beacon chains run from the border to the capital, about 5–20 km (3–12 mi) apart in hilly country.

**Read:** How many castles and how far apart; How many to draw

**Check:** Can you name what each castle controls, and is the map sparse in the core and dense on the border?

#### LATER ERA (1500s+)

Low, thick star forts with angled bastions spread beyond Italy in the 1530s–1540s. They were so expensive that only frontiers, capitals, main ports and key crossings get them. See *Gunpowder and the new fortifications*.

### Step 11: Draw roads and crossings

**Place:** bridges, fords, ferries, roads, inns, hospices and tolls.

**Rules:**

- Crossings first: roads bend to reach them and towns grow at them. A great river gets only a few bridges, often tens of km apart, with ferries and fords between.
- Main roads link cities through crossings and passes, along valleys and dry ridges; only an older empire's roads run straight. Village tracks run to the market town like wheel spokes.
- Put an inn or village every 15–30 km (10–19 mi) on main roads, a caravanserai (walled roadside inn) every 30–40 km (19–25 mi) in desert, and a hospice at each main pass. Tolls sit at bridges, gates, gorges, straits and borders.
- Walkers cover 25–35 km (15–22 mi) a day, ox carts 15–25 km (10–15 mi) and armies with baggage 13–20 km (8–12 mi).

**Read:** Drawing the route network

**Check:** For each road, can you say where it goes, what it carries and why it bends where it does?

**LATER ERA (1500S+)**

Post stations every 20–40 km (12–25 mi) carry letters up to about 150 km (95 mi) a day, and canals with pound locks begin to climb hills. See [Posts, roads and canals](#).

**Step 12: Lay out the church geography**

**Place:** cathedrals, abbeys, friaries, shrines and pilgrim roads, hospitals and universities.

**Rules:**

- **Dioceses** (one bishop's area): in an English-style realm, one per 5,000–10,000 km<sup>2</sup> (2,000–4,000 sq mi), so a cathedral means a main city. In an Italian-style land, one per 500–2,000 km<sup>2</sup> (190–770 sq mi), many in small towns.
- **Monasteries:** about 1 religious house per 150 km<sup>2</sup> (60 sq mi). Cistercians in empty, well-watered valleys; canons at town edges; friars inside towns of about 3,000–5,000 and up. Give big abbeys granges (outlying farms), most within about 25 km (15 mi).
- **Friaries measure towns:** usually none at 2,000–3,000 people, about 2 at 10,000, 4 at 20,000–40,000.
- **Pilgrim roads** follow existing roads, with stops about 20 km (12 mi) apart and a hospice at every pass.
- **Hospitals and universities:** a hospital in every market town; a leper house 1–2 km (about 1 mi) outside each town of 2,000+; one or two universities in a kingdom of 2–5 million.

**Read:** [Monasteries, friaries and military orders](#); [Drawing it all: symbols, density and scale](#)

**Check:** Is every Cistercian abbey in an empty valley, and every friary in a town big enough to feed it?

**Step 13: Place industry and resource sites**

**Place:** mines, salt works, quarries, cloth districts, fisheries and fair towns.

**Rules:**

- Industry goes to its heaviest input: the ore, the fuel or the water power, not the customer.
- Per kingdom, aim for 1–3 mining districts, 1–3 salt sources, one cloth region, one or two fair towns and several ports. Every realm needs salt, or a route that brings it in.
- A mining district holds 2–7 mining towns 10–40 km (6–25 mi) apart, up side valleys, plus a mint town. Most had 1,000–5,000 people; rare boom towns reached 10,000–20,000.

- Ironworks need charcoal and fast streams: draw many small forges, not one big town.

**Read:** [The basic rules: what pulls industry to a place](#); [Special town types and how to spot them](#)

**Check:** For every industry, can you point to the resource, the fuel and the way out?

**MAP TIP**

A kingdom map shows the capital, cities, towns and the main market towns, with villages only as texture. Add county borders, the main castles, cathedrals, great abbeys and mining districts. Leave mills, hamlets and gallows for the local map.

## Part C: Local area scale

A local map covers roughly  $5 \times 5$  km to  $50 \times 50$  km ( $3 \times 3$  to  $30 \times 30$  mi). Almost every square kilometre has a human feature.

### Step 14: Place the villages

**Place:** villages, hamlets and farmsteads, each village with its parish church.

#### Rules:

- Open-field land gives tight (nucleated) villages. Hills, forest and marsh give hamlets and lone farms. A road, valley stream or dyke gives villages in a line.
- Lowland villages stand 1.5–4 km (1–2.5 mi) apart: on terraces above the floods, along spring lines and on the edge between two kinds of land. In hills, parish churches stand about 7–10 km (4–6 mi) apart, with hamlets between.
- A typical village has 30–60 households (150–300 people), with fields within about 2 km (1.3 mi). Land use forms rings: gardens, then arable, then pasture, then wood and waste at the parish edge. Meadow follows the streams.

**Read:** [Sizes, territory and walking distances](#)

**Check:** Does every village touch water and have its fields within about 2 km (1.3 mi)?

| On a $20 \times 20$ km ( $12 \times 12$ mi) lowland map, England c. 1300 | Number |
|--------------------------------------------------------------------------|--------|
| Villages with parish churches                                            | ~25–30 |
| Mills                                                                    | ~30–45 |
| Moated sites                                                             | ~15–20 |
| Deer parks                                                               | ~10    |
| Market places (many tiny)                                                | ~5     |

**Read:** [Drawing a local-area map](#)

**Check:** Does every patch of land have a use, and is any empty space labelled as marsh, moor or forest?

### Step 16: Add ruins and older layers

**Place:** old roads, hillforts, castle ruins, deserted villages and moved towns.

#### Rules:

- Ruins near living towns are quarried away; they survive in empty country.
- Draw a few long, straight "old roads" linking cities, and in hilly country 1–3 hillforts per 100 km<sup>2</sup> (39 sq mi).

#### FANTASY TWIST

Where monsters roam, people crowd into fewer, walled villages of 300–1,000, with fields within 4–5 km (2.5–3 mi) of the walls. See [Dangerous wilderness and monsters](#).

### Step 15: Add the local features

**Place:** mills, manor houses, parks and forests, meeting places, gallows and boundaries.

#### Rules:

- About one mill per village: watermills on streams with a weir and a leat (a channel to the wheel); windmills on high ground in flat, dry country.
- The manor house stands beside the church, often moated, with a dovecote and fishponds.
- A royal forest is a legal hunting area: draw a green dash-dot boundary and a large italic name, with villages and fields inside, not solid trees. Deer parks are rounded enclosures 0.5–1 km (0.3–0.6 mi) across.
- Hundreds (districts within a county) about 12–15 km (7–9 mi) across meet at a mound, stone or ford. A gallows stands on a hill by the main road outside each town.

- In long-settled land, draw one or two castle ruins or earthworks for every active castle; on a new frontier, almost none.
- England has more than 3,000 known deserted villages, at least 1,500 abandoned c. 1350–1520. Draw them as a grey lone church labelled "(lost village)".

**Read:** [Ancient and ruined layers](#); [How settlements change over time](#)

**Check:** Is there at least one older layer, and is every feature possible by the date in your title box?

**MAP TIP**

Use three styles: working features, old features still in use (like an ancient road), and ruins drawn with the feature's own symbol, broken or grey, labelled "(ruin)". These styles match the guide's [master legend](#). For the fine detail of Steps 14–16, follow the [method for a local map](#) in chapter 06.

## Finishing: fantasy and the final check

### Step 17: Make the fantasy adjustments

**Place:** the places that magic, monsters and other peoples create, and the places they destroy.

**Rules:**

- Real rules first. Then ask which input each element changes (water, food, defence, travel cost, threats or resources) and write: "X exists, so Y changes."
- Add magic in order: resources (a crystal is ore; a ley line is a river), then towers and boom towns, density changes, new hubs (portals, aeries) and finally fortresses.
- Danger concentrates people: walled hilltop villages, refuge forts about a day's walk (24–32 km, 15–20 mi) apart, and a frontier band of 30–100 km (20–60 mi) between the safe core and the wild.
- Strong fertility magic (about twice the yield) gives density like Flanders: 50–75 people per km<sup>2</sup> (130–190 per sq mi).
- Magic can give a town its reason to exist, but not its water and food. Use no more than three or four fantasy elements per region.

**Read:** [The method: change the inputs, not the rules; Consistency checklist](#)

**Check:** Does every magical city still show where its water and food come from?

**LATER ERA (1500S+)**

For a 1500–1650 map, keep the villages and market towns. Grow the capital and ocean ports, turn interior castles into ruins or palaces, and add star forts where they pay, post roads, a drained polder and faith areas (soft colour tints, never a new border line). See [Drawing a world in transition](#).

### Step 18: Run the final sanity check

**Place:** nothing new. Fix what fails.

**Rules:**

- Run the "why is it here?" test on every dot. A village needs water, mixed land nearby and safety from floods. A town must also stand where routes meet. A city must also be on navigable water and be the best-connected place in a rich region.
- Count symbols: far more villages than towns, far more towns than cities.
- Every empty area needs a reason; every crowded area needs good land.
- Place labels last, biggest first, with 2–3 typefaces at most.

**Read:** [The "why is it here?" test; Sanity check: 20 questions for a finished map](#)

**Check:** Can you answer "yes" to all 20 sanity-check questions?

**MAP TIP**

Keep a short list beside your map: name, rank, site type and a one-line situation ("Kingsford: town; ford and confluence; where the north road meets the Silverwater"). It takes a few minutes and catches most mistakes.

## Printable checklist

### Step 0: Before you start

- ☐ Date chosen (c. 1100, 1300, 1450 or 1600) and written in the title box
- ☐ Realm type chosen: centralised, fragmented or in between
- ☐ Scale chosen, scale bar drawn, target counts for that scale written in the margin
- ☐ Amount of magic decided

### Part A: Continent and world

- ☐ 1. Navigable stretches, heads of navigation, fords and tidal limits marked
- ☐ 2. Land shaded by quality; every empty zone has a reason
- ☐ 3. Population = area × density, checked against real kingdoms
- ☐ 4. At least three kinds of state; every border follows something
- ☐ 5. Capitals in rich, defensible cores on navigable water
- ☐ 6. Only 3–8 great cities, almost all on the sea or a navigable river
- ☐ 7. Sea lanes, river spines and passes drawn; exports and imports noted

### Part B: Kingdom and region

- ☐ 8. Counties 50–70 km (30–45 mi) across; towns at route nodes
- ☐ 9. Market towns about 9–16 km (6–10 mi) apart
- ☐ 10. Every castle has a one-word job; sparse core, dense border
- ☐ 11. Few bridges on big rivers; inns every 15–30 km (10–19 mi)
- ☐ 12. Dioceses, monasteries, friaries, shrines and hospitals placed
- ☐ 13. Every industry has a resource, fuel and a way out; every realm has salt

### Part C: Local area

- ☐ 14. Villages 1.5–4 km (1–2.5 mi) apart in lowland, each on water
- ☐ 15. Mills, manors, parks, meeting places and gallows added
- ☐ 16. Ruins and older layers added; nothing too new for the date

### Finishing

- ☐ 17. Each fantasy element has an "X exists, so Y changes" line
- ☐ 18. "Why is it here?" test and the 20 sanity questions passed; labels placed last

## Quick summary

- Decide the date, the realm type, the scale and the amount of magic first; date the map and draw a scale bar.
- At 1:N, 1 mm on paper is  $N \div 1,000$  metres. On a kingdom map (1:1,000,000) a great city is a 2–4 mm symbol; on a local map (1:50,000–1:100,000) a village is a 2.5–9 mm outline.
- Each scale has its own filter: a continent map shows 3–8 great cities and the greatest fortresses; a kingdom map shows every town of 2,000+, the counties and the major castles; a local map shows every village, church, mill and moated site.
- Work from big to small and from nature to people: continent first, then kingdom, then local area.
- Population is area × density. Good lowland holds 30–50 people per km<sup>2</sup> (80–130 per sq mi).
- Capitals sit in rich, defensible cores on navigable water. Great cities of 50,000+ are rare and almost always fed by river or sea.
- Bulk goods follow water (land : river : sea ≈ 8 : 4 : 1). Roads bend to crossings and passes, with a stop every 15–30 km (10–19 mi).
- Spacing: villages 1.5–4 km (1–2.5 mi), market towns 9–16 km (6–10 mi), working castles 20–30 km (12–19 mi) in a peaceful core and 4–6 km (2.5–4 mi) on a contested border.
- Churches form a carpet, monasteries a scatter (about 1 per 150 km<sup>2</sup>, 60 sq mi), cathedrals and universities a handful. Industry sits on its heaviest input.
- Finish with ruins, then fantasy changes, then the "why is it here?" test and the 20 sanity questions.

## Sources and further reading

Almost every historical number in this chapter comes from the chapter section linked in each step or table. Each chapter lists its own sources at the end, and the shared figures are in [research/baseline-numbers.md](#).

The scale arithmetic in [What to draw at each scale](#) (sizes on paper, what one sheet covers, the England-sized example) is this guide's own calculation. It uses these references:

- Ordnance Survey, [A Beginner's Guide to Understanding Map Scales](#) (what a scale such as 1:25,000 or 1:50,000 means; "large scale" and "small scale"; what each OS scale shows).
- Wikipedia, [ISO 216](#) (A3 paper is 297 × 420 mm; A4 is 210 × 297 mm).
- *Encyclopaedia Britannica*, [Mediterranean Sea](#) (west–east extent from Gibraltar to the Gulf of Iskenderun about 4,000 km, 2,500 mi).
- Wikipedia, [Cartographic generalization](#) (selecting features and replacing them with symbols as the scale gets smaller).

For the date table in Step 0:

- Mate, Mavis. "The Rise and Fall of Markets in Southeast England". *Canadian Journal of History* 31:2 (1996). Summary at [medievalists.net](#): market revenues began to fall in the 1390s and fell further in the mid-15th century, with empty market stalls common in the Weald.

## CHAPTER 1

# WHERE SETTLEMENTS ARE BUILT (AND WHY)

Real people did not put towns in random places. Every village, town and city answered the same few questions: is there water, can we grow food, can we defend it, and can goods and people reach it? This chapter explains those questions, shows the classic places where settlements grow, and gives you numbers for spacing and a method for naming places, so that every dot on your map has a reason to be there.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Population numbers for each settlement type are in [Population and Settlement Sizes](#). Roads and ports are in [Trade Routes, Roads and Transport](#).

## IN THIS CHAPTER

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Site and situation

Classic site types

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## The six things every settlement needs

## RULE OF THUMB

Water, food, fuel, safety, access, health. A village needs the first three close by. A town also needs access. A city needs access by water.

| Need                       | What people looked for                                                                    | How close it must be                                                  | What it looks like on a map                                                   |
|----------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Fresh water                | River, stream, spring, reliable well, lake                                                | Inside or next to the settlement                                      | Every village touches blue: a river, a stream or a spring line                |
| Food                       | A <b>mix</b> of land: arable (ploughed fields), meadow (hay), pasture (grazing), woodland | Fields within a 15–30 minute walk (roughly 1–2.5 km, 0.6–1.5 mi)      | Villages sit where two kinds of land meet                                     |
| Fuel and building material | Wood, peat or coal; stone, clay or timber                                                 | Within a few km for daily fuel; farther for building stone (by water) | Woods near most villages; quarries near stone towns                           |
| Defence                    | Hills, river bends, islands, spurs (rock tongues), marsh around a dry spot                | The site itself                                                       | Towns on hills, in river loops, on islands                                    |
| Communications             | Fords, bridges, confluences, harbours, the mouth of a mountain pass, crossroads           | The site itself                                                       | Bigger places at river crossings and route junctions                          |
| Shelter and health         | Sunny slopes, out of the wind, above floods, away from "fever marshes"                    | The site itself                                                       | Villages on the sunny side of valleys and on the edge of marshes, not in them |

### Fresh water

**Rule:** No settlement is far from water. Villages need it every day for people, animals, brewing and washing.

**Why:** Water is heavy. Carrying it uphill every day is hard work. Deep wells were expensive to dig.

### Real examples:

- Old Sarum is an Iron Age hill fort near modern Salisbury, England, with a Norman castle and cathedral inside it. The clergy complained about three things: water was scarce and expensive (and the castle guards sometimes controlled access to it), the wind was fierce, and they quarrelled with the castle garrison. Pope Hon-

orius III gave permission to move (1217–18), and in 1220 the bishop laid the foundation stone of a new cathedral down in the river valley. New Salisbury grew by the river. Old Sarum slowly emptied.

- On chalk and limestone hills, water sinks into the rock. It comes out as springs where the porous rock sits on clay. Villages form a line along these springs (see [spring-line villages](#) below).

**Flood risk.** People wanted water close, but not in the house. The usual answer was a **river terrace**: a flat step of gravel a

few metres above the flood plain. Roman and medieval London stood on two low hills (Cornhill and Ludgate Hill) on the north bank of the Thames. The higher one, Cornhill, rises only to about 18 m (58 ft) above sea level. Across the river, on the south bank, was a patch of dry ground at Southwark. The flood plain itself was left as meadow.

### Food: a mix of land

**Rule:** A farming village needs four kinds of land, not one.

| Land type | Use                                                                     | Where it is                                |
|-----------|-------------------------------------------------------------------------|--------------------------------------------|
| Arable    | Grain for bread, ale and pottage (a thick stew of grain and vegetables) | Well-drained, fairly flat land             |
| Meadow    | Hay to feed animals in winter                                           | Flood plains and wet valley floors         |
| Pasture   | Summer grazing for oxen, sheep, cows                                    | Hills, heaths, rough ground, fallow fields |
| Woodland  | Fuel, building timber, pigs (acorns)                                    | Steeper or poorer land, parish edges       |

**Why:** Ploughs need oxen. Oxen need hay in winter. Hay needs meadow. Fields need manure from animals. So a village with only one kind of land fails.

### Numbers:

- Meadow was precious. Medieval English land surveys usually valued an acre of meadow well above an acre of arable, often several times as much.
- A person needs about 1 ha (2.5 acres) of arable land, or 2–3 ha (5–7 acres) of all land including pasture, meadow and woodland (see the shared baseline in [Population and Settlement Sizes](#)).
- So a village of 200 people needs about 2 km<sup>2</sup> (0.8 sq mi) of fields and 4–6 km<sup>2</sup> (1.5–2.3 sq mi) of land in total. Add common waste (unfarmed rough land shared by the village) and woods, and you get the typical English lowland **parish** (the area served by one village church) of about 14 km<sup>2</sup> (5.4 sq mi).

**Map result:** Villages love **edges**. They sit where a river valley meets a hillside, where a fen meets dry land, or where downland meets a clay vale. From one spot the village can reach meadow, fields and pasture.

#### RULE OF THUMB

Put villages on the boundary between two kinds of land, not in the middle of one.

the Scottish islands) or dung (steppe). Building material was local: timber and wattle in forest country, stone where there was stone, clay brick in the stone-poor plains of the Low Countries and the Baltic coast. Fine building stone for cathedrals and castles moved long distances, but almost always by water.

### Defence

**Rule:** The more dangerous the times, the higher and more enclosed the site.

### Fuel and building material

Every household burned fuel every day for cooking, heating, baking and brewing. Most villages had woodland or a fuel source within a few kilometres. Where wood was scarce, people burned peat (fens, the Netherlands, Ireland,

| Defensive site       | How it helps                              | Real examples                                                                                                    |
|----------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Hilltop or plateau   | Attackers must climb                      | Orvieto (Italy, on a volcanic rock plateau); Laon (France); Old Sarum (England)                                  |
| Spur or promontory   | Cliffs on three sides; wall only one side | Luxembourg (the Bock rock, acquired by Count Siegfried in 963); Edinburgh (castle rock with a ridge behind it)   |
| River bend (meander) | Water on three sides                      | Durham (founded 995 in a loop of the River Wear); Bern (founded 1191 in a loop of the Aare); Shrewsbury (Severn) |
| Island               | Water all round                           | Paris (Île de la Cité); Stockholm (first recorded 1252); Venice; Lindau (Lake Constance)                         |
| Dry spot in a marsh  | Marsh slows attackers                     | Ely (England, on an "isle" in the Fens)                                                                          |

**Mediterranean example:** In Latium, central Italy, many scattered farms were regrouped into fortified hilltop villages called *castelli*. This happened mainly in the 10th and early 11th centuries. Historians call it *incastellamento* ("castle-building"), a term made famous by Pierre Toubert (1973). Scholars still argue about exactly when and why it happened, but the result is clear: central Italy is full of hill-top villages. Raided coasts show the same effect (see [Coasts](#)).

**The trade-off:** A strong site is often a bad site for trade and water. Old Sarum lost to Salisbury for exactly this reason. In peaceful times, towns move down to the river.

### Communications

A place that people **must pass through** has a reason to grow: a ford, a bridge, a harbour, the foot of a pass. These are explained in [Classic site types](#). The routes themselves are in [Trade Routes, Roads and Transport](#).

### Shelter and health

- **Sunny slopes.** In the Alps the sunny side of a valley (*adret* in French) has a longer growing season, less snow and higher tree and crop limits than the shady side (*ubac*). Villages and fields are on the adret; the ubac is left to forest.

- **Out of the wind.** At Old Sarum, the writer Peter of Blois said the wind was sometimes so strong that the church services could not be heard.
- **Away from fever marshes.** Malaria ("ague", "marsh fever") existed in Europe's coastal marshes, including Kent, Essex, the Fens, the Italian Maremma and the Pontine Marshes. Mary Dobson (1997) showed that in the 17th and 18th centuries, marsh parishes in Kent and Essex often buried more people than they baptised.
- **Healthy air.** The Abbasid caliph al-Mansur founded Baghdad in 762. Historians list his reasons as its place on trade routes, its water supply and its mild climate. When he later built his Khuld Palace there, he put it on the highest part of the Tigris bank, for fresh air and away from the insects of the riverside.

#### MAP TIP

Before you draw any settlement, mark on your terrain map: rivers and springs, flood plains, marshes, sunny valley sides, good soil, and defensible spots (hills, river loops, islands). Most villages will fall on the edges between these zones. A dot with none of these features nearby needs a special reason, such as a mine or a monastery.

## Site and situation

These two words come from school geography, and they are the most useful tools in this chapter.

|           | Site                                                                  | Situation                                                                                      |
|-----------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Meaning   | The exact spot: the ground the settlement stands on                   | Its place in the wider world: routes, neighbours, resources, region                            |
| Questions | Is it dry? Is there water? Can it be defended? Is there room to grow? | Is it on a river people sail? At a crossing? Between two regions that trade? In rich farmland? |
| Decides   | Whether the place <b>survives</b>                                     | Whether the place <b>grows big</b>                                                             |

**Rule:** Thousands of places have a good site. Only a few have a great situation. Situation decides which village stays a village and which becomes a city.

**Real examples:**

- **London.** The *site* was modest: two low gravel hills by a marshy river. The *situation* was superb: the lowest point where the Thames could be bridged, near the old tidal limit (the highest point the tide reaches up a river), where sea ships, river boats and roads all met. That is why it became the largest city in England.
- **Venice.** The *site* was terrible: mud islands in a lagoon, no farmland, and little fresh water (rain cisterns, plus water brought by boat). The *situation* was excellent: safe from land armies, at the head of the Adriatic, close to the passes into Germany. Venice became one of Europe's largest cities (see [Population and Settlement Sizes](#)).
- **Old Sarum and Salisbury.** Same situation, different site. The better site (water, flat ground by the river) won.

**Why situation matters so much:** Moving bulk goods over-land was very expensive. In England around 1300, carrying

goods by land cost about 8 times as much as by sea and 2 times as much as by river (the shared baseline gives land : river : sea  $\approx 8 : 4 : 1$ ). So towns that needed to import grain, timber or stone had to sit on **navigable** water (water deep enough for boats). Nearly every town above about 10,000 people sits on a navigable river or the coast. The few exceptions had something special. Granada sat in an unusually rich irrigated plain (the *Vega*). Siena lived on banking and on the main pilgrim road to Rome.

**RULE OF THUMB**

Good site = the village survives. Good situation = the village becomes a town. Great situation on water = it becomes a city.

**MAP TIP**

For each large settlement, write two short notes: "site: ..." and "situation: ...". If you cannot write the situation note for a city, it should probably be a town.

## Classic site types

These are the standard "reasons to be here." Most real towns combine two or three of them.

| Site type                                 | What it is                                                                                        | Why a town grows there                                                 | Real examples                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ford town</b>                          | Where a river is shallow enough to wade or drive a cart across                                    | Roads must meet here; travellers stop                                  | Oxford ("oxen ford", first recorded 912); Frankfurt ("ford of the Franks", first recorded 794); Hertford; Stratford                                                                                                                                                                                                                                                                                                                                                       |
| <b>Bridge town</b>                        | Where a bridge replaced or added to a ford                                                        | Tolls, markets, controls traffic                                       | Cambridge; Innsbruck ("bridge over the Inn"); Saarbrücken                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Lowest bridging point</b>              | The bridge nearest the sea                                                                        | Roads on both banks AND ships from the sea; often also the tidal limit | London (Thames); York (Ouse); Exeter (Exe); Gloucester (Severn); Stirling (Forth: a map by Matthew Paris from the 1240s–1250s seems to show only one bridge in all Britain, at Stirling)                                                                                                                                                                                                                                                                                  |
| <b>Head of navigation / break of bulk</b> | The highest point big ships can reach, or a point where cargo must change boats ("break of bulk") | Goods must be unloaded, stored, sold, reloaded                         | Cologne (ships from the lower Rhine were too big for the river above, so cargo usually changed boats here; from 1259 a <i>staple right</i> , a town's legal right to make passing merchants stop, forced them to unload and offer their goods for sale for 3 days); Schaffhausen (the name probably means "boat houses": goods were unloaded above the Rhine Falls and carried around them); Seville (sea port about 80 km (50 mi) up the Guadalquivir from the Atlantic) |
| <b>Confluence town</b>                    | Where two rivers join                                                                             | Two river routes meet; a defensible tongue of land between them        | Koblenz (Latin <i>Confluentes</i> : Rhine and Moselle); Lyon (Rhône and Saône); Passau (Danube, Inn and Ilz); Belgrade (Sava and Danube)                                                                                                                                                                                                                                                                                                                                  |
| <b>River-bend (meander) town</b>          | Inside a tight loop of a river                                                                    | Water on three sides; only one side to wall                            | Durham (995); Bern (1191); Shrewsbury; Besançon (Doubs); Toledo (Tagus)                                                                                                                                                                                                                                                                                                                                                                                                   |

| Site type                   | What it is                                                                                                                              | Why a town grows there                                      | Real examples                                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hilltop town</b>         | On a hill, ridge or plateau                                                                                                             | Defence, health, view                                       | Italian <i>castelli</i> ; Orvieto; Carcassonne; Shaftesbury                                                                                                                                                                                                                 |
| <b>Island town</b>          | On a river, lake or sea island                                                                                                          | Defence; easy river crossing in two short bridges           | Paris (Île de la Cité); Stockholm; Venice; Lindau; Mont-Saint-Michel (tidal island)                                                                                                                                                                                         |
| <b>Spring-line village</b>  | A row of villages where springs come out at the foot (or near the top) of a hill <b>scarp</b> (a steep slope along the edge of a ridge) | Reliable water plus fields below and pasture above          | Lincoln Cliff (a limestone ridge over 80 km (50 mi) long, with more than 40 villages and towns along it: roughly one every 2 km (1.2 mi)); the foot of the South Downs (Fulking, Poyning, Edburton); the edges of the Mendips (Wells, Wookey Hole, Blagdon, Compton Martin) |
| <b>Gap town</b>             | Where a river cuts a gap through a line of hills                                                                                        | All routes across the hills funnel through the gap          | Lincoln (River Witham through the limestone ridge); Guildford (River Wey through the North Downs); Lewes (River Ouse through the South Downs)                                                                                                                               |
| <b>Pass town</b>            | At the foot of a mountain pass                                                                                                          | Travellers rest, change animals, hire guides; customs posts | Altdorf and Bellinzona (Gotthard: the first bridge across the Schöllenen gorge, built around 1220, opened the route); Chur (several Alpine passes); Susa (Mont Cenis); Aosta (Great St Bernard)                                                                             |
| <b>Harbour town</b>         | A sheltered bay, deep water, protection from storms                                                                                     | Sea trade, fishing, shipbuilding                            | Genoa; Bergen; Dubrovnik (Ragusa); Brest                                                                                                                                                                                                                                    |
| <b>Estuary town</b>         | Up a tidal river mouth, usually at the first good crossing                                                                              | Sheltered from storms and raiders; reaches inland by river  | Bordeaux (Gironde/Garonne, about 100 km inland); Hamburg (Elbe, about 110 km); Antwerp (Scheldt, about 80 km); Rouen (Seine) — all roughly 80–120 km (50–75 mi) from the open sea                                                                                           |
| <b>Oasis town</b>           | Around a desert spring, river or underground channel                                                                                    | The only water for days of travel; caravans must stop       | Sijilmasa (Morocco, founded 757, northern end of a trans-Saharan route, abandoned 1393); Damascus (watered by the Barada river and the Ghouta gardens); Yazd and Kerman (Iran, on <i>qanats</i> , underground water channels: see <a href="#">Desert</a> )                  |
| <b>Wet-point settlement</b> | A spot chosen for water in a dry land                                                                                                   | Water is the scarce thing                                   | Spring-line villages on chalk; oasis villages; wells on dry plateaus                                                                                                                                                                                                        |
| <b>Dry-point settlement</b> | A spot chosen for dry ground in a wet land                                                                                              | Dry land is the scarce thing                                | Ely (on a clay "isle" about 26 m (85 ft) high in the Fens; abbey founded 673); Glastonbury (Somerset Levels); Frisian <i>terpen</i> (man-made dwelling mounds)                                                                                                              |

### Notes on the most important types

**Lowest bridging point and head of navigation.** On many rivers the lowest bridge, the old tidal limit and the highest point sea ships could reach were all at the same place, or close together. That place usually became the region's chief town (London, York, Exeter, Gloucester). It is the single best spot on your map for a big town.

**Tidal coasts: ports sit inland.** On the tidal Atlantic and North Sea coasts, the big medieval ports often sat 50–120 km (30–75 mi) up an estuary (a wide tidal river mouth). There they were safe from storms and raiders, they had the first bridge, and the tide helped ships upstream. Later, as ships grew bigger or rivers silted up, small "outports" grew nearer the sea: Damme and Sluis for Bruges, Topsham for

Exeter, Sanlúcar (at the river mouth) for Seville. In the almost tideless Mediterranean, ports more often sat directly on a sheltered bay (Genoa, Marseille, Barcelona, Dubrovnik).

**Fords.** A ford needs firm gravel and water shallow enough for carts at normal levels. On big rivers fords are rare, so a crossing that serves several roads is a strong candidate for a town.

### Reading place names

Many real names tell you the site type. You can use the same trick for your fantasy names. (The everyday village endings, and how to build a whole region's names, are in [Naming settlements realistically](#).)

| Name element                | Meaning                           | Examples                                                    |
|-----------------------------|-----------------------------------|-------------------------------------------------------------|
| -ford, -furt, -voorde       | Ford                              | Oxford, Frankfurt, Erfurt                                   |
| -bridge, -brück, -bruck     | Bridge                            | Cambridge, Innsbruck, Saarbrücken                           |
| -mouth, -münde, Aber-       | River mouth                       | Plymouth, Monmouth, Warnemünde, Aberdeen                    |
| Confluentes, Coblenz, Condé | Confluence                        | Koblenz, Condé-sur-l'Escaut                                 |
| -wick, -wich, -vik          | Trading place, bay                | Ipswich, Reykjavík                                          |
| -ley, -sart, -rode, -roth   | Forest clearing                   | Many English villages; "Les Essarts" in France; Wernigerode |
| -ey, -holm                  | Island, or dry ground in wet land | Chertsey, Romsey, Stockholm                                 |

**FANTASY TWIST**

If griffon riders or teleport circles move people and letters, crossings matter less for **travellers**. They still matter for **goods**: grain, timber and stone still go by boat and cart. Keep the big towns at the crossings. See [Fantasy Variants](#).

**MAP TIP**

Give each town one or two site types and draw them clearly: the river loop around the meander town, the bridge symbol at the lowest crossing, the gap in the hills at the gap town. Then name it so the name matches the site, like "Oakford" at a ford or "Highbridge" at a bridge. Readers notice when geography and names agree.

## Settlement by terrain and biome

The same rules apply everywhere, but the answer changes with the land. The first eleven rows below cover Europe and its nearest neighbours, the steppe and the desert. The last

four cover other climates, with examples from Asia, Africa and the far north.

| Terrain                                           | Where settlements go                                                      | Typical pattern                                                                             | Density (see <a href="#">chapter 02</a> )                     |
|---------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Fertile river valleys and plains                  | River terraces, valley edges, crossings                                   | Many nucleated (tightly grouped) villages; towns on the river                               | Highest: 30–50 per km <sup>2</sup> (80–130 per sq mi) at peak |
| Rolling hills                                     | Valley floors, spring lines, sheltered slopes                             | Villages in valleys; hilltops empty or pasture                                              | Medium                                                        |
| Hill country (farmed valleys with upland grazing) | Valley floors and lower slopes, spring lines                              | Small villages and hamlets in the valleys; scattered farms; open grazing on the hills above | Lower: 8–15 per km <sup>2</sup> (21–39 per sq mi) at peak     |
| High uplands, moors and mountains                 | Sheltered valley floors, sunny terraces, pass feet                        | Hamlets and farmsteads; seasonal huts higher up                                             | Low: 2–5 per km <sup>2</sup> (5–13 per sq mi)                 |
| Coasts                                            | Sheltered bays, estuaries, river mouths, slightly inland on raided coasts | Chains of fishing villages and ports                                                        | Varies                                                        |
| Marsh and fens                                    | Islands of dry ground, fen edges, man-made mounds                         | Villages on the edge; few inside                                                            | Low inside, high on the edge                                  |
| Forest                                            | Clearings, river valleys through the forest                               | Strings of new villages pushing in                                                          | Low, rising as land is cleared                                |
| Steppe and grassland                              | River valleys, steppe edges                                               | Nomad camps; a few trade towns on rivers                                                    | Very low: under 1–2 per km <sup>2</sup> (under 3–5 per sq mi) |
| Desert                                            | Oases, wadis, <i>qanat</i> lines                                          | Isolated clusters; caravan stops                                                            | Near zero between oases                                       |
| Islands                                           | Sheltered side, inland on raided islands                                  | One main port; villages on fertile patches                                                  | Varies                                                        |

| Terrain                                               | Where settlements go                                                                        | Typical pattern                                                                                 | Density (see chapter 02)                                                                                                                                                                                                        |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cold north                                            | Sheltered fjords, coastal lowlands                                                          | Scattered farms; very few towns                                                                 | Very low                                                                                                                                                                                                                        |
| Tropical forest and monsoon uplands                   | Ridges and river banks; towns only on rivers and coasts                                     | Small villages that move their fields every few years (shifting cultivation); paths, not roads  | Low: about 6 per km <sup>2</sup> (14 per sq mi) as a 1950s world average; about 12–35 per km <sup>2</sup> (30–90 per sq mi) in modern northern Laos and Thailand; at most about 55–56 per km <sup>2</sup> (about 145 per sq mi) |
| Wet-rice lowlands and deltas                          | Mounds, dikes, river levees and canal banks between the paddies                             | Continuous rice fields with villages packed along the water lines                               | Very high: Song China averaged roughly 30–40 per km <sup>2</sup> (80–100 per sq mi) over the whole empire; Greater Angkor about 230–300 per km <sup>2</sup> (600–780 per sq mi) over its core (derived)                         |
| Savanna and Sahel (dry grassland south of the Sahara) | Desert-edge river ports, wells, seasonal rivers                                             | Trade towns where caravans meet boats; farming villages by water; herders moving with the rains | Low, higher along rivers                                                                                                                                                                                                        |
| Tundra and northern forest (taiga)                    | Winter villages by rivers and lakes; fishing waters, hunting grounds and pastures in summer | A few winter villages; families scatter to seasonal camps                                       | Extremely low                                                                                                                                                                                                                   |

### Fertile river valleys and plains

This is where most people lived. Villages sit on terraces just above the flood plain, every 1.5–4 km (1–2.5 mi). The flood plain is meadow. Towns grow at crossings and confluences. In England's Midlands (the "Central Province" of Roberts and Wrathmell 2000), large nucleated villages with open fields filled almost all the land.

### Rolling hills

Valleys hold the villages; hilltops hold sheep pasture and sometimes old hill forts. On chalk and limestone, the spring line decides everything. Parishes here often became **strip parishes**: long, thin parishes running from the hilltop pasture, through the village at the spring line, down to the meadows in the valley. Each village gets a slice of every land type.

### Uplands and mountains

- **Two kinds of upland.** Farmed **hill country** (valleys with upland grazing above them) held about 8–15 people per km<sup>2</sup> (21–39 per sq mi) at the peak around 1300. **High uplands, moors and mountains** held only about 2–5 per km<sup>2</sup> (5–13 per sq mi). See chapter 02.
- **Valley floors and terraces** hold permanent villages, usually on the sunny side and on alluvial fans (fan-shaped slopes of gravel where side streams enter the valley) that are safe from the main river's floods.
- **Altitude bands.** In the Valais (Switzerland), hamlets sit on sunny terraces at about 1,000 m (3,300 ft). Higher hay barns and chalets (*mayens*) sit at 1,500–1,600 m (4,900–5,200 ft). Summer pastures start around 2,000 m (6,600 ft).

- **Seasonal settlement.** Herds moved up in summer and down in winter (*transhumance*). This is very old in the Alps: archaeology traces it back to about 3000 BC, and written records mention summer pastures by 1204. In northern England and Scotland the summer huts used for the same purpose are called *shielings*.
- **The upper limit.** Permanent villages above about 2,000 m (6,600 ft) were very rare. Juf in Switzerland, at 2,126 m (6,975 ft), was settled by Walser colonists in 1292 and is usually called the highest permanent village in western Europe.
- **Passes** create towns at both ends (see the table above).

### Coasts

Fishing villages sit in every sheltered cove. Real ports need shelter plus a route inland, usually a river. On coasts with raiders, villages sit a few kilometres inland and uphill, with only boat sheds and a watchtower on the shore. In Sardinia, the village of Posada moved to the foot of its castle hill in the late 15th or early 16th century to escape coastal raids. On many Aegean islands, villages came back down to the shore only after the pirate threat ended, often in the 19th century.

### Marsh and fens

People lived **around** marshes, not in them. The fen edge was rich: fish, eels, reeds, peat, salt and summer grazing. Inside the marsh, settlements sat on islands of higher ground (Ely, Glastonbury) or on man-made mounds. In Friesland and Groningen alone there were about 1,200 *terpen* (dwelling mounds), the first built around 500 BC.

## Forest

In 1000–1300 Europe cleared huge areas of forest. Clearing land for farming was called **assarting** in England (from French *essarter*, to grub out). Clearings show up in place names (-ley, -field, -royd, -sart, -rode). In Germany, new forest villages were often laid out as **Waldhufendörfer** ("forest-strip villages"): a line of farms along a road or stream, each with a long strip of land running back into the forest. This form appeared around 1000 in the northern Black Forest and spread east with German settlers in the 12th–13th centuries.

**A "forest" was not always a forest.** A medieval English "forest" was a **legal** area for royal hunting that could include villages and fields; clearing land in it needed the king's permission. See [Royal forests](#).

## Steppe and grassland

### RULE OF THUMB

On the steppe, people live in small camps of a few tents that move several times a year. Label pasture zones and draw routes, not village dots. Fixed towns are rare and sit on rivers or at the steppe edge.

**Why:** Grass is thin, so herds must keep moving to fresh pasture. A big crowd of animals in one place would eat the grass bare in days.

**Fixed places.** Towns appeared only on big rivers and at the steppe edge, where farmers, nomads and merchants met. Sarai, the Golden Horde's capital, was founded by Batu Khan in the 1240s–1250s on the lower Volga and grew into a large city. Karakorum, the Mongol capital, got its walls in 1235 (see [Capitals, Realms and Borders](#)). Kiev stands on high river bluffs on the Dnieper, near the line where forest meets steppe.

**Camp sizes.** Nomad camps came in three sizes:

| Camp                                                             | What it is                                                                                                                             | Size                                 | Example                                                                                                                                                                    |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Herding camp                                                     | A few round felt tents ( <i>ger</i> or <i>yurt</i> ), one family in each, sharing the herding work                                     | Usually 2–10 families                | The Mongolian <i>khot ail</i> ("camp group") today                                                                                                                         |
| Ring camp                                                        | A circle of tents with the wagons and animals inside and the chief's tent in the middle; used by chiefs and at big seasonal gatherings | A whole clan or tribe                | The Mongol <i>güre'en</i> (also written <i>kuriyen</i> ). The word later meant any fortified camp, and then the Mongol army unit of "a thousand"                           |
| The ruler's camp ( <i>ordu</i> , the origin of the word "horde") | A moving town around the khan's great tent, with officials, guards and traders                                                         | Many thousands of people and animals | Ibn Battuta saw Khan Uzbek's camp in 1332: "a vast city on the move with its inhabitants, with mosques and bazaars in it, and the smoke of the kitchens rising in the air" |

Even a side branch of the court was large. When one of Uzbek's wives travelled to Constantinople in 1332, Ibn Battuta counted an escort of 5,000 troops under an amir (a commander), her own 500 horsemen, about 200 maidser-vants, about 400 carts, 2,000 horses, 300 oxen and 200 camels.

**How often and how far they move.** Medieval figures are scarce, so modern Mongolia is the best guide. (Many modern herders move by truck, which may make some moves longer than in the past.)

| Measure                              | Typical value                                                                          | Source                                                           |
|--------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Camps used in a year                 | Usually about 4–10; GPS loggers on 142 households found 2–14 camps in 9 months         | Teickner et al. 2020; an earlier survey (Ganbold 2015) found 4–8 |
| Distance from one camp to the next   | Typically about 12 km (7 mi); some families move about 36 km (22 mi) or more each time | Teickner et al. 2020                                             |
| Total distance in a year             | Most families under 170 km (105 mi); a few over 300 km (190 mi)                        | Teickner et al. 2020                                             |
| Dry desert pasture (Dzungarian Gobi) | About 9 moves a year; 70–123 km (43–76 mi) between summer and winter camps             | Michler et al. 2022                                              |

### Where the camps go:

- **Winter camps** sit in sheltered places out of the wind, such as valleys and the foot of hills, where the snow is

not too deep for the animals to graze. Each family returns to its own winter site year after year, so these sites have stone pens for the animals. They are the most "fixed" points of nomad life.

- **Summer camps** spread out over open pasture near water: rivers, lakes and wells.
- **Spring and autumn camps** sit between the two, often along the migration route.

The whole system needs a lot of land: steppe nomads lived at under 1–2 people per km<sup>2</sup> (under 3–5 per sq mi) (see [Population and Settlement Sizes](#)).

#### MAP TIP

On a nomad region, do not colour the pastures: a soft tint means a language or faith area. Instead, write italic area labels such as "Summer pastures" and "Winter pastures" across the land they cover. Mark a few winter camps in sheltered river valleys with the nomad-camp symbol: 2–3 tiny **open** half-circles (round tents). Do not use dots, because a tiny dot means a farmstead (or, on a kingdom map, a village). Draw the seasonal migration routes between the pastures as dotted lines with arrows, with wells and fords on the way. Add one or two fixed towns (a Sarai or a Karakorum) on a river, and frontier markets where the steppe meets farmland. If the ruler's camp matters to your story, draw it as a large ring of the same open half-circles with a label such as "Khan's Ordu (summer)".

## Desert

Water decides everything. Settlements form islands and lines: green islands at oases, lines along wadis (dry river-beds that flood sometimes) and along *qanats*. A *qanat* is a gently sloping tunnel that brings groundwater from the foot of the mountains to the fields. They are typical of Iran. Most Iranian qanats are under 5 km (3 mi) long, but some near Kerman measure about 70 km (43 mi). Desert cities such as Yazd and Kerman sit at the foot of mountains where qanats can be dug. Caravan stops sit one day's march apart, about 30–40 km (19–25 mi) (see [Trade Routes, Roads and Transport](#)).

## Islands

An island usually has one main port on the sheltered side, facing the nearest mainland or trade route. Villages sit on fertile patches. On raided islands the main village (the *chora* on Greek islands) sits inland on a hill, often under a castle, with a small harbour below.

## Cold north

Growing grain is hard, so people lived on scattered farms with hay meadows and animals, along sheltered fjords and coastal strips. Towns were very few. Norway had only a handful, such as Bergen, Trondheim and Oslo, all on the coast or a fjord. Medieval Iceland had no towns at all: foreign traders came in summer to seasonal trading places

such as Gásir. At the end of the 11th century, Bishop Gizur Ísleifsson counted about 4,560 farmers in Iceland who were rich enough to pay the assembly tax (the count is recorded in Ari's *Book of the Icelanders*). That gives you a sense of scale: a whole country of a few thousand farms, and not one town. Norse Greenland had around 500 farms in its Eastern Settlement and maybe 100 in its Western Settlement, deep in the fjords. Estimates of its peak population vary: recent ones say about 2,000–3,000, while some older ones were higher.

## Tropical forest and monsoon uplands

**Rule:** In hot, wet forest, farmers usually practise **shifting cultivation** (also called *swidden* or slash-and-burn): they clear and burn a patch of forest, farm it for a year or two, then let it grow back for many years while they clear another patch.

**Why:** Tropical forest soils lose their fertility fast once cleared. A long fallow (rest period) lets the forest rebuild it. So each village needs a large area of forest of different ages around it.

**Numbers:** Population stays low.

| Measure                           | Density                                                                                                     | Source                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| World average, 1950s              | About 6 per km <sup>2</sup> (14 per sq mi), for very roughly 200 million shifting cultivators               | FAO (the UN's food and farming agency), 1957 |
| Modern examples in Southeast Asia | About 12–35 per km <sup>2</sup> (30–90 per sq mi): 12 in northern Laos, 35 in northern Thailand             | Warner 1991 (FAO)                            |
| Ceiling for traditional systems   | About 55–56 per km <sup>2</sup> (about 145 per sq mi); above that, fallows get too short and the soil fails | Ruthenberg 1980, cited by FAO                |

**Pattern:** Small villages on ridges or river banks, linked by footpaths. Some villages move every few years to follow the fields. Towns grow only where boats can reach them: on big rivers and coasts.

### Wet-rice lowlands and deltas

**Rule:** Where monsoon rain or rivers can flood level fields, **wet rice** (rice grown in flooded fields called paddies) feeds far more people per km<sup>2</sup> than European grain land. These are the most crowded farming landscapes of the medieval world.

**Why:** A flooded paddy keeps its fertility year after year and can give one or even two harvests a year. But it needs water control: dikes, canals, reservoirs and terraces. Whoever builds and runs the water works shapes the map (for local water features, see [Villages, Farms and the Countryside](#)).

#### Numbers and examples:

- **Song China** (960–1279) held about 3.1 million km<sup>2</sup> (1.2 million sq mi) in 980. A census of 1102 recorded about 20.5 million households, and estimates of the population around 1100 range from about 90 to over 120 million. That is roughly 30–40 people per km<sup>2</sup> (80–100 per sq mi) over the whole empire (derived), with the densest crowds in the rice lands of the Yangzi valley and the south.
- **Greater Angkor** (Cambodia, peak in the 13th century) had about 700,000–900,000 people spread over about 3,000 km<sup>2</sup> (1,160 sq mi) of houses, ponds, embankments and rice fields (Klassen et al. 2021). That is about 230–300 people per km<sup>2</sup> (600–780 per sq mi) (derived). Lidar (laser scanning from the air) shows a grid of small house mounds and little ponds between the fields.

**Pattern:** An unbroken carpet of paddies. Villages sit on anything slightly higher than the fields: river levees, dikes, canal banks and man-made mounds. Rivers and canals are the roads.

### Savanna and Sahel

The savanna is open grassland with scattered trees; the Sahel is its dry northern edge, next to the Sahara. Rain falls in one short season, so water decides everything.

- **Towns** grow where the desert meets a great river, because there camel caravans from the Sahara meet river boats. **Timbuktu** (Mali) was founded as a seasonal Tuareg camp around 1100 and grew into a great trading and scholarly town about 13–15 km (8–9 mi) north of the Niger, with a river port at Kabara. **Gao** sits on the Niger itself. See the trans-Saharan routes in [Trade Routes, Roads and Transport](#).
- **Farming villages** sit by wells, seasonal rivers and the river flood plain.
- **Herders** move north into the short-lived Sahel pastures in the rainy season, and back south to moister land, rivers and harvested fields in the dry season (a pattern still followed today). It works like the steppe migrations above, but the timing follows rain, not snow.

### Tundra and northern forest

North of the grain line (the tundra, or treeless arctic plain, and the taiga, the great northern conifer forest) nobody farms. In northern Scandinavia, the Sámi lived by hunting wild reindeer, fishing and trapping, and kept small tame herds for transport. Large-scale reindeer herding came later, as many communities shifted to it between the 16th and 19th centuries.

- **The siida.** The traditional Sámi community was the *siida*, a group of up to about 12 families who shared a territory. It moved between several seasonal sites during the year, with a different task at each: fishing, hunting, trapping or grazing.
- **The winter village.** The centre of each *siida* was its winter village. Families and extended families gathered there for the winter, then split into smaller groups for the snowless part of the year. Archaeologists have excavated a chain of eight such winter villages along the small River Nukkumajoki near Lake Inari (northern Finland), dated to the late 16th and early 17th centuries. One of

them (Nukkumajoki 5) had 8 huts (*goahti*) and 7 hearths in a line (Halinen 2019).

- **Summer.** The small groups moved to their own seasonal sites: fishing waters, hunting grounds and pastures.

#### MAP TIP

Draw shifting fields as scattered clearings of different ages around small villages, with forest in between. Draw wet-rice land as a continuous fine grid along rivers and deltas, with villages strung along the dikes and canals. In the savanna, put a trade town where a desert route meets a river. In the arctic, mark a few winter villages by lakes and rivers with the village symbol, with dotted seasonal routes out to the summer camps (2–3 tiny open half-circles, as for nomads); leave the rest empty.

#### FANTASY TWIST

Dwarves change the mountain rule. A dwarf city can be large and deep inside a mountain, but it still needs food, so expect a busy human or dwarf market town at the mountain gate, on the nearest valley floor. Treat it like a pass town. See [Fantasy Variants](#).

#### MAP TIP

Before placing settlements, lightly shade your map by terrain type, in pencil or on a layer you hide later. Then use the table above as a checklist. Plains get dense villages. Hill country gets small villages and hamlets in the valleys; high uplands get only a few. Marshes get a ring of villages around the edge. Deserts get a few oases, each drawn as a green belt of gardens with small palm marks around the town or well, joined by caravan routes. Rice deltas get the densest carpet of all. Leave the high mountains, deep forest, open steppe and tundra mostly empty. These symbols match the guide's [master legend](#).

## Patterns and spacing

### Nucleated, dispersed and linear

| Pattern          | What it looks like                                                             | Where and why                                                        | Real examples                                                                                                       |
|------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Nucleated</b> | Houses grouped tightly around a church, green or crossroads; fields all around | Good arable land farmed in common open fields; or a need for defence | English Midlands "Central Province"; Italian <i>castelli</i> ; much of northern France                              |
| <b>Dispersed</b> | Single farms and small hamlets scattered over the land                         | Uplands, pastoral farming, forest clearings, wet or broken land      | Wales, Devon, Cornwall, the Alps, Scandinavia, much of south-east England                                           |
| <b>Linear</b>    | Houses in a line along a road, stream, dyke, spring line or shore              | Each farm needs access to one line feature                           | German <i>Waldhufendorf</i> and <i>Straßendorf</i> ("street village"); marsh villages along dykes; spring-line rows |

England is a useful model: Roberts and Wrathmell (2000) mapped a central belt of nucleated villages (their "Central Province") running from the north-east down through the Midlands to the south coast, with more dispersed settlement on both sides. A real kingdom usually has more than one pattern. Village layouts in detail are in [Villages, Farms and the Countryside](#).

### Central place theory, made simple

In 1933 the German geographer Walter Christaller studied towns in southern Germany. He noticed that towns of the same rank were spaced fairly evenly, and he built a model to explain why. The key ideas:

- **Central place:** a settlement that sells goods and services to the people around it.

- **Threshold:** the number of customers a service needs to survive. A smith needs few customers; a cathedral or university needs many.
- **Range:** how far people will travel for it. People walk a short way for bread and a long way for a lawyer.
- **Result:** many small centres close together, a few big centres far apart. Each market area is roughly a **hexagon**, because hexagons fill the land with no gaps.

In his main model (called **K = 3**), each step up the hierarchy is spaced  $\sqrt{3}$  (about 1.7) times farther apart than the step below, and serves about 3 times the area.

| Christaller's level (1930s Germany, as his table is usually reproduced) | Spacing between places of this level | Model population (1930s) |
|-------------------------------------------------------------------------|--------------------------------------|--------------------------|
| Market hamlet ( <i>Markort</i> )                                        | ~7 km (4.5 mi)                       | ~1,000                   |
| Township centre ( <i>Amtsort</i> )                                      | ~12 km (7.5 mi)                      | ~2,000                   |
| County seat ( <i>Kreisstadt</i> )                                       | ~21 km (13 mi)                       | ~4,000                   |
| District city ( <i>Bezirksstadt</i> )                                   | ~36 km (22 mi)                       | ~10,000                  |
| Small state capital ( <i>Gaustadt</i> )                                 | ~62 km (39 mi)                       | ~30,000                  |
| Provincial capital ( <i>Provinzstadt</i> )                              | ~108 km (67 mi)                      | ~100,000                 |
| Regional capital ( <i>Landstadt</i> )                                   | ~186 km (116 mi)                     | ~500,000                 |

**Careful:** These are model numbers for the 1930s. Medieval places at each level were much smaller (see [chapter 02](#)), but the **spacing pattern** is a good guide. The medieval English evidence (below) shows market places 9–16 km (6–10 mi) apart, close to Christaller's lower levels.

Christaller had two other versions. In the **transport version** (**K = 4**), smaller towns sit on the straight roads between bigger towns. In the **administrative version** (**K = 7**), each big centre's territory fully contains its smaller centres, as in a county or diocese. Real landscapes mix all three. Outside Europe, G. William Skinner (1964) found a similar pattern in rural China: each basic market town served about 18 villages.

**What it looks like on a map:** Put a city in the middle.

Around it, at roughly equal distances, put six towns.

Between and around those, put market towns. Then **distort** the pattern: stretch it along rivers and roads, squeeze it together on rich land, spread it out on poor land, and break it at mountains and coasts. Real maps never show perfect hexagons, but you can usually see the idea underneath.

### How far apart: the numbers

These spacings come from the shared baseline used across this guide.

| Feature                         | Typical spacing                                                                                                                                                         | Notes                                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Villages, low-land              | 1.5–4 km (1–2.5 mi)                                                                                                                                                     | England c. 1300: one parish per ~14 km <sup>2</sup> (5.4 sq mi), about 4 km (2.5 mi) church to church. Upland parishes cover 50–100+ km <sup>2</sup> (20–40+ sq mi).                                                              |
| Market towns, legal rule        | 6⅔ mi (10.7 km)                                                                                                                                                         | The lawyer Bracton (13th century): a new market should not be closer than this to an old one. It was rarely enforced.                                                                                                             |
| Market towns, in practice       | 9–16 km (6–10 mi)                                                                                                                                                       | 1,746 English markets were recorded by 1300 (one per 75 km <sup>2</sup> (29 sq mi), ~9 km (6 mi) apart), but many were tiny. In the late 16th century there were ~650 working markets in England and Wales, ~16 km (10 mi) apart. |
| Peasant walk to market          | 6–10 km (4–6 mi) one way                                                                                                                                                | There and back in one day, with time to trade.                                                                                                                                                                                    |
| Bridge towns on a lowland river | About every 20 km (12 mi) of river; gaps 10–35 km (6–22 mi)                                                                                                             | Thames, London to Oxford, c. 1300: 9 road bridges in about 180 km (112 mi) of river. Great rivers had far fewer bridges. See <a href="#">How often is a river bridged?</a>                                                        |
| Head ports on a coast           | About one per 100–150 km (60–95 mi) of coast                                                                                                                            | A rough figure from late medieval England's 15 customs head ports; smaller ports and landing places sit between them. See <a href="#">Port ranks</a>                                                                              |
| Towns of 5,000+                 | Flanders 35–45 km (22–28 mi); northern Italy 50–60 km (31–37 mi); France, Germany, Iberia 80–95 km (50–59 mi); Britain 130+ km (80+ mi); Poland 180–250 km (110–155 mi) | An upper bound derived from city data.                                                                                                                                                                                            |
| Cities of 10,000+               | Flanders and Italy 60–90 km (37–56 mi); France, Germany, Iberia 130–170 km (80–105 mi); Britain and Poland 260–400 km (160–250 mi)                                      | Same basis.                                                                                                                                                                                                                       |

**How many in one area?** If places are spaced  $d$  apart in a rough hexagon pattern, each one "owns" about  $0.87 \times d^2$  of land. Using that rule on the table above, here is what you

would expect in a square of  $100 \times 100$  km ( $62 \times 62$  mi), which is 10,000 km<sup>2</sup> (3,860 sq mi):

| Feature           | Region type                                       | Land per place                          | Number in 100 × 100 km |
|-------------------|---------------------------------------------------|-----------------------------------------|------------------------|
| Parish villages   | Lowland farmland                                  | ~14 km <sup>2</sup>                     | ~700                   |
| Parish villages   | Upland                                            | 50–100+ km <sup>2</sup>                 | 100–200                |
| Market places     | England at the c. 1300 peak (many tiny)           | ~75 km <sup>2</sup>                     | ~130                   |
| Market places     | Busy markets only (England and Wales, late 1500s) | ~230 km <sup>2</sup>                    | ~45                    |
| Towns of 5,000+   | Very urban (Flanders)                             | 1,100–1,750 km <sup>2</sup>             | 6–9                    |
| Towns of 5,000+   | Northern Italy                                    | 2,200–3,100 km <sup>2</sup>             | 3–5                    |
| Towns of 5,000+   | France, Germany, Iberia                           | 5,500–7,800 km <sup>2</sup>             | 1–2                    |
| Towns of 5,000+   | Britain / Poland                                  | 15,000+ / 28,000–54,000 km <sup>2</sup> | Under 1 / 0–1          |
| Cities of 10,000+ | Flanders, Italy                                   | 3,100–7,000 km <sup>2</sup>             | 1–3                    |
| Cities of 10,000+ | France, Germany, Iberia                           | 15,000–25,000 km <sup>2</sup>           | Usually 0–1            |
| Cities of 10,000+ | Britain, Poland                                   | 59,000–139,000 km <sup>2</sup>          | Usually none           |

These are rough, derived numbers. Use them as a sanity check, not a rule. Note that England had many markets but few big towns: market density and city density are separate questions.

### Chains along rivers, coasts and roads

Settlements line up along anything that carries traffic:

- **Rivers.** On the Rhine between Mainz and Koblenz (about 90 km (56 mi) by river), small towns follow each other every 5–20 km (3–12 mi): Bingen, Bacharach, Oberwesel, St Goar, Boppard. Many had toll castles above them. Bridges are a separate question: this stretch of the Rhine had ferries, not bridges, but on a settled lowland river such as the Thames a bridge town came about every 20 km (12 mi) (see [How often is a river bridged?](#)).
- **Coasts.** On the southern Baltic, Hanseatic ports (members of the Hanse, a league of north German trading towns) such as Wismar, Rostock and Stralsund sit about 50–70 km (30–45 mi) apart in a straight line, each on its own bay or river mouth. Not every port is equal: late medieval England had about one big customs **head port** per 100–150 km (60–95 mi) of coast, with smaller ports and fishing landings between (see [Port ranks](#)).
- **Roads.** On main roads, inns and stopping places appear every 15–30 km (10–19 mi), so a traveller always finds one within a day's travel. Roman road stations (*mansiones*) were about 25–30 km (16–19 mi) apart, and many later towns grew on them.

### Clusters and empty spaces

Settlement is never even. In England in 1086, Suffolk had about 23–25 people per km<sup>2</sup> (60–65 per sq mi), while much of the north and west had under 5 per km<sup>2</sup> (under 13 per sq

mi). Good land is crowded; poor land is nearly empty. A map with evenly spaced dots looks false.

#### MAP TIP

First decide which pattern each region has: nucleated, dispersed or linear. Then use the spacing table. Draw villages close together on good lowland, in lines along springs and rivers, and widely spaced in uplands. Put market towns about 9–16 km (6–10 mi) apart, so that every village can walk to a market and back in one day. Space larger towns by the regional row in the table. Finally, check the "how many" table. If your 100 × 100 km (62 × 62 mi) farming region has only five villages, it is far too empty. If it has three cities, it is too full, unless it is your Flanders.

## How settlements change over time

A real map is a snapshot of a long story. Showing that story makes a map feel alive.

### Reuse of ancient and Roman sites

Old sites were reused because they already had walls, roads, a bridge or a bishop. In Italy, southern France and the Rhineland, most Roman cities stayed cities: Cologne, Trier, Lyon, Milan. In Britain many Roman towns shrank to almost nothing after 400 and later revived (London, York, Winchester). Some never came back:

- **Silchester** (Roman *Calleva*) was abandoned sometime between about 550 and 650 and never became a town again. A small medieval village grew up beside it and was deserted around 1400. Today its Roman walls still

stand around open fields, with only a church and a farm inside.

- **Wroxeter** (Roman *Viroconium*, probably the fourth-largest town in Roman Britain) faded out in the late 7th or early 8th century. **Shrewsbury**, 8 km (5 mi) away in a strong river loop, became the main town of the area instead.

### Planted "new towns" and bastides

In the High Middle Ages lords founded towns on purpose, to earn rents, tolls and market dues and to control land. Planted towns are easy to spot: a **grid** or a regular street plan, a big **central market place**, and often a name with "New", "Ville-", "-ville", "Villa Nova" or "Bastide".

| Region                       | How many                                                                                                                                                                                                                                                   | Examples                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| England (1066–1370)          | About 170 planted towns (Beresford 1967)                                                                                                                                                                                                                   | Bury St Edmunds (grid laid out c. 1080 by Abbot Baldwin; Domesday Book records 342 houses built on former ploughland by 1086); New Salisbury (1220s); New Winchelsea (charter 1281, laid out from 1288) |
| Wales                        | More than 80 (Beresford)                                                                                                                                                                                                                                   | Edward I's castle towns in North Wales (see <a href="#">chapter 04</a> )                                                                                                                                |
| Gascony and southwest France | About 125 in English Gascony (Beresford); almost 700 <b>bastides</b> (planned new towns of southern France) between 1222 and 1372                                                                                                                          | Cordes (1222, usually called the first); Monpazier (1284, Edward I); Villeneuve-sur-Lot                                                                                                                 |
| German lands                 | Central Europe had only a few hundred towns around 1100. Very many were founded from 1120 onwards, mostly in the next 250 years. Estimates vary with the definition of "town", but by the late Middle Ages there were several thousand, most of them small | Freiburg im Breisgau (1120); Bern (1191); Lübeck (founded 1143, refounded 1159)                                                                                                                         |

Many planted towns failed. A lord could grant a charter, but he could not force traders to come if the situation was poor.

- **Monastic town:** Bury St Edmunds; St Albans; Ely. See [Religious, Cultural, Legal and Ancient Sites](#).

### Castle-gate and monastic towns

Castles and abbeys were big customers. They needed food, crafts, labour and services, so a town often grew at the gate.

- **Castle-gate town:** Ludlow (castle begun c. 1085; a planned town laid out in the 12th century). See [Castles, Forts and Military Outposts](#).

## Why some places grow and others shrink

| Cause                 | What happens                                                                                    | Real example                                                                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silting               | The harbour fills with mud; ships cannot reach the town                                         | Bruges: the Zwin inlet began silting in the 13th century. By the 14th century big ships could no longer reach Damme and used Sluis instead. In the 15th century trade suffered badly, and it moved to Antwerp (the Hanseatic merchants moved their office there in 1520) |
| The sea               | Storms and erosion destroy the town                                                             | Dunwich (England): storms in 1286–87, 1328, 1347 and 1362 destroyed much of it; a leading port became a village. Old Winchelsea was destroyed by storms in 1287–88 and rebuilt on a hill                                                                                 |
| A blocked river       | A weir (a low dam across a river) or a bridge cuts the town off from the sea                    | Exeter: weirs built in the 1270s–80s and in 1317 cut off its port; ships had to unload at Topsham                                                                                                                                                                        |
| A better site nearby  | People move a short distance                                                                    | Old Sarum to Salisbury (1220s)                                                                                                                                                                                                                                           |
| A new route           | A new bridge, road or pass shifts traffic                                                       | The first Schöllenen bridge (around 1220) opened the Gotthard pass and brought trade to central Switzerland                                                                                                                                                              |
| Plague and low prices | Too few people to farm all the land; the worst land is abandoned                                | The Black Death (1347–51) and later plagues                                                                                                                                                                                                                              |
| Conversion to pasture | Lords turn arable land into sheep runs                                                          | Many deserted villages in the English Midlands, mostly in the 15th century                                                                                                                                                                                               |
| Politics              | A ruler moves the court, a bishop moves his see (the seat of his diocese), a war ruins a region | See <a href="#">Capitals, Realms and Borders</a>                                                                                                                                                                                                                         |

## Deserted medieval villages

England has **more than 3,000** known deserted medieval villages, counting all dates and causes (estimates vary). Most were deserted between about 1350 and 1520, especially in the 15th century, when landlords turned arable land into sheep pasture. Others were lost to floods, sand, the sea, or later to parks around great houses. The plague usually weakened a village; sheep or a landlord finished it. The best-studied is **Wharham Percy** in Yorkshire, on the western edge of the chalk Wolds. It shrank after the Black Death, and its last villagers were evicted for sheep pasture in the early 1500s. Archaeologists excavated it every year from 1950 to 1990. Today such sites show as bumps in fields: house platforms, sunken lanes and a lonely church.

### LATER ERA (1500S+)

New planned towns appear again, now for war and for princely show: Valletta (Malta, 1566, after the Great Siege of 1565), Palmanova (Italy, 1593, a star-shaped fortress town), Freudenstadt (Germany, 1599), Charleville (France, 1606). Engineers also make new land and new routes. The Exeter Canal (1564–66) bypassed the blocking weirs and brought small ships back to the city. Ocean trade makes Atlantic ports such as Seville, Lisbon, Antwerp and then Amsterdam grow fast. Seville's river was hard for big ships, and silting made it worse by the 1620s, so more and more of the American trade moved downstream to Sanlúcar and the Bay of Cádiz. See [What Changes After 1500](#).

### MAP TIP

Add history to your map. Mark a few ruins, each drawn with its own symbol broken or grey and labelled "(ruin)": for example an "Old Sarum"-style hill fort above a newer valley town. Draw one or two planted towns with a grid plan and names like "Newbridge" or "Villeneuve". Leave a deserted village or two in sheep country, drawn as a grey church alone in a field and labelled "(lost village)", as in the [master legend](#). If your map is set after a plague or war, show fewer villages on poor land and more sheep pasture.

## Where people do not build

Empty space is as realistic as settled space. Here is where medieval people avoided building, and what you find there instead.

| Place                                                  | Why people avoid it                   | What you find there instead                                                        |
|--------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------|
| Active flood plain                                     | Floods every winter                   | Hay meadows, mills at the edge, causeways                                          |
| Fever marsh                                            | Malaria; no firm ground               | Fishing huts, salt pans, summer grazing; villages on the edge                      |
| High mountains (above ~2,000 m (6,600 ft) in the Alps) | Too cold for crops; snow for months   | Summer huts and pastures only                                                      |
| Shady valley sides                                     | Short growing season; long snow cover | Forest                                                                             |
| Raided coasts                                          | Pirates and slave raiders             | Watchtowers, beacons, boat sheds; villages a few km inland                         |
| Royal forests and hunting parks                        | Laws limit clearing and building      | Hunting lodges, keepers' houses, a few old villages                                |
| Poor soils (sandy heath, thin chalk, bare rock)        | Crops fail                            | Sheep walks, rabbit warrens, a few hamlets                                         |
| Contested borders                                      | War and raids                         | Castles and empty "march" land (see <a href="#">Capitals, Realms and Borders</a> ) |
| Deep old forest                                        | Clearing takes generations            | Charcoal burners, swineherds, hermits, slowly growing clearings                    |
| Desert away from water                                 | No water at all                       | Nothing, except caravan stops at wells                                             |
| Right next to an existing market                       | Competition and legal rules           | Villages, but not another market town                                              |

### RULE OF THUMB

On a realistic map, a lot of land is empty or only lightly used. Mountains, marsh cores, heaths, forests and borders should look thin.

### LATER ERA (1500s+)

Two changes. First, drainage begins to turn empty marshes and shallow lakes into farmland with new farms and villages (the Beemster lake in Holland, drained by 1612; the English Fens, 1630s–1650s). Second, Barbary corsair (North African pirate) raids make some Mediterranean coasts **more** dangerous: Genoa built about 85–90 watchtowers around the coast of Corsica between 1530 and 1620 (a 1617 list counts 86), and coastal villages stayed inland and uphill. See [What Changes After 1500](#).

### FANTASY TWIST

Monsters push people together. Where wolves, trolls or raiding beasts are real dangers, expect fewer lone farms, more walled villages, and settlements on defensible sites even in peaceful kingdoms. The "empty" lands become truly empty: no charcoal burners in the cursed forest. See [Fantasy Variants](#).

### MAP TIP

After you place settlements, look for "holes" in the pattern. Each hole should have a reason you could explain: marsh, mountain, forest, heath, a royal forest, a dangerous border. If a hole has no reason, add villages. If a crowded area sits on bad land, remove some.

## Naming settlements realistically

Names are the first thing a reader sees on a map. Real names follow patterns, and those patterns tell the story of who settled the land.

### RULE OF THUMB

Most village names are plain. They join a common ending ("farm", "homestead", "clearing") to a person's name, a tree, an animal or a direction. Each region uses a small set of endings again and again. Save dramatic names for the few places that earned them.

**Why:** The first settlers simply described the place or its owner: "Acca's farm", "the oak clearing", "the north farm". Once a name was in use and written down, it rarely changed. Domesday Book (1086) already names 13,418 places, covering most of England except the far north, so

most English village names are older than the castles and abbeys on your map.

### Common English endings

These endings come from Old English, the language of the Anglo-Saxons.

| Ending                     | Meaning                                                                                  | Notes                                                    | Examples                                                                              |
|----------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| -ton                       | Farmstead, estate, village (Old English <i>tūn</i> )                                     | The commonest of all: over 4,000 places                  | Acton ("oak farm"), Wootton ("wood farm"), Newton ("new farm"), Sutton ("south farm") |
| -ham                       | Homestead, village ( <i>hām</i> )                                                        | Often among the older names                              | Nottingham, Dagenham                                                                  |
| -wick, -wich               | Specialist farm (often dairy), or trading place ( <i>wic</i> , from Latin <i>vicus</i> ) | Inland usually a farm; on coasts and rivers often a port | Keswick ("cheese farm"), Ipswich, Norwich                                             |
| -worth                     | Enclosure                                                                                | —                                                        | Tamworth, Kenilworth                                                                  |
| -stead                     | Place, site                                                                              | —                                                        | Hampstead                                                                             |
| -cot, -cote                | Cottage, hut                                                                             | Small places                                             | Didcot, Ascot                                                                         |
| -bury, -borough, -burgh    | Fortified place ( <i>burh</i> ); later a town                                            | Use it at old forts, earthworks or fortified towns       | Canterbury, Peterborough                                                              |
| -chester, -caster, -cester | Roman fort or town ( <i>ceaster</i> , from Latin <i>castra</i> )                         | Only on Roman sites                                      | Chester, Lancaster, Gloucester                                                        |
| -ley, -leigh               | Wood clearing, woodland pasture ( <i>lēah</i> )                                          | Common in once-wooded country                            | Bradley ("broad clearing"), Henley                                                    |
| -field                     | Open land without trees                                                                  | —                                                        | Sheffield, Lichfield                                                                  |
| -den, -don                 | Valley ( <i>denu</i> ); hill ( <i>dūn</i> )                                              | Easy to confuse; check the terrain you drew              | Croydon (a valley name)                                                               |
| -hurst                     | Wooded hill                                                                              | Wooded country                                           | Lyndhurst, Sandhurst                                                                  |
| -stow                      | Meeting place, holy place                                                                | —                                                        | Bristol ("place at the bridge"), Padstow                                              |
| -minster                   | Mother church, monastery                                                                 | One per large district (see <a href="#">chapter 08</a> ) | Westminster, Axminster                                                                |

**The Norse (Viking) layer.** Danish and Norwegian settlers of the 9th–11th centuries added their own endings in northern and eastern England:

| Ending          | Meaning                             | How many                                                                                                              | Examples                  |
|-----------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------|
| -by             | Farm, village                       | More than 600 places in England (Baugh and Cable); Domesday already records 303 in the East Midlands alone (Cameron)  | Grimsby, Whitby, Derby    |
| -thorpe, -thorp | Secondary settlement, outlying farm | About 300                                                                                                             | Scunthorpe, Mablethorpe   |
| -thwaite        | Clearing, meadow                    | About 300; mostly in the north-west (Cumberland, Westmorland, Lancashire), where Norwegians settled, and in Yorkshire | Braithwaite, Applethwaite |
| -toft           | Building plot, homestead            | —                                                                                                                     | Lowestoft                 |
| -holm(e)        | Island, dry ground in wet land      | —                                                                                                                     | Axholme                   |
| Kirk-           | Church                              | —                                                                                                                     | Kirkby                    |

The Normans, by contrast, gave few new village names in England, because the land was already named. They named their new castles and abbeys instead: Richmond in York-

shire ("strong hill", founded 1071 by Alan Rufus, probably after a Richemont in Normandy), Belvoir ("fine view"), Beaulieu ("fine place").

## Names in other lands

| Region and language                                   | Common elements (meaning)                                                                                                                                                                                                                                                                                | Examples                                                                                                              |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Celtic: Wales, Scotland, Ireland, Cornwall, Brittany  | Aber- (Welsh) and Inver- (Gaelic): river mouth; Llan- and Kil-: church; Caer- and Dun-: fort; Pen-: head, hilltop, end; Tre- and Bally-: farm, settlement; Strath-: broad valley                                                                                                                         | Aberystwyth, Inverness, Llandaff, Kilkenny, Caernarfon, Dundee, Penzance, Tredegar, Ballymena, Strathclyde            |
| German lands                                          | -ingen ("people of"; mostly 6th–9th centuries), -heim (home), -dorf (village), -hausen (houses), -hof (farm), -weiler (hamlet), -stadt or -stedt (place, town), -burg (fortress); clearing names -rode, -rath, -reuth, -ried (high-medieval clearance); -hagen (hedged settlement)                       | Tübingen, Mannheim, Düsseldorf, Mühlhausen, Darmstadt, Magdeburg, Wernigerode, Bayreuth                               |
| East of the Elbe (Slavic names, later Germanized)     | -itz, -ow, -in                                                                                                                                                                                                                                                                                           | Chemnitz, Güstrow, Schwerin                                                                                           |
| France                                                | Gallo-Roman <i>-acum</i> ("estate of") became -ac in the south and -y or -ay in the north; -court (farm with a yard) and -ville (farm, later village) from the early Middle Ages; in Normandy, Viking-age -tot (house site; over 300 places), -beuf (booth, shed) and -fleur (stream running to the sea) | Cognac, Bergerac; Vitry; Harcourt, Abbeville; Yvetot, Elbeuf, Honfleur                                                |
| Italy                                                 | -ano (a Roman estate named after its owner, Latin <i>-anum</i> ), Castel- and Rocca (fortified village, rock fortress), Borgo (market settlement or new town), Pieve (rural mother church)                                                                                                               | Castelfranco, Rocca di Papa, Borgo San Sepolcro (today Sansepolcro), Pieve di Cadore                                  |
| Iberia, Arabic layer                                  | Al- ("the"); Medina- ( <i>madina</i> , city); Alcalá ( <i>al-qal'a</i> , castle); Alcázar ( <i>al-qasr</i> , palace-fortress); Guad- ( <i>wadi</i> , river valley); Almadén ( <i>al-ma'din</i> , "the mine")                                                                                             | Medinaceli, Medina del Campo, Alcalá de Henares, Guadalquivir ( <i>al-wadi al-kabir</i> , "the great river"), Almadén |
| Iberia, Christian resettlement ( <i>repoblación</i> ) | Villanueva ("new town"), Villafranca ("free town"), Castro- (castle)                                                                                                                                                                                                                                     | Villanueva de la Serena, Villafranca del Bierzo, Castrojeriz                                                          |
| Slavic lands                                          | -grad, -gorod, -gród (fort, town); -ice, -owice ("people of", from a founder's name); -ów, -ov (possessive: "X's place")                                                                                                                                                                                 | Novgorod ("new town"), Belgrade ("white town"), Kraków ("Krak's town")                                                |

## How names pile up in layers

- Rivers and big hills keep the oldest names.** In England the oldest place names are river names, many of them Brittonic (the Celtic language spoken before English). *Avon* simply means "river". Use your oldest language for rivers and mountains.
- Villages carry the language of the people who founded them.** Danish settlers left -by and -thorpe in eastern England; Norwegians left -thwaite in the north-west. Where the older language survived longer, as in Cornwall and Cumbria, many village names stayed Brittonic. So a border between two kinds of names is often an old settlement frontier, not a political border.
- Later settlements fill the gaps.** Clearing names (-ley, -thwaite, -rode, -reuth) mark land taken from the forest; in Germany the -rode and -reuth names belong mainly to the great clearances of the High Middle Ages (see [Forest](#)). Planted towns get "new" names: Newport, Villeneuve, Neustadt, Villanueva, Novgorod.
- Same-name villages get extra words.** Two neighbouring Suttons or Stokes need telling apart. Common extras:
  - the **lord's family**: Stoke Mandeville (the suffix is first recorded in 1284, when the Mandeville family held the

manor); Sutton Courtenay (after the Courtenay family took the manor in the 1170s);

- **size or position**: Great and Little, Upper and Lower, North and South;
- a **market**: Market Harborough, Chipping Norton (*chipping* = market);
- a **church or royal owner**: Bishop's Stortford, Abbots Langley, King's Lynn;
- a **river**: Stratford-upon-Avon, Newcastle upon Tyne.

- Owners change, names follow.** King's Lynn was "Bishop's Lynn" until 1537, when the town passed from the Bishop of Norwich to Henry VIII.

**Endings change at language lines.** Where two languages meet, the endings switch within a few villages, even inside one realm. In Belgium, north of the Romance–Germanic line, many Flemish village names end in Germanic -gem ("home", like German -heim) or -zele ("house, settlement", as in Brussels; spelled -zeele across the border in French Flanders): Wevelgem, Zottegem, Herzele, Bollezeele. South of the line, Romance names such as Soignies and Ottignies ("Otto's lands": a Frankish owner's name with a Romance ending) take over; in Lorraine, -ange (the French spelling of German -ingen: Hayange is German *Hayingen*) on the Ger-

man-speaking side faces -ville and -court on the French side, and two villages called Audun were told apart as Audun-le-Tiche ("German Audun") and Audun-le-Roman ("Romance Audun"). See [Language lines are not borders](#).

### A method for naming a region

1. **Write the region's history in two lines.** Who lived here first? Who came later? Who rules now? Choose 1–3 language layers.
2. **Name rivers and mountains** from the oldest layer.
3. **Pick 4–8 common endings** from the main settler layer, and build most village names from them. Join them to personal names, trees, animals, colours and directions. (This is a mapmaking suggestion, not a measured rule, but real regions work this way: in England, -ton alone names over 4,000 places.)
4. **Use site words only where the map shows the site:** -ford at a ford, -bridge at a bridge, -mouth at a river mouth, -ey at an island (see [Reading place names](#)).
5. **Save special elements for special places.** Fort words (-bury, Caer-, Dun-, -grad, Alcalá) go on places with a fort or an old earthwork. Market words (Market-, Chipping-) go on market towns. Church words (Kirk-, Llan-, -min-

ster, Pieve) go on places with an important church. Roman words (-chester) go only on Roman sites.

6. **Show history with patches.** Put a band of second-layer names along the coast where raiders settled. Use clearing names in old forest. Mix both sides' names in a march.
7. **Split duplicates with extra words** (Great/Little, a lord's name, a river).

#### FANTASY TWIST

Give each people its own set of endings, and let border zones mix them. A valley where dwarf names sit under older elf river names, with human "-ton" villages on the valley floor, tells the reader three chapters of history before they read a word of your book. See [Fantasy Variants](#).

#### MAP TIP

Keep a name sheet for each region: its 4–8 endings, its old river words and 10–20 first parts. Reuse them. Real maps repeat themselves (England has many Suttons, Nortons and Newtons), and that repetition is what makes a region feel like one place. Readers notice when the names in one corner of the map share an "accent".

## Placing settlements on your map: a method

Work from big to small, and from nature to people. Start with a finished terrain map: coasts, mountains, rivers, forests, marshes.

1. **Mark the water.** Draw all rivers. Decide which stretches are **navigable** (deep enough for boats) and where they stop: falls, rapids, shallows. Mark fords and the tidal limit on each major river.
2. **Mark the land quality.** Lightly shade good farmland (river valleys, plains, gentle hills), pasture land (uplands, heath), marsh, forest, mountain and desert, in pencil or on a layer you hide later.
3. **Place the great city or cities.** Look for the best situation in the richest region: the lowest bridging point or estuary of a big navigable river, or a superb harbour. Remember that a city of 50,000+ must be fed by water.
4. **Place regional towns.** Put them at the next-best nodes: confluences, heads of navigation, gap towns, pass towns, harbours, crossings of major roads. Space them by the regional row in the spacing table. Cities of 10,000+ are 60–90 km (37–56 mi) apart in a very urban region, 130–170 km (80–105 mi) in an average one, and 260–400 km (160–250 mi) in a thinly urban one.

5. **Place market towns.** Fill the farmland with market towns about 9–16 km (6–10 mi) apart, on rivers and road junctions where possible. Spread them more widely on poor land.
6. **Place villages.** Fill good land with villages 1.5–4 km (1–2.5 mi) apart, on terraces, spring lines and land edges. In uplands, use hamlets and single farms in valleys. Leave marsh cores, mountains and heaths nearly empty.
7. **Add special sites.** Castles ([chapter 04](#)), monasteries and shrines ([chapter 08](#)), mines, salt works and other special towns ([chapter 07](#)). These are the main reasons for a settlement in an "odd" place.
8. **Connect them.** Draw roads and river routes ([chapter 05](#)). Small adjustments are fine: shift a village to the road, or a town to the bridge.
9. **Add time.** Mark ruins, deserted villages, moved towns and planted towns.
10. **Run the test below on every settlement.**

### The "why is it here?" test

Ask these questions about every dot. A **village** must pass 1–3. A **town** must also pass 4. A **city** must pass all six.

| # | Question                                                                  | If the answer is "no"                                                               |
|---|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | Is there fresh water on or next to the site?                              | Move it to a river, stream, spring or lake shore.                                   |
| 2 | Is there a mix of farmland, meadow, pasture and wood within a few km?     | Move it to an edge between land types, or make it a special site (mine, monastery). |
| 3 | Is it safe from floods (on a terrace, rise or island)?                    | Move it up a few metres onto a terrace or rise.                                     |
| 4 | Do routes meet here (crossing, confluence, junction, harbour, gap, pass)? | Make it a village, or move it to the crossing.                                      |
| 5 | Is it on navigable water or the coast?                                    | Make it a town under ~10,000 people.                                                |
| 6 | Is it the best-connected place in a rich region?                          | Make it a regional town, not a great city.                                          |

Also check:

- **Defence:** In a dangerous region, is there a hill, a river loop or a wall? If not, why not?
- **Spacing:** Is it too close to a place of the same rank? Two market towns 3 km (2 mi) apart compete; one should be a village.
- **Name:** Does the name fit the site (ford, bridge, mouth, hill) and the region's language layer? See [Naming settlements realistically](#).

#### FANTASY TWIST

A magical reason, such as a ley-line nexus, a dragon's peace treaty or a portal, can replace test 4 or 6. It cannot replace tests 1–3. Even a wizard's city needs water and food, or a very good explanation of where they come from.

#### MAP TIP

Keep a short list next to your map: settlement name, rank, site type, and one-line situation ("Kingsford: town; ford and confluence; where the north road meets the Silverwater"). It takes a few minutes and catches most mistakes. More common errors are in [Common Mistakes and How to Fix Them](#). The full workflow is in [Step-by-step](#).

## Quick summary

- Every settlement needs **water, food, fuel, safety, access and health**. Villages need the first three nearby; towns also need access; cities need access **by water**.
- Villages sit on **edges** between land types (valley and hill, fen and dry land, downland and clay), on **river terraces** above the flood plain, and along **spring lines**.
- **Site** (the spot) decides if a place survives. **Situation** (its place in the network) decides if it grows big.
- The best town sites are **lowest bridging points, heads of navigation, confluences, gaps, pass feet, sheltered harbours and estuaries**. On tidal coasts, big ports often sat 50–120 km (30–75 mi) up an estuary.
- In dangerous times and places, people choose **hills, river loops, islands and spurs**.
- Spacing: villages **1.5–4 km (1–2.5 mi)**, market towns **9–16 km (6–10 mi)**, bridge towns on a lowland river **about 20 km (12 mi)**, head ports **about one per 100–150 km (60–95 mi) of coast**, towns and cities much farther apart, depending on how urban the region is.
- Christaller's **central place** model: many small places close together, a few big ones far apart, in rough hexagons. Distort it along rivers and roads.
- Settlement is **clustered on good land and thin on bad land**. Empty areas need a reason, and so do crowded ones.
- Outside Europe: **shifting cultivators** live thinly (about 6 per km<sup>2</sup> as a 1950s world average, 12–35 in modern northern Laos and Thailand, at most about 55–56), **wet-rice lands** are the most crowded farmland of all (Angkor about 230–300 per km<sup>2</sup> over its core), and **steppe nomads** use small camps of 2–10 families that move about 4–10 times a year, with fixed winter sites.
- Places change: Roman sites reused or abandoned, **planted towns** with grids, towns moved, harbours silted, and **more than 3,000 deserted villages** in England alone.
- **Names** come in layers: rivers keep the oldest names, villages use a few common endings of their founders' language (-ton, -by, -dorf, -ac), extra words (Great/Little, a lord's name, a river) split duplicates, and the endings switch at language lines (-gem and -ange on one side, -ville and -court on the other).

- Use the "why is it here?" test on every dot.

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## CHAPTER 2

# POPULATION AND SETTLEMENT SIZES

This chapter gives you the numbers: how many people live in a village, a town or a city, how many of each your realm should have, and how crowded the land can be. All population and size figures used elsewhere in this guide come from here. The baseline is Europe in the High and Late Middle Ages (c. 1000–1500), with the Mediterranean and Middle East added where they help, plus a short section on Asia, Africa and the Americas. The numbers are real historical estimates, given as ranges because scholars disagree.

## IN THIS CHAPTER

- The settlement ladder
- How many of each kind
- Population density
- Carrying capacity: how much land feeds people
- Realm populations and areas
- The biggest cities
- Beyond Europe: Asia, Africa and the Americas
- Step-by-step population calculator
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- Who lives where
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- Quick summary
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## The settlement ladder

### RULE OF THUMB

Almost everyone lives in villages. A "big city" in medieval Europe is anything above 10,000 people. A place of 50,000 is one of the great cities of the known world.

**Why:** Before modern farming, most families had to farm just to feed themselves. Across Europe 85–95% of people lived in the countryside. So only about one family in ten (at best one in seven) could live mainly from a craft, trade or

service. Towns therefore stayed small and few. A place grew big only when it had something special: a ruler's court, a bishop, a great port, a river junction, or an export industry (see [Where Settlements Are Built](#)).

**Numbers.** Population means people, not households. A household is about 4.5–5 people. Medieval tax records often count *hearths* (households) instead of people, so multiply by about 4.5 to get people.

| Tier               | People                                    | Households   | What it usually has                                                                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isolated farmstead | 5–15                                      | 1–2          | Its own fields and pasture. No church. The main pattern in uplands, Wales, Devon, the Alps, Scandinavia and forest clearings.                                                                                                                                                         |
| Hamlet             | 10–75                                     | 2–15         | A few houses. A chapel at most. Belongs to a nearby village's parish (the area served by one church).                                                                                                                                                                                 |
| Village            | 75–500 (English lowland: usually 150–300) | 15–100       | Parish church, manor house (the local lord's house), a watermill or windmill, a smithy, open fields and common pasture. Sometimes an alehouse.                                                                                                                                        |
| Small market town  | 500–2,000                                 | 100–400      | A chartered weekly market and a yearly fair, an open market place, burgage plots (long narrow house plots rented by townspeople), inns, specialist craftsmen, one parish church, often a hospital (a charity house for the sick, old or poor).                                        |
| Town               | 2,000–10,000                              | 400–2,000    | Usually walled. Several parishes, merchant and craft guilds (associations that controlled trade and crafts), friaries (houses of preaching friars, found in towns from the 1220s), hospitals, often a castle, a borough charter (a written grant of town rights) and its own council. |
| City               | 10,000–50,000                             | 2,000–10,000 | Cathedral (a bishop's main church), many parishes and friaries, suburbs outside the walls, strong guilds, law courts, a mint (where coins are made); sometimes a university.                                                                                                          |
| Great city         | 50,000+                                   | 10,000+      | A capital or a major port. Several markets, a university, quarters of foreign merchants, an organised water supply. Its grain must arrive by river or sea.                                                                                                                            |

### Real examples:

- **England in 1377** (after the plague). These estimates come from the 1377 poll tax, a tax on every person over 14.
  - London had about 35,000 people; York about 11,000; Bristol about 9,500; Coventry about 7,000; Norwich about 6,000.
  - About 10 more towns had roughly 3,500–5,500 (for example Lincoln, Salisbury, King's Lynn, Colchester and Boston).
  - Outside London, only York passed 10,000. About 5 towns had 5,000–10,000, about 24 had 2,000–5,000, and many more had 1,000–2,000.
  - Before the Black Death, all of these were about 1.5–2 times larger.
- **Universities were rare:** about 20 in all of Europe c. 1300 and about 60–80 by 1500. (Over 80 had been founded by 1500, but not all of them lasted.)

**Regional differences:** The ladder above fits England, northern France and Germany best. Elsewhere:

- **Italy:** *castelli* were fortified hilltop villages of 200–1,000 people. Italy also had hundreds of small bishoprics, so a cathedral does not prove a big city there. England had only 17 bishoprics (sees), so there a cathedral does mean an important city.
- **Southern Spain, Sicily and Apulia:** *agro-towns* had 2,000–10,000 people, but most of them were farmers who walked out to their fields every day. They are large in population but not real towns in function.
- **Germany and the Low Countries:** many places with a town charter ("legal towns") had fewer than 2,000 people.

- **Byzantine lands:** most villages were small, usually well under 100 households (estimate). Even Lampsakos, a small town on the Dardanelles, had only about 173 households in one medieval record, and about a third of them did town work rather than farming.

### How big to draw it

Medieval towns were small on the ground. Historians often assume about 100–200 people per hectare (40–80 per acre) inside a town. (A hectare is 100 × 100 m, about 2.5 acres.) One major city database uses 150 per hectare, and 75 for spacious "garden cities" such as Baghdad, Basra and Sanaa (Bosker et al. 2013). Bigger cities were denser than small ones (Cesaretti et al. 2016).

| Place                                                            | People                                         | Built-up area                                                                                                                                         | Width if roughly round                                                         |
|------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Village                                                          | 150–300                                        | 5–15 ha (12–37 acres), houses spread out with gardens                                                                                                 | 250–450 m (0.15–0.3 mi)                                                        |
| Small market town                                                | 500–2,000                                      | 5–20 ha (12–50 acres)                                                                                                                                 | 250–500 m (0.15–0.3 mi)                                                        |
| Town                                                             | 2,000–10,000                                   | 20–80 ha (50–200 acres)                                                                                                                               | 0.5–1 km (0.3–0.6 mi)                                                          |
| City                                                             | 10,000–50,000                                  | 80–300 ha (200–740 acres)                                                                                                                             | 1–2 km (0.6–1.2 mi)                                                            |
| Great city                                                       | 50,000–200,000                                 | 300–1,000+ ha (740–2,500+ acres)                                                                                                                      | 2–4 km (1.2–2.5 mi)                                                            |
| Packed island city (Tenochtitlan, Mexico, 1519)                  | 200,000–250,000 (some estimates up to 400,000) | 800–1,350 ha (8–13.5 km <sup>2</sup> , 3.1–5.2 sq mi)                                                                                                 | 3–4 km (2–2.5 mi)                                                              |
| Dispersed "garden" city (Greater Angkor, Cambodia, 13th century) | 700,000–900,000                                | About 3,000 km <sup>2</sup> (1,160 sq mi) of house mounds, ponds, embankments and rice fields around a walled core of 9 km <sup>2</sup> (Angkor Thom) | About 60 km (37 mi) for the whole zone; the walled core is 3 × 3 km (2 × 2 mi) |

Checks against real places:

- **Bristol c. 1300:** about 130 ha including its suburbs, but only 55 ha inside the walls.
- **Paris:** the wall of Philip Augustus (built 1190–1213) enclosed 253 ha.
- **Florence:** its last medieval wall (finished 1333) enclosed about 430 ha for roughly 100,000 people.

The village row is a rough estimate.

**The two extremes.** The last two rows show how far city density can stretch. Tenochtitlan packed about 150–300

people per hectare onto its island (derived; Smith 2005 estimates 212,500 people on 13.5 km<sup>2</sup>, about 160 per ha). That is at or above the top of the European range. Greater Angkor averaged only about 2–3 people per hectare over its whole zone: about 50 per ha inside the walls of Angkor Thom, but only about 1.5 per ha in the wide outer zone of houses, ponds and fields (Klassen et al. 2021). Most medieval cities sit between these two: 100–200 per ha for walled towns and about 75 per ha for Islamic "garden cities" such as Baghdad. See [Beyond Europe](#) for more on both cities.

### Map symbols and labels

| Tier              | Continent map                     | Kingdom map                                                    | Local map                        | Label size                       |
|-------------------|-----------------------------------|----------------------------------------------------------------|----------------------------------|----------------------------------|
| Farmstead         | —                                 | —                                                              | Tiny dot                         | None                             |
| Hamlet            | —                                 | —                                                              | 2–5 small house marks, no church | 6–7 pt italic, or none           |
| Village           | —                                 | A tiny dot, only for notable villages                          | House marks around a church      | 7–8 pt                           |
| Small market town | —                                 | Small open circle                                              | Houses around a market square    | 8–9 pt                           |
| Town              | Only the important ones, as a dot | Circle with a ring; battlements on the ring if walled          | Real outline with wall and gates | 9–10 pt                          |
| City              | Dot or circle                     | Circle with a thick battlemented ring (a larger walled symbol) | Full outline                     | 11–12 pt, bold or small capitals |
| Great city        | Large circle with a ring          | Large walled symbol                                            | Full outline with suburbs        | 13–14 pt CAPITALS                |

These symbols match the guide's [master legend](#). A realm capital takes a filled star in place of its tier symbol, so do not use a star for any other place.

#### FANTASY TWIST

Where monsters roam the countryside, people give up isolated farmsteads and crowd into fewer, larger, walled villages. Italy did this for human reasons in the 10th–12th centuries (*incastellamento*, the move into hilltop *castelli*). Shift some of your hamlet population into fortified villages of 300–1,000 people. See [Fantasy Variants](#).

**LATER ERA (1500S+)**

The ladder stays the same, but the top grows taller. By 1600 Istanbul had perhaps 400,000–700,000 people, and Paris, Naples and London had 200,000–300,000 each. Garrison towns built around artillery forts become a new type. See [What Changes After 1500](#).

**MAP TIP**

On a kingdom map at 1 cm = 10 km (about 1 inch = 16 mi), even a great city is only 2–4 mm across, so use a symbol, not an outline. Draw real outlines only on local maps (1 cm = 1 km, about 1 inch = 1.6 mi, or larger). Make label size follow the tier, not the story importance. If you want a small place to look important, give it a special icon (castle, cathedral, shrine) instead of a big label. Capital markings are covered in [Capitals, Realms and Borders](#).

## How many of each kind

**RULE OF THUMB**

85–95% of people are rural almost everywhere. For every city of 10,000+, expect about 10 towns, about 50–100 small market towns, and a few thousand villages.

**Why:** Each town needs a ring of farmland around it to feed it and to buy its goods. Small towns serve a few villages

each. Bigger towns serve the small towns. Only a few places serve a whole region.

### Urbanization by region

*Urbanization* means the share of people who live in towns. The number depends on where you draw the line: "towns of 10,000+" gives a low figure, while "all towns" gives a higher one.

| Region                            | c. 1300                                                                         | c. 1500                              | Notes                                                                                                                                                                                                      |
|-----------------------------------|---------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Europe                            | ~5% in cities of 10k+; ~9.5% in all towns (broad definition)                    | 5.6% in 10k+                         | —                                                                                                                                                                                                          |
| England                           | ~2% in 10k+ (only London); ~10% in towns of 2k+; 15–20% counting small boroughs | 3.1% in 10k+                         | Dyer argues for ~20% in all towns by 1300.                                                                                                                                                                 |
| France                            | ~3–4% in 10k+; ~7–9% in 2k+                                                     | ~4.2% in 10k+                        | Paris dominated everything.                                                                                                                                                                                |
| Low Countries                     | Flanders ~33–40% in towns of all sizes                                          | Belgium 21%, Netherlands 16% in 10k+ | Holland ~45% urban by 1500.                                                                                                                                                                                |
| Northern Italy                    | ~18% in 10k+ (Centre-North); ~20% in 5k+                                        | ~15–16% in 10k+                      | Italy had 48–79 cities of 10k+ in 1300 (estimates differ).                                                                                                                                                 |
| Iberia                            | ~8–10% in 10k+                                                                  | Spain ~6%, Portugal ~3% in 10k+      | Inflated by agro-towns.                                                                                                                                                                                    |
| Germany                           | ~2–3% in 10k+                                                                   | 3.2% in 10k+                         | Many small towns of 1,000–2,000.                                                                                                                                                                           |
| Denmark                           | ~10% in all towns                                                               | ~14% in all towns                    | About 80 towns c. 1350, most of them small.                                                                                                                                                                |
| Poland, Hungary                   | under 2% in 10k+                                                                | ~1–2% in 10k+                        | —                                                                                                                                                                                                          |
| Arab Middle East and North Africa | ~7% in 10k+                                                                     | ~8% in 10k+                          | Stayed roughly level (about 6–10%) from 800 to 1800. It was clearly more urban than Europe around 800–1000; Europe only caught up in share c. 1700–1800. (Rough figures from the Bosker et al. city data.) |

### Counting the cities

| Year    | Cities of 10k+                                     | Cities of 50k+ | Cities of 100k+ |
|---------|----------------------------------------------------|----------------|-----------------|
| c. 1000 | ~130 (Europe 81, Middle East and North Africa 48)  | ~11            | 4               |
| c. 1300 | ~230 (Europe 172, Middle East and North Africa 60) | ~22            | 8               |
| c. 1500 | ~250 (Europe 195, Middle East and North Africa 52) | ~31            | 8               |

(Bosker, Buringh & van Zanden 2013 dataset; it excludes Russia and Iran.)

Looking further down the ladder, the historian Paul Bairoch counted about 1,450 towns and cities in 14th-century Europe. He sorted them by what they did:

- **About 60%** were small towns of 2,000–6,000 people with purely local roles: a church or two, a local market, and craftsmen serving nearby villages.

- **About 300–330** were "regional centres" of 4,000–12,000. They usually had walls, trade, and often a bishop.
- **About 210** were large cities of more than about 8,000–12,000.

(Bairoch 1988, as summarised by Jedwab, Johnson & Koyama 2020.)

### England c. 1300 as a model kingdom (4.75 million people on 130,000 km<sup>2</sup>):

| Tier                                  | Approximate number                                                                                                                                                                      | One per ... people |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Parishes (each with a village church) | ~9,500                                                                                                                                                                                  | ~500               |
| Places with a market grant            | ~1,750, but many were village markets and only 39% still had a market c. 1600                                                                                                           | ~2,700             |
| Boroughs (places with town status)    | about 600 (Beresford & Finberg 1973), many of them tiny                                                                                                                                 | ~8,000             |
| Towns of 2,000–10,000                 | roughly 50–60 (estimate)                                                                                                                                                                | ~80,000–100,000    |
| Cities of 10,000+                     | ~6: London (60–100k) plus about five provincial cities of 10–15k. The Bosker et al. data list Winchester, Norwich, Coventry, Plymouth and Bristol; other historians would include York. | ~800,000           |

### How many people per city?

| Region c. 1300           | Cities of 10k+ | One city per ... people |
|--------------------------|----------------|-------------------------|
| Italy                    | 48–79          | 130,000–260,000         |
| Iberia                   | ~24            | 250,000–375,000         |
| Egypt                    | ~12            | 350,000–650,000         |
| Germany (modern borders) | ~24            | roughly 400,000–500,000 |
| France (modern borders)  | ~34            | ~600,000                |
| England                  | ~6             | ~800,000                |
| Poland                   | ~2             | ~1,000,000              |

(City counts from the Bosker, Buringh & van Zanden data, using modern borders. Italy's higher count, 79, comes from Malanima. Population totals are from the realm table below. Treat every figure as a rough estimate.)

### The rank-size rule and primate cities

The **rank-size rule** (also called Zipf's law) says: the 2nd city has about ½ the population of the largest, the 3rd about ⅓, the 4th about ¼, and so on. It describes many modern countries well. A **primate city** is a capital far bigger than the rule predicts, often 3 or more times the size of the 2nd city.

Medieval reality did not follow the rule neatly:

#### • Primate realms (strong kings, one capital):

- **England, 1377:** London was 3–4 times the size of York or Bristol, and 8–9 times the size of Colchester or Canterbury. The rule predicts York at about 17,000; it had about 11,000.
- **France, c. 1300:** Paris had perhaps 200,000–250,000 (estimates vary widely). The next cities (Rouen, Montpellier, Saint-Omer) had about 35,000 each, so Paris was roughly 6–7 times larger.

- **Flat-topped regions (many independent cities):** In Italy c. 1300, Venice (~110k), Milan (~100k), Genoa (~100k) and Florence (~95k) were all about the same size. The Low

Countries were similar, with Ghent and Bruges at about 40,000 each. (Some estimates put Ghent higher, at 50,000–65,000.)

• **Overall:**

- Economic historians find that European city sizes only came to fit the rank-size rule between 1500 and 1800 (Dittmar 2008).
- Before that, the pattern was bent: in most regions the biggest cities were *smaller* than the rule predicts. Primate kingdoms such as England and France were the exception.
- In the Arab world, by contrast, the largest city always held 12–38% of all city-dwellers. In Western and Central Europe it never held more than 12% (Bosker et al. 2008).

**RULE OF THUMB**

A strong, centralised kingdom has one capital 3–7 times bigger than its second city. A land of city-states or weak kings has several rival cities of similar size.

**LATER ERA (1500S+)**

In 1500–1650 some capitals pulled further ahead and reached 5–10 times the size of the second city. London was the extreme case: about 200,000 people by 1603, over 13 times the size of Norwich, the next-largest English city (Mortimer 2012, via Wikipedia). See [What Changes After 1500](#).

**MAP TIP**

Decide first what kind of realm you have, centralised (one strong king) or fragmented (many rival lords or city-states), then size the top cities. For a centralised kingdom, draw one great capital and then a gap down to a handful of towns. For a fragmented region, draw 3–5 rival cities of similar size, each with its own little territory. Do not give every country a 100,000-person capital: around 1300, only 8 cities in all of Europe, North Africa and the Middle East reached that size. (A rice-growing empire with canals, like Song China, is different; see [Beyond Europe](#).)

## Population density

**RULE OF THUMB**

Good lowland farmland supports 30–50 people per km<sup>2</sup> (80–130 per sq mi) at its peak. A whole realm, uplands included, averages about 7–40 per km<sup>2</sup> (18–100 per sq mi): 10–40 in settled western kingdoms, 7–15 in young eastern ones. Europe as a whole c. 1300 averaged about 20 per km<sup>2</sup> (52 per sq mi).

**Why:** Density depends on how much food the land can grow (see the next section). Above about 50 per km<sup>2</sup> (130 per sq mi), a region needs towns, specialised commercial farming or imported grain.

### By type of land

| Land type                                                     | per km <sup>2</sup> | per sq mi | Real examples                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|---------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich lowland, peak times                                      | 30–50               | 80–130    | Northern Italy 34–40 c. 1300; England's <i>whole-country</i> average was already ~36 in 1290, so its best lowland was higher                                                                                                                                                                                                                                         |
| Very urban, rich lowland                                      | 50–75               | 130–190   | Flanders and Holland in the 15th century (estimate). Exceptional districts went far higher: Norfolk, where most people lived in villages (Norwich was its only city), had 486,920 people in 1290 (Broadberry, Campbell & van Leeuwen 2011, Table 8B) on about 5,400 km <sup>2</sup> (2,100 sq mi). That is roughly 90 per km <sup>2</sup> (230 per sq mi) (derived). |
| Average mixed farmland                                        | 15–30               | 40–80     | Suffolk ~23–25 in 1086; England ~19 in 1377                                                                                                                                                                                                                                                                                                                          |
| Poorer land, young kingdoms                                   | 7–15                | 18–40     | Hungary ~7 (c. 1200) and ~9–11 (c. 1495); Crown of Poland 8.6 (1370) and 13 (1500)                                                                                                                                                                                                                                                                                   |
| Hill country (farmed valleys with upland grazing), peak times | 8–15                | 21–39     | Cumberland and Westmorland about 15–17 in 1290, as whole counties that include the Eden valley and the Solway plain, so the fells themselves held fewer. The same two counties had about 6–7 in 1377, and northern and western England had under 5 in 1086. (Derived from county estimates in Broadberry, Campbell & van Leeuwen 2011.)                              |
| High uplands, moors, mountains                                | 2–5                 | 5–13      | Scottish Highlands, high Alps, Pyrenees, northern Pennines (estimate; few direct counts exist). Upland parishes covered 50–100+ km <sup>2</sup> (20–40+ sq mi) each.                                                                                                                                                                                                 |
| Forest or steppe frontier                                     | 1–5                 | 3–13      | Lithuania, the Rus' forest zone (estimate)                                                                                                                                                                                                                                                                                                                           |

| Land type                                      | per km <sup>2</sup>                                                | per sq mi                                     | Real examples                                                                                                                                                                 |
|------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shifting (swidden) farming in tropical forest  | about 6 on average; 12–35 in busier districts; at most about 55–56 | about 14 on average; 30–90; at most about 145 | World average about 6 in the 1950s (FAO); modern Southeast Asia from 12 in northern Laos to 35 in northern Thailand (Warner 1991, FAO); ceiling about 55–56 (Ruthenberg 1980) |
| Steppe nomads                                  | under 1–2                                                          | under 3–5                                     | Herders need huge pastures                                                                                                                                                    |
| Desert                                         | close to 0                                                         | close to 0                                    | Except oases and irrigated valleys                                                                                                                                            |
| Irrigated river valley                         | 100+ on the farmed strip                                           | 260+                                          | Egypt c. 1340: perhaps 4–8 million people, almost all on the Nile floodplain and delta                                                                                        |
| Wet-rice or lake-garden basin (outside Europe) | 80–125 over a whole region; 230–300 in a city-and-rice-field zone  | 210–325; 600–780                              | Yangzi delta and Zhejiang ~80 (1102); Valley of Mexico ~100–125 (1519); Greater Angkor ~230–300 (13th century). All derived; see <a href="#">Beyond Europe</a>                |

### By realm and date

| Realm and date                                        | per km <sup>2</sup> | per sq mi |
|-------------------------------------------------------|---------------------|-----------|
| England 1086 (1.71m on 130,000 km <sup>2</sup> )      | ~13                 | ~34       |
| England 1290 (4.75m)                                  | ~36                 | ~94       |
| England 1377 / 1450                                   | ~19 / ~15           | ~50 / ~38 |
| France c. 1328                                        | ~35–40              | ~90–105   |
| Byzantium 1025 (about 12m on 1.675m km <sup>2</sup> ) | ~7                  | ~19       |
| Poland-Lithuania 1500                                 | 6.6                 | 17        |

**How to read this:** A realm's average hides big differences. In 1086 Suffolk was five times denser than northern England. Even at the 1290 peak, much of England's upland was nearly empty while East Anglia was crowded.

#### MAP TIP

Before placing villages, lightly shade your map in pencil (or on a layer you hide later) into three or four density zones: crowded lowland, ordinary farmland, thin hill country and empty land (mountain, marsh, forest, desert). Then place settlements to match. Lowland villages sit 1.5–4 km (1–2.5 mi) apart. In uplands, parish churches may be 7–10 km (4–6 mi) apart, with scattered farms and hamlets between them. Leave real gaps: empty land makes the full land look full.

## Carrying capacity: how much land feeds people

*Carrying capacity* means how many people a piece of land can feed.

#### RULE OF THUMB

Each person needs about 1 hectare (2.5 acres) of ploughland, or 2–3 hectares (5–7 acres) of all land once pasture, meadow and woodland are counted.

**Why:** Medieval harvests were small, and a lot of each harvest went back into the ground:

- **Seed-to-yield ratio:** Typically 3–5 : 1, with a range of 2–7 : 1 in the 14th century. You sowed 1 sack and harvested 3–5. Modern farms get 30 : 1 or more.
- **Seed:** A quarter to a third of every crop had to be kept as next year's seed.
- **Fallow:** Land left unplanted for a year so the soil could recover. It took ⅓ of the arable land under the three-field system and ½ under the two-field system.

- **Wheat yields:** English demesne wheat (grown on the lord's own farm) yielded about 8–12 bushels per acre (0.55–0.8 tonnes per hectare). A bushel is an old measure of volume, about 36 litres.
- **Result:** About 250–350 kg of food grain per hectare of arable land per year. A person ate about 200–250 kg of grain a year, plus barley for ale and oats for horses.

**Check against history:** England c. 1300 had about 11.5 million acres of arable, counting the fallow field. About 7.4 million acres of it were sown each year (Broadberry, Campbell et al. 2010). For 4.75 million people, that is about 2.4 acres (1.0 ha) of arable per person, of which only about 1.6 acres (0.6 ha) carried a crop in any one year. Counting all land, it is about 2.7 ha per person. Even so, in 1300–1319 the main crops and animal products supplied only about 1,300–1,350 kcal per person a day. That is after seed, fodder, storage losses and the waste of milling and brewing are taken out. Grain gave about 1,200–1,250 kcal of it, and meat and dairy about 120 (Broadberry, Campbell & van Leeuwen 2011). Grain calories rose to about 1,800 a day by the 1380s, because the plague had cut the number of mouths. The same authors reckon a well-fed population needs about 2,000 kcal a day, with at least 1,500 of it from these main

crops and animal products. Garden vegetables, poultry, fish and game filled only part of the gap. England was at its limit.

### Village territory

- **Family holdings:**
  - A peasant family needed very roughly 10–15 acres (4–6 ha) of arable to live from its land alone (estimates vary).
  - The standard English holding, the *virgate* or *yardland*, was in theory about 30 acres (12 ha), but its real size varied a lot from place to place. Many families held only half a virgate, and many held less, so they also worked for wages.
- **One example parish:** Milton Ernest (Bedfordshire) in the 14th century had about 1,600 acres (650 ha, or 6.5 km<sup>2</sup>): 1,405 acres of arable, 81 of meadow, 84 under the village itself and 28 of roads (Hutchings 1969).
- **Average size:** England c. 1300 had about 14 km<sup>2</sup> (5.4 sq mi) per parish. Upland parishes covered 50–100+ km<sup>2</sup> (20–40+ sq mi).

| Village size | Arable needed         | Total territory (lowland)           | Territory (upland)                  |
|--------------|-----------------------|-------------------------------------|-------------------------------------|
| 100 people   | ~100 ha (250 acres)   | 2–4 km <sup>2</sup> (0.8–1.5 sq mi) | 10–30 km <sup>2</sup> (4–12 sq mi)  |
| 250 people   | ~250 ha (620 acres)   | 5–10 km <sup>2</sup> (2–4 sq mi)    | 25–60 km <sup>2</sup> (10–23 sq mi) |
| 500 people   | ~500 ha (1,240 acres) | 10–20 km <sup>2</sup> (4–8 sq mi)   | rare in uplands                     |

For how villages and their fields are laid out, see [Villages, Farms and the Countryside](#).

### Feeding towns: the hinterland

A **hinterland** is the area of countryside that feeds a town. The model below assumes rural density of 30 per km<sup>2</sup> on

good land (10 on marginal land). It also assumes each country person grows enough surplus to feed 0.15–0.25 townspeople (0.1 on marginal land). Real hinterlands were bigger, because towns competed for the same farms.

| Town size | Overland radius, good lowland | Overland radius, marginal land | With water transport                                                       |
|-----------|-------------------------------|--------------------------------|----------------------------------------------------------------------------|
| 1,000     | 7–8 km (4–5 mi)               | ~18 km (11 mi)                 | Not needed                                                                 |
| 10,000    | 21–27 km (13–17 mi)           | ~55 km (35 mi)                 | A river or coast lets food come from 2–10 times farther along the waterway |
| 50,000    | 46–59 km (29–37 mi)           | ~125 km (78 mi)                | Almost always needs a river or sea supply (rare exceptions: Granada)       |
| 100,000   | 65–84 km (40–52 mi)           | ~180 km (110 mi)               | Only possible by water                                                     |

**Why big cities need water:** Moving bulk goods overland was very expensive. In England around 1300, land transport cost roughly 8 times as much as sea transport, and

river transport about 4 times as much (see [Trade Routes, Roads and Transport](#)). Real examples:

- **London c. 1300** (60,000–100,000 people) drew its grain mainly along the Thames, through river ports such as Henley, and along the Thames estuary. In bad years it

reached up the coast as far as the Humber. Firewood was worth carting overland only 18–20 km (11–12 mi). Charcoal, which is more valuable for its weight, came from 40+ km (25+ mi) away (Galloway 1999).

- **Constantinople, Cairo, Venice and Genoa** were all fed by ship or by the Nile. Venice and Genoa brought grain from Sicily and the Black Sea.

#### FANTASY TWIST

Magic that raises harvests changes everything downstream. If priests can bless fields from a 4 : 1 to an 8 : 1 yield, the same land feeds about twice as many people. You can then double densities, enlarge towns and shrink hinterlands. Cheap magical transport (portals, flying freight) would also free great cities from the "must be on water" rule. See [Fantasy Variants](#).

#### MAP TIP

Draw a faint pencil circle around each town at the overland radius from the table, and erase it when you are done (on the finished map, rings around a place belong to the town symbol). If two towns' circles overlap heavily, one of them should be smaller or should be on water. Put most places over 10,000 people on a navigable river or a coast. Inland exceptions existed, but they sat in very rich farmland with good roads: Coventry in England (about 12,000 c. 1300) and hilltop Siena in Italy (perhaps 50,000) had no navigable river. Places over 50,000 almost always need water transport.

## Realm populations and areas

#### RULE OF THUMB

A medium European kingdom c. 1300 had 1–5 million people. Only France had more than 10 million people under one king (the Holy Roman Empire had more people, but it was loosely ruled). Europe as a whole had about 70–80 million.

| Realm                            | c. 1000                               | c. 1300 (peak)                                           | c. 1500                               | Area                                                         | Density c. 1300                |
|----------------------------------|---------------------------------------|----------------------------------------------------------|---------------------------------------|--------------------------------------------------------------|--------------------------------|
| England                          | 1.5–2.0m (1.71m in 1086)              | 4.75m (1290); 4.81m (1348)                               | 2.2–2.4m (low point 1.9m in 1450)     | 130,000 km <sup>2</sup>                                      | ~36/km <sup>2</sup> (94/sq mi) |
| Scotland                         | —                                     | 0.5–1m                                                   | —                                     | ~79,000 km <sup>2</sup>                                      | ~6–13/km <sup>2</sup>          |
| France                           | 5–6.5m                                | 16–17m in the kingdom (1328); ~20m within modern borders | 15–16m (modern borders)               | 551,000 km <sup>2</sup> (modern)                             | ~35–40/km <sup>2</sup>         |
| Iberia (all realms)              | 4.6–7m                                | 6–9m                                                     | 7.8–9m                                | 583,000 km <sup>2</sup>                                      | ~10–15/km <sup>2</sup>         |
| German lands / Holy Roman Empire | 3–4m                                  | 11–14m                                                   | ~16m in the HRE; Germany proper 9–13m | 0.8–1.1m km <sup>2</sup>                                     | ~10–17/km <sup>2</sup>         |
| Italy                            | ~5m                                   | 10–12.5m                                                 | 9–11m                                 | 301,000 km <sup>2</sup>                                      | ~33–41/km <sup>2</sup>         |
| Denmark (medieval)               | —                                     | ~0.95m (before 1348)                                     | ~0.6m                                 | —                                                            | —                              |
| Poland                           | 1.0–1.25m                             | ~2m (1370)                                               | Crown 3.4m; 7.5m with Lithuania       | ~240,000 km <sup>2</sup> (1370)                              | ~8.6/km <sup>2</sup>           |
| Hungary                          | 1–1.5m                                | ~2m (1200) to ~3m (14th century)                         | 3m+ (1495)                            | 280,000–325,000 km <sup>2</sup>                              | ~7–11/km <sup>2</sup>          |
| Byzantium                        | ~12m (1025; some estimates up to 18m) | 3–5m (1282), 2m (1312)                                   | Gone (fell 1453)                      | 1.675m km <sup>2</sup> (1025)                                | ~7/km <sup>2</sup> (1025)      |
| Egypt                            | —                                     | perhaps 4–8m                                             | —                                     | Farmed strip only a few tens of thousands of km <sup>2</sup> | very high on the farmed strip  |
| <b>Europe total</b>              | 32–40m (Russell: 38.5m)               | 70–80m (Russell: 73.5m in 1340; some say up to 100m)     | 70–90m                                | —                                                            | ~20/km <sup>2</sup>            |

**Notes on uncertainty:**

- England's 1300 figure was once put anywhere from 3.7 to 6 million. The 4.75–4.81 million series (Broadberry, Campbell and van Leeuwen) is now standard.
- Europe totals for 1000 depend on what you count as Europe. Russell's 38.5 million includes Russia. Other series put it higher.
- Scotland's figure is little more than a guess: it may have grown from about half a million to a million by the mid-1300s.
- Byzantium's 1282 figure is often given as 5 million; some historians put it as low as 3 million.

**LATER ERA (1500S+)**

Europe reached about 100 million by 1600. England had about 4.1 million in 1600 and 5.3 million in 1650, finally passing its 1348 peak. See [What Changes After 1500](#).

**MAP TIP**

Measure your realm's area (count grid squares or hexes), then compare it with this table. A realm the size of England (130,000 km<sup>2</sup>, 50,000 sq mi) should have 2–5 million people. A realm the size of Hungary should have 2–3 million. If your numbers are ten times bigger, either your land is magically fertile or you have made a mistake.

**The biggest cities****RULE OF THUMB**

In any period, only about 4–8 cities in Europe, North Africa and the Middle East had 100,000+ people. Nearly all of them were capitals or great ports, and almost all were on the sea or a big river. (Granada, fed by its rich irrigated plain, is a rare exception.) China was in a different league: its Song capitals Kaifeng and Hangzhou were each probably bigger than any city in this table (see [Beyond Europe](#)).

Ranges are given in thousands. Where scholars disagree a lot, the central figure from Bosker, Buringh & van Zanden (2013) is in brackets.

| Rank | c. 1000                                             | c. 1300                                                           | c. 1500                                               |
|------|-----------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|
| 1    | Constantinople 150–400 (300)                        | Cairo 220–600 (220)                                               | Cairo 150–400 (180)                                   |
| 2    | Baghdad 125–500 (300), declining                    | Paris 75–270 (250); ~200 is the most common figure                | Istanbul 100–280 (280); a 1478 count suggests ~80–100 |
| 3    | Cairo-Fustat 135–200                                | Granada 50–150 (150)                                              | Paris 95–250 (200)                                    |
| 4    | Córdoba 100–450 (100)                               | Venice 90–120                                                     | Naples 100–150 (125)                                  |
| 5    | Seville 70–90                                       | Milan 100–150                                                     | Edirne ~100–127                                       |
| 6    | Tinnis and Damietta ~80 each                        | Genoa 85–100                                                      | Venice 100–150                                        |
| 7    | Palermo 60–75                                       | Constantinople ~100                                               | Milan ~100                                            |
| 8    | Alexandria ~60                                      | Florence 95–120                                                   | Tunis ~65–100                                         |
| 9    | Kairouan, Salerno ~50                               | Baghdad ~95 (after the 1258 sack)                                 | Aleppo 65–80                                          |
| 10   | Venice 17–45                                        | Seville ~90                                                       | Granada ~70; Valencia ~75                             |
| 11   | Antioch ~45; Thessaloniki, Regensburg, Damascus ~40 | London 60–100                                                     | Lisbon, Palermo, Fez, Damascus 55–65                  |
| 12   | Rome 30–35; Kiev 20–45                              | Fez, Marrakesh, Tunis, Naples, Aleppo, Córdoba 55–70; Ghent 40–65 | Genoa, Rome, Florence, Bursa ~55–60; London ~50       |

**How uncertain are these?** Very.

- An older standard database (Bairoch, Batou & Chèvre 1988) gave Córdoba 450,000 people c. 1000 and Palermo 350,000. Later scholars cut these to about 110,000 (Glick) and 60,000 (Epstein and Johns).

- For many Middle Eastern cities there are no counts at all. Historians estimate the population from the walled area (about 150 people per hectare), or from the number of mosques or public bathhouses (about 1,000 people each) (Bosker et al. 2013, data appendix).

Treat every number above 100,000 as a rough guess.

### Things to notice:

- Around 1000, the biggest cities were in the Byzantine and Islamic worlds. Before about 1000, the Middle East's total city population was larger than Europe's.
- By 1300, Italian and Flemish trading cities had caught up. The great cities of the Middle East held steady or shrank.
- Cities could fall fast: Baghdad after the Mongol sack of 1258; Constantinople after 1204, and again by 1453.

### LATER ERA (1500s+)

c. 1600: Istanbul 400–700k; Paris ~220–300k; Naples ~250–280k; London ~200k (roughly 300–400k by the 1650s–60s; estimates vary); Venice, Seville, Lisbon and Milan 110–150k.

### MAP TIP

On a continent-sized map of a "medieval Europe" world, mark only 3–8 great cities of 50,000+. Put almost all of them on coasts, at river mouths or on big navigable rivers. Give at least one of them a clear reason: an emperor's court, a strait, a delta or an export trade.

## Beyond Europe: Asia, Africa and the Americas

All the tiers and densities above come from Europe, North Africa and the Middle East. Other ways of farming gave very different numbers. Use this section if your realm grows rice, farms by clearing forest, feeds a city from lake gardens, or lives from cattle and long-distance trade.

### RULE OF THUMB

What people farm sets the numbers. Wet rice and lake gardens can feed a much denser countryside and much bigger cities than European grain fields. Shifting forest farming feeds far fewer people. From the 11th to the 13th century, the biggest cities in the world were probably in China, not in Europe.

### Why:

- **Wet rice** grows in flooded fields (paddies). A paddy can be farmed every year with no fallow year, so the same land carries more people. A rice empire with good canals can also move grain cheaply to its capital.
- **Shifting (swidden) cultivation** means clearing and burning a patch of forest, cropping it for one to three years, then leaving it to grow back for many years. At any time, most of the land is regrowing forest, so few people can live on it.
- **Lake gardens, reservoirs and canals** (Mexico, Angkor) let a city grow much of its food at its own edge or inside its own zone. In Mexico, canoes on the lakes did the job that rivers and the sea did for Europe's great cities.

## Density beyond Europe

| Place and date                                                                      | People                                                                                        | Area                                                                                | Density                                                                                                                                              | Notes                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Song China, c. 1100                                                                 | 90 to over 120 million (estimates vary); the 1103 census listed about 20.5 million households | About 3 million km <sup>2</sup> (1.2 million sq mi)                                 | ~30–40 per km <sup>2</sup> (80–105 per sq mi) on average (derived)                                                                                   | Like England or France c. 1300 on average, but packed in the rice lands of the south                                                                                                                                                      |
| Liangzhe circuit (Yangzi delta and modern Zhejiang), 1102                           | ~10 million (households × 5)                                                                  | Roughly 120,000–130,000 km <sup>2</sup>                                             | ~80 per km <sup>2</sup> (~210 per sq mi) (derived)                                                                                                   | Denser than Europe's most crowded large regions (Flanders and Holland 50–75), even with the Zhejiang hills included. Only single districts in Europe went higher, e.g. Norfolk, about 90 in 1290 on about 5,400 km <sup>2</sup> (derived) |
| Valley of Mexico, 1519                                                              | 1–1.2 million                                                                                 | 9,600 km <sup>2</sup> (3,700 sq mi), including about 1,500 km <sup>2</sup> of lakes | ~100–125 per km <sup>2</sup> (260–325 per sq mi) (derived)                                                                                           | Lake-bed gardens (chinampas) and irrigated fields                                                                                                                                                                                         |
| Greater Angkor, 13th century                                                        | 700,000–900,000                                                                               | About 3,000 km <sup>2</sup> (1,160 sq mi)                                           | ~230–300 per km <sup>2</sup> (600–780 per sq mi) averaged over the zone (derived)                                                                    | A whole city-and-rice-field zone, not a walled town                                                                                                                                                                                       |
| Shifting cultivation in tropical forest (world in the 1950s; modern Southeast Asia) | About 200 million people worldwide in the 1950s (very rough FAO estimate)                     | About 36 million km <sup>2</sup> (14 million sq mi) worldwide                       | About 6 per km <sup>2</sup> (14 per sq mi) as a 1950s world average (FAO 1957); about 12–35 per km <sup>2</sup> (30–90 per sq mi) in modern examples | 12 in northern Laos, 35 in northern Thailand (Warner 1991, FAO). Ruthenberg (1980, cited by FAO) calculates a ceiling of about 55–56 per km <sup>2</sup> (about 145 per sq mi).                                                           |

**How Song China grew.** Registered households roughly tripled between 980 and 1101, from 6.2 to 17.5 million (Deng 2013). In 1012 the emperor had 30,000 bushels of seed of quick-ripening Champa rice (first brought from Champa, in today's central Vietnam) sent from Fujian to farmers in

the lower Yangzi and Huai valleys (Ho 1956). Historians still argue about how much this new rice mattered. Deng (2013) points out that growing two rice crops a year on the same field was rare under the Song and became common only under the Ming (1368–1644).

## Cities beyond Europe

| City                                     | Date                | People                                                                                      | What it was like                                                                                                                                              |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kaifeng (Northern Song capital)          | c. 1100             | 600,000 to over 1 million (estimates vary)                                                  | Three rings of walls; a trade and industry centre where four major canals met                                                                                 |
| Hangzhou (Southern Song capital)         | 1270s               | A count of 1270 listed about 186,000 families; Gernet estimated well over 1 million by 1276 | Probably the largest city in the world at the time                                                                                                            |
| Greater Angkor (Khmer capital, Cambodia) | 13th century        | 700,000–900,000 in the whole zone                                                           | A walled temple-city core (Angkor Thom, 3 × 3 km) inside a huge spread of house mounds, ponds, embankments and rice fields (Klassen et al. 2021)              |
| Tenochtitlan (Aztec capital, Mexico)     | 1519                | 200,000–250,000 (estimates run from 200,000 to 400,000)                                     | An island city of 8–13.5 km <sup>2</sup> in a shallow lake, joined to the shore by three causeways and fed partly by chinampas in Lakes Xochimilco and Chalco |
| Great Zimbabwe (southern Africa)         | 13th–15th centuries | Under 10,000 (older estimates 18,000–20,000)                                                | Capital of a state rich in cattle and gold trade; stone-walled royal enclosures. Chirikure et al. (2017) argue that it never passed 10,000 people.            |
| Kilwa Kisiwani (East African coast)      | 14th century        | At least 10,000, perhaps 20,000                                                             | Island port that controlled the coast's gold trade; a great mosque and palace built of coral stone                                                            |

Most cities in the Americas were far smaller than Tenochtitlan. Of 87 Mesoamerican cities of the last centuries before the Spanish conquest with good survey data, the

median covered only 90 ha (Smith 2005); less than 1 km<sup>2</sup>, the area of a large European town.

**What this means for your map:**

- A **rice empire** can have a whole-realm average like France's (30–40 per km<sup>2</sup>), with deltas and river basins at 80 per km<sup>2</sup> or more, and one or two capitals of several hundred thousand people or more. The "only 4–8 cities over 100,000" rule applies to Europe and the Middle East, not to Song China.
- A **lake or canal capital** can be packed (Tenochtitlan) or spread out over a huge farmed zone (Angkor). Decide which before you draw it.
- A **tropical forest with shifting farming** has small villages, about 6 people per km<sup>2</sup> on average (12–35 in busier districts, never much above 55), and real towns only on rivers and coasts.
- A **trading or herding state** (Great Zimbabwe, Kilwa) can be rich and powerful with a capital of only 10,000–20,000 people.

**LATER ERA (1500S+)**

After falling during the Mongol conquest, China's population grew strongly under the Ming; estimates for c. 1600 run from about 160 million to 200 million. In Mexico the opposite happened. After the Spanish conquest (1519–21), epidemics and war cut the native population by at least half in the 16th century, and perhaps by as much as nine-tenths over wide areas (McCaa 2000; estimates vary widely). Tenochtitlan was rebuilt as Mexico City. See [What Changes After 1500](#).

**MAP TIP**

For a rice empire, shade the deltas and river basins as your densest zone, link the capital to them by canals or rivers, and draw many small villages on dikes and mounds among the paddies (see [Villages, Farms and the Countryside](#)). For an Angkor-type capital, draw a walled square core with large rectangular reservoirs, then a wide zone of tiny house-mound dots, small ponds and rice fields with no outer wall. (Avoid stipple here: the master legend keeps it for commons and heath.) For a Tenochtitlan-type capital, draw a dense island city, causeways to the shore and a band of narrow garden plots in the shallow lake. For shifting farmers, scatter small villages far apart with clearings of different ages around them. Terrain and biome patterns are in [Where Settlements Are Built](#), and canal and causeway transport in [Trade Routes, Roads and Transport](#).

**Step-by-step population calculator**

Use this to get believable numbers for any realm.

**Step 1: Measure the area.** Count grid squares or hexes and multiply by the area of one square or hex. A hex's area is about  $0.866 \times (\text{width across the flat sides})^2$ . For example, a 30 km hex is about 780 km<sup>2</sup>; a 30 mi hex is about 780 sq mi.

**Step 2: Split the land by quality and apply a density** (from the [Population density](#) tables). Typical values: rich lowland 30–40 per km<sup>2</sup>, average land 15–25, hill country 8–15, high

uplands, mountains, forest and marsh 2–5, desert 0. For rice basins and tropical forest farming, use the figures in [Beyond Europe](#).

**Step 3: Add up to get the total population.**

**Step 4: Choose an urban profile and split the people into tiers.**

| Share of people              | Frontier (Poland, Hungary, Scotland, Scandinavia) | Average (England, France, Germany c. 1300) | Highly urban (Flanders, Lombardy, Tuscany) |
|------------------------------|---------------------------------------------------|--------------------------------------------|--------------------------------------------|
| Cities 10,000+               | ~1%                                               | ~3%                                        | ~15%                                       |
| Towns 2,000–10,000           | ~3%                                               | ~5%                                        | ~12%                                       |
| Small market towns 500–2,000 | ~4%                                               | ~6%                                        | ~8%                                        |
| Villages, hamlets, farms     | ~92%                                              | ~86%                                       | ~65%                                       |

**Step 5: Turn shares into places.** Divide by a typical size: about 3,000–4,000 for towns and about 800 for market

towns. For villages, use about 250 in lowland and 100–150 in hill country. Size the top cities using the rank-size sec-

tion above. The capital of a strong kingdom usually holds about 1–2% of all its people (London c. 1300  $\approx$  1.3–2%; Paris  $\approx$  1.25%). In a fragmented realm, the largest city holds about 0.5% or less (Cologne, Germany's largest city c. 1300, held about 0.4–0.5%).

#### Step 6: Check spacing and food.

- Villages should end up 1.5–4 km (1–2.5 mi) apart in low-land.

#### Quick ratios per 1 million people (from the profiles above):

| Per 1 million people | Frontier                                 | Average              | Highly urban |
|----------------------|------------------------------------------|----------------------|--------------|
| Cities 10,000+       | 0–1 (largest often 8–15k)                | 1–2 (largest 15–30k) | 5–10         |
| Towns 2,000–10,000   | 8–12                                     | 12–18                | 30–40        |
| Small market towns   | 40–60                                    | 60–90                | 80–100       |
| Villages             | 3,000–4,500, plus many hamlets and farms | 2,500–3,500          | 2,000–2,500  |

Real check: Denmark before 1348 had about 950,000 people and about 80 mostly small towns, holding ~10% of the people. That fits the frontier profile.

- Market centres should be about 9–16 km (6–10 mi) apart in a crowded realm, and up to about 25 km (15 mi) apart in a thinly settled one.
- Most places over 10,000 should be on navigable water or a coast. An inland one needs very rich farmland around it.

If the spacing is wrong, adjust the densities or the shares.

#### Worked mini-example

A kingdom of **60,000 km<sup>2</sup> (about 23,000 sq mi)**: a fertile river plain, hilly country and a mountain border.

| Land                           | Area                         | Density                                      | People              |
|--------------------------------|------------------------------|----------------------------------------------|---------------------|
| Fertile lowland (40%)          | 24,000 km <sup>2</sup>       | 35 per km <sup>2</sup>                       | 840,000             |
| Hills and mixed land (35%)     | 21,000 km <sup>2</sup>       | 15 per km <sup>2</sup>                       | 315,000             |
| Mountains, forest, marsh (25%) | 15,000 km <sup>2</sup>       | 4 per km <sup>2</sup>                        | 60,000              |
| <b>Total</b>                   | <b>60,000 km<sup>2</sup></b> | <b>~20 per km<sup>2</sup> (52 per sq mi)</b> | <b>~1.2 million</b> |

Using the **average** profile:

| Tier               | Share | People     | Places                                                                                                                   |
|--------------------|-------|------------|--------------------------------------------------------------------------------------------------------------------------|
| Cities 10,000+     | 3%    | ~36,000    | Capital of ~22,000 (1.8% of the realm) on the main river; a second city of ~11,000 at the river mouth or on the coast    |
| Towns 2,000–10,000 | 5%    | ~60,000    | ~16 towns: 2 of 6,000–7,000, 4 of about 4,000, 10 of 2,500–3,000                                                         |
| Small market towns | 6%    | ~72,000    | ~90 market towns of about 800, plus 50–100 large villages with a weekly market                                           |
| Rural              | 86%   | ~1,030,000 | ~3,000–3,500 villages of about 250 (mostly in the plain), plus thousands of hamlets and farmsteads (mostly in the hills) |

#### Checks:

- **Villages:** about 2,900 villages in the 24,000 km<sup>2</sup> plain means one per ~8 km<sup>2</sup> (3 sq mi), or about 3 km (2 mi) apart. ✓
- **Market centres:** about 110 market centres on 60,000 km<sup>2</sup> means one per ~550 km<sup>2</sup> (210 sq mi). That is about 25 km (15 mi) apart on average, closer in the plain. This is wider than in England, because this realm is less crowded. ✓

- **Capital:** a city of 22,000 would need an overland hinterland 30–40 km (19–25 mi) in radius, so it goes on the navigable river. ✓
- **Big places:** 4 places of 5,000+ in 60,000 km<sup>2</sup> means they are about 130 km (80 mi) apart. That is like medieval Britain. ✓

**MAP TIP**

You do not have to draw all 3,000 villages. On a kingdom map, draw the capital, the city, all 16 towns and the 30–40 most important market towns. Then add villages only as texture in the plain, so the viewer *feels* the crowding. Save full village detail for local maps.

## Change over time: growth, famine and plague

**RULE OF THUMB**

The date you choose changes the map. 1300 is the crowded peak. 1350–1450 has a third to a half fewer people, with deserted villages and half-empty towns. 1500 is in the middle of recovery.

| Date         | Europe        | England                  | What your map shows                                                                                                     |
|--------------|---------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| c. 1000      | 32–40m        | ~1.5–2m                  | Few towns, much forest and waste, castles spreading                                                                     |
| c. 1200      | growing fast  | ~3–4m                    | New towns founded, markets granted, forests cleared                                                                     |
| c. 1300–1348 | 70–80m (peak) | 4.75–4.81m               | Every good field farmed; villages climbing into marginal hills; towns at their largest                                  |
| c. 1350–1450 | ~50–60m       | 2.5m (1377), 1.9m (1450) | Deserted and shrunken villages; marginal land returned to pasture and woodland; towns with empty plots inside the walls |
| c. 1500      | 70–90m        | 2.2–2.4m                 | Recovery under way in the south, slower in the north; sheep pastures where villages once stood                          |

### The three big events:

- Growth, 1000–1300:** Europe's population roughly doubled to tripled. England grew about 2.7 times between 1086 and 1315. In England, 57% of all markets that existed by 1516 were created in the 13th century.
- Great Famine, 1315–22:** Northern Europe lost about 10% of its people (range 5–15%). England fell 12%, from 4.69 million in 1315 to 4.12 million by 1325. Cattle plague then killed up to 80% of livestock in places. The Mediterranean was largely spared. Some regions, such as parts of Tuscany, were already shrinking before 1348.
- Black Death, 1347–51, and later plagues:**
  - Europe lost 30–60% of its people.
  - England fell from 4.81 million in 1348 to 2.6 million in 1351.
  - Florence fell from about 100,000–120,000 c. 1300 to 37,000 in the 1427 tax survey.
  - Denmark fell from about 950,000 to 550,000.
  - Egypt lost roughly half its people once the later outbreaks are counted (Borsch 2005).
  - Plague kept returning (1361–62, 1369, 1374–75 and later), so recovery took until c. 1500–1550 in much of

Europe. England only regained its 1348 peak c. 1600–1630.

### Effects you can draw:

#### • Deserted villages:

- England: Beresford and Hurst's 1971 survey listed well over 2,000 deserted medieval villages, and later estimates exceed 3,000.
- Most were abandoned or shrank between c. 1350 and 1520, especially in the 15th century.
- Plague alone rarely emptied a village. More often, the survivors drifted away and lords turned the fields into sheep pasture.
- Germany: the number of settlements fell from about 170,000 to 130,000 between 1300 and 1500 (–23.5%). In one district of northern Thuringia, only 33 of 179 earlier settlements survived to the mid-16th century.

- **Shrunken towns:** Walls built for the 1300 population enclosed gardens, orchards and ruins after 1350. Florence did not regain its 1300 population until the 19th century, so for centuries its 1333 walls held gardens and fields as well as houses.

- **Abandoned marginal land:** Fields high on hillsides, in heath or at forest edges went back to rough pasture or woods. You can still see their old ridges.

**FANTASY TWIST**

A magical plague, a dragon's long rampage or an undead war works on a map exactly like the Black Death. Expect empty villages, towns too small for their walls, and good land going back to wild. A ruin-filled landscape needs a disaster in its recent past. See [Fantasy Variants](#).

**LATER ERA (1500S+)**

Population grew again after c. 1500, and big towns grew fastest. Italy suffered new plagues in 1629–31 and 1656–57, each killing 20% or more of the people in the regions they hit. The Thirty Years' War (1618–48) devastated parts of Germany. See [What Changes After 1500](#).

**MAP TIP**

For a post-plague setting, mark a few deserted villages in the lowland: a grey church alone in a field, labelled "(lost village)". Sometimes a manor house (small open square) survives beside it in sheep pasture. Show towns with open green space inside their walls, and push the hill-farm line downhill. For a 1300 setting, do the opposite: farms right up the valley sides and suburbs spilling outside the town gates.

## Who lives where

**RULE OF THUMB**

Out of every 100 people, about 85 farm, 10–15 live in towns, 1–2 are clergy, about 1 is noble in Western Europe (many more in Poland and Hungary), and fewer than 1 is a full-time soldier.

| Group                                 | Share of people                                                                                                                                                                                                                                            | Where they are on the map                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peasants and rural workers            | 80–90% (65% in Flanders or Tuscany)                                                                                                                                                                                                                        | Villages, hamlets, farmsteads; also rural craftsmen such as smiths, millers and carpenters                                                                                                                                                                                                                                                                                                                                                             |
| Townspople                            | 5–20% (England 15–20% counting all small towns; Flanders 33–40%)                                                                                                                                                                                           | Market towns, towns, cities                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Clergy (priests, monks, nuns, friars) | ~1–2% (estimates vary). In England in the 1530s, monks, canons, friars and nuns alone came to about 1 adult man in 50, before counting parish priests.                                                                                                     | Every parish has a priest. Monks and nuns live in monasteries, often in the countryside; friars live in towns; cathedral clergy live in cities. England had nearly 900 religious houses with about 12,000 monks, canons, friars and nuns in the 1530s.                                                                                                                                                                                                 |
| Nobility and gentry                   | About 1% in France and Italy; in England titled nobles were far fewer, but the gentry (untitled landowners) added more. ~4–5% in Hungary and ~8–10% in Poland (early modern figures). Perhaps ~10% in Castile, concentrated in the north (rough estimate). | Manor houses and castles across the countryside; town houses in capitals. Where nobles are 10% of the population, many are poor and live like farmers.                                                                                                                                                                                                                                                                                                 |
| Full-time soldiers                    | Well under 1% in peacetime                                                                                                                                                                                                                                 | Castle garrisons (5–20 men in peace; see <a href="#">Castles, Forts and Military Outposts</a> ), royal guards, mercenaries. Even 500–600 garrisoned castles (England's number after the Conquest) with 5–20 men each add up to only about 2,500–12,000 men, well under 1% of the people (estimate). Big wartime armies existed for one season only. Edward I's English army at Falkirk in 1298 had about 15,000 men (some older estimates are higher). |

**Towns needed newcomers.** Medieval towns had more deaths than births. Death rates could be as high as 6% a year even in normal years. Towns grew only by drawing people in from the countryside, and a quarter to a half of a

growing town's people were migrants from nearby villages (Nicholas 2014, via Jedwab et al. 2020). This is why towns sit inside a ring of villages that feed them both food and people.

**MAP TIP**

Put a manor house (small open square) in most lowland villages (some English villages had two or more). Place castles much more sparsely. Religious houses were common but mostly small: England had nearly 900 in the 1530s, roughly one per 145 km<sup>2</sup> (56 sq mi). Friaries sit in towns, while abbeys (a church with a square cloister beside it) often sit in remote valleys (see [Religious, Cultural, Legal and Ancient Sites](#)). Specialists are rare: a university, a great cathedral school or a famous library belongs to only one or two cities in a kingdom.

## Gaming rules of thumb vs history

Many worldbuilders use S. John Ross's free article *Medieval Demographics Made Easy* (1993–2018) or the size table in the *Dungeons & Dragons 3.5 Dungeon Master's Guide* (2003). Both are useful starting points. Ross says plainly that his num-

bers are simplified and not meant to recreate any real country. Here is how they compare with the historical data in this chapter.

| Topic                      | Gaming rule                                                                                        | History                                                                                                                                   | Verdict                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Density of a settled realm | Ross: 30–120 per sq mi (12–46 per km <sup>2</sup> )                                                | 7–40 per km <sup>2</sup> for whole realms; Europe c. 1300 ~20 per km <sup>2</sup> (52 per sq mi); Flanders up to 75 per km <sup>2</sup>   | Good for crowded western kingdoms. Too high for frontier, upland or young realms, which can be 7–15 per km <sup>2</sup> (18–40 per sq mi).                                                                                             |
| Village size               | Ross: 20–1,000, typically 50–300. D&D: thorp 20–80, hamlet 81–400, village 401–900                 | Hamlet 10–75; village 75–500, typically 150–300                                                                                           | Ross matches well. D&D's "villages" are big villages or small market towns.                                                                                                                                                            |
| Town and city size         | Ross: town 1,000–8,000; city 8,000–12,000; big city 12,000–100,000. D&D: metropolis 25,001+        | Town 2,000–10,000; city 10,000+; 50,000+ was very rare                                                                                    | Broadly right. Remember that 25,000 was a first-rank city almost everywhere in Europe.                                                                                                                                                 |
| Largest city               | Ross: $\sqrt{(\text{population})} \times 15$ on average                                            | Strong kingdoms: 1–2% of realm population                                                                                                 | Too small for centralised capitals: it gives England c. 1300 about 33,000 (real London 60–100k), France about 60,000 (real Paris ~200,000) and Byzantium 1025 about 52,000 (real Constantinople ~300,000). Fair for fragmented realms. |
| Second city                | Ross: 20–80% of the largest                                                                        | England 1377 ~30%; France c. 1300 ~15–20%; Italy ~90–100%                                                                                 | Real realms fall outside both ends of the range. Choose by the type of realm.                                                                                                                                                          |
| Towns per city             | Ross: number of cities $\times 2d8$ (average 9)                                                    | England c. 1300: ~50–60 towns of 2k+ to ~6 cities of 10k+, about 9–10 to 1                                                                | Very close.                                                                                                                                                                                                                            |
| Farmland                   | Ross: 180 people per sq mi of cultivated land ( $\approx 1.4$ ha each, pasture and roads included) | ~1 ha arable per person; 2–3 ha of all land                                                                                               | Consistent. Ross's "cultivated land" includes pasture.                                                                                                                                                                                 |
| Urban density              | Ross: ~150 per hectare                                                                             | 100–200 per ha; 150 is a standard assumption                                                                                              | Agrees.                                                                                                                                                                                                                                |
| Castles                    | Ross: 1 active castle per 50,000 people                                                            | England had 500–600 occupied at once after 1066, when it had only about 1.7–3 million people; the German lands had ~14,000 castles in all | Far too few for most of medieval Europe. See <a href="#">Castles, Forts and Military Outposts</a> .                                                                                                                                    |
| Universities               | Ross: 1 per 27.3 million people                                                                    | ~20 in Europe c. 1300 ( $\approx 1$ per 3.5–4 million); 60–80 by 1500                                                                     | Far too few. Use about 1 per 3–4 million for 1300 and 1 per ~1 million for 1500.                                                                                                                                                       |
| Clergy                     | Ross: 1 clergyman per 40 people (2.5%)                                                             | ~1–2% (estimates vary)                                                                                                                    | Somewhat high but close.                                                                                                                                                                                                               |

**In short:**

- Ross's village sizes, towns-per-city ratio, farmland and town density all match history well.
- Ross underestimates big capitals, castles and universities.
- Ross's minimum density is too high for thinly settled lands.
- D&D's size names put the labels "village" and "town" on larger places than historians would.

**MAP TIP**

Use a gaming formula as a first draft. Then run the checks from this chapter: Are most 10,000+ cities on water, and does each inland one have rich farmland around it? Does the capital fit the type of realm? Are villages 1.5–4 km (1–2.5 mi) apart in lowland? Is the realm's total within the range of a real kingdom of the same size? For more pitfalls, see [Common Mistakes and How to Fix Them](#).

**Quick summary**

- Population means people, not households. A household is about 4.5–5 people.
- Tiers: farmstead 5–15; hamlet 10–75; village 75–500 (typically 150–300); small market town 500–2,000; town 2,000–10,000; city 10,000–50,000; great city 50,000+.
- 85–95% of people are rural. A typical kingdom c. 1300 had about 3% of its people in cities of 10,000+. Northern Italy had about 18% in cities of 10,000+, and Flanders had 33–40% in towns of all sizes.
- For every city of 10,000+, expect about 10 towns, 50–100 market towns and a few thousand villages.
- Good lowland supports 30–50 people per km<sup>2</sup> (80–130 per sq mi). Whole realms average 7–40 per km<sup>2</sup> (10–40 in settled western kingdoms). Europe c. 1300 averaged about 20 per km<sup>2</sup> (52 per sq mi).
- Each person needs about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land.
- A town of 10,000 needs a farmland radius of 21–27 km (13–17 mi) overland on good land. Most cities above 10,000, and almost all above 50,000, are fed by river or sea.
- Strong kingdoms have one primate capital (1–2% of the realm's people, 3–7 times bigger than the second city; 5–10 times in some realms after 1500). Fragmented regions have several rival cities of similar size.
- Only about 8 cities in Europe, North Africa and the Middle East had 100,000+ people c. 1300 or c. 1500.
- Outside Europe the numbers change with the way of farming: Song China averaged ~30–40 per km<sup>2</sup> with capitals of perhaps a million; Angkor spread 700,000–900,000 people over ~3,000 km<sup>2</sup>; Tenochtitlan packed 200,000–250,000 onto an island of 8–13.5 km<sup>2</sup>; shifting forest farming supports only about 6 per km<sup>2</sup> on average (12–35 in busier districts, at most about 55).
- 1300 is the crowded peak. After the Black Death (1347–51), 30–60% of people are gone, villages are deserted and towns are half-empty for 100–150 years.
- Gaming formulas are a good first draft, but they shrink great capitals and make castles and universities far too rare.

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## CHAPTER 3

# CAPITALS, REALMS AND BORDERS

A map is not only towns and roads. It is also a pattern of power: who rules where, from which city, and where one ruler's land stops and another's begins. This chapter explains where capitals were built and why, what you find in and around them, how big kingdoms, duchies and counties were, and what borders really looked like in the Middle Ages, including why language lines rarely match them. It ends with practical advice for drawing realms, capitals and borders on your map.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Castles and their garrisons are covered in [Castles, Forts and Military Outposts](#). City sizes and the "one giant capital" pattern are in [Population and Settlement Sizes](#).

## IN THIS CHAPTER

What makes a capital

Travelling courts and fixed capitals

What is in and around a capital

Political units and their sizes

Borders and frontiers

Language lines are not borders

Kinds of states and how they look on a map

How big can a realm be

Drawing realms, borders and capitals

Quick summary

Sources and further reading

## What makes a capital

## RULE OF THUMB

A capital sits in the richest, most loyal region of the realm, on navigable water, at a place that is easy to defend. Centrality matters less than you think.

### The five things a capital needs

| Need                   | Why                                                                                                                      | Real examples                                                                                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| A rich core region     | The court, officials and garrison must be fed. The ruler's own estates (the <i>royal domain</i> ) supply food and money. | Paris sits in the grain lands of the Île-de-France. London sits in the Thames valley. Prague sits in the middle of the fertile Bohemian basin.  |
| River or sea access    | A capital grows into a big city. Big cities need grain by water (see <a href="#">Population and Settlement Sizes</a> ).  | Paris (Seine), London (Thames), Prague (Vltava), Kraków (Vistula), Buda (Danube), Cairo (Nile), Constantinople (Bosporus).                      |
| A defensible core site | The ruler's family, treasure and archives must be safe.                                                                  | Kraków's Wawel hill, Prague Castle on its ridge, Buda's Castle Hill, Paris's island (Île de la Cité), Toledo in a deep loop of the Tagus river. |
| Loyalty                | The capital is in the dynasty's home region, where the ruler owns land and the local nobles are relatives or clients.    | The Capetian kings of France ruled from the Paris–Orléans region, their old family land.                                                        |
| Legitimacy             | People must accept the ruler. Old sacred places, saints' tombs and old royal sites give that acceptance.                 | Westminster Abbey (coronations since 1066); Reims (French coronations); Aachen (Charlemagne's church).                                          |

## Centre, fortress or frontier?

A ruler choosing a capital balances three pulls:

| Type                | Where                                                         | Strength                                     | Weakness                                         | Examples                                                                                                                                                                    |
|---------------------|---------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Central capital     | Near the middle of the realm                                  | Fair to all regions; short messenger times   | Often not on the best river; no natural defences | Madrid, where Philip II settled his court in 1561. Its position near the centre of the Iberian peninsula is often given as one reason, though historians debate the others. |
| Core-region capital | In the dynasty's richest home region, which may be off-centre | Rich, loyal, easy to feed                    | Far from outer provinces                         | Paris, London, Kraków                                                                                                                                                       |
| Forward capital     | Close to a frontier the ruler wants to push or defend         | The ruler is near the army and the conquests | Exposed to attack                                | Ottoman capitals: Bursa (taken 1326), then Edirne (1360s), then Constantinople (1453), each closer to the conquests in Europe                                               |

"Forward capital" is a modern geographers' term, but the pattern is old. The Castilian kings moved their main residences south as they conquered: Toledo (taken in 1085), then Seville (taken in 1248). The Babenberg rulers of Austria moved their seat east to Vienna in 1145, closer to the frontier with Hungary. (Austria became a duchy in 1156.)

### RULE OF THUMB

Expanding realms move their capital toward the frontier. Defensive or settled realms keep it in the old core.

## Two capitals in one: the palace and the merchant city

Many capitals were really two towns side by side: a palace quarter and a merchant city.

- **London and Westminster:** London was the merchant city. Westminster, about 3 km (2 mi) upstream, had the royal palace, the abbey and the law courts.
- **Prague, Kraków, Buda:** a royal castle on a hill (Hradčany, Wawel, Castle Hill) above or across the river from the merchant town (Prague's Old Town, Kraków's market town, Pest).
- **Islamic capitals:** rulers often built a new palace city beside the old one. The Fatimids founded al-Qahira (Cairo) in 969 next to the older city of Fustat. The caliph Abd al-Rahman III of Córdoba founded the palace city of Madinat al-Zahra in 936 (main building work perhaps from 940), about 5 km (3 mi) west of Córdoba.

### MAP TIP

On a local or regional map, draw the capital as two linked parts: a fortified palace on the high ground and a larger merchant town by the river, often with a bridge between them. On a kingdom map, one capital symbol (a filled star) is enough.

## Capital is not always the biggest city

The capital is usually the biggest city in a strong, centralised kingdom (London, Paris). In other realms it is not:

- The Holy Roman Empire (HRE) had no capital at all for most of the Middle Ages.
- The Valois dukes of Burgundy (1363–1477) moved their court between Dijon, Brussels, Lille and Bruges. Dijon, the old ducal capital, was much smaller than the great Flemish cloth cities of Ghent and Bruges.
- In Castile, the court moved between several cities, and the biggest city, Seville, was only one of its stops.

### MAP TIP

In a strong, centralised kingdom, make the capital the biggest city on the map. In a loose realm (an elective empire, a group of duchies under one ruler), put the court in a middling town and let a trading port or cloth city be bigger. That contrast tells the viewer how the realm works.

### FANTASY TWIST

A capital can also be chosen for magic: a ley-line crossing, a dragon-proof mountain, the only place a sacred tree grows. Keep the other four needs, though. Even a magical capital needs food, water transport and loyal land around it, unless you explain how magic replaces them. See [Fantasy Variants](#).

## Travelling courts and fixed capitals

**RULE OF THUMB**

Before about 1200, most western kings had no fixed capital. The court travelled. Fixed capitals appear when government needs paper, courts and coin more than it needs to eat the king's food on the spot.

**Why courts travelled**

- **The court went to the food.** Taxes and rents were often paid in food. Carting food was expensive (see [Trade](#)

**How it worked**

| Feature                                                             | Numbers                                                                                                 | Source or example                |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------|
| Spacing of German royal palaces ( <i>Pfalzen</i> ) and royal manors | About 30 km (19 mi), one day's travel for the royal train                                               | Wikipedia <i>Kaiserpfalz</i>     |
| Spacing of royal farms on routes                                    | 24–31 km (15–19 mi)                                                                                     | Wikipedia <i>Itinerant court</i> |
| Length of stay                                                      | Rarely more than a few weeks; a few days at transit stops                                               | German royal itineraries         |
| Late example                                                        | Emperor Charles V (reigned 1519–56) made 40 journeys and spent about a quarter of his reign on the road | Wikipedia <i>Itinerant court</i> |

Famous Pfalzen include Aachen (Charlemagne's favourite, 790s), Ingelheim (c. 790), Paderborn and Goslar (built

[Routes, Roads and Transport](#)), so the king and several hundred followers moved from one royal estate to the next and ate what was stored there.

- **Presence was power.** A king who was seen, held court, judged disputes and received oaths in each region kept the loyalty of the nobles there.
- **Elections, not inheritance (in Germany).** HRE kings were elected. They came from different regions, so no single city became the permanent base.

1030–1055 near the silver mines and hunting forests of the Harz).

## When and why capitals became fixed

| Realm              | Fixed capital                               | Date                                                                        | What made it fixed                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Byzantium          | Constantinople                              | From 330                                                                    | Salaried officials, coin taxes, huge walls. Never itinerant in the western sense.                                                                                                                                                                                                                                                                                                                  |
| Islamic caliphates | Baghdad (founded 762), Córdoba, Cairo (969) | 8th–10th centuries                                                          | Same: paid bureaucracy and coin taxes. New capitals were often planned from scratch.                                                                                                                                                                                                                                                                                                               |
| France             | Paris                                       | From the 1190s, firmly by the mid-1300s                                     | In July 1194 Philip II lost his travelling archives, seal and treasure to Richard the Lionheart at the battle of Fréteval. After that, the king kept his records in a fixed archive in Paris (the <i>Trésor des Chartes</i> ) instead of carrying them around. The Parlement (high court, 1302) and the Chambre des comptes (audit office, c. 1303) sat in the royal palace on the Île de la Cité. |
| England            | Westminster / London                        | From the 12th century, firmly by the mid-1300s                              | The Exchequer (treasury office) was based at Westminster by the reign of Henry II (1154–89). Magna Carta (1215), clause 17, said common pleas (ordinary lawsuits) must be held "in some fixed place". In practice that place became Westminster Hall. The Chancery (writing office) followed.                                                                                                      |
| Poland             | Kraków                                      | Seat of government from 1038; coronations from 1320                         | Royal castle and cathedral on Wawel hill.                                                                                                                                                                                                                                                                                                                                                          |
| Bohemia            | Prague                                      | Always the main seat; imperial capital under Charles IV (1346–78)           | Archbishopric 1344, New Town and university 1348, Charles Bridge begun 1357.                                                                                                                                                                                                                                                                                                                       |
| Hungary            | Buda (with moves)                           | Buda until 1315; Temesvár 1315–23; Visegrád 1323–1408; Buda again from 1408 | The seat followed the king's power base and his building projects.                                                                                                                                                                                                                                                                                                                                 |
| Castile / Spain    | Madrid                                      | 1561 (Valladolid briefly 1601–06)                                           | Before 1561 the court moved between Toledo, Valladolid, Burgos, Seville and others.                                                                                                                                                                                                                                                                                                                |
| Holy Roman Empire  | None                                        | —                                                                           | Remained itinerant. Dynasties used their own home cities (Prague, Vienna).                                                                                                                                                                                                                                                                                                                         |

The forces that fixed capitals were:

1. **Paperwork.** Archives, account rolls and registers became too bulky and too valuable to move.
2. **Law courts.** People with lawsuits needed a court they could find.
3. **Money taxes.** When taxes came in coin, food could be bought anywhere, so the court no longer had to travel to eat.
4. **A bigger staff.** Clerks, judges and accountants needed offices and houses.

### MAP TIP

If your kingdom is in an "early medieval" stage, do not draw one capital. Draw 5–15 royal residences (small crowns) scattered across the royal lands, about a day's travel apart along the main routes, and mark one or two as favourites. If your kingdom is "late medieval", draw one capital (a filled star) and a ring of royal residences around it (see the next section).

### LATER ERA (1500S+)

Fixed capitals win everywhere. Philip II fixed Madrid in 1561 and built the Escorial palace-monastery nearby (1563–84). Sigismund III moved the Polish court from Kraków to Warsaw in 1596, closer to Lithuania and the Baltic. German princes settled into fixed *Residenzstädte* (residence towns). See [What Changes After 1500](#).

### FANTASY TWIST

Fast flying mounts, teleport circles or magical messages reduce the need to travel. A realm with them can have a fixed capital much earlier and govern a bigger area. A realm of nomadic or migrating people (or a king who must personally renew a magical oath in every province) stays itinerant.

## What is in and around a capital

### RULE OF THUMB

A capital is not one building. It is a cluster: palace, great church, treasury, writing office, law courts, mint, and later a university, surrounded within a day's ride by royal residences and hunting forests.

### Inside the capital

| Institution                  | What it is                                                               | Real examples                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Palace or royal castle       | Where the ruler lives and holds court; often on high ground or an island | Palais de la Cité and the Louvre (Paris); Westminster Palace; Prague Castle; Wawel (Kraków); Buda Castle (first royal residence built 1247–65)                                                                                                                                                                 |
| Cathedral or great abbey     | The ruler's main church, often a bishop's or arch-bishop's seat          | Notre-Dame (Paris); St Vitus (Prague); Wawel Cathedral; Hagia Sophia (Constantinople)                                                                                                                                                                                                                          |
| Treasury                     | Stores coin, plate and crown jewels                                      | The Paris house of the Knights Templar (the Temple) managed the French royal money from the 12th century until Philip IV moved it to the Louvre in 1307. Charles IV kept the imperial regalia (crown, sceptre and other symbols of rule) in Karlštejn castle, founded in 1348 about 30 km (19 mi) from Prague. |
| Chancery and archives        | Writes and stores charters, letters and laws                             | Trésor des Chartes in Paris (set up after 1194)                                                                                                                                                                                                                                                                |
| Law courts                   | The high court of the realm                                              | Westminster Hall; the Parlement in the Palais de la Cité                                                                                                                                                                                                                                                       |
| Mint                         | Makes the royal coinage                                                  | The Tower of London mint; the Paris mint                                                                                                                                                                                                                                                                       |
| University (later)           | Trains lawyers, clerks and priests for government and church             | Many capitals founded one: Prague 1348, Kraków 1364, Vienna 1365                                                                                                                                                                                                                                               |
| Royal chapel or relic church | Holds relics that give the king holy status                              | Sainte-Chapelle, Paris (1248), built for the Crown of Thorns                                                                                                                                                                                                                                                   |

### How many mints?

#### RULE OF THUMB

Early medieval kingdoms struck coins in many towns. Later, strong kings closed most mints and kept a few they could control. In loose realms, the opposite happened: more and more lords got the right to mint.

A mint was often a small workshop of a few moneyers (coin makers), not a big building. England is the best-recorded case:

| England                                 | Number of mints                                   | Notes                                                                                                                                                                        |
|-----------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Æthelred II (978–1016)                  | About 70 (one standard catalogue lists 73)        | Mostly in boroughs (fortified towns, <i>burhs</i> ), from London and York down to small places such as Lydford in Devon. A single coin type was struck at 60 or more places. |
| Henry II, 1154–58                       | Nearly 50                                         | —                                                                                                                                                                            |
| Henry II's recoinage of 1158            | 30 during the recoinage, then only 12 kept open   | A <i>recoinage</i> is when old coins are called in and struck again as new ones.                                                                                             |
| Henry II's death, 1189                  | 9                                                 | —                                                                                                                                                                            |
| Henry III (1216–72), between recoinages | 4: London, Canterbury, Bury St Edmunds and Durham | The capital plus three old church centres. For a big recoinage (1247), many old mints were briefly reopened.                                                                 |

The Holy Roman Empire went the other way. From the 10th century, kings granted the right to mint to bishops, from the 11th century to lay princes, and later to towns. By the 12th century the empire was split into many regional coin areas, each with its own penny (*Pfennig*).

- **Early medieval or newly united realm:** a mint in most fortified towns.
- **Centralised kingdom of the High or Late Middle Ages:** a mint in the capital, one to three church or regional centres, and one at each big silver mine (see [Industry, Resources and Special Towns](#)).
- **Loose realm (an empire of princes, a land of city-states):** a mint in every princely seat, many bishops' cities and the big free towns.

**MAP TIP**

Mints are too small for a symbol on a kingdom map, but they are good detail for a town description or a regional map. Use the number of mints to show what kind of realm it is: many mints for an old or loose realm, three or four for a strong, centralised one.

**Ceremonial cities: coronation and burial**

The city where a ruler is **crowned** or **buried** is often not the city where he governs. The ceremonial city is usually an older place, where the dynasty or the church first became strong. It keeps that role for centuries, even after the government has moved elsewhere.

| Realm             | Coronation                                                                 | Burial                                             | Government                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| France            | Reims (nearly all kings; about 130 km / 80 mi from Paris)                  | Saint-Denis abbey, just north of Paris             | Paris                                                                                                                                           |
| England           | Westminster Abbey (since 1066)                                             | Westminster Abbey (rebuilt by Henry III from 1245) | Westminster / London (Winchester was an older royal and treasury centre)                                                                        |
| Holy Roman Empire | Aachen (about 30 kings crowned 936–1531), then Rome for the imperial crown | Speyer cathedral (Salian emperors) and others      | No capital. The Golden Bull (1356) fixed elections at Frankfurt, coronation at Aachen and the first diet (assembly) of each reign at Nuremberg. |
| Hungary           | Székesfehérvár (about 38 monarchs crowned, 1038–1527)                      | Székesfehérvár                                     | Esztergom, Visegrád, Buda (about 60 km / 37 mi from Székesfehérvár)                                                                             |
| Poland            | Gniezno (early), then Kraków's Wawel (from 1320)                           | Wawel Cathedral                                    | Kraków                                                                                                                                          |
| Scotland          | Scone                                                                      | Dunfermline abbey                                  | Edinburgh, from the 15th century                                                                                                                |
| Aragon            | Zaragoza                                                                   | Poblet monastery                                   | Barcelona, Zaragoza, Valencia (itinerant)                                                                                                       |

**RULE OF THUMB**

Governments move; sacred sites stay. A small town with a coronation church or royal tombs can be more important on your map than a large trading city.

Legitimacy can also be borrowed. After the Mongols destroyed Baghdad in 1258, the Mamluk sultans of Egypt brought a member of the Abbasid family to Cairo (1261) and kept him there as a "shadow caliph" to bless their rule.

**Around the capital: the royal landscape**

Within about one day's ride of a fixed capital you usually find:

| Feature                                                         | Distance from capital | Real examples                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A ring of royal residences: smaller palaces, manors and castles | 5–60 km (3–37 mi)     | London: Eltham and Sheen (~13 km / 8 mi), Windsor (~35 km / 22 mi). Paris: Vincennes (~7 km / 4 mi), Saint-Germain-en-Laye (~20 km / 12 mi), Fontainebleau (~55 km / 34 mi). Buda: Visegrád (~35 km / 22 mi). Prague: Karlštejn (~30 km / 19 mi). Straight-line distances, rounded. |
| Royal hunting forests and lodges                                | 10–100 km (6–60 mi)   | England: the New Forest (c. 1079); Windsor Forest; Woodstock and Clarendon hunting palaces. Kraków: Niepołomice hunting castle (built by Casimir the Great in the mid-1300s).                                                                                                       |
| The ruler's favourite abbeys and burial church                  | 0–20 km (0–12 mi)     | Westminster, Saint-Denis                                                                                                                                                                                                                                                            |
| Estates of courtiers and officials                              | Within 1–2 days' ride | Nobles' town houses in the capital and country manors in the home counties                                                                                                                                                                                                          |

**Royal forests** were legal hunting zones that could contain villages, fields and heath as well as woods (see [Royal forests](#)). Kings used them for hunting and to raise money from fines and licences.

#### MAP TIP

Around your capital, draw 3–6 small royal residences (small crowns) within about 60 km (37 mi), at least one big royal forest with a hunting lodge, a great abbey or burial church close by, and, if you want, a separate coronation town within 50–150 km (30–90 mi), marked with a small crown beside its town symbol. Draw each royal forest as a green dash-dot boundary with its name in large italics, and leave its villages and fields visible inside. These details make a capital feel real.

## Political units and their sizes

#### RULE OF THUMB

Kingdom > duchy (or principality) > county > barony or lordship > manor. (A *fief* is any land held from a lord in return for loyalty and service; a *vassal* is the person who holds it.) Each level is usually several times to a few dozen times smaller in area than the one above. But medieval units overlapped, were scattered and changed constantly. Do not expect a neat pyramid.

## The units, top to bottom

| Unit                                                       | Typical area                                                                                                                                        | Typical population (c. 1300)              | Seat                                                                    | Notes and examples                                                                                                                                                                                                                                                    |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Empire                                                     | 0.8–1.7 million km <sup>2</sup> (300,000–650,000 sq mi)                                                                                             | 5–15 million                              | One great capital, or none (HRE)                                        | HRE 0.8–1.1m km <sup>2</sup> ; Carolingian Empire c. 800 ~1.2m km <sup>2</sup> ; Byzantium in 1025, 1.675m km <sup>2</sup>                                                                                                                                            |
| Kingdom                                                    | 10,000–450,000 km <sup>2</sup> (4,000–175,000 sq mi)                                                                                                | 0.1–17 million                            | Capital or itinerant court                                              | Navarre ~10,000–12,000 km <sup>2</sup> (~100,000 people c. 1300); Scotland ~79,000; Portugal ~89,000; England 130,000; Hungary 280,000–325,000; France was the giant                                                                                                  |
| Duchy or principality (great fief)                         | 10,000–50,000 km <sup>2</sup> (4,000–19,000 sq mi)                                                                                                  | 0.5–2 million (estimates)                 | A ducal city with castle, cathedral and court                           | Rough areas, based on the matching modern regions: Normandy ~30,000 km <sup>2</sup> ; Brittany ~34,000 km <sup>2</sup> ; Flanders (a county, but as powerful as a duchy) ~10,000 km <sup>2</sup>                                                                      |
| County (shire)                                             | 1,000–10,000 km <sup>2</sup> (400–4,000 sq mi); England averaged ~3,300 km <sup>2</sup>                                                             | 20,000–300,000; England averaged ~120,000 | County town with castle, sheriff and county court                       | 39 English shires: Rutland 394 km <sup>2</sup> (152 sq mi) to Yorkshire ~15,700 km <sup>2</sup> (6,067 sq mi). Hungary had ~70 counties in the 15th century, averaging ~4,000–4,500 km <sup>2</sup> .                                                                 |
| Hundred or wapentake (English sub-county district)         | Average ~130–150 km <sup>2</sup> (50–60 sq mi) in 1086, ~150–200 km <sup>2</sup> (60–80 sq mi) by 1316; range 4.6–982 km <sup>2</sup> (2–380 sq mi) | 2,000–10,000                              | A meeting place, often open-air: a mound, a ford, a crossroads, a stone | 812 hundreds can be traced in Domesday Book (1086), from Worth in Kent (4.6 km <sup>2</sup> ) to Salford in Lancashire (982 km <sup>2</sup> ); roughly 10–12 parishes each in 1086; after mergers about 630 named hundreds remained by 1316, with 10–15 parishes each |
| Barony, lordship, honour (a great lord's group of estates) | Very variable: a few manors to 100+                                                                                                                 | —                                         | The <i>caput</i> (head manor), usually with a castle                    | A great baron's manors were often scattered across several counties                                                                                                                                                                                                   |
| Knight's fee (land expected to support one knight)         | 1–2 manors; no fixed size                                                                                                                           | —                                         | A manor house                                                           | England had only about 5,000–6,000 knights' fees (1166 survey)                                                                                                                                                                                                        |
| Manor                                                      | 2–10 km <sup>2</sup> (one village, part of one, or several)                                                                                         | 50–500                                    | Manor house, often moated                                               | The village of Elton (Huntingdonshire) had 37 recorded households in 1086 (roughly 170–190 people), split among three manors                                                                                                                                          |
| Parish                                                     | ~14 km <sup>2</sup> in the lowlands; 50–100+ km <sup>2</sup> in uplands                                                                             | 75–500                                    | Parish church                                                           | ~9,500 parishes in England c. 1300                                                                                                                                                                                                                                    |

## Units do not nest neatly

- **Manors and villages often differed.** In a 1279 survey of 134 Leicestershire villages, only 37 had a single lord. 63 villages (47%) were split among three or more lords.
- **Baronies were scattered.** In 1086 the people who held land directly from the king were called *tenants-in-chief*. A small group of great barons held a large share of the country; one common estimate is that about a dozen

men held around a quarter of it. Their manors lay in many counties.

- **Counties had detached pieces.** Many English counties had small "islands" of land inside neighbouring counties. An Act of 1844 removed many of them, but some survived into the 20th century.
- **Church lands cut across everything.** Bishops and abbots held roughly a quarter of England in 1086 (estimates vary), and a bishop's lands did not match his diocese.

## Who sits where

| Lord                             | Seat                    | What the seat looks like                                                                                                                                                                                                                            |
|----------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| King                             | Capital or main palaces | See above                                                                                                                                                                                                                                           |
| Duke or great count              | Main city of the duchy  | Big castle or palace, cathedral, mint, the duke's own abbey and burial church. Example: Rouen for Normandy, Dijon for Burgundy, Ghent and Bruges for Flanders.                                                                                      |
| Count or sheriff (royal officer) | County town             | Royal castle with the sheriff's office and gaol, the county court, a market. In England many shires are named after their county town: Warwickshire, Oxfordshire, Bedfordshire. In Hungary each county head ( <i>ispán</i> ) sat in a royal castle. |
| Baron                            | Caput of the barony     | A castle (see <a href="#">Castles, Forts and Military Outposts</a> ), often with a small borough and a priory founded by the family                                                                                                                 |
| Knight or manor lord             | Manor                   | Manor house with hall, barns and often a moat, next to the parish church                                                                                                                                                                            |

## How many units in one kingdom?

| Realm                     | Count                                                                                                                                                                                                                                                                                                                    | Source                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| England c. 1300           | 1 kingdom; 39 shires (plus special zones such as the Palatinate of Durham); about 630 hundreds (628 named in a 1316 inquest; 812 can be traced in Domesday Book, 1086); ~9,500 parishes; 5,000–6,000 knights' fees                                                                                                       | Domesday; 1166 survey; 1316 inquest; baseline                       |
| France c. 1200            | Royal domain plus 6 great lay fiefs: the duchies of Burgundy, Normandy and Aquitaine and the counties of Flanders, Champagne and Toulouse (the six lay "peers of France", fixed c. 1180), plus many lesser counties                                                                                                      | Peerage of France                                                   |
| Hungary                   | ~37 counties named for 1038 (many of them disputed), ~45–50 (c. 1074), ~70 (late 14th and 15th centuries)                                                                                                                                                                                                                | Wikipedia <i>Administrative divisions of the Kingdom of Hungary</i> |
| Carolingian Empire c. 800 | Somewhere between about 110 and 600 counties (estimates vary widely)                                                                                                                                                                                                                                                     | Wikipedia <i>Carolingian Empire</i>                                 |
| Holy Roman Empire 1521    | About 400 <i>imperial estates</i> (territories and cities with a place in the imperial diet). About 53 were ecclesiastical principalities (archbishops and bishops, three of the archbishops being electors) and 87 were free and imperial cities. The rest were secular princes, counts and lords, abbots and abbesses. | Worms register of 1521                                              |
| China (for comparison)    | About 1,230 counties ( <i>xian</i> ) under the Song (960–1279); about 1,385 under the Ming (1368–1644)                                                                                                                                                                                                                   | Skinner 1977                                                        |

### MAP TIP

For a kingdom the size of England (130,000 km<sup>2</sup>, 50,000 sq mi), draw 30–40 counties, each about 50–70 km (30–45 mi) across, each with a county town near its middle. Group them into 4–8 bigger regions held by great lords or old duchies. Draw duchy borders as medium dash-dot lines and county borders as thin ones. Leave 1–3 special zones: a march (hatching or a fading colour band), or a palatinate or church territory (its own colour).

## A different model: the bureaucratic empire (China)

### RULE OF THUMB

An empire run by salaried officials gives a much more regular map than feudal Europe: county seats of similar size, spread fairly evenly, each one a walled government town, with market towns between them and few private castles.

**Why it is different.** In imperial China the basic unit was the county (*xian*). A magistrate ran it. He was an official appointed by the central government, not a lord who inherited the land. A magistrate could not serve in his home county (the "rule of avoidance", strictly enforced from the 6th century), and he was moved to a new post after a few years (in Ming times usually two or three). Nobody owned a county, so counties were not split by inheritance, scattered by marriage or turned into private lordships. Their number also stayed surprisingly steady from one dynasty to the next.

| Feature                 | Numbers                                                                                                                                                                                                                                                                                                                                | Source                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Number of counties      | Song: about 1,230. Ming: about 1,385. A recent study argues that the Song also had hundreds of "administrative towns" with officials doing county-like work.                                                                                                                                                                           | Skinner 1977; Han and Sng 2023 |
| Average county size     | About 2,000–2,500 km <sup>2</sup> (770–970 sq mi) in Song China: about 1,230 counties on 2.5–3.1 million km <sup>2</sup> (derived; area estimates vary). That is a little smaller than an average English shire (~3,300 km <sup>2</sup> ).                                                                                             | Derived                        |
| Spacing of county seats | About 45–55 km (28–34 mi) apart on average (derived from the area above). Closer on crowded river plains, much farther apart in hills and on the frontier.                                                                                                                                                                             | Derived                        |
| City walls              | Of all administrative cities (county seats and above): 62% walled in 1368, 70% in 1393, 80% in 1469, 90% in 1540 and 95% in 1576                                                                                                                                                                                                       | Xue et al. 2021                |
| Below the county seat   | Periodic market towns. In Skinner's model, each "standard market town" served about 18 villages (6 close by, 12 farther out) and roughly 6,000–12,000 people. Markets met on fixed days of a ten-day cycle, for example days 1, 4 and 7. A bigger market nearby used other days (for example 2, 5 and 8), so traders could visit both. | Skinner 1964–65                |

Be careful with Skinner's market figures. They come from the 19th and 20th centuries, when China was far more crowded than medieval Europe. Use them for the pattern, not for exact numbers in 1000–1500.

#### MAP TIP

For a bureaucratic empire, draw county seats as walled towns of similar size (the walled-town symbol with the county-town dot in the centre), spread fairly evenly about 45–55 km (28–34 mi) apart on good farmland, closer on crowded plains and farther apart in the hills. Give each one a regular wall, often a rectangle on flat land. Group every few counties into a prefecture, with a bigger city as its seat. Fill the gaps with unwalled market towns, each in the middle of its ring of villages. Draw few or no private castles: in this kind of state, the walls belong to the government's towns.

#### FANTASY TWIST

Other peoples may divide land differently. Dwarf holds might be counted by mountain or mine, not area; elves by forest; a sea people by islands or harbours. Keep the same rule: one seat per unit, one unit per day or two of travel.

## Borders and frontiers

#### RULE OF THUMB

In settled farmland, borders were known lines, walked and argued over field by field. In empty land (mountains, forest, marsh, desert, steppe), borders were wide zones. Draw lines where people live and bands where they do not.

## What borders follow

| Feature                        | How often used        | Why                                                                                                                             | Real examples                                                                                                                                                                                         |
|--------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Old boundaries                 | Very often            | Borders are "sticky". Once fixed, they last for centuries.                                                                      | Many French dioceses follow Roman city territories. Many English shire borders lasted from the 10th or 11th century until 1974.                                                                       |
| Watersheds and mountain crests | Often, but not always | Mountains are empty, so a crest line divides little.                                                                            | Much of the Pyrenees. But the Val d'Aran, on the French side of the crest, belonged to Aragon, and Lower Navarre lay north of the mountains.                                                          |
| Pass states (an exception)     | In the Alps           | Whoever holds a pass wants both sides of it.                                                                                    | Savoy, Tyrol and the Swiss canton of Uri straddled the main Alpine passes instead of stopping at the crest.                                                                                           |
| Rivers                         | Sometimes             | Easy to name in a treaty and easy to see. But a river valley is one economic unit, so peoples on both banks are often the same. | Tweed (England–Scotland, Treaty of York 1237); Scheldt (France–Empire, from the partition of 843); the Danube (old Roman frontier)                                                                    |
| Forests, marshes, heaths       | Often in early times  | Thin population; hard to cross.                                                                                                 | The "Wildnis" (wilderness) left between the Teutonic Order's Prussia and Lithuania                                                                                                                    |
| Empty waste-land               | Frontier wars         | Raiding emptied the land between enemies.                                                                                       | The Castile–Granada frontier (1246–1492); the steppe "Wild Fields" north of the Black Sea                                                                                                             |
| Man-made dykes and walls       | Rare but striking     | Mark a claim or block raids                                                                                                     | Offa's Dyke on the Welsh border (traditionally credited to King Offa of Mercia, who reigned 757–796, though recent dating suggests parts are older); the Danevirke in Denmark; the Scots' Dike (1552) |

## Frontier zones vs border lines

- **Line borders** existed in settled land. Parish and manor bounds were marked with stones, crosses, ditches and old trees, and villagers walked them every year ("beating the bounds"). Treaties named rivers and landmarks. The Anglo-Scottish border fixed by the Treaty of York (1237) changed very little afterwards; the only later changes were Berwick and the Debatable Lands.
- **Zone frontiers** were the norm on the edges of Christian Europe, in mountains and in war zones. Here power

faded out over 20–100+ km (12–60+ mi). People paid tribute to both sides, or to neither.

### Marches: militarised border provinces

A **march** (German *Mark*, Italian *marca*) was a border province given to a strong lord to defend. The **margrave** (march count) or marcher lord had more military power than an ordinary count.

| March                           | Dates                            | Features                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spanish March (Marca Hispanica) | From c. 795; Barcelona taken 801 | A ring of counties (Barcelona, Girona, Urgell, Cerdanya and others) facing al-Andalus. The count of Barcelona was effectively independent by the late 980s; the march became Catalonia.                                                                                                                                                                                              |
| Welsh March                     | From the 1070s to 1536           | Dozens of small Norman lordships between England and Wales. Marcher lords ruled "like kings": own courts, own castles, private war, their own markets and boroughs. Britain's densest cluster of motte-and-bailey castles (castles on a man-made earth mound with a fenced yard). The lords' special powers were abolished in 1536 by the first of the Laws in Wales Acts (1535–42). |
| Anglo-Scottish Marches          | 13th century to 1603             | Three marches (East, Middle, West) on each side, each under a royal Warden. A special border law, first written down in 1249 by six English and six Scottish knights. Raiding clans ("reivers").                                                                                                                                                                                     |
| Brandenburg                     | From 1134                        | Albert the Bear was made margrave of the Northern March in 1134 and called himself Margrave of Brandenburg from 1157. It grew into a major principality.                                                                                                                                                                                                                             |
| Austria                         | From the 10th century            | The eastern march of Bavaria; Vienna became its seat in 1145, and Austria became a duchy in 1156.                                                                                                                                                                                                                                                                                    |

Other frontier words that survive as place names:

*Extremadura* (often explained as "the far edge of the Duero" lands), *Marche* (France and Italy), *Mark*, and *Krajina* (Slavic for "borderland").

#### RULE OF THUMB

A march is bigger, emptier and more fortified than an inland county of the same importance. Expect noticeably more castles per area (the Welsh March has Britain's densest cluster of earth-and-timber castles) and fewer, smaller towns.

## Buffer states, enclaves, exclaves and contested land

| Thing               | What it is                                                                                                      | Real examples                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Buffer state        | A small realm between two big ones. Each big neighbour prefers it independent rather than in the other's hands. | Navarre between France, Castile and Aragon (its southern part was conquered by Ferdinand of Aragon in 1512 and later joined to Castile); Savoy on the Alpine passes; Wallachia and Moldavia between Hungary, Poland and the Ottomans; Andorra (co-ruled since 1278 by the bishop of Urgell and the count of Foix; 468 km <sup>2</sup> / 181 sq mi) |
| Enclave / exclave   | A piece of one realm surrounded by another (an enclave from the outside view, an exclave from the owner's view) | Baarle (Low Countries): medieval land deals between the dukes of Brabant and the lords of Breda left dozens of tiny pockets that still exist (22 Belgian pieces inside the Netherlands today). Avignon and the Comtat Venaissin: papal land inside France from 1274 (Venaissin) and 1348 (Avignon) until 1791.                                     |
| Overseas bridgehead | A defended exclave across a sea                                                                                 | The Pale of Calais, English from 1347 to 1558. It ran roughly from Gravelines to Wissant, about 35 km (22 mi) of coast, with marshy, often-changing borders inland. It was supplied by sea.                                                                                                                                                        |
| Palatinate          | A border county where a great lord holds royal powers                                                           | The prince-bishops of Durham guarded England against Scotland                                                                                                                                                                                                                                                                                      |
| Contested land      | Claimed by both sides, ruled by neither                                                                         | The Debatable Lands on the Anglo-Scottish border: about 16 × 6.4 km (10 × 4 mi), disputed for about three centuries and a refuge for raiders, split in 1552 by the Scots' Dike. Berwick-upon-Tweed changed hands more than a dozen times before becoming English for good in 1482.                                                                 |

## Border markers and crossings

- **Markers:** boundary stones, crosses, ditches, banks, marked trees and posts. Treaty commissioners rode the line and ordered markers to be set up. For the Scots' Dike (1552), an earth bank a little over 5 km (3 mi) long, they ordered a stone with a cross at each end.
- **Crossings:** bridges, fords and passes on main roads, with a toll station, often a border castle on each side, and an inn or hospice. Tolls and customs could be collected at sea straits too: Denmark's Sound Toll at Helsingør (from 1429).
- **Meeting places:** on the Anglo-Scottish border, wardens met on fixed "truce days" at traditional spots on the line to settle complaints.

#### MAP TIP

Draw dash-dot border lines where settled farmland meets, and a hatched band (a "march") across mountains, forests and war zones. Put paired castles (small castle icons) facing each other where a main road crosses the border, with a toll mark (a bar and a "T") on the road between them. Add one or two odd features: an enclave, a tiny buffer state, a disputed strip. These make borders look historical rather than planned. Castle spacing along frontiers is covered in [Castles, Forts and Military Outposts](#).

#### FANTASY TWIST

Monsters behave like an extra enemy. A monster-haunted forest or mountain becomes a frontier zone even between friendly realms, with marcher lords, watchtowers and wide empty bands. A magical barrier (a ward line, a cursed river) can create a sharp line border even in wilderness.

## Language lines are not borders

### RULE OF THUMB

A language line is not a political border. It is usually older and it moves much more slowly. Land changes hands across a language line many times while the line itself stays put. When a language line does move, it creeps a few villages at a time over centuries.

### Why they differ

- **Villagers stay put.** When the lord or king changes, the villagers keep speaking what their grandparents spoke.

### Real examples

| Example                                      | What happened                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | What it shows                                                                            |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| The Romance–Germanic line in today's Belgium | It took shape as Frankish settlers moved in, roughly between the 4th and 10th centuries. After about the 10th century it hardly moved in what is now Belgium. (Farther west, the coast from Boulogne to Dunkirk became French-speaking only in the later Middle Ages.) Medieval states sat across it: the County of Flanders held Romance-speaking (Picard) Lille and Douai, and the Prince-Bishopric of Liège held Romance-speaking (Walloon) Liège and the Dutch-speaking County of Loon. The line became a legal boundary only in 1962–63. | A stable line, with realms on both sides of it                                           |
| Breton in Brittany                           | Breton reached its farthest east in the 9th century, roughly from the bay of Mont-Saint-Michel to Donges on the Loire (the "Loth line", named after the linguist Joseph Loth). Even then the region's chief cities, Rennes and Nantes, stayed Romance-speaking. After that the line moved west: to about Dinan–Pornichet by the 12th century, and to about Binic–Guérande by 1588. That is roughly 30–80 km (20–50 mi) in seven centuries (derived from these places).                                                                        | A slow retreat inside one realm, with the capital on the "wrong" side of the line        |
| Transylvanian Saxons (a language island)     | King Géza II of Hungary (1141–62) began bringing German-speaking settlers to the thinly settled south-eastern edge of his kingdom, to farm it and defend it. The <i>Diploma Andreanum</i> (1224) gave them self-government on their own "Royal Land". Their chief towns included Hermannstadt (Sibiu), Kronstadt (Braşov) and Bistritz (Bistriţa), and they built about 300 villages with fortified churches. In 1848 the Royal Land still held 271 villages, market towns and towns.                                                         | A language island planted by a ruler, lasting for centuries inside another language area |
| Mudéjar villages in Valencia                 | After James I of Aragon conquered Valencia (1230s–40s), whole villages of Muslims ( <i>Mudéjars</i> ) kept their Arabic speech and their faith under Christian lords. In the late 1300s they may still have been a majority of the countryside. See <a href="#">Religious, Cultural, Legal and Ancient Sites</a> .                                                                                                                                                                                                                            | The border moved at once; the language of the villages did not                           |

### MAP TIP

Draw languages or cultures as a separate layer: a soft colour tint only, never a line, so nobody mistakes a language edge for a border or a track. Let it cross borders, and let one realm hold two or three language areas. Sometimes put the capital on the "wrong" side of the line, like Rennes. Change the place-name endings across the line, for example German-style *-ingen* and *-heim* on one side and French-style *-ville* and *-court* on the other (see [Naming settlements realistically](#)). Add one to three language islands where a ruler planted settlers: on an empty frontier, in a mining district or around a new town.

Language follows who settled and farmed the land long ago, not who rules it now.

- **Realms are built across language lines.** Marriage, inheritance and war join lands with different languages into one realm, and split lands with one language between several.
- **Lines move by slow pressure.** A language spreads when towns, lords, the church and trade use it. Its rival survives longest in remote villages. A language can also jump in all at once, when a ruler brings in settlers.

### FANTASY TWIST

Long-lived peoples change their speech slowly, so elf or dwarf language lines can be even older and more stable than human ones. A dwarf hold or an elven wood inside a human kingdom is a language island: give it its own place names, even if a human king rules the land around it.

## Kinds of states and how they look on a map

### RULE OF THUMB

Do not colour all of Europe-like land in big, solid blocks. Medieval maps were patchworks: kingdoms, duchies, free cities, church lands, leagues and empires overlapped and interlocked.

| Kind of state                     | How it works                                                                                                                                            | How it looks on a map                                                                                                                      | Real examples                                                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feudal kingdom                    | A king over dukes, counts and barons who hold land in return for loyalty and service. The royal domain is only one part.                                | One outline, but inside it a patchwork of great fiefs. The royal domain is the core around the capital. Some fiefs are nearly independent. | France c. 1200: royal domain in the Île-de-France, surrounded by Normandy, Flanders, Champagne, Burgundy, Aquitaine and Toulouse                                                                                                                                                                            |
| City-state and its <i>contado</i> | A self-governing city rules the countryside around it (the <i>contado</i> , often the same area as its diocese) and later conquers neighbouring cities. | A city with a territory of roughly 20–50 km (12–30 mi) radius, many such states side by side, with borders that move often.                | Florence (took Arezzo in 1384, conquered Pisa in 1406). In 1427 the Florentine state had roughly 260,000 people, of whom the city held only about 37,000–38,000 (the <i>catasto</i> tax survey, studied by Herlihy and Klapisch-Zuber). Also Milan, Siena, Genoa.                                           |
| League of towns                   | Towns cooperate for trade and defence but keep their own small territories.                                                                             | A network, not an area: dots linked by sea routes, plus trading posts in foreign cities.                                                   | Hanseatic League: between about 70 and 170 member towns at different times, and nearly 200 places linked to it overall. Its power peaked around 1370 (Peace of Stralsund). Main trading posts ( <i>Kontore</i> ) in Novgorod, Bruges, London (the Steelyard) and Bergen; meetings (diets) usually in Lübeck |
| Confederation                     | Rural valleys and towns ally for defence and rule conquered land together.                                                                              | A cluster of small units; some areas shown as joint subject lands.                                                                         | Swiss Confederacy: 8 cantons by 1353, 13 by 1513. Aargau (1415) and Thurgau (1460) were ruled jointly as subject lands.                                                                                                                                                                                     |
| Composite empire                  | Many realms and hundreds of lords under one elected ruler with weak central power.                                                                      | A big outer border with hundreds of small internal units.                                                                                  | Holy Roman Empire: ~400 imperial estates in 1521, including 87 imperial cities and ~50 prince-bishops and archbishops                                                                                                                                                                                       |
| Church state                      | A bishop, abbot or the Pope rules land as a prince. The territory is not the same as the diocese.                                                       | Separate colour; often scattered pieces.                                                                                                   | Papal States in central Italy; prince-bishoprics of Cologne, Mainz, Trier, Liège, Münster, Würzburg, Salzburg; Durham in England                                                                                                                                                                            |
| Military-order state              | A crusading order of knights rules conquered land.                                                                                                      | A frontier territory with many castles and commanderies (houses of the order).                                                             | Teutonic Order in Prussia, seat at Marienburg (Malbork) from 1309                                                                                                                                                                                                                                           |
| Maritime empire                   | A trading city holds ports and islands along a sea route, later also land behind it.                                                                    | A home city plus a chain of coastal dots and islands, not a solid area.                                                                    | Venice: Crete (c. 1210–1669), Negroponte (to 1470), Modon and Coron (1207–1500), Corfu (from 1386), Cyprus (1489–1570), Dalmatian ports; mainland expansion from 1337–39 (Mestre, Treviso) to Brescia and Bergamo (1420s)                                                                                   |
| Nomadic empire                    | Herders on seasonal pastures; settled peoples nearby pay tribute.                                                                                       | A huge, vague area: core pastures, a few capital cities, and a ring of tributary states around them. Borders are wide zones.               | Mongols: Karakorum walled in 1235; the Golden Horde's Sarai on the lower Volga (founded by Batu in the 1240s–1250s), destroyed by Timur in 1395–96. Khans also kept moving camps and tent palaces. Rus' princes paid tribute but kept their own cities.                                                     |

### Drawing the unusual kinds

- **City-states:** many small territories, each 1–3 days across, centred on rival walled cities about 50–90 km (30–55 mi) apart.
- **Leagues:** do not colour an area. Underline each member town's name in the league's colour (members can sit inside other rulers' kingdoms) and draw dotted sea lanes between them.

- **Nomadic empires:** show summer and winter pastures, one or two built capitals on rivers or trade routes, and the tributary ring in a lighter tint of the empire's colour.
- **Maritime empires:** colour only the islands and harbour towns along the sea route; the land behind them keeps the local ruler's colour.

**MAP TIP**

For a continent map, use at least three kinds of state: one or two big feudal kingdoms, one fragmented zone (city-states or a composite empire) and one unusual kind (a league, a church state, a maritime empire or nomads). This mix is what makes a medieval map look medieval.

**FANTASY TWIST**

Different peoples suggest different state types. A dwarven realm may be a league of mountain holds linked by tunnels (a network, like the Hanse). An elven forest realm may look like a nomadic or church state with vague borders. A wizards' city may rule a contado like Florence.

## How big can a realm be

**RULE OF THUMB**

A ruler can govern directly up to roughly 500–700 km (300–450 mi) from the capital. At a messenger's normal pace of about 50–65 km (30–40 mi) a day, that is about 8–12 days' travel. Beyond that, the ruler needs viceroys (governors who rule in the king's name), great vassals, sea routes or a relay post, or the realm splits. This is a guide's heuristic, not a measured law.

### Why distance limits power

- Orders, tax money and news travel at the speed of a horse (see the travel table in [Trade Routes, Roads and Transport](#)).

- A rebellion far away can grow for weeks before the king even hears of it.
- Feudal armies often owed only about 40 days of service, so a host raised in the core could not campaign for long on a far frontier.

### Distance from the capital and what it means

Speeds from the shared baseline: royal messenger 50–90 km (30–56 mi) a day; army with baggage 13–20 km (8–12 mi) a day.

| Distance from capital | Royal messenger | Army        | What it means                                                                          |
|-----------------------|-----------------|-------------|----------------------------------------------------------------------------------------|
| 100 km (60 mi)        | 1–2 days        | 5–8 days    | Core: royal domain, residences, hunting forests. Direct control.                       |
| 300 km (190 mi)       | 3–6 days        | 15–23 days  | Normal kingdom: counties ruled by royal officers (sheriffs, bailiffs, <i>ispáns</i> ). |
| 600 km (370 mi)       | 7–12 days       | 30–45 days  | Edge of a large kingdom: needs great lords, marcher lords or a royal governor.         |
| 1,000+ km (600+ mi)   | 11–20 days      | 50–75+ days | Only possible for empires with sea routes, relay posts or semi-independent governors.  |

A real measure comes from news reaching Venice around 1500 (after Sardella and Braudel). Venice had one of the best news networks in Europe. Typical times were:

| From                                         | Time for news to reach Venice (c. 1500) |
|----------------------------------------------|-----------------------------------------|
| Milan, Genoa, Florence                       | About 1 week                            |
| Paris, Lyon, Marseille, Vienna, Buda, Naples | About 2 weeks                           |
| Antwerp, Cologne, Barcelona                  | About 3 weeks                           |
| London                                       | 3–4 weeks                               |
| Castile (Burgos, Valladolid), Constantinople | 4–5 weeks                               |
| Lisbon                                       | About 6 weeks                           |

## Realistic realm sizes

| Realm                                         | Area                                                 | Population                       | Farthest point from the capital (approx.)                                                  |
|-----------------------------------------------|------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------|
| Andorra (micro-state)                         | 468 km <sup>2</sup>                                  | tiny                             | Under 1 day                                                                                |
| Navarre                                       | ~10,000–12,000 km <sup>2</sup>                       | ~100,000 (c. 1300)               | 2–3 days                                                                                   |
| Scotland                                      | ~79,000 km <sup>2</sup>                              | 0.5–1m (c. 1300; estimates vary) | 5–8 days                                                                                   |
| Portugal (border almost unchanged since 1297) | ~89,000 km <sup>2</sup>                              | ~1.0m (1500)                     | 4–7 days                                                                                   |
| England                                       | 130,000 km <sup>2</sup>                              | 4.75m (1290)                     | London to Berwick ~500 km (310 mi) in a straight line, ~550 km (340 mi) by road: 6–11 days |
| Poland (1370)                                 | ~240,000 km <sup>2</sup>                             | ~2m                              | 6–10 days                                                                                  |
| Hungary                                       | 280,000–325,000 km <sup>2</sup>                      | 2–3m                             | 8–12 days                                                                                  |
| France (1328)                                 | smaller than modern France's 551,000 km <sup>2</sup> | 16–17m                           | 10–14 days to the Mediterranean or the Pyrenees                                            |
| Holy Roman Empire                             | 0.8–1.1m km <sup>2</sup>                             | 11–14m (c. 1300)                 | Weeks; no single capital                                                                   |
| Byzantium (1025)                              | 1.675m km <sup>2</sup>                               | ~12m                             | Weeks, but sea routes helped                                                               |

Around 1500, Europe held roughly 500 more or less independent political units (Tilly 1975), from tiny lordships to great kingdoms. Most were small.

Realms that grew too big had three choices. They split (the Carolingian Empire at Verdun in 843; the Mongol Empire into separate khanates by the 1260s). They were run through regents and governors (Charles V). Or they held their distant parts by sea (Venice, Byzantium).

### MAP TIP

Measure from your capital along the roads, not in a straight line. Mark the 5-day and 12-day messenger lines in pencil (about 250–300 km and 600–700 km, or 150–190 mi and 370–430 mi, at a normal messenger pace). Put direct royal counties inside the first line, great vassals and marches between the lines, and tributary or semi-independent lands beyond the second. If your kingdom is much bigger than that, add a second-tier capital (a viceroy's seat) or make the outer parts nearly independent.

### LATER ERA (1500S+)

Organised postal relays (Taxis, from about 1500, up to ~150 km or 95 mi a day) and faster shipping roughly double the range a capital can control. Empires like Spain and the Ottomans govern huge areas through viceroys and provincial governors. Borders are increasingly surveyed, mapped and marked by boundary commissions. See [What Changes After 1500](#).

### FANTASY TWIST

Every faster messenger extends the realm. A griffon post at 300 km (190 mi) a day is 3–6 times faster than a royal messenger, so orders reach 3–6 times farther in the same time. Instant magical messages remove the communication limit but not the army limit. Armies with baggage still march at 13–20 km (8–12 mi) a day unless magic moves them too.

## Drawing realms, borders and capitals

### Step by step

1. **Choose the realm type** and **find the core**: the best farmland on a navigable river, with a defensible hill or island for the capital.
2. **Add the royal landscape**: 3–6 residences, a royal forest, the burial church, perhaps a separate coronation town.
3. **Draw the outer border**: dash-dot lines in settled land, march bands across mountains, forests, marshes and war zones.
4. **Divide** into 4–8 duchies or great fiefs, then counties 50–70 km (30–45 mi) across, each with a seat near its middle. Add a few oddities (church land, an enclave, a free city, a disputed strip).
5. **Check distance**: anything more than about 12 days' messenger travel (roughly 700 km or 430 mi by road) from the capital needs a viceroy or becomes semi-independent.

## Symbols and line styles

### RULE OF THUMB

One symbol, one meaning. A filled star means the capital and nothing else. Borders are always dash-dot lines (a line of dashes with dots between them); roads, tracks and sea lanes are always solid or dotted lines. Then nobody mistakes a border for a road.

These symbols match the guide's [master legend](#). The "town symbol" below means the normal symbol for the place's size (village, town, city and so on), from [Map symbols and labels](#).

| Feature                                    | Symbol                                                                                   | Scale              |
|--------------------------------------------|------------------------------------------------------------------------------------------|--------------------|
| Realm capital                              | Filled star in place of the town symbol, bigger for a bigger city; name in bold capitals | Continent, kingdom |
| Coronation or royal burial town            | Small crown beside the town symbol, labelled "(coronations)" or "(royal tombs)"          | Kingdom            |
| Duchy or regional seat                     | Town symbol with a small pennant (flag) on top                                           | Kingdom            |
| County town                                | Town symbol with a dot in the centre                                                     | Kingdom            |
| Royal residence or hunting lodge           | Small crown on its own                                                                   | Kingdom, local     |
| Royal forest                               | Green dash-dot boundary with a large italic name; villages and fields inside             | Kingdom, local     |
| Moot (open-air meeting place of a hundred) | Small lone tree, labelled "Moot"                                                         | Local              |
| Toll at a border crossing                  | One bar across the road or river, with a "T"                                             | Kingdom, local     |
| Realm border                               | Thick dash-dot line, or a coloured band along the inside edge                            | Continent, kingdom |
| Duchy or great fief border                 | Medium dash-dot line                                                                     | Continent, kingdom |
| County border                              | Thin dash-dot line                                                                       | Kingdom            |
| Hundred, parish or manor bounds            | Thin dash-dot-dot line                                                                   | Local              |
| Boundary stone                             | The letters "BS" on the boundary line, as on Ordnance Survey maps                        | Local              |
| Disputed border                            | Two parallel dash-dot lines, or alternating colours                                      | Kingdom            |
| March or frontier zone                     | Hatching, or a fading colour band 10–50 km (6–30 mi) wide                                | Continent, kingdom |
| Enclave                                    | Same colour as its owner, with a thin outline                                            | Kingdom            |
| League member town                         | Town symbol as usual; name underlined in the league's colour                             | Continent, kingdom |
| Language or culture area                   | Soft colour tint only, never a line                                                      | Continent, kingdom |

Castles, beacons and watchtowers on the border use the symbols in [Castles, Forts and Military Outposts](#).

## Colour and labels

- Use pale fills for realms, so roads and towns stay readable. Give vassal duchies a lighter shade of their king's colour.
- Give church lands and free cities their own colours, and underline league towns' names in the league's colour. Do not merge them into the surrounding kingdom.
- Write realm names in widely spaced capitals across the whole territory. Write duchies in smaller italic capitals and counties in lower case.

- Label marches and frontier zones by name ("the West March", "the Debatable Land"). Named empty spaces make a map feel lived-in.

### MAP TIP

On a continent map, show only realms and great fiefs. On a kingdom map, add counties and the capital's royal landscape. On a local map, add hundreds, manors and parish bounds (thin dash-dot-dot lines) and boundary stones ("BS"). Do not show every level at every scale.

## Quick summary

- A capital needs a rich core region, navigable water, a defensible site, loyal land around it and a source of legitimacy. Centrality matters less.
- Expanding realms move their capital toward the frontier ("forward capital"); settled realms keep the old core.
- Before c. 1200 most western courts travelled. German kings moved between royal palaces and manors about 30 km (19 mi) apart. Fixed capitals came with archives, law courts and money taxes: Paris and Westminster in the 12th–14th centuries, Madrid only in 1561. The Holy Roman Empire never had one.
- Coronation and burial cities are often different from the governing city, and keep their role for centuries.
- Around a capital, draw 3–6 royal residences and hunting forests within about 60 km (37 mi). At their peak (late 12th and early 13th centuries), royal forests covered about one-third of southern England.
- Mints shrink as kings get stronger: about 70 in late Anglo-Saxon England, 9 in 1189, normally 4 under Henry III. Loose realms like the Holy Roman Empire had more and more.
- Typical sizes: kingdom 10,000–450,000 km<sup>2</sup>; duchy 10,000–50,000 km<sup>2</sup>; county 1,000–10,000 km<sup>2</sup> (England averaged ~3,300 km<sup>2</sup> and ~120,000 people); hundred ~130–150 km<sup>2</sup> in 1086 (~150–200 km<sup>2</sup> by 1316); manor 2–10 km<sup>2</sup>.
- A bureaucratic empire (China) looks more regular: about 1,230–1,385 counties, each with a walled seat roughly 45–55 km (28–34 mi) from the next, and periodic market towns between them.
- Borders are lines in settled land and wide zones (marches) in empty land. They follow old boundaries, crests, rivers, forests and wasteland. Marches, buffer states, enclaves and contested strips were normal: add some. Draw every border as a dash-dot line and keep the filled star for the capital only.
- Language lines are older and slower than borders. Draw them as a separate layer that crosses borders, and add a few language islands of planted settlers.
- Medieval political maps are patchworks: kingdoms, city-states, leagues, church states, maritime chains and nomadic empires side by side. Europe c. 1500 had about 500 political units.
- Direct rule reaches roughly 500–700 km (300–450 mi), about 8–12 days' messenger travel. Beyond that, rulers need viceroys, vassals, sea routes or relays.

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Population, area and travel-speed figures follow the guide's shared baseline (Broadberry, Campbell et al.; Russell 1972; McEvedy & Jones 1978; Hill 1961). Figures marked "estimate" or "derived" (duchy populations, county population ranges, travel-time zones, Chinese county sizes and spacing, the distance of the Breton retreat) are worked out by this guide from area, density or map distances and are not measured values.

## CHAPTER 4

# CASTLES, FORTS AND MILITARY OUTPOSTS

Castles are the most "fantasy" feature on a medieval map, and also the easiest to get wrong. Real castles were not scattered at random. Each one controlled something: a road, a river crossing, a town, a border or an estate. This chapter explains what fortifications were for, the main types and their dates, where they were built, how many there were and how far apart, how big their garrisons and armies were, how armies moved, camped and were fed, and how to draw all of this on your map.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Borders and marches as political units are covered in [Capitals, Realms and Borders](#). Roads, river transport and travel speeds are covered in [Trade Routes, Roads and Transport](#).

## IN THIS CHAPTER

What fortifications are for  
Types of fortification  
Where to put a castle  
How many castles and how far apart  
Castles and towns  
Garrisons and armies  
Military geography and logistics  
Camps and siege towns  
Outposts in wild and contested land  
Watchtowers, beacons and coastal warning  
Where battles happen  
Drawing military sites on your map  
Quick summary  
Sources and further reading

## What fortifications are for

## RULE OF THUMB

Every fortification answers one question: "What does it control?" If you cannot name the road, crossing, town, border or estate it watches, move it or delete it.

A castle was not only a military building. It was the lord's house, his office, his court, his prison, his store for rents paid in grain, and a barracks. Many historians (for example Charles Coulson) stress that walls and battlements were also a **status symbol**. A castle said "a lord rules here".

| Purpose                      | What it looks like                                                        | Typical site                                                     | Real example                                                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control a route              | A castle or tower right beside a pass, ford, bridge or narrow river       | The narrowest point of the route                                 | Krak des Chevaliers (Syria) guarded the Homs Gap between the coast at Tripoli and inland Homs. Château Gaillard (1196–98) guarded the Seine valley road to Rouen. |
| Protect a town or port       | Town walls and gates, often with a castle or citadel at one corner        | Around the town; the castle on the highest point or by the river | Carcassonne (two walls, about 3 km / 1.9 mi); Ávila (about 2.5 km / 1.6 mi); Visby (3.4 km / 2.1 mi)                                                              |
| Hold a border                | A chain of castles and towers, much denser than inland                    | Along the border, at every crossing                              | The Welsh March; the Arab–Byzantine frontier (the <i>thughur</i> ); the Christian–Muslim frontier in Spain                                                        |
| Dominate conquered land      | Big new castles in or beside the main towns, often with new settler towns | Inside existing towns, at ports, on rivers                       | William I's castles in English towns after 1066; Edward I's castles in North Wales (1277–1295); the Teutonic Order in Prussia (from 1231)                         |
| Lord's home and admin centre | A castle at the head manor (the <i>caput</i> ) of a barony                | Near the best farmland, a village and a church                   | Most of the 500–600 castles occupied at any one time in post-Conquest England                                                                                     |
| Refuge                       | Fortified churches, tower houses, walled villages, old hillforts          | Inside the village                                               | Over 150 Saxon villages in Transylvania with fortified churches (13th–16th centuries)                                                                             |
| Watch and signal             | Watchtowers, beacons, coastal towers                                      | Hilltops and headlands with long views                           | The Byzantine beacon line across Asia Minor (9th century)                                                                                                         |

**MAP TIP**

Before you place a castle, write down its job in one word: *road, bridge, town, border, estate, refuge* or *watch*. The job tells you the site. A "road" castle sits on the road; an "estate" castle sits in the middle of good farmland; a "border" castle sits on the border.

## Types of fortification

### Castles: from earth to stone

| Type                                     | Dates                                                                                       | What it is                                                                                                                                      | Real examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ringwork                                 | 10th–12th centuries                                                                         | A ring of earth bank and ditch with a timber fence (palisade) on top. No mound.                                                                 | Common in south-west England and south Wales after 1066                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Motte-and-bailey                         | From c. 1000 in the Loire valley and Normandy; in England after 1066                        | A <b>motte</b> (a man-made earth mound) with a timber tower on top, and a <b>bailey</b> (a fenced yard) beside it for halls, stables and stores | About 80% of Norman castles in England were of this type. Mottes were about 3–30 m (10–100 ft) high and 30–90 m (100–300 ft) across at the base. A bailey usually covered about 0.15–0.7 ha (0.4–1.7 acres): The Mount, Ilketshall St John (Suffolk), has a bailey platform of 53 × 30 m; Ewyas Harold (Herefordshire) has a bailey of about 0.7 ha. A large baronial castle such as Castle Acre (Norfolk) covered about 3 ha (7.5 acres) to the outer edge of its defences. See <i>Footprints</i> . An average motte, 5 m (16 ft) high and 15 m (49 ft) across at the top, is estimated to have taken 50 people about 40 working days. The biggest mottes took up to about 24,000 working days. |
| Stone keep (great tower, <i>donjon</i> ) | From c. 950 in France; in England from the 1070s                                            | A massive stone tower holding the lord's hall and rooms                                                                                         | Doué-la-Fontaine (France, c. 950, often called the oldest known keep); the White Tower, London (c. 1075–1100); Dover's great tower (1180s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Shell keep                               | Mostly 11th–12th centuries                                                                  | A stone ring wall replacing the timber fence on top of a motte, because a mound cannot carry a heavy tower                                      | Windsor (Henry II); Restormel (1270s–80s); Lewes; Launceston                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Curtain-wall castle                      | Late 12th–13th centuries                                                                    | A strong stone wall (the <b>curtain</b> ) with towers along it that let defenders shoot along the wall face                                     | Framlingham (late 1100s, finished by 1213; 13 wall towers and no keep); Château Gaillard (1196–98, taken 1204)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Concentric castle                        | Byzantine examples from the 11th century; crusader examples from the 1160s; Wales 1268–1295 | Two or more rings of walls. The inner wall is higher, so both rings can shoot at once.                                                          | Belvoir (Hospitallers, from 1168); Krak des Chevaliers (Hospitaller from 1142; its outer ring of walls was added in the early 1200s); Caerphilly (begun 1268); Harlech (1282–89, cost £8,190); Beaumaris (begun 1295, never finished)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Stone castles were slow and costly. Stone needed skilled masons, and they had to be paid. A keep's walls usually rose no more than about 3.7 m (12 ft) a year, so a great tower took around ten years (the keep at Scarborough is a typical case). Edward I spent at least £80,000 on his Welsh castles between 1277 and 1304. For Beaumaris alone his master

mason asked for 400 masons, 2,000 labourers, 100 carts, 60 wagons and 30 boats. Earth-and-timber castles, by contrast, could go up in a few weeks or months, within a single building season, using unpaid local labour. That is why invaders built them first.

## Small fortified homes

| Type                        | Dates                                                         | What it is                                                                                                                                                                                                   | Real examples and numbers                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tower house                 | 14th–17th centuries                                           | A tall stone house of 3–6 storeys for one family and its servants, often with a walled yard ( <i>barmkin</i> in Scotland, <i>bawn</i> in Ireland). Built where raiding was common and the king was far away. | Ireland: over 3,000 built (more than 2,000 still stand). In 1429 the English crown offered £10 to any man in the Pale (the English-ruled area around Dublin) who built one, though recent studies doubt this made much difference. Scotland about 800. England over 250, mostly near the Scottish border. Also common in the Basque lands, Italy and the Caucasus. |
| Peel tower and bastle house | Peels 14th–17th centuries; bastles mostly 16th–17th centuries | A <b>peel</b> is a small tower house on the Anglo-Scottish border. A <b>bastle</b> is a fortified farmhouse: animals below, family above.                                                                    | Northumberland, Cumbria and the Scottish Borders                                                                                                                                                                                                                                                                                                                   |
| Fortified manor house       | 13th–15th centuries                                           | A manor house with a moat, gatehouse and battlements. Strong enough to stop a raid, not an army.                                                                                                             | England has about 460 surviving <b>licences to crenellate</b> (royal permission to add battlements) for just over 500 sites, mostly for knights. Stokesay (1291), Cooling (1381), Bodiam (1385), Herstmonceux (1441).                                                                                                                                              |

## Community defences

| Type                           | Dates                                              | What it is                                                                                     | Real examples and numbers                                                                                                                                                                                                                                                             |
|--------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Town walls and gates           | All periods; a big wave in the 12th–14th centuries | Stone wall or earth bank and ditch, with towers and gatehouses. Gates also collected tolls.    | Carcassonne about 3 km (1.9 mi); Ávila about 2.5 km (1.6 mi); Visby 3.4 km (2.1 mi). England paid for walls with <b>murage</b> (a toll on goods entering a town), first granted to Shrewsbury in 1220; about 1,470 murage grants are recorded in England, Wales, Scotland and France. |
| Citadel                        | All periods                                        | A fortress inside or at the edge of a city. It defends the city, and also watches it.          | Tower of London; Cairo Citadel (begun by Saladin in 1176); Aleppo Citadel (rebuilt c. 1200)                                                                                                                                                                                           |
| Fortified bridge               | 12th–14th centuries                                | A bridge with a gate tower on it, to stop attackers and collect tolls                          | Pont Valentré, Cahors (1308–78; 138 m / 453 ft long, three towers); Monnow Bridge, Monmouth (bridge traditionally dated 1272, gate tower added c. 1300), the only British river bridge with its gate tower still standing on it                                                       |
| Fortified church or monastery  | 13th–17th centuries                                | A church with thick walls, a strong tower and often a ring wall. Villagers took refuge inside. | Transylvania: over 150 villages (13th–16th centuries); Slovenia: about 300 churches fortified against Ottoman raids (late 15th–17th centuries); Albi Cathedral, a brick fortress-church (1282–1480)                                                                                   |
| Reused hill-fort or Roman fort | Whenever old walls were handy                      | Old earthworks or Roman walls given a new castle inside                                        | Old Sarum: an Iron Age hillfort about 400 × 360 m (1,300 × 1,180 ft), a Norman castle by 1069 and a cathedral in 1092, abandoned for New Salisbury from 1220. Pevensey and Portchester: Norman castles built inside Roman forts.                                                      |

## Frontier and special types

| Type                          | Dates               | What it is                                                                                                                                                      | Real examples                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crusader castle               | 1100–1291           | Large stone castles on spurs and hills, held by small numbers of Western knights among a hostile population                                                     | Kerak (1142); Krak des Chevaliers (Hospitaller 1142–1271); Chastel Pèlerin at Atlit (Templar, 1218–1291); Montfort (Teutonic Order, 1220s–1271)                                                                                                                                                                                                  |
| Islamic <i>ribat</i>          | 8th–11th centuries  | A fortified lodge for volunteer frontier fighters, with a watch and signal tower. Built in chains along coasts and borders.                                     | Monastir (founded 796) and Sousse (rebuilt or enlarged 821) in Tunisia                                                                                                                                                                                                                                                                           |
| Islamic frontier fortress     | 8th–15th centuries  | Big walled fortresses and fortified towns in the frontier zone                                                                                                  | Gormaz, Spain (built or rebuilt 965–66 for the caliph of Córdoba; about 1 km / 0.6 mi of walls with 24–27 towers); Tarsus (main Arab base facing Byzantium, 4,000–5,000 troops in its garrison)                                                                                                                                                  |
| Military-order castle         | 12th–15th centuries | A castle that is also a monastery for knight-monks (an <i>Ordensburg</i> in German)                                                                             | Malbork/Marienburg (begun c. 1274, built over about 130 years; Teutonic Order headquarters from 1309; 21 ha / 52 acres inside its outer walls, the largest castle by area); Calatrava (Order of Calatrava, from 1158)                                                                                                                            |
| Siege castle (counter-castle) | All periods         | A fort built to block an enemy castle or town: either a temporary siege fort close by, housing the besiegers, or a permanent counter-castle within a day's ride | Malregard at Antioch (1097); Mons Peregrinus at Tripoli (1102–03; the city fell only in 1109); four castles ringing Fatimid Ashkelon (Castrum Arnaldi, Beth Gibelin, Ibelin and Blanchegarde, 1133–42; the city fell in 1153). In 12th-century England, siege castles stood about 180–270 m (200–300 yd) from the target, just beyond bow range. |

### MAP TIP

Match the castle type to your map's date. Around 1100, almost every castle is earth and timber. By 1300, the important ones are stone and the small ones are either rebuilt in stone or abandoned. From 1350, add tower houses and fortified manors, and let old castles decay.

### LATER ERA (1500S+)

Cannon broke tall thin walls (the Ottomans breached Constantinople's walls in 1453). From the late 1400s Italian engineers built walls **lower and thicker**, often of earth faced with brick. They added angled **bastions** (projecting platforms for guns). This *trace italienne* ("Italian outline"), or "star fort", spread across Europe in the 1530s–1540s. A famous example is Palmanova (1593), a whole star-shaped new town. Star forts were very expensive: Siena bankrupted itself on its walls in 1544. Henry VIII's **Device Forts** (1539–47, total cost £376,000) were low, round artillery forts guarding harbours and anchorages, such as Deal and Pendennis. Their garrisons were small: from 9 men (West Tilbury) to 29 (Camber). Draw one with the guide's coastal artillery fort symbol: a small **rosette** outline (rounded lobes, like the plan of Deal Castle, 1539–40, a round tower ringed by six rounded bastions). Never give it points: pointed outlines mean star forts. See [What Changes After 1500](#).

## Where to put a castle

### RULE OF THUMB

Put the castle where a few men can stop many: at the narrowest point of a route, on the strongest ground next to it, with water nearby.

| Site                              | Why it works                                                                                                        | Real example                                                                                                                              |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| River crossing (ford or bridge)   | Every road and every boat must pass                                                                                 | Stirling (lowest crossing of the Forth); Chepstow (begun 1067, on a cliff above the Wye)                                                  |
| Mountain pass or gap              | The only easy way between two regions                                                                               | Krak des Chevaliers (Horns Gap); Loulon at the northern exit of the Cilician Gates (Byzantine–Arab frontier)                              |
| Spur or promontory                | A ridge end with cliffs on two or three sides. Cut one ditch across the neck and the castle is strong on all sides. | Château Gaillard above the Seine; Kerak; Montfort                                                                                         |
| River bend (meander)              | Water on three sides, a short neck to defend                                                                        | Durham (castle begun 1072 in a loop of the Wear); Toledo in a loop of the Tagus                                                           |
| Confluence of two rivers          | Controls traffic on both rivers                                                                                     | Koblenz, where the Moselle meets the Rhine                                                                                                |
| Island or lake shore              | Water as a moat; controls river or lakeside road                                                                    | Pfalzgrafenstein, an island in the Rhine (1326–27); Chillon, on Lake Geneva beside the road to the Great St Bernard Pass                  |
| Coastal headland or harbour mouth | Sea on several sides; supply by ship; watches the harbour                                                           | Dunstanburgh (begun 1313); Harlech, with a water gate down to the shore                                                                   |
| Hill above a town                 | Overlooks and protects the town, and watches it                                                                     | Edinburgh; Prague; Carcassonne                                                                                                            |
| Flat wet lowland                  | No hills, so dig a moat or flood the ground                                                                         | Caerphilly (artificial lakes); Bodiam (moat); Muiderslot near Amsterdam (c. 1285). These are "water castles" (German <i>Wasserburg</i> ). |

**Hills or lowlands?** It depends on why the castle was built.

- In the German lands about two-thirds of castles were on high ground (*Höhenburg*, "height castle"). Many lords there built for safety and display on wooded spurs.
- In England most castles stood on low ground, next to villages and towns. Norman castles were built to control people and farmland, so they sat where the people were.

#### MAP TIP

Draw the terrain first, then look for the "choke points": fords, bridges, passes, narrow valleys, river bends and harbour mouths. Put a castle at the most important ones. Put estate castles in the middle of good farmland, beside a village. Do not put a castle on a remote peak far from any road or village. Nobody would pay for it.

#### FANTASY TWIST

If your world has flying mounts or flying monsters, high spurs lose some value and roofs matter more. Expect covered courtyards, roofed towers, heavy crossbows aimed upward, and aeries (bases for flying mounts) inside the strongest castles. See [Fantasy Variants](#).

## How many castles and how far apart

### Counts and densities

The numbers below count **every castle site ever built** in the period, including small mottes that were used for only a few decades. The column "typical gap" is the square root of

the area per castle. It is a rough guide to the distance from one castle to the next.

| Region                                                                           | Castles                                               | Area per castle                                 | Typical gap to next castle                                    |
|----------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|
| England and Wales overall                                                        | 1,761 sites (Gatehouse gazetteer)                     | ~86 km <sup>2</sup> (33 sq mi)                  | ~9 km (6 mi)                                                  |
| Peaceful eastern England (Norfolk, Essex, Kent, Suffolk, Lincolnshire)           | 22–45 per county                                      | 130–250 km <sup>2</sup> (50–95 sq mi)           | 11–16 km (7–10 mi)                                            |
| Welsh March (Herefordshire, Shropshire, Monmouthshire, Pembrokeshire, Glamorgan) | 60–127 per county                                     | 17–28 km <sup>2</sup> (7–11 sq mi)              | 4–6 km (2.5–4 mi)                                             |
| North-west Wales (Caernarfonshire), conquered 1283                               | 21                                                    | ~71 km <sup>2</sup> (27 sq mi)                  | ~8 km (5 mi), but a few very large royal castles              |
| Anglo-Scottish border (Northumberland, Cumberland)                               | 32–46 per county                                      | 110–130 km <sup>2</sup> (42–50 sq mi)           | ~11 km (7 mi), plus hundreds of tower houses not counted here |
| German-speaking lands                                                            | ~14,000                                               | ~35–40 km <sup>2</sup> (14–15 sq mi) (estimate) | ~6 km (4 mi)                                                  |
| Switzerland                                                                      | ~4,000 (many very small)                              | ~10 km <sup>2</sup> (4 sq mi)                   | ~3 km (2 mi)                                                  |
| Teutonic Prussia                                                                 | Several dozen brick convent castles plus smaller ones | ~370 km <sup>2</sup> (143 sq mi) (estimate)     | ~19 km (12 mi)                                                |
| Livonia (modern Latvia and Estonia)                                              | —                                                     | ~790 km <sup>2</sup> (305 sq mi) (estimate)     | ~28 km (17 mi)                                                |
| Western Europe as a whole                                                        | 75,000–100,000 built                                  | —                                               | —                                                             |

## Notes:

- County figures are from the Gatehouse gazetteer (England and Wales; its total area is about 151,000 km<sup>2</sup> / 58,000 sq mi). The German, Swiss and Western Europe totals follow the guide's shared baseline. The German area per castle assumes German-speaking lands of roughly 500,000–550,000 km<sup>2</sup>, so it is only a rough guide. The Prussia and Livonia figures are uncited estimates quoted on Wikipedia (its Livonia figure also covers Courland and Estonia); treat them as rough.
- **Not all at once.** In England only 500–600 castles were in use at any one time after the Conquest. That is about one-third of all sites. Active castles were therefore about one per 250–300 km<sup>2</sup> (about 95–115 sq mi), or roughly **16–17 km (10–11 mi) apart**.
- **Castle booms happen in times of weak rule.** In Provence there were 12 castles in 950, 30 by 1000 and over 100 by 1030. England had a civil war in 1139–53 (the "Anarchy"). The chronicler Robert of Torigni claimed that 1,115 uncensured castles were built during it, but elsewhere he gave a figure of only 126. Expect exaggeration in your own world's chronicles too.

## Control radius

- **Direct military reach:** only about 400 m (1,300 ft). This was a castle's immediate area of influence, roughly the reach of the defenders' bows and crossbows.
- **Practical control:** a castle's mounted men could ride out, patrol and return in one day. That gives a radius of about

15–30 km (10–20 mi). This is a working heuristic, not a measured figure.

- **Why it still matters:** an enemy cannot safely march past an untaken castle. Its garrison can cut his supply line, attack foragers and retake the land behind him. That is why so much medieval war was sieges.

## Borderlands and conquered land

Borders and newly conquered land have far more castles than the peaceful core of a realm. The Welsh March had roughly **5–10 times** as many castle sites per km<sup>2</sup> as eastern England. (A **march** is a border zone held by local lords with special powers to defend it.)

| Situation                                    | Spacing                                                                                                                                          | Real examples                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peaceful core of a kingdom                   | All sites 11–16 km (7–10 mi) apart; active castles about 20–30 km (12–19 mi) apart (estimate): one per county town plus scattered baronial seats | Eastern England (Norfolk, Essex, Suffolk)                                                                                                                                                                                                                                                                                |
| March or contested border (many small lords) | 4–6 km (2.5–4 mi) apart; small castles, many short-lived                                                                                         | Welsh March, 1070s–1280s                                                                                                                                                                                                                                                                                                 |
| Conquest chain along a coast or river        | 20–40 km (12–25 mi) apart, about one day's march; each castle reachable by water                                                                 | Edward I's coastal castles in Wales, from Flint to Aberystwyth: about 18–50 km (11–31 mi) apart in a straight line, most about 18–25 km (11–16 mi), all supplied by sea. The Teutonic Order's castle-towns along the Vistula (Toruń 1231, Chełmno, Grudziądz, Kwidzyn, Malbork, Elbląg 1237): 25–40 km (15–25 mi) apart. |
| Ring around a hostile city                   | About 25–50 km (15–30 mi) from the city and 10–25 km (6–15 mi) from each other, all within a day's ride of it                                    | The four crusader castles around Fatimid Ashkelon (1133–42), which blocked its raiders until the city fell in 1153. Three of them stood about 25–32 km (15–20 mi) from the city; the fourth, Castrum Arnaldi, was nearer 45 km (28 mi) away.                                                                             |
| Thinly settled frontier zone                 | Fortress towns 25–100 km (15–60 mi) apart, with open land between                                                                                | The Arab <i>thughur</i> facing Byzantium (Tarsus, Adana, Mopsuestia, Malatya); the Duero frontier in 10th-century Spain                                                                                                                                                                                                  |

**RULE OF THUMB**

Peaceful core: a working castle every 20–30 km (12–19 mi). Contested border: one every 4–6 km (2.5–4 mi). Conquest chain: one every day's march (20–40 km, 12–25 mi) along the main road, river or coast.

**MAP TIP**

Do not spread castles evenly. Cluster them along borders, at river crossings and around rich towns. Leave the safe middle of the kingdom fairly empty: a castle in each county town and a few baronial seats. On a frontier, draw a line of castles along the road or river the invaders used, about a day's march apart.

## Castles and towns

**RULE OF THUMB**

Castles and towns come in pairs. A lord who builds a castle wants a market and rents beside it; a town wants protection. Where one exists, look for the other.

Three common patterns:

1. **Castle planted in an existing town.** After 1066 the Normans built most royal castles inside or beside English towns, to control them. They demolished houses to make room: 166 at Lincoln, 113 at Norwich, 27 at Cambridge (Domesday Book).
2. **New town planted beside a new castle.** This is the conqueror's method. Edward I built Conwy, Caernarfon, Harlech and Beaumaris each with a new English borough beside it. At Conwy a royal charter of 1284 made the castle's constable (the officer in charge of the castle) also the town's mayor. The Teutonic Order did the same in Prussia: castle first, then a chartered town under its walls. Many of these towns used Kulm law, a model town charter first issued for Chełmno (Kulm) in 1233.
3. **Village gathered around a castle.** In Italy many villages moved into fortified hilltop sites (*incastellamento*). In

south-west France nearly 700 planned new towns (*bastides*) were founded in 1222–1372, many walled later. See [Where Settlements Are Built](#).

### How they sit together on the ground:

- The castle is at the **edge** of the town, not the middle: usually at the highest point, at a corner of the town wall, or by the river or bridge.
- The castle has its own gate to the outside. The lord could bring in troops even if the town rebelled.
- The town grows along the road leading from the castle gate, with a market place in front of the gate.
- In a capital, the pattern is often "castle on the hill, merchant town below" (see [Capitals, Realms and Borders](#)).

**MAP TIP**

On a local map, draw the castle on the high ground or river bank at one corner of the town wall, with the market street running away from its gate. On a kingdom map, nearly every town of 2,000+ people should have a castle, a citadel or at least walls (see [Population and Settlement Sizes](#)).

## Garrisons and armies

### Garrisons

#### RULE OF THUMB

Garrisons were tiny. The walls did the work. A castle with 30–60 men could often hold off an army ten or more times bigger for weeks or months, if it had food and water.

| Situation                        | Garrison           | Real examples                                                                                                                                                                                                          |
|----------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peacetime caretaker              | 5–20               | A constable (the officer in charge), a porter, watchmen and a few men-at-arms (armoured professional soldiers)                                                                                                         |
| New frontier castle in peacetime | 30–60              | Harlech 1284: 36 (a constable, 30 soldiers of whom 10 crossbowmen, a chaplain, a smith, a carpenter and a mason). Conwy 1284: 30 soldiers, 15 of them crossbowmen, plus a chaplain and craftsmen. Caernarfon 1284: 40. |
| Small castle under siege         | 20–150             | Caernarfon in 1403–04: a garrison of only about 30 men held out against Welsh besiegers of Owain Glyndŵr, who had French help.                                                                                         |
| Great castle in a major war      | 150 to about 1,200 | Dover 1216: 140 knights plus many more soldiers held out against the French prince Louis. Kenilworth 1266: a garrison usually estimated at about 1,200 held out for about six months (June to December).               |
| Great crusader fortress          | 1,500–2,200        | Krak des Chevaliers: about 2,000 at its peak, but only about 60 knights. Safed (rebuilt by the Templars 1240–43, described c. 1260): 1,700 in peace and 2,200 in war, of whom only 50 were knights.                    |
| Frontier fortress city           | 4,000–5,000        | Tarsus, the main Arab base facing Byzantium. In early Abbasid times (late 700s and early 800s) the whole Arab frontier zone held about 25,000 troops.                                                                  |

Most people in a big garrison were not knights. They were crossbowmen, sergeants, servants, craftsmen and (in the East) slaves and workers.

### Field armies and population

| Campaign              | Army (modern estimates)                                                                                                                                                      | Realm population                                    | Share of population                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------|
| Hastings, 1066        | William: 7,000–8,000 (some say 10,000–12,000). Harold: 5,000–13,000, most say 7,000–8,000.                                                                                   | England 1.5–2.0 million                             | ~0.4% (Harold)                       |
| Falkirk, 1298         | Edward I: about 15,000 (about 2,000–2,500 cavalry and about 12,000 paid foot soldiers, many of them Welsh). Some estimates are lower.                                        | England and Wales ~5 million                        | ~0.3%                                |
| Crécy, 1346           | English: 7,000–15,000 (Ayton: ~14,000)                                                                                                                                       | England ~4.8 million                                | ~0.3%                                |
| Siege of Calais, 1347 | English: a peak of about 32,000 in July 1347, the largest English army sent overseas before 1600                                                                             | England ~4.8 million                                | ~0.7%, for a few weeks               |
| Agincourt, 1415       | English: about 6,000–9,000 (Curry: about 8,500–9,000). French: about 12,000 (Curry) or 14,000–15,000 (Rogers, Barker, Sumption); up to ~24,000 if armed servants are counted | England ~2–2.5 million; France well over 10 million | English ~0.3–0.4%; French under 0.3% |

#### RULE OF THUMB

A medieval realm puts about 0.2–0.5% of its people into a field army. About 1% is an all-out effort that cannot last. A kingdom of 4 million people fields 10,000–20,000 soldiers, not 100,000. Chroniclers often claimed much bigger numbers; modern counts from pay records are usually several times smaller.

#### LATER ERA (1500S+)

Armies grew fast. Gunpowder forts needed bigger garrisons, and states began to keep permanent troops in **garrison towns**. Field armies of 20,000–50,000 became common in major wars of the 1500s and 1600s.

**MAP TIP**

You do not draw garrisons, but they tell you what a castle is. On a kingdom map, an ordinary castle is a post of a few dozen men, not a barracks for thousands. Put big troop numbers only in two or three great frontier fortresses and the capital's citadel. If you label armies or invasion routes, keep the sizes realistic: about 0.2–0.5% of the realm's people. A kingdom of 2 million fields about 4,000–10,000 soldiers.

## Military geography and logistics

### Speed and season

Army speeds follow the guide's shared baseline (details in [Trade Routes, Roads and Transport](#)):

| Mode                            | Per day                                                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Large army with baggage         | 13–20 km (8–12 mi)                                                                                                                                                                                       |
| Army living off the land        | 10–26 km (6–16 mi)                                                                                                                                                                                       |
| Forced march, a few days only   | 30–55 km (20–35 mi). Harold's march north to Yorkshire in 1066 covered about 300 km (185 mi) in perhaps 5–7 days (often said to be only four), about 43–60 km (27–37 mi) a day, probably partly mounted. |
| Mounted force with spare horses | 55–65 km (35–40 mi)                                                                                                                                                                                      |

**Campaign season.** Most campaigns ran from late spring, when there was grass for the horses, until after the harvest in autumn. Winter campaigns were rare. A vassal usually owed only **40 days** of military service a year, so feudal armies melted away during long sieges. There were exceptions: the Teutonic Order often raided Lithuania in winter, because then the rivers and marshes were frozen and could be crossed. Galley fleets mostly stayed in port in winter.

### Food and fodder

- A soldier needs about 1–1.5 kg (2–3 lb) of food a day, mostly bread or grain.
- A horse needs about 4.5 kg (10 lb) of grain a day if it can also graze. Without grazing it needs roughly double that, about 9–10 kg (20–22 lb) of grain and hay. On summer grass it can graze for much of its needs. Otherwise, hay and oats must be carried.

- An army of 10,000 men and 4,000 horses therefore needs about 10–15 tonnes of food and 18–40 tonnes of fodder **every day**. That is very roughly 30–100 cartloads a day (an estimate).
- A moving army can take only about 5–15% of the food in the country it passes through and still keep moving (Devereaux's estimate). So armies **keep moving**. An army that stops, for example for a siege, eats out the surrounding land within weeks.

**RULE OF THUMB**

Armies follow rivers and coasts. Bulk supply came by boat, so invasion routes, siege camps and frontier castles cluster along navigable water. Edward I placed every one of his royal castles in North Wales where ships could reach it. In 1294–95 Harlech was resupplied from Ireland by sea through its water gate.

## Military sites behind the front

| Site                    | What it is                                                    | Where to put it                                                                                | Real examples                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muster point            | Where the army gathers before a campaign                      | A royal town or castle near the frontier, on a main road or river, with grazing land around it | English armies gathered at Chester for North Wales, at Newcastle, Carlisle or Berwick for Scotland, and at Portsmouth or Southampton for France                                                                                                                                                                                                                                                                 |
| Armoury and royal store | Weapons, armour, arrows, siege engines, tents                 | In the capital's citadel, or a major castle on the main route to the frontier                  | The Tower of London was the English crown's main armoury and store                                                                                                                                                                                                                                                                                                                                              |
| Arsenal and naval base  | A state shipyard that builds, stores and fits out war galleys | A sheltered harbour or river port in or next to the capital or main port                       | Venice Arsenal (traditionally founded in 1104, though the exact date is uncertain; enlarged from 1320 and again in 1473; about 45 ha / 110 acres, roughly 15% of the city; up to 16,000 workers claimed for the early 1500s). Barcelona's royal shipyard, the Drassanes (ordered in 1285, enlarged 1328–90). The French royal galley yard at Rouen, the Clos des Galées (founded by Philip IV from c. 1292–94). |
| Invasion port           | Where a fleet gathers and sails, and where it lands           | The nearest good harbour to the enemy coast, with roads behind it to bring troops and food     | 1066: William's fleet sailed from Saint-Valery-sur-Somme and landed at Pevensey. 1346: Edward III sailed from Portsmouth and landed at Saint-Vaast-la-Hougue. 1415: Henry V sailed from Southampton to Harfleur. Calais, taken in 1347, stayed an English bridge-head until 1558.                                                                                                                               |

### MAP TIP

For each realm, mark one or two muster towns near each threatened border, one main armoury (usually the capital), and one naval base if it has a coast. Draw the obvious invasion route between two rival realms: the shortest sea crossing or the easiest river valley. Then fortify both ends of it.

### FANTASY TWIST

Magic that feeds armies (conjured food, bottomless wagons) removes the river rule. Armies could then march straight across dry land and besiege for months. That one change would make castles far less useful. If you allow it, make it rare and costly. See [Fantasy Variants](#).

## Camps and siege towns

### RULE OF THUMB

An army that stays in one place for more than a few weeks turns its camp into a town, with streets, a market, workshops and a defended edge, all fed by boat or by the main road. Most of these camps vanish when the war ends. A few become real towns.

**Why camps matter.** A field army of 10,000–30,000 people (see [Garrisons and armies](#)) is bigger than almost every town in its own kingdom. When it stops for a long siege or

for the winter, it needs shelter, food markets, smiths, carpenters and stables. It also needs a defended edge, because it can be attacked from the besieged town and by a relief army from outside. Merchants, servants and other camp followers come with it. So a long siege camp looks and works like a town of timber, thatch and canvas. Like the castles in [Military geography and logistics](#), it sits where boats or a main road can supply it.

## Kinds of camp

| Kind                    | How long it lasts          | What it looks like                                                                                                               | Size                                                                          | Real examples                                                         |
|-------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Overnight marching camp | One night to a few days    | Roman armies: rows of tents inside a bank and ditch. Medieval armies: tents and rough shelters, usually with only light defences | Roman camps from under 1 ha to 67 ha (165 acres); most 10–18 ha (25–45 acres) | Roman camps in Scotland, a day's march apart along the campaign roads |
| Winter camp             | One winter, several months | Naturally strong ground beside a river, with boats drawn up, workshops and trading                                               | Torksey: at least 55 ha (136 acres)                                           | The Viking "Great Army" at Torksey, England, winter 872–73            |
| Siege camp-town         | Months, up to about a year | Timber houses and huts around the besieged town, a market, stables, a fenced or ditched edge, and the commander's own quarters   | For an army of about 10,000–30,000                                            | Calais 1346–47; Neuss 1474–75                                         |
| Permanent siege town    | Built to last              | A small walled town on a planned grid, built to show the enemy that the besiegers will not leave                                 | Santa Fe: about 335 × 260 m (1,100 × 850 ft), about 9 ha (22 acres)           | Santa Fe near Granada, 1491                                           |

### Camps that became towns (or nearly did)

- **Torksey, 872–73 (England; earlier than this guide's baseline period).** The Viking Great Army spent the winter on higher ground beside the River Trent, partly surrounded by marsh. The camp covered at least 55 ha (136 acres), bigger than York at the time, and held several thousand people, including warriors, craftworkers and traders. They repaired boats, melted looted silver and gold into ingots, made and traded goods. After the army left, Torksey grew into an important Anglo-Saxon *burh* (fortified town) with a pottery industry. Archaeologists see the camp as a trigger for the later town.
- **Calais, 1346–47 (northern France).** Edward III besieged Calais for 11 months (4 September 1346 to 3 August 1347). His army built a temporary town on firm ground in the marsh, along the causeway road about 800 m (half a mile) from the walls. They called it *Villeneuve-la-Hardie* ("Bold New Town"). It had timber halls for the king and nobles, market halls, stables and thousands of huts of brushwood and thatch. Markets were held twice a week, and observers said its meat and cloth markets looked like those of Arras or Amiens. At its height it held more people than any English town except London. Supplies came by sea: 853 ships with 24,000 sailors served the siege at one time or another. It did not survive as a town: once Calais fell, Edward expelled the French townspeople and settled English colonists inside the captured city instead.
- **Neuss, 1474–75 (Rhineland, Germany).** Charles the Bold, Duke of Burgundy, surrounded Neuss, a fortified town of about 4,000 people, with an army of about 14,000 (estimates run from 13,000 to 20,000). He expected a siege of a few days or weeks. It lasted about ten months, from 29 July 1474 until a relief army sent by the Emperor and peace talks ended it in late May and June 1475. By mid-August 1474 the town was completely shut in.

Charles kept his court in the camp in great luxury. His courtier Olivier de La Marche is quoted as calling the camp "like a city", with craftsmen, merchants, barbers, inns and taverns, and even windmills. Disease still spread through it.

- **Santa Fe, 1491 (near Granada, Spain).** In the last campaign against the Muslim kingdom of Granada, Ferdinand and Isabella set up their royal camp in April 1491 on the open plain (the *Vega*), about 10 km (6 mi) west of the city and within sight of its walls. They began building a real walled town almost at once. In July 1491 a fire destroyed much of the tent camp. Older accounts say the town was built only after the fire, but recent research shows building had already begun. Chroniclers say the work took about eighty days. In October 1491 the royal court's historian, Peter Martyr, described it as nearly rectangular, about 400 by 312 paces (about 335 × 260 m, 1,100 × 850 ft), with battlemented walls, moats and towers, a square in the middle and a gate in each of the four sides. Two main streets crossed at that square, and the moat was filled from the plain's irrigation channels. Granada agreed to surrender in November 1491. In 1492 the camp became a town (*villa*), and its land was shared out among about 200 settler households. Santa Fe still exists; some of its gates still stand, and the old grid can still be seen.

### Roman marching camps as old earthworks

Roman armies on campaign built a defended camp at the end of each day's march. On a medieval map these camps are part of the older layer of the landscape: low banks in moorland or marks in fields (see [Religious, Cultural, Legal and Ancient Sites](#)).

- **How many:** nearly 500 are recorded in Britain, almost half of them in Scotland. No other Roman province has so many known, mostly because of aerial photography.

- **Shape:** the classic "playing-card" plan, a rectangle with rounded corners, made of one earth bank (**rampart**) and an outer ditch. Many real camps are less regular.
- **Gates:** most often 4 or 6, sometimes fewer (the Roman writer Hyginus describes a model camp with six). Each gate was a gap in the bank, usually guarded by a short bank and ditch set in front of it (a *titulus*), so attackers could not rush straight in.
- **Where:** beside Roman roads and forts, often just after a river crossing. Near some frontier forts, at river crossings and road junctions, are "gathering grounds" where armies camped again and again before a campaign.
- **Spacing:** camps built on the same campaign lie about a day's march apart: in Scotland mostly 11–24 km (7–15 mi), less across hills.

| Camp size                  | Area                     | Roughly how wide (estimate) | Examples (Scotland)                                                                                                 |
|----------------------------|--------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| Small                      | 1–5 ha (2.5–12 acres)    | 100–220 m (330–720 ft)      | Pennymuir II, 3.9 ha (9.7 acres)                                                                                    |
| Common                     | 10–18 ha (25–45 acres)   | 320–420 m (1,050–1,400 ft)  | Pennymuir I: almost 18 ha (44 acres), 520 × 343 m (1,710 × 1,125 ft), probably six gates                            |
| Large campaign series      | 25 ha (63 acres)         | about 500 m (1,650 ft)      | About 20 camps in two lines north of the Forth, from the Stirling area towards Strathclyde                          |
| Very large campaign series | 45–67 ha (110–165 acres) | 670–820 m (0.4–0.5 mi)      | Camps north of the Forth (45–54 ha) and on Dere Street, the main Roman road north through southern Scotland (67 ha) |

**How many soldiers?** Pennymuir I is said to have had room for two full legions, roughly 8,000–12,000 men. That is very roughly 500–700 soldiers per hectare (estimate). So an army of 10,000 in a tight, ordered camp needs about 15–20 ha (35–50 acres). A looser medieval camp, with baggage, horses and camp followers, spread far wider: the Viking camp at Torksey covered at least 55 ha for several thousand people.

#### MAP TIP

On a local map of a siege, draw the besieged town first. Then draw the camp as a rectangle or ring of huts and tents on dry ground beside the main road or the river landing, a few hundred metres to about 1 km (0.6 mi) from the walls, with a market square and the commander's quarters inside. Give it a name, as real soldiers did (*Villeneuve-la-Hardie*). For 10,000 soldiers, allow about 15–20 ha, roughly 400 × 450 m (1,300 × 1,500 ft), and more for horses and followers. On a kingdom map a siege camp is only a symbol (a closed dashed rectangle with rounded corners), and only on a map with a date. On a later map, leave old camps as the same outline drawn grey and labelled "Camp" or "Roman Camp", or, if the besiegers won and stayed, as a small grid-planned town with a religious or military name (*Santa Fe* means "Holy Faith").

#### LATER ERA (1500s+)

Santa Fe's plan, two main streets crossing at a central square with a gate at each end, is often seen as a forerunner of the grid towns that Spain planted in the Americas. See [What Changes After 1500](#).

#### FANTASY TWIST

Where monsters roam the wilds, every army halt needs a defended camp, as the Romans built every evening. Draw chains of earthwork camps (closed dashed rectangles) a day's march apart along old campaign roads, and let a few of them grow into towns.

## Outposts in wild and contested land

### Frontier posts

Real frontiers were zones, not lines (see [Capitals](#), [Realms](#) and [Borders](#)). Their military layout was usually:

- **Forward line:** small castles, towers and fortified villages at fords, passes and wells, held by settlers or military orders.
- **Rear line:** big fortress towns a day or two behind, where the army and stores were kept. The Arabs facing Byzantium called the forward zone the *thughur* ("openings") and the rear zone the *awasim* ("the defences").
- **No man's land between:** thinly settled, raided every summer.

Examples:

- **Iberia:** the Christian–Muslim frontier moved south over centuries, from the Duero valley (10th century) to the Tagus (Toledo taken 1085) and finally Granada (1492). Castile's own name comes from its castles. Fortress towns, castles and military orders (Calatrava from 1158, Santiago from 1170) held each new frontier. The Granada frontier lasted about 250 years (1240s–1492). Both sides lined it with castles and *atalayas* (watchtowers).
- **Prussia and the Baltic:** the Teutonic Order advanced from 1231 along the Vistula and the coast, building a castle and a town together at each step.

### LATER ERA (1500S+)

The best-known early modern frontier belt is the Habsburg **Military Frontier** (German *Militärgrenze*, Croatian *Vojna krajina*), facing the Ottomans in Croatia and Hungary. Austrian troops began to garrison Croatian border forts in 1522. From the 1550s (1553 is the usual founding date) it became a permanent system, partly paid for by the Austrian and Bohemian lands. By the end of the 1560s the Habsburg border-fort system had about 100–120 forts of all sizes along a border of almost 1,000 km (620 mi), from the Adriatic to Transylvania. In the later 1500s about 20,000–22,000 paid soldiers held them. That is very roughly one fort every 8–10 km (5–6 mi) (estimate). Between the forts were small outposts and palisades at river crossings: in 1601–03 the local lord in Međimurje (northern Croatia), György Zrínyi, set up nine outposts along the Mura to guard its crossings, several of them with 100 guards each. Behind the forts the Habsburgs settled Christian refugees from Ottoman lands as **armed peasants**: free land and no feudal dues in return for military service. The first came in the 1530s (to Žumberak), a large wave came in 1597–1600, and the *Statuta Valachorum* ("Vlach Statutes") of 1630 obliged every man over 16 to serve. New star forts anchored the belt, such as Karlovac (begun 1579 as a six-pointed star), the headquarters of the Croatian frontier. The Military Frontier lasted until 1881. On a 1500s–1600s map, draw such a frontier as a line of forts and posts at the crossings, with a band of soldier-villages behind it. See [What Changes After 1500](#).

### Toll castles on rivers

- **Where:** narrow, busy stretches of navigable river, especially in gorges where ships cannot avoid the bank.
- **Numbers:** the Rhine Gorge (between Bingen/Rüdesheim and Koblenz) has about 40 castles in 65 km (40 mi), one every 1.5–2 km (about 1 mi). Many of them collected tolls. Along the whole navigable Rhine, recorded toll stations grew from 19 at the end of the 1100s to 44 by 1300 and 62 by 1400.
- **How it worked:** Pfalzgrafenstein (built 1326–27 by King Louis IV) stands on an island in the Rhine at Kaub. A chain could block the channel. Ships paid at the customs house in Kaub. Ships that refused were seized, and their crews were held in the castle's dungeon until a ransom was paid.
- **Who owned them:** the archbishops of Mainz, Cologne and Trier, the Count Palatine and noble families. Tolls were a pillar of their power.

### Robber-knight castles

When royal power was weak, small lords used their castles to rob travellers, demand illegal tolls and hold merchants for ransom. Germans later called them *Raubritter* ("robber knights").

- During the German Interregnum (1250–73), toll stations multiplied.
- In 1254 Rhine cities and princes formed the Rhenish League. It took and destroyed robber castles: about 10–12 strongholds within three years.
- In 1282 King Rudolf of Habsburg destroyed more robber castles, among them Sooneck.

### Military-order houses

- **Frontier castles:** the orders held the hardest frontier castles: Krak des Chevaliers, Safed, Montfort, Calatrava, Malbork.
- **Commanderies in the home countries:** most of the orders' houses in western Europe were not forts. A **commandery** (or **preceptory**) was a manor farm with a chapel and a small community, run to send money, horses and food to the frontier. The Templars had nearly 1,000 commanderies and fortifications at their height,

and 15,000–20,000 members, only about one-tenth of them knights.

#### MAP TIP

On a frontier, mark the march with hatching or a fading colour band, the guide's symbol for a frontier zone (see *Capitals, Realms and Borders*). Put small forts at every ford, pass and well along its front edge, and two or three fortress towns a day or two's march behind them. The whole military zone, from the forward forts back to the fortress towns, is then about 30–80 km (20–50 mi) deep. Leave the land in front nearly empty. On a busy river gorge, crowd the cliffs with toll castles (small castle icons, with a toll bar and "T" across the river). Put one military-order commandery in the countryside of each inland county, drawn as a small church with a shield, not as a castle.

#### FANTASY TWIST

A wilderness full of monsters works like a hostile neighbour. Its edge becomes a frontier zone even inside one realm, with forward posts, refuge towers and fortified villages, and a rear line of strong towns. Where dwarves or others hold mountain passes, the pass fort may belong to them, not to the human kingdoms on either side.

## Watchtowers, beacons and coastal warning

#### RULE OF THUMB

Warning beats walls. A chain of fires on hilltops could carry a message hundreds of kilometres in an hour, faster than any rider.

**Beacons.** A beacon is a pile of fuel or an iron fire basket on a hill or tower, kept ready to light.

- **Byzantine beacon line (9th century, under Emperor Theophilos):** about 720 km (450 mi) from the fortress of Loulon near the Cilician Gates to Constantinople, with about nine main stations. Stations stood about 55–100 km (35–60 mi) apart on high peaks: closer in rugged country, farther in open country. Modern experiments suggest a message could cross the whole line in about one hour. Each of the 12 hours of the day stood for a set message. The hour at which the first fire was lit told the capital which message was meant.
- **Scottish Borders (Act of 1455):** one bale (fire) meant the English were coming; two bales together meant they were "coming indeed". Watches were kept at fords and passes.

- **Islamic coasts:** ribats were built at intervals along the North African shore so they could signal to each other. From Monastir's watchtower, signals were exchanged at night with neighbouring ribats. Sousse, the next big ribat up the coast, is about 20 km (12 mi) away.

**Spacing.** Beacon intervals depend on height and haze. In ordinary hilly country, plan for about **5–20 km (3–12 mi)** between beacons. Between high mountain peaks in dry air, 55–100 km (35–60 mi) is possible. These spacings are estimates.

**Watchtowers** stood at passes, fords, wells and frontier hilltops. They held a handful of men whose job was to see, signal and run, not to fight.

#### LATER ERA (1500S+)

Dense chains of coastal towers are mostly a 16th-century answer to Barbary and Ottoman raids. Genoa built about 85–90 towers around Corsica in 1530–1620 (a 1617 list counts 86). Each was usually about 12 m (39 ft) high and held 2–6 guards. Smoke, fire or a conch horn could put the whole island on alert within a few hours. Spain and the Kingdom of Naples built similar chains.

**MAP TIP**

Draw beacon chains as flame dots on hilltops leading from the border or coast to the nearest castle and on to the capital. Label a few hills "Beacon Hill" or "Watch Hill"; such names are common in Britain. On a raided coast, add a watchtower (a small triangle) on every headland that can see the next one.

## Where battles happen

**RULE OF THUMB**

Battles happen where armies must pass or where one side must save a besieged place. Pitched battles were rare and risky. Sieges and raids were normal war.

Historians such as John Gillingham point out that many successful commanders avoided pitched battle almost entirely. When battles did happen, they clustered in a few kinds of place:

| Where                                   | Why                                                                       | Real examples                                                                            |
|-----------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| River crossings                         | The attacker must cross; the defender waits on the far bank               | Stirling Bridge (1297); the fight at the Blanchetaque ford before Crécy (1346)           |
| Near a besieged castle or town          | The relieving army must come to the siege, and the besiegers must stop it | Lincoln (1141 and 1217); Muret (1213); Bannockburn (1314, near besieged Stirling Castle) |
| On the main road to a capital or a port | One army blocks the other's line of march                                 | Hastings (1066, on the road to London); Agincourt (1415, on the road to Calais)          |
| Near water in dry country               | Armies must drink                                                         | Hattin (1187), where the crusader army ran out of water                                  |

The defender usually picks ground with a slope in front and protected flanks: woods, villages, marsh or a river. At Crécy the English stood on a slope with their flanks resting on two villages. At Agincourt woods on either side narrowed the field.

**MAP TIP**

Name a battlefield after the nearest village and mark it with crossed swords and the year. Put it near a ford, a bridge, a road junction or a besieged castle, never in the middle of nowhere. Three or four famous battlefields in a kingdom's history are plenty.

## Drawing military sites on your map

### Suggested symbols

These symbols match the guide's [master legend](#).

| Feature                          | Symbol                                         | Notes                                                                                                                                                                                        |
|----------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Great royal castle or fortress   | Large castle icon (three towers)               | Label by name                                                                                                                                                                                |
| Ordinary castle                  | Small castle icon (one tower with battlements) |                                                                                                                                                                                              |
| Motte or small earth-work castle | Small mound with a tower tick on top           | Local maps only. A whole motte-and-bailey is only about 100–150 m (330–490 ft) long (estimate), about 1–1.5 mm at 1 cm = 1 km, so use a symbol, not a plan (see <a href="#">Footprints</a> ) |

| Feature                             | Symbol                                                                                                   | Notes                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tower house, peel or bastle         | Small solid square                                                                                       | Many in raided borderlands                                                                                                                           |
| Fortified or moated manor house     | Small open square; double outline if moated                                                              | An open square, so it does not look like a tower house                                                                                               |
| Walled town                         | Town symbol (a circle with a ring) with battlements on the ring                                          | A city has a thick battlemented ring                                                                                                                 |
| Citadel                             | Small castle icon at the edge of the town ring                                                           | Not a star: a filled star means a capital                                                                                                            |
| Fortified bridge                    | Bridge symbol with a small tower                                                                         |                                                                                                                                                      |
| Fortified church                    | Church with battlements                                                                                  |                                                                                                                                                      |
| Watchtower or coastal tower         | Small triangle                                                                                           | Draw chains of them; on a raided coast, one on each headland (coastal chains are mostly 1500s+)                                                      |
| Beacon                              | Flame dot                                                                                                | Draw chains of them from the border or coast to the capital                                                                                          |
| Military-order commandery           | Small church with a shield                                                                               | A manor farm, not a fort. Use a castle icon only for the orders' real frontier castles                                                               |
| Arsenal or naval base               | Anchor plus a castle tower                                                                               |                                                                                                                                                      |
| Siege castle                        | Small dashed castle near its target                                                                      | Short-lived; show only for a dated map                                                                                                               |
| Army or siege camp                  | Closed dashed rectangle with rounded corners, labelled with its name or "camp"                           | Dated maps only. Always a closed shape, never an open line. An old Roman or abandoned camp: the same outline drawn grey, labelled "Camp"             |
| Battlefield                         | Crossed swords and the year                                                                              |                                                                                                                                                      |
| Star fort or fortress town (1500s+) | Five- or six-pointed star outline, or a star-edged town outline; never filled                            | Later era only                                                                                                                                       |
| Coastal artillery fort (1500s+)     | Small round outline with rounded lobes (a rosette, like the plan of Deal Castle, 1539–40); never pointed | Later era only. At harbour mouths, anchorages and river mouths. Pointed outlines mean star forts: a fort with angled bastions takes the star outline |
| Ruin                                | The feature's own symbol, broken or grey, labelled "(ruin)"                                              |                                                                                                                                                      |

### Ruined and active castles

- In England only about one-third of all castle sites were in use at any one time. The rest were abandoned, never finished, or replaced.
- So a map dated 1300 should show many **abandoned mottes and ringworks** from the 1070s–1150s, especially in old border zones that are now peaceful.
- By 1450, many 12th-century castles were half-ruined. The crown kept up only the important ones.
- Old hillforts and Roman forts are often reused (Old Sarum, Pevensey). Others stay as grassy earthworks with

names like "Castle Hill" or "Camp" (old army camps are covered in [Camps and siege towns](#)). Ancient ruins in general are covered in [Religious, Cultural, Legal and Ancient Sites](#).

#### RULE OF THUMB

For every active castle, draw one or two ruins or earthworks in long-settled land. In a peaceful core, more ruins; in a newly conquered frontier, almost none (everything there is new).

## How many to draw

Per 1,000 km<sup>2</sup> (about 390 sq mi) of land, at about 1300:

| Zone                                            | All sites ever built                | Active castles   | Show on a kingdom map                                                                  | Show on a local map                      |
|-------------------------------------------------|-------------------------------------|------------------|----------------------------------------------------------------------------------------|------------------------------------------|
| Peaceful lowland core                           | 4–8                                 | 1–3              | The county town's castle, plus 1 major baronial castle per 2,000–3,000 km <sup>2</sup> | All active castles plus ruins            |
| Average (England and Wales overall)             | ~12                                 | ~4               | Royal and major castles only, about 1 per 1,000 km <sup>2</sup>                        | All                                      |
| Hilly lands with many small lords (German type) | ~25                                 | ~8–12 (estimate) | Major castles and toll castles                                                         | All, clustered on spurs and river cliffs |
| March or contested border                       | 35–60                               | 12–20            | All major castles; draw the border as a hatched band studded with castle icons         | All, including mottes and towers         |
| Raided border of the late Middle Ages           | 8–10 castles plus many tower houses | Most of them     | Castles and the main tower houses                                                      | Every tower house and bastle             |
| Thin conquered frontier (Prussia type)          | 1–3                                 | 1–3              | All of them, each paired with a town                                                   | All                                      |

The "active" column assumes about one-third of sites are in use at once, as in England. It is an estimate for other regions.

### Map scale guide:

- **Continent map:** show only capitals' citadels, the greatest frontier fortresses, famous fortress cities and naval arsenals. As a suggestion, about 1 per 20,000–50,000 km<sup>2</sup> (8,000–19,000 sq mi).
- **Kingdom map:** add royal castles, county-town castles, border fortresses and the main border chain.
- **Local map:** show everything: castles, ruins, mottes, tower houses, fortified churches, beacons and watchtowers.

#### MAP TIP

Check your castles with three questions. (1) What does each one control? (2) Can it be supplied by water or from rich farmland nearby? (3) Is the density right for the zone: sparse in the core, dense on the border? Common errors are covered in [Common Mistakes](#) and [How to Fix Them](#).

#### FANTASY TWIST

If mages can break walls like cannon, castles should evolve early into low, thick, star-shaped forts. If enemies can teleport or fly over borders, strong citadels inside cities matter more than border chains. Long-lived races (elves, dwarves) may keep very old fortresses in use, so their lands show fewer ruins and older styles.

## Quick summary

- Every fortification controls something: a route, a crossing, a town, a border, an estate, a refuge or a lookout. If it controls nothing, delete it.
- Types change with time: earth-and-timber mottes and ringworks (c. 1000–1150), stone keeps and curtain walls (1150–1250), concentric castles and town walls (1250–1350), then tower houses and fortified manors (1350–1500). From the 1500s come star forts and artillery forts.
- Best sites: fords, bridges, passes, spurs, river bends, confluences, headlands, and hills above towns. German castles were mostly on high ground; English castles mostly beside villages and towns.
- England and Wales had about 1,761 castle sites (one per ~86 km<sup>2</sup>, or 33 sq mi), but only 500–600 were in use at one time: an active castle every 16–17 km (10–11 mi).
- Border zones are roughly 5–10× denser: the Welsh March had one castle site per 17–28 km<sup>2</sup> (7–11 sq mi), a gap of 4–6 km (2.5–4 mi). In a peaceful core, working castles stand about 20–30 km (12–19 mi) apart. Conquest chains put a castle about every day's march (20–40 km, 12–25 mi) along a coast or river.
- Castles pair with towns: castle at the edge of town, on the high point or by the river, with the market at its gate.
- Garrisons were tiny: 5–20 in peace, 30–60 for a new frontier castle, a few hundred (rarely up to about 1,200) in a great siege. Only giant frontier fortresses held 2,000 or more.
- Field armies were about 0.2–0.5% of a realm's population: 10,000–15,000 for England in the 1300s, briefly 32,000 at most.

- Armies march 13–20 km (8–12 mi) a day, campaign from late spring to autumn, and follow rivers and coasts for supply. A stationary army eats out the land within weeks.
- Long sieges and winter quarters turn camps into temporary towns with markets (Calais 1346–47, Neuss 1474–75); a few became real towns (Torksey, Santa Fe 1491). Roman marching camps (nearly 500 known in Britain, mostly 10–25 ha, a day's march apart) survive as rectangular "camp" earthworks.
- Add muster towns near borders, an armoury in the capital, an arsenal at the main port, toll castles on river gorges and beacon chains from the border to the capital.
- Battles happen at crossings, near sieges and on roads to capitals and ports, and are named after the nearest village.

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Population figures, travel speeds, total castle counts, the control-radius heuristic and the Harlech and Safed garrisons follow the guide's shared baseline. Figures marked "estimate" (gaps between castles, active-castle densities outside England, cartloads per day, beacon spacing, army shares of population, distances between Edward I's and the Teutonic Order's castles, distances from Ashkelon to its ring of castles and from Santa Fe to Granada, measured in straight lines on a modern map; the widths of Roman camps, worked out from their areas; soldiers per hectare and camp sizes for an army of 10,000, worked out from the Pennymuir figures; and the spacing of forts on the Habsburg Military Frontier, worked out from its fort count and border length) are derived by this guide from the sources above and are not measured values. Bailey areas and the whole-castle length of a small motte-and-bailey match the footprints table in [chapter 08](#). Harold's march follows the baseline's 5–7 days; the traditional figure of four days would mean about 75 km (46 mi) a day.

## CHAPTER 5

# TRADE ROUTES, ROADS AND TRANSPORT

Roads, rivers and sea lanes are the blood vessels of your map. In the Middle Ages, moving goods over land was slow and expensive, and moving them by water was cheap. That one fact explains where cities grow, where roads go, which places become rich, and which stay poor. This chapter gives you travel speeds and costs, the spacing of inns, tolls and ports, and real trade networks you can copy.

The baseline period is the High and Late Middle Ages (c. 1000–1500), mainly in Europe. Site types such as ford towns, bridge towns and head-of-navigation towns are explained in [Where Settlements Are Built](#). How much land a town needs to feed itself is in [Population and Settlement Sizes](#). Castles that guard roads are in [Castles, Forts and Military Outposts](#). Pilgrim shrines are in [Religious, Cultural, Legal and Ancient Sites](#).

## IN THIS CHAPTER

How fast and how expensive: travel and transport

Roads

Crossing rivers: bridges, fords and ferries

Mountain passes

Inns, hospices and caravanserais

Roads without wheels: the Andes and Mexico

Tolls and customs

Rivers as roads

The sea: ships, harbours and sea lanes

Trade nodes: markets, fairs, staples and entrepôts

What is traded

Real trade networks to copy

Drawing the route network

Quick summary

Sources and further reading

## How fast and how expensive: travel and transport

## RULE OF THUMB

Over land, a traveller covers about 30 km (19 mi) a day and a loaded ox cart about 20 km (12 mi). By water, goods move at least as fast for a small part of the cost. Heavy, cheap goods go by water or stay at home.

## Speed per day

These are the guide's shared numbers. Use them for every journey on your map.

| Mode                              | km per day   | mi per day | Notes                                                                                                                                                                                                                  |
|-----------------------------------|--------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Walking traveller                 | 25–35        | 15–22      | Archbishop Sigeric's journey from Rome to Canterbury (c. 990): about 1,700 km (1,060 mi) in 80 stages, so an average stage was about 20 km (12 mi). A pilgrim's stage was often shorter than a hurried traveller's day |
| Peasant to market                 | 6–10 one way | 4–6        | Out, trade and home in one day                                                                                                                                                                                         |
| Ox cart                           | 15–25        | 10–15      | Slow, strong and cheap to feed                                                                                                                                                                                         |
| Horse cart or wagon               | 30–40        | 20–25      | Horses can double the speed of oxen                                                                                                                                                                                    |
| Pack horse or mule train          | 20–40        | 12–25      | Loads: horse 100–120 kg, mule 150–180 kg, donkey 80–100 kg                                                                                                                                                             |
| Camel caravan                     | 25–40        | 15–25      | About 230 kg per camel                                                                                                                                                                                                 |
| Cattle or sheep drove             | 16–30        | 10–20      | About 3 km/h (2 mph), grazing on the way: about 16 km (10 mi) a day in hills, up to 30 km (20 mi) in lowlands. Figures mostly from the 1700s–1800s; see <a href="#">Drove roads</a>                                    |
| Army with baggage                 | 13–20        | 8–12       | Forced march 30–55 km (20–35 mi) for a few days                                                                                                                                                                        |
| Rider, many days                  | 30–50        | 20–30      | —                                                                                                                                                                                                                      |
| Royal messenger with fresh horses | 50–90        | 30–56      | 30–40 mi normal, 52–56 mi when urgent                                                                                                                                                                                  |

| Mode                             | km per day                                    | mi per day    | Notes                                                                 |
|----------------------------------|-----------------------------------------------|---------------|-----------------------------------------------------------------------|
| River boat, downstream           | 40–100                                        | 25–60         | Estimate; travels by daylight                                         |
| River boat, upstream, towed      | 10–20                                         | 6–12          | Danube, Vienna to Passau: about 300 km (190 mi) in 20–31 days         |
| Galley along a coast             | about 55 (voyage average with stops)          | about 30–35   | Venice to Jaffa pilgrim galley: 48 days (1480)                        |
| Sailing ship (cog or round ship) | 40–100 voyage average; 90–220 while under way | 25–60; 55–135 | Candia (Crete) to Alexandria, about 600 km (370 mi), in 4 days (1323) |

### How long does it take to cross your map?

These are rough ranges from the table above. Add rest days, Sundays, bad weather and waiting for wind.

| Distance          | On foot   | Ox cart    | Courier with fresh horses | Galley       | Sailing ship  |
|-------------------|-----------|------------|---------------------------|--------------|---------------|
| 100 km (62 mi)    | 3–4 days  | 4–7 days   | 1–2 days                  | about 2 days | 1–3 days      |
| 300 km (190 mi)   | 9–12 days | 2–3 weeks  | 3–6 days                  | 5–6 days     | 3–8 days      |
| 1,000 km (620 mi) | 4–6 weeks | 6–10 weeks | 1.5–3 weeks               | 2.5–3 weeks  | 1.5–3.5 weeks |

A sailing ship can be faster than anything on land, or it can sit in port for two weeks waiting for a fair wind. Land travel is slower but more predictable.

### Cost: why water beats land

**Rule:** Per tonne and per kilometre, land transport cost several times more than river transport, and river transport cost more than sea transport.

**Why:** A pack horse carries about 100–120 kg. It needs a driver, food, a stable and a smith to shoe it. The Bremen cog

of 1380 (a cog is the main northern cargo ship) carried about 90–130 tonnes. A rule of thumb gave one sailor per 10 tonnes of cargo, so it needed a crew of only about 10–13 sailors, perhaps a few more in practice. One small ship moved as much as about 1,000 pack horses. For comparison, one modern estimate puts the goods that crossed the Brenner Pass, a main Alpine route, at about 3,000 tonnes a year in the 14th century. Treat that figure as a rough order of size. If it is right, a whole year's traffic over a main pass was the cargo of only 25–30 cogs.

### Numbers:

| Comparison                                                             | Ratio                               | Source and place                               |
|------------------------------------------------------------------------|-------------------------------------|------------------------------------------------|
| Land : river : sea, per ton-mile (the cost of moving one ton one mile) | about 8 : 4 : 1                     | England, 1290s–1340s (Masschaele)              |
| Water against land                                                     | Water costs about one-tenth of land | England (Campbell et al.; Clark)               |
| Sea : river : land                                                     | about 1 : 5 : 28–56                 | Roman Empire, Diocletian's Price Edict, 301 AD |

- **Grain by cart, England c. 1300:** each 10 mi (16 km) of carting added about 3.3% to the value of wheat. The cost doubled the price after about 480 km (300 mi). This is the best case: horses, carts and flat land.
- **Grain by wagon, Roman Empire:** carting wheat 100 Roman miles added 55% to its price. The cost doubled the price after about 270 km (170 mi).

- **By sea:** shipping a quarter (about 220 kg) of grain from Exeter to London, about 560 km (340 mi) by sea, cost the same as carting it 65–70 km (40–45 mi).

### Map advice:

- Bulk goods (grain, timber, stone, salt, wine) follow water.
- Bulk goods travel more than 30–50 km (19–31 mi) over land only when they are valuable, or in war or famine.

- Most towns above 10,000 people sit on navigable water or the coast (inland exceptions: see [chapter 02](#)). See [feeding towns](#).

**FANTASY TWIST**

A griffon or a teleport circle that carries 100 kg is a pack horse with wings. It changes the speed of news, letters and luxuries, not the price of bread. Only magic that moves *bulk* cheaply (wind-calling for ships, golem wagons, river spirits that tow barges) lets big cities grow away from water. See [Fantasy Variants](#).

**LATER ERA (1500S+)**

Organized relay posts with fresh horses at every stage made messages much faster. The Taxis family set up their first permanent relay line in 1490, from Innsbruck towards Mechelen in the Low Countries, with horse stations roughly every 37 km (23 mi). By about 1500 such posts could cover up to about 150 km (95 mi) a day. English royal post stages were about 32 km (20 mi) apart. Goods still moved at the old speeds.

## Roads

**RULE OF THUMB**

A medieval "road" was a right to pass, not a built thing. Only old Roman roads, causeways and bridges were engineered. Everything else was a strip of earth worn by use.

**Why:** In England, the "king's highway" was a legally protected route on which everyone had the right to travel. It did not mean a paved surface. Kings protected roads by law, not by building them.

### Kinds of roads

| Type                               | What it is                                                                                       | Where                                                              | How to draw it                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Old Roman road                     | A straight, engineered road on a raised bank (an <i>agger</i> ). Still used, often decayed       | Former Roman lands: Italy, Gaul, Spain, Britain, the Rhineland     | Straight thick solid line, labelled "Street" or "(Roman)"; grey where abandoned. Real names: "Street", "Strata", "Chaussée" |
| Royal highway (king's highway)     | A main route between cities and towns, under the ruler's protection                              | Between major towns                                                | Thick solid line                                                                                                            |
| Local road or lane                 | Village to market town, village to village, field access                                         | Everywhere                                                         | Thin solid line                                                                                                             |
| Hollow way (holloway, sunken lane) | A lane worn below the land around it by feet, hooves, wheels and rain. Some are 5 m (16 ft) deep | Soft rock (chalk, sandstone, loess) on slopes near towns and fords | Two thin parallel solid lines (local maps only)                                                                             |
| Causeway                           | A raised road across marsh or a flood plain, often with small bridges                            | Fens, flood plains, approaches to bridges                          | Solid road line on a bank across the marsh                                                                                  |
| Pack-horse track or bridle way     | A narrow track for animals, not carts; narrow pack-horse bridges                                 | Hills and mountains                                                | Thin dotted line                                                                                                            |
| Drove road                         | A wide grassy route for walking cattle or sheep to distant markets                               | From uplands to towns                                              | Broad dotted line (two rows of dots)                                                                                        |
| Caravan route                      | A track from well to well and from caravanserais to caravanserais                                | Desert and steppe                                                  | Dotted line with a larger dot at each stage                                                                                 |

### The Roman legacy

At its height the Roman road network had more than 400,000 km (250,000 mi) of roads, of which more than 80,500 km (50,000 mi) were stone-paved. Official lodges (*mansiones*) stood about 25–30 km (16–19 mi) apart. After Rome fell, surfaces decayed and many bridges collapsed, but the **routes** stayed in use because they linked the old Roman towns.

In 12th-century England, law books placed **four highways** under the king's special peace: Watling Street, Ermine Street and the Fosse Way (all Roman) and the Icknield Way (prehistoric). In Italy the Via Aemilia still formed the spine of the Po plain.

Medieval people did draw road maps. The Gough Map of Britain shows about 190 thin red lines between towns, nine out of ten marked with a distance in Roman numerals. Its date is uncertain. It is conventionally dated c. 1360–70, but a study of its handwriting (Smallwood 2010) suggests it

was made one or two decades after 1400, and parts were redone or added later in the 15th century (Solopova 2012; Delano-Smith et al. 2017; both summed up by Oksanen & Brookes 2025). It is a good model for a "period" style map: towns, rivers, and the main routes between them with distances. It also shows how much main road to draw on a kingdom map (see [How much main road to draw](#)).

### Who maintained roads

- **Landholders.** Repairs usually fell on local lords, landholders and communities, not on the king. England's Statute of Winchester (1285) added a security rule. On highways between market towns, the landholders had to clear ditches, trees and bushes for **200 feet (about 61 m) on each side**, so that robbers could not hide.
- **Towns.** Towns paid for paving their streets and the roads just outside the gates, often with a special toll called *pavage*.
- **Charity.** Building or mending roads and bridges counted as a pious work. For example, in 1474 Maud Heath gave land and houses to pay for a raised footway (a causeway) from Wick Hill across the flood plain of the River Avon to Chippenham in Wiltshire. A charity still looks after it from her gift, more than 500 years later.
- **Lords who took tolls** were supposed to keep the road or bridge in repair. Many did not.

### Drove roads

#### RULE OF THUMB

A drove moves at the speed of a grazing cow: about 16–30 km (10–20 mi) a day, with a field or pasture for the night at the end of each day's stage.

**Why:** Live animals walk themselves to market, so they can cross a kingdom over land far more cheaply than grain. But they must eat and drink on the way, or they arrive thin and sell for less. So drovers move slowly, let the animals graze, and keep to open hill tracks and broad green lanes. They avoid towns, crops and, where they can, tolls. A 19th-century saying about Welsh drovers was that "they'd lose a day goin' round sooner'n they'd pass a gate": they would rather lose a day going round than pay at a toll gate.

**When:** Cattle were driven from Wales to England from at least the 13th century. There are records of cattle driven from Wales to London, and sheep from Lincolnshire to York, in the early 14th century. Scottish drovers were being licensed by 1359. The great age of droving came later, in the 1600s–1800s, and most of the numbers below come from that time. Use them as the best guide we have for the Middle Ages too.

| Item                               | Number                                                                                                           | Where and when                             |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Speed                              | About 3 km/h (2 mph). About 16 km (10 mi) a day in steep hill country, up to 30 km (20 mi) a day in the lowlands | Scotland and Wales, 18th–19th c.           |
| Cattle herd                        | Usually 100–400 head, with 4–8 drovers and their dogs                                                            | Welsh droves, 19th c.                      |
| Sheep flock                        | 1,500–2,000 sheep, about 19–23 km (12–14 mi) a day                                                               | Welsh flocks driven to southern England    |
| Journey time                       | North Wales to Kent: about three weeks                                                                           | Same                                       |
| Overnight halts ( <i>stances</i> ) | About 16 km (10 mi) apart on average                                                                             | Scotland, evidence in a court case of 1848 |
| Width of a drove road              | 12–27 m (40–90 ft), much wider than a pack-horse track                                                           | Britain                                    |
| Hoof care                          | Cattle were shod before the long trek, and re-shod on the way                                                    | Wales, 19th c.                             |

**A Mediterranean example.** In Castile, the Mesta (a guild of sheep owners, with a royal charter from 1273) moved huge flocks between summer pastures in the north and winter pastures in the south. From 1436 to 1549 more than 2.5 million sheep made this journey each year. Flocks from León and Old Castile walked 550–750 km (340–470 mi) each way, along legally protected sheep walks (*cañadas*) that no one could farm, build on or block.

#### MAP TIP

Draw a drove road as a broad dotted line (two rows of dots) from upland breeding grounds (hills, moors, mountain pastures) to a lowland fair or a big city. Put a halt with a pasture and water every 16–25 km (10–15 mi), and a large livestock fair near the end. Keep the route off the main highway where you can: along ridges, across commons and around towns. Inns along it can have names like "Drovers' Arms".

## Road quality by season

- **Summer:** dry, hard roads; carts move well.
- **Autumn to spring:** clay roads turn to deep mud. Carts get stuck, so many carriers switch to pack animals. Floods close fords for days.
- **Northern winter:** in Russia and Scandinavia, winter was often the *best* time for land travel. Sledges ran fast over frozen rivers, lakes and marshes.

### MAP TIP

Do not draw roads as straight lines between cities. Real roads follow valleys and dry ridges, bend to reach fords and bridges, and avoid marshes and steep slopes. Only an older empire's roads run straight across country. If your world had one, draw its roads straight and let medieval roads reuse some of them and wander elsewhere.

### LATER ERA (1500S+)

England's Highways Act of 1555 made each parish choose unpaid surveyors of highways, and made parishioners give 4 days of labour a year to the roads (6 days from 1563). In France, Henry IV made the Duke of Sully *grand voyer* (chief officer of roads) in 1599. Main roads got better, but most were still dirt until the 1700s.

## Crossing rivers: bridges, fords and ferries

### RULE OF THUMB

Every crossing is a magnet. Roads bend to reach it, a town grows at it, and someone takes a toll there.

### Fords

A ford needs firm gravel and water shallow enough for carts at normal levels. Floods close it. Place names show fords: English *-ford* (Oxford), German *-furt* (Frankfurt), Dutch *-voorde*. On big lowland rivers good fords are rare, so a ford that serves several roads is a strong place for a town (see [ford towns](#)).

### Ferries

Where a river is too deep or wide for a ford and too expensive to bridge, there is a ferry. Ferry rights belonged to a

lord, a town or a monastery and earned money. For example, in 1330 Edward III gave the monks of Birkenhead Priory the right to run a ferry across the Mersey.

On very big rivers, ferries were normal. On the Rhine below Basel there were only a handful of fixed bridges in the Middle Ages. Cologne, the biggest city on the river, had no fixed Rhine bridge from Roman times until 1859.

### Bridges

**When were stone bridges built?** Early medieval bridges were mostly timber. A boom in stone bridges began in the 12th century.

| Bridge                              | Date                                                                 | Notes                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Stone Bridge, Regensburg (Danube)   | 1135–1146                                                            | A model for later stone bridges                                                                              |
| Judith Bridge, Prague               | 1158–1172                                                            | Badly damaged by a flood in 1342; replaced by the Charles Bridge, begun in 1357                              |
| Old London Bridge (Thames)          | 1176–1209                                                            | 19 arches and a drawbridge, about 282 m (926 ft) long, a chapel, and houses along it                         |
| Pont Saint-Bénézet, Avignon (Rhône) | First bridge 1177–1185 (probably timber); rebuilt in stone from 1234 | About 900 m (2,950 ft) and 22 arches when complete                                                           |
| Middle Bridge, Basel (Rhine)        | c. 1225                                                              | Part timber, part stone; the oldest bridge site still in use on the Rhine between Lake Constance and the sea |
| Krämerbrücke, Erfurt                | 1325 (in stone)                                                      | Lined with shops and houses                                                                                  |
| Ponte Vecchio, Florence             | 1345                                                                 | Shops on the bridge                                                                                          |

**How many?** For England, the historian David Harrison found that there were almost as many bridges in 1250 as in 1750. By the end of the Middle Ages most were solid, well-built structures that were repaired regularly, and most important crossings in a settled lowland country had a bridge. On the biggest rivers, bridges stayed rare. London Bridge was the only Thames bridge in London until Putney Bridge opened in 1729; the next bridge upstream was at Kingston, about 33 km (20 mi) up the river (see [the Thames example](#) below).

### Who paid?

- **Old duties** on certain lands or villages to repair a bridge.
- **Tolls.** In England a *pontage* grant let someone take a toll for a few years to repair a bridge. About 370 such grants were made between 1228 and the 1440s.
- **Rents.** Houses on London Bridge paid for its upkeep.
- **Charity and trusts.** Gifts and wills paid for bridges. The Rochester Bridge Trust (founded 1399) still owns and maintains bridges there.
- **Bridge chapels.** A small chapel on or beside the bridge collected gifts for repairs, and travellers prayed there for a safe journey. They were not uncommon in the Middle Ages, but most were lost after the Reformation, when an Act of 1547 dissolved chantries (endowed chapels for prayers). Only six survive in England: Wakefield (1342–56), St Ives (1426), Rotherham (1483), Bradford-on-Avon, Derby and Rochester. The last two stand beside their bridges, not on them.

**A bridge blocks ships.** Old London Bridge had such thick piers that the water level could differ by up to 1.8 m (6 ft) from one side to the other, making rapids. Sea ships unloaded *below* the bridge, so the port grew downstream of it.

### How often is a river bridged? The Thames example

#### RULE OF THUMB

On a navigable lowland river in a settled kingdom, put a bridge about every 20 km (12 mi) of river, with gaps of 10–35 km (6–22 mi). Each bridge has a town. Put ferries and fords in the gaps.

**Why:** A bridge is expensive, so it is built only where main roads must cross. Each one pulls the roads of a whole district towards it. A second bridge close by would split the traffic and the tolls, so bridges spread out along a river.

**Numbers:** The River Thames from London to Oxford is about 180 km (112 mi) long by river. By about 1300 it had nine road bridges, and a tenth (Marlow) was added in the 14th century. That is one bridge every 20–22 km (12–14 mi)

on average. In the gaps, people used ferries: in 1300 a ferrywoman called Sibille was paid three shillings to carry King Edward I across the Thames at Chertsey. After a bridge at Chertsey was licensed in 1410 and Abingdon's bridges were built in 1416–22, there were twelve, about one every 16 km (10 mi).

| Bridge town                          | Distance from London Bridge by river (approx.) | First record of a bridge                                                                                       |
|--------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| London                               | 0                                              | Stone bridge 1176–1209 (earlier bridges since Roman times)                                                     |
| Kingston                             | 33 km (20 mi)                                  | 12th century: a timber bridge on stone approaches                                                              |
| Chertsey                             | 51 km (32 mi)                                  | A ferry in 1300; a bridge licensed in 1410                                                                     |
| Staines                              | 57 km (35 mi)                                  | Repairs with wood from Windsor Forest ordered in 1222                                                          |
| Windsor                              | 69 km (43 mi)                                  | Probably 12th century; tolls on boats passing under it recorded by 1172; oaks granted for a new bridge in 1242 |
| Maidenhead                           | 80 km (50 mi)                                  | A wooden bridge c. 1280; "almost broken down" by 1297, when a toll (pontage) was granted for repairs           |
| Marlow                               | 91 km (57 mi)                                  | Perhaps 1309; certainly by the reign of Edward III (1327–77)                                                   |
| Henley                               | 104 km (65 mi)                                 | Recorded 1232; timber on stone piers, with chapels and granaries                                               |
| Reading (Caversham)                  | 120 km (75 mi)                                 | Recorded 1231, with a chapel of St Anne on it                                                                  |
| Wallingford                          | 145 km (90 mi)                                 | Mentioned 1141; first stone bridge in the 13th century                                                         |
| Abingdon                             | 166 km (103 mi)                                | Built 1416–22, with a second bridge at Culham and a causeway between them                                      |
| Oxford (Grandpont, now Folly Bridge) | 179 km (111 mi)                                | Stone bridge c. 1085                                                                                           |

Distances are measured on today's river and rounded, so treat them as rough. Around 1300 the gaps between bridges ranged from about 11 km (7 mi), from Windsor to Maidenhead, to about 34 km (21 mi), from Wallingford to Oxford before Abingdon had its bridge.

#### **A new bridge can make one town and break another.**

Abingdon's bridges (1416–22) were paid for by a town guild and two of its members, a London merchant and his wife. They replaced a ferry, so traffic that had crossed the Thames at Wallingford could now cross at Abingdon instead. The completion of the Abingdon bridge badly damaged trade at Wallingford, 21 km (13 mi) downstream.

**On the greatest rivers, bridges are much rarer.** Compare the Rhine below Basel, where Cologne had no fixed bridge for centuries (see [Ferries](#)).

#### **MAP TIP**

Use three symbols: a bridge (two short curved lines ")((" with the road running through), a ford (a short dashed line across the river) and a ferry (a short dotted line across the river with "F"). Put a settlement at almost every bridge on a main road. Put the main port of a river just **downstream** of the lowest bridge. On a settled lowland river, draw a bridge town about every 20 km (12 mi) of river, with ferries and fords between. On a great river, draw only a few fixed bridges, often many tens of km apart, and ferries in between.

#### **FANTASY TWIST**

A troll that charges travellers is just a strict toll collector. Traffic keeps using the bridge as long as the toll is cheaper than the detour. If the detour is short, the troll starves.

## Mountain passes

#### **RULE OF THUMB**

A few passes carry almost all the traffic across a mountain range. The lowest, gentlest one takes carts; the high ones take only mules. Each pass has a town at the foot on both sides and a hospice near the top.

**Why:** Snow, cold and steep climbs cost time and animals.

Traders choose the pass with the lowest summit, the shortest climb, a road for wheels and a safe valley.

| Pass             | Height                | Route                                                  | Medieval notes                                                                                                                                                                                                            |
|------------------|-----------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brenner          | 1,370 m<br>(4,495 ft) | Augsburg – Innsbruck –<br>Bolzano – Verona –<br>Venice | The lowest main Alpine pass. Mule trains and carts by the 12th century. Emperors (for example Frederick Barbarossa) used it to reach Italy. Perhaps about 3,000 tonnes of goods a year in the 14th century (one estimate) |
| Great St Bernard | 2,469 m<br>(8,100 ft) | Martigny – Aosta – Italy                               | The pilgrim road to Rome (Via Francigena). Hospice founded c. 1050 by St Bernard of Menthon to shelter pilgrims and protect them from bandits                                                                             |
| St Gotthard      | 2,106 m<br>(6,909 ft) | Lucerne – Uri – Ticino –<br>Milan                      | Became a through route after the first wooden bridge over the Schöllenen gorge, c. 1220. Only for people on foot and pack animals: the first carriage crossed in 1775                                                     |
| Mont Cenis       | 2,085 m<br>(6,841 ft) | Lyon – Savoy – Susa –<br>Turin                         | A main France–Italy route. Emperor Henry IV crossed it in the winter of 1076–77 on his way to Canossa, a famous dangerous winter crossing                                                                                 |

**Seasons.** High passes are snowbound for much of the year. The historic road over the Great St Bernard is open only from June to September, and winter snow there can be up to 10 m (33 ft) deep. Medieval travellers did cross in winter, but with local guides and real danger.

**Organization.** Valley villages organized the carrying trade. Local carriers took goods over their own stretch and handed them on to the next village's carriers. Lords and towns took tolls at gorges and narrows on the approach roads.

#### MAP TIP

For each mountain range, choose 1–3 main passes and 2–5 minor ones. Draw a chain of villages up each valley, a toll (one bar across the road with a "T") at the narrowest gorge, a hospice (a small house with a cross) near the summit, and a market town at the foot on each side, often where a lake or navigable river begins. Draw minor passes as thin dotted lines, label them "mule path" and give them no towns.

#### FANTASY TWIST

A dwarven tunnel through a range is a pass with no snow season. It would take most of the traffic from the passes nearby, and the towns at its two ends would grow rich. Whoever holds the tunnel holds the toll.

## Inns, hospices and caravanserais

#### RULE OF THUMB

Places to sleep sit one day's stage apart: 15–30 km (10–19 mi) on main roads in Europe, 30–40 km (19–25 mi) between caravanserais on plains and in deserts, and 10 km (6 mi) or less in mountains.

**Why:** A traveller must reach shelter by nightfall. Animals need water, fodder and rest. On busy roads, inns appear in every town and large village that is one stage from the last.

| Type of stop                           | Spacing                                                          | Example                                                         |
|----------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| Roman official lodge ( <i>mansio</i> ) | 25–30 km (16–19 mi)                                              | Roman roads across the empire                                   |
| Pilgrim stage                          | about 20 km (12 mi) on average                                   | Sigeric, c. 990: 80 stages from Rome to Canterbury              |
| Inn town on a main road                | 15–30 km (10–19 mi)                                              | Most European main roads                                        |
| Caravanserai on plains and desert      | 30–40 km (19–25 mi)                                              | Seljuk Anatolia and Iran                                        |
| Caravanserai in mountains              | 10 km (6 mi) or less                                             | Iran                                                            |
| Mongol relay station ( <i>yam</i> )    | 32–64 km (20–40 mi)                                              | Mongol Empire, 13th–14th centuries                              |
| Inca way station ( <i>tambo</i> )      | About one day's walk; from under 10 km to nearly 45 km (6–28 mi) | Andes, 15th century (see <a href="#">Roads without wheels</a> ) |
| Pass hospice                           | One near the top of each main pass                               | Great St Bernard                                                |

## Kinds of lodging

- **Town inns** clustered near the town gates, the market place and the bridge. In Chaucer's *Canterbury Tales* (c. 1390s) the pilgrims set off from the Tabard Inn in Southwark, on the main road leading south from London Bridge.
- **Monastic guest houses.** The Rule of St Benedict told monks to receive all guests as Christ. Monasteries on main roads ran large guest houses, and some built inns. The New Inn in Gloucester (mid-15th century) was built by the abbey there for pilgrims.
- **Hospitals and hospices** sheltered pilgrims and the poor, at town gates, river crossings and mountain passes. Roncesvalles, at the Spanish foot of a Pyrenees pass on the pilgrim road to Santiago, had a large pilgrim hospital.
- **Caravanserais** (Persian, "caravan palace") were fortified rural inns with one gate, a central courtyard, rooms, stables, water and often a mosque. Depending on the region, the same kind of building is called a *khan*, a *wikala* (Egypt) or a *funduq* (North Africa). A famous

example is Sultan Han (1229). The Seljuk sultan Kayqubad I built it about 40 km (25 mi) west of Aksaray, on the road to Konya. Rulers and pious endowments (*waqf*) paid for such buildings. In Seljuk Anatolia they gave travellers free food, fodder and lodging.

### MAP TIP

On a main road, put an inn (a small house on the road) or a village or town every 15–30 km (10–19 mi). In a desert, put a caravanserai (the same small house, labelled "Han" or "Caravanserai") or a well every 30–40 km (19–25 mi). Mark a hospice with a small house with a cross. If two towns on a road are more than 40 km (25 mi) apart, add a village or an inn between them. Label them: "Inn of the Seven Wells", "Han of ...", "Hospice of ...".

### FANTASY TWIST

Where monsters roam at night, every stop becomes a small fort with walls, a gate and guards, like a caravanserai. Stops may sit closer together than a full day, so that travellers always arrive before dark.

## Roads without wheels: the Andes and Mexico

### RULE OF THUMB

Where there are no carts and no horses, roads can be narrow, steep and stepped. Goods move on human backs, on llamas and in canoes. Way stations still sit about one day's walk apart, but news travels by relay runners.

**Why:** Before the Spanish arrived in the 1500s, the Americas had no horses, oxen or carts. The Inca of the Andes had llamas as pack animals; the Aztecs of Mexico had no pack animals at all. A road for walkers and llamas does not need gentle slopes, so it can climb straight up a mountainside on stairs. Without carts, a state that wants to move food and

news must organize people instead: way stations, storehouses and relay runners.

### The Inca road system (Andes, mainly 15th century)

The Inca road network (in Quechua, *Qhapaq Ñan*) ran from today's Colombia to Chile and Argentina. Much of it was built or rebuilt in the 15th century, after the Inca conquests began (c. 1438), but many routes were older roads that the Incas took over, some built centuries earlier by the Wari and Tiwanaku cultures.

| Feature                                   | Number                                                                                                                        | Notes                                                                                                                                                                                                              |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total length                              | Estimates vary: 23,000–40,000 km (14,000–25,000 mi), some up to 60,000 km (37,000 mi)                                         | Two main north–south roads, one in the mountains and one along the coast, joined by more than twenty cross roads. The mountain road alone, from Quito through Cusco to Mendoza, was about 5,650 km (3,500 mi) long |
| Road width                                | Usually 1–4 m (3–13 ft)                                                                                                       | The Cusco–Quito highway was always over 4 m (13 ft) wide, in places up to 16 m (52 ft)                                                                                                                             |
| Way stations ( <i>tambos</i> )            | 2,000 or more; usually about one day's walk apart, but from under 10 km to nearly 45 km (6–28 mi)                             | Lodging, kitchens and storehouses; sited by water and good ground, away from marsh and steep slopes                                                                                                                |
| Relay posts for runners ( <i>chaski</i> ) | About 2.5 km (1.6 mi) apart in one account; other sources give up to 7.5 km (4.7 mi). Four to six runners waited at each post | A message could travel up to 240–300 km (150–190 mi) a day                                                                                                                                                         |
| Llama                                     | Carries about 30 kg (66 lb) for about 20 km (12 mi) a day                                                                     | Up to 45 kg (99 lb) on short trips                                                                                                                                                                                 |
| Rope suspension bridges                   | The greatest spanned about 45 m (150 ft), over the Apurímac canyon                                                            | Made of woven grass; local villagers renewed them every one to three years (accounts differ) as labour service. One, the Q'eswachaka, is still rebuilt every June                                                  |
| State storehouses ( <i>qollqa</i> )       | Huánuco Pampa, a provincial centre, had 497 storehouses, about 37,100 m <sup>3</sup> in all                                   | Rows of storehouses on hillsides near provincial centres and along the roads                                                                                                                                       |

### Tenochtitlan: a city of causeways and canoes (Mexico, 1325–1521)

The Aztec capital Tenochtitlan stood on an island in Lake Texcoco. Its traditional founding date is 1325. Three main causeways (raised roads across the water) tied it to the shore: north to Tepeyac, south to Iztapalapa and west to Tlacopan. The western causeway was about 3 km (2 mi) long, by one classic account. The causeways had gaps crossed by bridges. Canoes could pass through the gaps, and the bridges could be removed to defend the city. Inside the city, a network of canals meant that every part could be reached on foot or by canoe. Two aqueducts, each over 4 km (2.5 mi) long, brought fresh water from the springs of Chapultepec.

Without pack animals, goods came across the lakes by canoe or on the backs of professional porters (*tlamemeh*). A porter carried about 23 kg (50 lb) for about 21–25 km (13–16 mi) a day, then handed the load on to the next porter.

#### MAP TIP

For a people without carts or horses, draw thin roads that climb straight up slopes (mark the steep parts with steps or small chevrons ">", as modern road atlases mark steep hills). Put a way station (the inn symbol: a small house on the road) about every day's walk, about 20–25 km (12–16 mi), closer in mountains. Add small relay huts every few km along the trunk roads (on a local map only), the bridge symbol labelled "rope bridge" at gorges, and a labelled group of storehouses near each provincial town. A lake capital can sit on an island, joined to the shore by two to four straight causeways with gaps for boats, and fed by canoe traffic across the lake.

#### FANTASY TWIST

This is a good model for any people without draft animals: dwarves in mountain halls, forest elves or a hill people who travel on foot. Their roads can be narrow and stepped, and their relay runners can carry news faster than a medieval royal messenger with fresh horses (see [Speed per day](#)). A flying mount does not change the model for bulk goods: it is still one small load per animal.

## Tolls and customs

#### RULE OF THUMB

Tolls are taken where traffic cannot avoid them: bridges, town gates, river narrows, mountain gorges, straits and borders.

**Why:** A lord or town needs money, and a narrow point makes it easy to stop every cart or boat. Merchants paid as long as the toll cost less than the detour.

#### Numbers and examples:

- **The Rhine.** In 1250 there were 12 toll stations on the Rhine between Mainz and Cologne alone, about one

every 15 km (9 mi). Stations seem to have been at least 5 km (3 mi) apart. Between about 800 and 1800, 79 different places took tolls on the Rhine and its tributaries. The number of toll stations shot up during the Great Interregnum (1250–1273), when there was no emperor to stop lords from adding new ones. The Pfalzgrafenstein (1326–27) is a toll castle built on an island in the middle of the Rhine near Kaub.

- **Straits.** From 1429 Denmark took the Sound Dues from every foreign ship passing Helsingør, at the entrance to the Baltic. The strait there is only about 4 km (2.5 mi) wide, so no ship could slip past. In 1479, 795 ships passed. By 1583 it was 5,400.

- **Bridges and towns.** Pontage (bridge tolls), murage (tolls to pay for town walls) and pavage (tolls for paving) were common in England.

- **Export customs.** From 1275 the English crown taxed every sack of exported wool. From 1363 all wool for export had to go through the wool staple at Calais (see [staple towns](#)).

#### MAP TIP

Mark a toll with one bar across the road or river and a "T" (two bars close together mean a lock). Put toll castles (the small castle icon) on river cliffs and islands at narrow points (see [Castles, Forts and Military Outposts](#)). Too many tolls on one route is a story: merchants will look for another road, and a rival town on that road will grow.

## Rivers as roads

### RULE OF THUMB

A navigable river is a highway with a one-way discount. Downstream is fast and cheap. Upstream is slow, hard work.

### How far up can boats go?

Every river has a **head of navigation**: the highest point boats can reach. Big sea ships stop lower down; small river boats go higher. Where cargo must move from one kind of boat to another, or onto carts, a town grows (see [head of navigation towns](#)). Around 1300, grain for London was loaded onto boats on the Thames as far upstream as Henley.

How far up English rivers were really navigable is debated. In 1991 Edwards and Hindle argued that rivers were navigable much farther upstream than once thought. On their map, most of England lay within about 15 miles (24 km) of navigable water. Other historians were more cautious. Langdon (1993) argued that some of their evidence showed only occasional river use, not regular traffic. Jones (2000) found that the navigable network shrank in the later Middle Ages, as weirs and other obstructions grew.

### Going upstream: towpaths

Boats went upstream by sail when the wind allowed, by poles and oars, or were **towed** by men or horses walking along a **towpath** on the bank. That is why upstream travel was only 10–20 km (6–12 mi) a day.

### Mills, weirs and locks

Mills need **weirs** (low dams) to raise the water. Fish traps were also built across rivers. Both blocked boats, so millers and boatmen fought for centuries.

- **Magna Carta (1215), clause 33**, ordered all fish-weirs removed from the Thames, the Medway and all of England except the sea coast.
- **Exeter.** In the 1270s or 1280s Isabella de Fortibus, Countess of Devon, built a weir for her mills that cut Exeter off from the sea. Trade was restored in 1290. Then in 1317 her cousin Hugh de Courtenay built another weir and a quay at Topsham, so boats had to unload there and pay his tolls. Exeter petitioned the king for about 250 years. In the end the city built its own canal (1564–66) around the weirs. It had the first pound locks in Britain (see below).

**Flash locks** were a gap or gate in a weir. It was opened to let boats shoot down on a rush of water, or to winch them up against it. Slow and dangerous.

**Pound locks** are a chamber with a gate at each end, as in modern canals. China had one by 984. In Europe the first appeared in the Low Countries in the late 14th century: a large lock basin at Vreeswijk (dated 1373 or 1385; sources differ) and the first true pound lock at Damme near Bruges (1396). Near Milan, Bertola da Novate built 18 pound locks on the Bereguardo canal in 1452–58.

**The Stecknitz Canal** (1391–98), the first canal in Europe to cross a watershed (the higher ground between two river systems), linked the Elbe at Lauenburg with Lübeck on the

Baltic, so that salt from Lüneburg could travel by water. It ran 97 km (60 mi) along two small rivers, with 11.5 km (7 mi) of dug canal across the watershed. It had 13 locks at first (17 later), most of them simple flash locks. Its barges carried about 7.5 tonnes of salt each and needed at least 10 days for the trip one way.

### Portages

A **portage** is a place where boats or cargo are carried or dragged overland between two rivers, or around rapids. On the Russian route "from the Varangians to the Greeks" (Baltic to Constantinople), merchants crossed watersheds between river systems by portage. On the Dnieper they also had to carry or drag their boats past seven rapids (later counts give nine), where the river fell 50 m (165 ft) in 66 km (41 mi), while nomads attacked. Russian place names keep the word *volok* (portage): Volokolamsk means "portage on the Lama".

### Staple rights

Some river towns forced passing merchants to stop and sell. From 1259 Cologne made every merchant passing on the Rhine unload and offer his goods for sale for three days. Vienna got a similar right on the Danube in 1221. Staple rights made river towns rich and annoyed everyone else.

### River ports

A river port needs a quay, warehouses, a crane and a customs post. The great treadwheel crane of Gdańsk (1442–44), wrecked in 1945 and rebuilt, can still be seen on the waterfront. Rivers freeze in northern winters, run low in summer droughts and flood in spring, so river traffic has seasons too.

#### MAP TIP

Draw navigable stretches of rivers a little thicker, from the head of navigation down to the sea, with a small boat at the head of navigation. Put a town at the head of navigation, at confluences and at every portage end. Draw portages as short dotted lines between two rivers, labelled with their length. Add weirs and mills (a wheel on the stream) on small rivers, and a toll castle at a narrow gorge on big ones.

#### LATER ERA (1500S+)

Pound locks spread, and more canals crossed watersheds. The Canal de Briare (1604–42) linked the Loire and the Seine; it was the first canal in Europe to cross a watershed using pound locks. In the Netherlands, horse-drawn passenger canal boats (*trekschuiten*) ran on timetables from 1632 (Amsterdam–Haarlem).

## The sea: ships, harbours and sea lanes

#### RULE OF THUMB

Sailing ships carry bulk cheaply; galleys carry people and luxuries quickly and safely. Both prefer to sail along coasts and island chains, and both stay in port in winter.

## Ship types

| Ship                            | Where and when                                                  | Size                                                                                              | Crew                                                                    | Best for                                                        |
|---------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------|
| Cog                             | North Sea and Baltic, 12th–14th c.                              | 15–25 m (49–82 ft) long, 30–200 t; Bremen cog (1380): about 23–24 m (75–79 ft), 90–130 t          | About 1 sailor per 10 t (a rule of thumb; real crews were often larger) | Bulk cargo: grain, timber, salt, fish, beer; the Hanseatic ship |
| Hulk                            | Northern Europe, 14th–15th c.                                   | By the 1300s carried as much as a cog                                                             | Small                                                                   | Bulk cargo; became the main Hanseatic ship in the 14th century  |
| Merchant great galley           | Mediterranean (Venice, also Genoa and Florence), c. 1290s–1530s | Up to about 46 m (150 ft); the Flanders galley about 37 m (121 ft) long; about 140–250 t of cargo | Up to about 250, including 150–180 oarsmen                              | Spices, silk, money and pilgrims; sailed in state convoys       |
| Round ship ( <i>nave, nef</i> ) | Mediterranean                                                   | Often hundreds of tonnes                                                                          | Small                                                                   | Bulk: grain, salt, wine, cotton, alum                           |
| Carrack                         | Mediterranean and Atlantic, 14th–16th c.                        | Large; 16th-century Portuguese carracks were often over 1,000 t                                   | Large                                                                   | Long voyages with big cargoes                                   |
| Caravel                         | Portugal and Spain, 15th c.                                     | 12–18 m (39–59 ft), 50–60 t                                                                       | Small                                                                   | Exploration and speed; too small for cheap cargo                |

**Why galleys carried only valuable goods:** a galley with 200 men aboard must stop for water and food every few days and pay all those wages. A cog of the same cargo needs about 10 men. So galleys carried spices, silk and passengers, and round ships carried grain and salt.

### Coastal hopping and open sea

Ships stayed near coasts when they could, for shelter, landmarks, water and trade at each port. They crossed open water on short, well-known lanes: across the Channel and the North Sea, across the Bay of Biscay, or from Crete to Egypt.

In the Mediterranean, the historian John Pryor showed that the main **trunk routes** (the main long-distance sea lanes) ran mostly along the northern coasts and island chains (the Adriatic coast, the Ionian islands, Crete, Rhodes, Cyprus). These routes hardly changed from antiquity to the 16th cen-

ture. Whoever held the islands and harbours along them controlled the sea. Venice held Crete (1211–1669) and the ports of Modon and Coron in southern Greece (1207–1500) as stations for its ships.

**How far apart were galley stops?** The Venice–Jaffa pilgrim galleys usually called at Poreč, Zadar, Korčula or Hvar, Dubrovnik, Corfu, Modon, Crete, Rhodes and Cyprus. Measured on a modern map, the stops were roughly 100–450 km (60–280 mi) apart, which is about two to eight days of travel. The whole voyage took 48 days in 1480, and could be done in as little as 29 days.

#### RULE OF THUMB

On a galley coast, put a watering port every 100–300 km (60–190 mi). Big ships can cross wider gaps, but they still prefer a friendly harbour every week or so.

### Sailing seasons

| Sea                  | Season                                                       | Real example                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mediterranean        | Main season roughly April to October; winter sailing avoided | Romans treated the sea as "closed" ( <i>mare clausum</i> ) from November to March, though bans were probably never enforced. 14th-century Genoese statutes banned winter sailing in the Black Sea, with heavy fines. Venetian convoys: Flanders galleys left in March–April and returned about eight months later; Alexandria and Beirut galleys left in late August or September and returned in November |
| North Sea and Baltic | Spring to autumn; Baltic ports freeze                        | Hanseatic rules discouraged or banned most winter sailing, roughly from mid-November to late February                                                                                                                                                                                                                                                                                                      |
| Indian Ocean         | One voyage each way per year, timed to the monsoon           | The south-west monsoon (summer) carries ships from Arabia and East Africa to India; the north-east monsoon (winter) brings them back. Merchants waited months in port between seasons                                                                                                                                                                                                                      |

## What makes a good harbour

| Feature                                     | Why it matters                    | Example                                                                                                                            |
|---------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Shelter from storms and the prevailing wind | Ships at anchor survive           | Bergen (deep fjord); Venice (lagoon behind sand bars)                                                                              |
| Deep water at low tide                      | Big ships can enter               | Atlantic ports often sat up estuaries for depth and shelter (see <a href="#">chapter 01</a> )                                      |
| Narrow, defensible entrance                 | A chain or tower closes it in war | Constantinople's Golden Horn was closed by a chain                                                                                 |
| Fresh water and food                        | Crews need supplies               | Every galley stop                                                                                                                  |
| A river or road into the land behind        | Goods can go on inland            | Bordeaux (Garonne), Rouen (Seine), Hamburg (Elbe)                                                                                  |
| No silting                                  | Harbours that fill with mud die   | Bruges lost its sea access as the Zwin inlet silted; it used outports at Damme and Sluis, and Antwerp took its place by about 1500 |

## Port ranks: how many ports on a coast?

### RULE OF THUMB

A kingdom's coast has a few big head ports, several smaller ports under each of them, a landing place in almost every sheltered cove or creek, and one or two famous anchorages where fleets wait for the wind.

**Why:** Overseas trade needs merchants, warehouses, money and a customs house, and these gather in a few places. Rulers made this stronger on purpose: it is easier to tax trade that must pass through a few ports. But fishing boats and small coasters (boats that carry goods along the coast)

need only a beach or a creek, so small landing places are everywhere.

**Numbers (late medieval England):** England's customs began with a tax on wool exports in 1275, and by the 1400s all foreign trade was taxed. It had to pass through designated **head ports**. After some changes in the 1200s and early 1300s their number settled at **15**, from Newcastle in the north-east round to Bristol in the south-west. That is roughly one head port for every 100–150 km (60–95 mi) of coast (a rough figure, measured on a modern map without the small inlets). London was by far the busiest: 462 ships arrived there in nine months in 1390. Hull, a big regional port, had about 80–90 arrivals a year in the late 1300s.

| Rank                              | What it is                                                                                                                                    | How many                                                     | Real example                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Head port                         | A big port with a customs house, quays, warehouses and resident merchants. All foreign trade for its stretch of coast must clear customs here | 15 for all of England                                        | London, Hull, Boston, King's Lynn, Southampton, Bristol, Newcastle                                                                                                                      |
| Member port                       | A smaller port inside a head port's district. It ships goods, but customs are run from the head port                                          | Several per head port                                        | In 1353–54 the customs district of Hull covered "all ports from Whitby inclusive to Grimsby", more than 100 km (60 mi) of coast                                                         |
| Creek or landing place            | A beach, creek or river landing used by fishing boats and small coasters. "Creek" was a customs word for such places                          | Every few km of sheltered shore or tidal river (an estimate) | Gatcombe, Newnham, Berkeley and Tewkesbury on the Severn, named as creeks of Gloucester in 1580                                                                                         |
| Anchorage (a "road" or roadstead) | Sheltered open water off the coast where ships anchor to wait for a fair wind or to gather into a fleet                                       | One or two on a busy coast                                   | The roadstead off Sluis in the Zwin, Flanders, where a French fleet of over 200 ships lay at anchor in 1340. The Downs, off Kent, which later held as many as 800 sailing ships at once |

**Promotion and decline.** Ranks changed. Throughout the Middle Ages, Gloucester and the whole Severn down to Avonmouth were under Bristol for customs. Only in 1580 did Gloucester become a customs port of its own, with its own creeks. Boston and King's Lynn were great wool ports in the 1200s and 1300s, but as wool exports fell, Hull, Southampton and Bristol became the biggest regional ports of the 1400s.

**How many ships?** The historian Wendy Childs estimates that England's whole commercial fleet was perhaps 1,000–2,000 vessels at times. In the 1300s about 300 ships a year sailed from England to Bordeaux for wine; most loaded 100–150 tons, and a few over 300 tons. In the later 1300s and the 1400s, most major ports could muster about 20 large ships over a decade. Hundreds of small coastal and fishing boats, often under 10 tons and up to 20 tons, made

up the rest. In war the king seized merchant ships as transports: fleets of 150–200 were common, and nearly 700 ships were used for the Crécy campaign (1346).

#### MAP TIP

On a kingdom map, draw a large anchor with a quay, with the name in capitals, for each head port, about one every 100–150 km (60–95 mi) of coast, at the best harbours and river mouths. Draw several small anchors for member ports between them, and leave most coves and creeks as unlabelled landing places. Add one named anchorage (an anchor drawn in open water, labelled "the Roads" or "the Downs") off the busiest stretch of coast or at the mouth of the main trade estuary.

#### LATER ERA (1500s+)

Anchorage became places to defend. Henry VIII guarded the Downs with three artillery forts, Deal, Walmer and Sandown, built in 1539–40 (see [The Later Era, 1500–1650](#)).

### Lighthouses and sea marks

Lighthouses were **rare** in medieval Europe. Most coasts had none. Ships relied on church towers, beacons, landmarks and local pilots. Lights stood only at the entrances to rich ports and at very dangerous points:

- **Genoa:** a light tower by 1128; the present Lanterna dates from 1543.
- **Hook Head, Ireland (1201–1240):** built by William Marshal to guide ships into Waterford Harbour and up to his port of New Ross, about 30 km (19 mi) upriver. Monks kept the light.
- **St Catherine's Oratory, Isle of Wight (1328):** built by a lord as a penance for stealing wine from a wreck. It is Britain's only surviving medieval lighthouse.

- **Cordouan (Gironde estuary):** a 14th-century tower built under Edward the Black Prince guarded the Bordeaux wine trade and charged ships "lighthouse dues"; the present tower dates from 1584–1611.

#### MAP TIP

Draw sea lanes as dotted lines from port to port, following coasts and island chains, with short open-sea crossings between them. Add arrows where the wind decides the direction, and label each lane with its season ("galleys, April–October"). Put lighthouses (a small tower with short rays; never a star, which means a capital) only at the mouths of rich estuaries and on dangerous capes near big ports. A port without a river, road or rich hinterland (the farmland and towns behind it) stays small.

#### FANTASY TWIST

Sea monsters push ships even closer to the coast and into armed convoys, and make every harbour mouth worth a watchtower. Magic that calms storms would open the winter season and make sea trade even cheaper, so port cities grow larger still.

#### LATER ERA (1500s+)

Ocean routes open. The Portuguese reached India by sea in 1498, and Seville got a monopoly of trade with Spanish America in 1503. Carracks and then galleons crossed oceans. Atlantic ports (Lisbon, Seville, Antwerp, then Amsterdam and London) grew fast. Baltic grain fed the Netherlands, and traffic through the Danish Sound rose about sevenfold between 1479 and 1583. See [The Later Era, 1500–1650](#).

## Trade nodes: markets, fairs, staples and entrepôts

#### RULE OF THUMB

Trade has a ladder. Every market town has a weekly market. Many have a yearly fair. A few towns in a large region host great fairs or force goods through a staple. Only a handful of great ports in a whole continent are true world markets.

| Node                                               | What happens there                                                                         | How often                             | Spacing or number                                                                            | Real examples                                                                                                             |
|----------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Weekly market                                      | Peasants sell food and buy tools, salt and cloth                                           | Weekly                                | Market towns 9–16 km (6–10 mi) apart in practice; England had 1,746 recorded markets by 1300 | Any market town; English place names with <i>Chipping</i> ("market"), such as Chipping Norton                             |
| Yearly fair                                        | Livestock, cloth, regional merchants                                                       | Once a year, a few days to some weeks | Many market towns had one                                                                    | St Ives (Huntingdonshire); Stourbridge near Cambridge (charter 1199, first fair 1211; later one of the biggest in Europe) |
| International fair cycle                           | Merchants from distant lands meet, trade and settle debts                                  | A cycle of fairs through the year     | One or two fair regions per continent at any time                                            | The Champagne fairs (12th–13th c.); later Geneva, Lyon, Frankfurt, Leipzig, Medina del Campo                              |
| Staple town                                        | The law forces certain goods to be unloaded, offered for sale or taxed there               | Permanent                             | One per river stretch or per export trade                                                    | Cologne (from 1259); Vienna (from 1221); the English wool staple at Calais (1363–1558)                                    |
| Entrepot                                           | A great port where goods from many regions are stored and shipped on                       | Permanent                             | Only a handful in all of Europe and the Mediterranean c. 1300                                | Bruges, Venice, Genoa, Constantinople, Alexandria, Lübeck; Antwerp from c. 1500                                           |
| Merchant colony ( <i>Kontor</i> , <i>fondaco</i> ) | Foreign merchants live and store goods in their own walled compound, under their own rules | Permanent                             | One per foreign nation in each big port                                                      | See below                                                                                                                 |
| Customs house                                      | Goods are weighed, sealed and taxed                                                        | Permanent                             | Every major port and border                                                                  | The Kaufhaus (warehouse and customs hall) at Konstanz, 1388                                                               |

### The Champagne fairs

From the 12th century, the Counts of Champagne protected merchants travelling to their lands, fixed weights and measures, and ran fair courts. Six fairs in four towns ran almost all year, and each lasted about six weeks: Lagny (from 2 January), Bar-sur-Aube (Lent), Provins (May), Troyes ("hot fair", summer), Provins (September) and Troyes ("cold fair", November). Each fair had set days for

cloth, then leather, then spices and goods sold by weight. It ended with a few days to settle accounts. Flemish cloth came south, and Italian silks, spices and money came north. In 1284 the heiress of Champagne married the future Philip IV of France, and Champagne became part of France. Royal wars and taxes then hurt the fairs, and Italian galleys began sailing directly to Flanders. By about 1350 the fairs were only regional markets.

## Merchant colonies: kontors and fondaci

| Colony                       | Place                                 | Dates                                                                                 | Notes                                                                                                                                                 |
|------------------------------|---------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peterhof                     | Novgorod (Russia)                     | c. 1200–1494 (reopened 1514, then faded)                                              | Palisaded yard with a stone church used as a warehouse; about 150–200 German merchants a season; furs and wax out, cloth, silver, salt and herring in |
| Steelyard ( <i>Stalhof</i> ) | London                                | Cologne merchants had a hall by the 1170s and royal privileges from 1175; closed 1598 | Walled compound of warehouses and halls on the Thames                                                                                                 |
| Bryggen (German Wharf)       | Bergen (Norway)                       | c. 1350–1754                                                                          | Timber warehouses on the quay; stockfish (dried cod) out, grain in                                                                                    |
| Hanseatic Kontor             | Bruges                                | 13th century to 16th                                                                  | No compound: merchants lodged with local innkeepers who acted as brokers; own house only from 1442                                                    |
| Fondaco dei Tedeschi         | Venice                                | 1228; rebuilt 1505–08                                                                 | The official house of the German merchants, by the Rialto Bridge, under Venetian supervision; the rebuilt house had 56 rooms for over 100 merchants   |
| Venetian fondaci             | Alexandria                            | 13th–15th c.                                                                          | Venetian warehouses and lodgings for the spice trade                                                                                                  |
| Genoese colonies             | Pera (Constantinople), Caffa (Crimea) | Pera from 1267; Caffa c. 1266–1475                                                    | Whole walled towns run by Genoa                                                                                                                       |
| Venetian quarter             | Constantinople                        | From 1082                                                                             | Quays, warehouses and churches, free of Byzantine customs                                                                                             |

### MAP TIP

In each great port, draw or label the foreign quarters: "Hanse Steelyard", "Genoese quarter", "Venetian fondaco". At fair towns, add the fair dates to the label ("Provins: May and September fairs"). A fair town sits where two economic regions meet, often on a political border or on neutral ground under a strong protector.

### FANTASY TWIST

If the roads are full of monsters, trade collapses into fewer, bigger events: one or two great fairs a year, reached by armed caravans. A neutral fair town protected by a powerful mage or temple could become the Champagne of your world.

### LATER ERA (1500S+)

Permanent exchanges began to replace the great fairs. Antwerp opened a purpose-built exchange (bourse) in 1531 and Amsterdam in 1611, so merchants could trade every day instead of a few weeks a year. Antwerp, then Amsterdam, became the leading entrepot of northern Europe.

## What is traded

### RULE OF THUMB

The higher the value per kilogram, the farther a good can travel over land. Cheap, heavy goods stay local unless they can float.

## Bulk goods and luxury goods

| Good                                            | Value per weight | How far over land                                                           | How far by water                                                                                                                                       |
|-------------------------------------------------|------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grain, flour                                    | Very low         | Usually under 30–50 km (19–31 mi); routine sales within about 17 km (10 mi) | Hundreds of km by river and sea, especially to big cities and in famine                                                                                |
| Firewood, timber, stone, brick                  | Very low         | A few km                                                                    | Far by water: Caen stone from Normandy built English cathedrals                                                                                        |
| Salt                                            | Low              | About 100 km (62 mi) or more on "salt roads" (Lüneburg to Lübeck)           | Across Europe: salt from the Bay of Bourgneuf (France) went to the Baltic                                                                              |
| Wine                                            | Medium           | Short distances                                                             | Very far: Bordeaux shipped about 100,000 tuns (a tun is a big cask of about 950 litres) in the record year 1308–09, much of it to England (James 1971) |
| Salted herring, stockfish                       | Medium           | Short to medium                                                             | Continent-wide                                                                                                                                         |
| Wool                                            | Medium to high   | Carried by pack horse and cart to ports                                     | England to Flanders and Italy                                                                                                                          |
| Cloth                                           | High             | Any distance                                                                | Any distance                                                                                                                                           |
| Metals (copper, tin, silver, iron)              | Medium to high   | Long distances                                                              | Long distances                                                                                                                                         |
| Furs, wax, amber                                | High             | Long distances                                                              | Long distances                                                                                                                                         |
| Spices, silk, dyes, gems, <a href="#">sugar</a> | Very high        | Across continents                                                           | Across oceans                                                                                                                                          |
| Live animals                                    | Walk themselves  | Hundreds of km along drove roads (see <a href="#">Drove roads</a> )         | —                                                                                                                                                      |

A cart journey that doubles the price of grain adds only a few per cent to the price of pepper, which was worth many

times more per kilogram. That is why spices crossed Asia by camel while grain rarely crossed a county by cart.

## Typical goods by region type

| Region type                                                                                     | Exports                                                                                                                                        | Imports                            |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Fertile lowland plain                                                                           | Grain, malt, ale or beer, flax, wool                                                                                                           | Salt, metal, wine, luxuries        |
| Hills and uplands                                                                               | Wool, sheep, cattle, cheese, hides, lead                                                                                                       | Grain, salt, cloth                 |
| Mountains                                                                                       | Silver, copper, iron, rock salt, timber, cattle, cheese; transit tolls                                                                         | Grain, wine, cloth                 |
| Northern forests                                                                                | Furs, wax, honey, timber, tar and pitch, potash                                                                                                | Salt, cloth, metal goods, wine     |
| Coasts and islands                                                                              | Salted fish, sea salt, ships and sailors                                                                                                       | Grain, timber, cloth               |
| Wine country (in Europe mostly south of about 50° N, though small vineyards grew farther north) | Wine, brandy (later)                                                                                                                           | Grain, cloth, fish                 |
| Mediterranean dry lands                                                                         | Olive oil, wine, dried fruit, <a href="#">sugar</a> (Cyprus, Sicily; see <a href="#">Sugar estates</a> ), alum, cotton (Syria), mastic (Chios) | Grain, timber, metals, cloth       |
| Cloth towns                                                                                     | Woollen and linen cloth                                                                                                                        | Wool, dyes, alum, food             |
| Desert and oasis                                                                                | Dates, salt; transit of gold and slaves                                                                                                        | Grain, cloth, metal goods          |
| The "Indies" (South and East Asia)                                                              | Pepper and spices, silk, cotton cloth, porcelain, gems                                                                                         | Silver, gold, horses, coral, cloth |

For mines, quarries, salt works and industrial towns, see [Industry, Resources and Special Towns](#).

**MAP TIP**

For each region on your map, write one line: "exports ..., imports ...". Then draw the route that carries each export to the nearest water or market. If two regions export the same thing to each other, one of them is wrong.

## Real trade networks to copy

**RULE OF THUMB**

A continent usually has one or two **bulk sea networks**, several **river spines**, and one or two **long luxury routes** over land. Copy a real one, rename it, and adapt it to your geography.

| Network                                                  | Peak                                                  | Shape on the map                                                                                                                    | Hubs                                                                                                                                                                                   | Main goods                                               |
|----------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Hanseatic League</b><br>(Baltic and North Sea)        | c. 1250–1500                                          | A chain of ports along a cold sea, fed by rivers; 70–170 member towns at different times, and nearly 200 in all; four great kontors | Lübeck, Hamburg (linked to Lübeck by a short land route of about 60 km (37 mi) across the neck of the Jutland peninsula), Visby, Danzig, Riga, Reval, Bergen, Bruges, London, Novgorod | Grain, timber, furs, wax, fish, salt, beer, cloth        |
| <b>Venice and Genoa</b><br>(Mediterranean and Black Sea) | c. 1100–1500                                          | State convoys and chains of island and coastal bases                                                                                | Venice, Genoa, Crete, Pera, Caffa, Chios, Alexandria, Beirut, Acre (until 1291)                                                                                                        | Spices, silk, cotton, alum, grain, slaves, salt          |
| <b>Silk Roads</b>                                        | Revived under the Mongols, c. 1250–1350               | A relay of oasis cities over about 6,400 km (4,000 mi); hardly anyone travelled the whole way                                       | Tabriz, Samarkand, Bukhara, Kashgar, Sarai, Tana, Khanbaliq (Beijing)                                                                                                                  | Silk, porcelain, spices, horses, silver                  |
| <b>Trans-Saharan</b>                                     | 8th to early 17th c.; Mali's peak in the 13th–14th c. | North–south lines from oasis to oasis; crossing took about two months                                                               | Sijilmasa, Taghaza (salt mine), Walata, Timbuktu, Gao, Agadez                                                                                                                          | Gold, salt, copper, cloth, slaves                        |
| <b>Indian Ocean monsoon</b>                              | Whole period                                          | Seasonal sea lanes; merchant quarters in every port                                                                                 | Aden, Hormuz, Cambay, Calicut, Malacca (from c. 1400), Kilwa, Quanzhou                                                                                                                 | Pepper, spices, cotton cloth, horses, gold, ivory        |
| <b>Rhine corridor</b>                                    | Whole period                                          | One river spine with many tolls and staples, linked to Italy by Alpine passes                                                       | Basel, Strasbourg, Mainz, Cologne, Dordrecht                                                                                                                                           | Wine, cloth, metals, stone, fish                         |
| <b>Russian rivers</b>                                    | 9th–15th c.                                           | River systems joined by portages                                                                                                    | Novgorod, Smolensk, Kiev, Bulghar                                                                                                                                                      | Furs, wax, honey, slaves, silver                         |
| <b>Amber Road</b>                                        | Prehistoric and Roman, Baltic to the Adriatic         | A long north–south land route                                                                                                       | Aquileia, Carnuntum; in the Middle Ages, amber came from Prussia, where the Teutonic Order controlled it                                                                               | Amber (later sold to rosary-makers in Bruges and Lübeck) |
| <b>Salt routes</b>                                       | 12th–16th c.                                          | Short land or canal routes from salt works to fish ports                                                                            | Lüneburg to Lübeck (about 100 km, 62 mi), for salting Baltic herring                                                                                                                   | Salt                                                     |
| <b>Wine trade</b>                                        | Gascony to England 1152–1453                          | Big seasonal wine fleets from one estuary                                                                                           | Bordeaux, La Rochelle; Rhine wine via Cologne                                                                                                                                          | Wine                                                     |
| <b>Pilgrim roads</b>                                     | Whole period                                          | Chains of hospices and inns to a shrine                                                                                             | Rome, Santiago de Compostela, Canterbury; Jerusalem by galley from Venice                                                                                                              | People and their money (see <a href="#">chapter 08</a> ) |

**Notes on using them:**

- **Silk Roads:** around 1340 the Florentine merchant Pego-lotti wrote that the road from Tana (on the Sea of Azov) to China was "perfectly safe, whether by day or by night", according to merchants who had used it. A strong empire across the route makes long land trade possible. When it breaks up, the route breaks into dangerous pieces.
- **Trans-Saharan:** in 1352 Ibn Battuta crossed from Sijilmasa to Walata in about two months. At Taghaza, 25 days out, the houses and mosque were built of blocks of salt. Near the end, water was brought out to the caravan from Walata, four days ahead. An average caravan may have had about 1,000 camels; some are said to have had as many as 12,000.

**MAP TIP**

Pick one template for each kind of sea or land on your continent: a Hanse-like network for a cold northern sea, a Venice-like network for a warm inland sea, a Silk-Road-like chain across a steppe or desert. Then join them at a few great entrepôts. Those meeting points are your richest cities.

**FANTASY TWIST**

Elves who travel by forest paths or dwarves with underground roads create networks that do not follow rivers. Treat them like any route: work out what they carry, how fast, and at what cost. A dwarven road that moves bulk ore cheaply is a river in stone.

## Drawing the route network

**Step by step**

1. **Water first.** Mark which rivers are navigable and how far (thicker lines). Choose head ports about every 100–150 km (60–95 mi) of coast (see [Port ranks](#)). Add coastal sea lanes and short open-sea crossings.
2. **Crossings.** Place fords, bridges and ferries: on a settled lowland river, a bridge town about every 20 km (12 mi) of river. Big rivers get few fixed bridges.
3. **Passes.** Choose the main passes over each range.
4. **Main roads.** Link cities and big towns through the crossings and passes. Follow valleys and dry ridges. Connect every main road to a river port or harbour. Aim for about 30 km of main road per 1,000 km<sup>2</sup> of settled land (see [How much main road to draw](#)).
5. **Secondary roads.** Link each market town to its neighbours, 9–16 km (6–10 mi) away, and to the nearest main road.
6. **Local tracks.** Link villages to their market town, 6–10 km (4–6 mi) away, like spokes of a wheel.
7. **Stops.** Add inns every 15–30 km (10–19 mi) on main roads, caravanserais every 30–40 km (19–25 mi) in deserts, hospices on passes, and watering ports every 100–300 km (60–190 mi) on galley coasts.
8. **Tolls.** Put tolls at bridges, gorges, straits and borders.
9. **Hubs.** Check that the biggest towns sit where routes meet (see below).

**Which places become hubs?**

- Where **land meets water:** ports, river ports at the head of navigation, and the lowest bridge on a river.
- **Confluences**, where two navigable rivers join.

- The **foot of a main pass**, on both sides.
- **Both ends of a portage.**
- **Border towns** between two economic regions, where goods change hands (fair towns, staple towns).
- **Crossroads** of two main roads, especially at a bridge.

As a rough guide (not a measured rule): a village has 2–3 roads, a market town 3–5, and a city 5–8 or more, plus water routes.

**How much main road to draw****RULE OF THUMB**

On a kingdom map of settled land, draw about 30 km of main road for every 1,000 km<sup>2</sup> (about 48 mi per 1,000 sq mi). A square of 100 × 100 km (62 × 62 mi) gets about 300 km (190 mi) of main road: for example three roads crossing it, or four or five shorter roads meeting at towns inside it.

**Why:** A kingdom map shows the main roads that a merchant or a royal messenger would plan a journey on. The lanes between villages are many times denser, but drawn at this scale they turn the map into a spider's web. They belong on a local map.

**Numbers:** A good medieval measure is the Gough Map of Britain (conventionally dated c. 1360–70; a handwriting study suggests one or two decades after 1400, with additions later in the 15th century; see [The Roman legacy](#)). It shows over 600 cities, towns and smaller settlements. About 190 red lines join about 180 of them in England and Wales. A GIS study (a study with digital map software) by

Eljas Oksanen and Stuart Brookes matched 99% of these lines to real historical routes.

| Measure                                                                                           | Value                                                               |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Red route lines on the map itself                                                                 | About 190                                                           |
| Length of the real roads behind the map's red lines, as reconstructed by Oksanen & Brookes (2025) | About 4,540 km (2,820 mi), stored as 455 segments in their database |
| Area of England and Wales                                                                         | About 151,000 km <sup>2</sup> (58,000 sq mi)                        |
| Main road per area                                                                                | About 30 km per 1,000 km <sup>2</sup> (48 mi per 1,000 sq mi)       |
| Share of the route length on Roman roads                                                          | 35.5%                                                               |
| Share on routes first recorded before 1086 (not Roman)                                            | 13%                                                                 |
| Share on routes first recorded after 1086                                                         | 16%                                                                 |
| Share known only from later (early modern) evidence                                               | 32.5%                                                               |
| Other (a line along the River Witham, or no known road)                                           | about 3%                                                            |

The Gough Map is a selection, not a full road atlas. It leaves out some important roads, such as the road from London to Colchester and East Anglia, and Watling Street between London and Canterbury. So 30 km per 1,000 km<sup>2</sup> is a good amount for a kingdom map, not the whole network. The village lanes (2–3 per village, see above) still exist, but draw them only on a local map.

**Roman roads live on.** About a third of the Gough Map's route length runs on Roman roads, nearly 1,000 years after Roman rule ended in Britain. Much of the rest first appears in the Middle Ages, when new roads grew up to serve new towns, markets and bridges.

#### MAP TIP

Spread the main roads unevenly: more around the capital and in rich, crowded lowland, fewer in hills, moors and forests. Put a town or an inn every 15–30 km (10–19 mi) along each main road. If your kingdom was once part of an older empire, draw about a third of the main-road length as long straight stretches on the old empire's roads, and let the rest bend between medieval towns and bridges.

### Symbols and labels

These symbols match the guide's [master legend](#). Routes are always solid or dotted lines. On the route network a short dashed line means a ford, and dash-dot lines are borders, never routes.

| Route                          | Suggested line                                                                        | Label example                          |
|--------------------------------|---------------------------------------------------------------------------------------|----------------------------------------|
| Major sea lane                 | Dotted line at sea, with arrows where the wind sets the direction, and a season label | "Galley route to Alexandria (Aug–Nov)" |
| Navigable river                | Thicker river line up to the head of navigation, with a small boat there              | "Navigable to Westbridge"              |
| Royal highway                  | Thick solid line                                                                      | "The King's Road", "Via Aurelia"       |
| Old Roman or older-empire road | Straight thick solid line; grey where abandoned                                       | "Ermine Street", "Fosse Way (Roman)"   |
| Regional road                  | Medium solid line                                                                     | "The Salt Road"                        |
| Local road                     | Thin solid line                                                                       | Usually unlabelled                     |
| Pilgrim road                   | The ordinary road line with small scallop marks along it                              | "Pilgrims' road to the shrine"         |
| Track, bridle way or mule path | Thin dotted line                                                                      | "Mule path, closed in winter"          |
| Drove road                     | Broad dotted line (two rows of dots)                                                  | "Drovers' road"                        |
| Caravan route                  | Dotted line with a larger dot at each stage                                           | "Caravan road, 9 stages"               |
| Portage                        | Short dotted line between two rivers, labelled with its length                        | "Portage, 12 km"                       |

| Point on the network | Symbol                                                   | Label example                          |
|----------------------|----------------------------------------------------------|----------------------------------------|
| Bridge               | Two short curved lines ")( with the road running through | Usually the town's name                |
| Ford                 | Short dashed line across the river                       | "Ford"                                 |
| Ferry                | Short dotted line across the river with "F"              | "Ferry"                                |
| Toll                 | One bar across the road or river with a "T"              | "Toll"                                 |
| Lock                 | Two short bars close together across the river or canal  | "Lock"                                 |
| Inn or caravanseraï  | Small house on the road                                  | "Inn of the Seven Wells", "Han of ..." |
| Hospice              | Small house with a cross                                 | "Hospice of ..."                       |
| Head port            | Large anchor with a quay                                 | Name in capitals                       |
| Member port          | Small anchor                                             | Name in ordinary letters               |
| Anchorage            | Anchor in open water                                     | "the Roads", "the Downs"               |
| Lighthouse           | Small tower with short rays                              | "Light"                                |

**Naming roads.** Real roads were named after where they go ("the London road"), who built them ("Via Appia", after Appius Claudius), or what they carry ("Salt Road", "Amber Road"). Give long distances in days ("12 days to the capital"), because that is how travellers thought.

#### MAP TIP

Check every road with three questions. Where does it go? What does it carry? Why does it bend here? If a road has no destination or cargo, delete it. Common errors (roads straight over mountains, too many bridges on a great river, big cities with no water access) are listed in [Common Mistakes](#).

## Quick summary

- Over land: about 30 km (19 mi) a day on foot, 15–25 km (10–15 mi) by ox cart, 16–30 km (10–20 mi) for a cattle drove, 50–90 km (30–56 mi) for a royal messenger.
- Water is much cheaper: land : river : sea  $\approx$  8 : 4 : 1 (England c. 1300). Bulk goods follow water; most towns above 10,000 people sit on it.
- Medieval roads were rights of way, not built roads. Roman roads were reused (about a third of the Gough Map's routes). Roads follow valleys, ridges, fords and bridges. On a kingdom map, draw about 30 km of main road per 1,000 km<sup>2</sup> (48 mi per 1,000 sq mi).
- Stone bridges spread from the 12th century. They were paid for by tolls (pontage), rents, gifts and bridge chapels. A settled lowland river has a bridge town about every 20 km (12 mi), like the Thames; the biggest rivers had few bridges and many ferries.
- A few passes carry most traffic across a range. The lowest takes carts (Brenner, 1,370 m or 4,495 ft). Hospices sit near the summits.
- Inns and stages are 15–30 km (10–19 mi) apart on main roads. Caravanserais are 30–40 km (19–25 mi) apart on plains and 10 km (6 mi) or less in mountains. Without wheels or horses (the Inca), way stations still sit a day's walk apart, and relay runners carry news 240–300 km (150–190 mi) a day.
- Tolls sit at choke points. The Rhine had 12 toll stations between Mainz and Cologne alone in 1250.
- Rivers have a head of navigation, towpaths, weirs, locks and portages. Millers and boatmen fought over the water.
- Cogs and hulks carry northern bulk. Galleys carry Mediterranean luxuries and pilgrims in convoys, with stops every 100–450 km (60–280 mi). Winter is the closed season.
- A coast has a few head ports (15 for all of late medieval England, roughly one per 100–150 km or 60–95 mi of coast), more member ports, landing places in most creeks, and one or two named anchorages.
- Trade nodes form a ladder: weekly markets, yearly fairs, great fairs and staples, then a few great entrepôts with foreign merchant colonies.

- Copy real networks (Hanse, Venice and Genoa, Silk Roads, trans-Saharan, Indian Ocean, rivers) and make hubs where land meets water.

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## CHAPTER 6

# VILLAGES, FARMS AND THE COUNTRYSIDE

Most people in the Middle Ages lived in villages, hamlets and farms. Between 85% and 95% of people lived in the country, so most of any realistic map is countryside. This chapter is for local-area maps: one parish, one valley, one barony or one county. It tells you what a village looks like and what is inside it. It shows how fields, woods and pastures are arranged, how much of a whole realm they cover, where mills, parks, ponds and barns go, and how far apart everything is.

The baseline is Europe c. 1000–1500, with England as the best-documented example. Where to put villages on a big map is covered in [Where Settlements Are Built \(and Why\)](#). How many people live in them is covered in [Population and Settlement Sizes](#).

## IN THIS CHAPTER

Village forms and layouts

Sizes, territory and walking distances

Fields and farming systems

Woods, forests and hunting grounds

How much of the map is fields and woods

Mills and water management

Other rural features

Reclaimed land: fens, polders and assarts

Local administration: parish, manor, hundred

Drawing a local-area map

Quick summary

Sources and further reading

## Village forms and layouts

## RULE OF THUMB

Good arable land with shared open fields gives **tight villages**. Hills, forest, marsh and pasture give **scattered farms and hamlets**. A line feature (a road, a valley stream or a dyke) gives **villages in a line**.

**Why:** People live as close as they can to the land they work most. On open-field land, every family has strips all over the parish, so the best place to live is in the middle, together. Where each family farms its own block of land (in hills, forest clearings or marsh), it is better to live on that block.

### The main village forms

| Form                                                 | What it looks like                                                                                          | Where (real examples)                                                                                                                                                                                                                             | How to draw it                                            |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Nucleated village</b><br>(houses in one cluster)  | 20–80 houses packed along a few lanes around the church and manor house; fields all around                  | English Midlands; northern France; much of Germany ( <i>Haufendorf</i> , "heap village"); Denmark                                                                                                                                                 | A tight cluster with a church, open fields around it      |
| <b>Green village</b>                                 | Two rows of houses facing each other across a long or square green                                          | Northern England: in County Durham many villages, perhaps most, show signs of deliberate planning, often as two rows round a green; German <i>Angerdorf</i> (a lens-shaped green with church and pond) in the lands settled from the 12th century | Two parallel rows around an open space                    |
| <b>Street (row) village</b>                          | One or two rows of houses along one street                                                                  | Planned villages everywhere; German <i>Straßendorf</i>                                                                                                                                                                                            | A line of houses along a road                             |
| <b>Polyfocal village</b><br>(several centres)        | Two or three small clusters (church end, manor end, green end) that grew together                           | Common in England                                                                                                                                                                                                                                 | Two or three knots joined by a lane                       |
| <b>Forest strip village</b> ( <i>Waldhufendorf</i> ) | Farms spaced along a valley road or stream. Behind each farm a long strip of land runs uphill to the forest | First in the northern Black Forest c. 1000; spread in the 12th–13th centuries to the Odenwald, Ore Mountains, Bohemia and Silesia. Chains of these villages run up to 25 km (15 mi) along valleys in Saxony                                       | Farms like beads on a string; parallel strips behind them |

| Form                                                     | What it looks like                                                                              | Where (real examples)                                                                                                                                                                                                                                 | How to draw it                                                                                                             |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Marsh and dyke village</b> ( <i>Marschhufendorf</i> ) | Farms in a line along a dyke or drainage ditch, with long narrow plots behind                   | North Sea marshes, Holland                                                                                                                                                                                                                            | A line along the dyke, thin strips at right angles                                                                         |
| <b>Round village</b> ( <i>Rundling</i> )                 | A horseshoe or circle of farms around a green, with one entrance                                | The German–Slavic border zone, mostly founded in the 12th century. Usually 5–7 farms at first; farms were divided until there were about 20 by the 16th century. The church stands outside the ring. About 93 survive, in the Wendland (Lower Saxony) | A small horseshoe of house marks round a green, with the church outside it (not a plain circle, which means a market town) |
| <b>Hamlet</b>                                            | 2–15 households, no church                                                                      | Everywhere. Main form in Devon, Cornwall, Wales, Ireland ( <i>clachan</i> ) and Scotland ( <i>fermtoun</i> )                                                                                                                                          | 2–5 small house marks with no church                                                                                       |
| <b>Dispersed farmsteads</b>                              | Single farms, each in its own fields                                                            | Uplands, Alps, Norway, forest clearings; Kent, Essex and the Weald                                                                                                                                                                                    | Scattered tiny dots joined by lanes                                                                                        |
| <b>Mediterranean hill village</b>                        | A dense, walled cluster of stone houses on a hilltop or spur, with castle and church at the top | Lazio and Tuscany ( <i>incastellamento</i> , the move into fortified hill villages, 10th–12th centuries); Provence ( <i>villages perchés</i> ); Catalonia                                                                                             | A dense blob on a hill, with a wall                                                                                        |
| <b>Agro-town</b>                                         | 2,000–10,000 people, mostly farmers, in one big settlement                                      | Sicily, Apulia, southern Spain                                                                                                                                                                                                                        | A town-sized symbol, but with few town functions                                                                           |

England is the classic case of mixed patterns. A central belt of nucleated villages ran from the north-east through the Midlands to the south coast, with more dispersed settlement on both sides (Roberts and Wrathmell 2000). In

France, the open-field villages of the north-east contrast with the *bocage* (small hedged fields and scattered hamlets) of the west.

### What is in a village

| Feature                          | How common                                                                              | Where it sits                                                                                           | Notes                                                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parish church</b>             | One per parish                                                                          | On a slight rise, often at one end of the village, next to the manor house                              | The biggest building, usually stone. Has a churchyard                                                                                 |
| <b>Manor house</b>               | One per manor; a village could have two or three manors                                 | Next to the church; often moated                                                                        | Hall, barns, stables, often a dovecote and fishpond. England has about 6,000 known moated sites, most dug c. 1250–1350                |
| <b>Mill</b>                      | About one per village (England c. 1300)                                                 | On the stream, often a little away from the houses, with its leat and pond; a windmill on rising ground | The lord's monopoly (see <a href="#">Mills</a> )                                                                                      |
| <b>Parsonage and glebe</b>       | One per parish                                                                          | Near the church                                                                                         | The priest's house and his own land (glebe)                                                                                           |
| <b>Smithy</b>                    | Most villages                                                                           | On the main street, the green or the road                                                               | Shoes horses and oxen, mends ploughs                                                                                                  |
| <b>Alehouse</b>                  | Common                                                                                  | Any house that brewed for sale                                                                          | Before c. 1400 usually an ordinary house showing an <i>ale-stake</i> (a pole with a bush). Inns are mainly in towns and on main roads |
| <b>Green</b>                     | Many villages, mainly in the north and east of England and in German <i>Angerdörfer</i> | In the middle                                                                                           | Grazing, gatherings, sometimes a pond and the pound                                                                                   |
| <b>Pond</b>                      | Most villages                                                                           | On the green or at the lowest point                                                                     | For animals, not drinking                                                                                                             |
| <b>Pound or pinfold</b>          | Most villages by the 16th century                                                       | At the edge of the village or green                                                                     | A walled or hedged pen for stray animals, run by the <i>pinder</i>                                                                    |
| <b>Lord's oven and winepress</b> | Common in France and Germany                                                            | Near the manor or in the centre                                                                         | In France these were <i>banalités</i> : the lord's mill, oven and press, which peasants had to use and pay for                        |
| <b>Tofts and crofts</b>          | All villages                                                                            | Along the streets                                                                                       | A <b>toft</b> is the house plot with its yard; a <b>croft</b> is the enclosed garden or paddock behind it                             |

**Real example: Wharram Percy (Yorkshire, England).** This deserted village on the chalk Wolds had about 40 peasant house plots in planned rows, two manor houses, a church and a green. It also had two watermills, and their mill-ponds were fished. At its height, in the late 13th and early 14th centuries, it had perhaps 200 people. About 30 houses were occupied around 1368 and at least 16 in 1436. Around 1500 the lord evicted the last four families and turned the land into sheep pasture.

**MAP TIP**

Pick one village form for each region of your map and stick to it. Draw the church (a small church with a tower) as the biggest symbol, the manor house (a small open square) beside it, and the mill (a wheel on the stream) a short way off. On a local map, show tofts as a row of narrow plot outlines along the street, each with a small house mark at the street end, and crofts as long narrow plots behind. Do not draw them as plain small rectangles: in the guide's legend, a chain of small rectangles means fish-ponds. Most villages need a green, a pond or a road junction as their centre.

**FANTASY TWIST**

If wolves, raiders or monsters roam the countryside at night, dispersed farmsteads become rare and dangerous. People crowd into nucleated or hilltop villages with a ditch and palisade, as Italian peasants did during *incastellamento*. Isolated farms survive only as fortified towers or inside a lord's protection. See [Fantasy Variants](#).

## Sizes, territory and walking distances

**RULE OF THUMB**

A lowland village has 30–60 households (150–300 people) and owns about 5–15 km<sup>2</sup> (2–6 sq mi) of land. Its farthest fields are a 20–35 minute walk from the houses.

**Why:** A person needs about 1 ha (2.5 acres) of arable land and 2–3 ha (5–7 acres) of all land to eat (see [Population and Settlement Sizes](#)). Peasants walked to their fields and back every day, often with an ox team and a plough. Every extra kilometre wastes working time.

### Village territory

| Settlement              | People                             | Households | Territory, lowland                                  | Farthest regular fields                                              |
|-------------------------|------------------------------------|------------|-----------------------------------------------------|----------------------------------------------------------------------|
| Hamlet                  | 10–75                              | 2–15       | 1–5 km <sup>2</sup> (0.4–2 sq mi), part of a parish | under 1 km (0.6 mi)                                                  |
| Small village           | 75–150                             | 15–30      | 2–7 km <sup>2</sup> (0.8–2.7 sq mi)                 | 1–1.5 km (0.6–1 mi)                                                  |
| Typical lowland village | 150–300                            | 30–60      | 5–15 km <sup>2</sup> (2–6 sq mi)                    | 1.5–2.5 km (1–1.5 mi)                                                |
| Large village           | 300–500                            | 60–100     | 10–20 km <sup>2</sup> (4–8 sq mi)                   | 2–3 km (1.2–1.9 mi)                                                  |
| Upland parish           | 100–500, in many hamlets and farms | 20–100     | 50–100+ km <sup>2</sup> (20–40+ sq mi)              | Each farm's infield is under 1 km away; summer pastures much farther |
| Agro-town               | 2,000–10,000                       | 400–2,000  | ~50–250 km <sup>2</sup> (20–100 sq mi), derived     | 5–10 km (3–6 mi), an hour or more each way                           |

These numbers are worked out from national totals. England c. 1300 had about 9,500 parishes on 130,000 km<sup>2</sup>, or about 14 km<sup>2</sup> (5.4 sq mi) per parish. A circle of 14 km<sup>2</sup> has a radius of about 2.1 km (1.3 mi). Lowland villages are therefore 1.5–4 km (1–2.5 mi) apart.

**Real example:** Milton Ernest (Bedfordshire) in the 14th century, according to a field survey by Hutchings (1969), had

about 1,600 acres (650 ha, 6.5 km<sup>2</sup>): 1,405 acres of arable, 81 of meadow, 84 under the village itself and 28 of roads. This was a "champion" (open-field) parish with almost no wood or waste.

**How far is too far?** The geographer Michael Chisholm (1962) collected evidence from many farming societies. Costs rise noticeably once fields are more than about 1 km

(0.6 mi) from the farmhouse. At 3–4 km (2–2.5 mi) people usually change the system: they build an outlying farm or field barn, or they turn the far land into pasture. Agrotowns of southern Italy and Sicily are the famous exception, where farmers walked long distances every day.

### Rings of land use around a village

Land use forms rough rings. The land that needs the most work and manure is closest to the houses.

| Ring                   | Distance from the houses         | What is there                                                                                              | Why                                       |
|------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1. House plots         | 0–100 m (0–330 ft)               | Tofts and crofts, vegetable gardens, orchards, hemp and flax, pigsties, the manor's dovecote and fishponds | Daily work; household waste is the manure |
| 2. Arable              | 0.2–2 km (0.1–1.2 mi)            | The open fields, or the infield in the uplands                                                             | Most of the year's labour                 |
| Meadow (not a ring)    | Wherever the flood plain is      | Hay                                                                                                        | Meadow follows water, not distance        |
| 3. Pasture and commons | 1–3 km (0.6–2 mi)                | Common pasture, heath, rough grazing                                                                       | Animals walk there themselves             |
| 4. Wood and waste      | Parish edge, 2–4 km (1.2–2.5 mi) | Coppice woods, wood-pasture, moor, marsh                                                                   | Visited now and then                      |
| 5. Summer pasture      | 5–50+ km (3–30+ mi)              | Shielings, alps, long-distance grazing                                                                     | Seasonal only                             |

In the Mediterranean the rings have different crops. In the hill villages of Lazio, studied by Pierre Toubert (1973), the order from the walls outwards was: vegetable gardens, then vineyards (and olives), then grain fields, then woods and pasture.

#### MAP TIP

For each village, imagine a circle of about 2 km (1.3 mi) radius. Put the open fields inside it, a ribbon of meadow along the stream, and woods, heath or moor in the gaps between the circles of neighbouring villages. Parish boundaries run through this outer land, often along streams, ridges or the edges of woods.

## Fields and farming systems

### Open fields: two-field and three-field

#### RULE OF THUMB

In open-field country there are no hedges between fields. Two or three huge fields surround each village, cut into hundreds of long, thin strips.

**How it worked:** Each family held strips scattered across all the fields, so everyone shared good and bad soil. All the strips in one field followed the same crop plan.

- **Three-field system:** one field of winter grain (wheat or rye), one of spring crops (barley, oats, peas, beans) and one left **fallow** (resting, grazed by the village animals). It spread in the 12th–13th centuries.
- **Two-field system:** half crop, half fallow. It was older, and it stayed normal in the dry Mediterranean, where a biennial (every-other-year) rotation was the rule.

| Element                | What it is                                                                 | Typical size                                                                                                                                                                                                                                                               |
|------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field                  | One of the 2–3 great unhedged fields                                       | Several hundred acres each                                                                                                                                                                                                                                                 |
| Furlong                | A block of parallel strips all running the same way; each had its own name | A parish had dozens. A "furlong" as a length is 220 yards (201 m)                                                                                                                                                                                                          |
| Strip (selion, "land") | The unit ploughed by one holder                                            | About 200 m (220 yd) long; 4.6–20 m (5–22 yd) wide; 0.1–0.4 ha (¼–1 acre)                                                                                                                                                                                                  |
| Ridge and furrow       | Strips ploughed into raised ridges with furrows between, for drainage      | Ridges 3–20 m (3–22 yd) apart; today up to 0.6 m (2 ft) high, once higher. Older ones curve in a gentle reverse-S, made by the long ox team swinging round at the end; later ones are straight                                                                             |
| Headland               | The strip at the end of a furlong where the plough team turned             | Later ploughed as a ridge of its own                                                                                                                                                                                                                                       |
| Family holding         | The land of one peasant family                                             | One family might hold about 70 strips totalling 20 acres (8 ha). The standard holding was the <b>virgate</b> (in the north and east of England counted as two <b>oxgangs</b> ): nominally 30 acres (12 ha), in practice often 24–32 acres, and varying from place to place |

**Who held what:** A survey of 104 English manors in the 13th century found that 45% of tenants had less than 3 acres (1.2 ha), 31% had half a virgate and 22% a full virgate. A family needed at least 10 acres (4 ha) to live from its land alone. Smallholders worked for wages or crafts.

**Where:** In England, open fields covered a broad belt from Yorkshire and Lincolnshire across the Midlands to south-

central England. Kent and Essex kept small, square, hedged fields. Open fields survive today at Laxton (Nottinghamshire, three great fields) and Braunton (Devon). Ridge and furrow is still visible under grass all over the Midlands, best seen in low sun or light snow.

### Meadow, pasture, commons and waste

| Land           | Use                                                        | Where                                                    | Notes                                                                                                                                                                                                                                |
|----------------|------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meadow</b>  | Hay, cut in early summer to feed oxen and horses in winter | Flat land along streams and rivers that floods in winter | Small but very valuable: Milton Ernest had only 81 acres (33 ha) of meadow against 1,405 acres of arable. After haymaking, many meadows were opened for common grazing (in England, "Lammas land", from 1 August)                    |
| <b>Pasture</b> | Grazing                                                    | Hillsides, heath, fallow fields, stubble after harvest   | Ploughs needed oxen: an 8-ox team for a heavy plough. The <i>carucate</i> , a land unit of about 120 acres (50 ha, four virgates), was the land one team could plough in a year. These units were used for tax, so real sizes varied |
| <b>Commons</b> | Shared rights over the lord's waste                        | Heath, moor, marsh, wood                                 | Rights could include pasture, pannage (pigs on acorns), estovers (wood), turbary (cutting turf or peat) and piscary (fishing). Numbers of animals were often limited ("stinted") per holding                                         |
| <b>Waste</b>   | Unploughed land: heath, moor, marsh, scrub                 | Poor soils and parish edges                              | Not useless: fuel, bracken, grazing, rabbits, reeds                                                                                                                                                                                  |

### Infield-outfield in the uplands

#### RULE OF THUMB

In hills and poor soils, one small, well-manured field near the houses was cropped every year. Bigger outfields farther out were cropped for a few years, then rested as grass.

- **Infield:** close to the farm or hamlet, given all the manure, cropped every year (oats and bere, a hardy barley).
- **Outfield:** cropped for a few years until the soil was tired, then left to grass for several years.

- **Where:** Scotland, Ireland, Wales, northern England, Scandinavia, and the sandy north of Germany and the Netherlands, where the infield (*Esch*) was built up with "plaggen": heath turf soaked in animal dung.
- **Runrig:** in Scotland, strips of the infield were often shared out among the families of a *fermtoun* (farm hamlet) and sometimes redistributed.

## The Mediterranean: terraces and irrigation

### RULE OF THUMB

In dry lands, water decides everything. Gardens and orchards cluster along canals and below springs. Grain grows on dry slopes. Vines and olives climb terraced hill-sides.

| Feature                  | What it is                                                                                                          | Real examples and numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terraces</b>          | Steps built into slopes with stone walls, to hold soil and water                                                    | All round the Mediterranean, for vines, olives and grain. Dating of terrace soils shows a major phase of building c. 1100–1600                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Acequia</b>           | An irrigation canal fed by a weir on a river (Spanish, from Arabic)                                                 | Valencia: 8 main canals from the river Turia water about 17,000 ha (170 km <sup>2</sup> , 65 sq mi) of <i>huerta</i> (irrigated garden land). The Water Tribunal that settles disputes between irrigators goes back to al-Andalus and still meets today. (Modern city growth has since covered much of the <i>huerta</i> .)                                                                                                                                                                                                 |
| <b>Noria</b>             | A big wheel turned by the river current that lifts water in pots or scoops                                          | Hama (Syria): wheels up to 20 m (66 ft) across. The Albolafia wheel in Córdoba is a medieval example (partly rebuilt today). A wheel turned by an animal walking in a circle is called a <i>saqiya</i>                                                                                                                                                                                                                                                                                                                      |
| <b>Qanat</b>             | A gently sloping tunnel that brings groundwater from the foot of hills to the surface, with a line of access shafts | Iran: most are under 5 km (3 mi) long, a few up to ~70 km (43 mi); access shafts every 20–35 m (65–115 ft); the "mother well" at the head is usually 20–200 m (65–650 ft) deep. About 50,000 worked in Iran in the mid-20th century. Palermo (Sicily) has a medieval qanat network, usually credited to its Arab rulers (9th–11th centuries). The oases of the Tafilalt and the Draa valley (Morocco) have relied on qanats, there called <i>khettara</i> , since the late 14th century; the Tafilalt had about 400 of them |
| <b>Coltura promiscua</b> | Grain grown between rows of trees and vines (vines trained up the trees)                                            | Central Italy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Chestnut woods</b>    | Chestnuts dried and ground as flour                                                                                 | Mountain villages of Corsica, the Apennines, the Cévennes                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

In Egypt, villages sat on mounds above the Nile flood, with basins of fields that the flood watered each year. In Syria, the Ghouta of Damascus was a belt of irrigated orchards around the city, fed by channels from the Barada river.

## Rice, tanks and lake gardens (Asia and the Americas)

### RULE OF THUMB

Wet rice needs flat, flooded fields. On slopes people build **terraces**; on dry plains they store the monsoon rain in **tanks** (reservoirs held back by an earth dam, called a *bund*); in shallow lakes they build **raised garden beds**. Each system ties a group of villages to one shared water source, so villages, temples and fields line up along the water.

**Why:** A rice field (a *paddy*) must hold standing water, so it needs a level surface with a low bank round it. Water has to pass from field to field down the slope, so farmers form groups that build and run the canals together. Temples or village councils often decide who gets water and when. A paddy can be cropped every year with no fallow, so rice lands feed far more people per square kilometre than European grain fields; for the numbers, see [Beyond Europe](#).

| Landscape                                      | Where and when                                                                                                                                                                                                                | Numbers and facts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | How to draw it                                                                                                                                                                                        |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subak</b> (a cooperative irrigation group)  | Bali (Indonesia). UNESCO dates the system back to the 9th century; networks of water temples have managed whole river basins since the 11th century                                                                           | About 1,200 subak (UNESCO; a 2019 estimate gives about 1,560). Each has 50–400 farmers who share water from one source; a subak covers from a few hectares to almost 800 ha (2,000 acres). It is built of canals, tunnels and weirs. Temples mark the water sources and the points where water is divided; the lake temple Pura Ulun Danu Batur is the chief water temple. The UNESCO site covers about 19,500 ha (195 km <sup>2</sup> , 75 sq mi)                                                                                            | Terraces in contour bands down a volcano's slopes, forest above them, small temples at springs, weirs and canal splits                                                                                |
| <b>Mountain rice terraces</b>                  | Honghe Hani terraces, Yunnan (south-west China), built up over about 1,300 years (UNESCO)                                                                                                                                     | The UNESCO site covers 16,603 ha (166 km <sup>2</sup> , 64 sq mi) with 82 villages. In places there are over 3,000 terraces between the forest edge and the valley floor                                                                                                                                                                                                                                                                                                                                                                      | Four bands down the slope: forest on top (the water source), then villages, then terraces, then the river                                                                                             |
| <b>Village tank cascade</b>                    | Dry zone of Sri Lanka; developed over nearly 2,000 years (FAO)                                                                                                                                                                | A chain of small tanks down one shallow valley; each tank spills into the one below, and the paddy lies below each dam. About 14,200 village tanks are known in Sri Lanka, on average one per 2.6 km <sup>2</sup> (1 sq mi) in the dry-zone provinces. One studied area today (Palugaswewa, 187 km <sup>2</sup> or 72 sq mi) has 78 tanks in 12 cascades and 37 villages; each tank watered up to about 75 ha (190 acres). A village had one main tank; the local saying "tank, stupa, village, temple" names the four parts of every village | A curved earth dam across a shallow valley, the water behind it, a sluice in the dam, a fan of paddy below; the village on dry ground beside it; several tanks in a chain                             |
| <b>Royal tank</b>                              | Parakrama Samudra, at Polonnaruwa (Sri Lanka). King Parakramabahu I (reigned 1153–1186) joined several tanks into one lake; its oldest part, Topa Wewa, dates from about 386 AD                                               | About 22 km <sup>2</sup> (2,200 ha, 8.5 sq mi) of water. The bund is about 14 km (8.5 mi) long and 12 m (40 ft) high. It waters about 7,300 ha (18,000 acres) of paddy                                                                                                                                                                                                                                                                                                                                                                        | A big lake behind a long dam beside the royal city, with canals fanning out below                                                                                                                     |
| <b>Chinampas</b> (raised beds built in a lake) | Lakes Xochimilco and Chalco, Valley of Mexico. The earliest securely dated beds are from 1150–1350; they spread over the lake beds after about 1400, when the Aztec Triple Alliance needed food for its capital, Tenochtitlan | Long, narrow beds built up from layers of plants and mud, with canoe canals between them. Britannica gives 6–10 m (20–35 ft) wide and 100–200 m (330–650 ft) long; other sources give smaller beds. Willows planted at the edges hold the soil                                                                                                                                                                                                                                                                                                | Narrow parallel green strips with thin blue canals between them, in the shallow edge of a lake, with canoe routes to the city                                                                         |
| <b>Angkor</b> (Khmer Empire)                   | Cambodia, 9th–16th centuries                                                                                                                                                                                                  | A low-density settlement spread over about 1,000 km <sup>2</sup> (390 sq mi) of water works (Evans et al. 2007): houses on low mounds with household ponds; local temples (a mound about 20 m square, with a moat and a small pond to the east); rice fields, embankments, canals and roads. Huge rectangular reservoirs ( <i>barays</i> ): the West Baray, begun in the 11th century, is about 7.8 × 2.1 km (4.8 × 1.3 mi). Whether the barays mainly watered fields or were mainly religious symbols is debated                             | A grid of rice fields and embankments dotted with small mounds, each with a pond; temple mounds with moats; straight canals and raised roads; one or two giant rectangular reservoirs near the centre |

**MAP TIP**

Draw rice terraces as tight bands that follow the contour lines, and keep forest on the hilltops above them as the water source. Draw a tank as a curved dam across a shallow valley, with the water behind it, a sluice in the dam and a fan of paddy below; put the village on dry ground at one side and its temple close by. Chain several tanks down one valley. Put water temples at springs, weirs and the points where canals split, and put houses on low mounds above the flooded fields.

**Orchards, vineyards and gardens**

- **Gardens and orchards** sit next to the houses (ring 1). Orchards give apples, pears, cherries and plums in the north, and figs, almonds and citrus in the south. Normandy and the west of England are cider country.

- **Vineyards** need sun and warm summers. The great medieval wine regions were around Paris, Champagne, Burgundy, the Rhine and Moselle, and Gascony (Bordeaux). Gascon wine was shipped to England in large amounts after 1152, when Henry of Anjou (King Henry II from 1154) married Eleanor of Aquitaine.
- **The northern limit was farther north than today's wine map.** Domesday Book (1086) lists over 40 vineyards in England, and 139 were recorded in 1509, when Henry VIII became king. In the 14th century, vineyards worked in Poland, for example in Silesia and at Kraków, Toruń and Zielona Góra. Most of these made thin wine for monks and lords. Put northern vineyards on south-facing slopes, near monasteries or castles. Commercial wine regions are in [Industry, Resources and Special Towns](#).

- **Olives** mark the true Mediterranean climate. They do not grow where winters are hard. If your region has olives, it has dry summers, terraces and probably biennial fallow.

**MAP TIP**

Draw open fields as large unhedged areas with faint hatching in different directions (each block of hatching is a furlong). Draw hedged, enclosed fields as a patchwork of small irregular shapes. Draw meadow as a pale green ribbon along streams, and heath or moor with tufts. In dry lands, put green gardens along canals and below qanat outlets, and leave the slopes above them brown. Draw a qanat as a straight line of tiny rings across dry country: each ring is a shaft mouth with its heap of spoil (dug-out earth) around it, and from the air a qanat looks like a line of holes or anthills. Put the village and its gardens where the line ends. Keep the rings much smaller than the small open circle used for a market town.

**LATER ERA (1500S+)**

In England, landlords enclosed many open fields for sheep pasture in the 15th and 16th centuries. How much was enclosed, and when, is debated. J. R. Wordie (1983) estimated that about 45% of England was already enclosed by 1500, 47% by 1600 and 71% by 1700. Other historians think more than half the farmland was still open in 1700. Either way, most remaining open fields went in the parliamentary enclosures of the 18th–19th centuries. "Floated" water meadows (channels that flooded meadows on purpose to get an early crop of grass) appear in the early 1600s; Rowland Vaughan described his in Herefordshire in 1610. Maize from the Americas spreads in Iberia and northern Italy in the 16th century. See [What Changes After 1500](#).

**FANTASY TWIST**

Magic that doubles harvests halves the land a village needs. Its fields shrink to about a 1.5 km (1 mi) circle and more land is left for woods and pasture. It also lets villages sit closer together. Keep the ring pattern anyway: gardens near, fields out, woods at the edge.

## Woods, forests and hunting grounds

**RULE OF THUMB**

A medieval wood is not wild. Almost every wood belongs to someone and is cut on a regular cycle. A medieval "forest" is a **legal** area for royal hunting, which may contain villages, fields and heath as well as trees.

### Managed woodland

| Practice                      | What it is                                                                                           | Numbers                                                                                                                                                                                                                                                                                          | On the map                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Coppice</b>                | Trees (hazel, ash, oak, hornbeam, maple, lime) cut to the ground; many new poles grow from the stump | Cycles depended on the tree and the product: birch for faggots (bundles of sticks) every 3–4 years, oak poles up to about 50 years. Medieval cycles were mostly short, often under 10 years (Rackham); they grew longer in later centuries. Products: firewood, poles, wattle, hurdles, charcoal | Compact woods with sharp edges, bounded by a bank and ditch to keep animals out after cutting |
| <b>Coppice with standards</b> | Coppice plus scattered tall trees left to grow into timber                                           | A 1544 English statute required 12 standards per acre (about 30 per hectare) to be left, and the wood to be fenced after cutting                                                                                                                                                                 | Same as coppice                                                                               |
| <b>Pollards</b>               | Trees cut well above the ground, usually about 2–4 m (6–13 ft) up, out of reach of grazing animals   | Cut every 2–6 years for leaf fodder, or 8–15 years for poles                                                                                                                                                                                                                                     | Scattered big trees in open grassland                                                         |
| <b>Wood-pasture</b>           | Grassland with pollards and old trees, grazed by cattle, sheep or deer                               | Most commons, parks and forests                                                                                                                                                                                                                                                                  | Trees scattered over pasture                                                                  |
| <b>Pannage</b>                | Pigs turned into woods in autumn to eat acorns and beechmast                                         | Domesday Book (1086) measured many woods by how many pigs they fed, e.g. "wood for 40 pigs". The New Forest still has a pannage season of at least 60 days                                                                                                                                       | Pig herders in oak and beech woods                                                            |

**Real example: Hatfield Forest (Essex).** A royal forest made between 1086 and 1225, 403 ha (996 acres) today. The ecologist Oliver Rackham called it "almost certainly unique": all the elements of a medieval forest survive there, with deer, cattle, coppice woods, pollards, scrub, timber trees, grass-land and fen.

### Royal forests

- **What:** an area under special **forest law**, which protected deer and wild boar ("venison") and the vegetation that fed them ("vert"). Clearing land, cutting trees or hunting without permission were offences.
- **How big:** At the peak (late 12th to early 13th century), royal forest covered about **one third of southern England**. At one stage in the 12th century all of Essex was

forest. In 1086 Sherwood covered perhaps a quarter of Nottinghamshire.

- **Dates:** The Normans brought forest law (the New Forest was made by William I). The Assize of the Forest came in 1184. The **Charter of the Forest (1217)** cut the forests back and protected commoners' rights. The forest bounds were cut back again around 1300.
- **Officials:** wardens, foresters, verderers (forest judges), regards (inspectors who looked for illegal clearing and building) and agisters (who ran the grazing and pannage).
- **Elsewhere:** German kings and princes claimed similar hunting rights (*Wildbann*). French kings kept great forests such as Fontainebleau and Compiègne.

### Parks, chases, warrens and lodges

| Feature              | What it is                                                                                             | Numbers and examples                                                                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deer park</b>     | A private enclosed hunting ground and venison farm, needing a royal licence to "empark"                | Domesday Book (1086): 36 parks. Rackham's estimate for c. 1300: about <b>3,200 parks</b> , covering nearly 2% of England, one per ~40 km <sup>2</sup> (15 sq mi) on average. Usually 40–80 ha (100–200 acres); some much larger. Hampton Court (Herefordshire) got a 15th-century licence for 1,000 acres. Mostly stocked with fallow deer                      |
| <b>Park pale</b>     | The boundary: a bank with a wooden fence on top and a ditch on the <b>inside</b>                       | Outlines usually have rounded corners. "Deer leaps" let deer in but not out                                                                                                                                                                                                                                                                                     |
| <b>Chase</b>         | An unfenced hunting ground held by a great lord, not the king (though a chase could pass to the Crown) | Cranborne Chase (Dorset, Wiltshire and Hampshire), held by great lords such as the earls of Gloucester and later the earls of March                                                                                                                                                                                                                             |
| <b>Rabbit warren</b> | Rabbits farmed for meat and fur                                                                        | The Normans brought rabbits in the late 11th or 12th century. In the 13th century one rabbit was worth more than a workman's daily wage. Warrens sit on dry, sandy, poor land such as the Breckland (Norfolk and Suffolk). Artificial burrows are long low mounds called <b>pillow mounds</b> . Thetford Warren Lodge (c. 1400) is a fortified warrener's house |
| <b>Hunting lodge</b> | A house for the lord's hunting visits, inside a forest or park                                         | Clarendon (Wiltshire): a hunting lodge stood in its park by 1130; Henry II and Henry III turned it into a royal palace                                                                                                                                                                                                                                          |

#### MAP TIP

Do not paint a royal forest as solid trees. Draw its boundary as a green dash-dot line with a large italic name ("Forest of ..."), and inside it show woods, heath, villages and fields. Do not use a dotted line: dotted lines mean tracks and other routes. Give a county 1–3 royal forests in the hunting country nearest the king's palaces. Draw deer parks as rounded pale (fence) lines 0.5–1 km (0.3–0.6 mi) across, with scattered trees inside, next to castles and great manor houses, and put a lodge inside. Mark a royal hunting lodge with a small crown. Put rabbit warrens on sandy heaths.

#### LATER ERA (1500S+)

Charcoal for blast furnaces and glassworks raised the value of coppice, and more woods were enclosed and managed. Many deer parks became landscaped parks around country houses. Forest law faded away by the mid-17th century.

## How much of the map is fields and woods

### RULE OF THUMB

In a settled lowland kingdom c. 1300, colour about **a third** of the land as ploughland and about **a tenth** as woods. Most of the rest is pasture, heath, moor and marsh. Hill country has far less ploughland. After a great plague, cut the ploughland by a quarter to a third and turn it into pasture and scrub.

**Why:** Ploughland follows people. Each person needed about 1 ha (2.5 acres) of arable land (see [Population and Settlement Sizes](#)), so the fields were largest when the popula-

tion peaked, c. 1300. Woods were cleared for new fields until the Black Death stopped it. One open-field parish can be almost all arable, like Milton Ernest ([above](#)), but a whole kingdom also has hills, heaths, marshes and wooded parish edges.

### England, the best-measured case

England covers about 13 million ha (32 million acres, 130,000 km<sup>2</sup>).

| Date | Arable land, fallow included        | Share of England | Sown that year                     | What was happening                                       |
|------|-------------------------------------|------------------|------------------------------------|----------------------------------------------------------|
| 1300 | 11.5 million acres (4.7 million ha) | ~36%             | 7.4 million acres (3.0 million ha) | Population near its medieval peak (about 4.5–5 million)  |
| 1380 | 8.7 million acres (3.5 million ha)  | ~27%             | 5.2 million acres (2.1 million ha) | After the Black Death (about 2.5 million people in 1377) |
| 1450 | 7.6 million acres (3.1 million ha)  | ~24%             | 4.6 million acres (1.8 million ha) | Population at its low point (about 1.9 million)          |
| 1600 | 8.9 million acres (3.6 million ha)  | ~28%             | 6.7 million acres (2.7 million ha) | Population growing again; less land left fallow          |

These figures come from Broadberry, Campbell and others (2010, Table 1); the shares are worked out from England's area. "Arable" here counts the fallow field too. Only the sown

part grew a crop in any one year, but on a map fallow looks like part of the open fields, so colour it as fields.

| Date       | Woodland and wood-pasture                              | Notes                                                                                                                                                |
|------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1086       | ~15% of England                                        | Rackham's estimate from Domesday Book                                                                                                                |
| c. 1349    | perhaps ~7%                                            | About half the 1086 figure, after two and a half centuries of clearing (Royal Forestry Society summary; a rough estimate)                            |
| After 1349 | Stopped falling; some land went back to scrub and wood | Pollen records from north-west Europe show birch and hazel spreading over abandoned grazing land by the late 14th century (Yeloff and van Geel 2007) |

Woods were spread very unevenly. Heavy clays, steep slopes and parish edges kept more wood; "champion" (open-field) parishes such as Milton Ernest had almost none.

**A "forest" is not tree cover.** At its peak, royal forest covered about a third of southern England, but this was a legal zone that included villages, fields and heath as well as woods (see [Royal forests](#)). Do not add it to the woodland share.

## A land budget for a whole realm

| Land                                                    | Share of a settled lowland kingdom c. 1300 | Basis                                                                                 |
|---------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------|
| Ploughland (open fields, fallow included)               | 35–40%                                     | England: ~36% in 1300                                                                 |
| Woodland and wood-pasture                               | 7–15%                                      | England: ~15% in 1086, perhaps ~7% by 1349                                            |
| Meadow                                                  | about 5%                                   | Milton Ernest: 81 of 1,600 acres. More along big rivers, less in dry or hilly country |
| Deer parks                                              | nearly 2%                                  | Rackham's ~3,200 parks (see <a href="#">above</a> ); parks overlap with wood-pasture  |
| Villages, gardens and roads                             | a few per cent                             | 7% in Milton Ernest (84 acres of village and 28 of roads); less where people are few  |
| Pasture, heath, moor, fen, marsh and other common waste | the rest, roughly a third to a half        | Worked out as what is left                                                            |

**Hill country is different.** An upland parish of 50–100 km<sup>2</sup> (20–40 sq mi) with 100–500 people needs only about 100–500 ha of arable, so only about 1–10% of its land is ploughed (worked out from this chapter's own numbers). The rest is rough grazing, moor and wood.

**After a plague:** England's arable shrank by about a quarter between 1300 and 1380, and by about a third by 1450. The lost fields became sheep and cattle pasture, and some went back to scrub and wood. Whole villages were deserted (see [chapter 01](#)).

### MAP TIP

On a kingdom map of settled lowland, colour about a third of the land as fields, in broad patches around each cluster of villages. Spread the woods in many separate patches (on parish edges, heavy clay, steep slopes and hunting grounds) adding up to about a tenth, not one solid block. Give the rest to pasture, heath, moor and marsh, mostly on poor soil, hills and wet ground. In hill country, shrink the fields to small patches in the valleys. Show royal forests as a named green dash-dot boundary, not as solid trees.

### LATER ERA (1500S+)

By 1600 English arable had climbed back to about 28% of the land, and much less of it lay fallow: 2.2 of 8.9 million acres in 1600, against 4.1 of 11.5 million in 1300. More of the sown land grew peas, beans and other crops. Enclosure also turned some fields into hedged pasture (see [above](#)).

## Mills and water management

### RULE OF THUMB

By 1300 there was roughly **one mill per village**. Water-mills sit on streams with a reliable flow; windmills stand on open, exposed high ground, especially in flat country with weak streams.

**Why:** Grinding grain by hand with a quern took hours every day. A mill saved that labour, and it made the lord a

steady income. Tenants had to grind at the lord's mill (in England, "suit of mill") and pay a toll in grain called **mul-ture**, usually around a twentieth of the grain (often quoted as 1/16 to 1/24). Lords fought hand-mills. At St Albans, in a long quarrel with the townspeople in the late 13th and 14th centuries, the abbot had their querns seized and used the stones to pave the floor of his parlour.

## How many mills

| Date                 | Mills in England                                                                | Notes                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1086 (Domesday Book) | <b>5,624</b> watermills (Hodgen 1939); later counts give ~ <b>6,000</b> or more | Mills in about 4,000 places; about one mill for every 40–50 recorded households. Some "mills" were extra pairs of millstones in one building      |
| c. 1300              | ~ <b>10,000–15,000</b> mills, most of them watermills                           | Estimates vary (see Holt 1988 and Langdon 2004). Either way, England had roughly as many mills as parishes. Windmills were now common in the east |
| After 1350           | Fewer                                                                           | After the Black Death there was less grain to grind, and many mills closed                                                                        |

## Kinds of mill

| Type                             | Where                                                                                                    | Dates and facts                                                                                                                                                                                                                                         | How to draw it                                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Watermill</b>                 | On streams and small rivers with a steady flow; a weir and leat create the fall of water the wheel needs | Vertical wheels in most of Europe: <i>undershot</i> (the water pushes the bottom of the wheel) or <i>overshot</i> (the water pours onto the top); small horizontal-wheeled "Norse" mills in Ireland, Scotland, Scandinavia and the mountains            | A wheel symbol on the stream, with its leat                                                                                                                      |
| <b>Windmill</b>                  | Flat, open or dry country: eastern England, Flanders, northern France, the Low Countries                 | Earliest certain reference in northern Europe: <b>1185</b> , at Weedley in Yorkshire. The first type was the <b>post mill</b> (the whole body turns on a post). Masonry <b>tower mills</b> appeared by c. 1300. Dutch drainage mills appeared from 1408 | An X of four sails on a dot or small mound, on high ground away from trees                                                                                       |
| <b>Tide mill</b>                 | Estuaries and sheltered coasts with a strong tide                                                        | Nendrum (Northern Ireland) c. 619, rebuilt c. 787; Ebbsfleet (Kent) 691–692 (dated by tree rings); Woodbridge (Suffolk) first recorded 1170. A pond fills at high tide through a one-way gate and drives the wheel as the tide falls                    | A mill wheel on a dam across a tidal creek                                                                                                                       |
| <b>Horse or donkey mill</b>      | Castles, besieged towns, dry places with no stream                                                       | Slow but independent of water                                                                                                                                                                                                                           | No special symbol; inside a castle or town                                                                                                                       |
| <b>Ship mill and bridge mill</b> | Big rivers with no space for weirs                                                                       | Floating mills on the Tiber, Danube and other large rivers; Paris had a bridge lined with mills                                                                                                                                                         | Small mill wheels in midstream, labelled "ship mills"; mill wheels on the bridge symbol. Do not draw boats: a small boat on a river marks the head of navigation |

## Leats, ponds and weirs

- **Weir:** a low dam across the river that raises the water and turns part of it aside.
- **Leat (mill race, head race):** an artificial channel that takes water from the weir to the mill. It can run from a few hundred metres to over a kilometre (0.6 mi). It runs along the valley side, falling more gently than the river, so the water arrives at the mill higher than the river.
- **Millpond:** stores water so the mill can run when the stream is low.
- **Tail race:** returns water to the river below the mill.
- **Conflicts:** mill weirs blocked boats and fish. Rivers with many mills were hard to navigate (see [Trade Routes, Roads and Transport](#)). Mills also drove fulling, forges and other industry (see [Industry, Resources and Special Towns](#)).

### MAP TIP

Give almost every village a mill. Draw the river, a weir, the leat running parallel to the river as a thin blue line, the mill as a wheel on the water, and the tail race rejoining the river. On a good stream, mills can follow each other every 1–2 km (0.6–1.2 mi). In flat, dry country, put windmills (an X of four sails on a dot) on the highest ground near the village, often on an old mound. Label them "Mill", "Mill Hill" or "Windmill Hill". Put tide mills on creeks near a coastal town.

### LATER ERA (1500S+)

Wind-pumped drainage made the great Dutch land reclamations possible (see below). Wind-driven sawmills appeared in Holland from the 1590s. Mills multiplied again for paper, gunpowder and metalworking.

## Other rural features

### RULE OF THUMB

Status features (moats, dovecotes, fishponds, parks, warrens) cluster around manor houses and monasteries.

Working features (barns, folds, kilns, crosses, paths) follow the fields, the flocks and the roads.

### Quick list

| Feature                           | What it is                                                                                                             | Where to put it                                                                                                         | Numbers and examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fishponds</b>                  | Ponds for breeding and storing freshwater fish for fast days                                                           | Near manors, monasteries and castles, in a valley floor fed by a stream; often a chain of 2–12 ponds joined by channels | About 2,000 are recorded in England, thought to be only a small part of the medieval total. Building peaked in the 12th century. Groups of up to 12 ponds in a line or a cluster; big storage ponds and small, shallow breeding ponds. Example: a grange at North Kelsey (Lincolnshire) had a pond 65 × 50 m (210 × 165 ft) and another about 75 m (250 ft) long                                                                                                                                                                                                                 |
| <b>Dovecotes</b>                  | Towers for breeding pigeons for meat and manure                                                                        | Next to manor houses, parsonages and monasteries                                                                        | Only manorial lords could keep them in medieval England. In France this was the <i>droit de colombier</i> . Medieval ones are usually round and stone-built. Kinwarton (Warwickshire, 14th century) has over 580 nesting holes                                                                                                                                                                                                                                                                                                                                                   |
| <b>Moated sites</b>               | A house platform (the "island") surrounded by a wet ditch                                                              | Clay lowlands; manor houses, granges, rectories                                                                         | About 6,000 known in England, most dug c. 1250–1350, with by far the most in central and eastern England; a status symbol as much as a defence. Small: the island is usually about 25–50 m (80–165 ft) across, inside a moat about 7–15 m (23–50 ft) wide, so the whole site is only about 40–80 m (130–260 ft) across. Lincolnshire examples: Heapham, a square island 24 m across in a moat 8–12 m wide; Withcall, islands of 40 × 30 m and 45 × 20 m. A few high-status sites are much bigger: South Kelsey Hall's main island is 110 × 100 m. See <a href="#">Footprints</a> |
| <b>Monastic grange</b>            | An outlying farm run by a monastery, often worked by lay brothers (monks who did the manual work) or by paid labourers | Next to the abbey or wherever it held land, up to a day's journey or more away                                          | First appeared in the 12th century, led by the Cistercians, and used until the English monasteries were dissolved in 1536–1541. Byland Abbey (North Yorkshire) had an industrial grange at Denby Dale, some 50–60 miles (80–95 km) away                                                                                                                                                                                                                                                                                                                                          |
| <b>Tithe barn</b>                 | A great barn for crops paid as tithe (one tenth) or rent                                                               | Next to a church, rectory or grange                                                                                     | About 200 medieval barns (built before 1550) survive in Britain; not all were tithe barns. Great Coxwell (Oxfordshire), built c. 1292 (roof timbers felled 1291–92) for Beaulieu Abbey's grange: 46 m long, 13 m wide, 15 m high (152 × 43 × 48 ft)                                                                                                                                                                                                                                                                                                                              |
| <b>Sheepfolds and sheep-cotes</b> | Pens and long sheep-houses for winter                                                                                  | On downs, wolds and moors; on monastic sheep granges                                                                    | In Norfolk, lords had the right to fold their sheep on tenants' fields at night to manure them                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Wells</b>                      | Village water                                                                                                          | Village centre or each toft                                                                                             | Clay villages used ponds and shallow wells; chalk villages needed deep ones                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Wayside crosses</b>            | Stone crosses at junctions, boundaries and moor paths                                                                  | Crossroads, parish edges, routes over moors (Dartmoor)                                                                  | Mostly 13th–14th century in Britain. Many hundreds survive, often only the base or a broken shaft; Dartmoor alone has about 130 known crosses. In Germany, small <i>Sühnekreuze</i> (penance crosses) mark a killing                                                                                                                                                                                                                                                                                                                                                             |
| <b>Limekilns</b>                  | Kilns that burn limestone into lime                                                                                    | Near limestone and fuel, by a creek or harbour, or at a big building site                                               | Mostly for mortar in the Middle Ages. Spreading lime on fields became practical first in coal-mining areas, where coal made lime cheap from the late 13th century; Fitzherbert describes it in his <i>Boke of Husbandry</i> (1523). See <a href="#">Industry, Resources and Special Towns</a>                                                                                                                                                                                                                                                                                    |
| <b>Pound</b>                      | Pen for stray animals                                                                                                  | Edge of village                                                                                                         | See <a href="#">above</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Summer pastures and transhumance

**Transhumance** means moving herds between winter and summer pastures every year. It shapes mountain and dry landscapes.

| Type                                              | Region                                                                                                                                                               | Distance                                                                                                                                        | Season                                                                                             | On the map                                                                                                                                                                                                                  |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shielings</b><br>(summer huts on hill pasture) | Scotland ( <i>àirigh</i> ), northern England, Wales ( <i>hafod</i> = summer house, <i>hendre</i> = winter house), Ireland ( <i>booley</i> ), Norway ( <i>seter</i> ) | Usually within a day's walk, uphill                                                                                                             | Scotland: families went up "about mid-summer" for "a month or six weeks" to make butter and cheese | A cluster of tiny squares (the huts) by a stream on high pasture. Scottish huts were mostly rectangular, 5.7–14 m (19–46 ft) long, of dry stone or turf. A 1995–2001 survey in Scotland recorded about 3,100 shieling huts  |
| <b>Alpine pastures</b>                            | Alps, Pyrenees, Carpathians                                                                                                                                          | From the valley village up to the high meadows                                                                                                  | Summer                                                                                             | Tiny squares (huts and cheese-making dairies) above the tree line                                                                                                                                                           |
| <b>Long-distance transhumance</b>                 | Castile (the <b>Mesta</b> , royal charter 1273)                                                                                                                      | 550–750 km (340–470 mi) from León and Old Castile to Extremadura and Andalusia; rarely more than 250 km (155 mi) from New Castile and La Mancha | South in autumn (arriving in October), north in April–May                                          | More than 2.5 million sheep moved each year in 1436–1549. The royal drove roads ( <i>cañadas reales</i> ) were given a fixed width only where they crossed farmland: 90 <i>varas</i> (Castilian yards), about 75 m (250 ft) |
| <b>Tratturi</b>                                   | Abruzzo to Apulia (southern Italy)                                                                                                                                   | The main route, L'Aquila to Foggia, is about 244 km (152 mi)                                                                                    | Winter in the Apulian lowlands, summer in the mountains                                            | Wide grassy drove roads ( <i>tratturi</i> ), drawn as broad dotted lines (two rows of dots); King Alfonso V of Aragon organised them as a royal toll system (the <i>Dogana</i> ) in 1447                                    |

Place names often record shielings: English and Scots "shiel" or "shield" (Galashiels, Pollokshields), Old Norse *sætr* ("-side", "-seat") and *ærgi* ("-ergh", "-argh"), and Gaelic *àirigh* ("Airie", "Arie").

### Boundaries, paths and hollow ways

- **Parish and manor boundaries** followed streams, ridges, old roads and banks, and were marked with stones, crosses, ancient trees and banks. Each spring (Rogation week) parishioners "**beat the bounds**": they walked the whole boundary and struck the markers with green boughs. Boys came along and were sometimes bumped on the boundary stones, so that they would remember the line all their lives.
- **Hedges and banks**: where fields were enclosed early, the hedges themselves are boundaries. Fields cleared from woodland have hedges full of woodland plants.
- **Hollow ways** (sunken lanes) are tracks worn deep by centuries of feet, hooves and rainwater, especially on chalk and soft sandstone. One estimate puts England's total at over 1,600 km (1,000 mi); in Germany some reach 5 m (16 ft) deep.

- **Footpaths and church paths**: every hamlet had a path to its parish church and mill. In big northern parishes, "corpse roads" carried the dead from distant hamlets to the parish churchyard.

#### MAP TIP

On a local map, add a manor house with a moat (a small open square with a double outline), a dovecote and a chain of fishponds (small rectangles along the stream) below it. Put a grange (a small open square topped with a cross, or the label "Grange") with a great barn in a monastery's outlying land. Mark the parish boundary with a thin dash-dot-dot line that follows streams and ridges. Where paths cross it, put a wayside cross (a tiny cross on a base) or a boundary stone (just the letters "BS" written on the line, as on Ordnance Survey maps; a tick would read as a standing stone, and a dot as a farmstead). Draw a few thin dotted tracks from the village up to summer huts on the hills, and a drove road (a broad dotted line, two rows of dots) in sheep country.

## Reclaimed land: fens, polders and assarts

#### RULE OF THUMB

In the High Middle Ages population pressure pushed farming into marsh, coast and forest. Reclaimed land has **straight lines**: dykes, drains and long narrow fields. Old land has **curved lines**.

| Type                                       | What it is                                                      | Real examples                                                                                                                                                                                                                                                                                                                | On the map                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Silt fen and salt-marsh reclamation</b> | Banks built against the sea or floods; marsh drained by ditches | The Fens (about 3,900 km <sup>2</sup> , 1,500 sq mi): medieval farmers embanked the silt land around the Wash. Romney Marsh (about 260 km <sup>2</sup> , 100 sq mi): land "inned" (enclosed by walls) in stages; much of Walland and Denge Marsh by the 14th century                                                         | Long sea banks (bold lines with short ticks on the water side), straight drains, villages in a line along the old banks, sheep on the marsh                                                              |
| <b>Peat fen</b>                            | Wet inland marsh                                                | The southern Fens stayed mostly undrained: fish, eels, wildfowl, reeds, turf, summer grazing. Monasteries (Ely, Thorney, Crowland, Ramsey, Peterborough) stood on dry "islands"; the Isle of Ely is about 60 km <sup>2</sup> (23 sq mi)                                                                                      | Islands of dry land with a church or abbey, surrounded by marsh and meres                                                                                                                                |
| <b>Polders and dykes</b>                   | Land below water level, enclosed by dykes and drained           | Netherlands: first polders in the 11th century; a cooperative to build and keep up a dam on the Rhine near Wijk bij Duurstede (south-east of Utrecht) c. 1122; the Rijnland water board in 1255; the first drainage windmill near Alkmaar in 1408. Drained peat shrinks and sinks, so dykes had to be raised again and again | A grid of drains inside a dyke. Draw the dyke as a bold line with short ticks on the water side (a plain thick line means a highway). Villages strung along the dykes; drainage windmills only from 1408 |
| <b>Assarts</b>                             | Fields cleared from woodland                                    | Peak in the 13th century, by single farmers, village groups, lords and monasteries (especially the Cistercians). In royal forests a licence or a fine was needed. Many went back to woodland after the Black Death                                                                                                           | Irregular hedged fields biting into woodland, with isolated farms                                                                                                                                        |

**Place-name clues:** English assart names include Stocking, Stubbings, Ridding, Royd and Breach. Older English clearing names end in *-ley* or *-leigh* (Old English *lēah*, "wood, clearing"), and Norse ones in *-thwaite*. French *essart* and *-sart* and German *-rode*, *-rath* and *-reuth* (as in Bayreuth) mean "cleared land". For more name endings and how to use them, see [Naming settlements realistically](#). German eastward settlement (*Ostsiedlung*, 12th–14th centuries) cleared forest in the same way, with Waldhufendörfer and standard farm holdings (the Frankish *Hufe*, about 24 ha or 60 acres).

#### MAP TIP

If your kingdom is in its growth phase, show new strip villages pushing up forest valleys and new dykes on the coast. If it is after a plague, show some assarts going back to scrub and a few deserted villages (see [chapter 01](#)).

#### LATER ERA (1500s+)

Big drainage projects begin. The Dutch drained whole lakes with windmills: the Beemster (drained by 1612) and the Schermer (by 1635). See [What Changes After 1500](#). In England, drainage of the Fens began in earnest in the 1630s, and the Dutch engineer Cornelius Vermuyden worked on its second phase in the 1650s. Straight new rivers and huge rectangular fields replaced the old marsh.

## Local administration: parish, manor, hundred

#### RULE OF THUMB

The **parish** is the unit of church and village life. The **manor** is the unit of lordship and rent. The **hundred** groups roughly 10–15 parishes for justice and tax. They often do not match.

| Unit                        | Area                                                                                                                                                                                                                         | What it does                                                                                                                                                                                                                                                                                                                    | Seat or meeting place                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Township or vill</b>     | Part of a parish in the north and uplands; equal to the parish in the lowlands                                                                                                                                               | Farming community, tax unit                                                                                                                                                                                                                                                                                                     | The village                                                            |
| <b>Parish</b>               | ~14 km <sup>2</sup> (5.4 sq mi) lowland; 50–100+ km <sup>2</sup> upland                                                                                                                                                      | Church, tithes, burial; ~9,500 in England c. 1300                                                                                                                                                                                                                                                                               | Parish church                                                          |
| <b>Manor</b>                | 2–10 km <sup>2</sup> (0.8–4 sq mi): one village, part of one, or several                                                                                                                                                     | Lord's land, rents, labour services, manor court                                                                                                                                                                                                                                                                                | Manor house                                                            |
| <b>Hundred / wapen-take</b> | Very variable: small in the south-east (Sussex, Kent), large in the Midlands and north. A rough average is ~150–200 km <sup>2</sup> (60–80 sq mi) and 10–15 parishes (about 630 named hundreds were listed in 1316; derived) | Court for local justice and order. It met 12 times a year by the 12th century; later it met every two weeks, until a 1234 ordinance set it at every three weeks. "Wapentake" (Old Norse for "weapon-taking") is the name in the Danelaw, the part of England once under Danish law (Yorkshire, Lincolnshire, the East Midlands) | Often an open-air site: a mound, a tree, a stone, a ford, a crossroads |
| <b>Lathe / rape</b>         | Groups of hundreds in Kent (7 lathes) and Sussex (4 rapes) at the Norman Conquest                                                                                                                                            | Between hundred and shire                                                                                                                                                                                                                                                                                                       | A royal centre or castle                                               |
| <b>Shire / county</b>       | See <i>Capitals, Realms and Borders</i>                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                 | County town                                                            |

**Equivalents elsewhere:** the Frankish *centena*, the Scandinavian *herred* or *häräd*, the French *châtellenie* (castle district), and the Italian *pieve* (a large rural district served by one baptismal church with several chapels).

**Moot sites:** A *moot* is an assembly. Assemblies met in the open at places everyone could find. Secklow Mound (in modern Milton Keynes) was the meeting place of Secklow hundred. Tynwald Hill (Isle of Man) and Thingwall (Wirral) come from the Norse *thing* (assembly). Place-name clues are -low or -law (mound), "Moot Hill", "Gallows Hill", "Court Hill", and names with Thing- or -ting.

#### MAP TIP

On a county map, divide the land into hundreds about 12–15 km (7–9 mi) across, smaller in rich, crowded country and larger in the hills. Draw their boundaries as thin dash-dot-dot lines. Name each hundred after its meeting place, and mark that place with a small lone tree labelled "Moot", even when the real meeting place was a mound, a stone or a ford (a small mound symbol would read as a barrow). The site is often on a road or at a ford near the middle of the hundred and away from any village. Gallows (a small "II" on a hill) often stand nearby.

## Drawing a local-area map

#### RULE OF THUMB

At local scale, almost every square kilometre has a human feature: a field, a path, a hedge, a mill, a wood with a name. Empty white space means marsh, moor or forest, and should be labelled as such.

### What to show at each scale

| Map size                 | Example               | What to show                                                                                                                                  |
|--------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ~5 × 5 km (3 × 3 mi)     | One or two parishes   | Individual houses, tofts and crofts, furlongs, the mill and its leat, ponds, pound, paths, named woods and fields                             |
| ~20 × 20 km (12 × 12 mi) | A hundred or a barony | Every village and hamlet, churches, mills, manor houses, moated sites, parks, granges, woods, commons, roads and footpaths, parish boundaries |
| ~50 × 50 km (30 × 30 mi) | A small county        | Villages as dots, market towns, castles, abbeys, forests, big parks, rivers and main roads. Leave out mills and hamlets                       |

### How many features: England c. 1300 averages

These are derived from national totals and the area of England (130,000 km<sup>2</sup>). Use them as a sanity check. Real fea-

tures cluster: parks and moats on lowland clays, warrens on sandy heath, windmills in the flat east.

| Feature                              | One per ...                               | Per 100 km <sup>2</sup> (10 × 10 km; 39 sq mi)      | Per 20 × 20 km map (400 km <sup>2</sup> ) |
|--------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------|
| Parish church and village (low-land) | ~14 km <sup>2</sup>                       | ~7                                                  | ~25–30                                    |
| Hamlets and farmsteads               | Varies by region                          | 0–5 (open-field lowland) to 20+ (dispersed uplands) | 0–100+                                    |
| Mills (c. 1300)                      | ~9–13 km <sup>2</sup>                     | ~8–11                                               | ~30–45                                    |
| Deer parks                           | ~40 km <sup>2</sup>                       | ~2–3                                                | ~10                                       |
| Moated sites                         | ~22 km <sup>2</sup>                       | ~4–5                                                | ~15–20                                    |
| Market places (c. 1300, many tiny)   | ~75 km <sup>2</sup>                       | ~1–2                                                | ~5                                        |
| Hundred meeting places               | ~150–200 km <sup>2</sup>                  | ~0.5–0.7                                            | ~2–3                                      |
| Fishponds                            | Recorded: ~65 km <sup>2</sup> (a minimum) | at least 1–2                                        | at least 6                                |

For castles, abbeys and roads, see [Castles, Forts and Military Outposts](#), [Religious, Cultural, Legal and Ancient Sites](#) and [Trade Routes, Roads and Transport](#).

### Suggested symbols

These symbols match the guide's [master legend](#).

| Feature                 | Symbol                                                                                                                                                |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Farmstead               | Tiny dot                                                                                                                                              |
| Hamlet                  | 2–5 small house marks, no church                                                                                                                      |
| Village                 | House marks around a church; on a kingdom map, a tiny dot (notable villages only)                                                                     |
| Parish church           | Small church with a tower (a small plain cross with no tower means a chapel)                                                                          |
| Manor house             | Small open square; double outline if moated (a real moated site is only about 40–80 m across, so on a 1 cm = 1 km map this is a symbol, not to scale) |
| Grange                  | Small open square topped with a cross, or the label "Grange"                                                                                          |
| Watermill               | Wheel on the stream, with the leat as a thin blue line                                                                                                |
| Windmill                | X of four sails on a dot or small mound                                                                                                               |
| Tide mill               | Mill wheel on a dam across a tidal creek                                                                                                              |
| Open fields             | Unhedged areas with blocks of hatching                                                                                                                |
| Enclosed fields         | Small irregular polygons                                                                                                                              |
| Meadow                  | Pale green band along streams                                                                                                                         |
| Vineyards               | Hatched strips on south-facing slopes                                                                                                                 |
| Wood                    | Tree symbols with a firm edge (managed wood)                                                                                                          |
| Wood-pasture            | Scattered single trees on grass                                                                                                                       |
| Deer park               | Rounded pale (fence) line with scattered trees inside                                                                                                 |
| Royal forest            | Green dash-dot boundary with a large italic name; villages and fields inside                                                                          |
| Commons, heath, moor    | Tufts or stipple                                                                                                                                      |
| Fishponds               | Small rectangles in a chain along a stream                                                                                                            |
| Shielings (summer huts) | Cluster of tiny squares on high pasture                                                                                                               |

| Feature                           | Symbol                                                               |
|-----------------------------------|----------------------------------------------------------------------|
| Track or footpath                 | Thin dotted line                                                     |
| Drove road                        | Broad dotted line (two rows of dots)                                 |
| Wayside cross                     | Tiny cross on a base, set on the road or path                        |
| Boundary stone                    | The letters "BS" on the boundary line                                |
| Moot (assembly) site              | Small lone tree, labelled "Moot"                                     |
| Gallows                           | Small "Π" on a hill                                                  |
| Deserted village                  | Grey church alone in a field, labelled "(lost village)"              |
| Dyke or sea bank                  | Bold line with short ticks on the water side                         |
| Qanat                             | Straight line of tiny rings (shaft mouths) ending in green gardens   |
| Rice terraces                     | Tight parallel lines that follow the contours                        |
| Tank (reservoir)                  | Curved dam line with water behind it and a fan of small fields below |
| Parish, manor or hundred boundary | Thin dash-dot-dot line along streams and ridges                      |

### A method for a local map

1. Draw relief and water: streams, springs, flood plains, marsh.
2. Choose the settlement pattern for each part (nucleated, dispersed, linear).
3. Place villages 1.5–4 km (1–2.5 mi) apart on good land, near water and on the edge of the flood plain. Place hamlets and farms in the uplands, forest and marsh.
4. Draw the parish boundaries (thin dash-dot-dot lines) through the land between villages.
5. Draw the rings: fields around each village, meadow along streams, pasture and wood at the parish edges.
6. Add one mill per village, with its leat or on its hill.
7. Add manor houses, moats, dovecotes, fishponds and one or two deer parks near the richest manors.
8. Add a royal forest or chase, rabbit warrens on heath, granges in monastic land, and shielings in the hills.
9. Add paths: village to church, mill, market town and neighbours; drove roads to summer pastures.
10. Add the hundred moot site, wayside crosses, boundary stones and a few names.

#### FANTASY TWIST

Other peoples would farm differently. Dwarves in the mountains could farm like Mediterranean terrace villages, with irrigation tunnels like qanats. Elves might manage forests as wood-pasture and coppice on very long cycles, with tightly enforced "forest law". In a land with flying monsters, open-field villages would need shelters in the fields and lookouts on the church tower. In each case keep the basic rules: water, walking distance and rings of land use. See [Fantasy Variants](#).

### Quick summary

- 85–95% of people lived in the countryside. On a local map, almost all the land is fields, pasture, wood or waste, and almost all of it has an owner and a name.
- Open-field arable gives tight nucleated or green villages; uplands, forest and marsh give hamlets and farmsteads; valleys and dykes give strip villages in a line.
- A typical lowland village has 30–60 households (150–300 people), a territory of 5–15 km<sup>2</sup> (2–6 sq mi), and lies 1.5–4 km (1–2.5 mi) from its neighbours.
- Land use forms rings: gardens and orchards, then arable within about 2 km (1.3 mi) of the houses, then pasture, then wood and waste at the parish edge. Meadow follows the streams. Beyond 3–4 km (2–2.5 mi), people build outlying farms or switch to pasture.
- Across a whole settled lowland kingdom c. 1300, about a third of the land is ploughland (England: ~36%, fallow included), 7–15% is woodland, and most of the rest is pasture, heath, moor and marsh. After a plague, ploughland shrinks by a quarter to a third.
- Open-field strips were about 200 m (220 yd) long and ¼–1 acre in size, ploughed into ridge and furrow. Systems had two or three great fields, one left fallow each year.

- Uplands used infield-outfield and summer shielings; the Mediterranean used terraces, canals, norias and qanats; wet-rice lands used terraces, tanks and shared canals with water temples (Bali, Sri Lanka, Angkor), and the Aztecs built raised beds in lakes; Castile moved over 2.5 million sheep each year, some of them 550–750 km (340–470 mi).
- Mills: 5,624–6,000+ in Domesday England (1086); about 10,000–15,000 by 1300 (estimates vary), roughly one per village. Windmills in England from 1185; tide mills from the 7th century.
- A medieval "forest" is a legal hunting area, not just trees. Royal forest covered about a third of southern England at its peak. There were about 3,200 deer parks c. 1300, usually 40–80 ha (100–200 acres) with rounded outlines.
- Many manors have moats (about 6,000 moated sites are known in England, most only 40–80 m across), dovecotes and fishponds. Monasteries have granges and great barns.
- Parish, manor and hundred are different units. Hundreds of very roughly 150–200 km<sup>2</sup> (60–80 sq mi) met at open-air moot sites.
- Reclaimed land (fens, polders, assarts) has straight lines and new names; old land has curved boundaries.

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## CHAPTER 7

# INDUSTRY, RESOURCES AND SPECIAL TOWNS

Most medieval towns grew because of farming and trade. Some towns grew for a different reason: there was silver in the hill, salt in a spring, herring in the sea or a fast stream to drive a mill. This chapter tells you what people dug, cut, boiled, wove and caught, where each industry sat on the land, and what kind of settlement it created. Use it to give your map "odd" towns in odd places, each with a clear reason to exist.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Settlement sizes follow [Population and Settlement Sizes](#). Roads, rivers, ports and transport costs are in [Trade Routes, Roads and Transport](#). Ordinary village mills and woodland use are in [Villages, Farms and the Countryside](#). The map symbols for this chapter are listed in [Suggested symbols](#).

## IN THIS CHAPTER

The basic rules: what pulls industry to a place

Mining and metals

Salt

Stone, clay and lime

Forest industries

Peat diggings

Textiles

Fishing

Shipbuilding, naval stores and arsenals

Metalworking, crafts, leather, drink and sugar

Where industry sits in and around a town

Special town types and how to spot them

Quick summary

Sources and further reading

## The basic rules: what pulls industry to a place

## RULE OF THUMB

Industry goes to the heaviest thing it needs. That is usually the ore, the fuel or the water power, not the customer.

**Why:** Moving goods over land was very expensive. In England c. 1300, moving goods by land cost about 8 times as

much as by sea and 2 times as much as by river (see [chapter 05](#)). So a smelter (a furnace that heats ore, the metal-bearing rock, to get the metal out) is built next to the mine and the woods, not next to the city. Only the finished, lighter product (silver bars, iron blooms, salt, cloth) travels.

Three things decide almost everything:

| Input                                               | Why it matters                                                   | What it means for the map                                              |
|-----------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Raw material</b> (ore, clay, stone, brine, fish) | Heavy, bulky, often worthless until processed                    | Industry sits on top of the resource                                   |
| <b>Fuel</b> (wood, charcoal, peat, coal)            | Smelting, salt boiling, glass, brick and lime use huge amounts   | Industry sits in or next to forest, peat bog or coal outcrop           |
| <b>Water power</b> (a fast stream with enough fall) | Drives bellows, hammers, fulling mills, ore crushers, mine pumps | Industry sits in hill valleys, on streams, behind small dams and ponds |

Then there are two more pulls:

- **Water transport.** Bulk goods follow rivers and coasts. A quarry or coal pit near a navigable river can sell far away; one 30 km (19 mi) inland usually sells only locally.

- **Skilled people and capital.** Fine cloth, armour and glass need trained workers and merchants with money. These stay in towns, even when the raw material comes from far away.

## The master table

| Resource                     | What it needs                                                                    | Where it goes on the map                                              | What settlement it creates                                                                              |
|------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Silver, lead-silver          | Ore veins; wood for pit props and smelting; water to drain mines and turn wheels | Old mountains and hill country, up side valleys                       | Chartered mining town, a mint, many mining hamlets around it                                            |
| Gold                         | Veins in mountains, or gold-bearing river gravel                                 | Mountains; river sands                                                | Mining town with a royal mint                                                                           |
| Copper                       | Ore and a lot of fuel for roasting                                               | Hills and mountains                                                   | Mining town with bare, smoke-killed land around it                                                      |
| Iron                         | Ore (common), charcoal, water power                                              | Wooded hills with fast streams                                        | Scattered bloomeries and forges; iron villages; an iron-trading town where the valley meets a big river |
| Tin                          | Tin-bearing gravels and granite moors                                            | Moorland valleys                                                      | Tinners' camps; "coinage towns" at the edge of the moor                                                 |
| Coal                         | Seams that reach the surface                                                     | Coalfields near a coast or navigable river                            | Pit hamlets; a coal port                                                                                |
| Salt from brine              | A salt spring plus a great deal of fuel                                          | Lowland springs                                                       | A rich salt town; "salt roads" leading away from it                                                     |
| Rock salt                    | Salt beds in the ground                                                          | Foothills and mountains                                               | Mining town with a royal official's castle                                                              |
| Sea salt                     | Flat shore, sun, dry summers                                                     | Lagoons and sheltered bays, mostly in the south                       | Salt-pan villages and a loading port                                                                    |
| Building stone               | A good outcrop close to water                                                    | Hill edges near rivers or the sea                                     | Quarry hamlet; stone shipped by boat                                                                    |
| Clay                         | Clay plus fuel                                                                   | River valleys, estuaries, glacial lowlands                            | Brickfields outside towns; potters' villages                                                            |
| Lime                         | Limestone or chalk plus fuel                                                     | Limestone edges, building sites, ports                                | Kilns, not towns                                                                                        |
| Timber                       | Forest plus a river that can float logs                                          | Upland forests above rivers                                           | Rafting villages; a timber market in a river town                                                       |
| Charcoal, glass, potash, tar | Woodland                                                                         | Deep forest                                                           | Temporary camps; glasshouses that move                                                                  |
| Peat (turf)                  | A fen or bog, plus a river or canal to a fuel-hungry town                        | Wet lowlands with few trees                                           | Turf commons beside fen villages; flooded pits that become lakes                                        |
| Wool                         | Grazing land                                                                     | Downs, limestone hills, uplands, dry plateaus                         | Sheep villages, monastic farms, wool market towns                                                       |
| Cloth                        | Wool, fulling water, dyes, skilled labour                                        | Towns; fulling mills in hill valleys                                  | Cloth towns; rural weaving villages                                                                     |
| Dye plants (woad, madder)    | Rich arable land                                                                 | Plains near cloth regions                                             | Woad towns and woad mills                                                                               |
| Herring                      | Seasonal shoals plus salt                                                        | Shallow seas and straits                                              | Seasonal beach markets and curing stations                                                              |
| Cod (stockfish)              | Cold, dry, windy spring weather                                                  | Arctic coasts                                                         | Fishing stations, drying racks, one big trading port                                                    |
| Ships                        | Oak, pine, tar, hemp, iron                                                       | Sheltered estuaries near forest and rivers                            | Shipyards; walled state arsenals                                                                        |
| Wine                         | Sunny slopes and a river for export                                              | South-facing river slopes                                             | Wine villages; an export port downriver                                                                 |
| Sugar cane                   | Frost-free land, irrigation water, a crushing mill, fuel for boiling, a port     | Irrigated coastal plains in the warm south (Cyprus, Sicily, Valencia) | A fortified estate centre with a mill and boiling house; cane villages; a loading port                  |
| Beer                         | Clean water and grain                                                            | Grain regions; towns                                                  | Brewing towns                                                                                           |

**MAP TIP**

For every industry you draw, check three things on the map: the resource, the fuel and the way out (a river, a coast or a road to one). If one is missing, move the industry or add what is missing.

## Mining and metals

### What a mining town is like

**RULE OF THUMB**

Mines are found in old, rugged hill country, up side valleys, far from the best farmland. The mining town must buy almost all its food.

- Little farmland. Food came in from lowland markets.

**Why:** Metal ores form in old, folded and broken rocks: the Harz, the Ore Mountains (Erzgebirge), the Slovak and Bohemian hills, Cornwall, the Pennines, the Alps. The valleys give water power. The hills give wood.

**Special laws.** Rulers wanted miners badly, so they offered freedoms:

- **Free mining.** In German lands this was called *Bergfreiheit* (mining freedom): any settler could dig for ore, even without owning land, if he paid a share (often a tenth) to the lord. The silver had to be sold to the lord's mint.
- **Own courts and law.** Mining codes (*Bergordnung*) set out claims, duties and miners' rights. Tinnern in Cornwall and Devon had their own stannary courts from King John's charter of 1201. Derbyshire lead miners had the Barmote Court.
- **Lower taxes and extra rights**, such as market, brewing and customs rights. Mining towns often had more self-government than ordinary towns.

### What it looks like:

- Streets on slopes. Older mining towns are often irregular, because they grew fast around mine shafts. Later ones founded by a ruler (such as the Ore Mountain towns of c. 1500) were often planned, with a grid of streets and a square market place.
- Heaps of waste rock (spoil heaps) on the hillsides.
- Smelting huts and ore-crushing mills along the stream below.
- A **mint**, because silver and gold were turned into coin on the spot. A strong kingdom often kept a mint at each big silver mine, as well as in the capital. See [How many mints?](#)
- A large, rich church for its size. Often a royal castle or official's house to watch the money.

**Size:**

| Type                  | Population    | Notes                                                                                                                                                                                                                                                                     |
|-----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mining camp or hamlet | 20–300        | Huts near shafts. Many vanish when the vein runs out.                                                                                                                                                                                                                     |
| Ordinary mining town  | 1,000–5,000   | The usual case. Most mining towns stayed small.                                                                                                                                                                                                                           |
| Boom town             | 10,000–20,000 | Rare. Two examples from just after 1500: Annaberg (Saxony), founded in 1496, grew from about 3,000 people c. 1500 to about 12,000 by 1540. Joachimsthal (Bohemia), founded in 1516, had about 20,000 people by 1534 and was then the second town of Bohemia after Prague. |

Boom towns are the main exception to the rule that towns over 10,000 need navigable water. Silver is light and valuable, so it pays for food carted in over the hills. Mining towns boom and bust. A rich vein can create a town in 20 years. Flooding, deep shafts and exhausted ore can empty it again. Drainage was the big limit. Medieval pumps could

not drain deep shafts, so miners dug long drainage tunnels (adits) out to the valley side.

**Silver**

Silver was the money metal of medieval Europe, so silver mines made kings rich.

| Mine                                        | Dates                                              | What happened                                                                                                                                                                                                                                                                            |
|---------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rammelsberg, Goslar</b> (Harz, Germany)  | Mining recorded c. 968; much older                 | Ore with silver, lead, copper and zinc. Henry II built an imperial palace at Goslar from 1005 because of the silver. Over 100 miners died in a collapse in 1376. Worked until 1988.                                                                                                      |
| <b>Freiberg</b> (Saxony)                    | Silver found c. 1168                               | Found at the village of Christiansdorf. The margrave of Meissen offered free mining; miners poured in, and within about 20 years the town of Freiberg ("free mountain") existed. Its first boom faded in the 14th century.                                                               |
| <b>Kutná Hora</b> (Bohemia)                 | Silver mined from c. 1260; great rush in the 1290s | Near Sedlec Abbey. King Wenceslas II issued a mining code and created the Prague groschen coin around 1300, struck at the central mint in the "Italian Court". In the late 1200s the district produced perhaps a third of Europe's silver. It became Bohemia's second town after Prague. |
| <b>Iglesias</b> (Villa di Chiesa, Sardinia) | 13th century                                       | Founded by the Pisan count Ugolino della Gherardesca. Ruled by its own law code, the <i>Breve</i> (a 1327 copy survives). Estimated at about a tenth of Europe's silver in the early 1300s.                                                                                              |

Population estimates for Kutná Hora vary hugely. Popular books give up to 60,000. That is very likely too high. Treat it as a large town of many thousands.

**Gold**

Medieval Europe had little gold. The main source in the 14th century was the Kingdom of Hungary, in the mountains of today's Slovakia and Transylvania.

- Around 1330, Hungarian mines produced about 1,400 kg of gold a year. That was more than 30% of the world's output (an estimate).
- **Kremnica** (Slovakia) was one of the most important of these gold mines. It got royal town privileges in 1328. From 1331 it was the seat of the royal official (the "chamber count") who ran all mines and mints in twelve counties. From 1335 its mint struck gold florins, later the famous Kremnica ducats. Kremnica's own yearly output is not well recorded; treat any exact figure with caution.
- Kremnica sits at 564 m (1,850 ft) in a mountain valley, about 20 km (12 mi) in a straight line from Banská

Bystrica. Several mining towns stood within a day's travel of each other.

- Gold also came from washing river sand, a small, scattered trade along certain rivers.

**Copper**

- **Falun** (Sweden): the "Great Copper Mountain". Copper was dug here by the mid-1200s, and possibly as early as about 1000. A document of 1288 records a share in the mine, which was run by a group of share-owning master miners. The first known royal charter dates from 1347.
- Copper ore was roasted in open fires, which gave off sulphur smoke and killed the plants for a long way around. Draw a bare zone around a big copper mine.
- Copper mixed with tin makes bronze (bells, cannon); mixed with zinc ore (calamine) it makes brass.

**LATER ERA (1500S+)**

Falun peaked in the 1600s, when it produced a large share of Europe's copper (popular accounts say about two-thirds of world output in the mid-1600s; treat this as a rough claim). New silver towns boomed too: Joachimsthal (1516) gave its name to the *thaler* coin and so to the word "dollar". After about 1550, cheap silver from Spanish America (Potosí, found in 1545) hurt many European mines.

**Iron****RULE OF THUMB**

Iron ore is common. Charcoal and water power are the real limits. Put ironworks where forest, fast streams and ore meet.

**Why:** Medieval iron was made with charcoal. Bloomeries (simple clay furnaces) burned roughly as much charcoal as ore, by weight. Estimates of the total vary widely, but they often fall around 10–20 kg of charcoal for each 1 kg of finished iron. Since charcoal is made from about four or more times its weight in wood, that means very roughly 50–150 kg of wood for 1 kg of iron. Wood is too bulky to move far, so ironworks moved to the wood.

| Region                                             | Features                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Erzberg</b> (Styria, Austria)                   | A whole mountain of iron ore, worked in open pits. First recorded in 1171. Two mining communities, Innerberg (today Eisenerz) and Vordernberg, smelted ore in water-powered "wheel works" ( <i>Radwerke</i> ) along the valley. The iron trade was given to privileged towns: Steyr for Innerberg iron (1287) and Leoben for Vordernberg iron (1314). |
| <b>Siegerland</b> (Germany)                        | Iron mining and smelting in wooded hills. Communities managed their oak and birch woods as <i>Hauberg</i> : coppice (trees cut to the stump so they regrow) cut every 16–20 years, with a year of rye sown after each cut. Formal forest ordinances date from 1562 and 1565.                                                                          |
| <b>Basque country</b> (Biscay and Gipuzkoa, Spain) | Water-powered ironworks ( <i>ferrerías</i> ) in steep, rainy, wooded valleys from the 1300s. Iron was exported to England and Flanders through ports such as Bilbao (founded 1300).                                                                                                                                                                   |
| <b>The Weald</b> (south-east England)              | Bloomeries from early times; the first blast furnace in England c. 1490. See <a href="#">Forest industries</a> .                                                                                                                                                                                                                                      |

**What to draw:** iron districts are not one big town. They are many small sites (a furnace, a hammer forge, a pond, charcoal hearths) scattered along every stream, plus one or two towns that trade and finish the iron.

**Tin**

- Most of Europe's medieval tin came from **Cornwall and Devon**. The Ore Mountains (Saxony and Bohemia) became the other big source, especially in the late Middle Ages.
- Most medieval tin was "streamed": washed out of gravels in valley floors, not mined deep. Tinnners turned moorland valleys into rows of spoil heaps and water channels.
- **Stannaries** (tin-mining districts) had their own law. King John's charter of 1201 let tinnners dig for tin on almost anyone's land and placed them under the Lord Warden of the Stannaries and his own courts, not the ordinary courts.
- All tin had to be brought to **coinage towns** to be weighed, tested and taxed. In Cornwall these were Lostwithiel, Bodmin, Liskeard, Truro and Helston. In Devon,

Tavistock, Ashburton and Chagford (1305) and Plympton (1328).

**MAP TIP**

Put coinage towns on the edge of the moor, where tin came down from the uplands and met a road or a tidal river. Do not put them in the middle of the moor.

**Lead**

- Main English fields: **Derbyshire** (the Peak, around Wirksworth) and the **Mendip Hills** (Somerset). Also the northern Pennines.
- Derbyshire had free mining: anyone who showed the *barmaster* (the mining official) good ore could open a mine and keep it while he worked it. Disputes went to the Barmote Court. Lead at Wirksworth and Bakewell is already recorded in Domesday Book (1086).
- Lead was smelted in **boles**: open hearths on windy hill-tops and ridges, using the wind as bellows. Many "Bole Hill" place names in the Peak mark them.

- Uses: roofs of churches and castles, gutters, water pipes, the strips that hold stained glass. Lead ore often holds silver, which was separated in a furnace.

## Coal

Coal was a small, local fuel in the Middle Ages, used mainly by smiths, lime burners and the poor in coal districts. It mattered in two places above all:

- **Newcastle upon Tyne** (England). Seams came to the surface along the coast and river banks, so the English called it "sea coal". Sea coal from the north-east was reaching London by 1228, and the coal trade by sea grew through the 1200s. In 1306 a royal proclamation banned London craftsmen from burning sea coal in their furnaces because of the smoke. By the early 1500s (before 1539) the Tyne shipped roughly 15,000 tonnes a year; medieval shipments were probably of this size or smaller.

- **Liège** (today's Belgium). The first texts on coal here date from the 12th century, and coal digging was regulated by 1229. Miners dug drainage tunnels called *areines*. A special court, the *Voir-Jurés du charbonnage*, judged mining disputes.

Medieval coal pits were small: "bell pits" (a shaft widened at the bottom, abandoned when it got dangerous) and shallow tunnels into hillsides.

### LATER ERA (1500S+)

Coal boomed when wood became scarce. Tyne shipments grew from about 15,000 tonnes a year before 1539 to about 240,000 tonnes in 1600 and 425,000 in 1634. Draw many colliery (coal-mine) hamlets along the Tyne, each with the mine symbol (crossed hammers), and a sea lane (a dotted line at sea) to London labelled "colliers" (coal-carrying ships).

## Mining districts on the map

| Scale      | What to show                                                                                                                                                                                                                                          |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continent  | A mining-district symbol (crossed hammers or a pick) on 3–6 mountain areas. Not every range has ore.                                                                                                                                                  |
| Kingdom    | 1–3 mining districts. In each, a cluster of 2–7 mining towns, each 10–40 km (6–25 mi) apart, plus a mint town (see <a href="#">How many mints?</a> ). The central Slovak mining towns and the towns of the Ore Mountains are real clusters like this. |
| Local area | Shafts, spoil heaps, adit mouths in the valley side, ore-crushing mills on the stream, smelting huts, charcoal hearths in the woods, the miners' town on a slope, a castle or official's house.                                                       |

### FANTASY TWIST

Dwarves who can drain or shore up deep mines remove the medieval depth limit, so their mines last for centuries instead of decades. Magic ores (mithril and the like) behave like gold or silver: valuable enough to pay for long, guarded roads, so a remote mine can still support a large town. A dragon or monster in the hills is a good reason why a known ore field is *not* worked. More in [chapter 10](#).

### MAP TIP

Draw a mining town up a side valley, with a road down to the nearest market town or river. Label the mine with a name (in German lands, mine names were often saints or wishes: "St Anne's Find", "Good Hope"). Add a few abandoned shafts (the mine symbol drawn grey, labelled "(old shaft)") and a ghost hamlet (grey house marks, labelled "(lost hamlet)"). Mining landscapes are full of failed attempts.

## Salt

### RULE OF THUMB

Everyone needs salt, and few places have it. Every realm on your map needs at least one salt source, or a trade route that brings salt in.

**Why:** People and animals need salt to live. Salt also preserved fish, meat, butter and cheese for winter and for fast days. A medieval region without salt had to import it, so salt towns became rich, and salt was taxed heavily.

### Three ways to make salt

| Method                 | How it works                                                                                                       | Fuel                             | Where                                             | Examples                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| <b>Brine springs</b>   | Brine (very salty water) comes up from salt beds underground. It is boiled in large pans until salt crystals form. | Huge: wood, peat or (later) coal | Inland lowlands and valleys over buried salt beds | Lüneburg, Droitwich, Halle, Salins                                    |
| <b>Rock salt mines</b> | Salt is dug as rock, or water is pumped into the mountain to dissolve it and the brine is boiled                   | Some (for boiling brine)         | Foothills and mountains                           | Wieliczka and Bochnia (Poland), Hallstatt and other Alpine salt towns |
| <b>Sea salt pans</b>   | Seawater is let into shallow clay pans and the sun evaporates it                                                   | Almost none                      | Flat, sheltered coasts with hot, dry summers      | Bay of Bourgneuf (France), Venetian lagoon (Chioggia), Ston (Croatia) |

In the cold, wet north, people also boiled seawater or burned salt-soaked peat (in Zeeland and Frisia) to get salt. This produced much less.

### Salt towns

#### Brine-spring towns:

- **Lüneburg** (northern Germany). Salt making dominated the town from the 12th century. A brine spring was surrounded by 54 boiling huts, each with 4 pans: 216 pans in all. The workers lived in their own walled "Salt Quarter" of the town; they numbered several hundred at the works' height. The salt went to Lübeck on the Baltic, mostly to salt herring. In the 16th century about 19,000 tonnes a year made that trip.
- **Droitwich** (England). Brine here is nearly saturated: about 7–10 times saltier than seawater (often quoted as "ten times"). Domesday Book (1086) shows the king holding the brine pits, and many manors across the West Midlands owning salt pans or rights to salt. Salt was carted along "saltways" to neighbouring counties and to the River Severn at Worcester.
- **Halle** (Saale, Germany). The name is usually linked to an old word for salt. Four brine wells stood around the market. The salt workers, the *Halloren*, were a famous guild. Halle was in the Hanseatic League by 1281.
- **Salins** (Franche-Comté, France). Two saltworks were working by 1115. Salt made Salins the second town of the region after Besançon, and the title "lord of Salins" was prized by the counts and dukes of Burgundy.

#### Rock-salt towns:

- **Wieliczka and Bochnia** (near Kraków, Poland). Rock salt mining began in the 13th century. King Casimir the Great issued a mining ordinance in 1368. In the 14th century the Kraków saltworks gave the Polish crown about a third of its income. At its height (16th to mid-17th century) Wieliczka had about 2,000 miners and produced over 30,000 tonnes a year.

- **Hallstatt** (Austria). Salt mined since prehistoric times. A tiny town squeezed between a lake and a mountain, reached only by boat or narrow paths until the 19th century.

#### Sea-salt districts:

- **Bay of Bourgneuf** (south of the Loire, France). "Bay salt" was cheap, grey and coarse. In the late Middle Ages, fleets of Hanseatic and later Dutch ships (the "Bay fleet") came every year to load it for the Baltic fisheries. The bay was a major salt producer in the 15th and 16th centuries.
- **Venetian lagoon and Chioggia**. Salt pans made Chioggia rich in the 12th–14th centuries. Venice tried to control the salt trade of the whole northern Adriatic.
- **Ston** (Croatia). Dubrovnik bought the Pelješac peninsula in 1333 for its salt pans. From 1358 it built walls more than 7 km (4.3 mi) long, with 40 towers, across the neck of the peninsula; 5.5 km (3.4 mi) survive. The twin planned towns of Ston and Mali Ston sit at each end. The salt pans were one of Dubrovnik's main sources of income.

### The fuel problem

Brine boiling burned enormous amounts of wood. Brine towns were often blamed for clearing the forests around them. Lüneburg Heath is the classic example, but the story is not proven: the heath probably began with much older clearances, back in the Bronze Age. When wood ran out, there were two answers: bring fuel in by water, or pipe the brine to the forest.

#### LATER ERA (1500S+)

Brine pipelines appear. From 1595 to 1607 a wooden pipeline was built to carry brine from Hallstatt down to Ebensee, where wood was plentiful. Sources give its length as 34–40 km (21–25 mi). From 1775, brine from Salins was piped 21 km (13 mi) to the new Royal Saltworks at Arc-et-Senans, built next to the large Forest of Chaux.

## Salt roads

- Salt moved overland farther than other bulk goods because everyone had to buy it. The **Old Salt Route** from Lüneburg to Lübeck was about 90–100 km (55–60 mi). In 1398 the Stecknitz Canal opened to carry salt by water.
- Around Droitwich, the Upper and Lower Saltways led east across the Midlands and south over the Cotswolds.
- Salt roads usually lead from the salt source to the nearest navigable river or port, then by water to the big markets. Many old roads are still called "Salt Way", *Salzstraße* or *Via Salaria*.

### FANTASY TWIST

Fire magic or magical heat removes the fuel problem of brine boiling. Then brine towns can grow without eating their forests, and a salt spring in a treeless land becomes valuable. A "salt mage" guild would be as powerful as a mint.

### MAP TIP

Place a brine town on lowland near a spring, with woods or peat bogs nearby (or a river to bring fuel). On a local map, mark the spring with a small blue dot labelled "Brine Pit"; on a kingdom map, add "(salt)" to the town's label instead. Draw 2–4 roads radiating from it, each labelled "Salt Road" or "Saltway" on the first stretch. Sea-salt pans need flat, sheltered shore and a warm, dry summer: draw a checkerboard of small rectangles along a lagoon. In the north, keep sea salt small and import "bay salt" by ship. Place names with "Hall", "Salz", "Sal" or "Saline" often, but not always, mark salt. In England, the "-wich" brine towns of Cheshire and Worcestershire (Droitwich, Nantwich, Middlewich, Northwich) are a real cluster.

## Stone, clay and lime

### Quarries

#### RULE OF THUMB

Ordinary buildings use local stone. Only fine stone for great churches and castles travels far, and almost always by water.

**Why:** Stone is the heaviest common building material. Moving it often cost more than quarrying it, so builders used the nearest stone that would do.

#### Real examples:

- **Caen stone** (Normandy). A fine, cream-coloured limestone. The quarries lay close to the River Orne, so ships could carry it across the Channel. Canterbury Cathedral was rebuilt in it from 1070.
- **Barnack stone** (Northamptonshire). Used for Peterborough and Ely cathedrals and many fenland churches, moved by river and fen waterways.
- **Purbeck marble** (Dorset). A dark, polishable limestone used for thin columns in many English cathedrals. Shipped from the coast.
- **Millstones**. Hard lava stone from the area between Mayen and Andernach (Eifel, Germany) was carried by cart to the Rhine at Andernach and shipped as far as England and Scandinavia. Rough sandstone (gritstone) from the Peak District made English millstones.

- **Slate** for roofs came from places with natural splitting rock, such as Cornwall and north Wales. Most medieval roofs were thatch, wooden shingles or tiles instead.

**What to draw:** a quarry is a notch in a hillside, with a track down to water and a small hamlet of quarrymen. For a big church project, masons set up a temporary lodge (workshop) at the quarry or at the building site.

### Brick

#### RULE OF THUMB

Where there is no good stone, people build in brick. Look for flat lowlands with clay.

**Why:** The lands around the southern Baltic, the North Sea coast, the Po plain and the Garonne valley have deep clay and few hard rocks.

| Brick region                                                                           | Examples                                                                                                                 |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Northern Germany, Poland, the Baltic states, Denmark, southern Sweden ("Brick Gothic") | Lübeck, Gdańsk, Toruń; the Teutonic Order's castle at Malbork (begun in the late 13th century), the largest brick castle |
| Low Countries                                                                          | Most towns of Holland and Flanders                                                                                       |
| Lombardy and the Po plain                                                              | Milan, Pavia, Cremona, Bologna                                                                                           |
| Southern France                                                                        | Toulouse ("the pink city"); Albi Cathedral (1282–1480)                                                                   |
| Eastern England (late medieval)                                                        | Hull; Tattershall Castle (1430s–40s)                                                                                     |

Brick and tile kilns stood outside the town walls, near clay pits and water, because of fire risk and smoke. Old clay pits became ponds.

### Lime kilns

- Lime (made by burning limestone or chalk) was needed for mortar, plaster and limewash, and sometimes spread on fields.
- Kilns stood next to limestone or chalk outcrops, or next to big building sites, or at ports where stone and fuel could both arrive by boat. Lime burners were among the first medieval users of coal.
- A lime kiln is a small industrial site, not a settlement. On a local map, mark kilns along a limestone edge with the word "Limekiln" (or "Limekilns" for a row), not with a picture. A small tower would read as a castle, and a small mound as a barrow (an ancient burial mound) or a motte (a castle mound).

### Pottery

- Everyday pots were made in **potters' villages** near good clay, with woodland or heath for fuel. Most sold within a day's travel by cart.
- A few kinds of pottery were traded across Europe: Rhineland stoneware (Siegburg, Langerwehe, Raeren) went down the Rhine to the North Sea and the Baltic; lustreware from Málaga and Manises (near Valencia) went around the Mediterranean and to England.

#### MAP TIP

On a kingdom map, mark only famous quarries (with the quarry symbol, a notch in the hillside, and a name like "Marble Quarries") and the brick-built towns of clay lowlands. On a local map, add clay pits and kilns outside the walls (labelled "Clay Pits", "Brick Kilns"), a quarry track to the river, and a lime kiln at a limestone edge. If you colour your town symbols, use a pale stone colour (buff or cream) in a region of stone and brick red in a clay lowland. Do not use grey: the guide's **master legend** keeps grey for ruins and abandoned places.

## Forest industries

Medieval forests were not empty wilderness. They were working landscapes full of woodcutters, charcoal burners, iron smelters, glassmakers, tar boilers and pig herders. Ordinary village woods (fuel, pannage (letting pigs feed on acorns), timber for houses) are in **chapter 06**; this section covers the industries.

### Timber

- Big building and ship timber came from forests close to floatable rivers. Logs were tied into rafts and floated downstream to towns.
- Venice drew timber from Alpine forests down rivers like the Piave and the Adige.
- English and Dutch builders imported Baltic oak boards ("wainscot") from Prussia and Livonia in the 14th–15th centuries.

### Charcoal burning

- Charcoal was made in **clamps**: a dome of stacked logs covered with turf and earth, burned slowly for 6–8 days (several weeks for very large ones).
- It takes roughly 4–8 kg of wood to make 1 kg of charcoal. A traditional clamp turns about a quarter of the wood's weight into charcoal on average; small, careless burns do worse.
- Charcoal is light but crumbles if carried far, so it was made in the woods near where it would be used. The burners lived in huts next to the clamp to watch it day and night.
- Woods for charcoal were usually **coppiced**: cut to the stump and allowed to regrow, giving poles every few years. Medieval cycles were mostly under 10 years; the

10–20-year cycles of charcoal woods (such as the Siegerland *Hauberg*) belong mostly to the 1500s and later (see [Managed woodland](#)).

### Iron smelting in the forest

**Bloomeries** were small furnaces that produced a spongy lump of iron (a bloom), which was then hammered. Early blooms weighed a few kilograms. Water-powered bellows and hammers, spreading in the High Middle Ages, allowed much bigger furnaces. Late medieval blooms could reach a few hundred kilograms.

**Blast furnaces** melt the iron completely and run for months. In Sweden, Lapphyttan is an early example, dated somewhere between about 1150 and 1350. In England the first one was built in the Weald (at Queenstock, Buxted) about 1490. Blast furnaces need a dam and a pond for a waterwheel, and a steady supply of charcoal.

How much forest does an ironworks need? In the Weald, historians estimate that a post-medieval blast furnace needed, for a sustainable supply, the wood from about a 5 km (3 mi) radius, in a landscape that was a quarter to a third woodland.

#### LATER ERA (1500S+)

Blast furnaces spread. The Weald had nearly 180 furnace and forge sites over its history and produced over 9,000 tonnes of iron a year in the 1590s. Cast-iron cannon were made there from 1543. Furnace ponds and "hammer ponds" (millponds for forge hammers) dot the map of the Weald to this day.

### Forest glass

- From about 1000 to 1700, northern European glass was made in the forest from sand and wood ash (potash). This is called *forest glass* (*Waldglas*).
- It used enormous amounts of wood: estimates are 150–200 kg of wood for 1 kg of glass, counting both fuel and ash. A large glasshouse might burn about 67 tonnes of wood a week, for about 40 weeks a year.

- Glasshouses moved to new woods when the old ones were cut out. The Spessart (Germany) is a good example of a glass forest. In the early 1400s just four glassworks in its Mainz-ruled part made around 230,000 pieces of blown glass a year, plus flat (window) glass. Other glass districts were Bohemia, the Black Forest, the Jura and the Weald.

#### LATER ERA (1500S+)

A 1615 English proclamation banned wood fuel in glass-making. In the same year Sir Robert Mansell got a monopoly on English glass and used sea coal instead of wood. Glassmakers moved to coalfields.

### Potash, tar and pitch

- **Potash**, made by burning wood and leaching the ash, was needed for glass, soap, bleaching linen and dyeing. The great forests of Prussia, Poland and Lithuania exported it through Baltic ports such as Gdańsk (Danzig).
- **Tar and pitch**, made by slowly burning pine wood and stumps in pits or kilns dug into slopes, waterproofed ships, ropes and roofs. Norway, Sweden, Finland and the eastern Baltic were the main sources.

#### FANTASY TWIST

If the forest is home to elves, spirits or monsters, charcoal burners, glassmakers and tar boilers become the frontier people of your world. They need permission, guards or a truce. A realm that cannot cut its forests cannot make iron or glass, and must import them.

#### MAP TIP

Draw a forest with small clearings, each labelled with its work: "Charcoal Hearths", "Glasshouse", a "Forge" by a stream with its pond (draw the forge as a watermill wheel on the stream), a woodcutters' hamlet. Add one track out of the forest to a river. Mark a few clearings as abandoned ("Old Glasshouse"). Forest industries leave no towns, only hamlets with names like Coalpit, Furnace, Hammer or, in German lands, Glashütte (glass hut).

## Peat diggings

#### RULE OF THUMB

Where wood is scarce in wet lowlands, people dig peat for fuel. Big peat diggings fill with water and turn into lakes. So some "natural" lakes on a real map are really old fuel pits.

**Why:** Peat (also called turf) is half-rotted plant matter that builds up over thousands of years in fens and bogs. Cut into blocks and dried, it burns well. Where the woods had been cleared and towns needed fuel (eastern England, Holland, Flanders, Frisia), peat was dug on a large scale and sent to town by boat. In low land a pit soon goes below the

water table, so it fills with water. Waves then eat away the soft peat banks, and the lakes grow.

Fen villages often had **turbary** (the right to cut turf or peat on a common; see [chapter 06](#)). Each village dug inside its own boundary, so the diggings follow parish lines.

### Real example: the Norfolk Broads (England)

- The Broads are 63 shallow lakes, most less than 4 m (13 ft) deep, joined by rivers. Today they have over 200 km (120 mi) of navigable waterways.
- They are flooded medieval peat pits. Peat was dug here for fuel from the 12th to the 14th centuries, by monasteries and villages, and sold to **Norwich** and **Great Yarmouth**. The records of Norwich Cathedral Priory show that large amounts of peat were dug, in what was then one of the most crowded parts of England.
- Each parish dug inside its own boundary, so the shapes of the broads still follow old parish boundaries. Banks of uncut peat (baulks) were left between neighbouring dig-

gings. Some still separate one broad from the next, and roads cross the wet ground on causeways.

- The water rose (a wetter climate and a rising sea level). By the end of the 14th century almost all the diggings had been given up and had flooded.
- People forgot. Until the 1950s most people thought the Broads were natural. The botanist Joyce Lambert and her colleagues showed that they have almost vertical sides and flat floors, like dug pits, and published the proof in 1960.
- Size: most broads are small, from a few hectares up to about 1.4 km<sup>2</sup> (0.5 sq mi) for the largest, Hickling Broad.

### Real example: the Haarlemmermeer (Holland)

Between Haarlem, Leiden and Amsterdam, peat cutting for fuel (much of it by people from Haarlem) and storms turned a few small lakes into one big inland sea. The Dutch called the growing lake the "water wolf".

| Date         | What the map shows                                                                                                                                                                                                                                                                                             |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13th century | Three or four separate peat lakes (the old Haarlemmermeer, the Spieringmeer, the Leidsemeer and the Oude Meer), with farmland and villages on the strips of land between them. One Dutch timeline gives about 9,100 ha (91 km <sup>2</sup> , 35 sq mi) for all of them around 1250; other estimates are lower. |
| 1477–1508    | Storms (such as the flood of 1477) break through the strips. By 1508 the old Haarlemmermeer and the Spieringmeer have joined.                                                                                                                                                                                  |
| Late 1500s   | The lakes have become one lake.                                                                                                                                                                                                                                                                                |
| c. 1650      | About 15,000 ha (150 km <sup>2</sup> , 58 sq mi).                                                                                                                                                                                                                                                              |
| 1848         | About 16,850 ha (169 km <sup>2</sup> , 65 sq mi). It was then pumped dry by 1852.                                                                                                                                                                                                                              |

Villages such as Nieuwerkerk and Rietwijk were swallowed, and the village of Aalsmeer lost more than half of its land.

#### LATER ERA (1500s+)

After about 1530, Dutch diggers began to dredge peat from under the water with a net on a long pole. They dried the wet peat on long, narrow strips of land left between the pits (*legakkers*). This made whole lake districts shaped like a comb of thin strips and long water channels, such as the Loosdrechtse and Vinkeveense Plassen in the north-west of the province of Utrecht. From the 1600s, Dutch investors pumped many lakes dry with windmills (the Beemster by 1612); see [Drainage and new land](#).

#### MAP TIP

Draw peat lakes with straight or blocky edges, not smooth round ones, and leave narrow banks (baulks) and causeways between them. Put a "turbary" or "Turf Fen" common beside each fen village (the commons symbol, tufts or stipple, with that label), and a river or canal carrying turf boats to the nearest town. The Broads lie within about 30 km (19 mi) of Norwich, along its rivers. For a map set a century or more after the digging, turn the pits into a chain of lakes, and mark a drowned village or two: a grey church standing in the water (the master legend's symbol for a lost village), labelled, for example, "Old Fenby (lost village)" or "Drowned Church".

## Textiles

Cloth was the biggest manufacturing industry of medieval Europe. It needs wool (or flax, cotton or silk), water power for fulling, dyes, and many skilled workers.

### Wool regions

#### RULE OF THUMB

Sheep go where land is too dry, too thin or too high for the plough: chalk downs, limestone hills, uplands and dry plateaus.

| Region                    | Features                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>England</b>            | Europe's main source of fine wool in the 13th–14th centuries. Customs records show England exporting on average about 25,000–40,000 sacks a year between the 1280s and the 1330s (five-year averages; some estimates for peak years reach 45,000). One sack weighed 364 lb (165 kg), the fleeces of about 200–260 sheep. So the exported wool alone came from something like 5–10 million sheep. Famous wool areas: the Cotswolds, Lincolnshire, the Welsh Marches, Yorkshire. |
| <b>Castile</b><br>(Spain) | Large flocks moved between summer pastures in the north (Old Castile and León) and winter pastures in the south (Extremadura and Andalusia), walking 550–750 km (340–470 mi) each way. The herders' guild, the <i>Mesta</i> , received its royal charter in 1273. Wool exports began in the 14th century; fine merino wool became a big export to Flanders and England in the late 15th century.                                                                               |
| Uplands everywhere        | Scotland, Wales, the Alps, the Pyrenees, the Apennines: wool mainly for local cloth.                                                                                                                                                                                                                                                                                                                                                                                           |

Many English wool estates belonged to Cistercian monasteries (see [chapter 08](#)), which ran large sheep farms called

granges. After the Black Death, many abandoned villages became sheep runs.

### Cloth towns

| Cloth centre                                                | When                | Notes                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Flanders:</b> Ghent, Bruges, Ypres, plus Lille and Douai | 12th–14th centuries | The densest cloth region in Europe. Fine cloth from imported English wool. Ghent had perhaps 40,000–65,000 people in the 14th century (estimates vary widely; see <a href="#">chapter 02</a> ). Bruges perhaps 35,000–46,000. Ypres perhaps 20,000–30,000 c. 1300 (older books say up to 40,000), but only about 10,000–11,000 by 1412, after its industry declined. |
| <b>Florence</b>                                             | 13th–15th centuries | The chronicler Giovanni Villani wrote that in the 1330s Florence had over 200 wool workshops making 70,000–80,000 cloths a year, and that over 30,000 people lived from the trade. Wool came from England and Spain.                                                                                                                                                 |
| <b>English cloth districts</b>                              | 14th–15th centuries | After about 1350 England made more of its own cloth. Production spread into hilly countryside with fast streams: the West Country (Stroud valleys, Wiltshire), East Anglia, the West Riding of Yorkshire.                                                                                                                                                            |

#### What a cloth town looks like:

- A **cloth hall**, where cloth was inspected and sold. Ypres's huge Cloth Hall was built in the 1200s.
- **Tenter grounds:** open fields beside the town, filled with wooden frames where wet cloth was stretched on hooks to dry ("tenterhooks").
- Dyers' streets along the river, often downstream.
- Weavers' quarters with long, narrow houses and many small workshops.

#### Fulling mills and water power

**Fulling** means beating woollen cloth in water with a special clay (fuller's earth) to clean it and thicken it. At first people

did it by trampling the cloth with their feet. A **fulling mill** used a waterwheel to lift wooden hammers.

- Fulling mills are recorded in Normandy in the late 11th century (about 1086). In England the first certain ones appear in a survey of 1185, on lands of the Knights Templar.
- A water-powered mill did the work of many fullers, so water fulling was much cheaper than fulling by foot.
- The historian Eleanora Carus-Wilson argued (1941) that fulling mills pulled the English cloth industry out of the old towns and into the countryside, where fast streams were. Later historians think she exaggerated, but the pattern of cloth villages along hill streams is real.

**MAP TIP**

Put fulling mills on fast streams in hilly country near sheep land. A cloth district on the map is a hill valley with a string of villages and fulling mills along the stream (draw each mill as a watermill wheel; English names for a fulling mill include "Walk Mill" and "Tuck Mill", and in Wales a *pandy*), a market town at the valley mouth where cloth is sold, and tenter grounds on the slopes.

## Dyes and mordants

| Dye or mordant | Colour or use                     | Where it came from                                                                                                                                                                                                         |
|----------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Woad</b>    | Blue                              | Thuringia: the five "woad towns" of Erfurt, Gotha, Tennstedt, Arnstadt and Langensalza; the Lauragais (the triangle between Toulouse, Albi and Carcassonne), which made Toulouse merchants rich; Picardy; parts of England |
| <b>Madder</b>  | Red                               | Zeeland and Flanders; many other lowland areas                                                                                                                                                                             |
| <b>Weld</b>    | Yellow                            | Widespread                                                                                                                                                                                                                 |
| <b>Kermes</b>  | Expensive crimson, from an insect | Mediterranean                                                                                                                                                                                                              |
| <b>Alum</b>    | Mordant (fixes colours)           | Phocaea in Asia Minor, run by Genoese merchants from the 1270s; after 1460–62, the papal mines at Tolfa near Rome                                                                                                          |

Woad needed rich, well-manured arable land and a woad mill to crush the leaves. Woad merchants in Erfurt helped found its university; in Toulouse they built grand town houses.

## Linen, cotton blends and silk

- **Linen** came from flax, grown on damp, rich soils. Retting (rotting the stems in ponds or streams) stank and polluted water. Main linen regions: Swabia and Lake Constance (Ulm, Konstanz, St Gallen), Westphalia and Flanders.
- **Fustian** (a cloth of linen and cotton) was made in southern German towns such as Ulm and Augsburg from the 14th century, with cotton imported through Venice.
- **Silk** came first from the Byzantine and Islamic worlds (Constantinople, Syria, Muslim Spain: Almería, later Granada). In Italy, Lucca led silk weaving in the 13th–14th centuries. After political trouble in Lucca in 1314,

many of its silk workers moved to Venice, Florence and Bologna.

**LATER ERA (1500s+)**

Protestant refugees from Flanders brought the "New Draperies" (lighter cloths) to English towns such as Norwich and Colchester. English cloth replaced raw wool as the country's great export.

**MAP TIP**

On a kingdom map, draw one or two textile regions, not more. Show sheep country (downs, hills, dry plateaus) with a few wool market towns; a cloth region with several close-set towns and mill streams; and, if you like, a dye district on good plains nearby ("Woad Country"). Silk belongs in rich southern or eastern cities that trade with the Byzantine or Islamic world. On a local map, add tenter grounds (an open field beside the town labelled "Tenters"), a cloth hall, dyers by the river below town and fulling mills on the side streams.

## Fishing

**RULE OF THUMB**

Fresh fish is eaten near the water. Preserved fish (salted, dried or smoked) travels across a continent. Big fisheries need salt, barrels and a market.

**Why:** The Church forbade meat on fast days: Fridays, often Wednesdays and Saturdays, and all of Lent. That is well over a third of the year. Inland towns needed huge amounts of preserved fish.

## Herring

- The great medieval herring fishery was off **Scania** (southern Sweden, then Danish). Every year from about 1200, a huge market was held on the beaches between the small towns of Skanör and Falsterbo. The official season is usually given as 24 August to 9 October, but in practice it could start in late July and run into late October.
- The Danish king gave Hanseatic towns their own plots of beach (a *fitte* or *vitte*) to gut, salt and pack fish. Counts vary, but around 25–30 towns had plots at the peak; the largest (Lübeck's and Danzig's) covered 6–10 hectares (15–25 acres) each. Lübeck dominated, bringing salt from Lüneburg and barrels.
- A good 14th-century year may have produced about 300,000 barrels of herring (an estimate). A contemporary claimed 300,000 people worked there during the season, surely an exaggeration. By one estimate, the market supplied about a third of the Danish king's income at its peak.
- The herring left Scania in the early 15th century, and the fishery moved to the North Sea.
- English herring centred on **Great Yarmouth**, with a large autumn herring fair.

## Stockfish and cod

- **Stockfish** is cod dried on wooden racks in cold, dry, windy northern springs, without salt. It keeps for years.
- The main source was **Lofoten** (northern Norway). Fishermen came every winter and spring to fishing stations. Traders held summer fairs at Vågan.
- The fish was shipped south to **Bergen**, which controlled the trade. Stockfish exports are estimated at about 4,000 tonnes a year in the 1300s. German Hanseatic merchants ran a trading post there (the *Kontor*, from about 1350–60). Nearly 1,000 Germans lived in its houses and warehouses on the main wharf, and more came in the summer sailing season.

## Fishing villages, weirs and curing

- **Fishing villages** sat on beaches where boats could be pulled up, in sheltered coves, and at river mouths. Many were seasonal stations with huts, not true villages.
- **Fish weirs** were fences of stakes or stone in rivers, estuaries and tidal shores that trapped fish. They often blocked boats. Magna Carta (1215, clause 33) ordered fish weirs removed from the Thames, the Medway and all of England, except on the sea coast. (Fishponds at manors and monasteries are in [chapter 06](#).)

- **Curing** needs salt, barrels (so coopers and timber), drying racks or smokehouses, and space on the shore.

### LATER ERA (1500S+)

Dutch "herring busses" (large decked fishing boats that salted the fish at sea) made Holland rich from the 1400s on. From the 1500s, Basque, Portuguese, French and English ships sailed to Newfoundland for cod.

### MAP TIP

Put herring markets on a beach or spit next to a narrow sea passage, with a small town or two and a large empty area labelled "fair ground" or "curing booths". Put stockfish stations along Arctic coasts, with their racks labelled "Fish Racks", and one big trading town with a German (or foreign) quarter at the end of the route. Draw fishing hamlets in every sheltered cove, roughly a few kilometres apart along good coasts, fewer along cliffs.

## Shipbuilding, naval stores and arsenals

### RULE OF THUMB

Shipyards need a sheltered shore, timber that can be floated in, and a town with skilled carpenters. A state arsenal is a walled factory inside a great port.

### What ships need:

| Material      | Use                      | Source                                                                      |
|---------------|--------------------------|-----------------------------------------------------------------------------|
| Oak           | Hull frames and planks   | Lowland forests near rivers; Baltic oak by sea                              |
| Pine and fir  | Masts, spars, decks      | Mountain and northern forests: the Alps, Scandinavia, the Baltic            |
| Tar and pitch | Waterproofing            | Pine forests of the north and east (see <a href="#">Forest industries</a> ) |
| Hemp          | Rope and caulking        | Damp farmland                                                               |
| Iron          | Nails, anchors, fittings | Iron districts                                                              |
| Sailcloth     | Sails                    | Linen and hemp regions                                                      |

**Small yards:** Most medieval ships were built on open beaches, creek sides and river banks near towns, not in special yards. Every port town had some.

### Great arsenals:

- **Venice Arsenal.** Begun around 1104 (the exact date is uncertain). Expanded with the *Arsenale Nuovo* (c. 1320) and again in 1473. It covered about 45 hectares (110 acres), around 15% of the city, and was surrounded by a wall about 3.2 km (2 mi) long. Popular accounts say it employed "some 16,000 people" in the early 1500s and could produce nearly a ship a day using standard parts. The regular skilled workforce was probably smaller; estimates vary.
- **Barcelona:** the royal shipyards (*Drassanes*), ordered by Peter III in 1285 and enlarged in 1328–90.

- **Seville:** the *Atarazanas* (shipyards), built under Alfonso X from 1252 outside the walls, by the river, with 17 long brick halls.
- **Rouen:** the French royal galley yard (*Clos des Galées*), founded in the 1290s (usually dated 1294) with Genoese help.

### MAP TIP

Put an arsenal at the edge of a big port city, on its own basin, behind its own walls. Draw a timber road or river from the forests to it. On a kingdom map, mark 1–3 royal shipyards or arsenals on the coast facing the realm's main enemy, each with an anchor plus a castle tower. Ordinary beach and creek yards need no symbol; on a local map, just label them "Shipyard". Add tar and timber ports in the northern forests if your realm builds ships.

## Metalworking, crafts, leather, drink and sugar

### Metal and craft towns

### RULE OF THUMB

Raw metal is made near the mine and the forest. Fine metal goods are made in towns with skilled workers, merchants and good roads.

| Town                               | Famous for                                         | Notes                                                                                                                                                                                                                                           |
|------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Milan</b>                       | Armour                                             | Probably the leading armour centre of the 15th century. Workshops such as that of the Missaglia family (famous in the 1400s) exported armour all over Europe. Workshops specialized in helmets, gauntlets and other parts. Brescia came second. |
| <b>Nuremberg</b>                   | Mail armour, metal goods, later firearms           | A trading city where many metal crafts met.                                                                                                                                                                                                     |
| <b>Passau and Solingen</b>         | Sword blades                                       | Passau's "running wolf" mark was already on its blades before 1300 and meant quality. Later, Solingen smiths copied it. Solingen lies in hilly, wooded country on the Wupper river, with many fast streams for grinding mills.                  |
| <b>Cologne</b>                     | Swords and mail (from the 12th century)            | A great Rhine trading city.                                                                                                                                                                                                                     |
| <b>Dinant</b><br>(Meuse valley)    | Brass and copper vessels, called <i>dinanderie</i> | Used zinc ore from near Aachen. Sacked by the Burgundians in 1466.                                                                                                                                                                              |
| <b>Toledo</b>                      | Sword blades                                       | Famous in the late Middle Ages and after.                                                                                                                                                                                                       |
| <b>Fabriano</b><br>(central Italy) | Paper                                              | Making paper by 1276, with water-powered mills. Paper making had reached Europe through Muslim Spain, where Xàtiva (near Valencia) made paper by the 12th century.                                                                              |

Metal craft towns sit on trade routes, close to (but not inside) iron or copper districts: Solingen near the Bergisch hills, Steyr at the edge of the Styrian iron district, Nuremberg between several mining regions.

### Tanneries and other dirty trades

- **Tanning** turned hides into leather by soaking them for many months in pits of water and crushed oak bark (tanbark), after removing hair with lime or dung. It smelled terrible and poisoned water.
- Tanneries needed running water, hides (from butchers and cattle markets) and oak bark (from coppiced woods, as in the Siegerland *Hauberg*).
- Towns often forced tanners to work outside the walls and downstream. In Bristol, tanners, dyers, fullers and weavers were grouped in the Redcliffe and Temple area across the river.

### Breweries

- Beer was brewed everywhere, but some towns brewed for export. Hopped beer, which keeps longer, spread from north German towns to the Low Countries and England in the 14th–15th centuries.
- **Hamburg** was called "the brewhouse of the Hanse". Around 1360 it had over 450 breweries, producing more than 25 million litres a year, most of it for export.
- **Einbeck** (Lower Saxony) exported strong beer from Antwerp in the west to Riga in the east, and from Stockholm in the north to Munich in the south. Its exports are recorded from 1351, and more than half of the town's houses had the right to brew. The word *bock* (a strong beer) is usually said to come from "Einbeck", as pronounced in Munich.

- Breweries need clean water, grain (barley, wheat or oats) and fuel. Put them inside towns, by wells or clean streams.

### Wine regions

#### RULE OF THUMB

Vineyards grow on sunny slopes above rivers. The river carries the wine to a port.

- **Bordeaux** (Gascony, ruled by English kings 1152–1453). The early 14th century was the peak of the Bordeaux wine trade. In good years Bordeaux shipped tens of thousands of tuns of wine (a tun is about 950 litres, or 252 gallons). The record usually quoted is about 100,000 tuns (102,724) in 1308–09; treat these as estimates from customs records. Hundreds of ships sailed from Gascony each year in the wine fleets. Wine from inland (the "high country") came down the Garonne and Dordogne rivers.
- **The Rhine and Moselle**. Rhenish wine was shipped down the Rhine through Cologne to the Low Countries, England and the Baltic.
- **Burgundy**. The Côte around Beaune. In 1395 Duke Philip the Bold banned the Gamay grape to protect the quality of Pinot Noir.
- Vines grew farther north than today. Domesday Book (1086) lists over 40 vineyards in southern England.

**MAP TIP**

Draw vineyards as hatched strips on south- or south-east-facing slopes along a river, with villages at the foot of the slope. Put a wine port where the river meets the sea or a big river, with warehouses and a customs house. Label the region's wine ("Claret", "Rhenish"). For beer, there is no need to draw anything; just note "famous breweries" for a town of 2,000 or more.

**Sugar estates (Mediterranean)****RULE OF THUMB**

Sugar is a southern crop with a factory attached. It needs a warm coastal plain, river water for irrigation, a mill, fuel and a port, all close together.

**Why:** Sugar cane needs a frost-free climate and a lot of water. Mediterranean summers are dry, so the fields must be irrigated from a river. Cane is heavy and bulky, so it is crushed near the fields, in a mill turned by water or by animals. The juice is then boiled in big copper cauldrons, which burns a lot of fuel. The finished sugar was a luxury, sold far away, so the estate needs a port.

Sugar came to the Mediterranean from the Islamic world. Crusaders met it in the Holy Land, and early in the 12th century Venice acquired villages near Tyre (in today's Lebanon) and set up sugar estates there. In the late Middle Ages the main Christian producers were Cyprus, Sicily and, from the 15th century, the coast of Valencia. Venice was Europe's chief sugar refining and trading centre in the 15th century.

**Real examples:**

- **Kolossi** (Cyprus). In 1210 King Hugh I gave the fief of Kolossi, about 60 villages from the fertile plain at the mouth of the Kouris River up into the hills, to the Knights Hospitaller. It became their main estate on the island. Beside the castle are the remains of a sugar factory, a 14th-century mill for crushing cane and an aqueduct that brought water to it. The tower that stands today was built in 1454. Kolossi is about 14 km (9 mi) west of the port of Limassol.
- **The great Cypriot owners.** Sugar was one of Cyprus's main exports. The three best-studied sugar works belonged to the island's biggest landowners: the Lusignan kings at Kouklia, the Hospitallers at Kolossi and the Venetian Cornaro family at Episkopi. The royal works at Kouklia-Stavros was built in the late 13th century and worked until the end of the 16th century.
- **Gandia** (Valencia, Spain). In the early 15th century, sugar-making methods were brought from Sicily with

the help of Sicilian masters. Around Gandia there were 4 sugar mills by 1433 and 14 by about 1500. In 1500 the industry there employed some 500 people and 220 animals.

**LATER ERA (1500S+)**

Sugar moves out into the Atlantic: to Madeira (cane brought in by the Portuguese in the 1420s; a large producer by the 1490s), the Canary Islands, and then Brazil and the Caribbean, on plantations worked by enslaved Africans. The Mediterranean sugar estates, with their higher costs, slowly decline.

**MAP TIP**

In a warm southern kingdom, draw one or two sugar districts on irrigated coastal plains: cane fields as small enclosed plots (the enclosed-fields symbol), thin blue irrigation channels from a river (drawn like a mill leat, thinner than a canal), and a sugar mill and boiling house fed by an aqueduct. Draw a water-driven mill as a water-mill wheel labelled "Sugar Mill". Put a fortified estate tower beside the works, drawn as a small solid square (the tower-house symbol); if a military order owns the estate, as the Hospitallers owned Kolossi, use the commandery symbol instead, a small church with a shield. (The older castle at Kolossi was wrecked by Mamluk raiders in the 1420s.) Put a loading port within a short cart trip, like Kolossi and Limassol. A sugar district is a cluster of several mills, not one big town; name the estate as well as the mills ("the Commandery").

## Where industry sits in and around a town

### RULE OF THUMB

Clean water upstream, dirty trades downstream. Fire and smoke outside the walls. Big land-hungry works in the suburbs and fields.

| Trade                                                       | Where                                                                      | Why                                 |
|-------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|
| Drinking water, breweries                                   | Upstream, or at wells and springs                                          | Needs clean water                   |
| Grain mills, fulling mills                                  | On the river at weirs, above and below town; on streams in the countryside | Need a fall of water                |
| Butchers and slaughterhouses                                | At the edge of town, near the river                                        | Blood and waste dumped in the water |
| Tanners, dyers, flax retting                                | Downstream and outside the walls                                           | Smell and pollution                 |
| Smiths, potters, brick and tile kilns, lime kilns, glass    | Outside the walls or at the edge                                           | Fire risk and smoke                 |
| Tenter grounds, rope making, timber yards, bleaching fields | Suburbs and open fields                                                    | Need a lot of space                 |
| Smelters, charcoal, glasshouses                             | At the mine and in the forest, not in town                                 | Fuel and ore are too heavy to move  |
| Shipyards, timber yards                                     | Waterfront, often at the edge of the port                                  | Need shore and space                |

Street names often record these trades: Tanner Street, Dyers' Lane, Shambles (butchers), and in Florence the *Corso dei Tintori* (dyers' street) by the Arno.

### MAP TIP

On a local map of a town and its surroundings, draw the river flowing through. Upstream: mills and a water intake. In town: markets, brewers, guild halls. Just below town: butchers and tanners. Outside the walls: kilns, tenter grounds and rope walks. Then draw a few kilometres more: fulling mills on side streams, quarries and clay pits on the slopes, charcoal hearths in the woods.

## Special town types and how to spot them

Most towns are market towns. These are the special ones.

Each has a "signature" a reader can spot on the map.

| Town type              | How to spot it on the map                                                                                                     | Typical size                                                    | Where it sits                                           | Real examples                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|
| <b>Mining town</b>     | Up a valley in the hills; spoil heaps, shafts, mint, big church; little farmland                                              | 1,000–5,000; boom towns 10,000–20,000                           | Ore-bearing hills, near wood and fast water             | Goslar, Freiberg, Kutná Hora, Kremnica, Iglesias    |
| <b>Salt town</b>       | Brine spring or salt mine; boiling houses; salt roads radiating out                                                           | Small to middle-sized, but rich for its size                    | Over salt beds; sea-salt pans on warm, sheltered coasts | Lüneburg, Droitwich, Halle, Salins, Wieliczka, Ston |
| <b>Port</b>            | Harbour, quays, warehouses, lighthouse or harbour chain                                                                       | Any size; the largest cities were ports                         | Estuaries, bays, river mouths                           | See <a href="#">chapter 05</a>                      |
| <b>Fishing village</b> | Beach or cove, drying racks or smokehouses (label them; do not draw boats, because a small boat marks the head of navigation) | 50–500 (a village or hamlet); seasonal stations swell in season | Sheltered coast, river mouths                           | Lofoten stations, Skanör and Falsterbo              |

| Town type                            | How to spot it on the map                                                                               | Typical size                                           | Where it sits                                    | Real examples                                                                  |
|--------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Cloth town</b>                    | Cloth hall, tenter grounds, fulling mills on nearby streams, dyers by the river                         | 2,000–10,000 usually; the Flemish giants 20,000–65,000 | Near wool, water power and trade routes          | Ghent, Bruges, Ypres, Florence, English cloth towns                            |
| <b>Fair town</b>                     | Big open fairground (often a meadow outside the walls), many inns and warehouses, several roads meeting | Often only a small or middle town                      | Where regions and routes meet                    | Troyes, Provins, Lagny, Bar-sur-Aube; Frankfurt; Stourbridge Fair at Cambridge |
| <b>Iron or craft town</b>            | Forges and grinding mills along streams; guild halls                                                    | 1,000–10,000; bigger trade cities                      | Edge of iron or wood districts, on a trade route | Steyr, Solingen, Dinant, Nuremberg                                             |
| <b>Wine town</b>                     | Vineyards on slopes behind, cellars, river quay                                                         | Small; the export port can be large                    | South-facing river valleys                       | Beaune; villages along the Rhine and Moselle; Bordeaux as port                 |
| <b>Arsenal or naval town</b>         | Walled shipyard basin, rope walks, timber stores                                                        | Part of a big port                                     | Sheltered coast facing the main sea enemy        | Venice, Barcelona, Seville                                                     |
| <b>Spa (bath) town</b>               | Hot or mineral springs, bath houses, many inns                                                          | Small                                                  | Over hot springs, often in hills                 | Bath, Aachen, Baden (Switzerland), Karlsbad (Karlovy Vary, privileges 1370)    |
| <b>University or pilgrimage town</b> | Colleges, great shrine church, hostels                                                                  | Varies                                                 | Varies                                           | See <a href="#">chapter 08</a>                                                 |

### Fair towns

A **fair** was a big market held once or a few times a year, often for days or weeks, where long-distance merchants met. A **market** was weekly and local (see [Trade nodes](#)).

- The **Champagne fairs** (eastern France, late 12th and 13th centuries) were six fairs a year in four towns (Lagny, Bar-sur-Aube, Provins and Troyes) where Flemish cloth met Italian merchants. Their calendar, trading phases and decline are described in [The Champagne fairs](#).
- Later great fairs: Frankfurt (autumn fair privileged in 1240, a spring fair added in 1330), Geneva and then Lyon in the 15th century, Medina del Campo in Castile.
- Fair towns are often **not** big cities. Their signature is a large open space, many inns and storage buildings, and good roads from several directions. Neighbouring Champagne fair towns were about 45–65 km (28–40 mi) apart in a straight line, two or three days by road, so merchants could move from one fair to the next.

### Spa towns (briefly)

Hot springs attracted bathers from Roman times. Medieval spa towns were small but busy with visitors: Bath in England, Aachen (Charlemagne's favourite residence), Baden in Switzerland (described as a lively resort by the Italian scholar Poggio Bracciolini in 1416) and Karlsbad in Bohemia. In the Islamic world, public bath houses (*hammams*) stood in every city, so baths alone do not make a spa town there.

### Suggested symbols

These symbols match the guide's [master legend](#).

Many small works (kilns, racks, forest clearings) are best shown with a word, not a picture. A new little picture can easily look like a symbol that already means something else: a small tower reads as a castle, a mound as a barrow, a boat as the head of navigation.

| Feature                      | Symbol                                                                                                                        | Show at which scale                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Mine or mining district      | Crossed hammers (or a pick)                                                                                                   | Continent (3–6 districts), kingdom, local |
| Mining town                  | Its usual town symbol, with crossed hammers beside it                                                                         | Kingdom, local                            |
| Old shaft or worked-out mine | Crossed hammers drawn grey, labelled "(old shaft)"                                                                            | Local                                     |
| Mint or coinage town         | No extra symbol: add "(mint)" or "(coinage town)" to the town's label                                                         | Kingdom, local                            |
| Salt pans                    | Checkerboard of small rectangles along the shore                                                                              | Kingdom, local                            |
| Brine spring or hot spring   | Small blue dot, labelled "Brine Pit" or "Hot Spring"; on a kingdom map, add "(salt)" or "(baths)" to the town's label instead | Kingdom (label only), local               |

| Feature                                                                           | Symbol                                                                                                                                                                      | Show at which scale          |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Salt road                                                                         | The ordinary road line, labelled "Salt Road" or "Saltway" on its first stretch                                                                                              | Kingdom, local               |
| Quarry                                                                            | Notch in a hillside, with a track to water                                                                                                                                  | Kingdom (famous ones), local |
| Lime, brick or tile kilns; clay pits                                              | Words only: "Limekiln", "Brick Kilns", "Clay Pits"                                                                                                                          | Local                        |
| Water-powered works (forge, furnace, ore-crushing mill, fulling mill, sugar mill) | Watermill wheel on the stream, labelled with the trade ("Forge", "Furnace", "Walk Mill", "Sugar Mill"); a hammer pond as one blue pond with a straight dam at its lower end | Local                        |
| Charcoal hearths, glasshouse, tar kilns                                           | A clearing in the wood symbol, labelled with the trade                                                                                                                      | Local                        |
| Peat diggings (turbary)                                                           | Commons symbol (tufts or stipple), labelled "Turf Fen"; flooded pits as blue lakes with straight, blocky edges                                                              | Local                        |
| Drowned village                                                                   | Grey church standing in the water, labelled "(lost village)"                                                                                                                | Kingdom (a few), local       |
| Tenter grounds, fair field, fish racks                                            | Open ground labelled "Tenters", "Fair Field" or "Fish Racks"                                                                                                                | Local                        |
| Fair town                                                                         | Its usual town symbol, with the fair months in the label ("Provins: May and September fairs")                                                                               | Kingdom                      |
| Vineyards                                                                         | Hatched strips on south-facing slopes                                                                                                                                       | Kingdom, local               |
| Port (for wine, coal, fish or timber)                                             | Large anchor with a quay (head port), or a small anchor (member port)                                                                                                       | Continent, kingdom           |
| Arsenal or royal shipyard                                                         | Anchor plus a castle tower                                                                                                                                                  | Continent, kingdom           |
| Ordinary shipyard                                                                 | The label "Shipyard" on the shore; no symbol                                                                                                                                | Local                        |
| Fortified estate tower (for example on a sugar estate)                            | Small solid square (tower house); a small church with a shield if a military order owns it                                                                                  | Local                        |

**FANTASY TWIST**

Flying mounts or teleport circles make light, valuable goods (gold, silver, silk, potions) even less tied to roads, but bulk goods (salt, grain, stone, timber) still follow water. So magic adds new special towns (an airship yard, a mage-crafting town at a crystal mine) without removing the old ones. See [chapter 10](#).

**MAP TIP**

Every special town needs its "why" visible on the map: a mine symbol, salt pans, a fairground, vineyards, a harbour. Use the symbols in the table above, and give each a name that hints at its trade (Silverdale, Saltwich, Clothford, Forgeham). On a kingdom map, aim for something like 1–3 mining districts, 1–3 salt sources, one cloth region, one or two fair towns and several ports. That is enough to make the economy feel real without crowding the map.

## Quick summary

- Industry goes to the heaviest input: ore, fuel (wood, charcoal, peat, coal) or water power. Only the light, finished product travels.
- Mining towns sit up valleys in old hill country, buy their food, have special privileges (free mining, own courts, a mint) and boom and bust. Most had 1,000–5,000 people; rare boom towns reached 10,000–20,000.
- Iron is limited by charcoal: very roughly 50–150 kg of wood per 1 kg of bloomery iron (estimates vary). A Wealden blast furnace needed the woods of about a 5 km (3 mi) radius.
- Every realm needs salt: brine springs (with huge fuel needs), rock-salt mines, or sea-salt pans on warm coasts. Salt roads run from the source to the nearest navigable water.
- Stone travels far only by water; clay lowlands build in brick. Kilns and clay pits sit outside the walls.
- Forests are industrial: charcoal hearths, forges with ponds, moving glasshouses (150–200 kg of wood per kg of glass), tar and potash.
- In wet, treeless lowlands people dig peat, and big diggings flood into blocky lakes: the 63 Norfolk Broads (dug 12th–14th centuries) and the ever-growing Haarlemmermeer.

- Cloth needs wool, fulling water and dyes. Fulling mills put cloth villages along fast hill streams; great cloth towns (Ghent perhaps 40,000–65,000; Florence) imported wool.
- Preserved fish travels far: Scania herring (a beach market every autumn), Lofoten stockfish through Bergen (about 4,000 tonnes a year in the 1300s).
- Shipyards are everywhere along shores; state arsenals (Venice, Barcelona, Seville) are walled factories in great ports.
- Clean water upstream, dirty trades (tanners, dyers, butchers) downstream and outside the walls; fire trades at the edge.
- Give every special town a visible reason on the map: a mine, pans, a fairground, vineyards, a harbour, or (in the warm south) a sugar mill among irrigated cane fields.

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## CHAPTER 8

# RELIGIOUS, CULTURAL, LEGAL AND ANCIENT SITES

Castles and markets show who rules and who trades. Churches, monasteries, shrines, gallows and ruins show what people believe, how they are judged and how old the land is. This chapter tells you where to put these sites, how many to draw, how far apart they stand, and why.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500), with examples from the Mediterranean and the Islamic world. Most counts come from England, because English records are the best. Village and town numbers are in [Population and Settlement Sizes](#). Coronation and royal burial churches are in [Capitals, Realms and Borders](#).

## IN THIS CHAPTER

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## The religious map at a glance

## RULE OF THUMB

Every village has a church. Every town has a hospital. Every big town has friars. Every region has a few monasteries. Every kingdom has a handful of cathedrals and one or two universities.

The Church was the biggest landowner and builder in medieval Europe. Bishops and abbots held roughly a

quarter of England in 1086 (estimates vary; see [chapter 03](#)). Clergy were only about 1–2% of the people, but their buildings were everywhere.

### How many of each, per area

These densities come from England, where the counts are good. England covers about 130,000 km<sup>2</sup> (50,000 sq mi).

| Feature                                       | Number in England                               | One per ...                             | Per 1,000 km <sup>2</sup><br>(386 sq mi) | Notes                                                |
|-----------------------------------------------|-------------------------------------------------|-----------------------------------------|------------------------------------------|------------------------------------------------------|
| Parish church                                 | ~9,500 (c. 1300)                                | ~14 km <sup>2</sup> (5.4 sq mi)         | ~73                                      | About 4 km (2.5 mi) church to church in the lowlands |
| Market (chartered)                            | 1,746 recorded by 1300                          | ~75 km <sup>2</sup> (29 sq mi)          | ~13                                      | Shown for comparison (baseline)                      |
| Religious house (monks, canons, nuns, friars) | ~900 (1530s)                                    | ~145 km <sup>2</sup> (56 sq mi)         | ~7                                       | Average only ~13 people each                         |
| Hospital or almshouse                         | 800–1,100 founded c. 1080–1540 (estimates vary) | ~120–160 km <sup>2</sup> (46–62 sq mi)  | ~6–8                                     | Not all open at the same time                        |
| Castle in use                                 | 500–600 at one time (England and Wales)         | ~250–300 km <sup>2</sup> (95–115 sq mi) | ~3–4                                     | See <a href="#">chapter 04</a>                       |
| Leper house                                   | ~300 (estimates vary)                           | ~430 km <sup>2</sup> (166 sq mi)        | ~2                                       | Mostly founded 1100–1350                             |
| Friary                                        | 183 (1530s)                                     | ~700 km <sup>2</sup> (270 sq mi)        | ~1.4                                     | Only in towns                                        |
| Cathedral                                     | 17 (from 1133)                                  | ~7,600 km <sup>2</sup> (2,950 sq mi)    | 0.13                                     | Plus 4 in Wales                                      |
| University                                    | 2 (Oxford, Cambridge)                           | ~65,000 km <sup>2</sup> (25,000 sq mi)  | 0.015                                    | Very rare everywhere                                 |

Take a sample county of about 2,500 km<sup>2</sup> (965 sq mi), a square 50 km (31 mi) on a side. At English densities it would have:

- about 180 parish churches;
- 15–20 religious houses;
- 3–4 friaries in its 1–3 biggest towns;
- 15–20 hospitals and almshouses over time;
- about 6 leper houses;

- 8–10 castles in use;
- about a one-in-three chance of a cathedral.

#### MAP TIP

Do not draw all of these at every scale. On a kingdom map, show cathedrals, great abbeys, major shrines and universities. On a local map, show every parish church, plus the abbeys, friaries, hospitals, gallows and ruins. See [Drawing it all](#).

## Parish churches and chapels

### RULE OF THUMB

One parish, one church, one priest. In good farmland the churches stand about 3–5 km (2–3 mi) apart. In hills they stand about 7–10 km (4–6 mi) apart, more in the emptiest uplands, with small chapels in between.

### Why every village has a church

A **parish** is the area served by one church and its priest. The people of the parish paid **tithes** (one-tenth of their crops

and young animals) to support the church. In return the parish church had the right to baptise their babies and bury their dead. Most English and French parishes were fixed by about 1200 and then changed very little for 600 years.

So a parish needed enough farmland to feed a priest from one-tenth of its harvest. That is why parishes are small in rich land and huge in poor land.

## Numbers

| Place and date                    | Parishes                                               | Average size                            | Notes                                                                                                                               |
|-----------------------------------|--------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| England c. 1300                   | ~9,500                                                 | ~14 km <sup>2</sup> (5.4 sq mi)         | Lowland parishes 5–15 km <sup>2</sup> (2–6 sq mi); upland parishes 50–100+ km <sup>2</sup> (20–40+ sq mi)                           |
| France (late Middle Ages to 1789) | roughly 40,000 (an estimate; counts vary, some higher) | ~13 km <sup>2</sup> (5 sq mi)           | Most of France's modern communes grew out of these parishes                                                                         |
| Northern and central Italy        | Older system: the <i>pieve</i>                         | A <i>pieve</i> covered several villages | A <i>pieve</i> was a rural "mother church" with the only font (baptism basin); smaller chapels around it served the villages        |
| Medieval towns in England         | Many tiny parishes                                     | Some under 1 ha (2.5 acres)             | London had over 100 parish churches inside its walls by the 1100s; Norwich about 56; Lincoln over 40; York ~45 in 1300 (39 by 1428) |

Old English towns like London, Norwich, York and Lincoln were full of tiny parishes, because rich townsmen founded many small churches before 1200. Norwich packed 56 parish churches into about 1.5 sq mi (4 km<sup>2</sup>) inside its walls. New planted towns of the 1200s usually got only one church, and sometimes only a chapel that belonged to an older village parish.

### Chapels of ease and other chapels

A **chapel of ease** is a smaller church inside a parish, built for people who live too far from the parish church. It usually had no right to bury the dead or baptise, so people still went to the mother church for those.

- **Where:** in hamlets and distant villages of big upland parishes, and in new settlements on cleared land. Many were founded from the late 1200s by the villagers themselves.
- **Private chapels:** inside castles, manor houses, hospitals and bridges (bridge chapels collected tolls and prayers for travellers).

- **Wayside crosses and shrines:** at crossroads, on parish boundaries and on moorland tracks. In Catholic Central Europe small roadside shrines are common.
- **Holy wells:** springs dedicated to a saint, common in Wales, Cornwall, Ireland and Brittany.

### Where the church stands in the village

- On the **highest dry spot** in or next to the village, often slightly apart from the houses.
- **Next to the manor house.** The lord often built the first church, and the two buildings stand side by side.
- On an **older sacred site.** At Rudston (Yorkshire) the Norman church stands beside a prehistoric standing stone about 7.6 m (25 ft) tall in its churchyard.
- In towns, near the **market place**, and at the gates.
- If the village later moved or died, the church often stayed behind, alone in the fields. A lonely church is a sign of a deserted or shrunken village (see [chapter 01](#)).

### Church towers as landmarks

Church towers and spires were the tallest things most people ever saw. Travellers and sailors steered by them.

| Building                                                                          | Height                         | How far you can see the top (flat land, clear day, eye at 1.7 m)                                                     |
|-----------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Small village tower                                                               | 15 m (50 ft)                   | up to ~18 km (11 mi)                                                                                                 |
| Large village or town tower                                                       | 25–40 m (80–130 ft)            | up to ~22–27 km (14–17 mi)                                                                                           |
| Boston "Stump", Lincolnshire (tower built c. 1450–1520)                           | 81 m (267 ft)                  | up to ~37 km (23 mi); a landmark for travellers in the Fens and ships in the Wash                                    |
| Lincoln Cathedral spire (tower finished 1311, spire added after; blown down 1548) | reputedly about 160 m (525 ft) | up to ~50 km (31 mi); possibly the tallest building in the world in its day, though some historians doubt the height |

The distances above are the geometric limit (the curve of the earth). Haze, trees and hills cut them a lot. In real life a village tower is a useful landmark for 5–10 km (3–6 mi). In flat lowland, where churches stand about 4 km (2.5 mi) apart, you can usually see several neighbouring towers from any one of them.

LATER ERA (1500s+)

Parishes barely change. In Protestant lands, side altars, chantries (endowed chapels for prayers for the dead) and many shrines go; England dissolved its chantries in 1547. Printed county maps such as Christopher Saxton's atlas of England and Wales (1579) mark towns and villages with small church-tower symbols, because the church was what defined a place.

MAP TIP

On a local map, give each village one church symbol: a small church with a tower. Use a spire for richer towns, and a small plain cross with no tower for chapels. In hill country, draw fewer churches and add 2–4 chapel symbols per parish in the outlying hamlets. In an old town, scatter many small church symbols inside the walls; in a planted grid town, draw only one.

Dioceses, cathedrals and archbishops

RULE OF THUMB

Old Christian lands around the Mediterranean have many small dioceses. Newly converted lands in the north and east have few huge ones. So in Italy a cathedral may sit in a town of 2,000 people; in England it means one of the realm's main cities.

A **diocese** (also **see** or **bishopric**) is the area ruled by one bishop. His main church is the **cathedral**, which holds his *cathedra* (throne). An **archbishop** is the senior bishop of a **province**, a group of dioceses.

Numbers and sizes

| Region                          | Dioceses                                                               | Average area                                                                     | Notes                                                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| England                         | 17 from 1133 (plus 4 in Wales)                                         | ~7,600 km <sup>2</sup> (2,950 sq mi)                                             | 2 archbishops: Canterbury and York. Lincoln diocese ran from the Thames to the Humber, over 200 km (125 mi).                                                           |
| France (within today's borders) | roughly 130–150 over 1200–1500 (the number changed over time)          | ~3,700–4,200 km <sup>2</sup> (1,450–1,650 sq mi)                                 | About 15 archbishops (Reims, Sens, Rouen, Tours, Lyon, Bourges, Bordeaux, Narbonne and others). Many small sees in the south; Pope John XXII created new ones in 1317. |
| Umbria (central Italy)          | 12                                                                     | ~850 km <sup>2</sup> (330 sq mi); range 100–2,500 km <sup>2</sup> (40–965 sq mi) | Terni's diocese covered only ~100 km <sup>2</sup> (40 sq mi). Città di Castello was a see with about 5,000 people.                                                     |
| Italy as a whole                | Hundreds                                                               | Often under 1,000 km <sup>2</sup> (390 sq mi)                                    | Italy still had about 325 dioceses before the mergers of 1986 (101 were suppressed then); many were of medieval origin. Southern Italy had the most.                   |
| Holy Roman Empire (c. 1500)     | Roughly 50 bishops and archbishops (counts vary with date and borders) | ~15,000–20,000 km <sup>2</sup> (5,800–7,700 sq mi) (estimate)                    | Many bishops were also princes ruling land (see chapter 03).                                                                                                           |
| Scandinavia                     | Few                                                                    | Very large                                                                       | Archbishoprics at Lund (1103/04), Nidaros/Trondheim (1150s) and Uppsala (1164).                                                                                        |
| Scotland                        | ~13                                                                    | Large                                                                            | No archbishop until St Andrews (1472) and Glasgow (1492).                                                                                                              |

Italian sees stayed even when the town shrank. In Tuscany, Grosseto fell to about 1,500 people after 1350, and

Why dioceses differ so much

The first bishops sat in Roman cities. Each Roman *civitas* (city with its own territory) got a bishop. Italy and southern France had many small Roman cities, so they got many small dioceses. Lands converted later (England, Germany, Scandinavia, Poland, Hungary) had few towns, so a bishop's diocese was drawn around a whole kingdom, tribe or county.

Cortona from about 10,000 to about 3,000, but both kept their bishops (Paganelli 2025).

### Cathedrals move to towns

Bishops wanted to sit in towns. After the Norman Conquest, the Council of London (1075) moved English sees out of villages: Selsey to Chichester, and Sherborne to Old Sarum. Dorchester-on-Thames moved to Lincoln (1072) and Thetford to Norwich (1090s). Old Sarum later moved again, down to the river at Salisbury (from 1220; see [chapter 01](#)). A small village with a "-minster" name or a ruined great church may be a former see.

### What a cathedral city looks like

- A cathedral much bigger than any parish church. Winchester Cathedral is about 170 m (560 ft) long. Cluny's great abbey church in Burgundy (not a cathedral) was the largest church in Europe until the 1500s.
- A walled **close** (precinct) around it, with houses for the canons (the cathedral clergy), the bishop's palace, a school and often a gate that was locked at night.

- In Islamic cities, the equivalent is the **Friday mosque** (the main congregational mosque), in the centre beside the main market; big cities later had several.
- In England, nine cathedrals were also Benedictine monasteries (Canterbury, Durham, Winchester, Norwich, Ely, Worcester, Rochester, Bath and Coventry), and Carlisle was served by Augustinian canons. This was rare elsewhere.
- Extra business: courts of the church, lawyers, pilgrims, craftsmen, and a big building site for centuries.

#### MAP TIP

On a kingdom map, mark each cathedral with a two-towered church or a mitre, and each archbishop's seat with a double cross beside the city symbol. For an English-style realm, use about one diocese per 5,000–10,000 km<sup>2</sup> (2,000–4,000 sq mi). For an Italian-style land, use one per 500–2,000 km<sup>2</sup> (190–770 sq mi), and let many cathedral towns be small. Draw diocese borders only on special church maps; they often follow old Roman or tribal borders, not the current political ones.

## Monasteries, friaries and military orders

#### RULE OF THUMB

Monks go to the countryside. Canons go to the edges of towns. Friars go inside towns. The more friaries a town has, the bigger it is.

## The orders and where they built

| Order                                                                                          | Founded                                                               | Where they built                                                                                                   | Typical size                                                                                 | England (1530s or total)                                                                                 |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Benedictines</b> (monks following the Rule of St Benedict, c. 530s)                         | Old houses from before 1000                                           | Often in or beside towns that grew up at their gates (Bury St Edmunds, St Albans, Peterborough)                    | 20–60 monks in big houses                                                                    | Part of ~260 houses of monks                                                                             |
| <b>Cluniacs</b> (reformed Benedictines)                                                        | Cluny, Burgundy, 910                                                  | Network of priories under one abbey; 314 houses paid allegiance to Cluny in the 1100s                              | Small priories of 5–20 monks                                                                 | A few dozen small priories                                                                               |
| <b>Cistercians</b> ("white monks")                                                             | Cîteaux, Burgundy, 1098                                               | Remote valleys with a river, "far from the concourse of men"; they drained land and built water channels and mills | Rievaulx under Abbot Aelred (1160s): 140 monks and 500 lay brothers                          | About 75–80 abbeys in England and Wales in all (13 of them in Wales)                                     |
| <b>Carthusians</b> (hermit monks)                                                              | Grande Chartreuse, French Alps, 1084                                  | The most isolated sites: mountains, woods, the foot of escarpments. Each monk lived alone in a cell with a garden  | 12–24 monks plus lay brothers; Mount Grace (1398) had 17 monks' cells and 6 for lay brothers | Only 9 charterhouses                                                                                     |
| <b>Augustinian canons</b> (priests living by a rule, serving parishes and hospitals)           | Spread from c. 1100                                                   | Near or in towns, at hospitals, at shrines, at castles                                                             | Often 6–20 canons                                                                            | Part of ~300 houses of canons                                                                            |
| <b>Premonstratensian canons</b> ("white canons", with a strict rule close to the Cistercians') | Prémontré, 1120                                                       | Rural, like Cistercians                                                                                            | Small                                                                                        | Part of the canons above                                                                                 |
| <b>Nuns</b> (all orders)                                                                       | —                                                                     | Mostly rural and small; often poorer than male houses                                                              | Average ~14 nuns                                                                             | 142 nunneries, ~2,000 nuns                                                                               |
| <b>Friars:</b> Dominicans, Franciscans, Carmelites, Austin Friars                              | 1210s–1250s; in England from 1221 (Dominicans) and 1224 (Franciscans) | Inside towns, usually on cheap land at the edge: near the walls, at a gate, or in a suburb                         | 10–60 friars                                                                                 | 183 friaries, ~3,000 friars                                                                              |
| <b>Military orders:</b> Templars (suppressed 1307–12), Hospitallers                            | 1110s–1120s                                                           | Commanderies (also called preceptories): farm estates with a chapel in good farmland, plus houses in big towns     | A few brothers and many servants                                                             | By about 1300 the Templars had at least 970 houses (commanderies and castles) across Europe and the East |

Across England in the 1530s there were nearly 900 religious houses with about 12,000 people: 4,000 monks, 3,000 canons, 3,000 friars and 2,000 nuns. That is only about 13 people per house on average. Most houses were small; a few great abbeys were very large.

### Cistercian valleys

Cistercians wanted three things: isolation, running water and flat land to build on. They founded abbeys in valleys of Yorkshire, Wales, Burgundy and Castile that had few people. At Rievaulx (1132) the monks moved the River Rye more than once in the 1100s to get enough level ground. The order grew fast: about 333 abbeys by 1152, when the order tried to halt new foundations, and nearly 750 at its height in the 1400s.

They often cleared or moved existing villages to get their "desert". So a Cistercian abbey on your map should sit in an empty valley, with no village at its gate.

### Granges: the monastery's farms

A **grange** is an outlying farm owned by a monastery. Early Cistercian rules said a grange should be within a day's walk of the abbey, so the **lay brothers** (working brothers who farmed it) could come back for Sunday and feast-day services. The rules gave no exact distance. A day's walk is about 25–35 km (15–22 mi) (see the travel speeds in [Trade Routes, Roads and Transport](#)). The rule was often broken.

- Fountains Abbey (Yorkshire) had 39 granges by the early 1300s. Its grange at Cowton lay about 20 mi (32 km) north of the abbey.
- Morker, a Fountains grange, held about 40 cows, 5 teams of oxen and 300–400 sheep in the late 1400s.
- Granges specialised: sheep runs, cattle farms, fisheries, iron or lead works, salt.

Farm layouts are in [Villages, Farms and the Countryside](#). Monastic mines and mills are in [Industry, Resources and Special Towns](#).

### Friaries tell you how big a town is

Friars lived by begging and preaching, so they needed many people nearby. In 1968 the historian Jacques Le Goff

suggested using the number of friaries as a rough measure of how big and important a French town was. The table below is a simplified rule of thumb in that spirit. Treat the sizes as rough guides, not exact figures.

| Number of friaries         | Town size (rough guide, France)                    |
|----------------------------|----------------------------------------------------|
| 0                          | Small towns of 2,000–3,000 people usually had none |
| 1                          | Towns of roughly 5,000 (estimate)                  |
| 2                          | Towns of about 10,000                              |
| 4 (one of each main order) | Big towns of 20,000–40,000                         |
| 5+                         | The great cities (London, Paris, Florence)         |

In England, friars reached further down the town ladder: many county towns of 3,000–5,000 had one or two friaries.

### Outside Latin Europe

- **Orthodox monasteries** favour mountains and islands. Mount Athos in Greece has 20 great monasteries; the Great Lavra was founded in 963. Meteora in Thessaly perched monasteries on rock pillars from the 1300s.
- **Islamic lands** had no monks, but they had **Sufi lodges** (*khanqah*, *zawiya*, *tekke*) in towns and along roads, and **ribats**: fortified posts on frontiers and coasts (North Africa, the Byzantine border) that later became hostels and Sufi centres.

#### LATER ERA (1500S+)

Protestant rulers close the monasteries: Sweden from 1527, Denmark in 1536, England and Wales in 1536–41, and many German states. Abbey churches become parish churches, country houses or roofless ruins; the land goes to nobles. In Catholic lands, new orders such as the Jesuits (1540) build colleges and churches inside towns, not in the countryside. See [What Changes After 1500](#).

#### FANTASY TWIST

A religious order of warrior-monks in a monster-haunted land behaves like the Templars or the Teutonic Order: fortified commanderies along dangerous roads and frontiers, each with farms to pay for it. Treat the fortified ones as castles on your map and use a castle symbol (see [chapter 04](#)); keep the small church with a shield for ordinary farm commanderies.

#### MAP TIP

Put about 1 religious house per 150 km<sup>2</sup> (60 sq mi) in a settled kingdom. Place Cistercian and Carthusian houses in empty, well-watered valleys away from villages. Put Benedictine abbeys in old towns or at the centre of a "monastery town". Put friaries in towns of about 3,000–5,000 and up, near the walls or gates, and nunneries in quiet rural spots. Give each big abbey 5–40 granges, most within about 25 km (15 mi) and a few much farther. Name some farms "Grange", "Temple ..." or "... Abbey Farm". On a local map, a great abbey is a walled precinct 350–600 m (0.2–0.4 mi) across, not a single church; see [Footprints](#).

## Pilgrimage: shrines, routes and hostels

#### RULE OF THUMB

Pilgrims use existing roads. A famous shrine pulls a chain of hospices, bridges and small towns along its route, about one day's walk apart.

A **shrine** is a holy place, usually the tomb or relics (holy remains) of a saint. A **pilgrimage** is a journey to a shrine for prayer, healing or forgiveness. Every region had small local shrines; a few became international.

## The great pilgrimages

| Destination                             | What drew people                                          | Key dates                                                       | Route and numbers                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Jerusalem</b>                        | Christ's tomb (Holy Sepulchre) and the holy places        | Crusader rule 1099–1187; pilgrim galleys from Venice afterwards | Venice to Jaffa by galley took 48 days in 1480 (baseline)                                                                                                                                                                                                                                                |
| <b>Rome</b>                             | Tombs of Saints Peter and Paul                            | First Jubilee (holy year) 1300                                  | Chronicler Giovanni Villani, who was there, claimed that Rome held about 200,000 pilgrims all through 1300, besides its own people. This was a guess, not a count                                                                                                                                        |
| <b>Santiago de Compostela</b> (Galicia) | Tomb of St James, "found" in the 800s                     | Guidebook in the <i>Codex Calixtinus</i> , c. 1140              | Four French roads (from Paris/Tours, Vézelay, Le Puy and Arles) join at Puente la Reina; about 800 km (500 mi) from Roncesvalles at the Pyrenees                                                                                                                                                         |
| <b>Canterbury</b>                       | Thomas Becket, killed 1170, made a saint 1173             | Shrine 1220–1538                                                | About 90 km (56 mi) from London                                                                                                                                                                                                                                                                          |
| <b>Cologne</b>                          | Relics of the Three Kings, brought from Milan in 1164     | 1164 onward                                                     | On the Rhine trade route                                                                                                                                                                                                                                                                                 |
| <b>Mecca</b> (Hajj)                     | The Kaaba; one pilgrimage is a duty for every able Muslim | Every year                                                      | Great caravans from Cairo, Damascus and Baghdad. The Darb Zubayda from Kufa (Iraq) was a desert road of wells, cisterns and stations. It was improved under the Abbasid caliphs (late 700s–early 800s) and named after Zubayda (died 831), wife of Caliph Harun al-Rashid, who paid for much of the work |

Nobody counted medieval pilgrims. Modern guesses for Santiago or Rome vary hugely, so treat any single number with caution.

### How long and how far

Archbishop Sigeric of Canterbury wrote down his route home from Rome in about 990. It had about 80 stages over about 1,700 km (1,100 mi): **about 20 km (12 mi) per stage**. It crossed the Alps by the Great St Bernard Pass. This is a good spacing for pilgrim stops on your map. It is shorter than a strong walker's day of 25–35 km (15–22 mi) (baseline), because pilgrims were often old, sick or slow.

### What a pilgrim road looks like

- **Hospices** (free lodgings for pilgrims and travellers) at passes, river crossings and empty stretches. The Great St Bernard hospice was founded around 1050 at 2,469 m (8,100 ft), right at the top of the pass. Roncesvalles, below the Pyrenees pass, had a famous pilgrim hospital from the 1100s.
- **Bridges** built for pilgrims, often by holy men or monasteries. Santo Domingo de la Calzada in Spain grew up around the bridge, road and hospital that St Dominic built in the 1000s. Hospital de Órbigo is named after its pilgrim hospital.
- **Pilgrim towns** along the road, with many inns, hospitals and a "street of the Franks" for foreign settlers. Puente la Reina is named after its pilgrim bridge.
- **At the shrine:** a great church, many inns, hospitals, souvenir sellers (pilgrim badges, scallop shells for Santi-

ago), money changers, and often a leper hospital on the approach road. At Canterbury, the leper hospital of Harbledown (founded c. 1084–85) stood just west of the city on the road from London. Passing pilgrims gave alms there, for example to see a slipper said to have been Becket's.

#### LATER ERA (1500S+)

Protestant lands end pilgrimage. Becket's shrine at Canterbury and the shrine of Walsingham were destroyed in 1538. Catholic pilgrimage continues and new shrines appear, but long international pilgrimages decline.

#### FANTASY TWIST

If miracles at a shrine are real and visible (healing, oracles), pilgrimage becomes a major industry. Make the shrine town 2–5× bigger than its farmland could support, with many inns and hospitals, and draw a pilgrim road with hospices every 20 km (12 mi) all the way to the nearest big cities.

**MAP TIP**

Pick one or two great shrines for your continent and several regional ones for each kingdom. Draw the pilgrim road along existing roads: keep the ordinary road line, add small scallop marks along it, and label it ("Pilgrims' Way", "Saint's Road"). Put a hospice at every mountain pass and dangerous crossing, a small pilgrim town every 2–4 days' walk, and a leper house on the last stretch before the shrine. Use a scallop shell or a small reliquary as the shrine symbol, never a star (a filled star means a capital).

## Hospitals, leper houses and almshouses

**RULE OF THUMB**

Hospitals stand at town gates, bridges and main roads. Leper houses stand outside the town, on a main road in, often within 1–2 km (about 1 mi) of the gate.

A medieval **hospital** was a religious charity house. Most did not treat illness as we do. They gave food, a bed and prayer to the poor, the old, the sick and travellers. An **almshouse** was a small home for poor old people. A **leper house** (leprosarium, lazaret house) housed people with leprosy, a disease that people feared and connected with sin.

### Numbers

| Place                                 | Number                                          | Notes                                                                    |
|---------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------|
| England, all hospitals and almshouses | 800–1,100 founded c. 1080–1540 (estimates vary) | Over 500 almshouses before the Reformation                               |
| England, leper houses                 | about 300 (estimates vary)                      | Most founded 1100–1350; many became almshouses later                     |
| Kingdom of France                     | ~2,000 leper houses                             | King Louis VIII's will (1227) left gifts to about 2,000 leper houses     |
| All of Christendom                    | 19,000 leper houses                             | A guess by the chronicler Matthew Paris for the early 1200s; not a count |

Most almshouses and small hospitals had **4–13 residents**. Thirteen was a favourite number (Christ and the twelve apostles). The Hospital of St Cross at Winchester, founded about 1132–36 for thirteen poor men, was one of the first. A few town hospitals were large, with 100 or more beds.

### Where they stood

- **Ordinary hospitals:** at a town gate, by a bridge, by a market or on a main road into town. Many towns had a "St John's Hospital" at the gate where travellers arrived.
- **Leper houses:** outside the walls, beside a main road where travellers would give alms, often near a stream. St Giles' leper hospital (founded 1101–09) stood on the road west from London, between the City and the gallows at Tyburn. In Paris, a leper house stood near the great gallows of Montfaucon on the hill outside the walls.
- **Islamic hospitals (bimaristans)** stood in the centre of the city, funded by a religious endowment (*waqf*). Nur al-

Din built one in Damascus in the 1150s; Sultan Qalawun's great hospital in Cairo was finished in 1284.

- **Pilgrim hospices:** see above.

Place names show where these were: "Spital", "Spittal", "Spitalfields", "Maudlin" (from Mary Magdalene, a common leper-house saint), "Lazar", or "Saint-Lazare".

**MAP TIP**

Give every market town one hospital, and every walled town 2–5. Put a leper house on a main road 1–2 km outside each town of 2,000+ and most market towns (England had about 300, roughly one per 430 km<sup>2</sup>); big cities may have one on each main road. On a local map, draw each as a small house with a cross and label it ("St John's Hospital", "the Lazar House"). In an Islamic city, place the hospital near the great mosque and the main market instead.

## Universities and schools

### RULE OF THUMB

Universities are as rare as great cities. A kingdom has none, one or two. Most sit in big cities with a bishop, a court or both; a few grew in middling towns that already had famous schools.

A **university** was a guild (association) of masters and students, licensed to teach and grant degrees. The scholars of the day called it a *studium generale*. There were about 20 in Europe c. 1300 and 60–80 by 1500 (baseline). Over 80 were founded before 1500, but some failed quickly.

### Major universities and when they began

| Region                                     | Universities (founding date)                                                                                                                                            |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Italy                                      | Bologna (by tradition 1088; organised in the late 1100s), Padua 1222, Naples 1224, Siena 1240, Rome 1303, Perugia 1308, Pisa 1343, Pavia 1361, Ferrara 1391, Turin 1404 |
| France                                     | Paris (c. 1150–1200), Montpellier (medicine from 1220), Toulouse 1229, Orléans 1235, Avignon 1303, Angers 1356, Caen 1432, Bordeaux 1441                                |
| Iberia                                     | Salamanca 1218, Lisbon/Coimbra 1290, Lleida 1300, Barcelona 1450, Zaragoza 1474, Valencia 1499                                                                          |
| Britain                                    | Oxford (teaching by 1096; grew fast after 1167), Cambridge 1209, St Andrews 1413, Glasgow 1451, Aberdeen 1495                                                           |
| Central Europe                             | Prague 1348, Kraków 1364, Vienna 1365, Pécs 1367, Heidelberg 1386, Cologne 1388, Erfurt (1379/1392), Leipzig 1409, Rostock 1419, Leuven 1425, Basel 1460, Tübingen 1477 |
| North                                      | Uppsala 1477, Copenhagen 1479                                                                                                                                           |
| Islamic world (colleges, not universities) | Al-Qarawiyyin mosque, Fez (859); al-Azhar mosque, Cairo (970s); Nizamiyya <i>madrasa</i> (college of law), Baghdad (1065–67)                                            |

The first universities grew out of famous cathedral and city schools in Italy, France and England. After 1340 rulers founded new ones as status symbols in their capitals (Prague, Kraków, Vienna; see [chapter 03](#)).

### How big

Student numbers are uncertain. Oxford had perhaps 1,500–2,000 students around 1300 (modern estimates vary); Paris had several thousand; most universities had a few hundred. The Baghdad Nizamiyya is said to have had about 3,000 students in 1096. A university town filled with colleges, halls, bookshops and inns, and with fights between students and townsfolk.

### Schools

- **Cathedral and monastery schools** in every cathedral city and big abbey.
- **Grammar schools** (Latin schools) in many towns by 1400. Many were run by a chantry priest or a guild.
- **Song schools and parish clerks** taught reading in some villages.

- **Islamic lands:** a *madrasa* (college) in every important city, and Qur'an schools attached to mosques. Like hospitals, they were funded by *waqf* endowments.

### LATER ERA (1500S+)

Universities multiply (Leiden 1575 in the Dutch Republic, and many more). Jesuit colleges appear in many Catholic towns after 1540. Protestant towns found grammar schools.

### MAP TIP

Give a kingdom of 2–5 million people one or two universities, placed in the capital and in the oldest cathedral city. Most universities sat in cities of 10,000+, but Oxford and Cambridge grew in towns of only a few thousand people, so avoid towns under about 3,000. Mark it with a book or a small "U" by the city name. Every town on a local map can have a grammar school, but you do not need to draw it.

## Legal and civic sites

### RULE OF THUMB

Gallows stand outside the town, on high ground, beside a main road, at the edge of the lord's land. Pillories and stocks stand in the market place. Courts meet at old, well-known landmarks.

### Gallows

Gallows were meant to be seen. They marked the lord's right to judge and kill (**high justice**) and warned travellers.

- **Where:** on a hill or rise, outside the town or village, near a main road, often at the boundary of the lordship or parish.
- **France:** a lord's gallows were called *fourches patibulaires*. By custom, the number of stone pillars showed his rank: 2 for a simple lord with high justice, 3 for a castellan (the lord of a castle and its district), 4 for a baron, 6 for a count, 8 for a duke, and as many as he liked for the king. This was the rule in principle; local customs varied.
- **Paris:** the great royal gallows of **Montfaucon** stood on a small hill outside the walls, with the leper house of Saint-Lazare in the same area. Viollet-le-Duc's 19th-century drawing shows it with 45 places for bodies, but it may make the gallows grander than it really was.
- **London:** **Tyburn** stood about 5 km (3 mi) west of Newgate prison, where two Roman roads met (today's Oxford Street and Edgware Road). Its first recorded hanging was in 1196. Close by stood "Oswulf's Stone", which gave its name to the local hundred (district), Ossulstone.
- **Anglo-Saxon England (before 1066):** archaeologist Andrew Reynolds showed that execution cemeteries were usually on **hundred boundaries**, beside **major roads**, and often on or next to **prehistoric barrows** (burial mounds).
- **Names:** Gallows Hill, Gallows Green, Hangman's Hill (England); *Galgenberg* (Germany); *Justice* (France).

### Pillories, stocks and lock-ups

- **Pillory** (holds the head and hands): in the market place, often on a platform. Used for cheating traders.
- **Stocks** (hold the feet): the Statute of Labourers of 1351 ordered **every town and village** in England to have stocks. They stood on the green, by the church or in the market.
- **Ducking stool** (a seat on a long pole, used to dip offenders into water) at a pond or river; a small lock-up or cage in some towns.

### Courts and meeting places

- **Open-air moots:** Early medieval assemblies met outdoors at a landmark: a mound, a prehistoric barrow, a standing stone, a big tree, a ford or a crossroads. By the 12th century the English hundred court met 12 times a year. Many hundreds are named after their meeting place (Ossulstone; names ending in "-low", from an old word for mound).
- **Things:** Norse and Germanic assemblies (the *thing*) met in the open, often at mounds, near water and land routes, and at territorial boundaries. Iceland's Althing met at Þingvellir from 930. Tynwald Hill on the Isle of Man is still used.
- **Inauguration sites:** Scottish kings were made at the mound of Scone.
- **Indoor courts:** manor courts in the lord's hall, shire courts in the county town or castle, church courts in the cathedral city.

### Town halls, guildhalls and market crosses

- **Town hall** (guildhall, moot hall, *palazzo comunale*, *Rathaus*): on or beside the main market place. Often with a covered market below and a council chamber above.
  - London's Guildhall was rebuilt in 1411–1440; a guildhall is first recorded there in 1127 or 1128.
  - Siena's Palazzo Pubblico (begun 1297) faces the Campo, where the city's three hills meet. Its tower, the Torre del Mangia, was built in the 1320s–1340s (sources give different dates). It was meant to be one of the tallest towers in Italy, a rival to the towers of Florence and of Siena's own cathedral.
- **Belfries:** Towns in Flanders and northern France built civic bell towers (Bruges, Ghent) to hold the town's bells, charters and archives. They were a symbol of town freedom, a rival to the church tower. UNESCO lists 56 of them.
- **Guildhalls of crafts and merchants:** in rich trading towns, near the market or the river (cloth halls in Flanders and England).
- **Market crosses:** a stone cross, later often a roofed shelter, marking the market place and the town's market rights. Proclamations were read there. Examples: Malmesbury (c. 1490), Chichester (1501).

**FANTASY TWIST**

Where the dead can rise, gallows and execution grounds are not left with bodies hanging. Expect burning places or burial under stones at crossroads, and a guard house or shrine nearby.

**MAP TIP**

On a local map, put one gallows symbol (a small "II") on a hill by the main road outside each town and at the edge of each great lord's land; label it "Gallows Hill". Put the town hall, market cross and pillory together in the market place. Mark the hundred or district meeting place at a mound, stone or ford roughly central to its area, often on an old road; draw it as a small lone tree labelled "Moot".

## Burial grounds

**RULE OF THUMB**

Christians are buried around their parish church. Jews, Muslims, plague victims and executed criminals are buried outside the town.

- **Churchyards:** around every parish church, usually walled or fenced. The rich were buried inside the church. Town churchyards filled up quickly.
- **Charnel houses:** buildings or crypts to store bones dug up to make room for new graves. Graves were often reused after about five years. A charnel house at St Bride's, Fleet Street, London, held an estimated 7,000 people's bones. Paris's Cemetery of the Holy Innocents had famous charnel galleries around its walls.
- **Plague cemeteries:** in 1348–49 London opened emergency cemeteries outside the walls at East Smithfield and West Smithfield. The London Charterhouse (a Carthusian monastery) was founded in 1371 on the West Smithfield burial ground.

- **Jewish cemeteries:** almost always outside the walls. In England, Jews could only bury their dead in London until 1177; after that, other communities got their own, such as York's "Jewbury" outside the walls. The Jewish cemetery at Worms lay outside the medieval wall; its oldest readable gravestone dates to 1058/59.
- **Muslim cemeteries:** outside the city gates, sometimes very large (Cairo's great cemeteries east and south of the city).
- **Criminals and outcasts:** at gallows sites and parish boundaries.

**MAP TIP**

You do not need to draw churchyards on most maps; the church symbol implies one. On a local or town-edge map, draw a Jewish cemetery or a Muslim cemetery as a small walled plot outside a gate, with a name label (for example "Jewbury", as at York) so it is not read as a friary or a garden. Add a plague pit or charnel chapel if your story has had a plague.

## Minorities and other faiths

**RULE OF THUMB**

Minorities live in their own quarter. At first it is usually near the ruler's castle or the market (for protection and trade). Later it is often forced to the town edge and walled off.

### Jewish communities

- **Where:** in towns, almost never in villages in northern Europe. Their quarter had many names: *Jewry* (England), *Judengasse* (Germany), *judería* or *call* (Iberia). It had a synagogue, a ritual bath, a bakery, and a cemetery outside the walls.
- **Near power:** Jews were under the king's protection in England and many other realms, so they often lived close to the castle or the market.
- **England:** communities in London, York, Lincoln, Norwich, Oxford, Cambridge, Winchester and other royal towns. There were no more than about 3,000 Jews by 1278. They were expelled in 1290.
- **Frankfurt:** Jews lived near the cathedral, at the centre of town life, until 1462. Then they were forced into the *Judengasse*, a single curved street about 330 m (1,080 ft) long and 3–4 m wide, just outside the old wall, with three gates. It started with about 110 people. It held about 260 in 1543 and about 2,700–3,000 by the early 1600s, packed into fewer than 200 houses.

- **Expulsions:** England 1290; France 1306 and finally 1394; many German cities in the 1400s; Spain 1492; Portugal 1496–97. After an expulsion, the quarter keeps its name ("Old Jewry") but not its people.

### Muslims under Christian rule

- **Iberia:** Muslims under Christian rule were called **Mudéjars**. They lived in their own quarter (*morería*) with a mosque, bath, cemetery, market and bakery, or in whole Muslim villages. They were especially numerous in the kingdoms of Valencia and Aragon, where they were mostly farmers. In Valencia most Mudéjars were Arabic-speaking Muslim peasants who farmed for Christian lords; they may have been a majority of the countryside in the late 1300s (see [Language lines are not borders](#)). In Aragon and Castile, by contrast, Mudéjars spoke the same Romance language as their Christian neighbours. Castile forced them to convert or leave in 1502; Valencia and Aragon in 1525–26.
- **Conquered mosques:** when Christians took a Muslim city, its great mosque usually became the cathedral: Toledo (taken 1085), Córdoba (1236), Seville (1248). Seville later replaced it with a new Gothic cathedral (decided 1401, begun 1402) but kept the minaret (the Giralda) as the bell tower.
- **Sicily:** under Norman rule (conquest 1061–1091, kingdom from 1130) Sicily had many Muslims, mosques and Arabic-speaking officials. Emperor Frederick II moved the remaining Muslims to **Lucera** in Apulia over about twenty years from 1224. Their number reached 15,000–20,000. The colony lasted until 1300, when it was sacked, most people were killed or enslaved, and its mosques were replaced with churches.

### Christians and Jews under Muslim rule

Christians and Jews were "protected peoples" (*dhimmi*) in Islamic lands. They kept their churches and synagogues, usually in their own quarters. In Cairo, many Coptic churches and an important synagogue stood in the old city of Fustat, beside the newer royal city.

### Orthodox, Armenian and foreign communities

- **Constantinople:** Italian merchants had their own quarters by the Golden Horn. The Venetian quarter dated from 1082. The Genoese were given Galata (Pera), across the water, in the late 1260s and held it as a walled colony from 1273 to 1453.
- **Southern Italy:** many Greek-speaking Orthodox monasteries survived in Calabria and Apulia under Latin bishops.

- **Lviv (Lwów, Poland):** the city had Armenian, Ruthenian (Orthodox) and Jewish quarters next to the Catholic one; the Armenian cathedral was built in 1363–1370.
- **Foreign merchants** often had their own walled compound (*fondaco*, *funduq*, *Kontor*) with a chapel. See [Trade Routes, Roads and Transport](#).

#### LATER ERA (1500S+)

Segregation becomes stricter. Venice created the first "ghetto" in 1516; in 1555 Venice had 923 Jews among 160,000 people. Rome followed in 1555. Spain expelled the Moriscos (Muslims forced to convert) in 1609–14, emptying hundreds of villages in Valencia and Aragon.

#### MAP TIP

On a city or town-edge map, give a trading town of 5,000+ a small minority quarter: a cluster of houses with a synagogue (a lamp symbol, never a star), or a mosque in a reconquered land (a dome with a crescent), near the castle or market. Put its cemetery outside a gate. On a kingdom map, a border region where two faiths meet can have villages of both: use the church symbol for one and the dome with a crescent for the other. Show recent expulsions with names like "Old Jewry" or "Moorish Quarter" and no symbol for a working temple.

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## Ancient and ruined layers

### RULE OF THUMB

Every medieval landscape is built on older ones. Prehistoric sites sit on high ground and ridges. Roman sites sit on straight roads and at river crossings. Medieval people used old things as quarries, fortresses, boundaries, meeting places and sources of legend.

## What a medieval traveller would see

| Old feature                                             | Age (in 1300 unless noted)                                                     | How common                                                                                                                                                                                                                                                  | Where                                                                                                                              | How medieval people used it                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Roman roads</b>                                      | 1,100–1,600 years                                                              | About 3,200 km (2,000 mi) of main paved roads in Britain, plus many minor roads. In the whole empire, over 80,000 km (50,000 mi) of stone-paved roads (out of 400,000 km in all), with official way stations ( <i>mansiones</i> ) every 25–30 km (16–19 mi) | Straight lines between old towns and forts, often on a raised bank ( <i>agger</i> ); the road itself usually 5–8 m (16–26 ft) wide | Still main roads (Watling Street, Fosse Way, Ermine Street). English law books of the 1100s put four great roads under the king's special protection. Many parish and county boundaries follow them.                                                                                                                                                                                                                                                                        |
| <b>Roman towns and forts</b>                            | 1,000–1,800 years                                                              | Every Roman province                                                                                                                                                                                                                                        | River crossings, coasts, road junctions                                                                                            | Most Mediterranean ones stayed towns. In Britain some died (Silchester, Wroxeter; see <a href="#">chapter 01</a> ). Forts became castles.                                                                                                                                                                                                                                                                                                                                   |
| <b>Roman marching camps</b> (an army's overnight camps) | ~1,100–1,250 years in Britain                                                  | Nearly 500 known in Britain, almost half of them in Scotland; most 10–18 ha (25–45 acres)                                                                                                                                                                   | Beside Roman roads, about a day's march apart: 11–24 km (7–15 mi)                                                                  | Low rectangular banks with a ditch, still visible on moors; in ploughed land most were levelled. At Chew Green, on Dere Street (the Roman road north from York) right beside today's English–Scottish border, a small medieval settlement (Kemylpeth, pottery of c. 1250–1450) and probably a chapel sat on the Roman earthworks, and the place was a meeting point for settling cross-border cases from 1249. See <a href="#">Roman marching camps as old earthworks</a> . |
| <b>Hillforts</b> (Iron Age earth-work forts)            | 1,300–2,000+ years                                                             | 4,147 in Britain and Ireland (Atlas of Hillforts, 2017): about one per 75 km <sup>2</sup> (29 sq mi) on average, but very uneven                                                                                                                            | Hilltops and ridge ends, especially in the west and south                                                                          | Some re-fortified as castles or towns (Old Sarum: castle and cathedral). Many used as sheep pasture or fair grounds.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Long barrows</b> (Neolithic burial mounds)           | ~4,300–5,100 years                                                             | About 300 earthen long barrows in eastern Britain, plus stone-chambered ones in the west; most 20–70 m (65–230 ft) long                                                                                                                                     | Ridges and hill brows                                                                                                              | Landmarks, boundary marks, legends. Wayland's Smithy (then in Berkshire, now Oxfordshire) is a chambered long barrow already named after a legendary smith in a charter of 955.                                                                                                                                                                                                                                                                                             |
| <b>Round barrows</b> (Bronze Age burial mounds)         | ~2,800–3,800 years                                                             | Many thousands in England; often in groups (barrow cemeteries)                                                                                                                                                                                              | Ridges and sky-lines, chalk downs                                                                                                  | New burials in the early Middle Ages; meeting places; gallows sites; boundary markers                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Stone circles and standing stones</b>                | ~3,800–4,600 years                                                             | About 1,300 circles in Britain and Ireland                                                                                                                                                                                                                  | Uplands, moors, river valleys                                                                                                      | Feared, Christianised or destroyed. At Avebury, villagers toppled and buried stones in the early 1300s. One man, a barber-surgeon, was found under a fallen stone, with coins of the 1320s. A church was built just west of the Avebury henge (a prehistoric ring of bank and ditch) in the 900s.                                                                                                                                                                           |
| <b>Deserted villages</b>                                | Recent: most emptied after 1350, so they matter most for maps set in 1350–1500 | About 3,000 known in England; 1,500+ emptied c. 1350–1520                                                                                                                                                                                                   | Clay lowlands and poor land; sheep country                                                                                         | House platforms, hollow lanes and a lone church in a sheep field (see <a href="#">chapter 01</a> )                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Abandoned castles</b>                                | Up to ~250 years in 1300; more by 1500                                         | England and Wales had about 1,700–1,800 castle sites (the Gate-house gazetteer lists 1,761), but only 500–600 in use at once                                                                                                                                | Hills, river crossings, old borders                                                                                                | Grass-covered mottes (earth mounds) and ruined towers; stone robbed for houses (see <a href="#">chapter 04</a> )                                                                                                                                                                                                                                                                                                                                                            |

## How old things were reused

- **As quarries.** Roman ruins were the easiest source of cut stone and brick. The Norman abbey church at St Albans was built largely of Roman brick and tile from the ruined Roman town of Verulamium, across the river. Many

churches near Roman sites (Wroxeter and its neighbours) are full of Roman stone.

- **As fortresses.** At Portchester, a Roman fort of about 285–290 still stood complete. The Normans built a castle keep in one corner (probably late 1000s) and an Augustinian priory inside the walls (founded 1128; its

church dates from the 1130s). In southern France, the Roman amphitheatre of Arles became a fortress with four towers and more than 200 houses inside it.

- **As holy ground.** Churches were built at or beside pagan sites (Rudston, Avebury). Some saints' shrines sat at old Roman cemeteries.
- **As boundaries and meeting places.** Barrows, stones and Roman roads made ideal parish edges and moot sites, and gallows sites (see [Legal and civic sites](#)).
- **As legends.** People explained old earthworks with giants, devils or old heroes. Names like "Grim's Ditch" (Grim being a name for the god Woden) and "Devil's Dyke" are common in England.

### Where ruins survive and where they disappear

#### RULE OF THUMB

Ruins near living towns get quarried away within a few generations. Ruins survive far from people, in hard stone, or where a taboo protects them.

- **Near towns:** Roman walls survive because the town still uses them; other Roman buildings vanish into new houses.
- **In empty country:** hillforts, barrows and stone circles survive, especially on moors, downs and heaths.
- **In the Middle East:** ancient cities often survive as **tells** (mounds of earlier settlements, built up over thousands of years). Roman and Byzantine ruins stand in the desert edge, where few people needed the stone.

#### LATER ERA (1500S+)

The Dissolution (1536–41) creates hundreds of ruined abbeys in England and Wales, often with a new country house built from their stones. Antiquarians begin to record ruins: William Camden's *Britannia* (1586) describes Roman sites county by county. Old castles that cannot take cannon are abandoned or slighted (deliberately wrecked) in the wars of the 1600s.

#### FANTASY TWIST

An older, vanished race (elves, giants, an ancient empire) leaves the same kinds of layers: roads, forts on hills, tombs on ridges, stones in circles. If its ruins are still whole near living towns, there must be a reason: fear, a curse, a monster, or a law. Otherwise people would have carted the stone away. See [Fantasy Variants](#).

#### MAP TIP

Give a kingdom map a few long, straight "old roads" that link cities in near-straight lines, with later towns along them. On local maps, add 1–3 hillforts per 100 km<sup>2</sup> (39 sq mi) in hilly country (ovals of rings on hilltops), scattered barrows along ridges (small filled mounds), a stone circle or two on moorland (small rings of dots), a Roman camp beside an old road (a grey, closed dashed rectangle with rounded corners), a ruined motte by an old crossing (the motte symbol drawn grey), and one or two deserted villages shown by a grey lone church labelled "(lost village)". Label them in the local style: "Old Fort", "Giant's Grave", "Kings' Stones", "Castle Hill", "Old Street".

## Drawing it all: symbols, density and scale

#### RULE OF THUMB

Draw what a traveller at that scale would notice. Cathedrals and great shrines on a continent map; abbeys and gallows on a local map; every church only when villages are shown.

### Suggested symbols

These symbols match the guide's [master legend](#).

| Feature           | Symbol idea                                                  | Show at which scale                 |
|-------------------|--------------------------------------------------------------|-------------------------------------|
| Archbishop's seat | Double cross beside the city symbol                          | Continent, kingdom                  |
| Cathedral         | Church with two towers, or a mitre                           | Kingdom, local                      |
| Parish church     | Small church with a tower; spire for rich towns              | Local                               |
| Chapel            | Small plain cross, no tower                                  | Local                               |
| Wayside cross     | Tiny cross on a base, set on the road or path                | Local                               |
| Abbey or priory   | Church with a square cloister beside it                      | Kingdom (great abbeys), local (all) |
| Friary            | Name only, or a small church inside the town outline         | Local and town maps                 |
| Grange            | Small open square topped with a cross, or the label "Grange" | Local                               |

| Feature                     | Symbol idea                                                                    | Show at which scale            |
|-----------------------------|--------------------------------------------------------------------------------|--------------------------------|
| Commandery (military order) | Small church with a shield                                                     | Kingdom, local                 |
| Great shrine                | Scallop shell or small reliquary (never a star: a filled star means a capital) | Continent, kingdom             |
| Pilgrim road                | The ordinary road line with small scallop marks along it (not a dotted line)   | Kingdom                        |
| Hospice at a pass           | Small house with a cross                                                       | Kingdom, local                 |
| Hospital or leper house     | Small house with a cross; leper house labelled "Lazar House"                   | Local                          |
| University                  | Book, or a "U" next to the city name                                           | Continent, kingdom             |
| Gallows                     | Small "Π" on a hill                                                            | Local                          |
| Moot or thing site          | Small lone tree, labelled "Moot" (or "Thing")                                  | Local                          |
| Mosque                      | Dome with a crescent                                                           | Kingdom (great mosques), local |
| Synagogue                   | Lamp (never a star)                                                            | Local, town edge               |
| Ruin                        | The feature's own symbol, broken or grey, labelled "(ruin)"                    | Kingdom (big ones), local      |
| Deserted village            | Grey church alone in a field, labelled "(lost village)"                        | Kingdom (a few), local         |
| Hillfort                    | Oval of rings or hachures on a hill                                            | Local                          |
| Barrow                      | Small filled mound; a long barrow as a longer mound                            | Local                          |
| Standing stone              | Small upright tick                                                             | Local                          |
| Stone circle                | Small ring of dots                                                             | Local                          |
| Old or Roman camp           | Grey closed dashed rectangle with rounded corners, labelled "Camp"             | Local                          |
| Roman road                  | Straight thick solid line labelled "Street" or "(Roman)"; grey where abandoned | Kingdom, local                 |

### Footprints: how big to draw them on a local map

#### RULE OF THUMB

A great abbey is not one building but a walled estate 350–600 m (0.2–0.4 mi) across, as big as a large hillfort or Malbork, the largest castle by area. A friary is one town block. A moated manor or a small motte is only a dot.

**Why so big?** A great abbey had to feed, house and employ hundreds of people. Its **precinct** (the walled area of the monastery) held much more than the church and the **cloister** (the square, covered walk with the monks' main rooms around it). Around them were an **inner court** with

guest houses and workshops, and an **outer court** with barns, mills, smithies, fishponds and orchards. At Fountains the River Skell ran through the precinct, and the farming and industry lay in the outer court south of the river. Rievaulx's outer court held a double water-mill and fishponds, and a great gatehouse stood at the entrance. Fountains and Rievaulx both built a wall around the whole precinct in the early 1200s.

Most of England's ~900 religious houses were small priories with much smaller precincts. The numbers below are for the great houses, plus other sites for comparison.

| Site                                         | Area                                  | Rough size across                                                                                                        | Drawn at 1 cm = 1 km     | Real examples                                                                                                                                                                              |
|----------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Great Cistercian abbey precinct              | 12–37 ha (30–92 acres)                | ~350–600 m (0.2–0.4 mi)                                                                                                  | 3.5–6 mm                 | Roche ~30 acres (12 ha); Kirkstall ~40 acres (16 ha); Fountains ~70 acres (28 ha), inside a wall almost 3.5 m (11 ft) high built in the first half of the 1200s; Rievaulx 92 acres (37 ha) |
| Great Benedictine abbey                      | ~15 ha (36 acres)                     | ~400 m (0.25 mi)                                                                                                         | ~4 mm                    | Glastonbury: today's abbey grounds cover 36 acres (15 ha); parts of the medieval precinct wall still stand                                                                                 |
| Cathedral close                              | Up to ~32 ha (80 acres)               | Up to ~570 m (0.35 mi)                                                                                                   | Up to ~6 mm              | Salisbury, laid out from 1220, has Britain's largest close: about 80 acres (32 ha). Most closes are smaller.                                                                               |
| Friary (in a town)                           | ~1.5–2 ha (4–5 acres)                 | ~120–140 m (400–460 ft)                                                                                                  | ~1.5 mm                  | Northampton Greyfriars: about 4 acres (1.6 ha); Canterbury Blackfriars: about 1.8 ha (4.5 acres)                                                                                           |
| Moated manor site                            | Island often under 0.15 ha (0.4 acre) | Island ~25–50 m (80–165 ft) across, inside a moat 7–15 m (23–50 ft) wide: the whole site is ~40–80 m (130–260 ft)        | Under 1 mm: use a symbol | Lincolnshire scheduled sites: islands of 24 m, 40 × 30 m and 50 × 30 m. A high-status site, South Kelsey Hall, has a main island of 110 × 100 m (about 1.1 ha)                             |
| Small motte-and-bailey castle                | Bailey ~0.15–0.7 ha (0.4–1.7 acres)   | Motte (earth mound) 30–90 m (100–300 ft) across at the base; whole castle roughly 100–150 m (330–490 ft) long (estimate) | ~1–1.5 mm                | The Mount, Ilketshall St John (Suffolk): motte 46 m across, bailey platform 53 × 30 m. Ewyas Harold (Herefordshire): bailey about 0.7 ha                                                   |
| Large baronial castle                        | ~3 ha (7.5 acres)                     | ~170 m (560 ft)                                                                                                          | ~2 mm                    | Castle Acre (Norfolk): about 3 ha to the outer edge of its defences. The small walled town beside it covered 3.8 ha (9.4 acres)                                                            |
| Hillfort reused as castle and cathedral town | ~11 ha (28 acres)                     | 400 × 360 m (1,300 × 1,180 ft)                                                                                           | 4 × 3.6 mm               | Old Sarum (see <a href="#">chapter 04</a> )                                                                                                                                                |
| The largest castle                           | 21 ha (52 acres)                      | ~450 m (0.3 mi)                                                                                                          | ~4.5 mm                  | Malbork, Teutonic Order headquarters (see <a href="#">chapter 04</a> )                                                                                                                     |
| Deer park                                    | 40–80 ha (100–200 acres)              | 0.5–1 km (0.3–0.6 mi)                                                                                                    | 5–10 mm                  | About 3,200 in England c. 1300 (see <a href="#">chapter 06</a> )                                                                                                                           |

"Rough size across" is the side of a square with the same area, calculated by this guide. Real precincts were irregular: they followed rivers, roads and valley sides. For a sense of scale, Fountains' precinct (28 ha) could hold the whole walled town of Castle Acre (3.8 ha) seven times.

#### MAP TIP

On a local map at 1 cm = 1 km, draw a great abbey as a walled outline 3.5–6 mm across, with a gatehouse on the road side and the church and cloister inside. At a larger scale, such as 4 cm = 1 km (1:25,000, where the precinct is 14–24 mm across), add the inner and outer courts, a mill on the stream, a chain of fishponds, barns and orchards. Draw a friary as a small walled plot against the town wall or at a gate. A moated manor or a small motte is under 1 mm at 1 cm = 1 km, so use a symbol: a small open square with a double outline for a moated manor, and a small mound with a tower tick for a motte (see [chapter 06](#) and [chapter 04](#)).

#### Density checklist for a kingdom of 50,000 km<sup>2</sup> (19,000 sq mi)

Use this for an English-style kingdom of about 1–2 million people c. 1300. Scale up or down by area.

| Feature                                                       | How many                                       | Where                                                                    |
|---------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------|
| Archbishop                                                    | 1                                              | Oldest or richest city, often not the capital                            |
| Cathedrals                                                    | 5–10 (English style) or 25–100 (Italian style) | Main cities; in the Italian style, many small towns                      |
| Parish churches                                               | ~3,500 in lowland; fewer in hills              | One per village; many in old towns                                       |
| Religious houses                                              | 300–350                                        | Monks in valleys, canons at towns, friars in towns, nuns in quiet places |
| Friaries                                                      | 60–75                                          | Only in towns of 3,000–5,000+; 4–5 in the biggest city                   |
| Hospitals and almshouses                                      | 300–400 over time                              | Gates, bridges, roads                                                    |
| Leper houses                                                  | 100–120                                        | 1–2 km (about 1 mi) outside towns on main roads                          |
| University                                                    | 0–1                                            | The biggest cathedral or capital city                                    |
| Great shrine                                                  | 1–3                                            | One national saint, plus regional ones                                   |
| Hillforts and barrows (if the land was settled in prehistory) | Hundreds                                       | Hilltops, ridges, downs                                                  |

### Place names that tell the story

Everyday village endings (-ton, -by, -ley and so on), the common church words (-minster, Kirk-, Llan-, Kil-) and the Roman word -chester are listed in [Naming settlements](#)

*realistically*, with a method for naming a whole region. The table below adds only the names that point to the sites in this chapter: saints, church owners, hospitals, gallows and ancient remains.

| Name element                       | Meaning                                                                            | Examples                                                           |
|------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Münster (German), Moutier (French) | Monastery or mother church (from Latin <i>monasterium</i> , like English -minster) | Münster; Marmoutier near Tours ("the greater monastery")           |
| -church                            | Church                                                                             | Whitchurch ("white church", probably one built of pale stone)      |
| San-, Santa-, Saint-, Sankt-       | Named after a saint's church or shrine                                             | Saint-Denis, Santiago                                              |
| Abbots-, Bishops-, Monks-, Nun-    | Owned by the church                                                                | Bishop Auckland (home of the Bishops of Durham), Monkton, Nuneaton |
| Temple-                            | Owned by the Templars                                                              | Temple Bruer, Templecombe                                          |
| Spital, Maudlin, Lazar             | Hospital or leper house                                                            | Spitalfields                                                       |
| Gallows, Galgen-, Justice          | Execution place                                                                    | Gallows Hill                                                       |
| Street, Strat-, Stret- (on a road) | Roman road                                                                         | Stratford, Stretton                                                |
| -low, Barrow, Tump                 | Burial mound or mound-shaped hill                                                  | Ludlow, Taplow                                                     |

#### MAP TIP

Check the balance at the end. If your kingdom has one splendid cathedral and nothing else religious, it will look empty. Most real kingdoms had a dense, even carpet of small churches, a scatter of monasteries, a few hospitals at every town and ruins of older ages on the hills. Common errors are listed in [Common Mistakes and How to Fix Them](#).

## Quick summary

- One parish church per village, about 4 km (2.5 mi) apart in the lowlands and about 7–10 km (4–6 mi) apart in the hills; England had ~9,500 parishes, one per ~14 km<sup>2</sup> (5.4 sq mi).
- Old towns had many tiny parishes (London over 100, Norwich about 56); planted towns had one.
- Dioceses are small and many in Italy and southern France (Umbria: ~850 km<sup>2</sup>, 330 sq mi, each), large and few in England (17; ~7,600 km<sup>2</sup>, 2,950 sq mi, each) and the north. In Italy a cathedral can sit in a small town.
- Religious houses: about 1 per 150 km<sup>2</sup> (60 sq mi) (England ~900 in the 1530s, average 13 people). Cistercians and Carthusians in empty, watered valleys; Benedictines in old towns; canons at town edges; friars inside towns. A great abbey's walled precinct covers 12–37 ha (30–92 acres), 350–600 m (0.2–0.4 mi) across; a friary about 1.5–2 ha (4–5 acres).
- Friaries roughly measure town size: none under ~3,000 people, 2 at about 10,000, 4 at 20,000–40,000.
- Granges were meant to lie within a day's walk of their abbey, but many lay farther (Fountains' grange at Cowton was about 32 km, 20 mi, away).
- Pilgrim roads follow existing roads, with stops about 20 km (12 mi) apart, hospices at passes, pilgrim towns and a leper house near the shrine.
- Hospitals stand at gates, bridges and roads; leper houses often 1–2 km (about 1 mi) outside towns on main roads.
- Universities are rare: ~20 in Europe c. 1300, 60–80 by 1500, mostly in big cities with a bishop or court.
- Gallows stand on high ground by a main road, outside town, at the lordship's edge; pillories and stocks in the market place; moots at mounds, stones, trees and fords.
- Jewish and Muslim quarters lie near the castle or market, often later pushed to the edge; their cemeteries lie outside the walls.
- Old layers matter: Roman roads, Roman marching camps, hillforts (4,147 in Britain and Ireland), barrows, stone circles, deserted villages and abandoned castles. Ruins near towns get quarried; ruins in empty land survive.

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- French Wikipedia: [Fourches patibulaires](#) (pillars by rank); [Léproserie](#) (Louis VIII's 2,000 leper houses).
- Digital Humanities Institute, University of Sheffield, *The Cistercians in Yorkshire: Granges and farming* (day's-walk rule; Fountains' 39 granges); [Fountains precinct](#) (precinct sizes of Fountains, Rievaulx, Kirkstall and Roche; Fountains' wall); [Rievaulx precinct](#) (92-acre precinct, wall, gatehouses, mills and fishponds).
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- English Heritage, [Castle Acre Castle and Bailey Gate: description](#) (castle about 3 ha; walled town 3.8 ha).
- Northamptonshire HER, [Greyfriars Franciscan Friary](#) (precinct about 4 acres); Kent HER, [Dominican Friary \(Blackfriars\), Canterbury, precinct boundary](#) (about 1.83 ha).
- Visit Somerset, [Glastonbury Abbey](#) (36 acres of grounds); Arundells, [The Close](#) (Salisbury Cathedral Close, about 80 acres).
- Fasti Ecclesiae Gallicanae project (a diocese-by-diocese record of bishops and canons for the dioceses of present-day France, 1200–1500): [HAL record of vol. XIV, Châlons-en-Champagne](#).
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- *The Past, Full of faith: new insights into the medieval parish churches of Norwich*.

Densities per 1,000 km<sup>2</sup>, the "sample county" and the "kingdom of 50,000 km<sup>2</sup>" checklist are this guide's own calculations from the English counts above, not historical counts. Tower visibility distances are geometric limits (curve of the earth only) calculated by this guide. In the footprints table, the "rough size across" and "drawn at 1 cm = 1 km" columns are this guide's own conversions from the recorded areas.

## CHAPTER 9

# WHAT CHANGES AFTER 1500

Your map may be set a little later than the classic Middle Ages: a world with cannon, printed books and ships that cross oceans. This chapter explains what changed on the ground between about 1500 and 1650, and how to show it on a map. Most rules in the rest of this guide still hold, because villages, market towns and travel speeds barely change. What changes is the top of the system: bigger capitals, new kinds of fortresses, ocean ports, post roads, canals, drained land and a new religious map.

The medieval baseline numbers (settlement sizes, density, travel speeds) are in [Population and Settlement Sizes](#) and [Trade Routes, Roads and Transport](#). This chapter only explains what is different.

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## What changes and what stays the same

### RULE OF THUMB

After 1500 the bottom of the map stays medieval and the top changes. Villages, fields and market towns look much the same. Capitals, ports, fortresses and long-distance links are new or much bigger.

**Why:** Most people (80–90% almost everywhere) still farm with the same tools and animals. On land, goods still move

by ox, horse and mule. So everything that depends on walking distance stays the same: how far apart villages are, how far apart market towns are, and how much land feeds a town. What changes is what rich states and merchants can now pay for: cannon and new walls, ocean ships, standing armies, big courts, printing presses and engineering projects.

### Timeline of the changes

| Date       | What happens                                                                                                                                                                                      | What it means on the map                                                                                      |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1490s      | Taxis relay post (from 1490); French siege guns in Italy (1494); Columbus (1492) and da Gama (1498)                                                                                               | Post stations along main roads; old walls suddenly weak; new ocean routes                                     |
| 1500–1540s | Bastioned fortifications develop in Italy and spread north in the 1530s–1540s                                                                                                                     | Star-shaped walls around key towns                                                                            |
| 1500–1600  | Population recovers and grows; Atlantic ports boom                                                                                                                                                | Bigger capitals and ports; Seville, Lisbon and Antwerp, then Amsterdam and London                             |
| 1517–1560s | Protestant Reformation; English monasteries closed 1536–41                                                                                                                                        | Abbey ruins, abbeys turned into houses, religious borders                                                     |
| 1560s–1600 | Fixed capitals (Madrid 1561); printed atlases and county maps; Spanish rules for planting colonial towns (1573)                                                                                   | Palace complexes; more accurate maps; grid towns around a plaza in the colonies                               |
| 1600–1650  | Chartered trading companies (English 1600, Dutch 1602); lakes drained (Beemster 1612); Japan's one-castle order (1615); French edict against interior castles (1626); Thirty Years' War (1618–48) | Trading posts overseas; new polder landscapes; far fewer working castles in strong states; devastated regions |

## What does not change

| Feature                          | Value (same as the medieval baseline)                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Village spacing, lowland         | 1.5–4 km (1–2.5 mi)                                                                                                                 |
| Market towns in practice         | 9–16 km (6–10 mi) apart; England had ~650 working markets in the late 1500s                                                         |
| Walking traveller                | 25–35 km (15–22 mi) a day                                                                                                           |
| Horse cart                       | 30–40 km (20–25 mi) a day                                                                                                           |
| Water vs land costs              | Bulk goods still go by water; land haulage stays several times dearer                                                               |
| Share living in towns of 10,000+ | Still under 10% in most countries; the big exceptions are the Netherlands (24% in 1600, 32% in 1650) and Belgium and Italy (15–19%) |

### MAP TIP

Do not redraw the whole countryside for a later setting. Keep your medieval village and market-town pattern. Then change the top layer: enlarge the capital and the main ports, add new forts, post stations, canals and drained land, and turn some castles and abbeys into ruins or country houses.

## Population and bigger cities

### RULE OF THUMB

Europe regains its 1300 peak during the 1500s and keeps growing until about 1600–1620. Almost all the extra growth in towns goes to capitals and ocean ports.

### How many people

- **Europe:** Jan de Vries's estimates for Europe without Russia and the Ottoman lands are roughly 62 million (1500), 70 million (1550), 78 million (1600) and 75 million (1650). Counting Russia and the Ottoman Balkans, the guide's baseline is about 100 million by 1600. Other historians give somewhat different totals, but the shape is the same: growth in the 1500s, a halt in the 1600s.

- **England:** about 2.35 million (1522), 4.1 million (1600) and 5.3 million (1650). England only passed its 1348 peak around 1600–1630.
- **The 1600s are harder.** Growth stops or reverses after about 1620. The Thirty Years' War (1618–48) devastated parts of Germany. Italian plagues in 1629–31 and 1656–57 each killed 20% or more of the people in the areas they hit. Milan lost about 60,000 of its 130,000 people in the plague of 1629–31. Naples lost 150,000 or more people, about half the city, in the 1656 plague.

### The big cities

Figures are estimates in thousands. Sources disagree, so treat each range as "somewhere in here".

| City      | c. 1500                                 | c. 1600                       | c. 1650                                               | Why it grew or shrank                                                 |
|-----------|-----------------------------------------|-------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------|
| Istanbul  | 100–280 (Ottoman count of 1478: 80–100) | 400–700                       | 400–700                                               | Ottoman capital, fed by sea from Egypt and the Black Sea              |
| Paris     | 95–250 (often ~200)                     | 220–300                       | 400–500                                               | Capital of the largest western kingdom                                |
| Naples    | 100–150                                 | 250–280                       | ~300 before the 1656 plague, then about half          | Spanish viceroy's capital; nobles and migrants from the whole kingdom |
| London    | ~50                                     | ~200                          | 375–400                                               | Capital and chief port; took over much of Antwerp's trade             |
| Amsterdam | 10–15                                   | 50–60 (105 in the 1622 count) | 150–175                                               | Took over Antwerp's trade after 1585; Baltic grain and Asian spices   |
| Seville   | 45–70                                   | 110–150 (peak, late 1500s)    | ~130 before the 1649 plague, which killed nearly half | Legal monopoly port for the Americas from 1503                        |

| City          | c. 1500                    | c. 1600                           | c. 1650                        | Why it grew or shrank                                                                                                           |
|---------------|----------------------------|-----------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Lisbon        | 55–65                      | 110–150                           | 130–170                        | Spice route to India; Brazil sugar                                                                                              |
| Antwerp       | ~40 (1496)                 | ~100 in 1566,<br>then ~42 in 1589 | ~50–60                         | Boom as Europe's market, then siege (1585) and blockade of the river Scheldt                                                    |
| Madrid        | under 20 (not yet capital) | 75–95                             | 100–130                        | Made the fixed capital in 1561 (about 18 then, about 80 by 1598)                                                                |
| Venice, Milan | 100–150 each               | 110–150 each                      | Venice ~100–120; Milan smaller | Large but no longer growing; the 1629–31 plague killed about a third of Venice (46 of 140) and almost half of Milan (60 of 130) |
| Rome          | 55–60                      | ~100                              | 100–120                        | Papal capital rebuilt as a showpiece for pilgrims                                                                               |

### Shifting rank: the Atlantic shift

The order of cities changes faster than in the Middle Ages. Three patterns are worth copying:

1. **Rise of ocean ports.** Seville, Lisbon, Antwerp, Amsterdam and London grow two to ten times. Seville grew from perhaps 45,000–70,000 people around 1500 to over 100,000 by the late 1500s, which made it the largest city in Spain.
2. **Rise and fall.** Antwerp grew from about 40,000 (1496) to 100,000 (1566). After the Spanish took it in 1585, the Dutch closed the river Scheldt to sea trade, and by 1589 only about 42,000 people were left. Its merchants and craftsmen moved to Amsterdam and London.
3. **Capital pulls life out of an old city.** When Madrid became the capital, the old regional centre Toledo, the largest city in the region until about 1600, declined (Ringrose 1973).

The Mediterranean does not collapse. Venice, Genoa, Naples and Istanbul stay huge. But the fastest growth is now on the Atlantic and North Sea coasts.

### Why capitals grow so much

A capital is now a **consumer city**. It holds the court, the law courts, the treasury, the nobles' town houses, foreign ambassadors, and thousands of servants and luxury craftsmen. By 1650 about one English person in thirteen lived in London (about 400,000 out of 5.3 million). In some realms the capital becomes 5–10 times bigger than the second city. England is the extreme case: London was more than ten times the size of the next English town. In other lands (Spain, Italy) several big cities stay close in size. See [Population and Settlement Sizes](#) for the "primate city" pattern.

#### MAP TIP

For a 1600-style map, draw the capital 2–4 times bigger than its medieval size, and the fastest-growing port or capital up to 5–10 times bigger. Show suburbs spilling far outside the old walls. Add one "fallen" city: a port with a silted or blockaded river, a smaller population and a grand but half-empty centre.

## Gunpowder and the new fortifications

### Why the old walls failed

Medieval walls were **high and thin**, to stop ladders. Cannon smashes high stone walls quickly. When Charles VIII of France invaded Italy in 1494, he brought about 18,000 men (other accounts say up to 25,000). He also brought a siege train of at least 40 guns on wheeled carriages (Parker). Castles that had once held out for months now fell in days. Round medieval towers also leave **dead ground**: spots at the foot of the wall that the defenders cannot shoot at.

#### RULE OF THUMB

After 1500, a wall that is tall is weak. Good walls are now low, very thick, backed with earth, sunk behind a wide ditch, and shaped so that every part can be shot at from another part.

### The bastion and the star fort

A **bastion** is an arrow-shaped platform that sticks out from the wall. Guns on each bastion fire along the face of the next bastion, so there is no dead ground. Seen from above, a ring of bastions looks like a star. This system is called the **trace italienne** ("Italian outline") because it was first built in Italy. Extra **outworks** stand in or beyond the ditch, for example the **ravelin** (a triangular island fort in front of the wall).

| Date         | Place                                                                   | Why it matters                                                                                                                   |
|--------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1501–c. 1503 | Nettuno (papal fort, usually credited to Antonio da Sangallo the Elder) | Early angled bastions; very thick, low walls without battlements                                                                 |
| 1508–1537    | Civitavecchia (Forte Michelangelo)                                      | Papal port fort; often called the first key example of the new style                                                             |
| 1509         | Padua                                                                   | The engineer Fra Giocondo replaced the old wall with a broad ditch swept by low gun positions; a huge imperial siege army failed |
| 1534–1537    | Fortezza da Basso, Florence                                             | Large pentagonal (five-sided) bastioned fortress by Antonio da Sangallo the Younger; often cited as an early model               |
| 1530s–1540s  | Spain, France, the Low Countries, England                               | The style spreads out of Italy                                                                                                   |
| 1564–c. 1570 | Turin citadel                                                           | Pentagon with five bastions (sources give finishing dates from 1566 to 1570)                                                     |
| 1566         | Valletta, Malta                                                         | New fortress capital, built after the Great Siege of 1565                                                                        |
| 1593         | Palmanova (Venice)                                                      | Planned nine-pointed star town; first ring of walls (finished 1623) about 7 km (4.3 mi) round                                    |
| 1593         | Bourtange (Dutch)                                                       | Small five-pointed fort guarding a road through a bog                                                                            |

## Sizes and garrisons

Sizes and garrisons below are rough estimates, except the named examples.

| Kind                                               | Typical size                                 | Garrison                                      | Where to put it                                             | Map symbol                                                                                                       |
|----------------------------------------------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Watchtower                                         | 10–15 m tall                                 | a handful of men                              | Every few km along raided coasts, each in sight of the next | Small triangle                                                                                                   |
| Coastal artillery fort                             | 30–100 m across                              | 10–30 in peace (Walmer 18, Camber 29 in 1540) | Harbour mouths, anchor-ages, river mouths                   | Small round outline with rounded lobes (a <b>rosette</b> , like the plan of Deal Castle, 1539–40), never pointed |
| Small fort ( <i>sconce</i> , Dutch <i>schans</i> ) | 100–250 m across, 4–5 bastions               | 50–300                                        | River crossings, dikes, roads through marshes               | Small star outline (the star-fort symbol), never filled                                                          |
| Citadel                                            | 400–700 m across, 4–6 bastions               | several hundred to 2,000+                     | At the edge of a big or rebellious city                     | Pentagon outline joined to the town (the bastioned-citadel symbol)                                               |
| Fortress town                                      | 1–2.5 km (0.6–1.5 mi) across, 6–12+ bastions | 1,000–5,000 in war                            | Frontiers, key river towns, big ports                       | Star-edged town outline, never filled                                                                            |

Bastions are usually spaced so that musket fire from one reaches the next, roughly 200–350 m apart (an approximation). This is why star forts look so regular.

**Cost.** Bastions were enormously expensive. Siena bankrupted itself in 1544 to pay for its new defences. Amsterdam's 22 bastions cost 11 million florins. Henry VIII spent £376,000 on his coastal forts (1539–47); one small block-house cost about £500, and the three forts of the Downs in Kent cost £27,092 together.

### RULE OF THUMB

Only rich states can afford bastions everywhere. Put full star forts at frontiers, the capital, the main ports and a few river crossings. Ordinary towns keep their medieval walls, perhaps with a few earth bastions added on the side facing the enemy.

## Citadels: forts that watch their own city

A **citadel** is a star fort built at the edge of a city, half inside and half outside the walls. It defends the city, but it also

controls the citizens. Its guns point both outward and into the streets.

- **Fortezza da Basso**, Florence (1534–37), built for the new Medici duke.
- **Turin** (from 1564), designed by Francesco Paciotto after Turin became the Savoy capital; meant to defend the city and also to overawe its people.
- **Antwerp** (1567–72), also by Paciotto, built on the Duke of Alba's orders during the Dutch Revolt. After the Spanish troops sacked the city in 1576, the citizens tore down part of it. It was rebuilt after 1585.

#### MAP TIP

Draw a citadel as a pentagon outline (the guide's bastioned-citadel symbol) joined to one corner of the town wall, with an open space (an *esplanade*) between the fort and the houses. If a ruler has just conquered a rebellious city, give it a citadel.

### Coastal defence

- **England:** Henry VIII's "Device Forts" (1539–47). These were dozens of forts, blockhouses and earth gun platforms at the anchorages and river mouths most open to invasion: the Downs in Kent, the Thames, the Solent and Falmouth. Twenty-four sites were already finished and manned by the end of 1540. Most were small, round, squat gun forts with tiny garrisons. Seen from above, the bigger ones look like a flower: Deal Castle (1539–40), about 70 m (230 ft) across, is a round central tower inside two rings of six rounded bastions; Walmer has four rounded bastions.
- **Kingdom of Naples:** viceroys' orders of 1532 and especially 1563 pushed the building of watchtowers along the whole coast against corsairs (sea raiders, here mainly from North Africa). An inventory of 1748 listed 379 coastal towers. They passed warnings by smoke by day and fire by night.
- **Genoa** built about 85–90 towers around Corsica between 1530 and 1620 (90 were planned in 1531; a list of 1617 counts 86). See [Watchtowers, beacons and coastal warning](#).

#### MAP TIP

On a raided coast, draw a chain of watchtowers (small triangles) 3–12 km (2–7 mi) apart, every one in sight of the next. Keep the villages a little inland and uphill. The tower spacing is an estimate. About 379 towers on roughly 2,000–2,400 km (1,250–1,500 mi) of Neapolitan mainland coast is one per 5–6 km (3–4 mi). That count comes from 1748, so it may include some towers built after 1600. About 85–90 towers on Corsica's roughly 1,000 km (620 mi) of coast is one per 11–12 km (about 7 mi). Put a real fort only at the harbour mouth, drawn as the coastal artillery fort symbol: a small rosette outline (rounded lobes, like Deal Castle's plan), never pointed. A harbour fort with angled bastions takes the star outline instead.

### Bigger armies, longer sieges

- **Army size:** Charles VIII's 18,000 men looked huge in 1494. By the 1630s the leading states kept armies of over 150,000 (Parker). These are paper strengths. The number of men actually present was often much lower.
- **Sieges become the main event.** A fortress with bastions could hold for months. Ostend held out for three years (1601–04) because it was supplied by sea; more than 100,000 people were killed, wounded or fell sick on both sides.
- **Military corridors.** Spain moved troops from Milan to the Netherlands along the **Spanish Road**, about 1,000 km (620 mi) through Savoy, Franche-Comté and Lorraine. A march took 32–60 days, usually 40–50, so about 20–25 km (12–16 mi) a day, with food bought at stopping places (*étapes*) along the way. More than 123,000 soldiers marched this route between 1567 and 1620, against only about 17,600 sent by sea.
- **Devastated land.** Large armies living off the land ruin whole regions. Show burnt villages and empty farms along their routes.

### Fortified lines on open frontiers

Where there are no mountains or rivers to block raiders, states built long lines:

- **Muscovy's Great Abatis Line** (nominally finished in 1566) stretched from Bryansk to Meshchera. It was made of felled trees laid as a barrier, ditches, earth banks, palisades and watchtowers, linking fortified towns such as Tula, Ryazan, Serpukhov and Kolomna. In the late 1500s about 35,000 local militia guarded it (some sources give more).
- **Muscovy's Belgorod Line** (1635–1654) was the second line, built further south along the southern edge of the forest steppe (the zone where woods give way to open grassland). It ran from the Vorskla River (today in

Ukraine) east to the Tambov area. It was about 800 km (500 mi) long with its bends, or about 600 km (370 mi) in a straight line. It linked fortress towns such as Belgorod, Kozlov and Tambov (both founded in 1635–36 for the line) with earth banks, felled-tree barriers and palisades (walls of wooden stakes). It used rivers, marshes and woods as part of the barrier. Eighteen of its fortress towns were new. Sources differ a little on when it was finished; one gives 1646.

### What happened to the old castles

| Fate                                         | Real examples                                                                                                                                                            | How to show it                                                                                             |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Updated with earth bastions or gun platforms | Common in frontier and war zones                                                                                                                                         | Castle icon inside a star outline (the star-fort symbol)                                                   |
| Rebuilt as a palace or residence             | Chambord (begun 1519) and the other Loire châteaux; Heidelberg's Renaissance wings (1550s and 1601–07)                                                                   | Castle icon with gardens, labelled "palace"; a royal palace or hunting lodge takes the small crown instead |
| Replaced by a country house nearby           | Elizabethan "prodigy houses" (1570–1620): Burghley (finished 1587), Hardwick Hall (1590–97)                                                                              | House inside a ring of park trees; the old castle nearby, drawn broken or grey                             |
| Kept as a prison, arsenal or mint            | Tower of London                                                                                                                                                          | Castle icon in the town, labelled with its new use ("prison", "mint")                                      |
| Deliberately wrecked ("slighted")            | In and after the English Civil War, Parliament slighted, or planned to slight, more than 100 buildings (castles, town walls and houses); Corfe (1646), Kenilworth (1649) | Castle icon drawn broken or grey, labelled "(ruin)"                                                        |
| Abandoned and used as a quarry               | Many minor castles in peaceful interiors                                                                                                                                 | Small castle icon, broken or grey, often near a village                                                    |

#### MAP TIP

In a 1500–1650 world, inland castles in peaceful regions should mostly be ruins, palaces or manor houses. New gun forts belong at frontiers, coasts, capitals and river crossings. Do not put a fresh star fort next to every village. See [Castles, Forts and Military Outposts](#) for the medieval castle network.

### When a strong ruler cuts the number of castles (France and Japan)

#### RULE OF THUMB

When one ruler finally controls the whole land, private castles stop being useful to him and become a danger. The number of working castles then falls sharply, often by a single order.

**Why:** A castle lets a local lord defy the ruler. A ruler with his own army and cannon no longer needs these castles for

• **The Habsburg–Ottoman frontier** in Hungary and Croatia was a chain of fortresses and soldier-farmer settlements from the mid-1500s. For the medieval frontier layout it grew from (a forward line of small posts and a rear line of fortress towns), see [Outposts in wild and contested land](#).

defence. So he keeps the forts on the frontier and the seats of his governors, and orders the rest pulled down.

- **France, 1626:** a royal edict, issued in Louis XIII's name under Cardinal Richelieu, ordered that all towns and castles "in the interior of our realm" that were not important for frontier defence "shall be razed and demolished". It aimed at the strongholds of rebellious nobles.
- **Japan, 1615:** after the Tokugawa family took Osaka Castle in 1615, the shogunate (the military government) issued the "one castle per province" order (*ikkoku ichijō rei*). Each daimyo (great lord) kept one castle, normally his own residence, and demolished the others. A few of the greatest lords were allowed more. In the same year, the Laws for the Military Houses (*Buke shohatto*) said that every repair had to be reported to the shogunate, and that new building or enlargement needed its permission. The order stayed in force until the lords' domains were abolished in 1871.

| Japan           | Before 1615                         | After 1615                              |
|-----------------|-------------------------------------|-----------------------------------------|
| Working castles | about 3,000 (a rounded figure)      | about 170 (more than nine in ten gone)  |
| Who holds one   | Any lord strong enough to build one | Only the great lords, usually one each  |
| Main job        | War: base and refuge                | Government: the seat of a lord's domain |
| New building    | Free                                | Only with the shogunate's permission    |

**Real examples:** Iwakuni Castle (western Japan) was built in 1601–08 and pulled down in 1615, when it was only seven years old. Himeji Castle, completely rebuilt in 1601–09, survived as a great lord's seat.

The castles that survived became the centres of **castle towns** (*jōkamachi*, "town below the castle"). The lord's officials, his warriors and the merchants who served them lived around the castle. These towns became the regional capitals of Japan for the next 250 years.

#### MAP TIP

For a 1600–1650 map of a strong, centralised realm, keep about one working castle per province (or per great lord), set inside the province's main town. Draw that town around the castle, as the province's capital. Show the other old castle sites as ruins (the castle icon drawn broken or grey) or just as a place name such as "Castle Hill". Keep extra forts only on the frontier and the coast. A realm still split between strong nobles keeps many castles.

#### FANTASY TWIST

If dragons or war-mages are your world's "artillery", the same logic applies: tall walls become weak, and defenders build low, thick, angled walls that cover each other. If the attackers **fly**, though, a star fort gives no protection. Expect roofed gun towers and underground shelters instead. See [Fantasy Variants](#).

## Ocean trade, ports and trading posts

### Bigger ships

| Ship                             | Main period                      | Typical size                                                                                                                         | Crew                                                         | Main use                         |
|----------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------|
| Cog (medieval)                   | 1200s–1400s                      | 30–200 tons; a few were larger (see chapter 05)                                                                                      | small                                                        | North Sea and Baltic cargo       |
| Carrack (Portuguese <i>nau</i> ) | late 1400s–1600                  | 400–2,000 tons; Portuguese India ships were often over 1,000; the <i>Madre de Deus</i> (captured 1592) was rated at about 1,600 tons | 600–700 aboard the <i>Madre de Deus</i>                      | India and spice route            |
| Galleon                          | mid-1500s–1700                   | up to about 1,000 tons on the Atlantic (many smaller); Manila galleons 1,700–2,000                                                   | —                                                            | Treasure fleets, warships        |
| Fluyt (Dutch)                    | from 1595 (first built at Hoorn) | medium cargo ship                                                                                                                    | about 12, where other ships of the same size needed about 30 | Cheap bulk cargo (grain, timber) |

Ship "tons" were measured in different ways (cargo space or the ship's own weight), so treat all these sizes as rough.

**Why it matters on the map:** big ships need deep water. Trade gathers in fewer, deeper harbours. Old river ports far inland lose out, and **outports** (small harbours near the sea) grow downstream. See [Where Settlements Are Built](#).

### Ocean routes and how long they took

- **Portugal to India (the *Carreira da Índia*):** about 6 months each way. Ships left Lisbon in February–April and came home in June–August of the following year, so a round trip, including the wait in India for the right wind, took about 14–18 months. There were 1,033 ship departures from Lisbon between 1497 and 1650, about 7 a year. Many

never came back: of 806 ships sent out between 1497 and 1612, only 425 returned to Portugal (many others stayed in Asian waters, and 66 were lost at sea).

- **Spain to the Americas:** from the mid-1560s (the convoy system was set up in 1564–66) ships sailed in armed convoys. Usually two fleets a year left Seville. A fleet had about 17 ships in 1550 and more than 50, much larger ones by 1600. The fleets met at Havana for the voyage home.
- **Across the Pacific:** the Manila galleon linked Acapulco (Mexico) and Manila (Philippines) from 1565.

Ocean routes follow the wind belts, not straight lines. They touch land at a few famous stopping places: islands, capes and harbours for water, food and repairs.

### Monopoly ports

A state could force all ocean trade through **one port**, to collect taxes and control it. That port then grows very fast.

- **Seville:** in 1503 the Crown gave it the monopoly of trade with the Americas and set up the **Casa de Contratación** (House of Trade) there. By the late 1500s Seville had over 100,000 people and was the largest city in Spain.
- **Lisbon:** the Crown's India House controlled the spice trade.

### Chartered companies and trading posts ("factories")

A **chartered company** is a merchant company given a monopoly of trade with one region by the state. It could build forts, keep soldiers and make treaties.

A **factory** in this period is not a workshop. It is a trading post run by a *fáctor* (agent): a walled compound with warehouses, offices and homes, on land the local ruler allows. Some grew into forts and then into colonial cities.

| Power or company                            | Founded | Examples of posts (date)                                              |
|---------------------------------------------|---------|-----------------------------------------------------------------------|
| Portuguese Crown (State of India from 1505) | —       | Elmina, West Africa (1482); Goa (1510); Malacca (1511); Hormuz (1515) |
| English East India Company                  | 1600    | Surat, India (permanent factory 1613)                                 |
| Dutch East India Company (VOC)              | 1602    | Batavia, now Jakarta (1619), its Asian capital                        |
| Dutch West India Company                    | 1621    | Elmina (taken from Portugal 1637)                                     |

**Scale:** between 1602 and 1796 the VOC sent 4,785 ships to Asia. The English company, its nearest rival, sent 2,690.

Merchants from abroad also have fixed quarters at home, for example the German merchants' hall in Venice, the *Fondaco dei Tedeschi* (rebuilt 1505–08).

#### RULE OF THUMB

A trading post sits beside an existing local port city, not in empty land. It prefers a spot that is easy to defend from the sea: an island, a peninsula or a river mouth. It stays small (often tens to a few hundred Europeans; an estimate) unless it becomes a colonial capital.

#### MAP TIP

On a continent map, mark ocean routes as long curving dotted lines (the sea-lane symbol), with arrows where the wind sets the direction, a sailing season ("out Feb–Apr") and two or three named ports of call. On the home coast, give the monopoly port the head-port symbol (a large anchor with a quay, name in capitals); on a local map add docks, a customs house and a "House of Trade". Abroad, mark trading posts with a small fort symbol next to a local port city, labelled with the company name. Do not use a flag: a small flag (pennant) marks a duchy seat.

### Planted colonial towns: the Spanish grid

Where a European power ruled the land itself, not just a trading post, it founded new towns. Spain founded the most in this period, and it planned them all to one pattern.

#### RULE OF THUMB

A planted colonial town is a grid of square blocks around one central **plaza** (open square). The main church and the government houses stand on or beside the plaza. Every such town in a colony looks alike.

**Why:** A grid is quick to lay out with a cord and stakes. It makes it easy to give each settler a plot, and the town can grow outward in any direction. One central square gives the church and the royal officials a place to be seen, and space for markets and festivals.

**The rules:** Philip II's ordinances of 1573 on discovery and new settlement (148 articles, later part of the "Laws of the Indies") wrote down a pattern that had already been used for decades. Lima, for example, was founded in 1535 on a grid of  $13 \times 9 = 117$  square blocks around its main square.

**Where the pattern came from:** Spain already had planned towns at home. The forerunner most often named is Santa Fe near Granada. In 1491, while besieging Granada, Ferdinand and Isabella turned their camp into a small walled town on a grid, with a square in the middle and a gate in each of the four sides (see [Camps that became towns](#)). Historians of town planning have often called it the direct model for the American towns. Others say it was too small and too military, and point instead to older planned grid towns of the medieval Crown of Aragon (Bielza de Ory 2002). Either way, the idea of a planted grid town around a central square came from medieval Spain.

| Rule (1573 ordinances) | What it says                                                                                                                                                                                                                                                                                         |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Where the plaza goes   | At the landing place in a port town; in the centre of an inland town                                                                                                                                                                                                                                 |
| Plaza shape            | A rectangle, its length at least 1.5 times its width ("best for fiestas in which horses are used")                                                                                                                                                                                                   |
| Plaza size             | Matched to the number of people expected. Not smaller than $200 \times 300$ feet; not larger than about $530 \times 800$ feet; $400 \times 600$ feet is "a good proportion". These are Castilian feet of about 28 cm (11 in), so roughly $56 \times 84$ m, $148 \times 223$ m and $111 \times 167$ m |
| Streets                | Twelve streets leave the plaza: one from the middle of each side and two from each corner                                                                                                                                                                                                            |
| Church                 | On a whole block of its own, near the plaza                                                                                                                                                                                                                                                          |
| Government             | The royal council and town council house, the customs house and the arsenal, near the church                                                                                                                                                                                                         |

Real towns did not always follow every rule, but the plaza and the grid are almost everywhere. In Lima the blocks were about 110–125 m (360–410 ft) on a side (sources differ), and the cathedral and the viceroy's palace stand on the main square.

#### MAP TIP

For a colony, draw small identical grid towns: a square of a few blocks around a plaza, with a church and a governor's or council house on the plaza. In a port town, put the plaza by the landing place. Draw the colonial capital to the same pattern, just with many more blocks. The regular grids should stand out against the irregular older towns of the local people. The grid is a local-map drawing: at kingdom scale, use the ordinary town symbols. For medieval planned towns with grids (*bastides*), see [Planted "new towns"](#) and *bastides*.

## Posts, roads and canals

### The post: relay stations

A **relay post** is a chain of stations where riders change horses (and sometimes riders) so that letters never stop.

- **Taxis post:** from about 1490 the Taxis family ran a line for Maximilian of Habsburg between Innsbruck and Mechelen in the Low Countries. Its centre moved to Brussels around 1500–01. The contract of 1505 added stations in Spain and France (Granada, Toledo, Blois, Paris, Lyon) and fixed delivery times for summer and winter.

- **Spacing:** stations every 20–40 km (12–25 mi).
- **Speed:** under the 1505 contract, Brussels to Innsbruck took  $5\frac{1}{2}$  days in summer and  $6\frac{1}{2}$  in winter, and Brussels to Paris 44 hours in summer (54 in winter). Relay posts covered up to about 150 km (95 mi) a day.
- **Public use:** the contract of 1516 let the Taxis carry private letters for profit, as long as court mail arrived on time. The emperor confirmed the Taxis monopoly in 1595, and from 1615 it was hereditary.
- **England:** there was a Master of the Posts from 1516, and post stages were about 32 km (20 mi) apart. Charles I

opened the royal post to the public in 1635. The General Post Office was founded in 1660.

#### MAP TIP

Along a post road, draw a small horn symbol (a post station) every 20–40 km (12–25 mi). Each station is an inn with large stables. Small towns on post roads grow; towns off them fall behind.

## Roads

Roads improve only a little. Some states make local communities repair them: England's Highways Act of 1555 made each parish give unpaid labour on its roads. Coaches for the rich become common in the later 1500s, but heavy goods still go by water wherever possible. The new thing is **planned military routes** such as the Spanish Road. Travel

speeds stay as in the baseline: see [Trade Routes, Roads and Transport](#).

## Canals with pound locks

A **pound lock** is a chamber with a gate at each end. A boat enters, the gates close, and the water level rises or falls gently. This lets a canal climb hills. Most medieval canals were flat, or used dangerous "flash locks" (a single gate opened to let a rush of water carry boats through). Pound locks were not completely new: Damme near Bruges had one by 1396, and one of Milan's canals (the Naviglio di Bereguardo) got 18 of them in 1452–58. What is new after 1500 is that they spread, and that whole canals start to climb over hills.

| Canal                    | Dates                                                                        | Size                                                                                                           | Why it matters                                                                                                        |
|--------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Exeter Canal, England    | 1564–66                                                                      | short: a few km (accounts differ; Wikipedia gives 5.2 km / 3¼ mi), 3 pound locks                               | First pound locks in Britain; brought small ships back to a city cut off by weirs                                     |
| Briare Canal, France     | 1604–42 (work stopped after Henry IV's murder in 1610 and restarted in 1638) | 57 km (35 mi), 36 locks, including a "staircase" (locks joined directly one after another) of 7 locks at Rogny | First summit-level canal in Europe with pound locks; joined the Loire and Seine river systems; 6,000–12,000 labourers |
| Dutch <i>trekvaarten</i> | 1630s–1660s (the first, Amsterdam–Haarlem, opened in 1632)                   | a network linking the cities of Holland                                                                        | Straight canals for horse-towed passenger barges; about 7 km/h (4.5 mph), about 30 passengers, on a timetable         |

A **summit-level canal** climbs over a watershed (the high ground between two river systems). It needs reservoirs on the summit to feed water to the locks.

#### MAP TIP

Draw canals as straight or gently bending blue lines, unlike rivers. Mark each lock with two short bars close together across the line (a single bar with a "T" means a toll), grouped where the land rises. A summit canal needs a reservoir lake near its highest point. Towns grow at the canal's junctions with rivers.

## Money, printing and news

### Exchanges and banking centres

An **exchange** (French *bourse*) is a building where merchants meet daily to trade goods, loans and bills of exchange (writ-

ten orders to pay a sum of money in another city, used instead of carrying coin).

| Date    | Place     | What                                                                                            |
|---------|-----------|-------------------------------------------------------------------------------------------------|
| 1531    | Antwerp   | First purpose-built exchange building                                                           |
| 1565–71 | London    | Royal Exchange, copied from Antwerp's; founded by Thomas Gresham, opened by Elizabeth I in 1571 |
| 1609    | Amsterdam | Exchange Bank ( <i>Wisselbank</i> ), a city-backed public bank                                  |
| 1611    | Amsterdam | Exchange building                                                                               |

Great banking families still matter: the Fuggers of Augsburg lent to the Habsburgs, and Genoese bankers financed the Spanish crown.

#### RULE OF THUMB

A continent has only one or two money centres at a time. They sit in the leading trading port of the day (Antwerp, then Amsterdam; Genoa; London), which is not always the capital.

### Printing towns

- By 1500, 236 towns in Europe had a printing press (a figure given by Braudel).
- Before 1650, printing took place in more than 1,300 places in Europe (Universal Short Title Catalogue data).
- Output grew roughly tenfold in the 1500s, to perhaps 150–200 million copies.

• **Big centres:** Venice (the largest c. 1500), Paris, Lyon, Antwerp, Basel, and the book-fair towns of Frankfurt and later Leipzig.

• **Ottoman lands:** printing in Arabic script was forbidden or discouraged until 1729. Jewish, Greek and Armenian presses did exist there.

A printing town needs paper (water-powered paper mills on nearby streams), customers (a university, a court, a bishop, lawyers) and good routes to send books out.

#### MAP TIP

Label one or two cities "book fair" or "printing centre" in your realm, usually a university or trade city. Put paper mills on fast streams within a day or two of it. Printed news, maps and pamphlets now travel along the post roads.

## Capitals, states and borders

### Fixed capitals and palace complexes

#### RULE OF THUMB

By 1600 most big states have one fixed capital, with a ring of royal palaces, hunting parks and monasteries within a day's ride.

- **Madrid:** Philip II made it the fixed capital in 1561. It was small before the court arrived: about 18,000 people in 1561. By 1598 it had about 80,000, and estimates for about 1600 range from 75,000 to 95,000.

• **El Escorial** (1563–84), about 45 km (28 mi) from Madrid: a huge palace, monastery, royal burial church and library in one building.

• **Other examples:** the Polish court moved from Kraków to Warsaw in 1596; German princes settled in fixed **residence towns** (*Residenzstädte*). For more, see [Capitals, Realms and Borders](#).

States also become **more centralized**: more paid officials, permanent ambassadors, standing garrisons, and governors in provincial capitals. On the map this means a governor's palace and a citadel in each provincial capital, and the capital itself as a city of offices and noble houses.

### Surveying and mapping

| Date | Development                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------|
| 1533 | Gemma Frisius explains <b>triangulation</b> : measure one baseline carefully, then use angles to fix far-away points |
| 1569 | Mercator's world map with his new projection for sailors                                                             |
| 1570 | Ortelius publishes the first modern atlas (53 maps) in Antwerp                                                       |
| 1579 | Christopher Saxton's atlas of the counties of England and Wales (35 maps, surveyed in 1574–78)                       |
| 1615 | Snellius measures a long chain of triangles across Holland                                                           |

### Borders becoming lines, slowly

Maps now show borders as lines, and rulers start to think of their realms as areas with edges. But on the ground, borders were still often defined as **lists of places**. The Treaty of the Pyrenees (1659) made the mountains the border between France and Spain and gave France the "villages" of

northern Cerdanya. Llívia was legally a town, not a village, so it stayed Spanish, and it is still a Spanish island inside France today. The exact line was only marked on the ground by the Treaties of Bayonne, 1856–68.

**MAP TIP**

For a 1600-style map you can draw firmer, unbroken borders for big kingdoms. Keep the dash-dot border line; a solid line means a road. Keep enclaves (the owner's colour with a thin outline), disputed strips (two parallel dash-dot lines) and patchwork areas, especially in a "Holy Roman Empire" type region. Put pairs of fortresses facing each other across important borders, and mark a few contested towns. See [Capitals, Realms and Borders](#) for medieval frontier zones.

## The countryside: enclosure, new crops and drainage

### Enclosure

**Enclosure** means turning open fields and common land (shared by the whole village) into private fields surrounded by hedges, walls or ditches. Often it also meant turning ploughland into sheep pasture.

- J. R. Wordie estimated that about 45% of England was already enclosed by 1500, 47% by 1600 and 71% by 1700. So the famous Tudor enclosures covered only a small area; the big new wave came in the 1600s (and continued in the 1700s).

- In the 1500s it was most visible, and most hated, in the Midlands, where landlords turned whole villages into sheep runs. Parliament passed laws against it ("tillage acts") between 1489 and 1597. Enclosure was a cause of Kett's Rebellion (1549) and the Midland Revolt (1607).
- In eastern Europe (Poland, Prussia, the Baltic lands) a different change happened: lords built large grain estates worked by serfs, and shipped grain down the rivers to Baltic ports for export.

### New crops

| Crop   | Arrives                                                            | Becomes a common field crop                                                        |
|--------|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Maize  | Spain within a few decades of 1492                                 | A common peasant food in southern Europe by the 1600s                              |
| Potato | Canary Islands 1567; Spain about 1570; Britain and Ireland 1588–93 | Ireland on a wide scale in the early 1600s; most of Europe only in the 1700s–1800s |
| Tomato | Spain soon after 1492; eaten in Italy by 1544                      | Garden crop in this period                                                         |

**RULE OF THUMB**

Before 1650, American crops are mostly garden plants. The big exception is maize in Iberia and Italy. Do not cover your 1600 map with potato fields.

### Drainage and new land

- **Dutch polders:** a **polder** is low land surrounded by dikes and kept dry by pumping. The Beemster lake in North Holland was drained between 1607 and 1612 with 43 windmills, creating about 7,200 ha (72 km<sup>2</sup> or 28 sq mi) of new farmland. The Purmer (1622), Wormer (1626) and Schermer (1635, about 48 km<sup>2</sup> or 18 sq mi) followed.
- **English Fens:** drainage began in earnest in the 1630s, led by the Earl of Bedford and his investors. A second phase in the 1650s, with the New Bedford River (completed 1652), finished the main work. The Great Level covered about 1,300 km<sup>2</sup> (500 sq mi). Local people, who lost their

fishing and fowling, fought it. The drained peat shrank, so by the late 1600s parts were flooding again.

**MAP TIP**

Draw a polder as a perfectly regular grid of straight roads and ditches inside a ring dike, with windmills (an X of four sails on a dot) along the edge and farms evenly spaced along the roads. It should look completely different from the irregular old villages around it.

### Estates and country houses

Rich families build **country houses** with parks, avenues, walled gardens and deer parks, often on former monastery land (see [Religion](#)). England's "prodigy houses" (1570–1620) were built to host the monarch on a royal tour. Big parks can swallow whole villages and old roads. For village layouts see [Villages, Farms and the Countryside](#).

**MAP TIP**

Give each county-sized area one to three country houses. Draw each as a house inside a ring of park trees, an oval park 1–3 km (0.6–2 mi) across (an estimate), with a straight avenue to the nearest road. Put some on old abbey sites and some beside a ruined castle. Show enclosed fields as small irregular polygons (hedged fields) in the sheep country, and keep open fields (unhedged areas with blocks of hatching) elsewhere.

## Industry: furnaces, mines and mills

### Blast furnaces

A **blast furnace** is a tall stone furnace with bellows driven by a waterwheel. It makes liquid cast iron, which can be poured into moulds, for example to make cannon. It needs iron ore, a lot of charcoal (so a lot of managed woodland) and a dammed stream.

- England's first blast furnace was at Queenstock, Buxted (about 1490–91), followed by Newbridge in Ashdown Forest (1496). Both are in the Weald of Sussex.
- They stayed few until about 1530. Then many were built: by the later 1500s the Weald had about 50 furnaces, and as many as 100 furnaces and forges together. The Weald industry peaked around 1590. The first British furnaces

outside the Weald appeared in the 1550s, and British output as a whole peaked around 1620.

- England cast its first iron cannon in 1543, at Buxted.
- The method came from the Namur region (today in Belgium), through Normandy, to the Weald.

**MAP TIP**

Put iron works in wooded valleys with dammed streams: a chain of **hammer ponds** (mill ponds for furnaces and forges) down one valley, with charcoal burners' clearings in the woods around them. Label them "Furnace Pond" or "Hammer Pond", so they are not read as fishponds (small rectangles in a chain). Iron districts are wooded, not bare.

### Mining booms

| Place                            | Start                    | Peak size                                                                                                           | What happened                                                                                                                                                                                                                                                               |
|----------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Joachimsthal (Jáchymov), Bohemia | Silver town founded 1516 | about 20,000 by 1534, the second town of Bohemia after Prague                                                       | Its coin, the <i>Joachimsthaler</i> , gave us the word "thaler" (and later "dollar"). Declined because of war (the Schmalkaldic War, 1546–47), the Counter-Reformation driving Protestant miners away, deep mines that flooded for lack of pumps, and cheap American silver |
| Potosí (in today's Bolivia)      | 1545                     | about 130,000 by 1603 (at least 120,000 of them Andean); higher figures of 160,000–200,000 are claimed but disputed | At 4,067 m (13,343 ft); produced about 60% of the world's silver in the late 1500s; worked by forced Indigenous labour (the <i>mita</i> , about 13,000 men a year)                                                                                                          |

**RULE OF THUMB**

A rich strike can turn an empty valley into one of a kingdom's biggest towns within 10–20 years, and then fade almost as fast. Draw boom towns big, crowded and unwallled, in rough terrain, with a mint, mine symbols (crossed hammers) around them, and roads carrying food in and silver out.

### Mills

- **Paper mills** spread with printing. They need fast, clean water and linen rags. England got its first paper mill in

the 1490s (John Tate's short-lived mill near Hertford, about 1495).

- **Gunpowder mills** grind saltpetre, sulphur and charcoal. They explode, so they sit away from towns, often as separate buildings spread along a stream. Example: Chilworth, Surrey, set up by the East India Company in the mid-1620s on the Tillingbourne stream.
- **Wind sawmills:** Cornelis Corneliszoon of Uitgeest patented his crankshaft for a wind-powered sawmill in 1593–94. It cut timber into planks about 30 times faster than sawing by hand. The Zaan district north of Amster-

dam filled with industrial windmills and is often called the first industrial area in the world.

### Shipyards and arsenals

State shipyards become huge. The **Venetian Arsenal** covered about 45 ha (110 acres), around 15% of Venice. Up to 16,000 workers is often quoted for its peak in the early 1500s, but normal numbers were much lower. Henry VIII founded royal dockyards at Woolwich and Deptford near London. Dutch yards on the Zaan and at Amsterdam built ships in series.

#### MAP TIP

Put shipyards at a sheltered deep-water port with timber arriving by river or sea. Draw them as a walled yard with slipways (sloping ramps where ships are built), next to a rope-walk (a very long, narrow shed for making rope). At kingdom scale, mark a state arsenal or royal dockyard with an anchor plus a castle tower. Put gunpowder mills a few kilometres outside a town on a stream, and paper mills on clean streams upriver of a printing town; draw both as watermill wheels on the stream, labelled "Powder Mills" or "Paper Mill". A wind-mill industrial district looks like a forest of windmill symbols (an X of four sails on a dot) along a waterway.

For medieval mines, mills and industrial towns, see [Industry, Resources and Special Towns](#).

## Religion: the Reformation on the map

### England and Wales: the dissolution of the monasteries (1536–1541)

- About 900 religious houses: about 260 for monks, 300 for canons, 142 nunneries and 183 friaries.
- About 12,000 religious people: 4,000 monks, 3,000 canons, 3,000 friars and 2,000 nuns.

- Monasteries held about a quarter of England's landed wealth.
- The small houses went first (243 were closed in 1536). The large ones "surrendered" by 1540.

### What happened to the buildings:

| Fate                                                 | Example                                                      | Map symbol                                                                           |
|------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Stripped of lead roofs and left as a ruin            | Fountains and Rievaulx (Yorkshire); Tintern (Wales)          | Abbey symbol (church with a square cloister) drawn broken or grey, labelled "(ruin)" |
| Turned into a country house                          | Lacock Abbey                                                 | House inside a ring of park trees, name ending in "Abbey" or "Priory"                |
| Abbey church bought by the town as its parish church | Tewkesbury                                                   | Large parish-church symbol in a small town                                           |
| Made a cathedral of a new diocese                    | Gloucester, Peterborough, Oxford, Bristol, Chester (1540–42) | Cathedral (church with two towers, or a mitre)                                       |
| Used as a quarry for local building stone            | Very common                                                  | A small grey fragment of the abbey symbol, or just the name                          |

Great shrines were destroyed (Walsingham and Becket's shrine at Canterbury, both 1538), and pilgrim routes emptied.

### Elsewhere in Europe

- **Protestant lands:** Scotland's Reformation came in 1560. Protestant German princes took over church lands, and in 1525 the Teutonic Knights' state became the secular Duchy of Prussia. Former abbeys became schools, hospitals, offices or princely residences.

- **Religious borders:** the Peace of Augsburg (1555) let each German ruler choose Lutheranism or Catholicism for his land. This created a patchwork, with churches of one faith just across the border from the other. In France the Edict of Nantes (1598) gave the Protestant Huguenots fortified "security towns" such as La Rochelle, which fell to the king after a siege in 1627–28. In the Dutch Republic, Catholics could worship only in hidden "house churches".
- **Catholic lands:** monasteries stay and even multiply. The **Counter-Reformation** (the Catholic reform movement)

adds new building: Jesuit colleges (the order was founded in 1540), new churches and new pilgrimage shrines.

**MAP TIP**

Make your religious map as clear as the political one. Where each ruler chose his land's faith (as in Germany after 1555), the faith border follows a political border, so the border line is already there. Give each faith a soft colour tint, not a new line, so it is never mistaken for a realm border. On the Protestant side, use ruined abbeys, country houses called "Abbey", and big town churches. On the Catholic side, keep the working monasteries and add a Jesuit college in the main towns. Put a few pilgrim shrines (scallop shell or reliquary symbols) just inside the Catholic side of the border. For medieval religious sites, see [Religious, Cultural, Legal and Ancient Sites](#).

## Comparison table: medieval vs 1500 to 1650

| Map feature              | Medieval baseline (c. 1000–1500)                                                                                                            | 1500–1650                                                                            |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Biggest city in the West | Paris ~200,000                                                                                                                              | Paris, Naples, London 200,000–400,000+; Istanbul 400,000–700,000                     |
| Capital                  | Often a travelling court; in strong kingdoms the capital is 3–7× the second city, while fragmented realms have rival cities of similar size | Fixed capital, 5–10× the second city in some realms, with palaces nearby             |
| Town walls               | High and thin, round or square towers                                                                                                       | Low, thick, earth-backed, bastioned (in rich or threatened towns only)               |
| Castles                  | Thousands; main military and lordly centre                                                                                                  | Ruins, palaces, prisons or manor houses; replaced by forts and country houses        |
| City control             | Castle at the town's edge                                                                                                                   | Bastioned citadel at the town's edge                                                 |
| Coastal defence          | Walled ports, a few towers                                                                                                                  | Lines of watchtowers; artillery forts at anchorages                                  |
| Field armies             | Usually 5,000–20,000                                                                                                                        | Total forces of 50,000–150,000+ for great states (paper strength)                    |
| Ships                    | Cogs of 30–200 tons; merchant galleys carrying about 140–250 tons                                                                           | Carracks and galleons of several hundred to 2,000 tons; Dutch fluyts                 |
| Long-distance trade      | Mediterranean, Baltic, overland caravans                                                                                                    | Plus ocean routes to Asia, Africa and the Americas                                   |
| Merchants abroad         | Foreign quarters and halls                                                                                                                  | Plus chartered companies with fortified trading posts                                |
| Messages                 | Royal messengers 50–90 km (30–56 mi) a day                                                                                                  | Relay post up to ~150 km (95 mi) a day; public letters                               |
| Canals                   | Short, flat, flash locks                                                                                                                    | Pound locks; summit canals; passenger barge canals                                   |
| Finance                  | Fairs, Italian banking houses                                                                                                               | Exchanges and public banks in the leading port                                       |
| Books                    | Hand-copied; monastery and university centres                                                                                               | Printed in 1,300+ places; book-fair towns                                            |
| Borders                  | Zones and marches                                                                                                                           | Lines on maps; still lists of villages on the ground                                 |
| Maps                     | Rare, schematic                                                                                                                             | Surveyed county maps, atlases, triangulation                                         |
| Countryside              | Open fields, commons, deserted villages                                                                                                     | More enclosure (47% of England by 1600, 71% by 1700); serf grain estates in the east |
| Wetlands                 | Mostly left wet                                                                                                                             | Polders and drained fens                                                             |
| Country seats            | Castles and manor houses                                                                                                                    | Country houses with parks                                                            |
| Monasteries              | Everywhere                                                                                                                                  | Ruined or converted in Protestant lands; flourishing in Catholic lands               |

| Map feature | Medieval baseline (c. 1000–1500) | 1500–1650                                |
|-------------|----------------------------------|------------------------------------------|
| Iron        | Bloomeries (small hearths)       | Blast furnaces, cast-iron cannon         |
| Mining      | Steady mining towns              | Sudden boom towns (Joachimsthal, Potosí) |

**MAP TIP**

Use this table as a checklist. Go down the left column, find each feature on your medieval map, and decide whether it stays, changes or is joined by the later version.

## Drawing a world in transition

**RULE OF THUMB**

A real 1600 landscape has three layers of age. Old things (castles, abbeys, medieval walls) are ruined, converted or patched. Middle-aged things (villages, fields, market towns) carry on unchanged. New things (star forts, canals, polders, country houses, post roads) are geometric and stand out.

9. **Add an overseas layer if you have a world map:** trading posts beside foreign port cities, labelled with company names.

### Symbols and labels

These symbols match the guide's [master legend](#).

### A checklist

1. **Start with your medieval map.** Keep villages, market towns and roads.
2. **Grow the top.** Enlarge the capital and one or two ocean ports. Shrink one old port whose river has silted up or been blockaded.
3. **Change the castles.** Interior castles become ruins or palaces. Draw a few country houses with parks nearby. In a strongly centralised realm, keep only about one working castle per province.
4. **Add new fortifications only where they pay:** a citadel in the capital or a rebellious city, star-fort towns on the dangerous frontier, gun forts at harbour mouths, towers along raided coasts.
5. **Mix old and new walls.** A frontier town can have a medieval wall on the safe side and new bastions on the side facing the enemy. This half-finished look is very realistic.
6. **Add the new links:** post stations every 20–40 km (12–25 mi) on the main roads, a straight canal or two, and ocean routes from your main port.
7. **Change the landscape where money went:** a geometric polder in a former lake or marsh, enclosed fields in the sheep country, a mining boom town in the hills, iron furnaces in wooded valleys.
8. **Draw the religious map:** a confessional border (a border between two faiths, shown with a colour tint for each faith), abbey ruins on one side, working monasteries on the other.

| Feature                       | Suggested symbol                                                                                                      | Label example                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Star fort or fortress town    | Five- or six-pointed star <b>outline</b> , or a star-edged town outline; never filled (a filled star means a capital) | "Fort St John", "Neufort"          |
| Bastioned citadel             | Pentagon outline joined to the town                                                                                   | "Citadel"                          |
| Ruined castle                 | Castle icon drawn broken or grey                                                                                      | "Old Castle (ruin)"                |
| Royal palace or hunting lodge | Small crown on its own                                                                                                | "Hartwood (royal lodge)"           |
| Country house                 | House inside a ring of park trees                                                                                     | "Elmswood House", "Lacey Abbey"    |
| Ruined abbey                  | Abbey symbol (church with a square cloister) drawn broken or grey                                                     | "Abbey (ruin)"                     |
| Post station                  | Small horn on the road                                                                                                | "Post"                             |
| Canal with locks              | Straight blue line; each lock as two short bars close together across it                                              | "Royal Canal"                      |
| Polder                        | Grid inside a dike, with mill symbols                                                                                 | "New Polder (drained 1612)"        |
| Monopoly port                 | Large anchor with a quay; name in capitals                                                                            | "SALTHAVEN"                        |
| Ocean route                   | Dotted line at sea, with arrows where the wind sets the direction                                                     | "To the Spice Isles (out Feb–Apr)" |
| Trading post                  | Small fort beside a foreign port city                                                                                 | "Company factory"                  |
| Coastal tower                 | Small triangles along the coast                                                                                       | none needed                        |
| Arsenal or royal dockyard     | Anchor plus a castle tower                                                                                            | "Royal Dockyard"                   |

**MAP TIP**

Date new things on your map or in its notes: "New Polder (drained 1612)", "Citadel (1567)". Dates make the history visible and help you keep the layers consistent.

**FANTASY TWIST**

In many fantasy worlds, magic plays the role of gunpowder: a new power that makes old defences useless. Use the real pattern: the rich centre adopts it first, old walls get patched, special forts appear on the frontier, and the remote countryside barely changes for generations. See [Fantasy Variants](#) and [Common Mistakes and How to Fix Them](#).

**Soon after 1650:** Several famous changes come just after this chapter's period. Vauban built France's great fortress belt (the *pré carré*) from the 1670s. The Canal du Midi (1666–81) linked the Atlantic and the Mediterranean. Versailles became the model palace capital from the 1680s. The potato and turnpike (toll) roads changed the countryside mostly in the 1700s.

## Quick summary

- After 1500 the countryside stays medieval; the change is at the top: capitals, ports, forts and long-distance links.
- Europe grows from about 62 to 78 million (without Russia and the Ottoman lands) between 1500 and 1600, then stalls. Capitals and ocean ports grow fastest: London went from about 50,000 to about 400,000 between 1500 and 1650.
- City ranks change quickly: Antwerp rose to about 100,000 and then collapsed; Amsterdam, London, Seville and Lisbon boomed.
- Cannon made high walls weak. Bastioned "star" fortifications appeared in Italy from the late 1400s, matured there c. 1500–1540, and spread north in the 1530s–40s. They were very expensive, so only frontiers, capitals, ports and key crossings get them.
- Citadels control cities; chains of towers and gun forts guard coasts; armies grow to over 150,000 for great states, and sieges can last years.
- Old castles become ruins, palaces, prisons or are replaced by country houses. Strong rulers order most castles destroyed: France in 1626; Japan in 1615, from about 3,000 castles to about 170.

- Ocean ships of several hundred to 2,000 tons link monopoly ports (Seville, Lisbon) with trading posts abroad; chartered companies (1600, 1602) build fortified "factories". Spain plants colonial towns on one standard grid around a central plaza (rules of 1573).
- Relay posts with stations every 20–40 km (12–25 mi) carry letters up to about 150 km (95 mi) a day; pound-lock canals start to climb hills (Briare, 1604–42).
- Exchanges (Antwerp 1531, London 1571, Amsterdam 1609–11) and printing (236 towns by 1500, 1,300+ places by 1650) create new kinds of special towns.
- Borders become lines on maps but are still lists of villages on the ground; surveying and atlases arrive.
- The countryside sees more enclosure, a few new crops (mainly maize), drained polders and fens, blast furnaces, and mining boom towns.
- The Reformation leaves ruined and converted abbeys and clear religious borders.

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Population, density and travel figures follow the guide's shared baseline (de Vries 1984; Broadberry, Campbell et al. 2015; Bosker, Buringh and van Zanden 2013; Chandler 1987). Values marked "estimate" (fort sizes, tower spacing, trading-post sizes) are rough guides derived for this chapter, not measured averages.

## CHAPTER 10

# FANTASY VARIANTS: MAGIC, MONSTERS AND OTHER RACES

The other chapters describe a world without magic. This chapter shows how to add magic, monsters and non-human peoples without breaking that realism. The method is simple: apply the real rules first, then ask which **input** a fantasy element changes (water, food, defence, travel cost, threats or resources) and follow the consequences. Every adaptation here is tied to a real historical case that worked the same way.

## IN THIS CHAPTER

The method: change the inputs, not the rules  
 Dangerous wilderness and monsters  
 Magic, effect by effect  
 Non-human peoples: archetypes with real analogues  
 Contact zones between peoples  
 Gods, undead and dangerous seas  
 What changes if: summary table  
 Suggested symbols  
 Consistency checklist  
 Quick summary  
 Sources and further reading

## The method: change the inputs, not the rules

## RULE OF THUMB

Fantasy does not cancel the rules of geography. It changes the numbers you feed into them. Change one input, then follow the chain of effects across the whole map.

Every settlement rule in this guide comes from six inputs. Each fantasy element changes one or more of them.

| Input              | The realistic rule (see chapter)                                                                                                                                                                                                            | Fantasy elements that change it                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Water</b>       | Every settlement needs reliable water. Desert towns exist only at oases, rivers or <i>qanats</i> (underground water channels). See <a href="#">Where Settlements Are Built</a> .                                                            | Water magic, gods, cursed or poisoned rivers            |
| <b>Food</b>        | About 1 ha (2.5 acres) of arable land per person. Good lowland holds 30–50 people per km <sup>2</sup> (80–130 per sq mi). A town of 10,000 needs farmland within 21–27 km (13–17 mi). See <a href="#">Population and Settlement Sizes</a> . | Fertility magic, weather control, healing (more mouths) |
| <b>Defence</b>     | Towns of 2,000+ are usually walled. A castle controls about 15–30 km (10–19 mi) around it. See <a href="#">Castles, Forts and Military Outposts</a> .                                                                                       | Monsters, flying enemies, magical wards, undead         |
| <b>Travel cost</b> | Land : river : sea cost $\approx$ 8 : 4 : 1. Towns over 10,000 sit on navigable water. See <a href="#">Trade Routes, Roads and Transport</a> .                                                                                              | Portals, flying mounts, dangerous seas, guarded roads   |
| <b>Threats</b>     | War, raids, famine, plague                                                                                                                                                                                                                  | Monsters, dragons, raiding peoples, blights, angry gods |
| <b>Resources</b>   | Ore, salt, stone and timber create special towns. See <a href="#">Industry, Resources and Special Towns</a> .                                                                                                                               | Magic crystals, ley lines, monster parts, dragon hoards |

**Why this works:** a reader cannot check your magic, but they can feel when a map is wrong. A city in a desert with no water looks wrong. A desert city with a great water-temple at its centre and a ring of gardens around it looks right, because the cause is visible.

#### MAP TIP

For each fantasy element, write one line: "X exists, so Y changes." For example: "Griffon riders exist, so castles have roofed towers and every lord keeps an aerie." If you cannot write the second half, the element is only decoration. That is allowed, but it will not make the map feel real.

## Dangerous wilderness and monsters

#### RULE OF THUMB

Danger concentrates people. The more dangerous the countryside, the fewer, bigger and better-defended the settlements, and the wider the empty land between them.

**Why:** a lone farm cannot defend itself against a troll or a raiding party. A wall is cheaper per person when many people share it. Fields must be close enough to a refuge

that workers can run back in time. All three push people together.

#### How danger reshapes settlement

The real world already gives us every level of danger. Pick the row that fits each region of your map.

| Threat level                                 | Real analogue                                                                                                    | Settlement pattern                                                             | Defences                                                                                                                  | How it looks on the map                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Low</b> (wolves, bandits)                 | England or France c. 1250, away from borders                                                                     | Normal mix: nucleated (clustered) villages on good land, lone farms in uplands | Working castles about 20–30 km (12–19 mi) apart (see <a href="#">chapter 04</a> ); villages unwall                        | The baseline map in the other chapters                               |
| <b>Seasonal raids</b> (a few times a decade) | Anglo-Scottish border, 14th–16th centuries                                                                       | Lone farms survive but become tower houses; villages huddle                    | Tower houses and fortified farmhouses. Late medieval England had over 250 tower houses, Scotland ~800, Ireland over 3,000 | Small solid squares (tower houses) scattered over farmland           |
| <b>Frequent raids or large monsters</b>      | Latium (central Italy) c. 920–1030; Wessex in the 870s–890s; the Provence coast in the 900s                      | Farms abandoned; people move into walled or hilltop villages of 300–1,000      | Walled villages; refuge forts within a day's walk; watchtowers; beacons                                                   | Fewer, bigger dots on hilltops; empty valley floors; coasts deserted |
| <b>Constant danger</b>                       | The "Great Wilderness" between the Teutonic Order and Lithuania (13th–14th centuries); Crusader frontier castles | Only fortified towns and garrison castles; fields only near walls              | Great garrisons. The Templar castle of Safed (1260s) held 1,700 people in peacetime and 2,200 in war                      | Islands of settlement in empty land                                  |

How far can fields be from a refuge? There is no measured figure. A useful heuristic is **about one hour's walk: 4–5 km (2.5–3 mi)**. That gives workers time to reach the walls when the alarm sounds.

#### Nucleated and walled villages

The best real case is Italian *incastellamento* (the "castle-ing" of the countryside). In Latium (the region around Rome), scattered farms were regrouped into fortified hilltop villages. Pierre Toubert (1973) put the main wave between about 920 and 1030; other historians see it continuing into the 12th century (see [Where Settlements Are Built](#)). Historians still argue how much of this was caused by raids and how much by lords wanting control. For your map, the res-

ult is what matters: hilltop villages everywhere, and few houses in the open.

Where raids are seasonal, lone farms do not disappear. They turn into towers. On the Anglo-Scottish border, families built small stone tower houses and fortified farmhouses so they could shut themselves and their animals in for a few days.

The most extreme refuge is underground. In Cappadocia (central Anatolia), the underground city of Derinkuyu reaches about 85 m (280 ft) deep. It was greatly enlarged during the Arab–Byzantine wars (roughly 780–1180) as a refuge from Arab raids. It may have sheltered up to about 20,000 people with their animals and food, although that is an estimate, not a count.

## Refuge networks and frontier forts

When King Alfred of Wessex faced Viking armies in the 870s–890s, he built a network of *burhs* (fortified towns). The Burghal Hidage is a list of these burhs and the land assigned to man them. It is usually dated to the early 900s, although some historians argue that it describes Alfred's own plan of about 880. Its two versions together list 33 burhs:

- Each hide (a unit of taxed land) supplied one man.
- Four men defended each pole (about 5 m or 16½ ft) of wall.
- Winchester was assessed at 2,400 hides, so 600 poles, or about 3 km (1.9 mi) of wall. This closely matches the length of its old Roman walls.
- The burhs were spread so that few country people lived more than about 24–32 km (15–20 mi), roughly one day's walk, from one.

## Watchtowers and beacons

A warning network buys time. Two real systems show the range of options:

| System                 | Date                | Length and spacing                                                                                                                                        | What it did                                                                                                                                                                                                                                                                                            |
|------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Byzantine beacon chain | 9th century         | About 720 km (450 mi), about nine mountain-top stations 55–100 km (35–60 mi) apart (details in <a href="#">Watchtowers, beacons and coastal warning</a> ) | Sent one of 12 pre-arranged messages, each tied to an hour of the day. Modern experiments suggest a message crossed the whole line in about an hour                                                                                                                                                    |
| Hungarian <i>gyepű</i> | 11th–13th centuries | A belt around the whole kingdom                                                                                                                           | An outer border zone deliberately kept empty, blocked with felled trees, hedges, ditches and banks of earth and stone. Gates were the only ways through. Guard settlements of border troops (often archers) kept watch. It failed against the Mongols in 1241, after which Hungary built stone castles |

How close should local watchtowers be? They must see each other's smoke or fire. In broken or wooded country that means a few kilometres apart. Mountain-top beacons can be 55–100 km (35–60 mi) apart. These are rough figures.

### LATER ERA (1500s+)

Muscovy's Great Abatis Line (complete by 1566) and the later Belgorod Line (1635–1654; sources differ, and one gives 1646 as the end date) are real versions of this: belts of felled trees, ditches, palisades (walls of wooden stakes), watchtowers and small forts that slowed raiders so villagers could flee and troops could gather. See [Fortified lines on open frontiers](#).

## Guarded roads

Roads through dangerous country need clearing, patrols and safe stops. All three existed:

- **Clearing:** England's Statute of Winchester (1285) ordered highways between market towns to be cleared for 200 feet (about 60 m) on each side. Ditches, under-

This is the best real model for a monster-haunted kingdom: **a planned net of refuges about a day's walk apart, each manned by the farms around it.**

Denmark shows a different version. In about 970–980, King Harald Bluetooth built five large circular "ring fortresses" (Trelleborg, Fyrkat, Aggersborg, Nonnebakken and Borgring) spread across his kingdom. They were royal strongholds that showed the king's power, more than refuges for farmers. A fantasy king can do the same: a few identical, planned forts placed across the realm at once.

Rivers can carry the threat inland. Against Viking ships, the Frankish king Charles the Bald ordered fortified bridges at towns on rivers in the Edict of Pîtres (864). The best known was at Pont-de-l'Arche on the Seine. If your monsters swim or sail, expect fortified bridges and river barriers below important towns.

growth and bushes had to go, so that no robber could hide close to the road. Big trees such as oaks could stay if the ground under them was kept clear. The gates of walled towns were to be shut from sunset to sunrise, and a night watch kept.

- **Patrols:** in the 12th century, towns and peasants in northern Spain formed brotherhoods (*hermandades*) to police the pilgrim road to Santiago. In 1476 the Catholic Monarchs (Isabella and Ferdinand) set up the Santa Hermandad, a kingdom-wide police force with power over crimes on roads and in open country. The Knights Templar were founded c. 1119–1120 to protect pilgrims on the roads of the Holy Land.
- **Safe stops:** the hospice on the Great St Bernard Pass in the Alps was founded in 1049 (the traditional date) to shelter travellers. Put such stops a day's travel apart: 15–30 km (10–19 mi) on foot roads, 30–40 km (19–25 mi) on caravan routes.

The opposite also happened. Muslim raiders from al-Andalus set up a base at Fraxinetum on the Provence coast

in about 889. From about 906 until they were driven out in 972–973, they held some or all of the western Alpine passes. The chronicler Flodoard wrote in 951 that they let travellers to Rome pass only after paying tribute. **A monster that holds a pass becomes a toll-collector.** Traffic either pays, fights, or finds another pass.

### Hunters, rangers and their lodges

Kings paid people to fight dangerous animals:

- Charlemagne's estate rules, the *Capitulare de villis* (c. 800), told stewards to hunt wolves and take wolf cubs in May. The French royal wolf-hunters (the *louveterie*) trace their origin to wolf-hunters (*luparii*) appointed by Charlemagne. French tradition dates this to 813.
- William of Malmesbury (writing c. 1125) says King Edgar (r. 959–975) took a tribute of 300 wolf skins a year from the Welsh kings. He wrote over a century after the events, and the story may be exaggerated.
- Edward I of England employed Peter Corbet (1281) to destroy wolves in five western counties.
- James I of Scotland (1427) ordered three wolf hunts a year.

Wolves were rare or gone in England by about 1500. They survived in Scotland until the 1600s or later. Organised hunting with royal support worked, but it took centuries.

#### MAP TIP

Put hunters' or rangers' lodges (a small house mark labelled "Lodge") at forest edges, at fords and passes, and where a road enters the wild. Space them about a day's patrol apart, 15–30 km (10–19 mi), which matches a castle's control radius. A monster-hunting order is a military order (like the Templars): give it castles on the frontier and a chapter house (its headquarters building) in a city.

### "Points of light": use it for frontiers, not whole kingdoms

"Points of light" is a game-design term from *Dungeons & Dragons* 4th edition (2008): a mostly wild world with a few small, isolated islands of civilization. It is a good picture of a **frontier**, but not of a whole realm. In real history, even dangerous kingdoms had a dense core where peasants walked to market in safety.

You can estimate the size of each "point". Suppose people farm only within 4–5 km (2.5–3 mi) of their walls. That circle covers about 50–80 km<sup>2</sup>. At 30–50 people per km<sup>2</sup>, each point holds roughly **1,500–4,000 people**. Larger towns are possible only where a river or a protected sea route brings food from farther away.

#### MAP TIP

Divide your map into three zones: safe core (no shading), frontier (light hatching, the guide's symbol for a march or frontier zone) and wild (dense cross-hatching). Do not use plain colour tints for these zones, because a soft tint means a language or culture area. In the core, use the normal rules. On the frontier, use fewer, bigger, walled settlements on hills, with watchtowers, refuge forts about a day's walk apart, and cleared roads with fortified stops. In the wild, draw only ruins, lairs and the occasional ranger lodge. A frontier band of 30–100 km (20–60 mi) is a good default.

## Magic, effect by effect

Take each kind of magic separately. Ask which input it changes, how strongly, and **who controls it**. The answer to the last question tells you where the new buildings go.

### Food and fertility magic

#### RULE OF THUMB

Fertility magic raises the ceiling on population. It does not remove the transport limit. Cities above about 50,000 still need water transport, unless magic also moves the grain.

**Why:** with a typical medieval yield of 4 : 1, a farmer keeps back 1 of every 4 units of grain as seed and eats or sells 3. If magic doubles the yield to 8 : 1, the farmer keeps 7. Net food more than doubles. The land fills up with people, and the surplus feeds bigger towns.

The best real comparison is the most densely farmed land in medieval Europe: Flanders and Holland in the 15th century. The figures below come from the model in [Population and Settlement Sizes](#). The "moderate" row is interpolated.

| Strength of the magic    | Treat the region like               | Rural density                                 | Share living in towns | Farmland radius for a town of 10,000                         |
|--------------------------|-------------------------------------|-----------------------------------------------|-----------------------|--------------------------------------------------------------|
| None                     | Good lowland c. 1300                | 30–50 per km <sup>2</sup> (80–130 per sq mi)  | 10–15%                | 21–27 km (13–17 mi)                                          |
| Moderate (+50% yield)    | Between the rows                    | 40–60 per km <sup>2</sup> (100–155 per sq mi) | 15–20%                | ~15–19 km (9–12 mi)                                          |
| Strong (about 2× yield)  | Flanders and Holland (15th century) | 50–75 per km <sup>2</sup> (130–190 per sq mi) | 20–40%                | ~12–15 km (7–9 mi)                                           |
| Very strong (3× or more) | Nothing in medieval Europe          | Higher still                                  | Higher still          | Small. The limit becomes water, fuel and transport, not food |

**Who controls it?** If priests bless the fields, the temple takes a share, as the real Church took a tithe (a tenth). Expect rich temples in every village and a great temple-estate near the capital. If the magic is in the soil (a blessed valley), that valley becomes the core of a realm, like the Île-de-France around Paris.

#### MAP TIP

Draw a blessed region exactly like Flanders: villages 1.5–2 km (about 1 mi) apart, market towns every 10 km (6 mi), and many walled towns of 5,000–20,000. Keep the edges of the blessing visible: the density should drop where the magic ends.

### Water magic and desert cities

Real desert cities show what water magic would do, because a *qanat* (an underground channel that brings groundwater from the hills) is a kind of engineered spring.

- **Marrakesh** was founded c. 1070 by the Almoravids on a dry plain. From the 11th century, *khettaras* (the Moroccan name for qanats) brought groundwater to it from the direction of the Atlas Mountains. That water turned the site into an imperial capital with large gardens.
- **Yazd and Kerman** in Iran grew at the foot of mountains where qanats could be dug (see [Where Settlements Are Built](#)).
- **Valencia's** irrigation farmers still settle water disputes at a weekly Water Tribunal, held every Thursday at a door of the cathedral. Water rights become law and politics.

**What changes:** each magical spring acts as an oasis. Settlement forms **dots** around each water source, not a carpet. The size of each dot depends on how much water the magic gives. Food is the next limit: irrigated gardens are very productive, but there are only so many of them.

**Who controls it?** If one mage or one temple makes the water, the whole city depends on them. That gives them enormous power. It also makes the city fragile: real desert towns died when their water failed. A water-mage's tower

in a desert city is the citadel, so draw it at the centre, walled.

#### MAP TIP

In a magically watered desert, draw: the water source (temple, tower, sacred well) in the centre; a ring of green gardens around it; then the town; then bare desert. Caravan stops still sit 30–40 km (19–25 mi) apart. Each one needs its own water: either a real well or a smaller magical one.

### Healing magic and population growth

#### RULE OF THUMB

Healing magic makes a population grow faster. It does not raise the ceiling. A healed realm fills its land in a few generations, then hits the food limit.

**Why:** in the Middle Ages births and deaths were nearly equal. Europe's population grew only 2–3 times between 1000 and 1300, which is about 0.2–0.4% a year. If healing removes most deaths from disease and childbirth, growth can jump.

| Yearly growth | Population doubles in | Real example                                                                                                                                                         |
|---------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.2–0.4%      | 175–350 years         | Europe 1000–1300                                                                                                                                                     |
| 1%            | ~70 years             | —                                                                                                                                                                    |
| 2–3%          | 23–35 years           | North America (the British colonies, then the United States). In 1798 Malthus noted that its population had doubled about every 25 years, because land was plentiful |

### What changes on the map:

- Faster frontier expansion: new villages cut into forests, and lords found planned new towns (see [Where Settlements Are Built](#)).
- More pressure on borders and more wars over land.
- No Black-Death-style collapse, so fewer deserted villages and ruins.
- Towns grow on their own. In real medieval towns, deaths outnumbered births, and towns grew only because people moved in from the countryside. Healers in a town would change that.

If healers are rare and expensive, only towns and nobles benefit. Then the towns grow and the countryside does not.

### Teleport circles and portals

#### RULE OF THUMB

A portal is a harbour without a sea. Treat it like a port, a bridge or a strait: a fixed point where traffic gathers and where the ruler can tax it.

### Real analogues:

- **Øresund (the Sound):** from 1429 every foreign ship passing between the North Sea and the Baltic had to stop at Helsingør and pay the Danish king. A castle (later rebuilt as Kronborg) guarded the strait.
- **Staple right:** from 1259, Cologne forced passing merchants to stop, unload and offer their goods for sale for three days before moving on. Any city that owns a portal will try the same.

**What grows around a portal:** a customs house, warehouses, inns, money-changers, a market, a guard fort, and very often a temple. The portal town becomes a hub, like a crossroads town. If one end is in another realm, expect a **twin town** at each end.

**Capacity decides everything.** Compare the portal with a ship. A typical cog (the main northern cargo ship) carried 30–200 tons (see [chapter 05](#)).

| If a portal moves...           | It acts like...                    | Effect on the map                                                                                            |
|--------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
| A few people and letters a day | A royal messenger service          | Capitals govern farther (see <a href="#">Capitals, Realms and Borders</a> ). Trade barely changes            |
| Cartloads of valuable goods    | A mountain pass or a caravan route | Luxury trade (spices, silk, gems) moves through portals. Portal towns get rich. Bulk goods still go by water |
| Shiploads of grain every day   | A great river or sea lane          | Inland cities can grow beyond 50,000 without rivers. The capital can sit wherever the portal network meets   |

**Effect on capitals:** a realm with portals will put its capital at the portal hub or build the network around the capital, the way Roman roads led to Rome. A portal into the capital is also a weak point. Guard it like a harbour mouth: Constantinople closed its harbour, the Golden Horn, with a chain.

### Flying mounts and airships

#### RULE OF THUMB

Flying moves people, messages and luxuries. It does not move bulk. A griffon carries about what a pack horse carries.

**Why:** a pack horse carries about 100–120 kg. A flying mount strong enough to carry a rider can carry a similar load. One cog carried 30–200 tons, which equals roughly **250 to 2,000** flying loads. Grain, timber, stone and wine stay on rivers and seas.

**What changes:**

| Change                  | Real analogue                                                                                                                                                   | Map result                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fast mes-sages          | Nur ad-Din ran a regular pigeon post between Baghdad and Syria by 1167. The Mamluk sultan Baybars (r. 1260–1277) rebuilt the state relay post, the <i>barid</i> | A chain of relay aeries (nests or stables for flying mounts) a day's flight apart. If a griffon flies about 300 km (190 mi) a day, as in <i>Capitals, Realms and Borders</i> , the capital can govern much farther |
| Trade over mountains    | Mule trains over Alpine passes carried mostly high-value goods                                                                                                  | Gems, spices and letters fly straight over the range. Pass towns lose travelers and post but keep bulk freight                                                                                                     |
| Aeries                  | Royal mews (falcon houses) and pigeon lofts sat in castles and towns                                                                                            | Aeries on crags and tower tops, with pasture or herds nearby to feed the animals                                                                                                                                   |
| Defence against the sky | No direct real case                                                                                                                                             | Roofed towers and wall-walks, bolt-throwers on towers, small courtyards, nets, lookouts on every peak, and keeps that can be held against an attack from above                                                     |

**How fast do fortresses change?** The best real comparison is gunpowder. After the French invasion of Italy in 1494, new artillery forts with low, thick walls and angled bastions spread across Italy within a few decades. If flying enemies are new in your world, show a mix: old open-topped castles beside new roofed ones.

**LATER ERA (1500s+)**

The bastion fort (*trace italienne*) is the real example of fortresses redesigned for a new threat. See [What Changes After 1500](#).

**Weather control**

**Real analogue:** around 815, Archbishop Agobard of Lyon wrote *On Hail and Thunder* against a popular belief in *tempestarii*, "storm-makers" who could raise or turn away hail. He complained that peasants paid them a share of their crops for protection. So even an imaginary weather-mage got paid like a tax collector.

**What changes:**

- **Fewer famines.** The Great Famine of 1315–17 was caused by years of heavy rain. Weather control prevents that kind of disaster, so there are fewer deserted villages.

- **Farming moves into marginal climates.** The Norse settled Greenland c. 985 and abandoned it in the 15th century, partly because the climate cooled. Weather magic would keep such colonies alive and push farming higher up mountains and farther north.
- **Safer seas.** Reliable winds make sea trade cheaper and more ports viable.
- **Weather as a weapon.** The Mongols tried to invade Japan by sea in 1274 and 1281. A storm hit the 1274 fleet as it withdrew (how much damage it did is debated). In 1281 a typhoon destroyed much of the second fleet at anchor. The Japanese called it *kamikaze*, the "divine wind". Coastal realms with storm-mages would be hard to invade by sea.

**Who controls it?** Like the lord's mill, which peasants were often forced to use and pay for, a weather-mage would be a lordly monopoly. Put weather towers on hilltops overlooking the most valuable farmland.

**Magical resources: crystals, ore and ley lines****RULE OF THUMB**

A magic crystal is ore. A ley line is a river. Place them, then apply the real rules for mines and rivers.

**Crystals as ore.** Real silver strikes show how fast a boom town grows:

| Real boom town       | Timeline                                                                                                                  | Size and features                                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kutná Hora (Bohemia) | Mining from c. 1260; royal mining code in 1300; central mint soon after 1300; main mining zone closed in 1543             | Grew fast and became the second town of Bohemia after Prague. Around 1300 its mines may have produced about a third of Europe's silver. (Some popular sources say 60,000 people. That is unlikely for a town that was smaller than Prague, so use a much lower figure.) |
| Novo Brdo (Serbia)   | Saxon miners brought in under King Milutin (1282–1321); named in 1326; taken by the Ottomans in 1455 after a 40-day siege | About 6,000 kg of silver a year by 1450. In the 15th century 4,000–5,000 people lived inside the walls and 8,000–10,000 more in satellite settlements around them                                                                                                       |
| Freiberg (Saxony)    | Silver found in 1168                                                                                                      | Grew into a major mining town                                                                                                                                                                                                                                           |

What a crystal boom brings: miners from far away with their own laws; a fortress and garrison; a mint; roads for food coming in; charcoal burners stripping the forests; and, when the lode runs out, decline. See [Industry](#), [Resources](#) and [Special Towns](#).

**Ley lines as rivers.** The modern idea of ley lines comes from Alfred Watkins in the 1920s. He saw them as old straight tracks. Only in the 1960s were they reimagined as lines of earth energy. Archaeologists reject them as real features, but in fantasy they are useful. Treat them like rivers:

- Towers and towns form **chains** along the line.

- A **nexus** (where lines cross) acts like a confluence (the place where two rivers meet): it attracts the biggest settlement and the strongest fortress.
- A place where a line is easy to reach acts like a ford: expect a tower, a toll and a fight over who owns it.

### Mage towers and academies

#### RULE OF THUMB

Place academies like universities (in big cities) and lone mage towers like monasteries (remote, but supplied).

| Type                            | Real analogue                                                                                                                                                                               | Numbers                                                  | Where                                                                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academy of magic                | Universities: about 20 in Europe c. 1300, 60–80 by 1500. England had only two (Oxford and Cambridge)                                                                                        | 1–3 per kingdom                                          | In cities of 10,000+: they need food, lodging, books, patrons and students                                                                                          |
| Order of mages with many houses | Cistercian monks: about 500 houses by 1300 and nearly 750 at their peak in the 1400s; roughly 75–80 abbeys of monks in England and Wales (about one per 2,000 km <sup>2</sup> or 770 sq mi) | Dozens per kingdom, if magic is as common as monasteries | Remote valleys, but with granges (outlying farms run by the monastery). Some granges were next door; others were tens of kilometres away. Their size varied greatly |
| Dangerous experiments           | Leper houses and plague hospitals, placed outside towns or on islands                                                                                                                       | A few per city                                           | Outside the walls, downwind (so smoke and smells blow away from the town), across water                                                                             |

### Magical defences

Wards (magical barriers) behave like walls. A wall's cost depends on its **length**, but the space inside depends on its **area**. So big wards are cheaper per person than small ones. This favours fewer, bigger warded towns, the same logic as the burhs, where each hide paid for a share of the wall.

Ask two questions:

1. **What does the ward stop?** If it stops monsters but not armies, towns still need walls, and castles stay. If it stops armies too, castles shrink and frontier forts become ward-stone posts that are watched and maintained.

2. **Who pays for its upkeep?** English towns paid for their walls with *murage*, a toll the king allowed them to charge. A warded town needs a ward-tax and a mage-guild, and it falls when the mages leave.

#### MAP TIP

Add magic in this order: (1) mark magical resources (crystals, ley lines, springs) like ore and rivers; (2) place the towers, academies and boom towns that use them; (3) adjust density for fertility or healing magic; (4) add the new hubs (portals, aeries); (5) adjust the fortresses. Do this after the realistic steps in [Step-by-step](#), not instead of them.

## Non-human peoples: archetypes with real analogues

Each common fantasy people matches a real way of life.

Use the real way of life to decide where they live, how many there are and how their land looks.

| People                                | Real analogue                                                                                    | Settlement type                                       | Density                                                   | Map signature                                                                         |
|---------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Dwarves</b>                        | Medieval mining communities (Saxon miners in Bohemia and Serbia); Capadocian underground refuges | Mountain holds plus trade towns at the mountain gates | Holds like mining towns: a few thousand to perhaps 15,000 | Gate towns on valley floors, mine symbols (crossed hammers), smelters, cleared slopes |
| <b>Elves</b>                          | Forest peoples of the Baltic: pagan Lithuanians and Prussians                                    | Small clearings, hill-forts, sacred groves            | 1–5 per km <sup>2</sup> (3–13 per sq mi) or less          | Big forest, few dots, no roads, wide border wilderness                                |
| <b>Halflings</b>                      | Free peasant communities of fertile lowlands: Dithmarschen, Flanders                             | Dense villages, mills, small market towns             | 30–75 per km <sup>2</sup> (78–190 per sq mi)              | Many small villages, few castles                                                      |
| <b>Orcs and other "horde" peoples</b> | Steppe nomads: Mongols, Cumans, Crimean Tatars                                                   | Seasonal camps, a few royal cities                    | Under 1–2 per km <sup>2</sup> (under 3–5 per sq mi)       | Pastures, camps, raid trails, frontier markets                                        |
| <b>Merfolk</b>                        | Sea nomads such as the Orang Laut of the Strait of Malacca                                       | Reefs and sheltered waters, shore markets             | —                                                         | Shore market towns, safe or unsafe straits                                            |
| <b>Giants and dragons</b>             | Raiders who are paid tribute (Vikings in England); predators that need huge ranges               | Lairs                                                 | —                                                         | Empty rings around lairs, tribute villages beyond                                     |

### Dwarves: mountain holds and gate towns

**Real model:** medieval mining towns. German ("Saxon") miners settled in Bohemia, Hungary and Serbia under special privileges. They had their own mining law, their own courts and their own churches. Novo Brdo and Kutná Hora (see the table above) show the size: a few thousand people inside the walls and more in the settlements around them.

**The food problem:** a hold of 10,000 people needs farmland within 21–27 km (13–17 mi) on good lowland, or about 55 km (35 mi) on marginal land. Mountains are marginal land. So a dwarf hold must buy food. That creates a **gate town**: a market town on the nearest valley floor, ideally on a river, where grain comes in and metal goes out. Treat it like a pass town or a mining town's market.

**Underground roads:** real tunnels were short. The tunnel said to link Derinkuyu with the next underground town, Kaymaklı, is about 8–9 km (5–5.6 mi) long. A long underground road between holds is a wonder of the world. Draw a few, not a network.

#### MAP TIP

For each dwarf hold, draw one gate town on the valley floor, a road climbing to the gate (a small, clearly rounded arched doorway set into the mountainside, so it is not read as a flat-topped gallows), a mine symbol (crossed hammers), smelter symbols on the stream below, and bare slopes where trees were cut for charcoal. Group holds into a league (like the Hanse) with one leading hold.

### Elves: forest realms

**Real model:** the forest and marsh lands of pagan Lithuania (Christian only from 1387) and the Baltic Prussians. Density was low: 1–5 people per km<sup>2</sup> (3–13 per sq mi). Their realm had no sharp line border. Between the Teutonic Order and Lithuania lay the "Great Wilderness", a forest belt emptied

by raids from both sides and crossed by war parties in summer and winter.

**What it looks like:** a large forest with few, small clearings; hill-forts and sacred groves; paths but no roads; and a wide empty border. Human villages pushing into the forest edge with new clearings are a source of conflict.

**Contact:** reclusive peoples may trade without meeting. Herodotus (5th century BC) described "silent trade" on the West African coast: Carthaginian traders left goods on the beach, lit a smoky fire and withdrew, and local people left gold beside the goods. Silent trade continued in the West African gold trade in the Middle Ages. Mark a few **trading stones** at the forest edge.

### Halflings: fertile farmland

**Real model:** self-governing peasant societies on rich land. Dithmarschen, a marshland on the North Sea coast of Germany, was a peasant republic from the 13th century until 1559. Its parishes ruled themselves. In 1500, at Hemmingstedt, its farmers used dykes and flooded ditches to defeat a Danish royal army.

**What it looks like:** density like Flanders (50–75 per km<sup>2</sup>) or good lowland (30–50 per km<sup>2</sup>). Villages 1.5–4 km (1–2.5 mi) apart, many mills, market towns 9–16 km (6–10 mi) apart. Very few castles. Their neighbours, however, build castles along the halfling border.

### Orcs and other "horde" peoples

Do not draw them as mindless. Model them on real steppe societies, which had laws, crafts, trade and some of the best logistics of the Middle Ages.

#### How steppe societies worked:

- **Herds:** life rested on sheep (meat, milk, wool, fuel) and horses (milk, riding, war). Each warrior rode with sev-

eral horses, so he could change to a fresh one. Estimates vary: 3–4 horses per Mongol soldier, or 5–8 per steppe warrior (Devereaux).

- **Few people, much land:** density was under 1–2 per km<sup>2</sup> (under 3–5 per sq mi). In 1206 the whole Mongol army probably numbered under 100,000 men (Timothy May, cited by Devereaux).
- **Seasonal movement:** camps moved in a regular yearly cycle, usually between summer pastures (uplands or open steppe) and sheltered winter camps (river valleys). For real camp sizes, how often camps moved and how far, see [Steppe and grassland](#). Use those figures for orc camps too.
- **Dependence on farmers:** nomads needed grain, metal and cloth that they could not produce. They got them by trade or by raiding (Khazanov 1984). Ming China refused the Mongol leader Altan Khan's requests for trade for years, and he raided instead; in 1550 he even burned the suburbs of Beijing. In 1571 the Ming gave him a title and trading rights, and border horse markets opened at Datong, Xuanfu and other garrisons. After that, raiding fell sharply.
- **Cities too:** nomad rulers built capitals and trade towns. Karakorum was built in the 1220s–1230s. Batu founded Sarai on the lower Volga in the 1240s–1250s. The Franciscan friar William of Rubruck, who passed by in 1253–54, described it as a new town that Batu was still building. The word "horde" comes from *ordu*, the ruler's camp and court.

**What it looks like:** open grassland with seasonal camps (small open half-circles for round tents, never the dashed rectangle that means an army camp); winter camps along rivers; a royal camp or capital; raid trails into farmland (like the Muravsky Trail used by Crimean Tatar raiders into Muscovy); frontier markets and defence lines on the farmers' side.

#### RULE OF THUMB

If a horde raids constantly, ask why the frontier markets are closed. Raids and trade are two ways of getting the same goods.

### Merfolk: coastal trade

**Real model:** the Orang Laut ("sea people") of the Strait of Malacca. For centuries they served the trading states of Srivijaya and later Malacca as pilots, fishermen and a naval force. They kept the straits safe for friendly ships and dangerous for others.

**What it looks like:** markets at sheltered bays and river mouths where land and sea peoples meet. The sea people

sell fish, pearls, coral and salvage. The land people sell metal, grain, timber and cloth. A land realm allied with merfolk controls the nearby strait. An enemy has to fight for every harbour.

### Giants and dragons: no-go zones and tribute

**Tribute:** a power you cannot defeat, you pay. England paid the Vikings huge sums in silver: 10,000 pounds in 991, 16,000 in 994, 36,000 in 1007 and 48,000 in 1012 (*Anglo-Saxon Chronicle*). A dragon that takes tribute creates a belt of paying villages, an official who collects the tribute, and perhaps a treaty stone or shrine where it is delivered.

**Food:** a big predator needs a big range. Here is a rough example. Suppose a dragon eats one cow a week, or 52 a year. A herd can lose perhaps 10–20% of its animals a year without shrinking (an assumption). The dragon then needs a herd of about 250–500 cattle. That is the cattle of many villages. A Domesday plough-team, for comparison, had eight oxen.

**No-go zones:** around a lair (a small skull with the creature's name), draw an empty ring, then a thin belt of tribute villages, then normal land. Giants in the mountains work the same way. Their lands look like a dangerous frontier, with watchtowers on the human side and ruined farms in between.

#### MAP TIP

Give each non-human people its own density and pattern. Then draw the **edges**: gate towns for dwarves, trading stones for elves, frontier markets for nomads, shore markets for merfolk, tribute villages for dragons. The edges are where most of the story happens.

## Contact zones between peoples

### RULE OF THUMB

Peoples who distrust each other trade at fixed places, under fixed rules, near the edge of one realm. Draw those places.

### Trading posts and merchant compounds

| Real example                                                             | Dates          | How it worked                                                                                                                                                                                                                                                                                                                      | Fantasy use                                                    |
|--------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Peterhof, the Hanseatic <i>kontor</i> (trading post) in Novgorod         | c. 1200–1494   | A palisaded compound of wooden houses around a stone church of St Peter, which was also the warehouse. About 150–200 German merchants per season, arriving as summer and winter groups. They lived under their own written rules (the <i>Schra</i> )                                                                               | A dwarf or human merchant compound inside an elven or orc town |
| Fondaco dei Tedeschi, Venice                                             | From 1228      | A building beside the Rialto where German merchants had to lodge and store goods, so Venice could watch and tax them. The word comes from the Arabic <i>funduq</i> (an inn and warehouse)                                                                                                                                          | A "strangers' inn" where foreign races must stay               |
| Galata (Pera), the Genoese colony across the harbour from Constantinople | 1267/1273–1453 | A self-governing colony with its own governor. The Genoese walled it despite a ban, and built the Galata Tower in 1348                                                                                                                                                                                                             | A foreign quarter that becomes a fortress                      |
| Caffa (Kaffa), the Genoese port in Crimea                                | 1266–1475      | Founded with the consent of the Golden Horde (the Mongol khanate that ruled the steppe). It traded grain, furs, slaves, silk and spices between the steppe and the Mediterranean. Some sources claim over 70,000 people, but figures for medieval Caffa are very uncertain and probably exaggerated; treat it as a large port town | A human port on the edge of orc or nomad lands                 |

Other kontors stood in London (the Steelyard), Bruges and Bergen (Bryggen, which closed in 1754).

### Frontier markets

- **China's frontier markets:** from the Song dynasty (960–1279), China traded tea for Tibetan horses at fixed markets in Sichuan and Gansu. The Ming briefly opened horse markets for the Mongols in 1551 but soon closed them again, and raids went on. In 1571 lasting border markets opened at Ming garrisons such as Datong and Xuanfu.
- **Silent trade:** for peoples who will not meet (see [Elves](#)).

Frontier markets sit **at a garrison or gate**, often open only on fixed days, with officials, interpreters and a strong guard.

### Mixed cities

Real mixed cities kept quarters. Toledo after its conquest by Castile in 1085 had Christians, Muslims and Jews. Norman Palermo (from 1072) had Latin, Greek and Arabic speakers. Many cities had a Jewish quarter and foreign merchant quarters, often walled or gated. In a fantasy city, give each people its own quarter with its own temple, market and gate, and put the shared market in the middle.

### Borders between peoples

The Anglo-Scottish border is a good model for a tense border between two peoples. In 1249, six English and six Scottish knights wrote down the "laws and customs of the march" (the border zone). Later agreements (1398) called for monthly "days of truce" where wardens met to settle crimes across the border. The meeting places were fixed: fords, streams and border villages such as Kershopefoot and Redden Burn. For border types in general, see [Capitals, Realms and Borders](#).

### FANTASY TWIST

A shared magical language, or telepathy, removes the need for interpreters and silent trade. It does not remove the need for rules, guards and fixed meeting places. Distrust, not language, is what makes contact zones.

### MAP TIP

Along each border between peoples, mark: two or three frontier markets at gates or fords; one treaty or truce meeting place; a merchant compound in the largest town on each side; and a mixed town where the main trade route crosses the border.

## Gods, undead and dangerous seas

### Gods with real power

**Real baseline:** the medieval Church was already a huge landowner. In England in 1086, bishops and abbeys held about a quarter (25–26%) of the land, measured by value. The Peace of God movement (first at Charroux in 989) declared churches, clergy and peasants off-limits to fighting. Later councils added pilgrims and merchants. The Truce of God (first at Toulouges in 1027) at first banned fighting on Sundays. Later truces stretched the ban from Wednesday evening to Monday morning, and to holy seasons. These rules were often broken. A god who **enforced** them would change the map:

- **Sanctuaries become real safe zones.** Expect villages and markets clustered around temples, and roads under divine protection busier than other roads.
- **Temple towns grow at sacred sites**, even where trade does not favour them (see the "why is it here?" test in [Where Settlements Are Built](#)). They still need food and water.
- **Temples own more land.** If temples hold a third or half of the realm, draw temple estates and granges across the best farmland.
- **Theocracies (states ruled by priests or religious orders) appear.** Real church states existed: the Papal States, and the Teutonic Order's state in Prussia.

Pilgrimage roads, hospices and shrines are covered in [Religious, Cultural, Legal and Ancient Sites](#).

### Undead, plagues and blights

A spreading undead curse or a magical blight works like the Black Death: 30–60% of people dead, towns too small for their walls, and, over the next 150 years, about 1,500 or more deserted villages in England alone. Real societies responded with isolation:

- **Quarantine:** in 1377 Ragusa (Dubrovnik) made travellers from plague areas spend 30 days on the islet of Mrkan or in the town of Cavtat before entering. The word "quarantine" comes from the later 40-day period (Italian *quaranta*, forty).
- **Plague hospitals on islands:** Venice opened its first *lazaretto* (plague hospital) on a lagoon island in 1423.
- **Outsiders outside:** leper houses stood outside town gates.

**What changes on the map:** if the dead can rise, graveyards move outside the walls and get their own walls and watch. Expect quarantine stations on roads and islands, guard

posts at the edge of blighted land, burned or salted villages, and a "dead land" that slowly returns to forest.

### Dangerous seas

If sea monsters or storms make the sea too risky, trade goes inland, even though water transport is otherwise cheaper.

- **Ming China:** grain for Beijing and the northern armies once went partly by sea, a route that suffered from storms and pirates. In 1411–1415 the Ming state rebuilt the Grand Canal with about 165,000 workers. When it reopened in 1415, sea transport of grain to the north ended. Thousands of canal barges (about 15,000 ships, by one count) carried the grain instead. The canal was slower but safer and easier to control. A dangerous sea creates canals and river ports.
- **Convoys:** Venice sent its long-distance trade in state-organised galley convoys (*mude*) from 1315 to 1533. The ships left in spring and came back in autumn. The Flanders convoy carried spices and cotton to England and Flanders and brought wool cloth home. A dangerous sea creates convoy ports, escort fleets and fixed sailing seasons.
- **Coast-hugging:** ships stay near shore and stop each night. A galley averaged about 55 km (34 mi) a day. Expect harbours of refuge about a day's sail apart.

**What changes on the map:** more canals and portages (places where boats or goods are carried overland between two waterways); fortified harbours with chains across the mouth; coastal towns set back from the shore (as on the raided Mediterranean coasts; see [Coasts](#)); lighthouses and watchtowers on headlands; and a few great convoy ports instead of many small ones.

#### MAP TIP

For each of these three, mark the **new places** they create: shrines and sanctuary towns for gods; walled graveyards, quarantine islands and blighted zones for undead; canals, convoy ports and fortified harbours for dangerous seas. Then mark the **places they destroy**: ruins, empty coasts, abandoned villages.

## What changes if: summary table

| If your world has...              | What changes on the map                                                     | Real analogue                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Monsters in the countryside       | Fewer lone farms; walled hilltop villages of 300–1,000; empty valley floors | Incastellamento, Latium, mainly c. 920–1030                                                         |
| Seasonal raiders or beasts        | Tower houses and fortified farms                                            | Anglo-Scottish border, 14th–16th centuries                                                          |
| A monster-haunted frontier        | Refuge forts a day's walk apart; beacons; emptied border belt               | Alfred's burhs; Hungarian <i>gyepű</i>                                                              |
| Monsters along roads              | Cleared verges; patrols; hospices a day apart                               | Statute of Winchester 1285; <i>hermandades</i> ; Great St Bernard hospice                           |
| A monster holding a pass          | Tolls, or traffic shifts to another pass                                    | Fraxinetum raiders in the Alps, 906–973                                                             |
| Fertility magic                   | Density like Flanders (50–75 per km <sup>2</sup> ); more and bigger towns   | Flanders and Holland                                                                                |
| Water magic                       | Desert cities as dots around magical springs                                | Qanat cities: Marrakesh, Yazd, Kerman                                                               |
| Healing magic                     | Faster growth; fast frontier settlement; towns grow by themselves           | Colonial North America (doubling in ~25 years)                                                      |
| Portals                           | Hub towns with customs, warehouses and forts; twin towns                    | Øresund toll (1429); Cologne staple right (1259)                                                    |
| Flying mounts                     | Aeries; relay posts; roofed towers; luxury trade over mountains             | Pigeon post (1167); Mamluk <i>barid</i>                                                             |
| Weather control                   | Fewer famines; farming in marginal climates; storm-defended coasts          | Agobard's <i>tempestarii</i> (c. 815); the storms that hit the Mongol fleets off Japan (1274, 1281) |
| Magic crystals                    | Boom towns with forts, mints and imported food                              | Kutná Hora; Novo Brdo                                                                               |
| Ley lines                         | Chains of towers; a big city at the nexus                                   | Settlements along rivers and at confluences                                                         |
| Mage academies and towers         | Academies in big cities; towers in remote valleys with granges              | Universities; Cistercian abbeys                                                                     |
| Wards                             | Fewer, bigger warded towns; ward-taxes                                      | Burghal Hidage; murage                                                                              |
| Dwarves                           | Mountain holds plus gate towns on valley floors                             | Saxon mining towns                                                                                  |
| Elves                             | Low-density forest realm; wide border wilderness; trading stones            | Lithuania; silent trade                                                                             |
| Halflings                         | Dense farm villages, few castles                                            | Dithmarschen; Flanders                                                                              |
| Steppe "hordes"                   | Seasonal camps; raid trails; frontier markets; defence lines                | Mongols; Ming horse markets (1571)                                                                  |
| Merfolk                           | Shore markets; controlled straits                                           | Orang Laut                                                                                          |
| Dragons or giants                 | Empty rings around lairs; tribute villages                                  | Danegeld (tribute paid to the Vikings, 991–1012)                                                    |
| Other peoples living side by side | Merchant compounds, strangers' inns, quarters, truce days                   | Hanseatic kontors; <i>fondaci</i> ; Anglo-Scottish march law                                        |
| Gods with real power              | Sanctuary towns; rich temples; theocracies                                  | Peace of God (989); Papal States                                                                    |
| Undead or blights                 | Walled graveyards outside towns; quarantine islands; deserted villages      | Black Death; Ragusa (1377); Venice (1423)                                                           |
| Dangerous seas                    | Canals; convoy ports; fortified harbours                                    | Ming Grand Canal (1415); Venetian <i>mude</i>                                                       |

**MAP TIP**

Pick no more than three or four rows from this table for any one region. Each row adds visible features. A region with every fantasy element at once looks crowded and loses its character. Give each region one main fantasy "problem" and show how its people solved it.

## Suggested symbols

**RULE OF THUMB**

One symbol, one meaning. Draw ordinary things (castles, churches, ports, tolls, hospices) with the normal symbols. Give magic its own shapes (diamonds for towers, ley lines and wards; a spiral for portals), so it never looks like a town, a border or a road.

These symbols match the guide's [master legend](#). Symbols marked † are fantasy-only. They are listed in the fantasy group of the master legend.

**Danger and defence:**

| Feature                       | Symbol                                                                                                                       | Notes                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Tower house or fortified farm | Small solid square                                                                                                           | Many in raided borderlands                             |
| Walled hilltop village †      | Village symbol on the hilltop (a tiny dot at kingdom scale); at local scale, a battlemented wall line around the house marks | Use the town symbol only if it really is a town        |
| Refuge fort or burh           | Town symbol with a battlemented ring; a fort with no town inside takes the small castle icon                                 | About a day's walk apart                               |
| Watchtower; beacon            | Small triangle; flame dot                                                                                                    | Draw chains, each in sight of the next                 |
| Hunters' or rangers' lodge †  | Small house mark labelled "Lodge"                                                                                            | A royal hunting lodge takes the small crown instead    |
| Frontier and wild zones       | Frontier: light hatching (the march symbol). Wild †: dense cross-hatching                                                    | Never a plain colour tint, which means a language area |
| Monster lair †                | Small skull with the creature's name                                                                                         | Leave an empty ring around it                          |

**Magic:**

| Feature                         | Symbol                                                                                                       | Notes                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Mage tower or weather tower †   | Slim tower topped with a small open diamond                                                                  | No battlements (castle) and no spire (church)                              |
| Academy of magic †              | Book beside the city name (the university symbol), labelled "Academy"                                        | Only in cities of 10,000+                                                  |
| Magic crystal mine              | Crossed hammers, labelled with the crystal                                                                   | The normal mine symbol                                                     |
| Ley line and nexus †            | Straight chain of small open diamonds in one colour (for example violet); a larger open diamond at the nexus | Not dotted, dashed or dash-dot, so it cannot be read as a road or a border |
| Ward stones †                   | Small solid diamonds spaced around the warded area                                                           | Not the upright tick, which means a standing stone                         |
| Magical spring or sacred well † | Small blue drop, with a green ring of gardens around the town                                                |                                                                            |
| Portal †                        | Small spiral; at a portal town, put it beside the town symbol                                                | Never a star (capital) or a circle (town)                                  |
| Aerie †                         | Small pair of wings on a crag or on a castle icon                                                            | Relay aeries in a chain, a day's flight apart                              |

## Peoples and contact:

| Feature                            | Symbol                                                                                             | Notes                                                                                                                                                  |
|------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dwarf hold gate †                  | Small arched doorway (n) in the mountainside                                                       | Draw the arch clearly rounded and set into the mountain, unlike the flat-topped gallows "П" on a hill. The gate town below uses the normal town symbol |
| Smelter †                          | Small chimney with a curl of smoke                                                                 | On the stream below the mine                                                                                                                           |
| Underground road †                 | Thin brown dotted line between hold gates, labelled "(underground)"                                | Draw a few, not a network                                                                                                                              |
| Nomad camp †                       | 2–3 tiny <b>open</b> half-circles (round tents); the ruler's camp as a large labelled ring of them | Never filled (a filled mound is a barrow) and never the dashed rectangle (an army camp)                                                                |
| Raid trail †                       | Red dotted line with arrowheads pointing into the farmland, named ("Muravsky Trail")               |                                                                                                                                                        |
| Frontier market or trading stone † | Small pair of scales; label the market days or "Trading Stone"                                     | At a gate, ford or garrison                                                                                                                            |
| Truce meeting place                | Small lone tree (the moot symbol, since it is also a meeting place), labelled "Truce Day"          | At a ford, stream or border village                                                                                                                    |

## Gods, undead and dangerous seas:

| Feature                             | Symbol                                                                                                        | Notes                                                          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Temple or sanctuary                 | The church symbols; a great sanctuary takes the great-shrine symbol (scallop or reliquary)                    | Never a star                                                   |
| Walled graveyard †                  | Small rectangle with a solid wall line and tiny crosses inside, labelled "Graveyard"                          | Outside the town walls                                         |
| Quarantine station or plague island | Small house with a cross, labelled "Lazaretto"                                                                | The master legend's hospital symbol                            |
| Blighted or dead land †             | Pale grey wash with small black dead-tree ticks, labelled ("the Blight"); watchtower triangles along its edge | Old villages inside drawn grey, labelled "(ruin)"              |
| Canal                               | Straight blue line                                                                                            | China's Grand Canal shows canals are not only a 1500s+ feature |

### MAP TIP

Put a small legend on the map itself, listing every fantasy symbol you used. Readers know what a castle icon means; they cannot guess what a spiral or a diamond means.

## Consistency checklist

If something exists in your world, other things must exist or change too. Check each line that applies.

- **If monsters roam the countryside**, then there are fewer lone farms, more walls, watchtowers, ranger lodges and cleared roads. Total population is lower than on a safe map.
- **If the whole world is "points of light"**, then most towns are small (about 1,500–4,000 people). Larger towns exist only where water brings food safely.

- **If fertility magic exists**, then density is higher and towns are bigger. Big cities still sit on water, unless magic also moves grain.
- **If healing magic exists**, then the frontier fills fast, there are more new towns and land wars, and fewer ruins from plagues.
- **If water magic makes desert cities**, then whoever makes the water holds power, and the city dies if the magic stops.

- **If portals exist**, then each portal has a town, a customs house and a guard. The capital is connected to them, or is in the wrong place.
- **If flying mounts exist**, then castles have roofed towers, lords keep aeries with food animals nearby, and messages are fast. Bulk freight still uses rivers.
- **If weather magic exists**, then someone controls it and takes payment, and famines are rare unless the weather-mages are at war.
- **If magic crystals exist**, then there are boom towns, mining law, mints and fortresses at the deposits, and ghost towns where deposits ran out.
- **If dwarves live under mountains**, then grain flows up to them through gate towns, and metal flows down.
- **If elves rule a forest**, then the forest has few roads, a wide border and a few fixed trading places.
- **If nomads raid farmland**, then there are frontier markets (or the reason they are closed), defence lines and burned villages near the steppe.

- **If dragons exist**, then there are lairs with empty land around them, tribute arrangements, and lost herds.
- **If gods enforce sanctuary**, then temples are rich and towns cluster around them.
- **If the dead can rise**, then graveyards are outside the walls and guarded, and there are quarantine places.
- **If the sea is deadly**, then canals and convoy ports replace many small harbours.
- **If two peoples meet**, then there are markets, strangers' inns or compounds, and agreed meeting places on the border.

**MAP TIP**

After finishing the map, read down this list and check every "if" that applies to your world. Most fantasy maps fail one or two lines, usually the food supply of a magical city or the missing edge where two peoples meet. More common errors are in [Common Mistakes and How to Fix Them](#).

## Quick summary

- Apply the real rules first. Then ask which input each fantasy element changes: water, food, defence, travel cost, threats or resources.
- Danger concentrates people: fewer lone farms, walled hill-top villages, refuge forts about a day's walk (24–32 km, 15–20 mi) apart, watchtowers, cleared roads.
- "Points of light" describes a frontier. Each point, if fields stay within 4–5 km (2.5–3 mi) of the walls, holds about 1,500–4,000 people.
- Fertility magic raises the population ceiling (treat a strongly blessed land like Flanders: 50–75 people per km<sup>2</sup>). Healing raises the growth rate, not the ceiling.
- Portals act like harbours and straits: hubs, tolls, warehouses, forts. Their capacity compared with a ship (a cog carried 30–200 tons) decides whether they change bulk trade.
- Flying mounts carry about a pack horse's load (100–120 kg). They change messages, travel and luxury trade, not bulk freight, and they force castles to defend against attack from above.
- Magic crystals are ore (boom towns, forts, mints). Ley lines are rivers (chains of towers, a big city at the nexus).
- Place academies like universities (big cities, a few per kingdom) and mage towers like monasteries (remote, but with farms).
- Model each non-human people on a real way of life: dwarves on mining towns, elves on Baltic forest peoples, halflings on free peasant lowlands, orcs on steppe nomads with real logistics, merfolk on sea nomads, dragons on tribute-taking raiders.
- Peoples meet at fixed places under fixed rules: kontors, strangers' inns, frontier markets, truce days.
- Gods, undead and deadly seas each create new places (sanctuaries, quarantine islands, canals) and destroy others (empty coasts, deserted villages).
- Draw ordinary features with the guide's master legend. Give fantasy features their own non-clashing symbols (diamonds for magic, a spiral for portals, a skull for lairs) and explain them in a legend on the map.

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## CHAPTER 11

## COMMON MISTAKES AND HOW TO FIX THEM

Most fantasy maps go wrong in the same few ways. Experienced mapmakers on forums, blogs and in mapping guides point out the same problems again and again: cities with no water, towns sprinkled at random, roads that go nowhere and a castle on every hill. This chapter lists 30 common mistakes with human-made features. For each one it explains why it looks wrong, how to fix it, and when it can be right, because almost every rule was broken somewhere by real people for a good reason.

Each mistake has three parts: **Why it looks wrong**, **Fix** and **Exception** (a real case where the "mistake" was true). The baseline is Europe c. 1000–1500, with other regions where they help. The full explanations live in the other chapters, which are linked in each section.

## IN THIS CHAPTER

Water and food: the mistakes that break a city

How many settlements, and where

Capitals and borders

Roads, routes and ports

Castles and town walls

Economy and purpose

Time: ruins, layers and eras

Drawing and labels

Sanity check: 20 questions for a finished map

Quick summary

Sources and further reading

## Water and food: the mistakes that break a city

## RULE OF THUMB

Before you draw a town, point to its water and to its fields. If you cannot point to both, move the town.

## 1. A city with no water

**Why it looks wrong:** People need water every day for drinking, animals, brewing, washing and crafts such as dyeing and tanning. Water is heavy, and deep wells were expensive. A town symbol far from any river, lake, spring or coast makes a careful reader ask how anyone lives there.

**Fix:** Put every village on a stream, a lake shore or a spring line (a row of springs at the foot of chalk or limestone hills). Put towns on rivers. If the site must be dry, such as a hilltop fortress, draw the spring, cistern (a tank for rainwater) or aqueduct that serves it. See [Fresh water](#).

**Exception:** Siena in Tuscany had no usable river nearby. It still grew to about 50,000 people (estimates vary: 50,000–70,000) before the Black Death of 1348. It did this with springs and the *bottini*: underground tunnels that collect

groundwater. They are first recorded in 1226, and in 1343 their water reached the main square, the Piazza del Campo. The whole system is about 25 km (16 mi) long. Venice collected rainwater in public cisterns and brought extra water by boat. Constantinople used aqueducts and huge cisterns. If you skip the river, you must draw the engineering.

**A note on rivers:** Check the rivers your towns sit on. Rivers join as they flow downhill. They almost never split, except in deltas and marshes. The fantasy author K. M. Alexander lists the splitting river among the "river sins" of fantasy maps: rivers only split for good in deltas. Put towns at confluences (where two rivers join), not at imaginary forks.

## 2. A desert city with no oasis or river

**Why it looks wrong:** A desert gets too little rain for farming without irrigation. A city in open sand, with no blue line or green patch near it, has nothing to drink and nothing to eat.

**Fix:** Give every desert town one of four water sources, and draw the green belt of gardens and palms that it feeds:

| Water source                | How it works                                                                               | Real example                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| A river crossing the desert | River floods or canals water a narrow strip                                                | Cairo on the Nile; Baghdad on the Tigris                                                          |
| An oasis                    | Springs or shallow groundwater                                                             | Sijilmasa in the Tafilalt oasis (Morocco), the northern end of the Sahara caravan trade           |
| A mountain-foot river       | Snow-fed stream spread into irrigated orchards                                             | Damascus and the Ghouta, an oasis of gardens and orchards watered by canals from the Barada river |
| Qanats                      | Gently sloping tunnels, with a line of shafts above them, bring groundwater from the hills | Yazd and Kerman (Iran)                                                                            |

**Exception:** Taghaza, a salt-mining village in the Sahara. When Ibn Battuta passed it in 1352, he found no trees, brackish (slightly salty) water and houses built of salt slabs roofed with camel skins. The miners lived on dates brought from Sijilmasa and the Draa valley, millet brought from the Sahel (the dry lands south of the Sahara) and camel meat. Sijilmasa was 25 days away by caravan. A desert place with no food of its own works only if it produces something valuable and lies on a caravan route.

#### FANTASY TWIST

A magic spring, a water-making temple or a bound water spirit can justify a desert city. But then the water source becomes the most important and most fought-over place in the city. Draw it, wall it, and put the palace or main temple next to it.

### 3. No farmland around cities

**Why it looks wrong:** In the Middle Ages, 85–95% of people lived in the countryside, and most were farmers. Each farmer could feed only a few town-dwellers. A city surrounded by forest, swamp or "wilderness" right up to its walls has nothing to eat.

**Fix:** Draw a ring of fields, villages and mills around every town. Put gardens and orchards next to the walls, grain farther out, and pasture and woodland at the edge. As a rule, a town of 1,000 needs farmland within about 7–8 km (4–5 mi) on good lowland, a town of 10,000 within 21–27 km (13–17 mi), and anything over 50,000 needs river or sea supply. The full table, including poor land, is in [Feeding towns: the hinterland](#).

**Exception:** Cities with good water transport could eat grain grown hundreds of kilometres away (see mistake 4). Mining camps, salt works and frontier fortresses in barren land imported their food, like Taghaza.

### 4. A huge city with no water transport

**Why it looks wrong:** Moving grain overland was slow and costly. In England c. 1300, carrying goods by land, river and

sea cost roughly 8 : 4 : 1 per ton-mile (Masschaele 1993). In other words, a cart cost about twice as much as a river boat and eight times as much as a ship. For the Roman world, the historian A. H. M. Jones wrote that it was "cheaper to ship grain from one end of the Mediterranean to the other than to cart it 75 miles" (120 km). So towns above 10,000 people almost always stood on navigable water or the coast. Cities above 50,000 always needed river or sea supply.

**Fix:** Put every city over 10,000 on a navigable river, a lake or a sheltered coast. If you want a big inland capital, give it a canal or a river port, and draw it.

**Exception:** Some big cities managed without a natural waterway, but each had a special answer:

- **Milan** (100,000–150,000 c. 1300) dug the Naviglio Grande, a 50 km (31 mi) canal from the river Ticino. It was begun in 1177, reached Milan in 1258 and was fully navigable from 1272.
- **Siena** (about 50,000) lived on the farmland of its own hilly countryside and on grain carted in overland.
- **Granada** (estimates vary: 50,000–150,000 c. 1300) fed itself from the irrigated plain (*vega*) around it.
- **Damascus and Aleppo** were fed by irrigated gardens and by pack camels. In the Middle East and North Africa, pack camels had largely replaced wheeled carts by late antiquity, because they were cheaper to run (Bulliet 1975). This made long overland hauls more practical there than in Europe.

LATER ERA (1500S+)

Canals with pound locks (a chamber with a gate at each end, which raises or lowers boats) appeared in the Low Countries in the late 14th century. Eighteen were built on a Milanese canal, the Naviglio di Bereguardo, in 1452–58. After 1500, canals and canalised rivers (rivers deepened and fitted with locks) spread, so an inland city can grow if you draw its canal. Madrid, made the permanent seat of the Spanish court in 1561, grew large without a navigable river, but feeding it was a constant problem. See [What Changes After 1500](#).

MAP TIP

Draw water first, then farmland, then towns. Shade a farmland ring around each town using the radii above. Then check: does every town over 10,000 touch navigable water? If not, add a canal, an irrigated plain or a caravan route, or make the town smaller.

How many settlements, and where

RULE OF THUMB

Most dots on a realistic map are small. For every city there are several towns, dozens of market towns and hundreds of villages.

Use this table to check the numbers on your map. It is for settled lowland in the High Middle Ages, c. 1300. The full method is in [Population and Settlement Sizes](#).

| Per 10,000 km² (a square 100 × 100 km, 62 × 62 mi) | Count                                                 | Typical spacing                                                                         | Basis                                                                             |
|----------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| People                                             | 200,000–400,000                                       | 20–40 per km² (50–105 per sq mi)                                                        | Baseline densities                                                                |
| Villages with a church                             | 500–700                                               | 1.5–4 km (1–2.5 mi)                                                                     | England: about 14 km² per parish                                                  |
| Market places                                      | 45–130                                                | 9–16 km (6–10 mi)                                                                       | Many were only villages with a weekly market; only 39% still had a market c. 1600 |
| Towns of 2,000–10,000                              | 3–8 (estimate)                                        | Towns of 5,000+: 80–95 km (50–59 mi) in France and Germany                              | Derived from urbanization rates                                                   |
| Cities of 10,000+                                  | 0–1 (France, Germany); 1–3 (Flanders, northern Italy) | 130–170 km (80–105 mi) in France and Germany; 60–90 km (37–56 mi) in Flanders and Italy | Derived from Buringh's city data                                                  |
| Great city of 50,000+                              | Usually none; at most one per kingdom                 | —                                                                                       | Only ~22 in all of Europe, the Middle East and North Africa c. 1300               |

5. Too few settlements

**Why it looks wrong:** Many fantasy maps show five towns in an area the size of England. Real England c. 1300 had about 9,500 parishes, more than 600 boroughs (towns with special legal rights) and about 1,746 places with a market grant, all on 130,000 km² (50,000 sq mi). A traveller in settled lowland passed a village every 1.5–4 km (1–2.5 mi).

A related mistake is to space villages a day's journey apart, about 25–35 km (15–22 mi). That is a traveller's day on foot, not the distance between villages.

**Fix:** At kingdom scale you do not need to draw every village. Draw the towns, and use a farmland texture or tiny dots to stand for the villages. At local scale, draw villages 1.5–4 km (1–2.5 mi) apart in good land. See [Villages, Farms and the Countryside](#).

**Exception:** Empty land was real in uplands, steppe, forest and war zones. In 1086, northern and western England had under 5 people per km² (13 per sq mi). One reason was William I's "Harrying of the North" (winter 1069–70), a campaign of deliberate destruction; Domesday Book (1086) still recorded much of Yorkshire as "waste". Poland-Lithuania in 1500 averaged 6.6 people per km² (17 per sq mi), and steppe nomads lived at under 1–2 per km² (3–5 per sq mi).

6. Too many settlements, or too many big ones

**Why it looks wrong:** This is the opposite error: a city every 30 km (19 mi), each with 20,000 people or more. Around 1300 there were only about 230 cities of 10,000+ in Europe, the Middle East and North Africa together, and only 8 of 100,000+.

**Fix:** Keep the ratio. For every city of 10,000+, draw about 10 towns of 2,000–10,000 (see chapter 02), dozens of market towns and hundreds of villages. Big symbols should be rare.

**Exception:** Flanders and northern Italy really were crowded. Flanders had 33–40% of its people in towns, towns of 5,000+ only 35–45 km (22–28 mi) apart, and cities of 10,000+ 60–90 km (37–56 mi) apart. A rich trading region can look busy, but it is the exception on a continent.

## 7. Evenly spaced settlements

**Why it looks wrong:** Towns set at equal distances look like a board game. Real settlements cluster along rivers, coasts, spring lines, good soils and roads, and leave gaps on heath, forest, marsh and upland.

**Fix:** Space by the land, not by the ruler. Draw chains of villages along valleys and coasts and leave gaps on hills. Central place theory (Christaller, 1933) is the idea that each market town serves the countryside around it, so market towns end up at fairly even distances. But it predicts a regular net only on a flat, uniform plain. Real rivers and hills break the pattern. See [Patterns and spacing](#).

**Exception:** Planned colonisation made regular patterns. Nearly 700 *bastides* (planned new towns) were founded in south-western France between 1222 and 1372, most on grid plans. During the German eastward settlement (12th–14th centuries), lords laid out planned villages. Their houses stood in rows along a road or a forest clearing, each with a long strip of land behind it. Even these follow rivers and ridges.

## 8. Everything on the coast

**Why it looks wrong:** Mapmakers often draw the coastline first and then line it with cities. But most people were farmers living inland on the best soils. A ring of ports around an empty middle looks like a colony, not a kingdom.

**Fix:** Put ports only where there is a good harbour: estuaries, bays, river mouths. Put most villages and many towns inland along rivers. The coast may have the biggest cities; the interior has most of the people.

**Exception:** Where mountains come down to the sea, life really was coastal: Norway, Dalmatia, and Norse Greenland (settled c. 985; the last written record is from 1408, and the colony died out in the 15th century). Both Greenland settlements lay at the heads of fjords. A sea-trading power such as Venice could rule a chain of islands and coastal forts (the

*Stato da Màr*, "State of the Sea") with almost no inland territory.

## 9. Nothing on the coast

**Why it looks wrong:** A long coast with no ports or fishing villages ignores the cheapest transport there was. In England c. 1300, sea transport cost about one-eighth as much as land transport.

**Fix:** Every good harbour near farmland or a river mouth gets a port. Add fishing villages along sheltered shores.

**Exception:** Some coasts were avoided because they were:

- **Unhealthy:** marshy, malarial coasts such as the Tuscan Maremma were thinly settled.
- **Harbourless:** long, straight dune coasts, such as the Landes of Gascony, had few ports.
- **Raided:** on Mediterranean coasts exposed to raiders, people lived in hilltop villages a few kilometres inland and used a small landing place on the shore.

### LATER ERA (1500S+)

Barbary corsair raids in the 16th century led the Spanish rulers of southern Italy and Spain to build chains of coastal watchtowers. Draw them as small triangles (the guide's watchtower symbol) on headlands within sight of each other.

## 10. Empty interiors

**Why it looks wrong:** The middle of the continent is blank, or filled with one "Great Forest", while the edges are crowded. In reality, interiors held some of the richest farmland: the Paris Basin, the Po valley, the Hungarian plain.

**Fix:** Follow rivers inland and put a chain of towns on every navigable river, with village networks across the plains. Leave an interior empty only for a reason you can name: desert, high mountains, cold forest or marsh.

**Exception:** Real empty interiors existed: the Sahara and Arabian deserts, the highlands of Iceland, the Scandinavian mountains and the northern forests of Russia (1–5 people per km<sup>2</sup>, or 3–13 per sq mi). Even there, draw the routes, oases and stations that cross them.

## 11. City sizes wrong for the era

**Why it looks wrong:** Fantasy capitals of 500,000 or a million people, or ordinary towns of 20,000+, break the scale of the period.

| Place, c. 1300                          | People (estimates vary)   |
|-----------------------------------------|---------------------------|
| Cairo, the largest city in the region   | 220,000–600,000           |
| Paris                                   | usually given as ~200,000 |
| Venice, Milan, Florence, Genoa          | 85,000–150,000 each       |
| London                                  | 60,000–100,000            |
| York, Bristol (1377, after the plague)  | ~11,000 and ~9,500        |
| The next ten or so English towns (1377) | about 3,500–5,500 each    |

**Fix:** In England and France c. 1300, the largest city held only about 1–2% of the realm's people (this guide's estimate from baseline numbers). Size the capital from the realm's population, then make the next cities much smaller. See [The rank-size rule and primate cities](#).

**Exception:** Imperial capitals were in a class of their own. Constantinople c. 1000 had 150,000–400,000 people, and Cairo c. 1300 even more. They were fed by sea and by the Nile, and by taxes from a whole empire.

#### LATER ERA (1500S+)

Capitals grew fast. By c. 1600, Istanbul had 400,000–700,000 people, London about 200,000 and Paris 220,000–300,000.

#### MAP TIP

Count your symbols. If you have more city symbols than town symbols, or no village marks at all, rebalance using the table at the top of this section. Leave gaps where the land is poor, and say why in a short label ("moor", "fen", "royal forest").

## Capitals and borders

#### RULE OF THUMB

Capitals sit where the wealth is, not where the middle is.  
Borders follow old claims and the land, not a ruler.

The full treatment is in [Capitals, Realms and Borders](#). Here are the three most common errors.

### 12. The capital at the exact centre

**Why it looks wrong:** A capital placed at the geometric centre of the realm looks planned and artificial. Real capitals grew where the wealth, water, trade and the ruler's home lands were, and that was often near an edge. Paris lay in the north of the French kingdom, London in the south-east of England, Kraków near Poland's southern edge, and Vienna near Austria's eastern border with Hungary. All four stood on navigable rivers.

**Fix:** Put the capital in the richest core region, on a navigable river or coast, near the ruling family's home lands, or facing the main frontier. See [Centre, fortress or frontier?](#).

**Exception:** Madrid became the permanent seat of the Spanish court in 1561, partly because it lay near the centre of the peninsula (historians still debate the reasons). Remember, too, that many medieval kings had no fixed capital at all. Their courts travelled.

### 13. Straight borders without a reason

**Why it looks wrong:** Medieval borders grew out of estates, parishes, river stretches, watersheds (the high ground that divides two river systems), forests and old church districts. Few medieval rulers could survey a long straight line, and land was claimed piece by piece. A ruler-straight border across hills and valleys looks modern.

**Fix:** Make borders wiggly in settled land. Use crests, river stretches, marsh and forest edges, and old district lines. In empty land, use a wide frontier zone instead of a line. See [Borders and frontiers](#).

**Exception:** A strong state with surveyors could draw a straight line. The Roman frontier in Germany was the Upper Germanic Limes: a palisade and ditch with watchtowers, built in the mid-2nd century AD. One stretch ran almost dead straight for 81 km (50 mi), across valleys and hills, from near Walldürn to the Haghof south of Welzheim. In 1494, the Treaty of Tordesillas split newly found lands between Spain and Portugal along a meridian (a north–south line). The line lay 370 leagues, about 2,100 km (1,300 mi), west of the Cape Verde Islands. It was a straight line drawn over land that nobody had seen.

#### 14. Borders that ignore the land, or follow every river

**Why it looks wrong:** A border that crosses a mountain range at a random point looks arbitrary. But the opposite error is also common: making every river a border. Big navigable rivers more often joined their valleys than divided them. For most of its course the Rhine had German lands on both banks, and the Danube ran through Austria and Hungary. Smaller units, like English counties, did often use rivers as boundaries.

**Fix:** Use crests and watersheds for long stretches. Use a river where the border is old or the river is a real obstacle. Example: the Anglo-Scottish border follows the River Tweed for part of its length. Its line was settled by the

Treaty of York (1237) and has changed little since, apart from Berwick and the Debatable Lands.

**Exception:** Realms that controlled a pass usually held both sides of it. Savoy held lands on both sides of the western Alps, Tyrol on both sides of the Brenner Pass, and Navarre on both sides of the western Pyrenees. If a realm lives from a pass, its border runs beyond the pass, not along the crest.

##### MAP TIP

For every border segment, ask "what is this line following?" Mark the answer in your notes: crest, river, forest, old county line or treaty. If the answer is "nothing", bend the line to the nearest real feature. Draw borders as dash-dot lines, and frontier zones as soft, wide bands of hatching or fading colour rather than hard lines.

## Roads, routes and ports

##### RULE OF THUMB

Roads follow the easy ground between places that matter. Ports need shelter. Every journey takes days, so count them.

Full details are in [Trade Routes](#), [Roads and Transport](#).

#### 15. No roads at all

**Why it looks wrong:** A map with towns but no roads suggests that nobody travels or trades. Medieval England had a real road network. The Gough Map of Britain (conventionally dated c. 1360–70; a handwriting study (Smallwood 2010) suggests one or two decades after 1400, with parts redone or added later in the 15th century) shows about 190 thin red route lines between towns. Nine out of ten have the distance between the towns written on them. A modern study traced the real roads behind these lines: about 4,540 km (2,820 mi) of route across England and Wales, split into 455 segments. About a third of that length runs on Roman roads, still in use 1,000 years after they were built. See [How much main road to draw](#).

**Fix:** Connect every town to its neighbours. Draw main roads between cities and along valleys. At kingdom scale you can leave out minor tracks. As a guide from the Gough Map, draw about 30 km of main road for every 1,000 km<sup>2</sup> of settled land (about 48 mi per 1,000 sq mi).

**Exception:** Where water is the road, roads matter less: river routes in Russia, the Venetian lagoon, the Fens of eastern

England and archipelagos all relied on boats. Steppe and desert had tracks and caravan routes rather than built roads.

#### 16. Roads that go nowhere, or run ruler-straight

**Why it looks wrong:** A road that stops in empty land has no reason to exist. A road that runs straight across hills, swamps and rivers suggests engineering that medieval roads did not have.

**Fix:** Every road should end at something: a settlement, port, ferry, ford, bridge, pass, mine, quarry, monastery or the map edge. Medieval roads bend. They follow valleys, ridges and dry ground and go to river crossings. Long straight roads are Roman, or later.

**Exception:** Dead-end roads were real when they led to one valuable place, such as a mine, a quarry, a monastery, a pilgrimage shrine or a castle on a rock spur. Draw that destination at the end of the road.

#### 17. Roads over mountains with no pass

**Why it looks wrong:** A road straight over the highest part of a range cannot be used by people on foot or with pack animals. Mountain crossings were few, slow and often closed in winter.

**Fix:** Run the road up a valley to a low saddle (a pass). Put a hospice (a shelter for travellers, often run by monks or canons), toll station or small castle at or near the top, and towns at the foot of the pass on both sides.

| Pass (Alps)      | Height             | Medieval note                                                                                            |
|------------------|--------------------|----------------------------------------------------------------------------------------------------------|
| Brenner          | 1,370 m (4,495 ft) | The lowest of the major passes in the Eastern Alps                                                       |
| Great St Bernard | 2,469 m (8,100 ft) | Hospice for travellers founded c. 1049–1050                                                              |
| Gotthard         | 2,106 m (6,909 ft) | Opened as a trade route c. 1220–1230, when the first wooden bridge was built across the Schöllenen gorge |

A new pass can make a region rich. The Gotthard route ran through Uri, and control of it helped Uri, Schwyz and Unterwalden, the valleys at its northern end, to form the core of the Swiss Confederacy.

**Exception:** Some travellers used high, hard passes because they were shorter or avoided enemies and tolls. Mark such passes "summer only".

## 18. Bridges everywhere

**Why it looks wrong:** On many maps every town on a big river has a bridge, and roads cross wide rivers wherever they like. In reality stone bridges were costly and valuable, and on the widest rivers they were few. London Bridge (stone, finished 1209) was the only road crossing of the Thames below Kingston upon Thames until Putney Bridge opened in 1729. That is more than 500 years with one bridge for the whole of London. Upstream, where the river is narrower, bridges were more common. Above Kingston the Thames had a bridge town about every 20 km (12 mi) by c. 1300, and more were added later, such as Chertsey (licensed 1410). On smaller rivers, too, bridges were more common than many people think: in England, most important crossings had a bridge by the 13th century (see [The Thames example](#)).

**Fix:** Cross wide rivers at fords (shallow places), ferries and a few bridges at important towns. A bridge attracts roads, trade and tolls, so it usually has a town or castle beside it. Small streams can have many small bridges.

**Exception:** On a local map, a busy town on a small river may have several bridges. Some rich cities also built several bridges across their own river.

## 19. Desert or mountain trade routes with no water or stops

**Why it looks wrong:** A caravan route drawn as one line across 500 km (310 mi) of sand, with nothing on it, cannot be used. Pack animals need water, rest and fodder every day.

**Fix:** Put a stop at every day's stage: 30–40 km (19–25 mi) apart on plains and desert, and 10 km (6 mi) or less in mountains. Each stop needs a well, cistern or oasis. Along trade roads these stops were often *caravanserais*: walled inns built around a courtyard for travellers and their animals.

**Real example:** The Darb Zubayda pilgrim road ran about 1,300 km (800 mi) from Kufa in Iraq to Mecca. It had 27 main stations, on average about 50 km (30 mi) apart, with smaller rest stops between them, plus wells, cisterns, reservoirs and dams. It was built and kept up mainly under the Abbasid caliphs, between 750 and 1258. Trans-Saharan caravans moved from oasis to well. Ibn Battuta took 25 days from Sijilmasa to Taghaza in 1352.

**Exception:** A route can cross a waterless stretch of a few days if travellers carry water in skins and know the way. Draw it as a dangerous "dry stretch" and write the number of days on it.

## 20. Ports on cliffs or exposed coasts

**Why it looks wrong:** Ships need deep, sheltered water out of the wind and waves, and a beach, quay or river mouth where they can unload. A port on top of a cliff, or on a straight coast battered by waves, cannot work.

**Fix:** Put ports in estuaries (wide, tidal river mouths), river mouths, bays behind headlands, behind islands or spits (long, narrow ridges of sand or shingle), or in drowned valleys (fjords and rias). Where a coast is mostly cliffs, put the port in the gap where a valley cuts through. Dover sits in the valley of the little River Dour, between white cliffs.

**Real warnings:** Exposed and silting harbours really did fail:

- **Dunwich** (Suffolk) was a leading English port in the 13th century. Storm surges in 1286 and 1287, and a great storm in 1328, ate into the town. Sand and shingle carried along the coast pushed the river mouth, and so the harbour, about 4 km (2.5 mi) to the north. Another storm in 1347 swept about 400 houses into the sea. With no harbour, the town was largely abandoned.
- **Bruges** slowly lost its access to the sea as the Zwin inlet silted up from the late 13th century onwards. By about 1500 the channel was nearly unusable, and the trade of the Low Countries moved to Antwerp.

**Exception:** A cliff-top town can have a port below it. Bonifacio in Corsica (founded as a fortress in 828) stands on limestone cliffs about 70 m (230 ft) high. Its harbour lies below, in a narrow, fjord-like inlet sheltered behind a long promontory. Draw the town on top and the quay below.

## 21. Ignoring travel time and scale

**Why it looks wrong:** Many maps have no scale, or a scale that breaks the story: an army crosses a 2,000 km (1,250 mi) empire in a week, or a "nearby" town is 300 km (190 mi) away. Readers notice.

Use the speeds in [Speed per day](#): for example, a walker 25–35 km (15–22 mi) a day, an army with baggage 13–20 km (8–12 mi), a royal messenger 50–90 km (30–56 mi), and a sailing ship 40–100 km (25–60 mi) as a voyage average.

**Fix:** Draw a scale bar first. Measure the important journeys and write them down. A kingdom 500 km (310 mi) wide is 15–20 days' walk across, and 25–40 days for an army. Popular gaming rules are often too fast. *Dungeons & Dragons* (5th edition) has travellers on foot cover 24 miles (39 km) a day at a normal pace and 30 miles (48 km) at a fast pace. That is a good day for a rider on a road, but too fast for walkers over many days, and much too fast for armies and carts.

**Exception:** Messages could move much faster with relays. The Mongol relay system was claimed to cover 200–300 km (125–190 mi) a day. Travel downstream and by sea was also much faster than by land.

### FANTASY TWIST

Teleport circles, flying mounts or fast magic messengers shrink travel time only for the few who can use them. Grain, stone and armies still move at cart and boat speed, so towns still need farmland and waterways. See [Fantasy Variants](#).

### MAP TIP

Draw main roads as solid lines (thicker for more important roads) and tracks as thin dotted lines. Put an inn (a small house on the road) or a village every 15–30 km (10–19 mi) on main roads. In desert, put a caravanserai (the same small house, labelled "Caravanserai" or "Han") every 30–40 km (19–25 mi), and draw the caravan route as a dotted line with a larger dot at each stage. Put a hospice (a small house with a cross) at each high pass. Write travel days on long routes: for example, 1,300 km (800 mi) by camel caravan at 25–40 km (15–25 mi) a day takes about 5–7 weeks. Check every port: is it in a bay, an estuary, a river mouth or a valley gap?

## Castles and town walls

### RULE OF THUMB

A castle controls something. Ask what it controls, who pays for it and who lives in it.

Full details are in [Castles, Forts and Military Outposts](#).

## 22. Castles on random hills

**Why it looks wrong:** Castles were built to control something: a town, a river crossing, a road, a pass, a harbour, a frontier, or the farms and tenants of a lord's estate. A castle on a random hill in the wilderness controls nothing and cannot feed its garrison. Also, not every castle stood on a hill. England and Wales had about 741 motte-and-bailey castles. A motte is a man-made earth mound with a wooden tower on top; the bailey is a fenced yard beside it. Mottes were typically built on low ground, many of them in towns or by rivers, and streams were dammed or diverted to fill their moats.

**Fix:** For each castle, write down what it controls. Put castles in or beside towns, at bridges and fords, at the mouths of passes, at harbours, along frontiers and at the centre of a lord's lands.

**Exception:** Hilltop castles were common where the hills overlooked a route. The Rhine Gorge, 65 km (40 mi)

between Bingen/Rüdesheim and Koblenz, has about 40 hilltop castles. Many of their lords lived on tolls from river traffic. The toll castle of Pfalzgrafenstein (its tower built 1326–27 for King Louis IV) even stands on an island in the river at Kaub. The hills were chosen because of the river, not at random.

## 23. Too many castles

**Why it looks wrong:** A castle on every hill and by every village, and every one a great stone fortress. Stone castles were very expensive, and it was usual for one to take the best part of ten years to finish. Every castle also needed a garrison and land to pay for it.

**Fix:** Keep big royal fortresses rare: one or two per region, at key towns, passes and frontiers. Peacetime garrisons were small, often 5–20 men. Fill the rest with smaller lordly castles, tower houses (tall, small stone towers lived in by minor lords) and fortified manor houses. Use the density table under [mistake 24](#).

**Exception:** Where lordship was split among many small nobles, castles really were everywhere. The German-speaking lands had about 14,000 castles over the whole period, and Switzerland, where no ruler controlled who could build one, about 4,000. Most were small towers, not great fort-

resses, and not all were in use at the same time. Late medieval border zones filled with tower houses: about 800 in Scotland and over 3,000 in Ireland.

## 24. Too few castles

**Why it looks wrong:** A realm of millions has three castles, one in each corner. Some popular gaming rules make this

worse. *Medieval Demographics Made Easy* (S. John Ross, first written 1993 and revised several times since) suggests one working castle per 50,000 people. For Norman England, with about 1.7–2 million people, that gives only 35–40 castles. In fact, England and Wales had 500–600 castles in use at any one time after 1066: more than ten times as many.

| Situation                                                                      | One castle per                                                      | Average spacing                    | Real basis                                                                                                |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Whole realm after a conquest: castles in use at one time (11th–12th centuries) | 250–300 km <sup>2</sup> (95–115 sq mi), or about 3,000–4,000 people | about 16–17 km (10–11 mi)          | England and Wales after 1066: 500–600 in use on about 150,000 km <sup>2</sup> (58,000 sq mi)              |
| March or contested border (all castle sites)                                   | 17–28 km <sup>2</sup> (7–11 sq mi)                                  | 4–6 km (2.5–4 mi)                  | Welsh March (see <a href="#">chapter 04</a> )                                                             |
| Peaceful core of a kingdom (castles in use)                                    | —                                                                   | 20–30 km (12–19 mi) (estimate)     | Eastern England: a castle in each county town plus a few baronial seats (see <a href="#">chapter 04</a> ) |
| Royal castle chain along a frontier                                            | —                                                                   | 20–40 km (12–25 mi) along the line | Edward I's North Wales castles, all supplied by sea                                                       |
| Fragmented lordship (many small towers, all periods together)                  | ~10 km <sup>2</sup> (4 sq mi)                                       | 3–4 km (2–2.5 mi)                  | Switzerland: ~4,000 sites on 41,000 km <sup>2</sup> (16,000 sq mi)                                        |
| Popular gaming rule                                                            | 50,000 people                                                       | —                                  | Less than one-tenth of what Norman England actually had                                                   |

**Fix:** Do not spread the national average evenly. It mixes very different zones: the Welsh March had roughly 5–10 times as many castle sites per km<sup>2</sup> as peaceful eastern England. Put most castles along borders, at river crossings and around rich towns, and keep the safe middle of the realm thinner.

**Exception:** Peaceful, centralised realms needed fewer working castles. Many early earth-and-timber castles were abandoned within a century or two. So a late medieval map can show fewer working castles and more castle ruins.

## 25. Every town walled

**Why it looks wrong:** Many fantasy maps put a wall around every dot. Walls were expensive to build and to maintain. From 1220 onwards, English towns usually paid for them with a royal grant of *murage* (the right to charge tolls on goods to pay for wall-building). Many of the early grants went to towns near the Welsh border, which faced a real threat of raids.

### Numbers:

- **England and Wales:** Mike Salter's survey of walled towns (2013) describes 99 defended towns in England and 33 in Wales. It also lists 54 more English places with slighter or less certain defences. So England had roughly 100–150 defended towns.
- **Compare:** England had more than 600 boroughs and about 1,700 places with market grants. So only about

one English borough in four to six was defended, and most small market towns had no walls at all.

- **Many "defended" towns were not stone-walled.** They had only a bank and ditch, with gates or bars (barriers across the street) where tolls were collected.
- **South-western France:** most of the nearly 700 bastides started without walls. In the lands of the Count of Toulouse, the Treaty of Paris (1229) at first forbade new fortifications. Many bastides were walled only during the Hundred Years' War (1337–1453).

**Fix:** Wall the big cities, the regional capitals, the frontier towns and the ports facing raids. Leave most market towns open, or give them a ditch and gates.

**Exception:** In rich or contested regions most real towns were walled: northern Italy, Flanders and much of the Holy Roman Empire. In conquered frontier zones such as North Wales, Edward I built walled towns next to his castles, for example at Conwy and Caernarfon.

### LATER ERA (1500S+)

Gunpowder changed walls. Low, thick walls with angled bastions (pointed platforms for cannon) appeared in Italy in the later 15th century and spread beyond Italy in the 1530s–40s. These "star forts" and garrison towns replace castles as the main strongholds. See [What Changes After 1500](#).

**FANTASY TWIST**

Flying enemies or monsters change the rules. If dragons or flying mounts are common, walls matter less than towers, roofs and shelters, and castles move to places that are hard to see from the air. If monsters roam the countryside, more villages get palisades (fences of sharpened stakes) and fewer people live on isolated farms. See [Fantasy Variants](#).

**MAP TIP**

Next to each castle symbol, write one word for what it controls: "ford", "pass", "port", "town", "march" (border province). Use a large castle icon (three towers) for great royal castles, a small castle icon (one tower) for lords' castles, and the same icon broken or grey, labelled "(ruin)", for ruins. Put battlements on a town's ring only if it was walled: walled cities and frontier towns get them, unwalled towns get a plain ringed circle, and market towns a small open circle. Do not use a plain dot for a market town; at kingdom scale a tiny dot means a village.

## Economy and purpose

**RULE OF THUMB**

Every region feeds itself first. Every town has a reason to be where it is, and you should be able to say it in one sentence.

Full details are in [Industry, Resources and Special Towns](#).

### 26. Single-product economies

**Why it looks wrong:** "The mining kingdom", "the wine province" or "the fishing town" where nothing else happens. In the Middle Ages almost every region grew its own grain first, and its speciality sat on top of farming. A whole region doing only one thing would starve.

**Fix:** Give every region farmland plus one or two specialities. Give every town several trades: a market, smiths, millers, tanners, brewers and the local speciality. If a place does specialise, draw how its food arrives, by road, river or sea.

**Exception:** Specialised places existed when trade fed them:

- **Bergen** (Norway) exported stockfish (dried cod) from northern Norway and imported grain. From about 1350 the trade was run by Hanseatic (north German) mer-

chants from their own quarter on the harbour, the Bryggen.

- **The Flemish cloth towns** imported English wool and grain from abroad.
- **Kutná Hora** (Bohemia) grew from a silver rush that began c. 1260. From 1300 it held the royal mint, and it became the second most important town in the kingdom after Prague.
- **Taghaza** (Sahara) lived only on salt.

### 27. Isolated towns with no reason to exist

**Why it looks wrong:** A town far from all others, off every road and river, with no special resource. Readers ask how it got there and who buys from it.

**Fix:** Give every isolated settlement one strong reason, and draw its sign: crossed hammers for a mine, a small plain cross for a chapel, a scallop shell for a great shrine, a castle icon for a fortress, an anchor for a harbour, or a blue "W" for a well. The usual reasons are a mine or salt, a pilgrimage shrine, a monastery, a fortress guarding a pass, a river crossing, an oasis or a harbour.

**Exception:** Real remote places, each with its reason:

| Place                            | Reason                             | Date                                                                                                                                                  |
|----------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Taghaza (Sahara)                 | Salt mine                          | Visited by Ibn Battuta in 1352                                                                                                                        |
| Kutná Hora (Bohemia)             | Silver                             | Mining from c. 1260; boom in the late 13th and 14th centuries                                                                                         |
| Santiago de Compostela (Galicia) | Pilgrimage to the tomb of St James | Tomb reported in the 9th century; pilgrims from beyond the Pyrenees from the mid-11th century; a highly organised mass pilgrimage by the 12th century |
| Great St Bernard hospice (Alps)  | Shelter at a high pass             | Founded c. 1049–1050                                                                                                                                  |
| Mount Athos (Greece)             | Monasteries                        | Great Lavra, the first large monastery, founded 963                                                                                                   |

FANTASY TWIST

A rare magic resource, such as a mana spring or a dragon-bone quarry, works just like a silver mine. It creates a boom town in a bad place, a road to reach it, a castle to guard it, and food caravans to feed it.

MAP TIP

Write a one-line "why it's here" note for every town on the map. If you cannot, move the town to a river, crossing or resource, or delete it.

Time: ruins, layers and eras

RULE OF THUMB

A real landscape is old. Every map should show some things built long ago, and nothing built too early.

28. No ruins or older layers

**Why it looks wrong:** If everything was built at once, the map looks like a stage set. Real medieval landscapes were full of older things:

| Older layer                                                           | Numbers (Britain and Ireland)                                                                                                                                                |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iron Age hillforts (earth-walled hilltop enclosures, c. 700 BC–AD 43) | 4,147 in Britain and Ireland (Atlas of Hillforts, 2017; see <a href="#">Ancient and ruined layers</a> ), most of them in Britain (Scotland about 1,700, England about 1,220) |
| Roman roads, forts, walls and towns                                   | Many roads still in use in the Middle Ages                                                                                                                                   |
| Early castles abandoned or replaced                                   | Of about 1,700–1,800 castle sites in England and Wales (the Gatehouse gazetteer lists 1,761; see <a href="#">chapter 04</a> ), only 500–600 were in use at any one time      |
| Deserted villages                                                     | More than 3,000 known in England from all periods; at least 1,500 abandoned c. 1350–1520, most in the 15th century when fields were turned into sheep pasture                |

*Medieval Demographics Made Easy* suggests this many ruined castles: (population ÷ 5 million) × the square root of the realm's age in years. Take a realm of about 5 million people where castle-builders have lived for 400–900 years. The formula gives only about 20–30 ruins. That is far fewer than the real landscape had. More than 1,000 castles were built in England and Wales in the 150 years after 1066. Only 500–600 were in use at any one time, so hundreds of early castles already stood abandoned by c. 1200.

**Fix:** Add a layer of ruins and older features: old forts on hill-tops, an old straight road that new towns ignore, a deserted village, a ruined abbey or castle, an old border dyke (a long earth bank and ditch). Let some towns stand on top of older ones. See [Religious, Cultural, Legal and Ancient Sites](#).

**Exception:** Truly new land, such as a frontier colony or fresh forest clearings, has few ruins. Iceland, settled from c. 870, had almost no earlier human layer.

FANTASY TWIST

Long-lived races and fallen magical empires multiply ruins. If elves or an ancient empire ruled for thousands of years, add more ruins, older roads and abandoned cities, and let new towns be built from their stone.

29. Mixing eras without reason

**Why it looks wrong:** Star forts next to new earth-and-timber castles, windmills in an early medieval world, or canal locks everywhere in 1100. Each feature has a date, and well-read readers know many of them.

| Feature                                                                    | First appears in Europe                              | Common by                                                       |
|----------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|
| Watermill                                                                  | Roman times                                          | About one per village in England by c. 1300 (over 10,000 mills) |
| Windmill (post mill: the whole mill body turns on a post to face the wind) | 1185, first certain record (Weedley, Yorkshire)      | 13th–14th centuries                                             |
| Friaries in towns                                                          | 1210s in Italy; 1220s in England and northern Europe | Late 13th century                                               |
| Universities                                                               | Bologna, late 11th century                           | About 20 c. 1300; 60–80 by 1500                                 |
| Gunpowder cannon                                                           | First European picture, 1326                         | Decisive in sieges by the mid-15th century                      |
| Canal pound locks                                                          | Low Countries, late 14th century                     | 15th century in northern Italy                                  |
| Bastioned "star" forts                                                     | Italy, late 15th century                             | Spread beyond Italy in the 1530s–40s                            |

**Fix:** Choose a date for your map. Use only features invented by that date. Old features can survive (layers are good), but new ones cannot appear early.

**Exception:** Real landscapes did mix eras. Roman roads were still used 1,000 years later. At Berwick-upon-Tweed, new walls with bastions were begun in 1558, while the old medieval castle was left outside them. Along one stretch the new works were never finished, so the medieval walls and towers were repaired and kept in use instead. In a fantasy world, uneven technology can be a deliberate choice, such as one dwarven city with cannon in a world of knights. If you do it, make it a story point and explain it.

#### LATER ERA (1500S+)

If your map is set after 1500, see [What Changes After 1500](#) for star forts, postal relays, ocean ports and much bigger capitals.

#### MAP TIP

Put a date in the map's title box ("The Kingdom of X, Year 1312"). Then use three styles: working features in normal ink; old but still used features, such as a Roman road labelled "Street" or "(Roman)"; and ruins, drawn as the feature's own symbol broken or grey and labelled "(ruin)". Do not use dashed lines for ruins: in this guide, dashes mean fords and camp outlines.

## Drawing and labels

### 30. Label clutter

**Why it looks wrong:** Every village is labelled in the same large font. Labels cross rivers and coasts, names are all in upper case, and the map uses five decorative fonts. The

map becomes hard to read, and the important places disappear.

**Fix:** Follow the classic rules of the Swiss cartographer Eduard Imhof (1975) and standard lettering practice:

| Rule                                | What to do                                                                                                                                                                                                                                                                                          |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Show importance                     | Make labels bigger and bolder for bigger places. Use UPPER-CASE letters only for realms and the largest features, because upper case is harder to read (one set of lecture notes puts it at about 13% less readable).                                                                               |
| Point labels (towns)                | Classic best position: above and to the right of the symbol. If that space is taken, try the other corners or directly above or below. A 2024 study with nearly 800 readers found they actually preferred labels placed directly above the point, so both work. Keep the label close to its symbol. |
| Line labels (rivers, roads)         | Follow the line on a fairly straight stretch, just above it. Water names in italics, traditionally blue.                                                                                                                                                                                            |
| Area labels (realms, forests, seas) | Keep the name inside the area, spread across it and curved to fit its shape. Gaps between letters should be no more than about 4 times the letter height.                                                                                                                                           |
| Fonts                               | Use 2–3 typefaces at most. Change size, weight or italics instead of adding fonts.                                                                                                                                                                                                                  |
| Size                                | Nothing smaller than about 6 point in print.                                                                                                                                                                                                                                                        |
| Spacing                             | Names should be neither densely clustered nor evenly spread (Imhof).                                                                                                                                                                                                                                |
| What to label                       | At kingdom scale, label cities, towns and major features. Leave most villages unlabelled.                                                                                                                                                                                                           |

**Exception:** Dense labelling is fine on a local map, or in a gazetteer (a list of places) beside the map. Old maps such as the Gough Map crowd names together, so a crowded "antique" look can be a deliberate style.

MAP TIP

Place labels last, from the biggest to the smallest: realms, seas and mountain ranges first, then cities, rivers and towns, and villages only if there is room. If a label must cross something, let it cross land texture, not a river, coast or border line. Print the map at its final size and read it at arm's length.

Symbols for the fixes in this chapter

RULE OF THUMB

One symbol, one meaning. If a reader can read a symbol two ways (a star for a capital and also for a shrine), change one of them.

The fixes above name many symbols. Here they are in one place. These symbols match the guide's [master legend](#). Scale: K = kingdom or region, L = local area.

| Feature                | Symbol                                                                                         | Scale | Mistakes   |
|------------------------|------------------------------------------------------------------------------------------------|-------|------------|
| Village                | Tiny dot (notable villages only); a farmland texture stands for the rest                       | K     | 5, 11      |
| Small market town      | Small open circle, no battlements                                                              | K     | 25         |
| Town                   | Circle with a ring; battlements on the ring only if walled                                     | K     | 6, 25      |
| City                   | Circle with a thick battlemented ring                                                          | K     | 6, 25      |
| Realm capital          | Filled star in place of the tier symbol; name in bold capitals                                 | K     | 12         |
| Realm border           | Thick dash-dot line                                                                            | K     | 13, 14     |
| March or frontier zone | Hatching, or a fading colour band 10–50 km (6–30 mi) wide                                      | K     | 13, 14     |
| Royal forest           | Green dash-dot boundary with a large italic name                                               | K, L  | 11         |
| Main road              | Solid line, thicker for more important roads                                                   | K     | 15, 16, 21 |
| Track or mule path     | Thin dotted line                                                                               | K, L  | 15, 21     |
| Roman road             | Straight thick solid line labelled "Street" or "(Roman)"; grey where abandoned                 | K, L  | 15, 28, 29 |
| Caravan route          | Dotted line with a larger dot at each stage                                                    | K     | 19         |
| Navigable river        | Thicker river line up to the head of navigation, with a small boat there                       | K     | 4          |
| Canal                  | Straight blue line                                                                             | K, L  | 4          |
| Bridge / ford / ferry  | "(" with the road through it / short dashed line across the river / short dotted line with "F" | K, L  | 18         |

| Feature                     | Symbol                                                           | Scale               | Mistakes   |
|-----------------------------|------------------------------------------------------------------|---------------------|------------|
| Toll                        | One bar across the road or river, with a "T"                     | K, L                | 17, 22     |
| Inn or caravanserai         | Small house on the road; label "Caravanserai" or "Han" in desert | K, L                | 19, 21     |
| Hospice at a pass           | Small house with a cross                                         | K                   | 17, 21     |
| Port                        | Anchor (a large anchor with a quay for a main port)              | K                   | 8, 9, 20   |
| Great royal castle          | Large castle icon (three towers)                                 | K                   | 23, 25     |
| Lord's castle               | Small castle icon (one tower with battlements)                   | K, L                | 22–24      |
| Tower house                 | Small solid square                                               | K (border zones), L | 23         |
| Watchtower or coastal tower | Small triangle                                                   | K (chains), L       | 9          |
| Star fort (1500s+ only)     | Star outline, never filled                                       | K, L                | 25, 29     |
| Mine                        | Crossed hammers (or a pick)                                      | K, L                | 26, 27     |
| Chapel                      | Small plain cross, no tower                                      | L                   | 27         |
| Abbey or priory             | Church with a square cloister beside it                          | K (great abbeys), L | 27         |
| Great shrine                | Scallop shell or reliquary, never a star                         | K                   | 27         |
| Any ruin                    | The feature's own symbol, broken or grey, labelled "(ruin)"      | K (big ones), L     | 24, 28, 29 |
| Deserted village            | Grey church alone in a field, labelled "(lost village)"          | K (a few), L        | 28         |
| Hillfort                    | Oval of rings or hachures on a hill                              | L                   | 28         |

These three features use the master legend's symbols:

| Feature                                       | Suggested symbol                                                                                                                     | Mistakes |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| Well or spring                                | Blue "W" for a well, "Spr" for a spring (the abbreviations used on Ordnance Survey maps)                                             | 1, 2, 27 |
| Qanat                                         | Straight line of tiny rings, much smaller than a market-town circle, ending at the village gardens (see <a href="#">chapter 06</a> ) | 2        |
| Old border dyke (a long earth bank and ditch) | Line with short ticks on the ditch side, grey if no longer used, labelled with its name                                              | 28       |

#### MAP TIP

Copy only the symbols you actually use into a small legend box in a corner of the map. Before you add a new symbol, check that its shape is not already used for something else.

## Sanity check: 20 questions for a finished map

Answer yes or no. Every "no" is a place to look again.

- Does every settlement touch water (a river, stream, spring, lake, coast or a drawn well or cistern)?
- Does every desert town have a river, oasis or qanat, with green around it?
- Does every town have farmland and villages around it?
- Does every city over 10,000 sit on navigable water, or have a canal, irrigated plain or caravan route explained?
- Are there far more villages than towns, and far more towns than cities?
- Are settlements clustered along rivers, coasts and good soil, with gaps on hills, forest and marsh?
- Does the interior have towns and farms, unless it is desert, mountain or cold forest?
- Are your largest cities realistic for the era (only a few over 50,000, and at most one over 100,000 per kingdom)?

9. Can you say why the capital is where it is, and is the reason not just "it's in the middle"?
10. Does every border segment follow something (crest, river, forest, old district line, treaty)?
11. Do roads connect all towns, and does every road end somewhere meaningful?
12. Do mountain roads cross at passes, with a hospice, toll point or fort?
13. Are big rivers crossed only at a few bridges, plus fords and ferries?
14. Do desert and steppe routes have a stop with water about every 30–40 km (19–25 mi)?
15. Are all ports in sheltered water: a bay, estuary, river mouth, fjord or valley gap?
16. Have you drawn a scale bar and checked travel times for the main journeys?
17. Can you say what each castle controls?
18. Are walls drawn only around cities, frontier towns and exposed ports, not around every market town?
19. Does each region have food plus a speciality, and does each remote town have a clear reason to exist?
20. Are there ruins and older layers, and is every feature possible for the date on your map?

## Quick summary

- Point to the water and the fields before you draw a town. Desert towns need a river, an oasis or qanats.
- A town of 10,000 needs farmland 21–27 km (13–17 mi) around it. Cities above 10,000 need navigable water, or a canal, a rich irrigated plain or caravans.
- Most dots are small. Per 10,000 km<sup>2</sup> of good land c. 1300: 500–700 villages, dozens of market places, 3–8 towns and 0–1 cities (1–3 in Flanders or northern Italy).
- Cluster settlements along rivers, coasts and good soil, with gaps on poor land. Do not use a ruler and do not line only the coast.
- Capitals sit in the richest region, not in the geometric centre. Borders follow land and old claims and are rarely straight.
- Roads end at something and cross mountains at passes. Bridges over big rivers are few, and desert routes need water every 30–40 km (19–25 mi).
- Ports need shelter. Dunwich and Bruges show what happens when a harbour fails.
- Draw a scale bar. Walkers make 25–35 km (15–22 mi) a day and armies only 13–20 km (8–12 mi).
- Every castle controls something. After 1066 England and Wales had about one castle in use per 250–300 km<sup>2</sup> (95–115 sq mi), and contested borders far more (a castle site every 4–6 km, or 2.5–4 mi, in the Welsh March). Only about one English borough in four to six was walled.
- Give every region food plus a speciality, and every remote town a reason.
- Add ruins and older layers, pick a date, and keep later inventions off the map.
- Label from big to small, above-right of (or directly above) points, with 2–3 fonts at most. Give each symbol one meaning only, as in the master legend.

## Sources and further reading

### Books and articles

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- K. M. Alexander, *We Need to Talk About Your Rivers* (2024; the "river sins", including rivers that split).
- Roll20 compendium, *D&D 5th edition: Movement* (travel pace: 18, 24 or 30 miles a day).
- ProFantasy, Pär Lindström: *Creating an Overland Map*, Part 3.
- EN World forum: *Worldbuilding: How far should things be?*
- S. John Ross, *Medieval Demographics Made Easy* (PDF).
- University of Northern British Columbia, *Map lettering lecture notes* (PDF) (Imhof's positioning rules).
- Ordnance Survey, *Abbreviations used on OS 1:25 000 and 1:50 000 scale mapping* ("W" for well, "Spr" for spring).
- Bobák, Petr, Ladislav Čmolík and Martin Čadík. *From Top-Right to User-Right: Perceptual Prioritization of Point-Feature Label Positions* (arXiv, 2024).

**History web pages consulted**

- Wikipedia: Bottini of Siena; Siena; Taghaza; Zubaydah Trail; Naviglio Grande; Lock (water navigation); Dunwich; Bonifacio, Corse-du-Sud; Schöllenen Gorge; Great St Bernard Pass; Upper Germanic-Rhaetian Limes; Motte-and-bailey castle; Rhine Gorge; Pfalzgrafenstein Castle; Bastide; List of town walls in England and Wales; Murage; Deserted medieval village; Hillfort; Windmill; Cannon; Bastion fort.

- Wikipedia, more pages checked for this chapter: Old London Bridge; Gough Map; Brenner Pass; Gotthard Pass; Castle; Castles in Great Britain and Ireland; Zwin; Bruges; Sijlmasa; Ghouta; Qanat; Harrying of the North; Treaty of York; Eastern Settlement (Norse Greenland); Bryggen; Kutná Hora; Camino de Santiago; Great Lavra.
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- Gatehouse Gazetteer, Salter, *Medieval Walled Towns* (2013); castle statistics by county (1,761 castle sites in England and Wales: 1,252 in England, 509 in Wales).
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- University of Edinburgh, *Experts map hillforts of UK and Ireland* (4,147 sites: 1,695 in Scotland, 1,224 in England); Smithsonian Magazine, *Explore Ancient British Isles Hill Forts with a New Online Atlas* (2017).
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Population, density, spacing, travel, transport-cost and castle figures follow the guide's shared baseline (Bosker, Buringh & van Zanden 2013; Broadberry, Campbell et al. 2015; Masschaele 1993; Dyer; the *Gazetteer of Markets and Fairs*). Figures marked "estimate" are derived by this guide and are not measured values.

## CHAPTER 12

# WORKED EXAMPLE: POPULATING A KINGDOM

---

This chapter builds one invented kingdom from an empty terrain map to a full kingdom map and one local map, and then redraws it at four dates. It uses the guide's own numbers, shows every calculation in a table, and links each number to the chapter that owns it. Treat it as a template: put in your own areas and land types and redo the arithmetic. The kingdom, its places and its history are invented; the real examples quoted come from the other chapters and from the sources at the end.

The general method is in [Step by Step](#). The coordinates of every place, road, river and border are in [worked-example-layout.json](#) (canvas 1000 × 800 units, 2 units = 1 km).

**IN THIS CHAPTER**

The kingdom at a glance

Step 1: Land and population

Step 2: Cities, towns and villages

Step 3: The capital

Step 4: Duchies, counties and the North March

Step 5: Castles and fortifications

Step 6: Roads, rivers and the sea

Step 7: The Church

Step 8: Industry and special towns

Step 9: Ruins and older layers

Step 10: Zoom in on the Leyford area

Step 11: A fantasy variant

Step 12: The same kingdom at four dates

Quick summary

Sources and further reading

# THE KINGDOM OF DARAVEL

c. 1300 · schematic map of the worked example (chapter 12)

Shown: the capital, 3 cities, all 38 towns and 26 of the ~180 market towns; the ~7,700 villages appear only as texture.

## KEY

### Settlements

- ★ Capital, 40,000
- City (3), 10–12,000
- Town (38), 2–8,000
- ◉ Walled town
- ⊙ County town
- ⊙ Duchy, earldom seat
- Market town (26)
- ⋯ Villages (texture)

### Church

- ⛪ Cathedral (13)
- ⚡ Archbishop
- ⛪ Abbey, priory
- ⚡ Military order
- ⛪ Great shrine

### Castles

- 🏰 Great castle (5)
- 🏰 Royal, county castle
- 🏰 Frontier castle (10)
- 🏰 Baronial castle (25)
- 🏰 Royal residence
- 🏰 Beacon chain (10)

### Industry

- ⚙ Silver, lead mines
- 🏠 Salt pans
- ⚙ Quarry, lime
- ⚙ Fulling mills
- 🍷 Vineyards
- 🚢 Royal shipyard
- ⚙ Other industry

Sept. Fair: months in label

### Routes

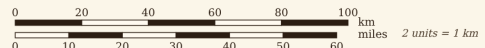
- Royal highway
- Regional road
- Old road, "Street"
- Pilgrims' Way
- Mule path
- Sea lane
- Bridge
- Ford
- Ferry
- Pass
- Head, member port
- Lighthouse
- Land and water
- Navigable river
- Head of navigation
- Stream
- Mountains
- Hills, downs, wolds
- Forest
- Royal forest bound
- Marsh, fen
- Heath
- Borders
- Realm
- Duchy, earldom
- Church liberty
- Frontier band
- Debatable Land
- Step 10 local map

Sept. Fair: months in label



### Older layers (grey)

- 🏰 Ruin, labelled "(ruin)"
- 🏰 Lost village
- 🏰 Abandoned motte
- 🏰 Hillfort
- 🏰 Old dyke



Daravel c. 1300, drawn from the layout file with the symbols of the *master legend*. It shows the capital, the 3 cities, all 38 towns and 26 of the ~180 market towns; the ~7,700 villages appear only as texture. Also shown: the 13 cathedrals, the White Keep, the 31 county castles, the 13 other royal and frontier castles, 25 baronial castles, the 9 ports (4 head ports), the 4 bridges on the Ambre and the beacon chain. The two passes take the master legend's saddle mark ("")(" means a bridge).

## The kingdom at a glance

**RULE OF THUMB**

Draw the terrain first and decide the date. Every number below depends on the land and on the year.

**Daravel** is a centralised feudal kingdom in the year **1300**, the crowded peak before the Great Famine and the Black Death (see [Change over time](#)).

| Feature    | Daravel                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Area       | About 116,500 km² (45,000 sq mi): about 390 km (240 mi) east to west and 380 km (235 mi) north to south. England is 130,000 km².          |
| Coast      | East and south, on the Grey Sea                                                                                                           |
| Estuary    | The Ambre estuary, 83 km (52 mi) from the open sea to its head                                                                            |
| Main river | The Ambre, navigable for river boats 158 km (98 mi) above the estuary; the Lisk joins it at Liskmeet                                      |
| Mountains  | The Fells along the whole west border. The crest is the border. One cart pass (the Wyndgap) and one mule path (the Carrow Gap).           |
| Forest     | The Harnwood, about 11,000 km² in the north                                                                                               |
| Marsh      | The Holm Fens north of the estuary, the Saltings salt marsh south of it, the Wend Levels                                                  |
| Other land | The Whitridge (a limestone ridge cut by the Ambre), Brackenheath (sandy heath), the Wend Downs and Brim Wolds (chalk and limestone hills) |
| Neighbours | Thelland to the north (hostile), Morvane to the west across the Fells (at peace)                                                          |

**MAP TIP**

Before any dot goes down, mark the navigable stretch of each river, the tidal limit, the fords, the pass and the land types. This is step 1 of the method in [Where Settlements Are Built](#).

## Step 1: Land and population

**Method:** measure each land type, multiply by a density from [Population density](#), and add up (calculator, steps 1–3).

| Land type                                                   | Area                                         | Density used (range)                         | People                                      |
|-------------------------------------------------------------|----------------------------------------------|----------------------------------------------|---------------------------------------------|
| Fertile lowland (Ambre vale, coastal plains)                | 44,500 km <sup>2</sup> (17,200 sq mi)        | 35 per km <sup>2</sup> (30–40)               | 1,557,500                                   |
| Mixed farmland and rolling hills (downs, wolds, foot-hills) | 33,500 km <sup>2</sup> (12,900 sq mi)        | 20 (15–25)                                   | 670,000                                     |
| Heath and poor sandy land                                   | 3,000 km <sup>2</sup> (1,200 sq mi)          | 10 (7–15)                                    | 30,000                                      |
| High uplands and mountains (the Fells)                      | 20,500 km <sup>2</sup> (7,900 sq mi)         | 5 (2–5)                                      | 102,500                                     |
| Forest (Harnwood, Kingswood)                                | 11,500 km <sup>2</sup> (4,400 sq mi)         | 4 (1–5)                                      | 46,000                                      |
| Marsh and fen                                               | 3,500 km <sup>2</sup> (1,400 sq mi)          | 5 (2–5)                                      | 17,500                                      |
| <b>Total</b>                                                | <b>116,500 km<sup>2</sup> (45,000 sq mi)</b> | <b>~21 per km<sup>2</sup> (54 per sq mi)</b> | <b>~2.4 million (range 1.9–2.8 million)</b> |

**Checks:**

- A realm the size of England should have 2–5 million people ([Realm populations](#)). Daravel, at 2.4 million, is less crowded than England in 1290 (~36 per km<sup>2</sup>) and close to the European average of about 20 per km<sup>2</sup> c. 1300. ✓
- **Food:** at about 1 ha (2.5 acres) of ploughland per person, Daravel needs about 24,000 km<sup>2</sup> of arable, under a third of its 78,000 km<sup>2</sup> of lowland and mixed land. At 2–3 ha

(5–7 acres) of all land per person it needs 48,500–73,000 km<sup>2</sup> ([Carrying capacity](#)). It has room to grow. ✓

**MAP TIP**

Shade the map lightly in these six land types before you place anything. The shading is your population map: crowded vale, ordinary hills, thin Fells, nearly empty forest and fen.

## Step 2: Cities, towns and villages

**Method:** choose an urban profile and split the people into tiers, then divide by a typical size ([calculator](#), [steps 4–5](#)).

Daravel uses the **average** profile (England, France and Germany c. 1300).

| Tier                     | Share | People     | Typical size                       | Number of places       |
|--------------------------|-------|------------|------------------------------------|------------------------|
| Cities of 10,000+        | ~3%   | ~73,000    | sized by rank (below)              | 4                      |
| Towns of 2,000–10,000    | ~5%   | ~121,000   | ÷ 3,000–4,000                      | 30–40 → <b>38</b>      |
| Small market towns       | ~6%   | ~145,000   | ÷ ~800                             | <b>~180</b>            |
| Villages, hamlets, farms | ~86%  | ~2,084,000 | ÷ 250 (lowland) or 100–150 (hills) | <b>~7,700 villages</b> |

Quick-ratio check for an average realm, per million people: 1–2 cities, 12–18 towns, 60–90 market towns, 2,500–3,500 villages. For 2.4 million that is 2–5, 29–44, 145–220 and 6,000–8,500. Daravel fits every row. ✓

### The top of the ladder

The capital of a strong kingdom holds about 1–2% of its people and is 3–7 times bigger than the second city ([Rank-size rule](#)). The rank-size rule predicts the 2nd place at ½ of the largest, the 3rd at ⅓, and so on.

| Rank  | Place                                                                                                                           | People                      | Rank-size prediction | Why it is here                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------|------------------------------------------------|
| 1     | <b>Hallowbridge</b> (capital)                                                                                                   | 40,000 (1.65% of the realm) | —                    | Lowest bridge on the Ambre, at the tidal limit |
| 2     | Liskmeet                                                                                                                        | 12,000                      | 20,000               | Confluence; old imperial city                  |
| 3     | Wendmouth                                                                                                                       | 11,000                      | 13,300               | River-mouth harbour                            |
| 4     | Brimhaven                                                                                                                       | 10,000                      | 10,000               | Wool port                                      |
| 5     | Tenterford                                                                                                                      | 8,000                       | 8,000                | Cloth town on a fast hill stream               |
| 6     | Ridgegate                                                                                                                       | 7,000                       | 6,700                | Gap town; second bridge                        |
| 7     | Norburgh                                                                                                                        | 6,500                       | 5,700                | March fortress and muster town                 |
| 8     | Wyndfoot                                                                                                                        | 5,000                       | 5,000                | Pass town below the Wyndgap                    |
| 9     | Holmstow                                                                                                                        | 4,500                       | 4,400                | Shrine on a fen island                         |
| 10–13 | Saltwich 4,000 (brine), Hawkridge 4,000 (march fortress), Cheapford 3,800 (ford, head of navigation), Silverhope 3,500 (silver) |                             |                      |                                                |

Hallowbridge is 3.3 times Liskmeet: a primate capital, like London c. 1377 at 3–4 times York. Below it the rule fits closely. ✓

**The other 29 towns** (2,000–3,600 each, 74,700 people in all):

| Region             | Towns                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Crownlands         | Kingsmoat 3,000; Ashenden 2,800; Ambremouth 2,400                                                                                |
| Duchy of Lisk      | Forgeham 2,800; Brockhurst 2,700; Wythmoor 2,600; Harnbury 2,500; Sedgeworth 2,400; Kingsferry 2,200                             |
| Duchy of the Fells | Carrowdale 2,400; Leadgill 2,200; Fellstead 2,000                                                                                |
| North March        | Skelbridge 2,800; Coldhaven 2,200; Scarwick 2,100; Wolfhope 2,000                                                                |
| Earldom of Brim    | Woldby 3,300; Gullhaven 3,000; Eastwold 2,800; Stowford 2,500; Brimstead 2,300                                                   |
| Duchy of Wend      | Wendbury 3,600; Merrow 3,100; Meremouth 2,800; Abbotsmere 2,800; Downholme 2,700; Bramling 2,400; Seacombe 2,300; Saltcove 2,000 |

## Villages

Rural people per zone are the zone's total minus the townspeople living there. Village sizes follow [step 5 of the](#)

[calculator](#); spacing uses the hexagon rule from [How far apart](#) (each place "owns"  $0.87 \times \text{spacing}^2$ ).

| Zone            | Rural people      | Share in villages | Typical village | Villages      | Land per village      | Spacing          |
|-----------------|-------------------|-------------------|-----------------|---------------|-----------------------|------------------|
| Fertile lowland | ~1,300,000        | 90%               | 250             | ~4,700        | ~9.5 km <sup>2</sup>  | ~3.3 km (2 mi)   |
| Mixed farmland  | ~613,000          | 75%               | 200             | ~2,300        | ~14.6 km <sup>2</sup> | ~4.1 km (2.5 mi) |
| Heath           | ~29,000           | 70%               | 150             | ~140          | ~21 km <sup>2</sup>   | ~5 km (3 mi)     |
| Fells           | ~85,000           | 50%               | 125             | ~340          | ~60 km <sup>2</sup>   | ~8.3 km (5 mi)   |
| Forest          | ~44,000           | 50%               | 125             | ~180          | ~64 km <sup>2</sup>   | ~8.6 km (5 mi)   |
| Marsh           | ~13,000           | 60%               | 125             | ~60           | ~58 km <sup>2</sup>   | ~8 km (5 mi)     |
| <b>Total</b>    | <b>~2,084,000</b> | <b>~83%</b>       |                 | <b>~7,700</b> |                       |                  |

The other ~355,000 country people live in hamlets (10–75 people) and farmsteads (5–15), mostly in the Fells, the forest and the fen edge.

#### Spacing checks (Patterns and spacing):

- Lowland villages ~3.3 km apart (rule: 1.5–4 km, 1–2.5 mi); Fell parishes ~60 km<sup>2</sup> (rule: 50–100+ km<sup>2</sup>). ✓
- 222 market centres (market towns, towns and cities) give one per ~525 km<sup>2</sup>, about 25 km (15 mi) apart. Add 100–200 villages with a weekly market (chapter 02's mini-example adds 50–100 for half as many people), and markets in the vale are 14–16 km (9–10 mi) apart: a 6–10 km walk at most. ✓
- The 8 places of 5,000+ are about 130 km (80 mi) apart, like Britain; the 4 cities about 180 km (115 mi), between France and Britain. ✓

- All four cities are on navigable water or the coast. Inland Tenterford stays under 10,000 (the "why is it here?" test). ✓

#### LATER ERA (1500s+)

By 1600 draw the capital 2–4 times bigger (80,000–160,000) with suburbs far outside its walls, and grow Brimhaven or Wendmouth fastest if ocean trade arrives (Population and bigger cities).

#### MAP TIP

On the kingdom map, draw the capital, the 3 cities, all 38 towns and only 30–40 of the ~180 market towns (the schematic map at the top shows 26, to keep its labels readable). Show villages as texture in the vale, not as 7,700 dots.

## Step 3: The capital

**Hallowbridge** stands at the lowest bridging point of the Ambre, at the tidal limit, 83 km (52 mi) up the estuary. On tidal coasts big ports sat 50–120 km (30–75 mi) inland (Notes on site types). It meets all five needs from [What makes a capital](#):

| Need                | Hallowbridge                                                                                                                                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich core region    | The lower Ambre vale: about 10,000 km <sup>2</sup> of lowland in the Crownlands (the royal domain)                                                                                                             |
| River or sea access | Sea ships to its quays; river boats 158 km (98 mi) up the Ambre. A city of 40,000 needs an overland hinterland of roughly 40–55 km (25–35 mi), so water supply is essential ( <a href="#">Feeding towns</a> ). |
| Defensible core     | The White Keep on a low bluff on the north bank, with the river in front                                                                                                                                       |
| Loyalty             | The Crownlands are the dynasty's home lands                                                                                                                                                                    |
| Legitimacy          | Royal burial abbey at Westhallow; coronation in the old city of Liskmeet                                                                                                                                       |

It is **not central** ([Common mistakes, no. 12](#)). The far north-west corner is 322 km (200 mi) away in a straight line,

about 350–400 km by road: 4–8 days for a royal messenger, a normal kingdom ([How big can a realm be](#)).

#### The royal landscape (Around the capital):

| Feature                               | Place                                                                                | Distance                                | Rule                                                       |
|---------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------|
| Palace quarter and royal burial abbey | Westhallow                                                                           | 4 km (2.5 mi) upstream                  | Like Westminster, ~3 km from London; burial church 0–20 km |
| Royal residences                      | Elmhurst manor, Redwater Castle, Kingsmoat castle, Kingswood lodge                   | 14, 31, 37 and 43 km (9, 19, 23, 27 mi) | 3–6 residences within ~60 km                               |
| Royal forest                          | Kingswood Chase (a great lord's chase that passed to the Crown; the old name stayed) | about 20–70 km (12–43 mi)               | 10–100 km                                                  |
| Coronation city                       | Liskmeet cathedral                                                                   | 119 km (74 mi)                          | 50–150 km; Reims is ~130 km from Paris                     |

**MAP TIP**

Draw Hallowbridge as two linked parts: the walled merchant city with the White Keep at its upstream corner, and the palace and abbey of Westhallow a little upstream. Put the port quays below the bridge, never above it (Bridges).

## Step 4: Duchies, counties and the North March

**Method:** divide the realm into 4–8 great fiefs, then into counties of 50–70 km (30–45 mi) across, each with a seat near its middle, and add 1–3 special zones (Political units).

People below come from each region's land types × the densities of step 1.

| Region                    | Area                                         | People              | Counties  | Seat         | Character                                  |
|---------------------------|----------------------------------------------|---------------------|-----------|--------------|--------------------------------------------|
| Crownlands (royal domain) | 13,900 km <sup>2</sup> (5,400 sq mi)         | ~400,000            | 5         | Hallowbridge | Lower Ambre vale                           |
| Duchy of Lisk             | 17,600 km <sup>2</sup> (6,800 sq mi)         | ~420,000            | 6         | Liskmeet     | Middle Ambre, Lisk valley, south Harnwood  |
| Duchy of Wend             | 24,100 km <sup>2</sup> (9,300 sq mi)         | ~650,000            | 7         | Wendmouth    | South coast, downs, heath                  |
| Earldom of Brim           | 9,400 km <sup>2</sup> (3,600 sq mi)          | ~240,000            | 3         | Brimhaven    | Wolds and north-east coast; small but rich |
| Duchy of the Fells        | 25,000 km <sup>2</sup> (9,700 sq mi)         | ~330,000            | 5         | Wyndfoot     | Mountains, pass, mines, cloth valley       |
| North March               | 23,900 km <sup>2</sup> (9,200 sq mi)         | ~365,000            | 6         | Norburgh     | Frontier facing Thelland                   |
| Liberty of Holmstow       | 2,600 km <sup>2</sup> (1,000 sq mi)          | ~13,000             | 1         | Holmstow     | The abbot's church territory in the fens   |
| <b>Kingdom</b>            | <b>116,500 km<sup>2</sup> (45,000 sq mi)</b> | <b>~2.4 million</b> | <b>33</b> |              |                                            |

**Checks:** duchies of 9,400–25,000 km<sup>2</sup> (rule: 10,000–50,000; Flanders was ~10,000 km<sup>2</sup>). 33 counties of ~3,500 km<sup>2</sup> (1,370 sq mi), about 59 km (37 mi) across, with ~73,000 people each (England: ~3,300 km<sup>2</sup> and ~120,000; the guide suggests 30–40 counties for an England-sized realm). ✓ The North March has ~15 people per km<sup>2</sup> against 27–29 in the Crownlands and Wend: emptier, as a march should be (Marches).

**Smaller units:** about 750 hundreds (district courts) of ~155 km<sup>2</sup>, each with 10–11 parishes and an open-air moot site (Local administration); 150–200 baronies, each with a castle at its head manor (*caput*); one to three manors per village.

**Borders (Borders and frontiers):**

- **West:** the crest of the Fells, with paired castles 17 km (11 mi) apart on either side of the Wyndgap pass.
- **North:** hill crests and the Harnwood in the west, the River Skel in the east, with a hatched **frontier band 30 km (19 mi) deep** (the guide's band is 10–50 km).

• **Oddities:** the **Debatable Land**, a disputed strip 16 × 6 km (10 × 4 mi), and the Liberty of Holmstow.

• **Inside:** duchy borders follow the Whitridge, the foothills, the fen and forest edges, never straight lines (Common mistakes, no. 13).

**MAP TIP**

Draw the realm border as a thick dash-dot line, duchy borders as medium dash-dot lines and county borders as thin ones (the schematic map leaves the county borders off to stay readable). Borders are always dash-dot lines, so no one mistakes them for roads (master legend). Mark each seat with a small pennant and each county town with a dot in the middle of its town symbol. Label the North March and the Debatable Land by name, fill the Debatable Land with stripes in the two claimants' colours, and give the Liberty its own colour.

## Step 5: Castles and fortifications

**Method:** apply the densities per 1,000 km<sup>2</sup> from [How many to draw](#) zone by zone. About one-third of all castle sites are in use at once.

| Zone                                                  | Area                    | Rate used                                                             | Sites ever built | Active c. 1300 | Gap between active castles |
|-------------------------------------------------------|-------------------------|-----------------------------------------------------------------------|------------------|----------------|----------------------------|
| Peaceful core (Crownlands, Lisk, Wend, Brim, Liberty) | 67,600 km <sup>2</sup>  | 6 sites, 2 active per 1,000 km <sup>2</sup> (rule: 4–8 and 1–3)       | ~410             | ~135           | ~24 km (15 mi)             |
| Duchy of the Fells (mostly mountain)                  | 25,000 km <sup>2</sup>  | 4 sites, 1 active (low end of the core rule)                          | ~100             | ~25            | ~34 km (21 mi)             |
| North March                                           | 23,900 km <sup>2</sup>  | One site per 110–130 km <sup>2</sup> , like the Anglo-Scottish border | ~200             | ~65            | ~20 km (12 mi)             |
| <b>Total</b>                                          | 116,500 km <sup>2</sup> |                                                                       | <b>~710</b>      | <b>~225</b>    |                            |

Core castles 20–30 km (12–19 mi) apart match the guide's peaceful core ([Borderlands](#)). Daravel has ~1.9 active castles per 1,000 km<sup>2</sup> against ~4 in England and Wales: fewer people, and no Welsh-style march of tiny lordships. A gam-

ing rule of one castle per 50,000 people would give only ~48 ([Gaming rules](#)).

**What the ~225 active castles are** ([Types of fortification](#)):

| Type                                                                       | Number | Where and why (its one-word job)                                                                                                            |
|----------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Royal citadel: great stone tower and curtain wall                          | 1      | White Keep, Hallowbridge: <i>town, bridge</i> , royal armoury                                                                               |
| County-town castles: stone keeps, shell keeps on old mottes, curtain walls | 31     | One at the edge of every county town except the capital and Holmstow. Norburgh, Hawkridge and Skelbridge are rebuilt as concentric castles. |
| Pass fortress (concentric)                                                 | 1      | Wyndgap Castle: <i>pass</i>                                                                                                                 |
| Forward line on the north border                                           | 10     | Small castles and towers at fords and passes, 29–43 km (18–27 mi) apart, about a day's march                                                |
| Estuary castle                                                             | 1      | Ness Castle at the estuary mouth: <i>harbour</i>                                                                                            |
| Royal residence castle                                                     | 1      | Redwater Castle                                                                                                                             |
| Baronial castles and fortified manor houses                                | ~180   | At the head manors of baronies: <i>estate</i>                                                                                               |

- **Garrisons:** 5–20 men per ordinary castle, 30–60 at the 14 frontier castles: 1,500–5,000 men in peace, about 0.06–0.2% of the people ([Garrisons and armies](#)).
- **Field army:** 0.2–0.5% of 2.4 million = **5,000–12,000** soldiers; an all-out 1% (~24,000) lasts only weeks.
- **Behind the front:** muster towns at Norburgh and Wyndfoot, the armoury in the White Keep, the fleet at Brimhaven ([Military geography](#)).
- **Walls:** stone walls only at the capital, the cities, Ridgewater, Norburgh, Hawkridge, Skelbridge and Wyndfoot; a ditch and gates at Tenterford and Saltwich; the rest open ([Common mistakes, no. 25](#)).
- **Beacons:** 10 beacons about 16 km (10 mi) apart from Skelbridge to the capital (rule: 5–20 km): warning within the hour, against 2–3½ days for a rider ([Beacons](#)).

- **Ruins:** the other ~485 sites are abandoned mottes and ringworks.

### LATER ERA (1500S+)

Put bastioned star forts at Skelbridge and Norburgh, a citadel at Hallowbridge and low gun forts at the estuary mouth. Turn most inland castles into ruins, palaces or manor houses ([Gunpowder and the new fortifications](#)).

**MAP TIP**

On the kingdom map draw the White Keep, the 31 county castles, the 13 other royal and frontier castles, about one major baronial castle per 2,000–3,000 km<sup>2</sup> (about 25) and a few grey ruined mottes. Give the five great fortresses (the White Keep, Wyndgap and the three concentric castles) the large three-tower icon and every other castle the small one-tower icon; draw each beacon as a flame dot and each abandoned motte as a grey mound with a tower tick. Leave the other ~155 baronial castles for local maps.

## Step 6: Roads, rivers and the sea

**Water first** (Rivers as roads):

| Waterway      | Stretch                                         | Length          | Traffic                                                    |
|---------------|-------------------------------------------------|-----------------|------------------------------------------------------------|
| Ambre estuary | Mouth to Hallowbridge                           | 83 km (52 mi)   | Sea ships (cogs)                                           |
| Ambre         | Hallowbridge – Ridlegate – Liskmeet – Cheapford | 158 km (98 mi)  | River boats; towpaths; Cheapford is the head of navigation |
| Lisk          | Liskmeet to Forgeham                            | ~55 km (34 mi)  | River boats carrying iron and timber                       |
| Brim          | Brimhaven to Norburgh                           | ~100 km (60 mi) | Small boats: they supply the march fortress                |
| Wend          | Wendmouth to Wendbury                           | ~60 km (37 mi)  | Small boats                                                |

**Main roads** (the three royal highways, the old imperial road and the Brim Road; Roads):

| Road                              | Route                                                                               | Length          |
|-----------------------------------|-------------------------------------------------------------------------------------|-----------------|
| The King's Way                    | Hallowbridge – Leyford – Ridlegate – Liskmeet – Cheapford – Wyndfoot – Wyndgap pass | 285 km (177 mi) |
| The North Road                    | Hallowbridge – Eastwold – Norburgh – Skelbridge                                     | 170 km (106 mi) |
| The South Road                    | Hallowbridge – Kingsmoat – Bramling – Wendmouth                                     | 173 km (107 mi) |
| Old Street (an old imperial road) | Liskmeet – Wythmoor – Abbotsmere – Wendmouth, dead straight                         | 167 km (104 mi) |
| The Brim Road                     | Eastwold – Woldby – Brimhaven                                                       | 109 km (68 mi)  |

About 900 km of main road needs an inn every 15–30 km (10–19 mi): **30–60 stopping places**, most in existing towns and villages (**Inns**). Regional roads (Downs Road, Fells Road, Harn Road, Salt Roads, Pilgrims' Way) link every market town to its neighbours. With them and Fell Street, the roads on the kingdom map add up to about 3,200 km (2,000 mi) inside the realm, close to the guide's 30 km of main road per 1,000 km<sup>2</sup> (about 3,500 km for Daravel; **How much main road to draw**).

- **Crossings:** the Ambre has only 4 fixed bridges: Hallowbridge, Ridlegate (71 km upstream), Liskmeet (51 km further) and Wyndfoot (96 km further). Between them: fords at Cheapford and Ambreford, ferries at Kingsferry and Thornwick (**Crossing rivers**). This is chapter 05's pattern for a great river: a few fixed bridges, tens of km apart. A settled lowland river like the Thames had a

bridge town about every 20 km (12 mi), which would put about 8 bridges between Hallowbridge and Cheapford instead of 2 (**How often is a river bridged?**).

- **Passes:** the Wyndgap is a low cart pass with a hospice near the summit, a toll, a castle and a chain of villages down to Wyndfoot. The Carrow Gap is a summer mule path with no towns (**Mountain passes**).
- **Tolls:** pontage and port customs at Hallowbridge; bridge tolls at Ridlegate and Liskmeet; the Wyndgap; the Skel bridge into Thelland (**Tolls**).
- **Ports:** about 600 km (370 mi) of open coast needs 4–6 head ports at one per 100–150 km. Daravel has 4 head ports (the customs ports): Hallowbridge (below the bridge), Brimhaven (royal shipyard), Wendmouth and Gullhaven (herring). Its 5 member ports are Ambremouth (outport and pilots; the only lighthouse stands on

Ness Castle at the estuary mouth, about 15 km (10 mi) to the north), Meremouth, Coldhaven, Seacombe and Saltcove. Sea lanes hug the coast; one open-sea crossing runs east, sailed from March to November only, as in the North Sea ([The sea](#); [Sailing seasons](#)).

**Travel times** at the guide's [speeds per day](#): walker 25–35 km, ox cart 15–25 km, royal messenger 50–90 km, river boat 40–100 km downstream and 10–20 km upstream, sailing ship 40–100 km.

| From Hallowbridge to | Distance                                  | On foot   | Ox cart    | Messenger  | By water                                     |
|----------------------|-------------------------------------------|-----------|------------|------------|----------------------------------------------|
| Ridgegate            | 70 km (44 mi)                             | 2–3 days  | 3–5 days   | 1–1½ days  | Upstream 3½–7 days                           |
| Liskmeet             | 120 km (75 mi)                            | 3½–5 days | 5–8 days   | 1½–2½ days | Upstream 6–12 days; back downstream 1–3 days |
| Norburgh             | 127 km (79 mi)                            | 3½–5 days | 5–9 days   | 1½–2½ days | —                                            |
| Brimhaven            | 167 km (104 mi) road; 182 km (113 mi) sea | 5–7 days  | 7–11 days  | 2–3½ days  | Sailing ship 2–5 days                        |
| Skelbridge (border)  | 170 km (106 mi)                           | 5–7 days  | 7–11 days  | 2–3½ days  | —                                            |
| Wendmouth            | 173 km (107 mi) road; 373 km (232 mi) sea | 5–7 days  | 7–12 days  | 2–3½ days  | Sailing ship 4–9 days                        |
| Wyndfoot             | 218 km (135 mi)                           | 6–9 days  | 9–15 days  | 2½–4½ days | —                                            |
| Wyndgap pass         | 285 km (177 mi)                           | 8–11 days | 11–19 days | 3–6 days   | —                                            |

An army with baggage (13–20 km a day) needs 9–13 days to reach the Skel bridge, so the North March must hold out alone for about two weeks.

#### MAP TIP

Draw the navigable Ambre thicker up to Cheapford, with a small boat at the head of navigation. Draw the 4 bridges as ")((" with the road running through, each ford as a short dashed line across the river and each ferry as a short dotted line with an "F" ([Symbols and labels](#)). Draw head ports as a large anchor with a quay and write their names in capitals; member ports get a small anchor. Sea lanes and the mule path are dotted lines; the season goes in the label ("2–5 days by sea, Mar.–Nov."). Write the travel days on the long roads ("Norburgh 4 days"). Every road must end somewhere: a town, a port, a pass, a mine or a shrine ([Drawing the route network](#)).

## Step 7: The Church

**Method:** use the English densities and rules in [Religious](#), [Cultural](#), [Legal](#) and [Ancient Sites](#).

| Feature                | Daravel                                                                                                                                                                        | Rule from chapter 08                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Parish churches        | ~8,000 (7,700 villages plus ~300 in towns and cities; one per ~14.5 km <sup>2</sup> , 5.6 sq mi)                                                                               | ~73 per 1,000 km <sup>2</sup> would give ~8,500                             |
| Dioceses (cathedrals)  | 13, about 9,000 km <sup>2</sup> (3,500 sq mi) each                                                                                                                             | English style: one per 5,000–10,000 km <sup>2</sup>                         |
| Archbishop             | 1, at Liskmeet: the oldest city, not the capital                                                                                                                               | Oldest or richest city                                                      |
| University             | 1, at Liskmeet                                                                                                                                                                 | 1–2 for a kingdom of 2–5 million, in the capital or oldest cathedral city   |
| Religious houses       | ~730 (one per ~160 km <sup>2</sup> )                                                                                                                                           | about 1 per 150 km <sup>2</sup>                                             |
| Hospitals open c. 1300 | ~250: one in each market town and open town, 2–5 in each of the 11 towns and cities with walls or a ditch                                                                      | Every market town one, every walled town 2–5; at gates, bridges, main roads |
| Leper houses           | ~270 (chapter 08's English density, one per ~430 km <sup>2</sup> ): one outside each town of 2,000+ and most market towns, one on each main road out of the capital and cities | 1–2 km outside the gate                                                     |

**Cathedral cities:** Liskmeet (archbishop), Hallowbridge, Wendmouth, Brimhaven, Ridgeway, Norburgh, Wyndfoot, Holmstow (a monastic cathedral, like Ely), Hawkridge, Wendbury, Merrow, Carrowdale and Woldby. In an Eng-

lish-style realm a cathedral marks an important place: 9 of the 13 are cities or towns of 4,000+, and the other 4 are county towns of 2,400–3,600 ([Dioceses](#)).

**Religious houses by order** (England's monks, canons and nuns scaled to Daravel's area; friaries counted town by town and commanderies county by county; [Monasteries](#)):

| Order                         | Number | Where                                                                                                 |
|-------------------------------|--------|-------------------------------------------------------------------------------------------------------|
| Benedictine and Cluniac monks | ~165   | Old towns: Westhallow (royal), Holmstow (shrine), Abbotsmere (monastery town)                         |
| Cistercian monks              | ~60    | Empty, watered valleys: Harnvale, Greywater, Coldwater, Stillwater, Mirefield                         |
| Carthusians                   | 1      | Sallowhope, deep in the Fells. England had only two charterhouses by 1300 (Witham 1178, Hinton 1227). |
| Augustinian and other canons  | ~270   | Town edges, hospitals, castles                                                                        |
| Nunneries                     | ~130   | Quiet rural sites                                                                                     |
| Friaries                      | ~75    | In towns: 6 in the capital, 3 per city, 2 per large town, 1–2 in most county towns                    |
| Military-order commanderies   | ~25    | One farm estate per inland county                                                                     |

Each big abbey has 5–40 granges (outlying farms), most within about 25 km (15 mi).

**The pilgrimage:** the shrine of the invented national saint, St Aldwen, is at **Holmstow Abbey** on its fen island. The Pilgrims' Way runs 87 km (54 mi) from the capital to the shrine: 67 km (42 mi) to the pilgrim town of Stowford, then about 20 km (12 mi) by causeway or boat. That is 4–5 stages of about 20 km (12 mi), with a hospice at Stowford and a leper house on the last stretch ([Pilgrimage](#)).

#### MAP TIP

At kingdom scale, show only the 13 cathedrals (a church front with two towers; a double cross beside Liskmeet for the archbishop), the great abbeys (a church with a square cloister beside it), Temple Ambre (a small church with a shield), the shrine (a scallop shell, never a star) and the university. A small crown beside Liskmeet labelled "(coronations)" and beside Westhallow labelled "(royal tombs)" marks the two royal churches. Save the 8,000 parish churches and the friaries for local and town maps.

## Step 8: Industry and special towns

**Method:** for every industry, find the resource, the fuel and the way out ([The basic rules](#)). The guide suggests 1–3 mining districts, 1–3 salt sources, one cloth region, one or two

fair towns and several ports per kingdom ([Special town types](#)).

| Industry                                  | Resource, fuel and way out                                                                                      | Special place (people)                                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Silver and lead (Mining)                  | Veins, wood and fast streams in the Fells; road down to Wyndfoot                                                | Silverhope with the royal mint (3,500), Leadgill (2,200), Orsdale (~900), 25–28 km (16–17 mi) apart; mining towns usually have 1,000–5,000 |
| Brine salt (Salt)                         | Brine springs; fuel by water (timber, fen peat); three Salt Roads, one to an estuary landing 26 km (16 mi) away | Saltwich (4,000)                                                                                                                           |
| Sea salt                                  | Salterns boiling sea water on the Saltings; small output, as in the north                                       | Marsh-edge hamlets                                                                                                                         |
| Stone and lime (Stone)                    | Whitridge limestone right beside the Ambre                                                                      | Chalkhythe quarries and kilns                                                                                                              |
| Iron, charcoal, glass (Forest industries) | Ore, coppice and streams of the Harnwood; iron goes down the Lisk                                               | Forgeham (2,800), forge ponds, a glasshouse                                                                                                |
| Wool and cloth (Textiles)                 | Sheep on the Fells, downs and wolds; fulling mills on the Tenter Water                                          | Tenterford (8,000), Cheapford wool fair in September (3,800), Brimhaven wool exports                                                       |
| Fish (Fishing)                            | Autumn herring; Saltwich salt                                                                                   | Gullhaven herring fair, late September to November (3,000); fishing hamlets a few km apart                                                 |
| Ships, wine, brick                        | Harnwood oak via the Brim; south slopes; estuary clay                                                           | Royal shipyard at Brimhaven; monastic vineyards at Abbotsmere; brickfields at Hallowbridge                                                 |

Every region still grows its own grain first; the mining towns buy food from the vale through Wyndfoot ([Common mistakes, no. 26](#)).

The fair dates copy real ones. Stourbridge Fair near Cambridge, one of the biggest English fairs, was held around Holy Cross Day (14 September), after the harvest. The free herring fair at Great Yarmouth began at Michaelmas (29 September) and lasted about 40 days, the main herring season on that coast ([Fishing](#)).

#### MAP TIP

Give each special place a visible reason: crossed hammers at Silverhope, salt pans (a small checkerboard) along the shore of the Saltings, "(salt)" in Saltwich's label (a brine town takes the label, not the pan symbol, at kingdom scale), a quarry notch at Chalkhythe, a mill wheel on the Tenter Water, an anchor with a castle tower for the royal shipyard at Brimhaven. A fair gets no symbol of its own: write the months in the town's label ("Cheapford, wool fair, Sept."). Label the three roads out of Saltwich "Salt Road".

## Step 9: Ruins and older layers

#### RULE OF THUMB

Every medieval landscape stands on older ones. Old things survive far from towns; near towns their stone is carted away ([Ancient and ruined layers](#)).

| Layer                | In Daravel                                                                                                                              | Real model                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Older empire's roads | Old Street and Fell Street: straight, still used                                                                                        | Watling Street                                             |
| Old towns and forts  | Liskmeet and Hawkridge stand on old imperial sites; <b>Wendchester</b> is a walled ruin in the fields about 10 km (6 mi) from Wendmouth | Wroxeter, 8 km from Shrewsbury ( <a href="#">Reuse</a> )   |
| Hillforts            | Hundreds; show 6. <b>Old Harrow</b> above Wyndfoot has a ruined church: the bishop moved down to the river town.                        | Old Sarum and Salisbury                                    |
| Barrows and stones   | Barrows on the Wend Downs ridge; the Grey Wives stone circle in the Fells                                                               | Avebury                                                    |
| Old border           | The Thelling Dyke, 25 km (16 mi) inside today's border                                                                                  | Offa's Dyke                                                |
| Abandoned castles    | ~485 mottes and ringworks; show about 10                                                                                                |                                                            |
| Deserted villages    | Few in 1300: Old Harnvale (cleared by Cistercians), Lostwick (royal forest), Skelby (burnt in raids)                                    | Most came after 1350 ( <a href="#">Deserted villages</a> ) |

**MAP TIP**

Draw ruins in grey with their own symbol and label them "(ruin)": Wendchester is a grey walled-town symbol, Old Harrow a grey church inside hillfort rings. A deserted village is a grey church alone in a field, labelled "(lost village)". Keep dotted lines for paths and draw the Thelling Dyke as a grey line with short ticks on the ditch side (here the Thelland side), labelled with its name. If you redate the map to 1450, add a few more "lost village" symbols in the clay vale (hundreds of villages are gone by then, but a kingdom map shows only a few), shrink the towns inside their walls and turn hill fields back into pasture ([Step 12](#)).

## Step 10: Zoom in on the Leyford area

The local map covers 10 × 10 km (6 × 6 mi), 100 km<sup>2</sup> (39 sq mi), of rich vale on the south bank of the Ambre, 38 km (24 mi) west of the capital on the King's Way. At 35 people per km<sup>2</sup> it should hold about 3,500 people. It is centred on the

market town of **Leyford**, where the King's Way crosses the Ley Brook.

**Features** compared with the English averages per 100 km<sup>2</sup> in [How many features](#):

| Feature                       | Leyford area                                          | English average per 100 km <sup>2</sup>   |
|-------------------------------|-------------------------------------------------------|-------------------------------------------|
| Market town                   | 1 (Leyford)                                           | 1–2 market places                         |
| Villages with a parish church | 9, plus Leyford's church                              | ~7 (this is the richest part of the vale) |
| Hamlets                       | 5                                                     | 0–5 in open-field lowland                 |
| Mills                         | 11 (10 watermills, 1 windmill)                        | ~8–11                                     |
| Manors                        | 9 lay manors and 1 monastic grange; 5 of them moated  | ~4–5 moated sites                         |
| Deer parks                    | 2 (about 1.1 km and 0.8 km across)                    | ~2–3                                      |
| Fishponds                     | 2 groups (6 ponds)                                    | at least 1–2                              |
| Hundred meeting place         | 1 (Ley Low, a mound by the King's Way)                | ~0.5–0.7                                  |
| Other                         | Hospital, gallows, 2 wayside crosses, a rabbit warren | —                                         |

**Households** (4.5–5 people each; **Village territory**):

| Place                                                           | Households | People                                         |
|-----------------------------------------------------------------|------------|------------------------------------------------|
| Leyford (market town: market cross, 2 inns, hospital, bridge)   | 160        | 720–800                                        |
| Stanmere (moated hall, dovecote, fishponds, park)               | 75         | 340–375                                        |
| Thornwick (two manors, Ambre ferry)                             | 70         | 315–350                                        |
| Kettlebury, Hollowell, Ashlade (each with a moated manor)       | 60, 55, 50 | 740–825                                        |
| Crossby (windmill), Oxlade (grange nearby)                      | 50, 45     | 425–475                                        |
| Upper Ley, Brackenhurst (heath-edge commons)                    | 40, 35     | 340–375                                        |
| 5 hamlets (Mill End, Fennick End, Heathcot, Woodside, Coldhams) | 40         | 180–200                                        |
| Millers, warrener, park keepers, grange lay brothers            | 12         | 55–60                                          |
| <b>Total</b>                                                    | <b>692</b> | <b>~3,100–3,450 (31–35 per km<sup>2</sup>)</b> |

**How it is laid out:**

- **Spacing:** villages 1.5–3.4 km (1–2 mi) apart (rule: 1.5–4 km), all on terraces above the flood plain, each church beside its manor house (**Village forms**). The farthest, Brackenhurst, is 5.5 km (3.4 mi) from Leyford market.
- **Rings** (**Rings of land use**): tofts and crofts by the houses; two or three open fields within 1.1–1.7 km (0.7–1 mi); meadow on the Ambre flood plain and along the brooks; four coppice woods at the parish edges; Ley Heath (common pasture and warren) on the sandy south.
- **Mills:** three brooks run north into the Ambre; the Ley's four mills are 1.6–2.8 km apart; the navigable Ambre has

no mill weir; Crossby, with no stream, has the windmill (**Mills**).

- **Roads:** the King's Way, cleared 200 ft (61 m) each side under the Statute of Winchester; lanes from every village to Leyford like spokes; drove tracks to the heath; a lane to the Thornwick ferry.

**MAP TIP**

At this scale every square kilometre has an owner and a name. Draw parish boundaries as thin dash-dot-dot lines along brooks and wood edges, put a mill on every village's stream, and keep empty white space only for the heath, which you label.

**Step 11: A fantasy variant**

Apply the real rules first, then change one input and follow the chain (**The method**). Here are two changes.

**A. Monsters in the Harnwood** (**Dangerous wilderness**):

| Feature        | Realistic Daravel                              | With monsters                                                                                                             |
|----------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Forest people  | ~46,000 in ~180 open villages and many hamlets | Perhaps half stay (an estimate), packed into 25–75 walled villages of 300–1,000 at the forest edge; the interior is empty |
| Refuges        | None needed                                    | Refuge forts a day's walk (24–32 km, 15–20 mi) apart round the forest's ~390 km (240 mi) edge: about 12–16 forts          |
| Forest roads   | The Lisk Road (154 km) and Fell Street         | Cleared about 60 m (200 ft) each side, fortified stops every 15–30 km, patrols                                            |
| Rangers        | —                                              | Lodges at the forest edge, fords and road entries, 15–30 km (10–19 mi) apart                                              |
| Iron and glass | Forgeham smelts with Harnwood charcoal         | Charcoal burners need guards; iron costs more or is imported                                                              |

**B. Dwarves under the Fells** (**Dwarves**):

- A hold of about 10,000 under Silverhope would need farmland within about 55 km (35 mi) on marginal land. The Fells cannot supply it, so the hold buys food.

- At 200–250 kg of grain per person a year that is 2,000–2,500 tonnes: 17,000–25,000 pack-horse loads of 100–120 kg, or 45–70 loads a day.

- Grain comes by boat to Cheapford, the head of navigation, then 90 km (56 mi) by road to a **gate town** at Fellbridge on the valley floor, which grows from a market town into a main Fells town. Metal and coin flow back down.
- A dwarven tunnel under the Wyndgap is a pass with no snow season (**Mountain passes**): it takes the traffic and the toll. Real tunnels were short (8–9 km between two Cappadocian underground towns), so draw one, not a network.

**FANTASY TWIST**

Change only these two inputs. The vale, rivers, capital, counties and 7,700 villages stay where the real rules put them, and that contrast makes the fantasy parts feel real.

**MAP TIP**

Draw the edge of the haunted Harnwood as a second hatched frontier band, its empty interior in dense cross-hatching (wild land), and the refuge forts along the edge as small castle icons. Draw Fellbridge with a road climbing to a hold gate (a small rounded arch,  $\cap$ ) in the mountainside, and mark the tunnel mouths on both sides of the Wyndgap with the same arch, joined by a thin brown dotted line labelled "(underground)".

## Step 12: The same kingdom at four dates

**RULE OF THUMB**

A new date changes the numbers, not the geography. Rivers, passes, bridges, the capital's site and most village sites stay where they are. The number of people, towns, markets, fields, woods and working castles changes, and some places become ruins.

**Why:** Each kind of place follows the population, but at its own speed (**Change over time**):

- **Villages come first and last longest.** Most lowland villages already existed by 1100 (Domesday Book, 1086, names 13,418 places), so growth mostly made them big-

ger. New villages appeared mainly in forest, fen, heath and upland.

- **Markets, towns, castles and abbeys multiply** in the 1100s and 1200s. In Kent, market places grew from under 20 in 1200 to more than 80 by 1350, if every charter became a working market (McLain 1997). England's religious houses grew from 61 in 1066 to about 400 by 1154 (Webber 2006).
- **After the Black Death** towns shrink inside their walls, the worst fields go back to grass or scrub, and villages on poor land are given up, mostly in the 1400s.
- **After 1500** most of the new growth goes to the capital and the ports (**What changes after 1500**).

**The scaling factors.** England is the best-measured realm, so use its curve unless you have a reason not to:

| England                      | 1086         | 1290–1300                 | 1400                                     | 1450              | 1600              |
|------------------------------|--------------|---------------------------|------------------------------------------|-------------------|-------------------|
| People                       | 1.71 million | 4.75 million (1290)       | 2.08 million                             | 1.90 million      | ~4.1 million      |
| Share of the 1290 peak       | 36%          | 100%                      | 44%                                      | 40%               | ~86%              |
| Arable land, fallow included | —            | 11.5 million acres (1300) | 8.7 million acres (1380)                 | 7.6 million acres | 8.9 million acres |
| Woodland and wood-pasture    | ~15%         | ~7% (c. 1349)             | stopped falling; scrub on abandoned land | —                 | —                 |

Sources: people from Broadberry, Campbell and van Leeuwen (2011, Table 6), with 1600 from **chapter 09**; arable land from Broadberry and others (2010, Table 1); woodland from Rackham's Domesday estimate and a Royal Forestry Society summary (**How much of the map is fields and woods**).

England fell further than most of Europe. Europe as a whole went from about 70–80 million in 1300 to about 50–60 million by 1400–1450, a loss of a quarter to a third

(**Realm populations**). So for 1400, an England-like Daravel keeps about 1.05 million people and a Daravel with the European average keeps 1.6–1.8 million. The table uses about 1.2 million.

### Daravel at four dates

The 1300 column is the kingdom of steps 1–11. The other columns are estimates made with the factors above; treat every number as a range.

| Tier or feature                                 | 1100                                               | 1300                               | 1400                                                             | 1600                                                                                            |
|-------------------------------------------------|----------------------------------------------------|------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| People                                          | ~1.0 million (0.9–1.4 million)                     | ~2.4 million                       | ~1.2 million (1.0–1.8 million)                                   | ~2.4 million (2.0–2.8 million)                                                                  |
| Density                                         | ~9 per km <sup>2</sup> (22 per sq mi)              | ~21 (54)                           | ~10 (27)                                                         | ~21 (54)                                                                                        |
| Capital (Hallowbridge)                          | 10,000–15,000 (1–1.5% of the realm)                | 40,000                             | 20,000–25,000 (about half, like London between c. 1300 and 1377) | 80,000–160,000 (2–4 times its medieval size)                                                    |
| Other cities of 10,000+                         | 0–1 (Liskmeet, the old imperial city)              | 3                                  | 0–1                                                              | 2–4, the ports first                                                                            |
| Towns of 2,000–10,000                           | 10–15                                              | 38                                 | 10–15 (most small towns drop below 2,000)                        | 30–40                                                                                           |
| Market centres (market towns, towns and cities) | 50–80                                              | ~222, plus 100–200 village markets | ~200; most village markets lapse                                 | 130–160 market places in all                                                                    |
| Villages                                        | 6,000–6,500, most of them smaller (100–150 people) | ~7,700                             | 7,000–7,500, many half empty; 10–25% gone by 1500                | 6,000–7,000                                                                                     |
| Ploughland, fallow included                     | ~10,000 km <sup>2</sup> (~9%)                      | ~24,000 km <sup>2</sup> (~21%)     | 16,000–18,000 km <sup>2</sup> (a quarter to a third less)        | 18,000–21,000 km <sup>2</sup> (less fallow, better yields)                                      |
| Woods                                           | 15–20%                                             | ~10%                               | ~11–12%: scrub and birch on abandoned edges                      | ~10%                                                                                            |
| Castles in use                                  | 300–400, almost all earth and timber               | ~225; the important ones stone     | 150–200, plus tower houses in the North March                    | the royal fortresses, 2–3 star forts, gun forts at the estuary; most others are ruins or houses |
| Religious houses                                | ~100, almost all Benedictine                       | ~730 (75 friaries)                 | ~700                                                             | ~730 if the realm stays Catholic; none working if it turns Protestant                           |

### How the counts were made:

- **People:** 1300 × England's share of the peak (1100: about 40%), or the European loss for 1400. Density is people ÷ 116,500 km<sup>2</sup>.
- **Capital:** the guide's rule that a strong kingdom's capital holds 1–2% of its people ([Rank-size rule](#)); in 1100 the court still travels, so take the low end. London fell from 60,000–100,000 c. 1300 to about 35,000 in 1377.
- **Villages:** in 1100, about 90% of the vale villages, 80% of the hill villages and a third of the villages in the Fells, forest, fen and heath (~720 in 1300). For 1500, Germany lost about 23.5% of its settlements between 1300 and 1500 (about 170,000 down to 130,000), and England has more than 3,000 known deserted villages ([Deserted villages](#)).
- **Towns:** in England the towns of 1377 were only half to two-thirds of their pre-plague size, and about 30 towns

of 2,000–10,000 remained for 2.5 million people ([The settlement ladder](#)). At that rate Daravel's 3 cities fall into the town tier, and all but one of its 29 small towns (2,000–3,600 people) fall below 2,000. The 1100 count is an estimate in proportion to the people and the markets.

- **Markets in 1100:** about a fifth of the 1300 number, following Kent's growth from under 20 market places in 1200 to more than 80 by 1350.
- **Markets in 1600:** only about 39% of the market places of c. 1300 still worked c. 1600 in England. 39% of Daravel's 320–420 market places is about 130–160.
- **Ploughland:** about 1 ha (2.5 acres) per person at the peak; after the plague, England's arable fell by a quarter (1380) to a third (1450).
- **Castles and abbeys:** see [Types of fortification](#), [Timeline of the changes](#) and [the dissolution of the monasteries](#). The 1100, 1400 and 1600 castle counts are estimates.

## What to redraw at each date

| Date | Add                                                                                                                                                                                                                                                                                                                                     | Remove, shrink or turn grey                                                                                                                                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1100 | An earth-and-timber motte at most county towns and at many manors; old Benedictine abbeys; a bigger Harnwood and more woods in the clay vale; a travelling court with several royal residences ( <i>What exists by which date</i> )                                                                                                     | Two-thirds or more of the market towns and towns; all friaries (they arrive in the 1220s) and the Cistercian valley abbeys; stone county castles (only the White Keep stands in stone); the Tenter Water fulling mills (first recorded in England in 1185); most villages in the Fells, fen and heath |
| 1300 | Everything in steps 1–11: farms high up the valley sides, suburbs outside the gates                                                                                                                                                                                                                                                     | Only the older layers of step 9                                                                                                                                                                                                                                                                       |
| 1400 | Grey "(lost village)" churches on poor land (200–700 of the 7,700 villages are gone or shrunk to hamlets by now, about 800–1,900 by 1500; at kingdom scale draw only a few); sheep pasture on old ploughland; scrub at the forest edges; gardens and empty plots inside the walls; tower houses and fortified manors in the North March | Most village markets; some small towns drop to market-town size; castles no longer kept up turn grey                                                                                                                                                                                                  |
| 1600 | A capital 2–4 times bigger with suburbs far outside the walls; star forts at Skelbridge and Norburgh, a citadel at Hallowbridge, gun forts at the estuary mouth; a post station every 20–40 km (12–25 mi) on the main roads (about 23–45 on 900 km); country houses in parks                                                            | Most inland castles (ruins, palaces or manor houses); if Protestant, the abbeys become grey ruins labelled "(ruin)" or houses called "Abbey"; St Aldwen's shrine is destroyed and the Pilgrims' Way empties; more than half the market places of 1300                                                 |

### LATER ERA (1500s+)

The 1600 column is the medieval map with a new top layer, not a new map. Villages, market spacing and walking distances stay the same; the capital, the ports, the forts and the post roads change (*What changes and what stays the same*).

### MAP TIP

Keep one base map with the terrain, rivers, roads and village dots. Draw each date as a separate overlay, and change the symbols, not the coastline: smaller or fewer town symbols in 1100 and 1400, grey ruins and lost villages from 1400, star forts and post horns in 1600. Put the date in the title cartouche, because a reader cannot judge the map without it.

## Quick summary

- **Land × density:** 116,500 km<sup>2</sup> (45,000 sq mi) gives about 2.4 million people (1.9–2.8 million), ~21 per km<sup>2</sup> (54 per sq mi).
- **Average urban profile:** 4 cities, 38 towns, ~180 market towns and ~7,700 villages, 3.3 km (2 mi) apart in the vale.
- **Primate capital:** Hallowbridge, 40,000 (1.65% of the realm, 3.3 times the second city), at the lowest bridge, 83 km (52 mi) up the estuary.
- **Royal landscape:** palace and burial abbey 4 km (2.5 mi) away, 4 more residences within 43 km (27 mi), a royal forest, coronation 119 km (74 mi) away in the archbishop's city.
- **Politics:** 5 great fiefs plus the Crownlands, 33 counties of ~3,500 km<sup>2</sup> (1,370 sq mi), ~750 hundreds, a march with a 30 km (19 mi) frontier band, a Debatable Land, a church liberty.
- **Castles:** ~710 sites, ~225 in use: 20–30 km (12–19 mi) apart in the core, a forward line every 29–43 km (18–27 mi), a beacon chain, a field army of 5,000–12,000.
- **Routes:** water first; ~900 km of main road with an inn every 15–30 km; 4 bridges on the Ambre; one cart pass; Liskmeet is 3½–5 days' walk from the capital.
- **Church:** ~8,000 parishes, 13 dioceses, ~730 religious houses (~75 friaries), 1 university, 1 great shrine.
- **Industry:** one mining district, two salt sources, one cloth region, two fairs (months in the label), a quarry, an iron forest and 9 ports (4 head ports), each with a visible reason.
- **Older layers and fantasy:** add old roads, grey ruins, lost villages and abandoned mottes; then change one or two fantasy inputs and follow the consequences.
- **Four dates:** about 1.0 million people in 1100, 2.4 million in 1300, 1.2 million in 1400 and 2.4 million again in 1600. Keep the geography; change the counts, the ploughland, the woods, the castles and the ruins.
- **Symbols:** use the *master legend*: a filled star for the capital, dash-dot lines for borders, solid or dotted lines for routes, ")((" for bridges, grey for ruins.

## Sources and further reading

All numbers come from the chapters of this guide and the shared baseline ([research/baseline-numbers.md](https://research.baseline-numbers.md)). Their full source lists are in chapters 01 to 11. The works below are the ones this example leans on most.

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- English Heritage, the Carthusians (Mount Grace Priory): <https://www.english-heritage.org.uk/visit/places/mount-grace-priory/history-and-stories/carthusians/>
- Historic England, *The Carthusian Monastery at Hinton Charterhouse*: <https://historicengland.org.uk/research/results/reports/92-1995>
- Hundred (county division) (Wikipedia): [https://en.wikipedia.org/wiki/Hundred\\_\(county\\_division\)](https://en.wikipedia.org/wiki/Hundred_(county_division))

## CHAPTER 13

# QUICK REFERENCE CHEAT SHEET

This sheet puts the guide's working numbers on one page, to keep open while you draw. Every figure is copied from the chapter that owns it, and the link under each table takes you to the full explanation. Numbers describe Europe c. 1300 unless a table says otherwise. The ranges are real: scholars disagree, so pick a value inside the range that fits your region.

## IN THIS CHAPTER

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## The numbers you will use most

### RULE OF THUMB

Almost everyone lives in villages. A "big city" in medieval Europe is anything above 10,000 people. A place of 50,000 is one of the great cities of the known world.

| What                                      | Number                                                                                                        |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| People per household (hearth)             | 4.5–5                                                                                                         |
| Share of people living in the countryside | 85–95%                                                                                                        |
| Good lowland at its peak                  | 30–50 people per km <sup>2</sup> (80–130 per sq mi)                                                           |
| Whole realm, uplands included             | about 7–40 per km <sup>2</sup> (18–100 per sq mi); Europe c. 1300 about 20 per km <sup>2</sup> (52 per sq mi) |
| Land per person                           | about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land                                       |
| Village                                   | 75–500 people (usually 150–300); 1.5–4 km (1–2.5 mi) apart in lowland                                         |
| Small market town                         | 500–2,000 people; 9–16 km (6–10 mi) apart                                                                     |
| Town of 10,000                            | needs farmland within 21–27 km (13–17 mi) on good land, or water transport                                    |
| For every city of 10,000+                 | about 10 towns, 50–100 small market towns and a few thousand villages                                         |

| What                                                           | Number                                                                                |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Capital of a strong kingdom                                    | 1–2% of the realm's people; 3–7 times the second city                                 |
| Walker / ox cart / army with baggage / royal messenger         | 25–35 / 15–25 / 13–20 / 50–90 km a day (15–22 / 10–15 / 8–12 / 30–56 mi)              |
| Transport cost, land : river : sea                             | about 8 : 4 : 1 (England c. 1300)                                                     |
| Inns on main roads                                             | every 15–30 km (10–19 mi)                                                             |
| Working castles                                                | 20–30 km (12–19 mi) apart in a peaceful core; 4–6 km (2.5–4 mi) on a contested border |
| Peacetime castle garrison                                      | 5–20 men                                                                              |
| Field army                                                     | 0.2–0.5% of the realm's people                                                        |
| Religious houses                                               | about 1 per 150 km <sup>2</sup> (60 sq mi)                                            |
| Cities of 100,000+ in Europe, the Middle East and North Africa | 4 (c. 1000), 8 (c. 1300), 8 (c. 1500)                                                 |

Source: 02 The settlement ladder, 02 How many of each kind, 02 Counting the cities, 02 The rank-size rule and primate cities, 02 Population density, 02 Carrying capacity, 02 Feeding towns, 01 How far apart: the numbers, 05 Speed per day, 05 Cost: why water beats land, 05 Inns, hospices and caravanserais, 04 Borderlands and conquered land, 04 Garrisons and armies, 08 How many of each, per area and 08 Monasteries, friaries and military orders.

**MAP TIP**

Write the date in the title box and draw a scale bar before anything else. Every number on this sheet changes with the date (see 02 Change over time) and only means something once you know your scale.

## Settlement tiers

**RULE OF THUMB**

Make the label size follow the tier, not the story. To make a small place look important, give it a special icon (castle, cathedral, shrine), not a bigger label.

| Tier               | People                                   | Households   | What it usually has                                                                                                            | Symbol: continent / kingdom / local                                                                              | Label                            |
|--------------------|------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Isolated farmstead | 5–15                                     | 1–2          | Own fields and pasture; no church                                                                                              | — / — / tiny dot                                                                                                 | none                             |
| Hamlet             | 10–75                                    | 2–15         | A few houses; a chapel at most; belongs to a village's parish                                                                  | — / — / 2–5 small house marks                                                                                    | 6–7 pt italic, or none           |
| Village            | 75–500 (English lowland usually 150–300) | 15–100       | Parish church, manor house, mill, smithy, open fields, common pasture                                                          | — / tiny dot (notable villages only) / house marks around a church                                               | 7–8 pt                           |
| Small market town  | 500–2,000                                | 100–400      | Weekly market, yearly fair, market place, burgh plots, inns, craftsmen, one parish church, often a hospital                    | — / small open circle / houses around a market square                                                            | 8–9 pt                           |
| Town               | 2,000–10,000                             | 400–2,000    | Usually walled; several parishes, guilds, friaries, hospitals, often a castle, a borough charter, its own council              | dot (important ones only) / circle with a ring (battlements on the ring if walled) / outline with wall and gates | 9–10 pt                          |
| City               | 10,000–50,000                            | 2,000–10,000 | Cathedral, many parishes and friaries, suburbs, guilds, law courts, mint; sometimes a university                               | dot or circle / circle with a thick battlemented ring (a larger walled symbol) / full outline                    | 11–12 pt, bold or small capitals |
| Great city         | 50,000+                                  | 10,000+      | Capital or major port; several markets, university, foreign merchants' quarters, organised water supply; grain by river or sea | large circle with a ring / large walled symbol / full outline with suburbs                                       | 13–14 pt<br>CAPITALS             |

A capital takes a filled star in place of its tier symbol (see the [master legend](#)).

Source: [02 The settlement ladder](#) and [02 Map symbols and labels](#).

**How big to draw it** (towns held about 100–200 people per hectare, or 40–80 per acre):

| Place             | People         | Built-up area                    | Width if roughly round  |
|-------------------|----------------|----------------------------------|-------------------------|
| Village           | 150–300        | 5–15 ha (12–37 acres)            | 250–450 m (0.15–0.3 mi) |
| Small market town | 500–2,000      | 5–20 ha (12–50 acres)            | 250–500 m (0.15–0.3 mi) |
| Town              | 2,000–10,000   | 20–80 ha (50–200 acres)          | 0.5–1 km (0.3–0.6 mi)   |
| City              | 10,000–50,000  | 80–300 ha (200–740 acres)        | 1–2 km (0.6–1.2 mi)     |
| Great city        | 50,000–200,000 | 300–1,000+ ha (740–2,500+ acres) | 2–4 km (1.2–2.5 mi)     |

Source: [02 How big to draw it](#).

### Regional exceptions:

| Region                         | What is different                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Italy                          | <i>Castelli</i> (fortified hilltop villages) of 200–1,000 people. Hundreds of small bishoprics, so a cathedral does not prove a big city. |
| Southern Spain, Sicily, Apulia | <i>Agro-towns</i> of 2,000–10,000 people who are mostly farmers: big in people, not real towns in function.                               |
| Germany and the Low Countries  | Many chartered "legal towns" had fewer than 2,000 people.                                                                                 |
| England                        | Only 17 bishoprics, so a cathedral does mean an important city.                                                                           |
| Byzantine lands                | Most villages well under 100 households (estimate).                                                                                       |

Source: [02 The settlement ladder](#).

#### LATER ERA (1500s+)

The ladder stays the same, but the top grows taller: by 1600 Istanbul had perhaps 400,000–700,000 people, and Paris, Naples and London 200,000–300,000 each. See [Later-era changes](#).

#### FANTASY TWIST

Where monsters roam the countryside, shift hamlet people into fortified villages of 300–1,000, as Italy did in its *incastellamento*. See [Fantasy adjustments](#).

#### MAP TIP

On a kingdom map at 1 cm = 10 km (about 1 inch = 16 mi), even a great city is only 2–4 mm across. Use symbols at that scale and draw real outlines only on local maps (1 cm = 1 km, about 1 inch = 1.6 mi, or larger).

## Population density by land type

#### RULE OF THUMB

Above about 50 people per km<sup>2</sup> (130 per sq mi), a region needs towns, specialised commercial farming or imported grain.

| Land type                | per km <sup>2</sup> | per sq mi | Real examples                                                                     |
|--------------------------|---------------------|-----------|-----------------------------------------------------------------------------------|
| Rich lowland, peak times | 30–50               | 80–130    | Northern Italy 34–40 c. 1300; England's whole-country average already ~36 in 1290 |
| Very urban, rich lowland | 50–75               | 130–190   | Flanders and Holland in the 15th century (estimate)                               |

| Land type                                                     | per km <sup>2</sup>                                                | per sq mi                                     | Real examples                                                                                                            |
|---------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Average mixed farmland                                        | 15–30                                                              | 40–80                                         | Suffolk ~23–25 in 1086; England ~19 in 1377                                                                              |
| Poorer land, young kingdoms                                   | 7–15                                                               | 18–40                                         | Hungary ~7 (c. 1200), ~9–11 (c. 1495); Crown of Poland 8.6 (1370), 13 (1500)                                             |
| Hill country (farmed valleys with upland grazing), peak times | 8–15                                                               | 21–39                                         | Cumberland and Westmorland about 15–17 in 1290 as whole counties (valleys and coastal plain included); about 6–7 in 1377 |
| High uplands, moors, mountains                                | 2–5                                                                | 5–13                                          | Scottish Highlands, high Alps, Pyrenees, northern Peninsulas (estimate)                                                  |
| Forest or steppe frontier                                     | 1–5                                                                | 3–13                                          | Lithuania, the Rus' forest zone (estimate)                                                                               |
| Shifting (swidden) farming in tropical forest                 | about 6 on average; 12–35 in busier districts; at most about 55–56 | about 14 on average; 30–90; at most about 145 | 1950s world average about 6; modern Southeast Asia from 12 in northern Laos to 35 in northern Thailand                   |
| Steppe nomads                                                 | under 1–2                                                          | under 3–5                                     | Herders need huge pastures                                                                                               |
| Desert                                                        | close to 0                                                         | close to 0                                    | Except oases and irrigated valleys                                                                                       |
| Irrigated river valley                                        | 100+ on the farmed strip                                           | 260+                                          | Egypt c. 1340                                                                                                            |
| Wet-rice or lake-garden basin (outside Europe)                | 80–125 over a region; 230–300 in a city-and-rice-field zone        | 210–325; 600–780                              | Yangzi delta ~80 (1102); Valley of Mexico ~100–125 (1519); Greater Angkor ~230–300 (13th century)                        |

These are typical values, not limits. Exceptional districts went far above them: Norfolk, a mostly rural county, had about 487,000 people in 1290 (a model estimate) on about 5,400 km<sup>2</sup> (2,100 sq mi), roughly 90 per km<sup>2</sup> (230 per sq mi). It still fits the rule of thumb: it had very intensive farming, one of England's largest cities (Norwich) and two busy ports (Lynn and Yarmouth). Keep such a figure for one exceptional district, not a whole realm.

Source: [02 By type of land](#) (including Norfolk, from [Broadberry, Campbell & van Leeuwen 2011](#), Table 8B: 486,920 people in 1290) and [02 Density beyond Europe](#); on Norfolk's farming, Campbell, "Agricultural progress in medieval England: some evidence from eastern Norfolk", *Economic History Review* 36 (1983).

**Quick values for the population calculator** (people = area × density):

| Land                                      | per km <sup>2</sup> | per sq mi    |
|-------------------------------------------|---------------------|--------------|
| Rich lowland                              | 30–40               | about 78–104 |
| Average land                              | 15–25               | about 39–65  |
| Hill country                              | 8–15                | about 21–39  |
| High uplands, mountains, forest and marsh | 2–5                 | about 5–13   |
| Desert                                    | 0                   | 0            |

A hex's area is about  $0.866 \times (\text{width across the flat sides})^2$ : a 30 km hex is about 780 km<sup>2</sup>, a 30 mi hex about 780 sq mi. Multiply per km<sup>2</sup> by 2.59 to get per sq mi.

Source: [02 Step-by-step population calculator](#).

**Whole realms:**

| Realm and date                         | per km <sup>2</sup> | per sq mi |
|----------------------------------------|---------------------|-----------|
| England 1086                           | ~13                 | ~34       |
| England 1290 (peak)                    | ~36                 | ~94       |
| England 1377 / 1450 (after the plague) | ~19 / ~15           | ~50 / ~38 |
| France c. 1328                         | ~35–40              | ~90–105   |
| Byzantium 1025                         | ~7                  | ~19       |
| Poland-Lithuania 1500                  | 6.6                 | 17        |
| Europe c. 1300                         | ~20                 | ~52       |

Source: [O2 By realm and date](#).

**Food: how far a town's farmland reaches** (assumes 30 people per km<sup>2</sup> on good land, 10 on marginal land; real radii are bigger where towns compete):

| Town size | Overland radius, good lowland | Overland radius, marginal land | With water transport                                    |
|-----------|-------------------------------|--------------------------------|---------------------------------------------------------|
| 1,000     | 7–8 km (4–5 mi)               | ~18 km (11 mi)                 | Not needed                                              |
| 10,000    | 21–27 km (13–17 mi)           | ~55 km (35 mi)                 | Food can come 2–10 times farther along a river or coast |
| 50,000    | 46–59 km (29–37 mi)           | ~125 km (78 mi)                | Must have river or sea supply                           |
| 100,000   | 65–84 km (40–52 mi)           | ~180 km (110 mi)               | Only possible by water                                  |

Source: [O2 Feeding towns: the hinterland](#).

**MAP TIP**

Before placing villages, lightly shade the map into crowded lowland, ordinary farmland, thin hill country and empty land. Then draw a faint circle at the overland radius around each town. If two circles overlap heavily, shrink one town or put it on water. Do both in pencil or on a layer you hide later, and remove them from the finished map: in the [master legend](#) a soft tint means a language area and a ring around a place belongs to the town symbol.

**Urbanization and how many of each tier****RULE OF THUMB**

For every city of 10,000+, expect about 10 towns, about 50–100 small market towns and a few thousand villages.

| Region                            | c. 1300                                                                         | c. 1500                                                  |
|-----------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------|
| Europe                            | ~5% in cities of 10k+; ~9.5% in all towns                                       | 5.6% in 10k+                                             |
| England                           | ~2% in 10k+ (only London); ~10% in towns of 2k+; 15–20% counting small boroughs | 3.1% in 10k+                                             |
| France                            | ~3–4% in 10k+; ~7–9% in 2k+                                                     | ~4.2% in 10k+                                            |
| Low Countries                     | Flanders ~33–40% in towns of all sizes                                          | Belgium 21%, Netherlands 16% in 10k+; Holland ~45% urban |
| Northern Italy                    | ~18% in 10k+ (Centre-North); ~20% in 5k+                                        | ~15–16% in 10k+                                          |
| Iberia                            | ~8–10% in 10k+ (inflated by agro-towns)                                         | Spain ~6%, Portugal ~3% in 10k+                          |
| Germany                           | ~2–3% in 10k+                                                                   | 3.2% in 10k+                                             |
| Denmark                           | ~10% in all towns                                                               | ~14% in all towns                                        |
| Poland, Hungary                   | under 2% in 10k+                                                                | ~1–2% in 10k+                                            |
| Arab Middle East and North Africa | ~7% in 10k+                                                                     | ~8% in 10k+                                              |

Source: [02 Urbanization by region](#).

**Urban profiles** (choose one per realm):

| Share of people              | Frontier (Poland, Hungary, Scotland, Scandinavia) | Average (England, France, Germany c. 1300) | Highly urban (Flanders, Lombardy, Tuscany) |
|------------------------------|---------------------------------------------------|--------------------------------------------|--------------------------------------------|
| Cities 10,000+               | ~1%                                               | ~3%                                        | ~15%                                       |
| Towns 2,000–10,000           | ~3%                                               | ~5%                                        | ~12%                                       |
| Small market towns 500–2,000 | ~4%                                               | ~6%                                        | ~8%                                        |
| Villages, hamlets, farms     | ~92%                                              | ~86%                                       | ~65%                                       |

| Per 1 million people | Frontier                                 | Average              | Highly urban |
|----------------------|------------------------------------------|----------------------|--------------|
| Cities 10,000+       | 0–1 (largest often 8–15k)                | 1–2 (largest 15–30k) | 5–10         |
| Towns 2,000–10,000   | 8–12                                     | 12–18                | 30–40        |
| Small market towns   | 40–60                                    | 60–90                | 80–100       |
| Villages             | 3,000–4,500, plus many hamlets and farms | 2,500–3,500          | 2,000–2,500  |

To turn shares into places, divide by a typical size: towns about 3,000–4,000, market towns about 800, villages about 250 in lowland and 100–150 in hill country. The capital of a strong kingdom holds about 1–2% of all its people;

in a fragmented realm the largest city holds about 0.5% or less.

Source: [02 Step-by-step population calculator](#).

**A model kingdom: England c. 1300** (4.75 million people on 130,000 km<sup>2</sup>, 50,000 sq mi):

| Tier                                  | Approximate number                             | One per ... people |
|---------------------------------------|------------------------------------------------|--------------------|
| Parishes (each with a village church) | ~9,500                                         | ~500               |
| Places with a market grant            | ~1,750 (many only village markets)             | ~2,700             |
| Boroughs (places with town status)    | about 600, many tiny                           | ~8,000             |
| Towns of 2,000–10,000                 | roughly 50–60 (estimate)                       | ~80,000–100,000    |
| Cities of 10,000+                     | ~6: London (60–100k) plus about five of 10–15k | ~800,000           |

Source: 02 How many of each kind and 02 The rank-size rule and primate cities.

**MAP TIP**

On a kingdom map you do not draw every village. Draw the capital, the cities, all the towns and the 30–40 most important market towns, then add villages as texture so the viewer feels the crowding. Count your symbols at the end: far more villages than towns, far more towns than cities.

## Spacing

**RULE OF THUMB**

Many small places close together, a few big places far apart. Stretch the pattern along rivers and roads, squeeze it on rich land and spread it on poor land.

| Feature                                      | Typical spacing                                                                                                                                                         | Notes                                                                                |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Villages, lowland                            | 1.5–4 km (1–2.5 mi)                                                                                                                                                     | One parish per ~14 km <sup>2</sup> (5.4 sq mi); about 4 km (2.5 mi) church to church |
| Parish churches in hills                     | about 7–10 km (4–6 mi), more in the emptiest uplands                                                                                                                    | Upland parishes cover 50–100+ km <sup>2</sup> (20–40+ sq mi)                         |
| Peasant walk to market                       | 6–10 km (4–6 mi) one way                                                                                                                                                | There and back in one day                                                            |
| Market towns, legal rule                     | 6⅔ mi (10.7 km)                                                                                                                                                         | Bracton (13th century); rarely enforced                                              |
| Market towns, in practice                    | 9–16 km (6–10 mi); up to about 25 km (15 mi) in a thinly settled realm                                                                                                  | 1,746 English markets by 1300; ~650 working markets in the late 1500s                |
| Towns of 5,000+                              | Flanders 35–45 km (22–28 mi); northern Italy 50–60 km (31–37 mi); France, Germany, Iberia 80–95 km (50–59 mi); Britain 130+ km (80+ mi); Poland 180–250 km (110–155 mi) | An upper bound from city data                                                        |
| Cities of 10,000+                            | Flanders and Italy 60–90 km (37–56 mi); France, Germany, Iberia 130–170 km (80–105 mi); Britain and Poland 260–400 km (160–250 mi)                                      | Same basis                                                                           |
| Rival city-states                            | 50–90 km (30–55 mi)                                                                                                                                                     | Each territory 1–3 days across                                                       |
| Inns and stops on main roads                 | 15–30 km (10–19 mi)                                                                                                                                                     | Roman <i>mansiones</i> 25–30 km (16–19 mi); pilgrim stages about 20 km (12 mi)       |
| Caravanserais                                | 30–40 km (19–25 mi) on plains and desert; 10 km (6 mi) or less in mountains                                                                                             | One day's stage                                                                      |
| Bridge towns on a settled lowland river      | about 20 km (12 mi) of river; gaps 10–35 km (6–22 mi)                                                                                                                   | Thames c. 1300: one bridge every 20–22 km (12–14 mi)                                 |
| Head ports                                   | about 1 per 100–150 km (60–95 mi) of coast                                                                                                                              | 15 for all of late medieval England                                                  |
| Galley watering ports                        | 100–300 km (60–190 mi)                                                                                                                                                  | Venice–Jaffa stops 100–450 km (60–280 mi) apart                                      |
| Main roads on a kingdom map                  | about 30 km per 1,000 km <sup>2</sup> (about 48 mi per 1,000 sq mi)                                                                                                     | From the Gough Map of Britain                                                        |
| Rhine toll stations, Mainz–Cologne, 1250     | about 15 km (9 mi)                                                                                                                                                      | At least 5 km (3 mi) apart                                                           |
| Active castles, England and Wales after 1066 | one per 250–300 km <sup>2</sup> (about 95–115 sq mi); about 16–17 km (10–11 mi) apart                                                                                   | 500–600 in use at once                                                               |
| Working castles, peaceful core               | 20–30 km (12–19 mi) (estimate)                                                                                                                                          | A castle per county town plus a few baronial seats                                   |

| Feature                                 | Typical spacing                                                                   | Notes                             |
|-----------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------|
| Castle sites, march or contested border | 4–6 km (2.5–4 mi)                                                                 | Welsh March, 1070s–1280s          |
| Castles in a conquest chain             | 20–40 km (12–25 mi), one day's march                                              | Each reachable by water           |
| Beacons                                 | about 5–20 km (3–12 mi) in hilly country; 55–100 km (35–60 mi) between high peaks | Estimates                         |
| Religious houses                        | about 1 per 150 km <sup>2</sup> (60 sq mi)                                        | Monks in valleys, friars in towns |
| Leper houses                            | 1–2 km (about 1 mi) outside each town of 2,000+                                   | On a main road                    |
| Mills on a good stream                  | every 1–2 km (0.6–1.2 mi)                                                         | About one mill per village        |
| Mining towns in one district            | 10–40 km (6–25 mi)                                                                | 2–7 towns plus a mint town        |
| Champagne fair towns                    | 45–65 km (28–40 mi)                                                               | Two or three days by road         |

Source: 01 How far apart: the numbers, 08 Parish churches and chapels, 05 Inns, hospices and caravanserais, 05 How often is a river bridged?, 05 Port ranks, 05 Coastal hopping and open sea, 05 How much main road to draw, 05 Tolls and customs, 04 How many castles and how far apart, 04

Watchtowers, beacons and coastal warning, 08 Monasteries, friaries and military orders, 06 Mills and water management, 07 Mining districts on the map, 07 Fair towns, 03 Drawing the unusual kinds.

**How many in a square of 100 × 100 km (62 × 62 mi, 10,000 km<sup>2</sup> or 3,860 sq mi):**

| Feature           | Region type                                                 | Land per place                                               | Number                           |
|-------------------|-------------------------------------------------------------|--------------------------------------------------------------|----------------------------------|
| Parish villages   | Lowland farmland                                            | ~14 km <sup>2</sup>                                          | ~700                             |
| Parish villages   | Upland                                                      | 50–100+ km <sup>2</sup>                                      | 100–200                          |
| Market places     | England at the c. 1300 peak (many tiny)                     | ~75 km <sup>2</sup>                                          | ~130                             |
| Market places     | Busy markets only (late 1500s)                              | ~230 km <sup>2</sup>                                         | ~45                              |
| Towns of 5,000+   | Flanders / northern Italy / France, Germany, Iberia         | 1,100–1,750 / 2,200–3,100 / 5,500–7,800 km <sup>2</sup>      | 6–9 / 3–5 / 1–2                  |
| Towns of 5,000+   | Britain / Poland                                            | 15,000+ / 28,000–54,000 km <sup>2</sup>                      | under 1 / 0–1                    |
| Cities of 10,000+ | Flanders, Italy / France, Germany, Iberia / Britain, Poland | 3,100–7,000 / 15,000–25,000 / 59,000–139,000 km <sup>2</sup> | 1–3 / usually 0–1 / usually none |

Source: 01 How far apart: the numbers.

#### LATER ERA (1500s+)

Add post stations every 20–40 km (12–25 mi) on main roads and, on raided coasts, towers 3–12 km (2–7 mi) apart, each in sight of the next. See 09 Posts, roads and canals and 09 Coastal defence.

#### MAP TIP

If two market towns are only 3 km (2 mi) apart, make one of them a village. If two towns on a road are more than 40 km (25 mi) apart, add a village or an inn between them.

## Travel per day and transport costs

#### RULE OF THUMB

Over land a traveller covers about 30 km (19 mi) a day and a loaded ox cart about 20 km (12 mi). By water, goods move at least as fast for a small part of the cost.

| Mode                              | km per day                                    | mi per day    | Notes                                                                 |
|-----------------------------------|-----------------------------------------------|---------------|-----------------------------------------------------------------------|
| Walking traveller                 | 25–35                                         | 15–22         | Pilgrim stages were shorter: about 20 km (12 mi)                      |
| Peasant to market                 | 6–10 one way                                  | 4–6           | Out, trade and home in one day                                        |
| Ox cart                           | 15–25                                         | 10–15         | Slow, strong, cheap to feed                                           |
| Horse cart or wagon               | 30–40                                         | 20–25         | Horses can double the speed of oxen                                   |
| Pack horse or mule train          | 20–40                                         | 12–25         | Loads: horse 100–120 kg, mule 150–180 kg, donkey 80–100 kg            |
| Camel caravan                     | 25–40                                         | 15–25         | About 230 kg per camel                                                |
| Cattle or sheep drove             | 16–30                                         | 10–20         | About 3 km/h (2 mph), grazing on the way; mostly 1700s–1800s evidence |
| Army with baggage                 | 13–20                                         | 8–12          | —                                                                     |
| Army living off the land          | 10–26                                         | 6–16          | —                                                                     |
| Forced march, a few days only     | 30–55                                         | 20–35         | —                                                                     |
| Mounted force with spare horses   | 55–65                                         | 35–40         | —                                                                     |
| Rider, many days                  | 30–50                                         | 20–30         | —                                                                     |
| Royal messenger with fresh horses | 50–90                                         | 30–56         | 30–40 mi normal, 52–56 mi when urgent                                 |
| Inca relay runners                | 240–300                                       | 150–190       | No horses or wheels                                                   |
| River boat, downstream            | 40–100                                        | 25–60         | Estimate; daylight travel                                             |
| River boat, upstream, towed       | 10–20                                         | 6–12          | Vienna to Passau: about 300 km (190 mi) in 20–31 days                 |
| Galley along a coast              | about 55 (voyage average with stops)          | about 30–35   | Venice to Jaffa: 48 days (1480)                                       |
| Sailing ship (cog or round ship)  | 40–100 voyage average; 90–220 while under way | 25–60; 55–135 | Can wait two weeks in port for wind                                   |

Source: [05 Speed per day](#); for army speeds, [04 Speed and season](#); for relay runners, [05 Roads without wheels](#).

**How long to cross your map** (add rest days, Sundays, bad weather and waiting for wind):

| Distance          | On foot   | Ox cart    | Courier with fresh horses | Galley       | Sailing ship  |
|-------------------|-----------|------------|---------------------------|--------------|---------------|
| 100 km (62 mi)    | 3–4 days  | 4–7 days   | 1–2 days                  | about 2 days | 1–3 days      |
| 300 km (190 mi)   | 9–12 days | 2–3 weeks  | 3–6 days                  | 5–6 days     | 3–8 days      |
| 1,000 km (620 mi) | 4–6 weeks | 6–10 weeks | 1.5–3 weeks               | 2.5–3 weeks  | 1.5–3.5 weeks |

Source: [05 How long does it take to cross your map?](#)

**Transport costs:**

| Comparison                       | Ratio or cost                                                                   | Where and when                                        |
|----------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| Land : river : sea, per ton-mile | about 8 : 4 : 1                                                                 | England, 1290s–1340s                                  |
| Water against land               | water about one-tenth of land                                                   | England                                               |
| Sea : river : land               | about 1 : 5 : 28–56                                                             | Roman Empire, Price Edict of 301 AD                   |
| Carting wheat                    | +3.3% of its value per 10 mi (16 km); price doubled after about 480 km (300 mi) | England c. 1300 (best case: horses, carts, flat land) |
| Wagon haulage of wheat           | +55% per 100 Roman miles; price doubled after about 270 km (170 mi)             | Roman Empire                                          |
| Grain by sea, Exeter to London   | about 560 km (340 mi) by sea cost the same as 65–70 km (40–45 mi) by cart       | England                                               |
| Bulk goods overland              | more than 30–50 km (19–31 mi) only if valuable, or in war or famine             | General rule                                          |

Source: 05 Cost: why water beats land.

**LATER ERA (1500s+)**

Relay posts with fresh horses at every stage carried letters up to about 150 km (95 mi) a day. Goods still moved at the old speeds. See 09 The post: relay stations.

**FANTASY TWIST**

A griffon or a teleport circle that carries 100 kg is a pack horse with wings. It changes the speed of news and luxuries, not the price of bread. See 10 Flying mounts and airships.

**MAP TIP**

Write travel days on long routes ("12 days to the capital"). That is how travellers thought, and it shows at once if your map is too big or too small for the story.

**Political units and sizes****RULE OF THUMB**

Kingdom > duchy > county > barony > manor, each several to a few dozen times smaller than the one above. But medieval units overlapped and were scattered: do not expect a neat pyramid.

| Unit                  | Typical area                                                                                   | Typical population (c. 1300)              | Seat                                                   |
|-----------------------|------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------|
| Empire                | 0.8–1.7 million km <sup>2</sup> (300,000–650,000 sq mi)                                        | 5–15 million                              | One great capital, or none (Holy Roman Empire)         |
| Kingdom               | 10,000–450,000 km <sup>2</sup> (4,000–175,000 sq mi)                                           | 0.1–17 million                            | Capital or travelling court                            |
| Duchy or principality | 10,000–50,000 km <sup>2</sup> (4,000–19,000 sq mi)                                             | 0.5–2 million (estimates)                 | Ducal city with castle, cathedral and court            |
| County (shire)        | 1,000–10,000 km <sup>2</sup> (400–4,000 sq mi); England averaged ~3,300 km <sup>2</sup>        | 20,000–300,000; England averaged ~120,000 | County town with castle, sheriff and court             |
| Hundred or wapentake  | ~130–150 km <sup>2</sup> (50–60 sq mi) in 1086; ~150–200 km <sup>2</sup> (60–80 sq mi) by 1316 | 2,000–10,000                              | Open-air meeting place: mound, ford, crossroads, stone |
| Barony, lordship      | a few manors to 100+, often scattered over several counties                                    | —                                         | <i>Caput</i> (head manor), usually with a castle       |
| Knight's fee          | 1–2 manors; no fixed size                                                                      | —                                         | Manor house                                            |
| Manor                 | 2–10 km <sup>2</sup>                                                                           | 50–500                                    | Manor house, often moated                              |
| Parish                | ~14 km <sup>2</sup> in lowland; 50–100+ km <sup>2</sup> in uplands                             | 75–500                                    | Parish church                                          |

Source: [03 The units, top to bottom](#).

**How many units:** England c. 1300 had 1 kingdom, 39 shires, about 630 hundreds (812 traceable in 1086), ~9,500 parishes and 5,000–6,000 knights' fees. France c. 1200 had the royal domain plus 6 great lay fiefs. Hungary had about 70 counties in the 15th century. The Holy Roman Empire in 1521 had about 400 imperial estates. Song China had about

1,230 counties, each about 2,000–2,500 km<sup>2</sup> (770–970 sq mi), with seats about 45–55 km (28–34 mi) apart. Europe c. 1500 had roughly 500 political units.

Source: [03 How many units in one kingdom?](#), [03 A different model: the bureaucratic empire](#), [03 Realistic realm sizes](#).

#### Real realms for comparison:

| Realm             | Area                                                 | People                  | Farthest point from the capital |
|-------------------|------------------------------------------------------|-------------------------|---------------------------------|
| Andorra           | 468 km <sup>2</sup>                                  | tiny                    | under 1 day                     |
| Navarre           | ~10,000–12,000 km <sup>2</sup>                       | ~100,000 (c. 1300)      | 2–3 days                        |
| Scotland          | ~79,000 km <sup>2</sup>                              | 0.5–1 million (c. 1300) | 5–8 days                        |
| Portugal          | ~89,000 km <sup>2</sup>                              | ~1.0 million (1500)     | 4–7 days                        |
| England           | 130,000 km <sup>2</sup>                              | 4.75 million (1290)     | 6–11 days                       |
| Poland            | ~240,000 km <sup>2</sup> (1370)                      | ~2 million              | 6–10 days                       |
| Hungary           | 280,000–325,000 km <sup>2</sup>                      | 2–3 million             | 8–12 days                       |
| France            | smaller than modern France's 551,000 km <sup>2</sup> | 16–17 million (1328)    | 10–14 days                      |
| Holy Roman Empire | 0.8–1.1 million km <sup>2</sup>                      | 11–14 million (c. 1300) | weeks; no single capital        |
| Byzantium (1025)  | 1.675 million km <sup>2</sup>                        | ~12 million             | weeks, helped by sea routes     |

Europe as a whole had 32–40 million people c. 1000, 70–80 million c. 1300 and 70–90 million c. 1500.

Source: [03 Realistic realm sizes](#) and [02 Realm populations and areas](#).

### Distance from the capital:

| Distance            | Royal messenger | Army        | What it means                                                           |
|---------------------|-----------------|-------------|-------------------------------------------------------------------------|
| 100 km (60 mi)      | 1–2 days        | 5–8 days    | Core: royal domain, residences, hunting forests                         |
| 300 km (190 mi)     | 3–6 days        | 15–23 days  | Counties ruled by royal officers                                        |
| 600 km (370 mi)     | 7–12 days       | 30–45 days  | Edge of a large kingdom: great lords, marcher lords, governors          |
| 1,000+ km (600+ mi) | 11–20 days      | 50–75+ days | Only empires with sea routes, relay posts or semi-independent governors |

Direct rule reaches roughly 500–700 km (300–450 mi), about 8–12 days at a messenger's normal pace (a heuristic, not a measured law).

Source: 03 [How big can a realm be.](#)

### Kinds of state and how they look:

| Kind                              | How it looks on the map                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Feudal kingdom                    | One outline with a patchwork of great fiefs inside; royal domain around the capital                                                     |
| City-state and its <i>condado</i> | A city with a territory of roughly 20–50 km (12–30 mi) radius; many side by side                                                        |
| League of towns                   | A network: member towns keep their normal symbols, with names underlined in the league's colour, linked by sea routes; no coloured area |
| Confederation                     | A cluster of small units, some jointly ruled subject lands                                                                              |
| Composite empire                  | A big outer border with hundreds of small internal units                                                                                |
| Church state                      | Its own colour, often in scattered pieces                                                                                               |
| Military-order state              | A frontier territory with many castles and commanderies                                                                                 |
| Maritime empire                   | A home city plus a chain of coastal dots and islands; only these take its colour                                                        |
| Nomadic empire                    | Huge vague area: pastures, a few capitals, a ring of tributary states in a lighter tint of the empire's colour                          |

Source: 03 [Kinds of states and how they look on a map.](#)

#### MAP TIP

For an England-sized kingdom (130,000 km<sup>2</sup>, 50,000 sq mi), draw 30–40 counties, each 50–70 km (30–45 mi) across with its town near the middle, grouped into 4–8 duchies or great fiefs. On a continent map use at least three kinds of state: one or two feudal kingdoms, one fragmented zone and one unusual kind.

## Capitals checklist

#### RULE OF THUMB

A capital sits in the richest, most loyal region of the realm, on navigable water, at a place that is easy to defend. Centrality matters less than you think.

| # | Check               | Rule and numbers                                                           |
|---|---------------------|----------------------------------------------------------------------------|
| 1 | Rich core region    | The court, officials and garrison must be fed from the ruler's own estates |
| 2 | River or sea access | A capital grows into a big city, and big cities need grain by water        |

| #  | Check                    | Rule and numbers                                                                                                                                                                        |
|----|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3  | Defensible core site     | Hill, ridge, island or river loop for the palace, treasure and archives                                                                                                                 |
| 4  | Loyal land               | In the dynasty's home region, which may be off-centre (Paris, London, Kraków)                                                                                                           |
| 5  | Legitimacy               | An old sacred or royal site: coronation church, saint's tomb                                                                                                                            |
| 6  | Type                     | Central, core-region, or forward (moved toward a frontier the ruler is pushing)                                                                                                         |
| 7  | Size                     | Strong kingdom: 1–2% of the realm's people and 3–7 times the second city. Loose realm: the largest city holds about 0.5% or less, and may not be the capital                            |
| 8  | Fixed or travelling      | Before about 1200 most western courts travelled: draw 5–15 royal residences about a day apart (about 30 km, 19 mi)                                                                      |
| 9  | Inside the capital       | Palace or castle, cathedral or great abbey, treasury, writing office and archives, law courts, mint, later a university                                                                 |
| 10 | Palace and merchant city | Often two linked parts, such as Westminster about 3 km (2 mi) from London                                                                                                               |
| 11 | Royal landscape          | 3–6 royal residences within about 60 km (37 mi); hunting forests 10–100 km (6–60 mi) away; burial church 0–20 km (0–12 mi); a coronation town 50–150 km (30–90 mi) away if you want one |
| 12 | Mints                    | Strong kingdom: the capital, 1–3 church or regional centres and one at each big silver mine. Loose realm: a mint in every princely seat                                                 |
| 13 | University               | One or two per kingdom of 2–5 million, in the capital and the oldest cathedral city                                                                                                     |
| 14 | Reach                    | Anything beyond about 12 days' messenger travel (roughly 700 km or 430 mi by road) needs a viceroy or becomes semi-independent                                                          |

Source: 03 What makes a capital, 03 Travelling courts and fixed capitals, 03 What is in and around a capital, 03 How many mints?, 02 The rank-size rule and primate cities, 08 Universities and schools.

#### LATER ERA (1500s+)

Fixed capitals win everywhere (Madrid from 1561), and some capitals grow to 5–10 times the second city. See 09 Why capitals grow so much.

#### FANTASY TWIST

A capital chosen for magic (a ley-line crossing, a dragon-proof mountain) still needs food, water transport and loyal land, unless you show how magic replaces them.

#### MAP TIP

In a strong kingdom, make the capital the biggest city on the map. In a loose realm, put the court in a middling town and let a trading port or cloth city be bigger: the contrast tells the viewer how the realm works.

## Fortifications by date and where they go

#### RULE OF THUMB

Every fortification controls something: a road, a crossing, a town, a border, an estate, a refuge or a lookout. If you cannot name it, move or delete the fort.

| Type             | Dates                                              | What it is                                                                     | Where it goes                                                                |
|------------------|----------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Ringwork         | 10th–12th centuries                                | Earth bank and ditch with a timber fence; no mound                             | Early castles; common in south-west England and south Wales after 1066       |
| Motte-and-bailey | From c. 1000 (Loire, Normandy); England after 1066 | A motte (man-made earth mound) with a timber tower, and a bailey (fenced yard) | Beside towns, villages and crossings; about 80% of Norman castles in England |
| Stone keep       | From c. 950 in France; England from the 1070s      | Massive stone tower; about ten years to build                                  | Royal and great baronial castles                                             |
| Shell keep       | Mostly 11th–12th centuries                         | Stone ring wall on top of a motte                                              | Rebuilt mottes                                                               |

| Type                          | Dates                                                            | What it is                                              | Where it goes                                                                           |
|-------------------------------|------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Curtain-wall castle           | Late 12th–13th centuries                                         | Strong wall with flanking towers                        | Important royal and baronial sites                                                      |
| Concentric castle             | Byzantine 11th century; crusader from the 1160s; Wales 1268–1295 | Two or more rings of walls                              | Great royal and crusader fortresses on conquest frontiers                               |
| Tower house                   | 14th–17th centuries                                              | Tall stone house of 3–6 storeys                         | Raided borderlands: Ireland over 3,000, Scotland about 800, England over 250            |
| Peel tower and bastle house   | Peels 14th–17th; bastles mostly 16th–17th centuries              | Small border tower; fortified farmhouse                 | Anglo-Scottish border                                                                   |
| Fortified manor house         | 13th–15th centuries                                              | Moat, gatehouse, battlements; stops a raid, not an army | Lowland estates; about 460 English licences to crenellate                               |
| Town walls and gates          | All periods; big wave 12th–14th centuries                        | Wall or bank and ditch with gates                       | Cities, regional capitals, frontier towns, raided ports; English murage tolls from 1220 |
| Citadel                       | All periods                                                      | Fortress at the edge of a city, watching it too         | Capitals and big or rebellious cities                                                   |
| Fortified bridge              | 12th–14th centuries                                              | Bridge with a gate tower                                | Main-road crossings; collects tolls                                                     |
| Fortified church              | 13th–17th centuries                                              | Thick walls, strong tower, often a ring wall            | Raided villages (Transylvania, Slovenia)                                                |
| Reused hillfort or Roman fort | Whenever handy                                                   | New castle inside old walls                             | Old Sarum, Pevensey, Portchester                                                        |
| Crusader castle               | 1100–1291                                                        | Large stone castles on spurs and hills                  | Hostile frontier, few defenders                                                         |
| Islamic <i>ribat</i>          | 8th–11th centuries                                               | Fortified lodge with a signal tower                     | Chains along coasts and borders                                                         |
| Military-order castle         | 12th–15th centuries                                              | Castle and monastery in one                             | Crusading frontiers (Malbork, Calatrava)                                                |
| Siege castle                  | All periods                                                      | Fort built to block an enemy castle or town             | 180–270 m (200–300 yd) from the target, just beyond bow range                           |
| Watchtower and beacon         | All periods                                                      | A handful of men who see, signal and run                | Passes, fords, wells, hilltops, headlands                                               |

Source: 04 Types of fortification.

### Where to put a castle:

| Site                              | Why it works                                            |
|-----------------------------------|---------------------------------------------------------|
| River crossing (ford or bridge)   | Every road and every boat must pass                     |
| Mountain pass or gap              | The only easy way between two regions                   |
| Spur or promontory                | Cliffs on two or three sides; one ditch across the neck |
| River bend (meander)              | Water on three sides                                    |
| Confluence                        | Controls traffic on both rivers                         |
| Island or lake shore              | Water as a moat; controls the river or lakeside road    |
| Coastal headland or harbour mouth | Supply by ship; watches the harbour                     |
| Hill above a town                 | Protects the town and watches it                        |
| Flat wet lowland                  | Dig a moat or flood the ground (a "water castle")       |

In the German lands about two-thirds of castles stood on high ground; in England most stood on low ground beside villages and towns.

Source: 04 Where to put a castle.

**How many to draw, per 1,000 km<sup>2</sup> (about 390 sq mi), c. 1300:**

| Zone                                           | All sites ever built                | Active castles   | Kingdom map                                                                     | Local map                        |
|------------------------------------------------|-------------------------------------|------------------|---------------------------------------------------------------------------------|----------------------------------|
| Peaceful lowland core                          | 4–8                                 | 1–3              | County-town castle plus 1 major baronial castle per 2,000–3,000 km <sup>2</sup> | All active castles plus ruins    |
| Average (England and Wales)                    | ~12                                 | ~4               | Royal and major castles, about 1 per 1,000 km <sup>2</sup>                      | All                              |
| Hilly land with many small lords (German type) | ~25                                 | ~8–12 (estimate) | Major and toll castles                                                          | All, on spurs and river cliffs   |
| March or contested border                      | 35–60                               | 12–20            | A castle-studded band                                                           | All, including mottes and towers |
| Raided border, late Middle Ages                | 8–10 castles plus many tower houses | Most of them     | Castles and the main tower houses                                               | Every tower house and bastle     |
| Thin conquered frontier (Prussia type)         | 1–3                                 | 1–3              | All, each paired with a town                                                    | All                              |

On a continent map, show only about 1 fortress per 20,000–50,000 km<sup>2</sup> (8,000–19,000 sq mi). England and Wales had 1,761 castle sites in all, but only 500–600 in use at once; draw one or two ruins or earthworks for every active castle in long-settled land.

Source: 04 How many to draw, 04 Counts and densities, 04 Ruined and active castles.

**Garrisons and armies:**

| Situation                        | People                                                                         |
|----------------------------------|--------------------------------------------------------------------------------|
| Peacetime caretaker garrison     | 5–20                                                                           |
| New frontier castle in peacetime | 30–60 (Harlech 1284: 36)                                                       |
| Small castle under siege         | 20–150                                                                         |
| Great castle in a major war      | 150 to about 1,200 (Kenilworth 1266)                                           |
| Great crusader fortress          | 1,500–2,200 (Safed: 1,700 in peace, 2,200 in war)                              |
| Frontier fortress city           | 4,000–5,000 (Tarsus)                                                           |
| Field army                       | 0.2–0.5% of the realm's people; about 1% is an all-out effort that cannot last |

Source: 04 Garrisons and armies.

**LATER ERA (1500S+)**

Low, thick bastioned "star" forts appeared in Italy from the late 1400s and spread north in the 1530s–1540s. Only frontiers, capitals, main ports and key crossings got them. See 09 Gunpowder and the new fortifications.

**MAP TIP**

Next to each castle, write its job in one word: *road*, *bridge*, *town*, *border*, *estate*, *refuge* or *watch*. Put the castle at the edge of its town, on the high point or by the river, with the market at its gate.

**Trade goods and route types****RULE OF THUMB**

The higher the value per kilogram, the farther a good can travel over land. Cheap, heavy goods stay local unless they can float.

| Good                               | Value per weight | How far over land                                                           | How far by water                |
|------------------------------------|------------------|-----------------------------------------------------------------------------|---------------------------------|
| Grain, flour                       | Very low         | Usually under 30–50 km (19–31 mi); routine sales within about 17 km (10 mi) | Hundreds of km by river and sea |
| Firewood, timber, stone, brick     | Very low         | A few km                                                                    | Far by water                    |
| Salt                               | Low              | About 100 km (62 mi) or more on "salt roads"                                | Across Europe                   |
| Wine                               | Medium           | Short distances                                                             | Very far                        |
| Salted herring, stockfish          | Medium           | Short to medium                                                             | Continent-wide                  |
| Wool                               | Medium to high   | By pack horse and cart to ports                                             | England to Flanders and Italy   |
| Metals (copper, tin, silver, iron) | Medium to high   | Long distances                                                              | Long distances                  |
| Cloth, furs, wax, amber            | High             | Any or long distances                                                       | Any or long distances           |
| Spices, silk, dyes, gems, sugar    | Very high        | Across continents                                                           | Across oceans                   |
| Live animals                       | Walk themselves  | Hundreds of km along drove roads                                            | —                               |

Source: 05 Bulk goods and luxury goods.

#### Route types:

| Route                          | What it is                                                                  | Where                                    |
|--------------------------------|-----------------------------------------------------------------------------|------------------------------------------|
| Old Roman road                 | Straight, engineered, often decayed; still used                             | Former Roman lands                       |
| Royal highway                  | Main route between cities under the ruler's protection                      | Between major towns                      |
| Local road or lane             | Village to market town, village to village                                  | Everywhere                               |
| Hollow way                     | Lane worn below the land, up to 5 m (16 ft) deep                            | Soft rock on slopes near towns and fords |
| Causeway                       | Raised road across marsh or flood plain                                     | Fens, flood plains, bridge approaches    |
| Pack-horse track or bridle way | Narrow track for animals, not carts                                         | Hills and mountains                      |
| Drove road                     | Wide grassy route, 12–27 m (40–90 ft) wide, halts about 16 km (10 mi) apart | From uplands to towns and fairs          |
| Caravan route                  | Track from well to well and caravanserai to caravanserai                    | Desert and steppe                        |
| Navigable river                | Highway with a one-way discount: fast downstream, towed upstream            | Up to the head of navigation             |
| Portage                        | Boats or cargo carried overland between two rivers or round rapids          | Watersheds, rapids                       |
| Sea lane                       | Coast-hugging route with short open-sea crossings                           | Coasts and island chains                 |

As a rough guide, a village has 2–3 roads, a market town 3–5 and a city 5–8 or more. Each mountain range gets 1–3 main passes and 2–5 minor ones.

Source: 05 Kinds of roads, 05 Drove roads, 05 Mountain passes, 05 Which places become hubs?.

**Trade nodes, from small to great:**

| Node                                               | How often                  | How many                                                      |
|----------------------------------------------------|----------------------------|---------------------------------------------------------------|
| Weekly market                                      | Weekly                     | Market towns 9–16 km (6–10 mi) apart                          |
| Yearly fair                                        | Once a year, days to weeks | Many market towns had one                                     |
| International fair cycle                           | A cycle through the year   | One or two fair regions per continent at a time               |
| Staple town                                        | Permanent                  | One per river stretch or export trade                         |
| Entrepot (great storage-and-reshipping port)       | Permanent                  | Only a handful in all of Europe and the Mediterranean c. 1300 |
| Merchant colony ( <i>Kontor</i> , <i>fondaco</i> ) | Permanent                  | One per foreign nation in each big port                       |

Source: 05 Trade nodes: markets, fairs, staples and entrepots.

**Ships, ports and seasons:**

| Item                                                  | Number                                                                                                                                                         |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cog (North Sea and Baltic, 12th–14th centuries)       | 30–200 t; the Bremen cog (1380) about 90–130 t; about 1 sailor per 10 t                                                                                        |
| Merchant great galley (Mediterranean, c. 1290s–1530s) | about 140–250 t of cargo; up to about 250 crew                                                                                                                 |
| Caravel (15th century)                                | 50–60 t                                                                                                                                                        |
| Head ports                                            | 15 for all of England; about 1 per 100–150 km (60–95 mi) of coast; several member ports under each; landing places in most creeks; one or two named anchorages |
| Mediterranean sailing season                          | roughly April to October                                                                                                                                       |
| North Sea and Baltic                                  | most winter sailing banned, roughly mid-November to late February                                                                                              |
| Indian Ocean                                          | one voyage each way per year, timed to the monsoon                                                                                                             |

Source: 05 Ship types, 05 Port ranks, 05 Sailing seasons.

**LATER ERA (1500S+)**

Ocean routes open (India by sea 1498; Seville's monopoly of the Americas trade 1503), and Atlantic ports grow fast. See 09 Ocean trade, ports and trading posts.

**MAP TIP**

For each region, write one line: "exports ..., imports ...". Then draw the route that carries each export to the nearest water or market. The biggest cities sit where networks meet: where land meets water, at confluences, at both ends of a pass or portage.

**Rural features****RULE OF THUMB**

At local scale almost every square kilometre has a human feature. Empty white space means marsh, moor or forest, and should be labelled as such.

**What is in a village:**

| Feature          | How common                                        | Where it sits                                          |
|------------------|---------------------------------------------------|--------------------------------------------------------|
| Parish church    | One per parish                                    | Slight rise, often at one end, next to the manor house |
| Manor house      | One per manor (a village could have two or three) | Next to the church; often moated                       |
| Mill             | About one per village (England c. 1300)           | On the stream with its leat; windmill on rising ground |
| Smithy           | Most villages                                     | Main street, green or road                             |
| Green, pond      | Many villages                                     | In the middle; pond at the lowest point                |
| Pound            | Most villages by the 16th century                 | Edge of the village or green                           |
| Tofts and crofts | All villages                                      | House plots along the streets, gardens behind          |

Source: [06 What is in a village.](#)

**Village territory:**

| Settlement              | People                   | Territory, lowland                     | Farthest regular fields                                |
|-------------------------|--------------------------|----------------------------------------|--------------------------------------------------------|
| Hamlet                  | 10–75                    | 1–5 km <sup>2</sup> (0.4–2 sq mi)      | under 1 km (0.6 mi)                                    |
| Small village           | 75–150                   | 2–7 km <sup>2</sup> (0.8–2.7 sq mi)    | 1–1.5 km (0.6–1 mi)                                    |
| Typical lowland village | 150–300                  | 5–15 km <sup>2</sup> (2–6 sq mi)       | 1.5–2.5 km (1–1.5 mi)                                  |
| Large village           | 300–500                  | 10–20 km <sup>2</sup> (4–8 sq mi)      | 2–3 km (1.2–1.9 mi)                                    |
| Upland parish           | 100–500, in many hamlets | 50–100+ km <sup>2</sup> (20–40+ sq mi) | Infields under 1 km away; summer pastures much farther |
| Agro-town               | 2,000–10,000             | ~50–250 km <sup>2</sup> (20–100 sq mi) | 5–10 km (3–6 mi)                                       |

Beyond 3–4 km (2–2.5 mi), people build an outlying farm or turn the land to pasture.

Source: [06 Village territory.](#)

**Rings of land use around a village:**

| Ring                   | Distance from the houses         | What is there                                            |
|------------------------|----------------------------------|----------------------------------------------------------|
| 1. House plots         | 0–100 m (0–330 ft)               | Tofts and crofts, gardens, orchards, dovecote, fishponds |
| 2. Arable              | 0.2–2 km (0.1–1.2 mi)            | Open fields, or the infield in the uplands               |
| Meadow                 | Wherever the flood plain is      | Hay; follows water, not distance                         |
| 3. Pasture and commons | 1–3 km (0.6–2 mi)                | Common pasture, heath, rough grazing                     |
| 4. Wood and waste      | Parish edge, 2–4 km (1.2–2.5 mi) | Coppice, wood-pasture, moor, marsh                       |
| 5. Summer pasture      | 5–50+ km (3–30+ mi)              | Shielings, alps                                          |

Source: [06 Rings of land use around a village.](#)

**Land budget of a settled lowland kingdom c. 1300:**

| Land                                      | Share                                                               |
|-------------------------------------------|---------------------------------------------------------------------|
| Ploughland (open fields, fallow included) | 35–40% (England ~36% in 1300: 11.5 million acres, 7.4 million sown) |
| Woodland and wood-pasture                 | 7–15%                                                               |
| Meadow                                    | about 5%                                                            |
| Deer parks                                | nearly 2%                                                           |
| Villages, gardens and roads               | a few per cent                                                      |
| Pasture, heath, moor, fen and marsh       | the rest, roughly a third to a half                                 |

Hill country ploughs only about 1–10% of its land. After a great plague, cut the ploughland by a quarter to a third.

Source: [06 A land budget for a whole realm](#) and [06 England, the best-measured case](#).

### How many features, England c. 1300 averages:

| Feature                             | One per ...              | Per 100 km <sup>2</sup> (39 sq mi)           | On a 20 × 20 km (12 × 12 mi) map |
|-------------------------------------|--------------------------|----------------------------------------------|----------------------------------|
| Parish church and village (lowland) | ~14 km <sup>2</sup>      | ~7                                           | ~25–30                           |
| Hamlets and farmsteads              | varies                   | 0–5 (open fields) to 20+ (dispersed uplands) | 0–100+                           |
| Mills                               | ~9–13 km <sup>2</sup>    | ~8–11                                        | ~30–45                           |
| Deer parks                          | ~40 km <sup>2</sup>      | ~2–3                                         | ~10                              |
| Moated sites                        | ~22 km <sup>2</sup>      | ~4–5                                         | ~15–20                           |
| Market places (many tiny)           | ~75 km <sup>2</sup>      | ~1–2                                         | ~5                               |
| Hundred meeting places              | ~150–200 km <sup>2</sup> | ~0.5–0.7                                     | ~2–3                             |
| Fishponds (recorded minimum)        | ~65 km <sup>2</sup>      | at least 1–2                                 | at least 6                       |

Source: [06 How many features: England c. 1300 averages](#).

### Footprints on a local map:

| Site                            | Size across                                     | At 1 cm = 1 km           |
|---------------------------------|-------------------------------------------------|--------------------------|
| Deer park                       | 0.5–1 km (0.3–0.6 mi); 40–80 ha (100–200 acres) | 5–10 mm                  |
| Great Cistercian abbey precinct | ~350–600 m (0.2–0.4 mi)                         | 3.5–6 mm                 |
| The largest castle (Malbork)    | ~450 m (0.3 mi)                                 | ~4.5 mm                  |
| Large baronial castle           | ~170 m (560 ft)                                 | ~2 mm                    |
| Friary in a town                | ~120–140 m (400–460 ft)                         | ~1.5 mm                  |
| Small motte-and-bailey castle   | roughly 100–150 m (330–490 ft) long (estimate)  | ~1–1.5 mm                |
| Moated manor site               | ~40–80 m (130–260 ft)                           | under 1 mm: use a symbol |

Source: [08 Footprints: how big to draw them on a local map](#).

#### LATER ERA (1500S+)

About 45% of England was enclosed by 1500, 47% by 1600 and 71% by 1700 (Wordie's estimate). Show hedged fields in sheep country. Polders behind dykes are medieval (Low Countries, from the 11th century), but whole lakes drained by windmills are new (Beemster, by 1612), and so are the drained English Fens. See [09 The countryside: enclosure, new crops and drainage](#).

#### MAP TIP

For each village, imagine a circle of about 2 km (1.3 mi) radius: open fields inside, a ribbon of meadow along the stream, woods and heath in the gaps between neighbouring circles. Draw parish boundaries through that outer land, along streams, ridges and wood edges.

## Industry siting

#### RULE OF THUMB

Industry goes to the heaviest thing it needs: the ore, the fuel or the water power, not the customer. Only the light, finished product travels.

| Resource                     | Where it goes on the map                                        | What settlement it creates                                                               |
|------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Silver, lead-silver          | Old mountains and hill country, up side valleys                 | Chartered mining town, a mint, mining hamlets around it                                  |
| Gold                         | Mountains; river sands                                          | Mining town with a royal mint                                                            |
| Copper                       | Hills and mountains                                             | Mining town with bare, smoke-killed land around it                                       |
| Iron                         | Wooded hills with fast streams                                  | Scattered bloomeries and forges; an iron-trading town where the valley meets a big river |
| Tin                          | Moorland valleys                                                | Tinners' camps; "coinage towns" at the edge of the moor                                  |
| Coal                         | Coalfields near a coast or navigable river                      | Pit hamlets; a coal port                                                                 |
| Salt from brine              | Lowland springs, with wood or peat nearby                       | A rich salt town; salt roads leading away                                                |
| Rock salt                    | Foothills and mountains                                         | Mining town with a royal official's castle                                               |
| Sea salt                     | Lagoons and sheltered bays with dry summers                     | Salt-pan villages and a loading port                                                     |
| Building stone               | Hill edges near rivers or the sea                               | Quarry hamlet; stone shipped by boat                                                     |
| Clay and brick               | River valleys, estuaries, stoneless lowlands                    | Brickfields outside towns; potters' villages                                             |
| Lime                         | Limestone edges, building sites, ports                          | Kilns, not towns                                                                         |
| Timber                       | Upland forests above floatable rivers                           | Rafting villages; a timber market in a river town                                        |
| Charcoal, glass, potash, tar | Deep forest                                                     | Temporary camps; glasshouses that move                                                   |
| Peat                         | Wet lowlands with few trees                                     | Turf commons; flooded pits that become lakes                                             |
| Wool                         | Downs, limestone hills, uplands, dry plateaus                   | Sheep villages, monastic farms, wool market towns                                        |
| Cloth                        | Towns; fulling mills in hill valleys                            | Cloth towns; weaving villages along mill streams                                         |
| Dye plants (woad, madder)    | Rich plains near cloth regions                                  | Woad towns and woad mills                                                                |
| Herring                      | Shallow seas and straits                                        | Seasonal beach markets and curing stations                                               |
| Cod (stockfish)              | Arctic coasts                                                   | Fishing stations, drying racks, one big trading port                                     |
| Ships                        | Sheltered estuaries near forest and rivers                      | Shipyards; walled state arsenals in great ports                                          |
| Wine                         | South-facing river slopes                                       | Wine villages; an export port downriver                                                  |
| Sugar cane                   | Irrigated, frost-free coastal plains (Cyprus, Sicily, Valencia) | Fortified estate with mill and boiling house; a loading port                             |

Fuel numbers: very roughly 50–150 kg of wood per 1 kg of bloomery iron; 150–200 kg of wood per 1 kg of forest glass;

a Wealden blast furnace needed the woods of about a 5 km (3 mi) radius.

Source: [07 The master table](#), [07 Iron](#), [07 Forest industries](#).

### In and around a town:

| Trade                                              | Where                                             | Why                            |
|----------------------------------------------------|---------------------------------------------------|--------------------------------|
| Drinking water, breweries                          | Upstream, or at wells and springs                 | Clean water                    |
| Grain and fulling mills                            | At weirs above and below town; on country streams | A fall of water                |
| Butchers                                           | Edge of town, near the river                      | Waste into the water           |
| Tanners, dyers, flax retting                       | Downstream and outside the walls                  | Smell and pollution            |
| Smiths, potters, brick, tile and lime kilns, glass | Outside the walls or at the edge                  | Fire and smoke                 |
| Tenter grounds, rope walks, timber yards           | Suburbs and open fields                           | Space                          |
| Smelters, charcoal, glasshouses                    | At the mine and in the forest                     | Fuel and ore too heavy to move |
| Shipyards                                          | Waterfront, at the edge of the port               | Shore and space                |

Source: [07 Where industry sits in and around a town.](#)

### Special towns:

| Town type          | Typical size                                        | Signature on the map                                                                                                              |
|--------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Mining town        | 1,000–5,000; boom towns 10,000–20,000; camps 20–300 | Up a valley; spoil heaps, shafts, mint, big church, little farmland                                                               |
| Salt town          | Small to middle-sized, rich for its size            | Brine spring or salt mine; salt roads radiating out                                                                               |
| Fishing village    | 50–500; seasonal stations swell in season           | Beach or cove; drying racks or smokehouses as labels ("Fish Racks"). Do not draw boats: a small boat marks the head of navigation |
| Cloth town         | 2,000–10,000; the Flemish giants 20,000–65,000      | Cloth hall, tenter grounds, fulling mills, dyers by the river                                                                     |
| Fair town          | Often only a small or middle town                   | Big open fairground, many inns, several roads meeting                                                                             |
| Iron or craft town | 1,000–10,000                                        | Forges and grinding mills along streams                                                                                           |
| Wine town          | Small; the export port can be large                 | Vineyards on slopes behind, river quay                                                                                            |
| Spa town           | Small                                               | Hot springs, bath houses, many inns                                                                                               |

Per kingdom, aim for 1–3 mining districts (each 2–7 mining towns plus a mint town), 1–3 salt sources, one cloth region, one or two fair towns and several ports. On a continent map, put mining symbols on 3–6 mountain areas.

Source: [07 Special town types and how to spot them](#) and [07 Mining districts on the map.](#)

#### LATER ERA (1500s+)

Blast furnaces, coal and boom towns: the Weald had about 50 furnaces by the later 1500s, and Joachimsthal reached about 20,000 people by 1534. See [07 What a mining town is like](#) and [09 Industry: furnaces, mines and mills.](#)

#### MAP TIP

For every industry, point to three things on the map: the resource, the fuel and the way out (a river, a coast or a road to one). If one is missing, move the industry or add what is missing. Every realm needs salt, or a route that brings it in.

## Religious sites siting

#### RULE OF THUMB

Every village has a church. Every town has a hospital. Every big town has friars. Every region has a few monasteries. Every kingdom has a handful of cathedrals and one or two universities.

**How many, per area (England):**

| Feature               | Number in England              | One per ...                            | Per 1,000 km <sup>2</sup> (386 sq mi) |
|-----------------------|--------------------------------|----------------------------------------|---------------------------------------|
| Parish church         | ~9,500 (c. 1300)               | ~14 km <sup>2</sup> (5.4 sq mi)        | ~73                                   |
| Religious house       | ~900 (1530s)                   | ~145 km <sup>2</sup> (56 sq mi)        | ~7                                    |
| Hospital or almshouse | 800–1,100 founded c. 1080–1540 | ~120–160 km <sup>2</sup> (46–62 sq mi) | ~6–8                                  |
| Leper house           | ~300                           | ~430 km <sup>2</sup> (166 sq mi)       | ~2                                    |
| Friary                | 183 (1530s)                    | ~700 km <sup>2</sup> (270 sq mi)       | ~1.4                                  |
| Cathedral             | 17 (plus 4 in Wales)           | ~7,600 km <sup>2</sup> (2,950 sq mi)   | 0.13                                  |
| University            | 2                              | ~65,000 km <sup>2</sup> (25,000 sq mi) | 0.015                                 |

Source: 08 [How many of each, per area.](#)

**Where each one goes:**

| Feature                        | Where                                                                                        | How many                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Parish church                  | Highest dry spot in the village, next to the manor house; in towns near the market and gates | One per parish; 7–10 km (4–6 mi) apart in hills                                                                                     |
| Chapel of ease                 | Hamlets and distant villages of big upland parishes; new clearings                           | 2–4 per upland parish                                                                                                               |
| Cathedral                      | Main cities (English style) or many small towns (Italian style), with a walled close         | English style 1 per 5,000–10,000 km <sup>2</sup> (2,000–4,000 sq mi); Italian style 1 per 500–2,000 km <sup>2</sup> (190–770 sq mi) |
| Archbishop                     | Oldest or richest city, often not the capital                                                | 1 per kingdom of about 50,000 km <sup>2</sup>                                                                                       |
| Benedictine abbey              | In or beside old towns, often a "monastery town" at its gate                                 | Part of the 1 per 150 km <sup>2</sup> (60 sq mi)                                                                                    |
| Cistercian or Carthusian house | Empty, well-watered valleys, with no village at the gate                                     | Same                                                                                                                                |
| Augustinian canons             | Near or in towns, at hospitals, shrines and castles                                          | Same                                                                                                                                |
| Nunnery                        | Quiet rural spots                                                                            | Same                                                                                                                                |
| Friary                         | Inside towns of about 3,000–5,000 and up, near the walls or a gate                           | 4–5 in the biggest city                                                                                                             |
| Grange (monastery farm)        | Most within about 25 km (15 mi) of the abbey, a few much farther                             | 5–40 per big abbey                                                                                                                  |
| Military-order commandery      | A manor farm with a chapel in good farmland                                                  | One per inland county                                                                                                               |
| Hospital or almshouse          | Town gates, bridges, main roads                                                              | One per market town; 2–5 per walled town                                                                                            |
| Leper house                    | 1–2 km (about 1 mi) outside the town on a main road, often near a stream                     | One per town of 2,000+; big cities one per main road                                                                                |
| Pilgrim hospice                | Every mountain pass and dangerous crossing                                                   | Pilgrim stops about 20 km (12 mi) apart                                                                                             |
| Great shrine                   | Its own town on existing roads, with inns and hospitals                                      | 1–2 per continent; 1–3 per kingdom                                                                                                  |
| University                     | Capital or oldest cathedral city; avoid towns under about 3,000                              | 0–2 per kingdom of 2–5 million                                                                                                      |
| Gallows                        | On a hill by the main road outside each town; at the edge of a great lord's land             | One per town and great lordship                                                                                                     |
| Moot (assembly) site           | A mound, stone or ford roughly central to its district, often on an old road                 | One per hundred                                                                                                                     |

| Feature                   | Where                                                 | How many                           |
|---------------------------|-------------------------------------------------------|------------------------------------|
| Jewish or Muslim cemetery | Outside a town gate                                   | One per community                  |
| Minority quarter          | Near the castle or market of a trading town of 5,000+ | One per such town, if you want one |

Source: 08 Parish churches and chapels, 08 Dioceses, cathedrals and archbishops, 08 Monasteries, friaries and military orders, 08 Pilgrimage, 08 Hospitals, leper houses and almshouses, 08 Universities and schools, 08 Legal and

civic sites, 08 Burial grounds, 08 Minorities and other faiths, 04 Military-order houses.

**Friaries tell you how big a town is** (rough guide, France):

| Number of friaries         | Town size                                   |
|----------------------------|---------------------------------------------|
| 0                          | Small towns of 2,000–3,000 usually had none |
| 1                          | Roughly 5,000 (estimate)                    |
| 2                          | About 10,000                                |
| 4 (one of each main order) | 20,000–40,000                               |
| 5+                         | The great cities                            |

In England, many county towns of 3,000–5,000 had one or two.

Source: 08 Friaries tell you how big a town is.

**Checklist for an English-style kingdom of 50,000 km<sup>2</sup> (19,000 sq mi), about 1–2 million people:**

| Feature                                          | How many                                       |
|--------------------------------------------------|------------------------------------------------|
| Archbishop                                       | 1                                              |
| Cathedrals                                       | 5–10 (English style) or 25–100 (Italian style) |
| Parish churches                                  | ~3,500 in lowland; fewer in hills              |
| Religious houses                                 | 300–350                                        |
| Friaries                                         | 60–75                                          |
| Hospitals and almshouses                         | 300–400 over time                              |
| Leper houses                                     | 100–120                                        |
| University                                       | 0–1                                            |
| Great shrine                                     | 1–3                                            |
| Hillforts and barrows (if settled in prehistory) | Hundreds                                       |

Source: 08 Drawing it all: symbols, density and scale.

#### LATER ERA (1500s+)

Protestant lands close their monasteries (England and Wales 1536–41) and end pilgrimage: draw ruined abbeys and country houses called "Abbey" on one side of the faith divide, working monasteries and Jesuit colleges on the other. Show each faith as a soft colour tint, never as a new line. See 09 Religion: the Reformation on the map.

#### MAP TIP

On a kingdom map show only cathedrals, great abbeys, great shrines and universities. Save parish churches, friaries, hospitals and gallows for local maps.

## City sizes by era

### RULE OF THUMB

In any period only about 4–8 cities in Europe, North Africa and the Middle East had 100,000+ people. Nearly all were capitals or great ports on the sea or a big river.

| Year    | Cities of 10k+                                     | Cities of 50k+ | Cities of 100k+ |
|---------|----------------------------------------------------|----------------|-----------------|
| c. 1000 | ~130 (Europe 81, Middle East and North Africa 48)  | ~11            | 4               |
| c. 1300 | ~230 (Europe 172, Middle East and North Africa 60) | ~22            | 8               |
| c. 1500 | ~250 (Europe 195, Middle East and North Africa 52) | ~31            | 8               |

Source: [02 Counting the cities](#).

**The biggest cities** (thousands of people; central estimate in brackets where ranges are wide):

| Rank | c. 1000                                             | c. 1300                                                           | c. 1500                                               |
|------|-----------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|
| 1    | Constantinople 150–400 (300)                        | Cairo 220–600 (220)                                               | Cairo 150–400 (180)                                   |
| 2    | Baghdad 125–500 (300), declining                    | Paris 75–270 (250); ~200 is the most common figure                | Istanbul 100–280 (280); a 1478 count suggests ~80–100 |
| 3    | Cairo-Fustat 135–200                                | Granada 50–150 (150)                                              | Paris 95–250 (200)                                    |
| 4    | Córdoba 100–450 (100)                               | Venice 90–120                                                     | Naples 100–150 (125)                                  |
| 5    | Seville 70–90                                       | Milan 100–150                                                     | Edirne ~100–127                                       |
| 6    | Tinnis and Damietta ~80 each                        | Genoa 85–100                                                      | Venice 100–150                                        |
| 7    | Palermo 60–75                                       | Constantinople ~100                                               | Milan ~100                                            |
| 8    | Alexandria ~60                                      | Florence 95–120                                                   | Tunis ~65–100                                         |
| 9    | Kairouan, Salerno ~50                               | Baghdad ~95 (after the 1258 sack)                                 | Aleppo 65–80                                          |
| 10   | Venice 17–45                                        | Seville ~90                                                       | Granada ~70; Valencia ~75                             |
| 11   | Antioch ~45; Thessaloniki, Regensburg, Damascus ~40 | London 60–100                                                     | Lisbon, Palermo, Fez, Damascus 55–65                  |
| 12   | Rome 30–35; Kiev 20–45                              | Fez, Marrakesh, Tunis, Naples, Aleppo, Córdoba 55–70; Ghent 40–65 | Genoa, Rome, Florence, Bursa ~55–60; London ~50       |

Source: [02 The biggest cities](#).

**A middle-sized kingdom: England in 1377** (after the plague; before it, all were about 1.5–2 times larger):

| Place               | People                   |
|---------------------|--------------------------|
| London              | ~35,000                  |
| York                | ~11,000                  |
| Bristol             | ~9,500                   |
| Coventry            | ~7,000                   |
| Norwich             | ~6,000                   |
| About 10 more towns | roughly 3,500–5,500 each |

Source: 02 The settlement ladder.

### Beyond Europe:

| City                             | Date                | People                                                                 |
|----------------------------------|---------------------|------------------------------------------------------------------------|
| Kaifeng (Northern Song capital)  | c. 1100             | 600,000 to over 1 million (estimates vary)                             |
| Hangzhou (Southern Song capital) | 1270s               | about 186,000 families in a 1270 count; perhaps over 1 million by 1276 |
| Greater Angkor                   | 13th century        | 700,000–900,000 over about 3,000 km <sup>2</sup> (1,160 sq mi)         |
| Tenochtitlan                     | 1519                | 200,000–250,000 (estimates run to 400,000) on 8–13.5 km <sup>2</sup>   |
| Kilwa Kisiwani                   | 14th century        | at least 10,000, perhaps 20,000                                        |
| Great Zimbabwe                   | 13th–15th centuries | under 10,000 (older estimates 18,000–20,000)                           |

Source: 02 Cities beyond Europe.

#### MAP TIP

On a continent map of a "medieval Europe" world, mark only 3–8 great cities of 50,000+, almost all on coasts, at river mouths or on big navigable rivers, each with a clear reason to be great: a court, a strait, a delta or an export trade.

## What exists by which date

#### RULE OF THUMB

Pick a date for the map. Old things may survive (layers are good), but nothing may appear before its date.

| Feature                                       | First appears                                                      | Common by                                                                |
|-----------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------|
| Watermill                                     | Roman times; 5,624–6,000+ in England in 1086                       | About one per village in England c. 1300 (~10,000–15,000 mills)          |
| Tide mill                                     | Nendrum c. 619; Ebbsfleet 691–692                                  | Estuaries and sheltered coasts                                           |
| Stone keep                                    | c. 950 in France; England from the 1070s                           | Important castles by 1300                                                |
| Motte-and-bailey castle                       | c. 1000 (Loire, Normandy); England after 1066                      | 11th–12th centuries; many abandoned by 1300                              |
| Three-field system                            | —                                                                  | Spread in the 12th–13th centuries                                        |
| Stone bridges                                 | Boom from the 12th century: Regensburg 1135–1146; London 1176–1209 | Most important English lowland crossings bridged by the late Middle Ages |
| Windmill (post mill)                          | 1185, Weedley (Yorkshire); tower mills by c. 1300                  | 13th–14th centuries                                                      |
| Fulling mill                                  | Normandy c. 1086; England 1185                                     | Cloth valleys, 14th–15th centuries                                       |
| Gotthard pass as a trade route                | First Schöllenen bridge, c. 1220                                   | —                                                                        |
| Murage (tolls for town walls)                 | 1220 (Shrewsbury)                                                  | About 1,470 grants recorded                                              |
| Friaries                                      | 1210s in Italy; England 1221 (Dominicans) and 1224 (Franciscans)   | Late 13th century                                                        |
| Bastides (planned towns of south-west France) | 1222 (Cordes)                                                      | Nearly 700 by 1372                                                       |
| Pontage (bridge tolls)                        | 1228                                                               | About 370 grants to the 1440s                                            |
| Concentric castle                             | Byzantine 11th century; crusader 1160s                             | Wales 1268–1295                                                          |

| Feature                                            | First appears                                                | Common by                                                                                            |
|----------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Moated manor sites                                 | —                                                            | Most dug c. 1250–1350                                                                                |
| Universities                                       | Bologna (traditionally 1088)                                 | About 20 c. 1300; 60–80 by 1500                                                                      |
| Paper mill                                         | Xàtiva (Spain) by the 12th century; Fabriano (Italy) by 1276 | —                                                                                                    |
| Gunpowder cannon                                   | First European picture, 1326                                 | Decisive in sieges by the mid-15th century                                                           |
| Tower houses                                       | 14th century                                                 | 14th–17th centuries                                                                                  |
| Polder (drained land behind a dyke)                | Low Countries, 11th century; Rijnland water board 1255       | Sea banks and reclaimed marsh spread in the High Middle Ages (Low Countries, the Fens, Romney Marsh) |
| Navigable canal                                    | Naviglio Grande (Milan): begun 1177, fully navigable 1272    | Short canals in Lombardy and the Low Countries; long canals mostly after 1500                        |
| Pound lock                                         | China by 984; Vreeswijk 1373 or 1385; Damme 1396             | 18 on a Milanese canal, 1452–58                                                                      |
| Canal over a watershed                             | Stecknitz Canal, 1391–98 (mostly flash locks)                | —                                                                                                    |
| Drainage windmill                                  | 1408 (Holland)                                               | —                                                                                                    |
| Hopped beer                                        | North German towns                                           | Low Countries and England, 14th–15th centuries                                                       |
| Sound Dues (Danish strait toll)                    | 1429                                                         | —                                                                                                    |
| Blast furnace                                      | Lapphyttan (Sweden), c. 1150–1350; England c. 1490           | Later era (see below)                                                                                |
| Relay post (Taxis)                                 | 1490                                                         | Later era (see below)                                                                                |
| Printing press                                     | —                                                            | 236 towns by 1500                                                                                    |
| Bastioned "star" fort ( <b>later era</b> )         | Italy, late 1400s                                            | Beyond Italy in the 1530s–1540s                                                                      |
| Exchange building ( <b>later era</b> )             | Antwerp 1531                                                 | London 1571; Amsterdam 1611                                                                          |
| Pound locks in Britain ( <b>later era</b> )        | Exeter Canal, 1564–66                                        | —                                                                                                    |
| Dutch fluyt cargo ship ( <b>later era</b> )        | 1595                                                         | —                                                                                                    |
| Lake drained as a polder ( <b>later era</b> )      | Beemster, by 1612                                            | Purmer 1622, Wormer 1626, Schermer 1635                                                              |
| Summit canal with pound locks ( <b>later era</b> ) | Briare Canal, 1604–42                                        | —                                                                                                    |
| Potato as a field crop ( <b>later era</b> )        | Spain about 1570                                             | Ireland early 1600s; most of Europe only 1700s–1800s                                                 |

Source: 06 How many mills, 06 Kinds of mill, 04 Types of fortification, 05 Bridges, 05 Mountain passes, 05 Mills, weirs and locks, 05 Tolls and customs, 07 Fulling mills and water power, 07 Metal and craft towns, 07 Breweries, 07 Iron smelting in the forest, 01 Planted "new towns" and bastides, 08 Monasteries, friaries and military orders, 08 Major universities, 29. Mixing eras without reason, 09 Timeline of the changes, 09 Bigger ships, 09 Exchanges and banking centres, 09 Printing towns, 09 Canals with pound locks, 09 Drainage and new land, 09 New crops, 06

Fields and farming systems, 06 What is in a village (moated sites), 06 Reclaimed land, 4. A huge city with no water transport (Naviglio Grande).

#### MAP TIP

Put the date in the title box ("The Kingdom of X, Year 1312"), then run down this table. Use three styles: working features, old features still in use (like an ancient road), and ruins in grey.

## Later-era changes (1500–1650)

### RULE OF THUMB

After 1500 the bottom of the map stays medieval and the top changes. Villages, fields and market towns look much the same; capitals, ports, fortresses and long-distance links are new or much bigger.

| Map feature                                  | Medieval (c. 1000–1500)                                                               | 1500–1650                                                                                     |
|----------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Village spacing; market towns; travel speeds | 1.5–4 km (1–2.5 mi); 9–16 km (6–10 mi); walker 25–35 km (15–22 mi) a day              | Unchanged                                                                                     |
| Capital                                      | Often a travelling court; 3–7 times the second city in strong kingdoms                | Fixed capital, 5–10 times the second city in some realms; draw it 2–4 times its medieval size |
| Biggest city in the West                     | Paris ~200,000                                                                        | Paris, Naples, London 200,000–400,000+; Istanbul 400,000–700,000                              |
| Town walls                                   | High and thin                                                                         | Low, thick, earth-backed, bastioned (rich or threatened towns only)                           |
| Castles                                      | Thousands; main military and lordly centre                                            | Ruins, palaces, prisons or manor houses; replaced by forts and country houses                 |
| Coastal defence                              | Walled ports, a few towers                                                            | Watchtower chains 3–12 km (2–7 mi) apart; artillery forts at anchorages                       |
| Field armies                                 | Usually 5,000–20,000                                                                  | Total forces of 50,000–150,000+ for great states (paper strength)                             |
| Ships                                        | Cogs of 30–200 tons; galleys carrying about 140–250 tons                              | Carracks and galleons of several hundred to 2,000 tons; Dutch fluyts                          |
| Messages                                     | Royal messengers 50–90 km (30–56 mi) a day                                            | Relay posts every 20–40 km (12–25 mi), up to ~150 km (95 mi) a day                            |
| Canals                                       | Short and flat, mostly flash locks; the first pound locks (Damme 1396, Milan 1452–58) | Pound locks spread; summit canals; passenger barge canals                                     |
| Books                                        | Hand-copied                                                                           | Printed in 1,300+ places by 1650                                                              |
| Countryside                                  | Open fields, commons, deserted villages                                               | More enclosure (47% of England by 1600); serf grain estates in the east                       |
| Wetlands                                     | Mostly left wet; sea banks and small polders on the coasts                            | Whole lakes drained as polders (Beemster by 1612); inland fens drained                        |
| Monasteries                                  | Everywhere                                                                            | Ruined or converted in Protestant lands; flourishing in Catholic lands                        |
| Iron and mining                              | Bloomeries; steady mining towns                                                       | Blast furnaces; sudden boom towns (Joachimsthal, Potosí)                                      |

Source: 09 What does not change and 09 Comparison table: medieval vs 1500 to 1650.

**Big cities after 1500** (thousands):

| City      | c. 1500                    | c. 1600                       | c. 1650                                      |
|-----------|----------------------------|-------------------------------|----------------------------------------------|
| Istanbul  | 100–280                    | 400–700                       | 400–700                                      |
| Paris     | 95–250 (often ~200)        | 220–300                       | 400–500                                      |
| Naples    | 100–150                    | 250–280                       | ~300 before the 1656 plague, then about half |
| London    | ~50                        | ~200                          | 375–400                                      |
| Amsterdam | 10–15                      | 50–60 (105 in the 1622 count) | 150–175                                      |
| Seville   | 45–70                      | 110–150                       | ~130 before the 1649 plague                  |
| Lisbon    | 55–65                      | 110–150                       | 130–170                                      |
| Madrid    | under 20 (not yet capital) | 75–95                         | 100–130                                      |

Source: 09 [The big cities](#).

**New fortifications:**

| Kind                         | Typical size                                 | Garrison                  | Where                                    |
|------------------------------|----------------------------------------------|---------------------------|------------------------------------------|
| Watchtower                   | 10–15 m tall                                 | a handful                 | Every few km along raided coasts         |
| Coastal artillery fort       | 30–100 m across                              | 10–30 in peace            | Harbour mouths, anchorages, river mouths |
| Small fort ( <i>sconce</i> ) | 100–250 m across, 4–5 bastions               | 50–300                    | River crossings, dykes, marsh roads      |
| Citadel                      | 400–700 m across, 4–6 bastions               | several hundred to 2,000+ | Edge of a big or rebellious city         |
| Fortress town                | 1–2.5 km (0.6–1.5 mi) across, 6–12+ bastions | 1,000–5,000 in war        | Frontiers, key river towns, big ports    |

Bastions sit roughly 200–350 m apart. When one ruler controls the whole land, castles are cut sharply: France's edict of 1626 ordered interior castles razed; Japan went from about 3,000 working castles to about 170 after 1615.

Source: 09 [Sizes and garrisons](#) and 09 [When a strong ruler cuts the number of castles](#).

**MAP TIP**

Do not redraw the countryside for a later setting. Keep your medieval villages and market towns, then change the top layer: enlarge the capital and one or two ocean ports, shrink one silted or blockaded port, add star forts only where they pay, post roads, a canal, a drained lake and faith areas (soft colour tints, never a new border line).

**Fantasy adjustments****RULE OF THUMB**

Fantasy does not cancel the rules of geography. It changes the numbers you feed into them. Change one input (water, food, defence, travel cost, threats or resources) and follow the chain of effects.

**How danger reshapes settlement:**

| Threat level                     | Real analogue                                           | Settlement pattern and defences                                                     |
|----------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| Low (wolves, bandits)            | England or France c. 1250                               | The normal map; working castles about 20–30 km (12–19 mi) apart; villages unwallled |
| Seasonal raids                   | Anglo-Scottish border, 14th–16th centuries              | Lone farms become tower houses; villages huddle                                     |
| Frequent raids or large monsters | Latium c. 920–1030; Wessex 870s–890s                    | Walled or hilltop villages of 300–1,000; refuge forts within a day's walk; beacons  |
| Constant danger                  | The "Great Wilderness" of the Baltic; crusader frontier | Only fortified towns and garrison castles; fields only near walls                   |

Source: [10 How danger reshapes settlement](#).

**Key fantasy numbers:**

| If your world has...         | What changes                                        | Numbers                                                                                                                                                  |
|------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monsters in the countryside  | Fewer lone farms; fields close to a refuge          | Fields within about one hour's walk, 4–5 km (2.5–3 mi), of the walls (heuristic)                                                                         |
| A "points of light" frontier | Isolated walled settlements                         | Each point about 50–80 km <sup>2</sup> of fields and 1,500–4,000 people                                                                                  |
| A monster-haunted frontier   | Refuge forts manned by the farms around them        | Refuges about a day's walk, 24–32 km (15–20 mi), apart; a frontier band of 30–100 km (20–60 mi)                                                          |
| Fertility magic              | Higher density, bigger towns                        | +50% yield: 40–60 per km <sup>2</sup> (100–155 per sq mi); about 2× yield: 50–75 per km <sup>2</sup> (130–190 per sq mi), like Flanders and Holland      |
| Healing magic                | Faster growth, same ceiling                         | 2–3% a year doubles the people in 23–35 years                                                                                                            |
| Flying mounts                | Fast messages and luxuries, not bulk                | One load about 100–120 kg; a cog's 30–200 tons is roughly 250–2,000 flying loads; a griffon post at 300 km (190 mi) a day is 3–6 times a royal messenger |
| Portals                      | Hub towns with customs, warehouses and forts        | Only shiploads of grain every day free inland cities from the water rule                                                                                 |
| Mage academies               | Placed like universities                            | 1–3 per kingdom, in cities of 10,000+                                                                                                                    |
| Dwarves                      | Mountain holds that buy food through gate towns     | Holds of a few thousand to perhaps 15,000; a hold of 10,000 needs farmland within 21–27 km (13–17 mi) on lowland or about 55 km (35 mi) on marginal land |
| Elves                        | Low-density forest realm, wide border wilderness    | 1–5 per km <sup>2</sup> (3–13 per sq mi) or less                                                                                                         |
| Halflings                    | Dense farm villages, few castles                    | 30–75 per km <sup>2</sup> (78–190 per sq mi); villages 1.5–4 km (1–2.5 mi) apart                                                                         |
| Steppe "hordes"              | Seasonal camps, raid trails, frontier markets       | Under 1–2 per km <sup>2</sup> (under 3–5 per sq mi)                                                                                                      |
| Dragons                      | Empty ring around the lair, tribute villages beyond | One cow a week needs a herd of about 250–500 cattle (rough example)                                                                                      |
| Undead or blights            | Like the Black Death                                | 30–60% dead; walled graveyards outside towns; quarantine islands                                                                                         |

Use no more than three or four of these in any one region. A fantasy danger frontier (30–100 km, 20–60 mi) is wider than a real march band (10–50 km, 6–30 mi), but both are drawn with the same hatching; truly wild land beyond it takes dense cross-hatching. For the symbols, see the fantasy group of the [master legend](#).

Source: [10 Dangerous wilderness and monsters](#), [10 Suggested symbols](#), [10 Magic, effect by effect](#), [10 Non-human peoples](#), [10 Gods](#), [undead and dangerous seas](#), [10 What](#)

[changes if: summary table](#), and for the griffon post [03 How big can a realm be](#).

**MAP TIP**

For each fantasy element, write one line: "X exists, so Y changes" ("griffon riders exist, so castles have roofed towers and every lord keeps an aerie"). Then draw the edges where peoples meet: gate towns, trading stones, frontier markets, tribute villages.

## Master legend and label hierarchy

### RULE OF THUMB

One symbol, one meaning. A filled star means a capital and nothing else. Borders are always dash-dot lines; routes are always solid or dotted lines; colour fills and tints are for areas only.

and chapters 02–11 now use it. If a map tip anywhere seems to use a symbol differently, this legend wins. Scale: C = continent, K = kingdom or region, L = local area. Many small works (kilns, racks, shipyards) take a word, not a picture: a new little picture easily looks like a symbol that already means something else.

The chapters suggest symbols for their own topics. This legend gives one symbol per feature for the whole guide,

### Settlements and power:

| Feature                                                                                    | Symbol                                                                                                                                                                                        | Scale      |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Farmstead                                                                                  | Tiny dot                                                                                                                                                                                      | L          |
| Hamlet                                                                                     | 2–5 small house marks, no church                                                                                                                                                              | L          |
| Village                                                                                    | Tiny dot (K, notable villages only); house marks around a church (L)                                                                                                                          | K, L       |
| Village without a parish church (pagan, Sámi, steppe or other non-Christian peoples)       | Tiny dot (K); a cluster of hut marks with no church (L)                                                                                                                                       | K, L       |
| Walled hilltop village (Italian <i>castello</i> , or a refuge village in a dangerous land) | Village symbol on the hilltop (a tiny dot at K); a battlemented wall line around the house marks (L)                                                                                          | K, L       |
| Small market town                                                                          | Small open circle                                                                                                                                                                             | K          |
| Town                                                                                       | Circle with a ring; battlements on the ring if walled                                                                                                                                         | K          |
| Town with only a bank, ditch and gates                                                     | Circle with a plain ring, no battlements (K); at L, the town edge drawn as an earthwork (a line with short ticks on the ditch side) with gaps for the gates                                   | K, L       |
| City                                                                                       | Circle with a thick battlemented ring (a larger walled symbol)                                                                                                                                | C (dot), K |
| Great city                                                                                 | Large circle with a ring (C); large walled symbol (K)                                                                                                                                         | C, K       |
| Realm capital                                                                              | Filled star in place of the tier symbol, sized by tier; name in bold capitals                                                                                                                 | C, K       |
| Dispersed capital (Angkor type)                                                            | Filled star on a walled square core with large rectangular reservoirs ( <i>barays</i> ); around it a wide zone of tiny house-mound dots and small ponds among rice fields, with no outer wall | K, L       |
| Duchy or regional seat                                                                     | Tier symbol with a small pennant (flag) on top                                                                                                                                                | K          |
| County town                                                                                | Tier symbol with a central dot                                                                                                                                                                | K          |
| League member town (Hanse type)                                                            | Normal tier symbol; name underlined in the league's colour                                                                                                                                    | C, K       |
| Mint or coinage town                                                                       | No extra symbol: add "(mint)" or "(coinage town)" to the label                                                                                                                                | K, L       |
| Mining town                                                                                | Its tier symbol with crossed hammers beside it                                                                                                                                                | K, L       |
| Coronation or royal burial town                                                            | Small crown beside the tier symbol, labelled "(coronations)" or "(royal tombs)"                                                                                                               | K          |
| Royal residence or royal hunting lodge                                                     | Small crown on its own (royal places only)                                                                                                                                                    | K, L       |
| Lord's or ranger's lodge (not royal)                                                       | Small house mark labelled "Lodge"                                                                                                                                                             | L          |
| Planted colonial grid town (1500s+)                                                        | Ordinary tier symbol (K); a small square grid of blocks around a plaza, with church and council house on the plaza (L)                                                                        | K, L       |
| Town after a great plague                                                                  | Open green space inside the walls                                                                                                                                                             | L          |
| Nomad camp                                                                                 | 2–3 tiny <b>open</b> half-circles (round tents), never filled; the ruler's camp ( <i>ordu</i> ) as a large labelled ring of them                                                              | K, L       |
| Manor house                                                                                | Small open square; double outline if moated                                                                                                                                                   | L          |
| Moot (assembly) site                                                                       | Small lone tree, labelled "Moot" (or "Thing" for a Norse or Germanic assembly)                                                                                                                | L          |

| Feature                    | Symbol                                | Scale |
|----------------------------|---------------------------------------|-------|
| Border truce meeting place | Small lone tree, labelled "Truce Day" | L     |
| Gallows                    | Small "Π" on a hill                   | L     |

**Military:**

| Feature                                    | Symbol                                                                                                                                                       | Scale                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Great royal castle or fortress             | Large castle icon (three towers)                                                                                                                             | C (greatest only), K                         |
| Ordinary castle                            | Small castle icon (one tower with battlements)                                                                                                               | K, L                                         |
| Toll castle                                | Small castle icon, with a toll bar and "T" across the river or road                                                                                          | K, L                                         |
| Refuge fort or burh (no town inside)       | Small castle icon; a burh with a town inside uses the town symbol with a battlemented ring                                                                   | K, L                                         |
| Motte or small earthwork castle            | Small mound with a tower tick on top (grey if abandoned)                                                                                                     | L (K for notable or abandoned ones, in grey) |
| Tower house, peel or bastle                | Small solid square                                                                                                                                           | K (border zones), L                          |
| Citadel                                    | Small castle icon at the edge of the town ring                                                                                                               | K, L                                         |
| Fortified bridge                           | Bridge symbol with a small tower                                                                                                                             | L                                            |
| Fortified church                           | Church with battlements                                                                                                                                      | L                                            |
| Watchtower or coastal tower                | Small triangle                                                                                                                                               | K (chains), L                                |
| Beacon                                     | Flame dot                                                                                                                                                    | K, L                                         |
| Siege castle                               | Small dashed castle near its target                                                                                                                          | L (dated maps)                               |
| Army or siege camp                         | <b>Closed</b> dashed rectangle with rounded corners, labelled with its name or "camp"; grey for an old or Roman camp. A closed shape, never an open line     | K, L (dated maps)                            |
| Arsenal, naval base or royal shipyard      | Anchor plus a castle tower (C, K); at L, a walled yard with slipways next to a rope-walk                                                                     | C, K, L                                      |
| Battlefield                                | Crossed swords and the year                                                                                                                                  | K, L                                         |
| Star fort or fortress town (1500s+ only)   | Five- or six-pointed star <b>outline</b> or a star-edged town outline, never filled                                                                          | K, L                                         |
| Bastioned citadel (1500s+ only)            | Pentagon outline joined to the town                                                                                                                          | K, L                                         |
| Castle refitted with gun bastions (1500s+) | Castle icon inside a star outline                                                                                                                            | K, L                                         |
| Castle rebuilt as a palace (1500s+)        | Castle icon with gardens, labelled "palace"; a royal palace takes the small crown instead                                                                    | K, L                                         |
| Castle kept as a prison or mint            | Castle icon in the town, labelled with its new use ("prison", "mint")                                                                                        | K, L                                         |
| Coastal artillery fort (1500s+)            | Small round outline with rounded lobes (a rosette, like the plan of Deal Castle, 1539–40), never pointed; a fort with angled bastions takes the star outline | K, L                                         |

A walled town needs no military symbol of its own: use the town or city symbol with battlements. A fortified or moated manor house uses the manor-house square (double outline if moated). The coastal artillery fort is a rosette outline

(rounded bastions, like Deal Castle), never pointed; see [09 Coastal defence](#).

**Religious and charitable:**

| Feature                                            | Symbol                                                                                                                   | Scale                   |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Archbishop's seat                                  | Double cross beside the city symbol                                                                                      | C, K                    |
| Cathedral                                          | Church with two towers, or a mitre                                                                                       | K, L                    |
| Parish church                                      | Small church with a tower; spire for rich towns                                                                          | L                       |
| Chapel                                             | Small plain cross, no tower                                                                                              | L                       |
| Wayside cross                                      | Tiny cross on a base, set on the road or path                                                                            | L                       |
| Abbey or priory                                    | Church with a square cloister beside it                                                                                  | K (great abbeys),<br>L  |
| Friary                                             | Name only, or a small church inside the town outline                                                                     | L                       |
| Grange                                             | Small open square topped with a cross, or the label "Grange"                                                             | L                       |
| Military-order commandery                          | Small church with a shield (a farm commandery); a fortified commandery or an order's frontier castle takes a castle icon | K, L                    |
| Great shrine                                       | Scallop shell or reliquary, never a star                                                                                 | C, K                    |
| Hospital, almshouse, hospice                       | Small house with a cross; leper house labelled "Lazar House"                                                             | K (pass hospices),<br>L |
| Cemetery outside the town (Jewish, Muslim, plague) | Small walled plot outside a gate, with a name label (e.g. "Jewbury")                                                     | L                       |
| University                                         | Book, or "U" beside the city name                                                                                        | C, K                    |
| Mosque                                             | Dome with a crescent                                                                                                     | K (great mosques), L    |
| Synagogue                                          | Lamp, never a star                                                                                                       | L                       |

**Economy and countryside:**

| Feature                                                                                            | Symbol                                                                                                                 | Scale                  |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------|
| Head port                                                                                          | Large anchor with a quay; name in capitals                                                                             | C, K                   |
| Member port                                                                                        | Small anchor                                                                                                           | K                      |
| Anchorage                                                                                          | Anchor in open water, labelled "the Roads"                                                                             | K                      |
| Ordinary shipyard                                                                                  | The label "Shipyard" on the shore; no symbol                                                                           | L                      |
| Inn or caravanserai                                                                                | Small house on the road; label "Han" or "Caravanserai" in desert                                                       | K, L                   |
| Fair town                                                                                          | Tier symbol; fair months in the label ("Provins: May and September fairs")                                             | K                      |
| Mine or mining district                                                                            | Crossed hammers (or a pick)                                                                                            | C, K, L                |
| Salt pans                                                                                          | Checkerboard of small rectangles along the shore                                                                       | K, L                   |
| Quarry                                                                                             | Notch in a hillside, with a track to water                                                                             | K (famous ones), L     |
| Lime, brick or tile kilns; clay pits                                                               | Words only: "Limekiln", "Brick Kilns", "Clay Pits"                                                                     | L                      |
| Watermill                                                                                          | Wheel on the stream; leat as a thin blue line                                                                          | L                      |
| Water-powered works (forge, furnace, ore-crushing mill, fulling mill, sugar, paper or powder mill) | Watermill wheel on the stream, labelled with the trade ("Forge", "Furnace", "Walk Mill", "Paper Mill", "Powder Mills") | L (K for notable ones) |
| Hammer or furnace pond                                                                             | One blue pond with a straight dam at its lower end, labelled "Hammer Pond" or "Furnace Pond"                           | L                      |
| Tide mill                                                                                          | Mill wheel on a dam across a tidal creek                                                                               | L                      |
| Ship mill or bridge mill                                                                           | Small mill wheels in midstream, labelled "ship mills"; or mill wheels on the bridge symbol. Never boats                | L                      |

| Feature                                                | Symbol                                                                                                                             | Scale            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Windmill                                               | X of four sails on a dot or small mound                                                                                            | L                |
| Forest works (charcoal hearths, glasshouse, tar kilns) | A clearing in the wood symbol, labelled with the trade                                                                             | L                |
| Peat diggings (turbary)                                | Commons symbol labelled "Turf Fen"; flooded pits as blue lakes with straight, blocky edges                                         | L                |
| Tenter grounds, fair field, fish racks                 | Open ground labelled "Tenters", "Fair Field" or "Fish Racks"                                                                       | L                |
| Royal forest                                           | Green dash-dot boundary with a large italic name; villages and fields inside                                                       | K, L             |
| Deer park                                              | Rounded pale (fence) line with scattered trees inside                                                                              | L                |
| Wood                                                   | Tree symbols with a firm edge                                                                                                      | K, L             |
| Wood-pasture                                           | Scattered single trees on grass                                                                                                    | L                |
| Open fields                                            | Unhedged areas with blocks of hatching                                                                                             | L                |
| Enclosed fields                                        | Small irregular polygons                                                                                                           | L                |
| Tofts and crofts                                       | A row of narrow plot outlines along the street, each with a small house mark at the street end; crofts as long narrow plots behind | L                |
| Meadow                                                 | Pale green band along streams                                                                                                      | L                |
| Commons, heath, moor                                   | Tufts or stipple                                                                                                                   | L                |
| Upper limit of hill farming                            | No line of its own: the field symbols stop and the moor tufts begin. Push this edge downhill after a plague                        | L                |
| Fishponds                                              | Small rectangles in a chain along a stream                                                                                         | L                |
| Shielings (summer huts)                                | Cluster of tiny squares on high pasture                                                                                            | L                |
| Vineyards                                              | Hatched strips on south-facing slopes                                                                                              | K, L             |
| Dyke or sea bank                                       | Bold line with short ticks on the water side                                                                                       | K, L             |
| Polder (Low Countries from the 11th century)           | Grid of drains inside a dyke; villages along the dykes; drainage mills only from 1408                                              | K, L             |
| Rice paddies                                           | Fine grid of small fields along rivers and deltas, with villages on the dykes and on mounds among them                             | K, L             |
| Rice terraces                                          | Tight parallel lines that follow the contours                                                                                      | K, L             |
| Chinampas (lake gardens)                               | Narrow parallel green strips with thin blue canals between them, in the shallow edge of a lake                                     | K (as a band), L |
| Shifting cultivation                                   | Small villages far apart, with scattered clearings of different ages in the forest around them                                     | K, L             |
| Country house (1500s+)                                 | House inside a ring of park trees                                                                                                  | K, L             |
| Overseas trading post (1500s+)                         | Small fort beside a foreign port city, labelled with the company                                                                   | C                |

**Water supply and irrigation:**

| Feature                    | Symbol                                                                                                                                      | Scale             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Well                       | Blue "W" (the Ordnance Survey abbreviation)                                                                                                 | L                 |
| Spring                     | Blue "Spr" (the Ordnance Survey abbreviation)                                                                                               | L                 |
| Brine spring or hot spring | Small blue dot labelled "Brine Pit" or "Hot Spring"; on a kingdom map, add "(salt)" or "(baths)" to the town's label instead                | K (label only), L |
| Cistern                    | The word "Cistern" in blue; no picture                                                                                                      | L                 |
| Aqueduct                   | Thin blue line labelled "Aqueduct", drawn on a row of small arches where it crosses a valley                                                | K (great ones), L |
| Qanat                      | Straight line of tiny rings (shaft mouths with spoil heaps), much smaller than a market-town circle, ending in green gardens                | K, L              |
| Oasis                      | Green belt of gardens and small palm marks around the town or well                                                                          | K, L              |
| Irrigation channel         | Thin blue line drawn like a mill leat, thinner than a canal                                                                                 | L                 |
| Tank (reservoir)           | Curved dam line with water behind it and a fan of small fields below; Angkor-type <i>barays</i> as large straight-sided rectangles of water | K, L              |

**Routes and crossings (always solid or dotted):**

| Feature                                                    | Symbol                                                                                                                                             | Scale |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Royal highway                                              | Thick solid line                                                                                                                                   | K     |
| Regional road                                              | Medium solid line                                                                                                                                  | K     |
| Local road                                                 | Thin solid line                                                                                                                                    | L     |
| Hollow way (sunken lane)                                   | Two thin parallel solid lines                                                                                                                      | L     |
| Causeway                                                   | Solid road line on a bank across marsh, flood plain or shallow lake; where it joins an island city to the shore, leave gaps with bridges for boats | K, L  |
| Roman or older-empire road                                 | Straight thick solid line labelled "Street" or "(Roman)"; grey where abandoned                                                                     | K, L  |
| Steep or stepped section (roads without carts, Andes type) | Steps or small chevrons ">" on the road line                                                                                                       | K, L  |
| Track, bridle way, mule path                               | Thin dotted line                                                                                                                                   | K, L  |
| Drove road                                                 | Broad dotted line (two rows of dots)                                                                                                               | K, L  |
| Seasonal migration route (nomads, herders, Sámi)           | Dotted line with arrows showing the direction of each season's move; label the season                                                              | K, L  |
| Caravan route                                              | Dotted line with a larger dot at each stage                                                                                                        | C, K  |
| Pilgrim road                                               | The ordinary road line with small scallop marks along it                                                                                           | K     |
| Raid trail                                                 | Red dotted line with arrowheads pointing into the farmland, named (for example the Muravsky Trail, used by Crimean Tatar raids in the 1500s)       | C, K  |
| Sea lane                                                   | Dotted line at sea, with arrows where the wind sets the direction, and a season label                                                              | C, K  |
| Navigable river                                            | Thicker river line up to the head of navigation, with a small boat there                                                                           | C, K  |
| Portage                                                    | Short dotted line between two rivers, labelled with its length                                                                                     | K     |
| Bridge                                                     | Two short curved lines ")((" with the road running through; label "rope bridge" where it is one                                                    | K, L  |
| Ford                                                       | Short dashed line across the river                                                                                                                 | K, L  |
| Mountain pass                                              | Small saddle mark (a shallow "U" between two hill ticks) on the route, with the pass name and height; never ")((", which is a bridge               | K, L  |
| Ferry                                                      | Short dotted line across the river with "F"                                                                                                        | K, L  |
| Toll                                                       | One bar across the road or river with a "T"                                                                                                        | K, L  |

| Feature               | Symbol                                                                                                                                                                   | Scale |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Canal                 | Straight blue line. Medieval canals were short (Naviglio Grande, Milan, 1177–1272; Stecknitz Canal, 1391–98); long canals with many pound locks belong mostly after 1500 | K, L  |
| Lock                  | Two short bars close together across the river or canal                                                                                                                  | L     |
| Lighthouse            | Small tower with short rays; never a star                                                                                                                                | K, L  |
| Post station (1500s+) | Small horn on the road                                                                                                                                                   | K     |

### Boundaries (always the dash-dot family):

| Feature                    | Symbol                                                                                                      | Scale                            |
|----------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------|
| Realm border               | Thick dash-dot line, or a coloured band along the inside edge                                               | C, K                             |
| Duchy or great fief        | Medium dash-dot line                                                                                        | C, K                             |
| County                     | Thin dash-dot line                                                                                          | K                                |
| Hundred, parish, manor     | Thin dash-dot-dot line                                                                                      | K (hundreds, on a county map), L |
| Boundary stone             | The letters "BS" on the boundary line, as on Ordnance Survey maps (which also use "BP" for a boundary post) | L                                |
| Disputed border            | Two parallel dash-dot lines, or alternating colours                                                         | K                                |
| March or frontier zone     | Hatching, or a fading colour band 10–50 km (6–30 mi) wide                                                   | C, K                             |
| Enclave                    | Owner's colour with a thin outline                                                                          | K                                |
| Language or culture area   | Soft colour tint only, never a line                                                                         | C, K                             |
| Faith area (mainly 1500s+) | Soft colour tint only, never a new line                                                                     | C, K                             |

The march band is the frontier itself. A whole military zone, from the forward forts back to the fortress towns a day or two behind them, can be 30–80 km (20–50 mi) deep, so fortress towns may lie behind the hatched band. A fantasy

danger frontier is a wider band than a real march: 30–100 km (20–60 mi).

### Area colours and tints:

| Feature                                                               | How to colour it                                                                                                                                         | Scale |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Realm                                                                 | One pale flat colour per realm, so roads and towns stay readable                                                                                         | C, K  |
| Vassal duchy                                                          | A lighter shade of its king's colour                                                                                                                     | C, K  |
| Church land, palatinate, free city                                    | Its own colour; do not merge it into the kingdom around it                                                                                               | K     |
| League of towns                                                       | No area colour: underline the member towns' names in the league's colour                                                                                 | C, K  |
| Nomadic empire                                                        | Its colour on the core pastures; the tributary ring in a lighter tint of the same colour                                                                 | C     |
| Maritime empire                                                       | Its colour only on its islands and harbour towns; the land behind keeps the local ruler's colour                                                         | C, K  |
| Language, culture or faith areas                                      | A separate layer of soft tints in a hue family that no realm uses (for example, realms in warm colours and languages in cool ones), or a small inset map | C, K  |
| Frontier and wild land                                                | Hatching (march) or dense cross-hatching (fantasy wild land); never a plain tint                                                                         | C, K  |
| Drafting aids: density zones, hinterland circles, messenger-day lines | Pencil or a hidden layer only; remove before finishing                                                                                                   | —     |

**Antiquities and older layers:**

| Feature                                                | Symbol                                                                                         | Scale                        |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------|
| Any ruin                                               | The feature's own symbol, broken or grey, labelled "(ruin)"                                    | K (big ones), L              |
| Deserted village                                       | Grey church alone in a field, labelled "(lost village)"                                        | K (a few), L                 |
| Drowned village                                        | Grey church standing in the water, labelled "(lost village)"                                   | K (a few), L                 |
| Old border dyke or linear earthwork (Offa's Dyke type) | Line with short ticks on the ditch side, grey if no longer used, always labelled with its name | K, L                         |
| Hillfort                                               | Oval of rings or hachures on a hill                                                            | L (K for big or famous ones) |
| Barrow                                                 | Small filled mound; long barrow as a longer mound                                              | L                            |
| Standing stone                                         | Small upright tick                                                                             | L                            |
| Stone circle                                           | Small ring of dots                                                                             | L                            |
| Old or Roman camp                                      | Grey closed dashed rectangle with rounded corners, labelled "Camp"                             | L                            |

**Fantasy (only if your world has them):**

| Feature                          | Symbol                                                                                                             | Scale |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|-------|
| Wild, monster-held land          | Dense cross-hatching (the frontier in front of it takes the march hatching)                                        | C, K  |
| Monster lair                     | Small skull with the creature's name; leave an empty ring around it                                                | K, L  |
| Blighted or dead land            | Pale grey wash with small black dead-tree ticks, labelled ("the Blight"), with watchtower triangles along its edge | C, K  |
| Mage tower or weather tower      | Slim tower topped with a small open diamond (no battlements, no spire)                                             | K, L  |
| Academy of magic                 | Book beside the city name (the university symbol), labelled "Academy"                                              | C, K  |
| Ley line and nexus               | Straight chain of small open diamonds in one colour (for example violet); a larger open diamond at the nexus       | C, K  |
| Ward stones                      | Small solid diamonds spaced around the warded area                                                                 | L     |
| Magical spring or sacred well    | Small blue drop, with a green ring of gardens around the town                                                      | K, L  |
| Portal                           | Small spiral beside the town symbol                                                                                | C, K  |
| Aerie                            | Small pair of wings on a crag or on a castle icon                                                                  | K, L  |
| Dwarf hold gate                  | Small rounded arch (n) in the mountainside                                                                         | K, L  |
| Smelter                          | Small chimney with a curl of smoke                                                                                 | L     |
| Underground road                 | Thin brown dotted line between hold gates, labelled "(underground)"                                                | K     |
| Frontier market or trading stone | Small pair of scales, labelled with the market days or "Trading Stone"                                             | K, L  |
| Walled graveyard                 | Small rectangle with a solid wall line and tiny crosses inside, labelled "Graveyard"                               | L     |

Source: 02 Map symbols and labels, 02 Beyond Europe, 02 Change over time, 03 Symbols and line styles, 03 Drawing the unusual kinds, 03 Colour and labels, 04 Suggested symbols, 04 Frontier posts, 05 Symbols and labels, 05 Kinds of roads, 06 Suggested symbols, 07 Suggested symbols, 08 Suggested symbols, 08 Burial grounds, 09 What happened

to the old castles, 09 Planted colonial towns, 09 Symbols and labels, 10 Suggested symbols, 11 Symbols for the fixes in this chapter. Ordnance Survey abbreviations ("BS", "BP", "W", "Spr") are cited in chapters 03 and 11.

**Look-alike symbols: keep them apart.**

| Shape                       | Use it only for                                                                                             | Not for                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Star                        | Filled = realm capital; outline = star fort (1500s+)                                                        | Great city (large ringed circle), citadel (castle icon at the town edge), shrine (scallop), synagogue (lamp), lighthouse (tower with rays), portal (spiral) |
| Ring or dot in a circle     | Ring = town; central dot = county town                                                                      | Motte (mound with a tower tick), moot (lone tree), hinterland circles (pencil only)                                                                         |
| Small square                | Solid = tower house; open = manor house                                                                     | Farmstead (tiny dot); tofts (narrow plots with a house mark)                                                                                                |
| Small rectangles in a chain | Fishponds                                                                                                   | Hammer ponds (one pond, straight dam, labelled)                                                                                                             |
| Triangle                    | Watchtower                                                                                                  | Beacon (flame dot), barrow (filled mound), standing stone (tick)                                                                                            |
| Cross                       | Church or chapel                                                                                            | Windmill (X of sails on a dot)                                                                                                                              |
| Bars across a line          | One bar with "T" = toll; two bars = lock                                                                    | Bridge ("")                                                                                                                                                 |
| Small boat                  | Head of navigation                                                                                          | Ferry (dotted line with "F"), ship mills (wheels in midstream), fishing villages (labelled racks)                                                           |
| Dotted line                 | Routes and tracks                                                                                           | Borders and royal forests (dash-dot), pilgrim roads (shell marks), ley lines (diamonds)                                                                     |
| Dashed line                 | Fords; closed camp outlines                                                                                 | Duchy borders (dash-dot); sea lanes, drove roads, portages (dotted)                                                                                         |
| Line with ticks             | Dyke or sea bank (ticks on the water side); old earth-work (ticks on the ditch side, grey if old, labelled) | Highway (plain thick line)                                                                                                                                  |
| Soft colour tint            | Language, culture or faith areas                                                                            | Realms (pale flat fills), frontiers (hatching), density zones (pencil)                                                                                      |
| Crown                       | Royal places only: alone = residence or lodge; beside a town = coronation or burial town                    | Duchy seat (pennant), lord's lodge ("Lodge")                                                                                                                |
| Arch shape                  | Ń in a mountainside = dwarf hold gate                                                                       | Gallows ("Π", flat-topped, on a hill by a road)                                                                                                             |

### Label hierarchy:

| Feature                 | Lettering                                                        | Size               |
|-------------------------|------------------------------------------------------------------|--------------------|
| Realm                   | Widely spaced CAPITALS across the whole territory                | Largest on the map |
| Duchy or great fief     | Smaller italic capitals                                          | Below realm        |
| County                  | Lower case                                                       | Below duchy        |
| Great city              | CAPITALS                                                         | 13–14 pt           |
| City                    | Bold or small capitals                                           | 11–12 pt           |
| Town                    | Regular                                                          | 9–10 pt            |
| Small market town       | Regular                                                          | 8–9 pt             |
| Village                 | Regular; label only notable ones at kingdom scale                | 7–8 pt             |
| Hamlet                  | Italic, or none                                                  | 6–7 pt             |
| Rivers, lakes, seas     | Italic, traditionally blue, along a straight stretch of the line | By importance      |
| Marches and empty zones | Named in the area ("the West March", "moor", "fen")              | Area label         |

Rules: nothing under about 6 pt in print; 2–3 typefaces at most; upper case only for realms and the largest features; put point labels above and to the right of the symbol, or directly above; keep area labels inside the area, curved to its shape, with letter gaps no more than about 4 times the letter height; place labels last, biggest first.

**Older layers:** Ordnance Survey maps have long used separate typefaces for antiquities of different periods (pre-

Roman, Roman and medieval), so ruins read as ruins without extra symbols. Copy the idea: pick one extra typeface for your older layers, and use it only for them. Saxton's printed county atlas of 1579 shows the period look: towns and villages as small church-tower symbols.

Source: 30. Label clutter, 03 Colour and labels, 02 Map symbols and labels, 08 Church towers as landmarks.

**MAP TIP**

Print the legend beside your map and test it: cover the labels and ask a friend what each symbol means. Any symbol they read two ways needs changing.

## Numbers reconciled between chapters

Where chapters gave different values, this sheet uses the owner chapter's value. Most of the old disagreements are now fixed in the chapters; the last column says what the

other chapters say now (and what earlier drafts said), and the last two rows are small differences that remain.

| Topic                                        | Value used on this sheet                                                                                                                 | Owner chapter                       | What other chapters say                                                                                                     |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Cog capacity                                 | 30–200 t; Bremen cog (1380) about 90–130 t                                                                                               | 05 Ship types                       | 09 ("30–200 tons; a few were larger") and 10 now agree; earlier drafts said 100–300 t and 40–200 t                          |
| Merchant galley cargo                        | about 140–250 t                                                                                                                          | 05 Ship types                       | 09 agrees                                                                                                                   |
| Hundreds (English districts)                 | ~130–150 km <sup>2</sup> (50–60 sq mi) in 1086, with 812 traceable; ~150–200 km <sup>2</sup> (60–80 sq mi) by 1316, with about 630 named | 03 The units, top to bottom         | 06 gives the 1316 figure only                                                                                               |
| English arable c. 1300                       | 11.5 million acres including fallow (about 36% of England), 7.4 million sown                                                             | 06 England, the best-measured case  | 02 now agrees: about 2.4 acres (1.0 ha) of arable per person (an earlier draft said 10 million acres)                       |
| Peasant walk to market                       | 6–10 km (4–6 mi) one way                                                                                                                 | 05 Speed per day                    | 01 now agrees (earlier 4–7 mi)                                                                                              |
| Active castles, England and Wales after 1066 | one per 250–300 km <sup>2</sup> (about 95–115 sq mi); about 16–17 km (10–11 mi) apart                                                    | 04 Counts and densities             | 08 and 11 now agree (earlier drafts printed "100–115 sq mi" and a spacing of 15–18 km)                                      |
| Castle sites, England and Wales              | 1,761 (Gatehouse gazetteer)                                                                                                              | 04 Counts and densities             | 08 and 11 now say about 1,700–1,800 (1,761)                                                                                 |
| Castle spacing on a contested border         | 4–6 km (2.5–4 mi)                                                                                                                        | 04 Borderlands and conquered land   | 04's county table and 11 now agree (an earlier draft gave 4–5 km)                                                           |
| Swidden (shifting) farming density           | about 6 per km <sup>2</sup> (14 per sq mi) on average; 12–35 (30–90) in busier districts; at most about 55–56 (about 145)                | 02 By type of land                  | 01 now agrees (earlier drafts gave only 6 or only 10–35)                                                                    |
| Parish churches in hills                     | about 7–10 km (4–6 mi) apart                                                                                                             | 08 Parish churches and chapels      | 02 and 00 Step 14 now agree (00 earlier said 8–15 km)                                                                       |
| Good lowland density                         | 30–50 per km <sup>2</sup> (80–130 per sq mi)                                                                                             | 02 Population density               | 10 now agrees (it earlier converted it as 78–130 per sq mi)                                                                 |
| Hill and upland density                      | Hill country 8–15 per km <sup>2</sup> (21–39 per sq mi); high uplands, moors and mountains 2–5 (5–13)                                    | 02 By type of land                  | 00 Step 2 and 01's terrain table now agree (they earlier gave one "uplands and mountains" value of 2–10, or 5–25 per sq mi) |
| Inns on main roads                           | 15–30 km (10–19 mi)                                                                                                                      | 05 Inns, hospices and caravanserais | 01 and 11 now agree (they earlier converted it as 9–19 mi)                                                                  |
| Towns over 10,000 on water                   | Most, with inland exceptions: Coventry (~12,000), Siena (perhaps 50,000), Granada, silver boom towns                                     | 02 Feeding towns                    | 05 now points to 02's exceptions; 07 adds the boom towns                                                                    |
| Towns per city of 10,000+                    | about 10                                                                                                                                 | 02 How many of each kind            | 11 now agrees (earlier roughly 5–15)                                                                                        |
| Ghent in the 14th century                    | 40,000–65,000 (estimates vary widely)                                                                                                    | 02 The biggest cities               | 07 now agrees (earlier 55,000–70,000)                                                                                       |
| Market towns, in practice                    | 9–16 km (6–10 mi)                                                                                                                        | 01 How far apart                    | 02 and 10 now agree (earlier 10–16 km)                                                                                      |

| Topic                      | Value used on this sheet                                                                                            | Owner chapter                                 | What other chapters say                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Main road on a kingdom map | about 30 km per 1,000 km <sup>2</sup> (48 mi per 1,000 sq mi)                                                       | <a href="#">05 How much main road to draw</a> | 11 now agrees (about 190 red lines on the map; about 4,540 km of real road traced behind them; 48 mi per 1,000 sq mi) |
| Corsican watchtowers       | about 85–90 (a 1617 list counts 86)                                                                                 | <a href="#">04 Watchtowers</a>                | 01 and 09 now agree                                                                                                   |
| Religious houses, England  | ~900 in the 1530s: one per ~145 km <sup>2</sup> (56 sq mi); for drawing, about 1 per 150 km <sup>2</sup> (60 sq mi) | <a href="#">08 How many of each, per area</a> | 02 and 08 agree                                                                                                       |
| Joachimsthal in 1534       | about 20,000                                                                                                        | <a href="#">07 What a mining town is like</a> | 07 and 09 agree                                                                                                       |

**MAP TIP**

When two numbers disagree, use the range from the chapter linked in the "Owner" column, and pick a value inside it that suits your region.

## Sanity checklist: 15 questions

**MAP TIP**

Run this check when the map is finished. Every "no" is a place to look again before you add a single label.

| #  | Question                                                                                                                            | If the answer is "no"                                          | Read                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| 1  | Does every settlement touch fresh water (river, stream, spring, lake, coast, or a drawn well or cistern)?                           | Move it to water, or draw the engineering that brings water    | <a href="#">01 Fresh water</a>                          |
| 2  | Does every desert town have a river, oasis or qanat, with green gardens around it?                                                  | Add the water source and its green belt                        | <a href="#">2. A desert city with no oasis or river</a> |
| 3  | Does every town have villages and farmland within its radius (21–27 km, 13–17 mi, for a town of 10,000 on good land)?               | Draw the farmland ring, or shrink the town                     | <a href="#">02 Feeding towns</a>                        |
| 4  | Is every city over 10,000 on navigable water or a coast, or explained by a canal, irrigated plain, rich farmland or silver?         | Move it to water or make it smaller                            | <a href="#">4. A huge city with no water transport</a>  |
| 5  | Are there far more villages than towns and far more towns than cities (about 10 towns and 50–100 market towns per city of 10,000+)? | Rebalance the symbols                                          | <a href="#">02 How many of each kind</a>                |
| 6  | Is each realm's population within the range of a real realm of the same area?                                                       | Change the densities or the area                               | <a href="#">02 Realm populations and areas</a>          |
| 7  | Are settlements clustered on good land, rivers and coasts, and does every empty area have a reason you can name?                    | Add villages to unexplained holes; thin out crowds on bad land | <a href="#">01 Where people do not build</a>            |
| 8  | Is each capital in a rich, defensible core on navigable water, and sized for its kind of realm?                                     | Move or resize it                                              | <a href="#">Capitals checklist</a>                      |
| 9  | Does every border segment follow something (crest, river, forest, old district line, treaty), with wide marches across empty land?  | Bend it to a real feature, or widen it into a zone             | <a href="#">03 Borders and frontiers</a>                |
| 10 | Does every road end somewhere, bend to crossings, cross mountains only at passes, and do big rivers have only a few bridges?        | Redraw the road; replace extra bridges with fords and ferries  | <a href="#">05 Drawing the route network</a>            |
| 11 | Are there stops every 15–30 km (10–19 mi) on main roads, and water every 30–40 km (19–25 mi) on desert routes?                      | Add inns, villages, wells or caravanserais                     | <a href="#">05 Inns, hospices and caravanserais</a>     |
| 12 | Can you name what each castle controls, and are castles sparse in the core and dense on the borders?                                | Move it to a choke point or delete it                          | <a href="#">04 What fortifications are for</a>          |
| 13 | Does every region have food plus a speciality, every special town a visible reason, and every realm a source of salt?               | Add the resource, fuel and way out, or a trade route           | <a href="#">07 The basic rules</a>                      |

| #  | Question                                                                                                                                    | If the answer is "no"                                                | Read                                      |
|----|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------|
| 14 | Are there ruins and older layers, and is every feature possible by the date in your title box?                                              | Add old roads, hillforts and ruins; remove features that are too new | <a href="#">What exists by which date</a> |
| 15 | Does each fantasy element have an "X exists, so Y changes" line, and does every magical city still show where its water and food come from? | Add the consequence, or cut the element                              | 10 Consistency checklist                  |

For the order in which to use everything on this sheet, follow [OO Step by Step: From Blank Map to Living World](#).